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(Revision of IEEE Std 802.1AS™-2025)

Draft Standard for Local and metropolitan area networks—

Timing and Synchronization for Time-Sensitive Applications

Unapproved draft, prepared by the

Maintenance Task Group of IEEE 802.1

Sponsored by the

LAN/MAN Standards Committee
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The text proper of this draft begins with the title page (1). The cover pages (a), (b), (c) etc. are for 802.1 WG information, and will be removed prior to Sponsor Ballot.

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Editors' Foreword

This draft standard is a revision. The scope of changes to the base standard is thus strictly limited, as detailed in the PAR.

Information on participation in this project, and in the IEEE 802.1 Working Group can be found here.

Participation in 802.1 standards development

Comments on this draft are encouraged. **NOTE: All issues related to IEEE standards presentation style, formatting, spelling, etc. are routinely handled between the 802.1 Editor and the IEEE Staff Editors prior to publication, after balloting and the process of achieving agreement on the technical content of the standard is complete.** Readers are urged to devote their valuable time and energy only to comments that materially affect either the technical content of the document or the clarity of that technical content. Comments should not simply state what is wrong, but also what might be done to fix the problem.

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Project Authorization Request, Scope, Purpose, and Criteria for Standards Development (CSD)

The complete PAR, as approved by IEEE NesCom 5 June 2023, can be found at:

https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10637

The 'Scope of the Proposed changes' and the 'Need for the Project' specify the changes to be made by this revision (see below).

Scope of the Proposed changes:

This standard specifies protocols, procedures, and managed objects to support the synchronization requirements of time-sensitive network applications. These applications include audio, video, and time-sensitive control using IEEE 802 Local Area Networks and similar media. Synchronized time is maintained during normal operation and during network reconfiguration following the addition, removal, or failure of network components. This standard includes a profile of IEEE Std 1588™.

Need for the Project:

This revision project is needed in order to incorporate approved amendments and corrigenda, to incorporate technical and editorial corrections to existing functionality; to maintain consistency in the consolidated text; and restructure to improve readability and maintainability.

Criteria for Standards Development:

There is no CSD statement since this maintenance project is not intended to provide any new functionality

Notes from the Editor:

1) All changes in the draft that originated from IEEE Std 802.1ASds-2026 are shown in **green color.** Deleted text is not marked. The title of Clause 19 appears in **green color** because the entire clause was added based on ASds. Figures modified according to ASds have their figure titles shown in **green color.**
2) All changes in the draft that originated from IEEE Std 802.1ASed-2026 are shown in **violet color.** Deleted text is not marked. The title of Clause 20 appears in **violet color** because the entire clause was added based on ASed. Figures modified according to ASed have their figure titles shown in **violet color.**
3) The changes related to maintenance items 394, 396, and 397 are shown in magenta. Maintenance item 395 is not market in color, as text was not changed, clause 10.3.18 was moved to a new clause 10.2.14.

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Abstract: Protocols, procedures, and managed objects for the transport of timing over local area networks are defined in this standard. It includes the transport of synchronized time, the selection of the timing source (i.e., best timeTransmitter), and the indication of the occurrence and magnitude of timing impairments (i.e., phase and frequency discontinuities).

Keywords: best timeTransmitter, frequency offset, Grandmaster Clock, Grandmaster PTP Instance, IEEE 802.1AS™, phase offset, PTP End Instance, PTP Relay Instance, synchronization, syntonization, time-aware system

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Introduction

This introduction is not part of IEEE Std 802.1AS-2025, IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications.

The first edition of IEEE Std 802.1AS was published in 2011. A first corrigendum, IEEE Std 802.1AS-2011/Cor 1-2013, provided technical and editorial corrections. A second corrigendum, IEEE Std 802.1AS-2011/Cor 2-2015 provided additional technical and editorial corrections.

The second edition, IEEE Std 802.1AS-2020, added support for multiple gPTP domains, Common Mean Link Delay Service, external port configuration, and fine timing measurement for IEEE 802.11 transport. Backward compatibility with IEEE Std 802.1AS-2011 was maintained. A corrigendum, IEEE Std 802.1AS-2020/Cor 1-2021, provides technical and editorial corrections.

The third edition, IEEE Std 802.1AS-2025, is a roll-up of IEEE Std 802.1AS-2020 with the corrigendum IEEE Std 802.1AS-2020/Cor 1-2021 and amendments IEEE Std 802.1ASdr-2024, IEEE Std 802.1ASdn-2024, and IEEE Std 802.1ASdm-2024. Backward compatibility with IEEE Std 802.1AS-2011 and IEEE Std 802.1AS-2020 is maintained.

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IEEE Standard for Local and Metropolitan Area Networks—

Timing and Synchronization for Time-Sensitive Applications

1. Overview

1.1 Scope

This standard specifies protocols, procedures, and managed objects used to ensure that the synchronization requirements are met for time-sensitive applications, such as audio, video, and time-sensitive control, across networks, for example, IEEE 802 and similar media. This includes the maintenance of synchronized time during normal operation and following addition, removal, or failure of network components and network reconfiguration. It specifies the use of IEEE 1588™ specifications where applicable in the context of IEEE Std 802.1Q™.[6] Synchronization to an externally provided timing signal [e.g., a recognized timing standard such as Coordinated Universal Time (UTC) or International Atomic Time (TAI)] is not part of this standard but is not precluded.

1.2 Purpose

This standard enables systems to meet the respective jitter, wander, and time-synchronization requirements for time-sensitive applications, including those that involve multiple streams delivered to multiple end stations. To facilitate the widespread use of packet networks for these applications, synchronization information is one of the components needed at each network element where time-sensitive application data are mapped or demapped or a time-sensitive function is performed. This standard leverages the work of the IEEE 1588 Working Group by developing the additional specifications needed to address these requirements.

[6] Information on references can be found in Clause 2.

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2. Normative references

The following referenced documents are indispensable for the application of this standard (i.e., they must be understood and used; therefore, each referenced document is cited in text, and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

IEEE Std 754™-2008, IEEE Standard for Floating-Point Arithmetic.[7, 8]

IEEE Std 802®-2024, IEEE Standard for Local and Metropolitan Area Networks: Overview and Architecture.

IEEE Std 802.1AC™-2016, IEEE Standard for Local and metropolitan area networks—Media Access Control (MAC) Service Definition.

IEEE Std 802.1AX™-2020, IEEE Standard for Local and Metropolitan Area Networks—Link Aggregation.

IEEE Std 802.1Q™-2022, IEEE Standard for Local and Metropolitan Area Networks—Bridges and Bridged Networks.

IEEE Std 802.3™-2022, IEEE Standard for Ethernet.

IEEE Std 802.11™-2016, IEEE Standard for Information technology—Telecommunications and information exchange between systems—Local and metropolitan area networks—Specific requirements, Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications.

IEEE Std 1588™-2019, IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems.

IEEE Std 1588e™-2024, IEEE Standard for Precision Clock Synchronization Protocol for Networked Measurement and Control Systems—Amendment 5: MIB and YANG Modules.

IERS Bulletin C (see https://www.iers.org/IERS/EN/Publications/Bulletins/bulletins.html).

IETF RFC 3410, Introduction and Applicability Statements for Internet Standard Management Framework, Case, J., R. Mundy, D. Partain, and B. Stewart, Dec. 2002.[9]

IETF RFC 3418 (STD 62), Management Information Base (MIB) for the Simple Network Management Protocol (SNMP), Presuhn, R., ed, Dec. 2002.

IETF RFC 7950, The YANG 1.1 Data Modeling Language, Bjorklund, M., ed, Aug. 2016.

ISO 80000-3:2006, Quantities and units — Part 3: Space and time.[10]

[7] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://standards.ieee.org).

[8] The IEEE standards or products referred to in Clause 2 are trademarks owned by The Institute of Electrical and Electronics Engineers, Incorporated.

[9] IETF RFCs are available from the Internet Engineering Task Force (https://www.ietf.org/).

[10] ISO publications are available from the International Organization for Standardization (https://www.iso.org/) and the American National Standards Institute (https://www.ansi.org/).

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ITU-T Recommendation G.984.3 (Nov. 2009), Amendment 2, Gigabit-capable Passive Optical Networks (G-PON): Transmission convergence layer specification—Time-of-day distribution and maintenance updates and clarifications.[11]

ITU-T Recommendation G.9960 (Nov. 2018), Unified high-speed wire-line based home networking transceivers—System architecture and physical layer specification [with ITU-T G.9961, commonly referred to as "ITU-T G.hn"].

ITU-T Recommendation G.9961 (Nov. 2018), Unified high-speed wireline-based home networking transceivers—Data link layer specification [with ITU-T G.9960, commonly referred to as "ITU-T G.hn"].

MoCA® MAC/PHY Specification v2.0, MoCA-M/P-SPEC-V2.0-20100507, Multimedia over Coax Alliance (MoCA).[12]

[11] ITU-T publications are available from the International Telecommunications Union (https://www.itu.int/).

[12] MoCA specifications are available from the Multimedia over Coax Alliance (https://www.mocalliance.org/).

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3. Definitions

For the purposes of this standard, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[13]

3.1 accuracy: The mean of the time or frequency error between the clock under test and a reference clock over an ensemble of measurements (see IEEE Std 1588-2019).

3.2 Bridge: Either a MAC Bridge or a VLAN-aware Bridge, as specified in Clause 5 of IEEE Std 802.1Q-2022.

3.3 clock: A physical device that is capable of providing a measurement of the passage of time since a defined epoch.

3.4 default: The value of an implementation parameter or capability selection, in the absence of an overriding profile standard requirement or explicit configuration.

3.5 device: An entity implementing some functionality, e.g., a clock, a time-aware system, a port.

3.6 direct communication: A communication of IEEE 802.1AS information between two PTP Instances with no intervening PTP Instance.

3.7 end station: A device attached to a local area network (LAN) or metropolitan area network (MAN) that acts as a source of, and/or destination for, traffic carried on the LAN or MAN.

3.8 epoch: The origin of a timescale.

3.9 event message: A message that is timestamped on egress from a PTP Instance and ingress to a PTP Instance.

NOTE—See 8.4.3.[14]

3.10 fractional frequency offset: The fractional offset, y, between a measured clock and a reference clock as defined by the following:

$$y = \frac{f_m - f_r}{f_r}$$

where f_m is the frequency of the measured clock and f_r is the frequency of the reference clock. The measurement units of f_m and f_r are the same.

3.11 frequency offset: The offset between a measured frequency and a reference frequency as defined by $f_m - f_r$, where f_m is the frequency of the measured clock and f_r is the frequency of the reference clock. The measurement units of f_m and f_r are the same.

3.12 full-duplex Ethernet: IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in point-to-point full-duplex.

3.13 general message: A message that is not timestamped.

[13] *IEEE Standards Dictionary Online* is available at https://dictionary.ieee.org/. An IEEE Account is required for access to the dictionary, and one can be created at no charge on the dictionary sign-in page.

[14] Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

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3.14 gPTP communication path: A segment of a generalized precision time protocol (gPTP) domain that enables direct communication between two PTP Instances.

NOTE—See 8.1.

3.15 grandmaster-capable PTP Instance: A PTP Instance that is capable of being a Grandmaster PTP Instance.

3.16 Grandmaster Clock: In the context of a single gPTP domain, the synchronized time of a PTP Instance that is the source of time to which all other PTP Instances in the domain are synchronized.

3.17 Grandmaster PTP Instance: A PTP Instance containing the Grandmaster Clock.

NOTE—None of the PTP Ports of a Grandmaster PTP Instance is in the TimeReceiverPort state.

3.18 local area network (LAN): A network of devices, whether indoors or outdoors, covering a limited geographic area, e.g., a building or campus.

3.19 half-duplex Ethernet: IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex.

3.20 local clock: A free-running clock, embedded in a respective entity (e.g., PTP Instance, CSN node), that provides a common time to that entity relative to an arbitrary epoch.

3.21 message timestamp point: A point within an event message serving as a reference point for when a timestamp is taken.

3.22 message type: The message type of a message is the name of the respective message, e.g., Sync, Announce, Timing Measurement frame.

3.23 precision: A measure of the deviation from the mean of the time or frequency error between the clock under test and a reference clock (see IEEE Std 1588-2019).

3.24 primary reference: A source of time and/or frequency that is traceable to international standards. *See also:* **traceability.**

3.25 profile standard: A standard specifying the capabilities, options, and parameter values of one or more base standards for use in a specific target environment or application.

3.26 PTP End Instance: A PTP Instance that has exactly one PTP Port.

3.27 PTP Instance: An instance of the IEEE 802.1AS protocol, operating in a single time-aware system within exactly one domain. A PTP Instance implements the portions of IEEE Std 802.1AS indicated as applicable to either a PTP Relay Instance or a PTP End Instance.

NOTE—As used in IEEE Std 802.1AS, the term *PTP Instance* refers to an IEEE 1588 PTP Instance that conforms to the requirements of IEEE Std 802.1AS.

3.28 PTP Link: Within a domain, a network segment between two PTP Ports using the peer-to-peer delay mechanism of IEEE Std 802.1AS. The peer-to-peer delay mechanism is designed to measure the propagation time over such a link.

NOTE—A PTP Link between PTP Ports of PTP Instances is also a gPTP communication path (see 3.14).

3.29 PTP Relay Instance: A PTP Instance that is capable of communicating synchronized time received on one PTP Port to other PTP Ports, using the IEEE 802.1AS protocol.

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NOTE—A PTP Relay Instance could, for example, be contained in a Bridge, a router, or a multi-port end station.

3.30 recognized timing standard: A recognized standard time source that is a source external to IEEE 1588 precision time protocol (PTP) and provides time that is traceable to the international standards laboratories maintaining clocks that form the basis for the International Atomic Time (TAI) and Coordinated Universal Time (UTC) timescales. Examples of these sources are National Institute of Standards and Technology (NIST) timeservers and global navigation satellite systems (GNSSs).

3.31 reference plane: The boundary between a PTP Port of a PTP Instance and the network physical medium. Timestamp events occur as frames cross this interface.

3.32 residence time: The duration of the time interval between the receipt of a time-synchronization event message by a PTP Instance and the sending of the next subsequent time-synchronization event message on another PTP Port of that PTP Instance. Residence time can be different for different PTP Ports. The term *residence time* applies only to the case where syncLocked is TRUE.

NOTE 1—See 10.2.5.15 for the definition of the variable syncLocked.

NOTE 2—If a PTP Port of a PTP Instance sends a time-synchronization event message without having received a time-synchronization event message, e.g., if syncLocked is FALSE or if sync receipt timeout occurs (see 10.7.3.1), the duration of the interval between the most recently received time-synchronization event message and the sent time-synchronization event message is mathematically equivalent to residence time; however, this interval is not normally called a *residence time*.

3.33 sdoId: An attribute that is the primary mechanism for providing isolation of PTP Instances operating under a PTP profile specified by one Qualified Standards Development Organization (QSDO) from PTP Instances operating under a PTP profile specified by a different QSDO (see 7.1.3 and 7.1.4 of IEEE Std 1588-2019).

3.34 stability (of a clock or clock signal): A measure of the variations over time of the frequency error (of the clock or clock signal). The frequency error typically varies with time due to aging and various environmental effects, e.g., temperature.

3.35 synchronized clocks: Absent relativistic effects, two clocks are synchronized to a specified uncertainty if they have the same epoch and their measurements of the time of a single event at an arbitrary time differ by no more than that uncertainty.

NOTE—See 8.2.2.

3.36 synchronized time: The time of an event relative to the Grandmaster Clock.

NOTE—If there is a change in the Grandmaster PTP Instance or its time base, the *synchronized time* can experience a phase and/or frequency step.

3.37 syntonized clocks: Absent relativistic effects, two clocks are syntonized to a specified uncertainty if the duration of a second is the same on both, which means the time as measured by each advances at the same rate within the specified uncertainty. The two clocks might or might not share the same epoch.

3.38 time-aware system: A device that contains one or more PTP Instances and/or PTP services (e.g., Common Mean Link Delay Service).

NOTE 1—See 11.2.16 for a description of the Common Mean Link Delay Service.

NOTE 2—A time-aware system can contain more than one PTP Instance in the same domain and/or different domains.

3.39 timescale: A measure of elapsed time since an epoch.

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3.40 timestamp measurement plane: The plane at which timestamps are captured. If the timestamp measurement plane is different from the reference plane, the timestamp is corrected for ingressLatency and/or egressLatency. *See:* **reference plane**.

NOTE—For timestamping on egress and ingress, see 8.4.3.

3.41 traceability: See 3.1.81 in IEEE Std 1588-2019.

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4. Acronyms and abbreviations

Ack	acknowledgment
ADEV	Allan deviation
ARB	arbitrary
BC	Boundary Clock
BTC	best timeTransmitter clock
BTCA	best timeTransmitter clock algorithm
CID	Company identification (allocated by the IEEE)
CMLDS	Common Mean Link Delay Service
CSN	coordinated shared network
CTC	channel time clock
DTSF	dependent time selection function
EPON	IEEE 802.3™ Ethernet Passive Optical Network, as specified in IEEE Std 802.3-2022
FDE	point-to-point full-duplex Ethernet
FTM	fine timing measurement
FTT	fault-tolerant timing (with time integrity)
G.hn	ITU-T G.9960 and ITU-T G.9961
GM	Grandmaster
GMT	Greenwich mean time
GNSS	global navigation satellite system
GPS	global positioning (satellite) system
gPTP	generalized precision time protocol
HDE	half-duplex Ethernet
IERS	International Earth Rotation and Reference Systems Service
IP	Internet Protocol
ISS	Internal Sublayer Service
ITSF	independent time selection function

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LAN	local area network
LCI	location configuration information
LLC	logical link control
MAC	media access control
MACsec	media access control security
MD	media-dependent
MIB	Management Information Base
MLME	IEEE 802.11 MAC layer management entity
MPCP	IEEE 802.3 multipoint control protocol
MPCPDU	IEEE 802.3 MPCP data unit
MVTSA	mid-value time selection algorithm
NETCONF	Network Configuration Protocol
NMS	Network Management System
NQ	not qualified
NTP	network time protocol[15]
OLT	IEEE 802.3 optical line terminal
ONU	IEEE 802.3 optical network unit
OSSP	organization-specific slow protocol
OUI	Organizationally Unique Identifier
P2P	peer-to-peer
PICS	Protocol Implementation Conformance Statement
POSIX®	portable operating system interface (see ISO/IEC 9945:2003 [B27][16])
PPS	pulse per second
PTP	IEEE 1588 precision time protocol
PTPDEV	PTP deviation
QSDO	Qualified Standards Development Organization

[15] Information available at https://www.ietf.org/rfc/rfc1305.txt.
[16] The numbers in brackets correspond to the numbers in the bibliography in Annex K.

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RTT	round-trip time
SI	international system of units
SMI	Structure of Management Information
SMIv2	Structure of Management Information version 2
SNMP	Simple Network Management Protocol
STA	station
TAI	International Atomic Time
TAGP	time agreement generation and preservation
TC	Transparent Clock
TDEV	time deviation
TDM	time division multiplexing
TDMA	time division multiple access
TLV	type, length, value
TM	timing measurement
TSF	time selection function
UCT	unconditional transfer
UML®	Unified Modeling Language™ [17]
UTC	Coordinated Universal Time
VLAN	virtual local area network
WLAN	wireless local area network

[17] UML® is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

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5. Conformance

This clause specifies the mandatory and optional capabilities provided by conformant implementations of this standard.

5.1 Requirements terminology

For consistency with existing IEEE and IEEE 802.1 standards, requirements placed upon conformant implementations of this standard are expressed using the following terminology:

a) ***shall*** is used for mandatory requirements.

b) ***may*** is used to describe implementation or administrative choices ("may" means "is permitted to," and hence, "may" and "may not" mean precisely the same thing).

c) ***should*** is used for recommended choices (the behaviors described by "should" and "should not" are both permissible but not equally desirable choices).

The Protocol Implementation Conformance Statement (PICS) proforma (see Annex A) reflects the occurrences of the words shall, may, and should within the standard. The words shall, may, and should, as used in Annex A itself, reflect the use of the PICs and not conformance to the standard.

The standard avoids needless repetition and apparent duplication of its formal requirements by using ***is***, ***is not***, ***are***, and ***are not*** for definitions and the logical consequences of conformant behavior. Behavior that is permitted but is neither always required nor directly controlled by an implementer or administrator, or whose conformance requirement is detailed elsewhere, is described by ***can***. Behavior that never occurs in a conformant implementation or system of conformant implementations is described by ***cannot***. The word ***allow*** is used as a replacement for the phrase "support the ability for," and the word ***capability*** means "can be configured to."

5.2 Protocol Implementation Conformance Statement (PICS)

The supplier of an implementation that is claimed to conform to this standard shall complete a copy of the PICS proforma provided in Annex A and shall provide the information necessary to identify both the supplier and the implementation.

5.3 Time-aware system requirements and options

5.3.1 Time-aware system requirements

An implementation of a time-aware system shall support at least one IEEE 1588 precision time protocol (PTP) Instance.

5.3.2 Time-aware system options

An implementation of a time-aware system may support a fault-tolerant timing with time integrity (FTT) entity, as specified in 20.2.

If YANG is supported with a remote management protocol and if an FTT entity (see 20.2) is supported, the YANG data model ieee802-dot1as-ftt in 17.6.3 shall be supported.

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5.4 PTP Instance requirements and options

5.4.1 PTP Instance requirements

An implementation of a PTP Instance shall:

a) Implement the generalized precision time protocol (gPTP) requirements specified in Clause 8.
b) Support the requirements for time-synchronization state machines (10.1.2, 10.2.1, 10.2.2, 10.2.3, 10.2.4, 10.2.5, and 10.2.6).
c) Support at least one PTP Port.
d) On each supported PTP Port, implement the PortSyncSyncReceive state machine (10.2.8).
e) Implement the ClockTimeReceiverSync state machine (10.2.13).
f) Implement the SiteSyncSync state machine (10.2.7).
g) Implement the state machines related to signaling gPTP capability (10.4).

NOTE 1—5.4.1 g) does not include the GptpCapableIntervalSetting state machine (10.4.4), despite its name containing “GptpCapable”.

NOTE 2—The GptpCapableTransmit and GptpCapableReceive state machines are required; however, they can be disabled by setting the managed object gptpCapableStateMachinesEnabled (see 14.10.56) to FALSE.

h) For receipt of all messages and for transmission of all messages except Announce (see 10.6.3), support the message requirements as specified in 10.5, 10.6, and 10.7.

NOTE 3—The support of Signaling messages is mandatory [5.4.1 h)] because the state machines related to the signaling of gPTP capability are mandatory [5.4.1 g)].

i) Support the performance requirements in B.1 and B.2.4.

5.4.2 PTP Instance options

An implementation of a PTP Instance should:

a) Support the performance requirements in B.2.2 and B.2.3.

An implementation of a PTP Instance may:

b) Support the following media-independent timeTransmitter capability on at least one PTP Port:
    1) Implement the PortSyncSyncSend state machine (10.2.12).
    2) Implement the PortAnnounceTransmit state machine (10.3.16).
    3) Implement the AnnounceIntervalSetting state machine (10.3.17).
    4) For transmit of the Announce message, support the message requirements as specified in 10.5, 10.6, and 10.7.
c) Support the following for Grandmaster PTP Instance capability:
    1) Support the media-independent timeTransmitter capability specified in item b) of 5.4.2.
    2) Support the requirements for a grandmaster-capable PTP Instance (10.1.3).
    3) Implement the ClockTimeTransmitterSyncSend state machine (10.2.9).
    4) Implement the ClockTimeTransmitterSyncOffset state machine (10.2.10).
    5) Implement the ClockTimeTransmitterReceive state machine (10.2.11).
d) Support more than one PTP Port as a PTP Relay Instance (5.4.4).

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e) Support more than one PTP Instance; such support allows for more than one domain (7.3.2).
f) Support the following external port configuration capability on at least one port:
    1) Implement specifications for externalPortConfigurationEnabled value of true (10.3.1).
    2) Implement the PortAnnounceInformationExt state machine (10.3.14).
    3) Implement the PortStateSettingExt state machine (10.3.15).
g) Implement the SyncIntervalSetting state machine (10.2.14).
h) Implement the GptpCapableIntervalSetting state machine.
i) Implement one or more of the application interfaces specified in Clause 9; A PTP Instance that claims to support application interfaces shall state which application interfaces are supported.
j) Support timing and synchronization management as specified in Clause 14.
k) Support the use of a remote management protocol. A PTP Instance that claims to support remote management shall:
    1) State which remote management protocol standard(s) or specification(s) are supported (see A.19).
    2) State which standard(s) or specification(s) for managed object definitions and encodings are supported for use by the remote management protocol (see A.19).
    3) If the Simple Network Management Protocol (SNMP) is supported as a remote management protocol, support the managed object definitions specified as Structure of Management Information version 2 (SMIv2) Management Information Base (MIB) modules in Clause 15.
    4) If YANG is supported with a remote management protocol, support the YANG data model ieee802-dot1as-gptp in Clause 17.
    5) If YANG is supported with a remote management protocol, and if hot standby is supported, support the YANG data model ieee802-dot1as-hs in Clause 17.
l) Implement both BTCA and external port configuration on domains other than domain 0; if both possibilities are implemented on domains other than domain 0, the default value of externalPortConfigurationEnabled shall be FALSE.
m) Implement hot standby as specified in Clause 18, 14.8, 9.3.3.4, 9.4.3.4, 9.5.3.4, and 9.6.2.6.
n) Implement the Drift_Tracking TLV as specified in 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.13, 11.2.14, and 11.4.4.

5.4.3 PTP Instance defaults and recommendations

This standard identifies specific capabilities and parameter values as defaults. If these defaults are not further qualified by reference to the use of other capabilities or by profile standards, the defaults are intended to facilitate deployment and to provide interoperability in target environments and applications with time-aware system implementations conformant to this standard (including its 2011 and 2020 editions). Mandatory requirements that this standard does not identify as defaults, or call out in some other way, are not subsetted or modified by profile standards.

Changes to capability support or parameter defaults arising from conformance to a profile standard should be identified in the PCS for that profile standard. Changes to defaults by pre-configuration of parameter values by the supplier of a protocol implementation should be identified by Additional Information (see A.3.2) in a completed PICS for this standard.

For domain 0, an implementation of a PTP Instance shall, by default, support the following best timeTransmitter clock algorithm (BTCA) requirements:

a) Have the BTCA as the default mode of operation, with externalPortConfigurationEnabled FALSE.
b) Implement the BTCA (10.3.1.1, 10.3.1.2, 10.3.2, 10.3.3, 10.3.4, 10.3.5, 10.3.6, 10.3.8, and 10.3.10).

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c) Implement specifications for an externalPortConfigurationEnabled value of FALSE (10.3.1).
d) Implement the PortAnnounceReceive state machine (10.3.11).
e) Implement the PortAnnounceInformation state machine (10.3.12).
f) Implement the PortStateSelection state machine (10.3.13).

For domains other than domain 0, an implementation of a PTP Instance shall, by default, support the following best timeTransmitter clock algorithm (BTCA) requirements:

g) Implement at least one of the possibilities for externalPortConfigurationEnabled (i.e., FALSE, meaning the BTCA is used, and TRUE, meaning external port configuration is used) on domains other than domain 0.
h) If BTCA is used then:
    1) Implement the BTCA (10.3.1.1, 10.3.1.2, 10.3.2, 10.3.3, 10.3.4, 10.3.5, 10.3.6, 10.3.8, and 10.3.10).
    2) Implement specifications for an externalPortConfigurationEnabled value of FALSE (10.3.1).
    3) Implement the PortAnnounceReceive state machine (10.3.11).
    4) Implement the PortAnnounceInformation state machine (10.3.12).
    5) Implement the PortStateSelection state machine (10.3.13).

5.4.4 PTP Relay Instance requirements

An implementation of a PTP Relay Instance shall:

a) Support more than one PTP Port.
b) Support the media-independent timeTransmitter capability specified in item b) of 5.4.2.

5.5 MAC-specific timing and synchronization methods for full-duplex point-to-point (FDE) IEEE 802.3 links

An implementation of a time-aware system with full-duplex point-to-point IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

a) Support full-duplex operation, as specified in 4.2 and Annex 4A of IEEE Std 802.3-2022.
b) Support the requirements as specified in Clause 11 [with the exception of requirements more specifically addressed in this subclause (5.5)].
c) Support two-step capability on receive as specified in 11.2.13.
d) Support two-step capability on transmit as specified in 11.2.14.
e) Provide the CMLDS if the time-aware system implements more than one domain, as specified in 11.2.16 and 11.2.17.

An implementation of a PTP Instance with full-duplex point-to-point IEEE Std 802.3-2022 MAC services to physical ports may:

f) Support asymmetry measurement mode as specified in 10.3.12, 10.3.13, 10.3.16, 11.2.13, 11.2.14, 11.2.18, and 14.10.46.
g) Support one-step capability on receive as specified in 11.2.13.
h) Support one-step capability on transmit as specified in 11.2.14.
i) Support the OneStepTxOperSetting state machine specified in 11.2.15.
j) Support propagation delay averaging, as specified in 11.2.18.3.4.

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k) Provide the CMLDS if the time-aware system implements only one domain, as specified in 11.2.16 and 11.2.17.

l) Implement the LinkDelayIntervalSetting state machine (11.2.20).

5.6 MAC-specific timing and synchronization methods for IEEE Std 802.11-2016

An implementation of a time-aware system with IEEE 802.11 MAC services to physical ports shall:

a) Support the requirements as specified in Clause 12.

b) Support at least one of:

1) The media-dependent timeTransmitter state machines (12.5.1), or

2) The media-dependent timeReceiver state machine (12.5.2).

An implementation of a PTP End Instance with IEEE 802.11 MAC services to physical ports shall:

c) Support at least one of TIMINGMSMT as specified in IEEE Std 802.11-2016 or FINETIMINGMSMT as specified in IEEE Std 802.11-2016.

An implementation of a PTP Relay Instance with IEEE 802.11 MAC services to physical ports shall:

d) Support the requirements of TIMINGMSMT as specified in IEEE Std 802.11-2016.

An implementation of a PTP Relay Instance with IEEE 802.11 MAC services to physical ports should:

e) Support FINETIMINGMSMT as specified in IEEE Std 802.11-2016.

NOTE—In order to maintain backward compatibility with existing TM-based PTP End Instances that support only timing measurement (TM), the PTP Relay Instance is required to support TM. PTP End Instances are allowed to support TM or fine timing measurement (FTM), or both to permit PTP End Instances compliant with this standard to implement only the FTM standard, which requires a PTP Relay Instance that supports FTM.

5.7 MAC-specific timing and synchronization methods for IEEE 802.3 EPON

An implementation of a time-aware system with IEEE 802.3 Ethernet Passive Optical Network (EPON) MAC services to physical ports shall:

a) Support the requirements as specified in IEEE Std 802.3-2022 for multipoint MAC Control (64.2 and 64.3 of IEEE Std 802.3-2022) and multipoint physical coding sublayer (PCS) and physical medium attachment (PMA) extensions (Clause 65 of IEEE Std 802.3-2022).

b) Support the requirements as specified in Clause 13.

5.8 MAC-specific timing and synchronization methods for coordinated shared network (CSN)

An implementation of a time-aware system with CSN MAC services to physical ports shall:

a) Support the requirements as specified in Clause 16.

b) Support at least one MoCA port (16.6.2) or ITU-T G.hn port (16.6.3).

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5.9 MAC-specific timing and synchronization methods for half-duplex IEEE 802.3 (HDE) links

An implementation of a time-aware system with half-duplex IEEE Std 802.3-2022 media access control (MAC) services to physical ports shall:

a) Support half-duplex operation, as specified in Clause 4 of IEEE Std 802.3-2022.

b) Support the requirements as specified in Clause 19.

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6. Conventions

6.1 General

This clause defines various conventions and notation used in the standard, i.e., naming conventions, service specification method and notation, and data type definitions.

6.2 Service specification method and notation

The method and notation for specifying service interfaces is described in Clause 7 of IEEE Std 802.1AC-2016.

6.3 Lexical form syntax

A lexical form refers to the following:

— A name
— A data type

The conventions illustrated in the following list regarding lexical forms are used in this standard:

a) Type names: e.g., ClockQuality (no word separation, initial letter of each word capitalized).
b) Enumeration members and global constants: e.g., ATOMIC_CLOCK (word separation underscored, all letters capitalized).
c) Fields within PTP messages, instances of structures, and variables: e.g., secondsField, clockQuality, clockIdentity (two-word field names at a minimum, no word separation, initial word not capitalized, initial letter capitalization on subsequent words).
d) Members of a structure: e.g., clockQuality.clockClass (no word separation, structure name followed by a period followed by the member name).
e) Data set names: e.g., defaultDS, parentDS, portDS, currentDS, timePropertiesDS (no word separation, initial word not capitalized, initial letter capitalization on subsequent words, end marked by the letters DS).
f) Data set members: e.g., defaultDS.clockQuality.clockClass (data set name followed by a period followed by a member name followed by a period followed by a member name).
g) PTP message names: e.g., Sync, Pdelay_Req (word separation underscored, initial letter of each word capitalized).

When a lexical form appears in text, as opposed to in a type, or a format definition, the form is to be interpreted as singular, plural, or possessive as appropriate to the context of the text.

6.4 Data types and on-the-wire formats

6.4.1 General

The data types specified for the various variables and message fields define logical properties that are necessary for correct operation of the protocol or interpretation of IEEE 1588 precision time protocol (PTP) or IEEE 802.11 message content.

NOTE—Implementations are free to use any internal representation of data types if the internal representation does not change the semantics of any quantity visible via communications using the IEEE 802.1AS protocol or in the specified operations of the protocol.

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6.4.2 Primitive data types specifications

All non-primitive data types are derived from the primitive types in Table 6-1. Signed integers are represented in twos complement form.

Table 6-1—Primitive data types

Data type	Definition
Boolean	TRUE or FALSE
EnumerationN	N-bit enumerated value
UIntegerN	N-bit unsigned integer
IntegerN	N-bit signed integer
Nibble	4-bit field not interpreted as a number
Octet	8-bit field not interpreted as a number
OctetN	N-octet field not interpreted as a number, with N > 1
Float64 (see NOTE)	IEEE 754 binary64 (64-bit double-precision floating-point format)
NOTE—The Float64 data type was called Double in the 2011 edition of this standard. The semantics of the data type has not changed.	

6.4.3 Derived data type specifications

6.4.3.1 ScaledNs

The ScaledNs type represents signed values of time and time interval in units of 2^{-16} ns.

typedef Integer96 ScaledNs;

For example: –2.5 ns is expressed as:

0xFFFF FFFF FFFF FFFD 8000

Positive or negative values of time or time interval outside the maximum range of this data type are encoded as the largest positive or negative value of the data type, respectively.

6.4.3.2 UScaledNs

The UScaledNs type represents unsigned values of time and time interval in units of 2^{-16} ns.

typedef UInteger96 UScaledNs;

For example: 2.5 ns is expressed as:

0x0000 0000 0000 0002 8000

Values of time or time interval greater than the maximum value of this data type are encoded as the largest positive value of the data type.

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6.4.3.3 TimeInterval

The TimeInterval type represents time intervals, in units of 2^{-16} ns.

typedef Integer64 TimeInterval;

For example: 2.5 ns is expressed as:

0x0000 0000 0002 8000

Positive or negative time intervals outside the maximum range of this data type are encoded as the largest positive and negative values of the data type, respectively.

6.4.3.4 Timestamp

The Timestamp type represents a positive time with respect to the epoch.

```
struct Timestamp
{
        UInteger48 seconds;
        UInteger32 nanoseconds;
};
```

The seconds member is the integer portion of the timestamp in units of seconds.

The nanoseconds member is the fractional portion of the timestamp in units of nanoseconds.

The nanoseconds member is always less than 10^9.

For example:

+2.000000001 seconds is represented by seconds = 0x0000 0000 0002 and nanoseconds= 0x0000 0001

6.4.3.5 ExtendedTimestamp

The ExtendedTimestamp type represents a positive time with respect to the epoch.

```
struct ExtendedTimestamp
{
        UInteger48 seconds;
        UInteger48 fractionalNanoseconds;
};
```

The seconds member is the integer portion of the timestamp in units of seconds.

The fractionalNanoseconds member is the fractional portion of the timestamp in units of 2^{-16} ns.

The fractionalNanoseconds member is always less than $(2^{16})(10^9)$.

For example:

+2.000000001 seconds is represented by seconds = 0x0000 0000 0002 and fractionalNanoseconds = 0x0000 0001 0000

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6.4.3.6 ClockIdentity

The ClockIdentity type identifies a PTP Instance.

typedef Octet8 ClockIdentity;

6.4.3.7 PortIdentity

The PortIdentity type identifies a port of a PTP Instance.

```
struct PortIdentity
{
        ClockIdentity clockIdentity;
        UInteger16 portNumber;
};
```

6.4.3.8 ClockQuality

The ClockQuality represents the quality of a clock.

```
struct ClockQuality
{
        UInteger8 clockClass;
        Enumeration8 clockAccuracy;
        UInteger16 offsetScaledLogVariance;
};
```

6.4.4 Protocol data unit (PDU) formats

6.4.4.1 General

The data types defined in 6.4.2 and 6.4.3 shall be mapped onto the wire according to the mapping rules for the respective medium, e.g., IEEE Std 802.3-2022 and IEEE Std 802.11-2016, and the terms of 6.4.4.

IEEE 802.1AS PDUs consist of the messages defined or referenced in Clause 10, Clause 11, Clause 12, and Clause 13, based on the data types defined in 6.4.2 and 6.4.3. The internal ordering of the fields of the IEEE 802.1AS PDUs is specified in 6.4.4.3 to 6.4.4.5.

6.4.4.2 Numbering of bits within an octet

Bits are numbered with the most significant bit being 7 and the least significant bit being 0.

NOTE—The numbering and ordering of bits within an octet of a PDU, described here, is independent of and unrelated to the order of transmission of the bits on the underlying physical layer.

6.4.4.3 Primitive data types

Numeric primitive data types defined in 6.4.2 shall be formatted with the most significant octet nearest to the beginning of the PDU followed in order by octets of decreasing significance.

The Boolean data type TRUE shall be formatted as a single bit equal to 1 and FALSE as a single bit equal to 0.

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Enumerations of whatever length shall be formatted as though the assigned values are unsigned integers of the same length, e.g., Enumeration16 shall be formatted as though the value had type UInteger16.

6.4.4.4 Arrays of primitive types

All arrays shall be formatted with the member having the lowest numerical index nearest to the beginning of the PDU followed by successively higher numbered members, without any padding. In octet arrays, the octet with the lowest numerical index is termed the most significant octet.

When a field containing more than one octet is used to represent a numeric value, the most significant octet shall be nearest to the beginning of the PDU, followed by successively less significant octets.

When a single octet contains multiple fields of primitive data types, the bit positions within the octet of each of the primitive types as defined in the message field specification shall be preserved. For example, the first field of the header of PTP messages is a single octet composed of two fields, one of type Nibble bit 4 through bit 7, and one of type Enumeration4 bit 0 through bit 3 (see 11.4.2 and 10.6.2).

6.4.4.5 Derived data types

Derived data types defined as structs shall be formatted with the first member of the struct nearest to the beginning of the PDU followed by each succeeding member, without any padding. Each member shall be formatted according to its data type.

Derived data types defined as typedefs shall be formatted according to its referenced data type.

6.5 State diagram conventions

State diagrams follow the conventions specified in IEEE Std 802.1Q-2022 Annex E.

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7. Time-synchronization model for a packet network

7.1 General

This clause provides a model for understanding the operation of the generalized precision time protocol (gPTP), which specifies the operation of time-aware systems on a packet network. Although this standard is based on the precision time protocol (PTP) described in IEEE Std 1588-2019 (and, indeed, is a proper profile of IEEE Std 1588-2019 in particular configurations), there are differences, which are summarized in 7.6.

NOTE—Although this standard has been written as a stand-alone document, it is useful to understand the IEEE 1588 architecture as described in Clause 6 of IEEE Std 1588-2019.

7.2 Architecture of a time-aware network

7.2.1 General

A time-aware network consists of a number of interconnected time-aware systems that support the gPTP defined within this standard. These time-aware systems can be any networking device, including, for example, Bridges, routers, and end stations. A set of time-aware systems that are interconnected by gPTP-capable network elements is called a *gPTP network.* Each instance of gPTP that the time-aware systems support is in one *gPTP domain*, and the instances of gPTP are said to be part of that gPTP domain. A time-aware system can support, and therefore be part of, more than one gPTP domain. The entity of a single time-aware system that executes gPTP in one gPTP domain is called a *PTP Instance*. A time-aware system can contain multiple PTP Instances, which are each associated with a different gPTP domain. There are two types of PTP Instances:

a) PTP End Instance, which, if not a Grandmaster PTP Instance, is a recipient of time information, and
b) PTP Relay Instance, which, if not a Grandmaster PTP Instance, receives time information from the Grandmaster PTP Instance (perhaps indirectly through other PTP Relay Instances), applies corrections to compensate for delays in the local area network (LAN) and the PTP Relay Instance itself, and retransmits the corrected information.

This standard specifies gPTP in terms of media-independent and media-dependent elements for different media. The supported media are as follows:

c) IEEE 802.3 Ethernet using full-duplex point-to-point links as specified in Clause 11.
d) IEEE 802.3 EPON links as specified in Clause 13.
e) IEEE 802.11 wireless as specified in Clause 12.
f) Generic coordinated shared networks (CSNs, e.g., MoCA and ITU-T G.hn) as specified in Clause 16.
g) IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex as specified in Clause 19.

This standard specifies time distribution mechanisms that are tolerant to some faults if the network paths are redundant. One of these time distribution mechanisms updates the distribution path in reaction to changes (see 10.3.1.2), whereas the other one relies on redundant PTP Instances on top of the redundant network paths (see Clause 18).

This standard specifies multiple mechanisms for providing fault tolerant synchronized time:

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h) The best timeTransmitter clock algorithm (BTCA) dynamically updates the synchronization tree of a gPTP domain in reaction to topology changes, e.g., failures, see 10.3.1.2.

i) The use of multiple gPTP domains that allows the use of multiple timescales. It also allows independent instances of the BTCA providing an independent synchronization tree for each domain. Furthermore, it allows to provide fault tolerance over multiple statically configured gPTP domains, e.g., as per item j) and k), where the BTCA is not used.

j) Hot standby (see Clause 18) that provides fault tolerance for GM and path failures via a protection switching mechanism over two statically configured gPTP domains, where the BTCA is not used.

k) Fault-tolerant timing with time integrity (see Clause 20.) which allows comparison and selection of time from two or more PTP Instances receiving time over dependent and independent statically configured synchronization trees.

7.2.2 gPTP domains

gPTP requires that all systems in the gPTP domain be time-aware systems, i.e., the protocol does not transfer timing over systems that are not time-aware. A time-aware system uses media-dependent mechanisms to distinguish between time-aware and non-time-aware systems, as specified in the media-dependent clauses of this standard.

NOTE—A synchronization tree is calculated by the BTCA or statically configured to achieve the best synchronization result in a network. A forwarding spanning tree such as IEEE 802.1Q RSTP can differ because it serves different purposes.

7.3 Examples of time-aware networks

7.3.1 Time-aware network consisting of a single gPTP domain

Figure 7-1 illustrates an example time-aware network consisting of a single gPTP domain, using all the above network technologies [i.e., item c) through item f) of 7.2.1], where end stations on several local networks are connected to a Grandmaster PTP Instance on a backbone network via an EPON access network.

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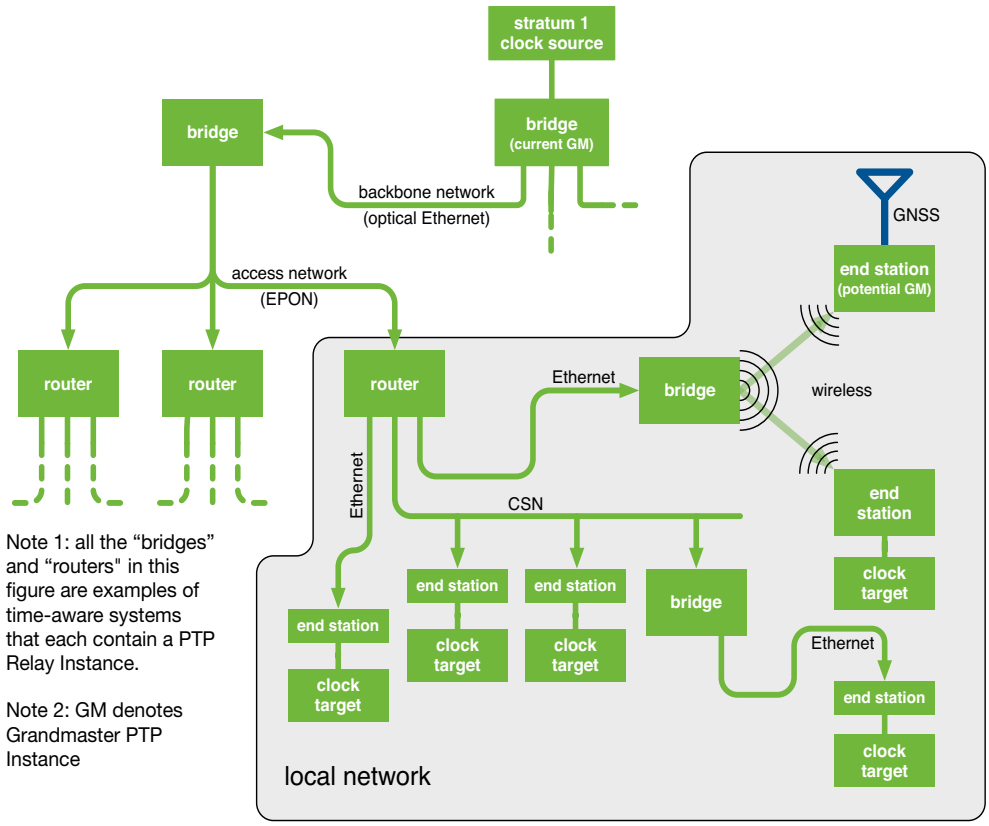


Figure 7-1—Time-aware network example

Any PTP Instance with clock sourcing capabilities can be a potential Grandmaster PTP Instance, and a selection method (the *best timeTransmitter clock algorithm*, or BTCA) ensures that all of the PTP Instances in the gPTP domain use the same Grandmaster PTP Instance.[18] The BTCA is largely identical to that used in IEEE Std 1588-2019, but somewhat simplified. In Figure 7-1 the BTCA process has resulted in the Grandmaster PTP Instance being on the network backbone. If, however, the access network fails, the systems on a local network automatically switch over to one of the potential Grandmaster PTP Instances on the local network that is as least as "good" as any other. For example, in Figure 7-2, the access network link has failed, and a potential Grandmaster PTP Instance that has a GNSS reference source has become the active Grandmaster PTP Instance. As a result, now two gPTP domains exist where there used to be one.

[18] There are, however, short periods during network reconfiguration when more than one Grandmaster PTP Instance might be active while the BTCA process is taking place.

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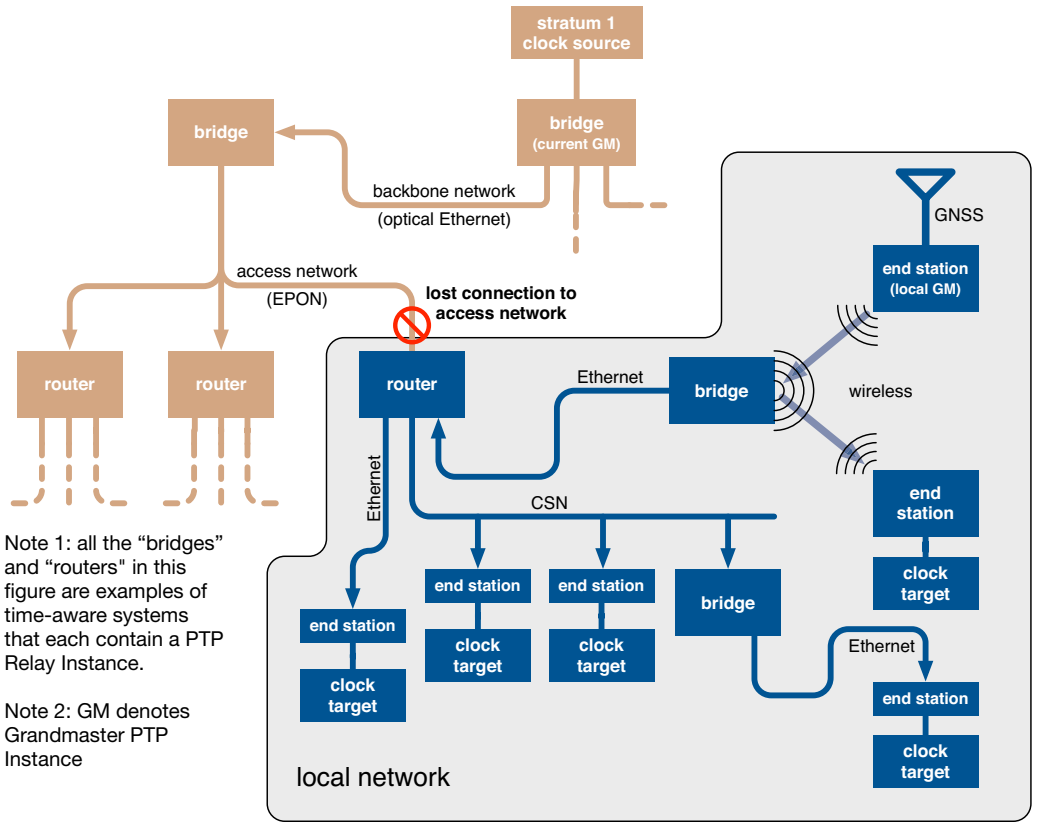


Figure 7-2—Time-aware network of Figure 7-1 after an access network link failure

7.3.2 Time-aware network consisting of multiple gPTP domains

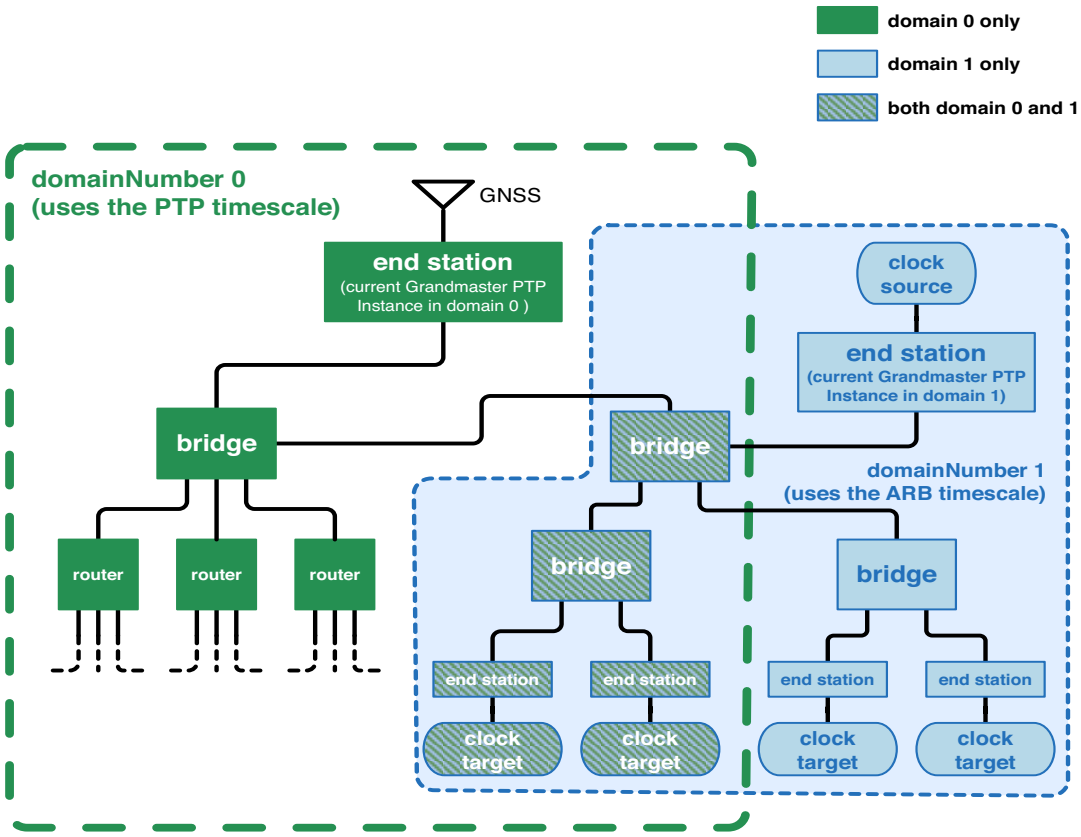
Figure 7-3 illustrates an example time-aware network consisting of multiple gPTP domains that could be used in an industrial application. Specifically, in this example the network has two timescales/domains, where domain 0 uses the PTP timescale and domain 1 uses the arbitrary (ARB) timescale (see 8.2). Notice that not all PTP Instances in domain 1 (within the blue shorter-dashed area) have domain 0 active in this example, even though every time-aware system supports domain 0 for backward compatibility with the 2011 edition of this standard. In addition, it is required that all PTP Instances belonging to the same domain have direct connections among them in their physical topology (e.g., time cannot be transported from one PTP Instance in domain 0 to another PTP Instance in domain 0 via a time-aware system that does not have domain 0 active). In addition, the time-aware systems for which both domains are active are depicted by slanted internal hatching, representing two independent, active PTP Instances.

As in the single-domain case, any of the network technologies of 7.2.1 can be used. The Grandmaster PTP Instance of each domain is selected by the BTCA; in this case, a separate, independent instance of the BTCA is invoked in each domain.

7.3.3 Time-aware networks using BTCA

Figure 7-4 shows an example network realizing two redundant synchronization trees from a single GM, each with its own GM. The two GMs receive timing from the single clock source.

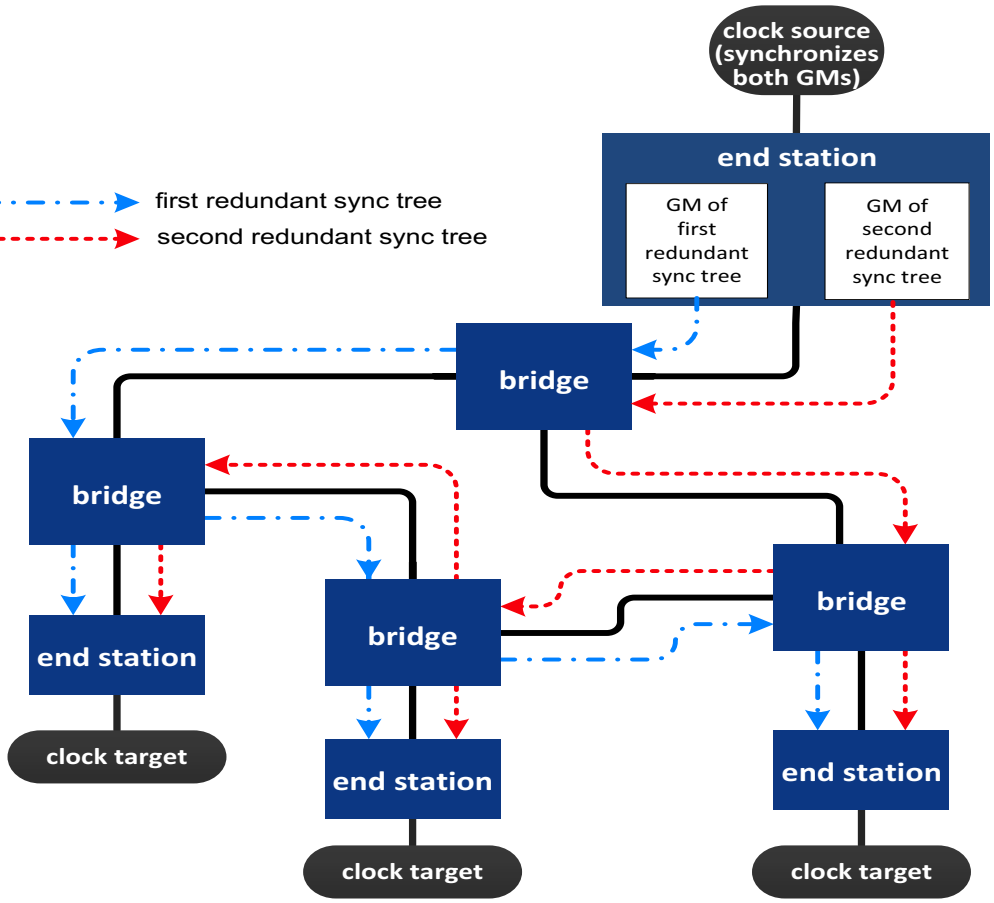
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NOTE —All the “bridges” and “routers” in this figure are examples of time-aware systems that contain at least one PTP Relay Instance, and the end stations are time-aware systems that contain at least one PTP End Instance.

Figure 7-3—Time-aware network example for multiple gPTP domains

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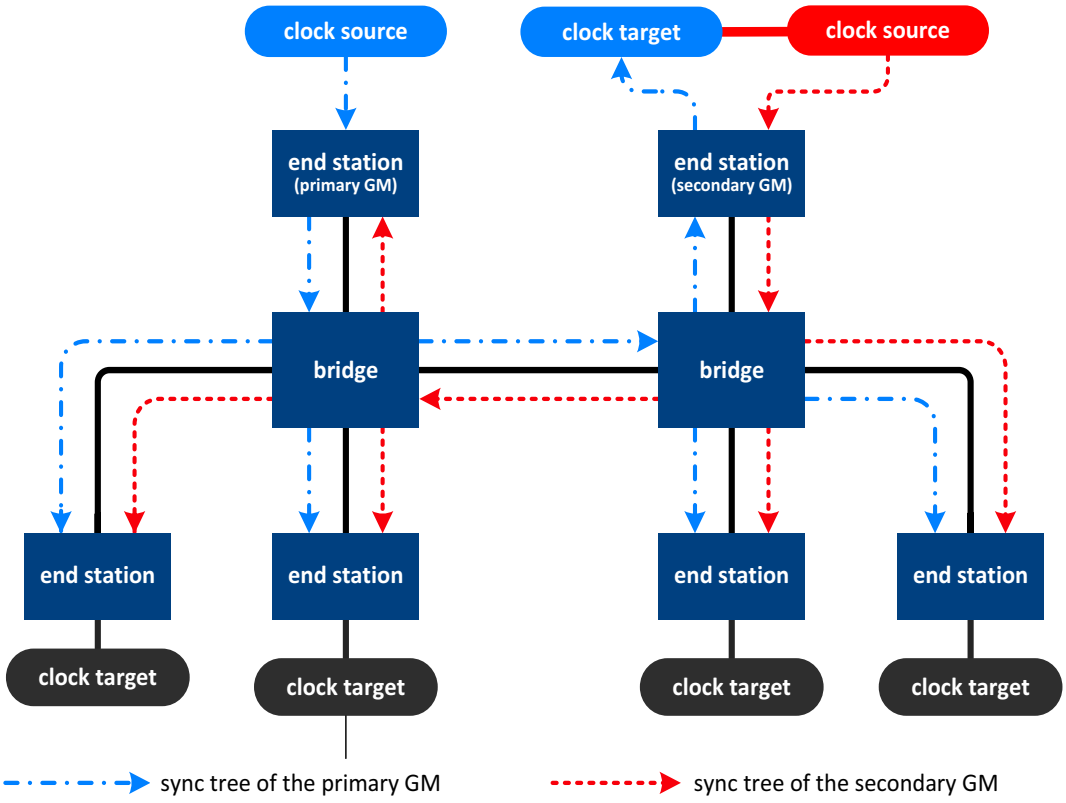


NOTE 1—The methods used for merging the redundant Sync messages received at each end station are not specified in this standard.
NOTE 2—All the "bridges" in this figure are examples of time-aware systems that contain PTP Relay Instances, and the end stations are examples of time-aware systems that contain PTP End Instances.
NOTE 3—GM denotes Grandmaster PTP Instance

Figure 7-4—Time-aware network example for synchronization path redundancy, with one clock source providing time to two domains

Figure 7-5 shows an example network with two redundant GMs, one as primary GM and the other as secondary GM, where each GM has one of the two redundant synchronization trees originating from it. This example supports hot-standby operating mode. In this mode, the secondary GM has to be synchronized to the primary GM, because it is part of the synchronization tree of the primary GM as shown in the figure. The secondary GM does not start sending Sync messages until it is synchronized to the primary GM.

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NOTE 1—The methods used for merging the redundant Sync messages received at each end station are not specified in this standard.
NOTE 2—All the "bridges" in this figure are examples of time-aware systems that contain PTP Relay Instances, and the end stations are examples of time-aware systems that contain PTP End Instances.
NOTE 3—GM denotes Grandmaster PTP Instance.

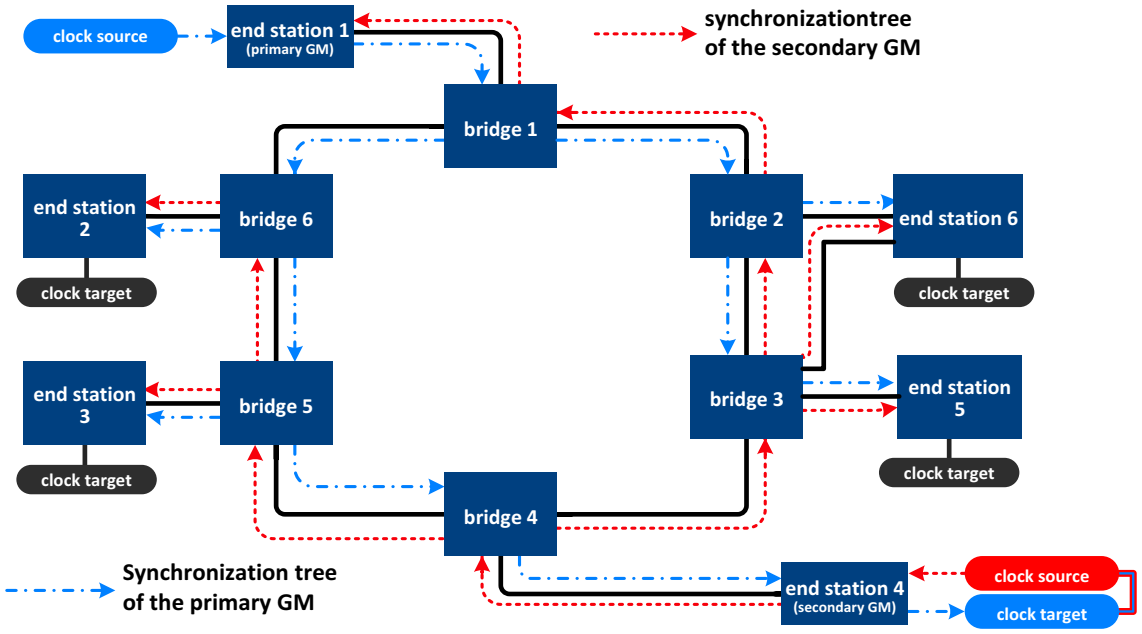
Figure 7-5—Time-aware network example for GM redundancy with one primary GM and one hot-standby GM, which are separated in two gPTP domains

7.3.4 Time-aware network with hot standby

Figure 7-6 shows an example of a time-aware network with hot standby (see Clause 18). The network has GM redundancy and path redundancy to all of the Bridges and end stations 1, 4, and 6 (path redundancy for end stations 2, 3, and 5 is possible, but is not illustrated to avoid clutter in the figure). In addition, for simplicity the optional split functionality (see 18.6.4) is assumed to be disabled.

End station 1 is the primary GM (see Clause 18), and its synchronization tree is illustrated by the arrows of the primary grandmaster synchronization tree. End station 4 is the secondary GM, and its synchronization tree is illustrated by the arrows of the secondary grandmaster synchronization tree. Except for the links to end stations 2, 3, and 5, the network can tolerate the failure of any single link and the Bridges and end stations remain synchronized by either the primary or the secondary GM. For example, if the link between bridge 3 and end station 6 fails, end station 6 is still synchronized by the primary domain via bridge 2. If the link between bridge 5 and bridge 6 fails, bridge 6 is still synchronized by the primary domain and bridge 5 is synchronized by the secondary domain. If the secondary GM fails, all the Bridges and end stations are still synchronized by the primary GM.

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NOTE 1—All the "bridges" in this figure are examples of time-aware systems that contain PTP Relay Instances, and the end stations are examples of time-aware systems that contain PTP End Instances.
NOTE 2—GM denotes Grandmaster PTP Instance.

Figure 7-6—Time-aware network example for hot standby with both GM and partial path redundancy

If the primary GM fails, all the Bridges and end stations are still synchronized by the secondary GM. In this case, if syncLocked is TRUE for any node in the path from end station 1 to end station 4, the primary PTP Instance of end station 4 stops receiving Sync messages and isSynced changes to FALSE; the hotStandbySystemState of end station 4 is then changed to NOT_REDUNDANT (see Clause 18). If syncLocked is FALSE, and assuming the Bridges and end stations other than the GMs are not grandmaster capable, the time difference between the primary and secondary domain eventually exceeds the primarySecondaryOffsetThresh (see Clause 18) and the hotStandbySystemState of end station 4 changes to NOT_REDUNDANT. Irrespective of the syncLocked state, the secondary GM no longer takes timing from the primary domain.

If the link between bridge 4 and bridge 5 fails, all the PTP Instances except for bridge 4 and end station 4 are then synchronized by the primary domain. Bridge 4 and end station 4 are synchronized by the secondary domain; in addition, bridges 1, 2, and 3, and end stations 1, 4, 5, and 6 receive timing on the secondary domain. Since end station 4 does not receive timing on the primary domain, its hotStandbySystemState changes to NOT_REDUNDANT, and the secondary domain GM does not take timing from the primary domain.

7.3.5 Time-aware network with fault-tolerant timing and time integrity

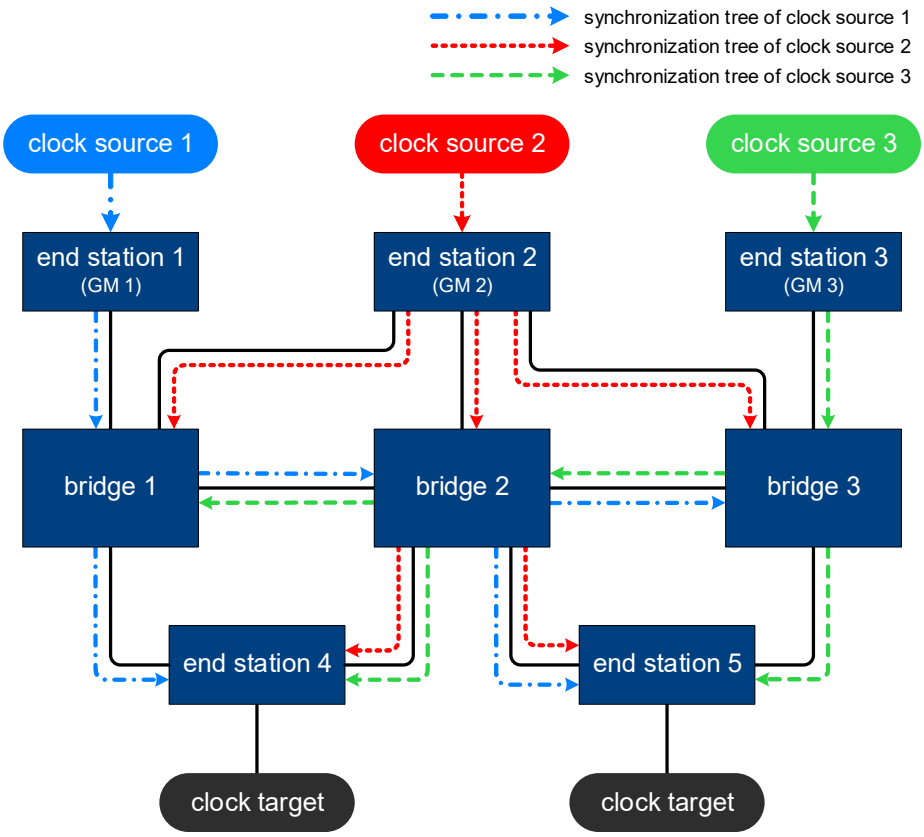
Figure 7-7 shows an example of a time-aware network with fault-tolerant timing and time integrity (see Clause 20). The network has GM redundancy and path redundancy to all the Bridges and to end stations 4 and 5 to enhance availability and integrity of time in the network.

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In this example, three independent clock sources (1, 2, and 3) provide time to three GMs (1, 2, and 3, respectively). These clock sources are synchronized to each other using a time agreement generation and preservation function (see I.4). The synchronization trees distribute the time from each of the three domains (from the three GMs) to all Bridges and to non-GM end stations 4 and 5. FTT entities (see 20.2) in the Bridges and in end stations 4 and 5 select a time from their three inputs. The time selected by each FTT entity has integrity (see 20.1.2.2). The selected time is provided to the local clock targets for use by mechanisms that require synchronized time (e.g., enhancements for scheduled traffic, see 8.6.8.4 in IEEE Std 802.1Q-2022). For example, at end station 4, the times received from GM2 and GM3 share a dependency (see 20.1.2.4) because the synchronization trees from GM2 and GM3 share a common Bridge node (bridge 2). Conversely, the synchronization tree from GM1 to end station 4 does not have any commonalities with the synchronization trees from GM2 and GM3 to end station 4, so the time from GM 1 is independent of the times from GM 2 and GM 3 at end station 4 (see 20.1.2.5). Because of these relationships, the FTT entity of end station 4 first selects between the times from GM 2 and GM 3 using a dependent time selection function (DTSF, see 20.2.3), and then selects between the DTSF instance's selected time and the time from GM 1 using an independent time selection function (ITSF, see 20.2.3) to produce an output time with time integrity. The FTT entity in bridge 2 sees three independent times and, thus, only uses its ITSF to select between them to produce an output time with time integrity.

The network shown in Figure 7-7 can tolerate failure of any one link or device (i.e., maintain the availability of time) as well as detect erroneous time on any one synchronization tree (i.e., maintain the integrity of time). For example, if the link between any two Bridges (bridge 1 to 2 or bridge 2 to 3) fails, all non-GM end stations and Bridges receive time over at least two synchronization trees and select a time using the ITSF. If bridge 1 experiences a fault and sends erroneous time downstream over the GM 1 synchronization tree, all the non-GM end stations and Bridges detect the erroneous time in comparison to the non-faulty time received over the other two synchronization trees. For the example shown in Figure 7-7, if any one Bridge is faulty and introduces errors in the relayed synchronization tree(s), all the other non-faulty Bridges and non-GM end stations are able to detect the erroneous times via the FTT entity because the topology allows for at least one non-faulty synchronization tree in the network.

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NOTE 1— All the "bridges" in this figure are examples of time-aware systems that contain PTP Relay Instances and an FTT entity. The methods used to terminate time from PTP Relay Instances is not specified in this standard.
NOTE 2—All the end stations in this figure are examples of time-aware systems that contain PTP End Instances and an FTT entity.
NOTE 3—The number of GMs and domains used in an implementation can vary based on the fault tolerance requirements.

Figure 7-7—Time-aware network example for fault-tolerant timing with time integrity

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7.4 Time synchronization

7.4.1 General

Time synchronization in gPTP is done the same way (in the abstract) as is done in IEEE Std 1588-2019: a Grandmaster PTP Instance sends information including the current synchronized time to all directly attached PTP Instances. Each of these PTP Instances must correct the received synchronized time by adding the propagation time needed for the information to transit the gPTP communication path from the Grandmaster PTP Instance. If the PTP Instance is a PTP Relay Instance, then it must forward the corrected time information (including additional corrections for delays in the forwarding process) to all the other attached PTP Instances.

To make this all work, there are two time intervals that must be precisely known: the forwarding delay (called the *residence time*), and the time taken for the synchronized time information to transit the gPTP communication path between two PTP Instances.

7.4.2 Delay measurement

Each type of LAN or gPTP communication path has different methods for measuring propagation time, but they are all based on the same principle: measuring the time that a well-known part of a message is transmitted from one device and the time that the same part of the same message is received by the other device, then sending another message in the opposite direction and doing the same measurement as shown in Figure 7-8.

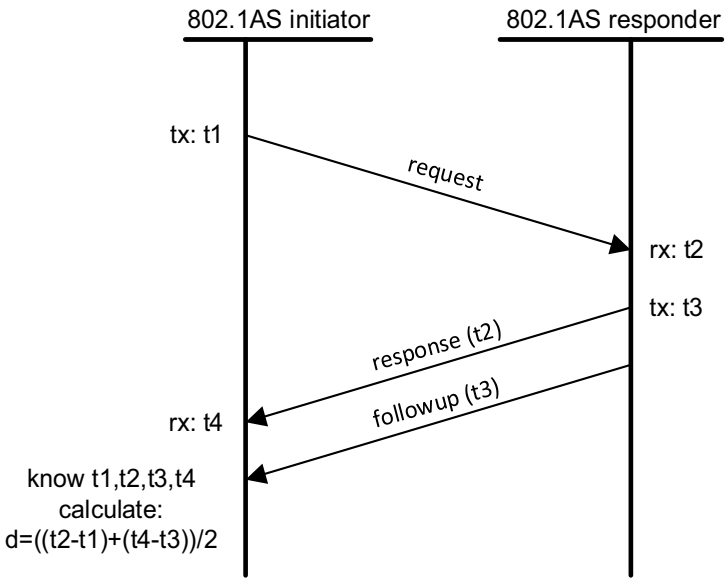


Figure 7-8—Conceptual medium delay measurement

This basic mechanism is used as the foundation for media-dependent measurement mechanisms specified in the media dependent clauses of this standard (see 7.2.1).

7.4.3 Logical syntonization

The time-synchronization correction previously described is dependent on the accuracy of the delay and residence time measurements. If the clock used for this purpose is frequency locked (syntonized) to the Grandmaster Clock, then all the time interval measurements use the same time base. Since actually adjusting

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the frequency of an oscillator (e.g., using a phase-lock loop) is slow and prone to gain peaking effects, PTP Relay Instances can correct time interval measurements using the Grandmaster Clock frequency ratio.

Each PTP Instance measures, at each PTP Port, the ratio of the frequency of the PTP Instance at the other end of the link attached to that PTP Port to the frequency of its own clock. The cumulative ratio of the Grandmaster Clock frequency to the local clock frequency is accumulated in a standard organizational type, length, value (TLV) attached to the Follow_Up message (or the Sync message if the optional one-step processing is enabled). The frequency ratio of the Grandmaster Clock relative to the local clock is used in computing synchronized time, and the frequency ratio of the neighbor relative to the local clock is used in correcting the propagation time measurement (NRR is not shown in Figure 7-8).

The Grandmaster Clock frequency ratio is measured by accumulating neighbor frequency ratios for two main reasons. First, if there is a network reconfiguration and a new Grandmaster PTP Instance is elected, the nearest neighbor frequency ratios do not have to be newly measured as they are constantly measured using the Pdelay messages. This results in the frequency offset relative to the new Grandmaster Clock being known when the first Follow_Up message (or first Sync message if the optional one-step processing is enabled) is received, which reduces the duration of any transient error in synchronized time during the reconfiguration. This is beneficial to many high-end audio applications. Second, there are no gain peaking effects because an error in frequency offset at one PTP Relay Instance, and resulting residence time error, does not directly affect the frequency offset at a downstream PTP Relay Instance.

7.4.4 Energy efficiency

Sending PTP messages at relatively high rates when there is otherwise little or no traffic conflicts with the goal of reducing energy consumption. This standard specifies a way to request that a neighbor PTP Port reduce the rate of sending Sync (and Follow_Up if optional one-step processing is not enabled), peer delay, and Announce messages and also to inform the neighbor not to compute neighbor rate ratio and/or propagation delay on this link. A time-aware system could do this when it enters low-power mode, but this standard does not specify the conditions under which this is done; it specifies only the actions a time-aware system takes.

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7.5 PTP Instance architecture

The model of a PTP Instance and its interfaces to higher-layer applications are shown in Figure 7-9.

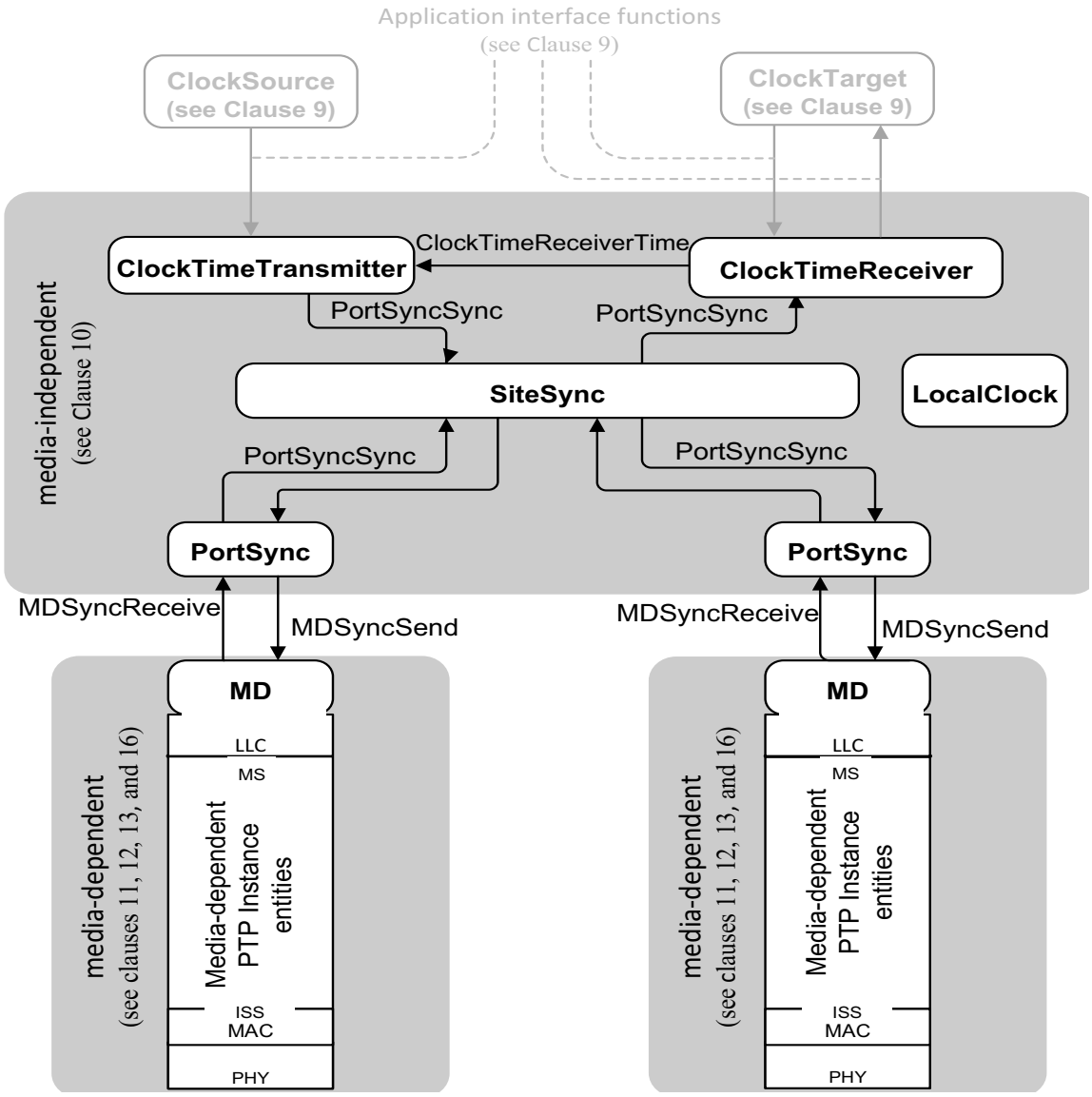


Figure 7-9—Model for a PTP Instance and its interfaces to higher-layer applications

A PTP Instance consists of the following major parts:

a) If the PTP Instance includes application(s) that either use or source time information, then they interface with the gPTP information using the application interfaces specified in Clause 9.

b) A single media-independent part that consists of ClockTimeTransmitter, ClockTimeReceiver, and SiteSync logical entities, one or more PortSync entities, and a LocalClock entity. The BTCA and forwarding of time information between logical ports and the ClockTimeReceiver and ClockTimeTransmitter is done by the SiteSync entity, while the computation of PTP Port-specific delays needed for time-synchronization correction is done by the PortSync entities. The media-independent layer can be extended by hot standby (see Clause 18).

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c) Media-dependent ports, which translate the abstract "MDSyncSend" and "MDSyncReceive" structures received from or sent to the media-independent layer and corresponding methods used for the particular LAN attached to the port.

7.6 Differences between gPTP (IEEE Std 802.1AS) and PTP (IEEE Std 1588-2019)

a) gPTP assumes all communication between PTP Instances is done only using IEEE 802 MAC PDUs and addressing, while IEEE Std 1588-2019 supports various layer 2 and layer 3-4 communication methods.

b) gPTP specifies a media-independent sublayer that simplifies the integration within a single timing domain of multiple different networking technologies with radically different media access protocols. gPTP specifies a media-dependent sublayer for each medium. The information exchanged between PTP Instances has been generalized to support different packet formats and management schemes appropriate to the particular networking technology. IEEE Std 1588-2019, on the other hand, has introduced a new architecture based on media-independent and media-dependent sublayers (see 6.5.2, Figure 5, and Figure 6 of IEEE Std 1588-2019); however, this architecture is optional. The architecture of IEEE Std 1588-2008 [B12], which is not based on media-independent and media-dependent layers, has been retained for Internet Protocol (IP) version 4, IP version 6, Ethernet LANs, and several industrial automation control protocols. The intent in IEEE Std 1588-2019 is that the new architecture, based on media-independent and media-dependent layers, will be used for IEEE 802.11 networks, IEEE 802.3 EPON, and CSN using the specifications of gPTP, and that the architecture must be used for transports that define native timing mechanisms if those native timing mechanisms are used.

c) In gPTP there are only two types of PTP Instances: PTP End Instances and PTP Relay Instances, while IEEE Std 1588-2019 has Ordinary Clocks, Boundary Clocks, end-to-end Transparent Clocks, and P2P Transparent Clocks. A PTP End Instance corresponds to an IEEE 1588 Ordinary Clock, and a PTP Relay Instance is a type of IEEE 1588 Boundary Clock where its operation is very tightly defined, so much so that a PTP Relay Instance with Ethernet ports can be shown to be mathematically equivalent to a P2P Transparent Clock in terms of how synchronization is performed, as shown in 11.1.3. In addition, a PTP Relay Instance can operate in a mode (i.e., the mode where the variable syncLocked is TRUE; see 10.2.5.15) where the PTP Relay Instance is equivalent to a P2P Transparent Clock in terms of when time-synchronization messages are sent. A time-aware system measures link delay and residence time and communicates these in a correction field. In summary, a PTP Relay Instance conforms to the specifications for a Boundary Clock in IEEE Std 1588-2019, but a PTP Relay Instance does not conform to the complete specifications for a P2P Transparent Clock in IEEE Std 1588-2019 because:

1) When syncLocked is FALSE, the PTP Relay Instance sends Sync according to the specifications for a Boundary Clock, and
2) The PTP Relay Instance invokes the BTCA and has PTP Port states.

d) PTP Instances communicate gPTP information only directly with other PTP Instances. That is, a gPTP domain consists ONLY of PTP Instances. Non-PTP Relay Instances cannot be used to relay gPTP information. In IEEE Std 1588-2019, it is possible to use non-IEEE-1588-aware relays in an IEEE 1588 domain, although this slows timing convergence and introduces extra jitter and wander that must be filtered by any IEEE 1588 clock.

e) For FDE and HDE links, gPTP requires the use of the peer-to-peer delay mechanism, while IEEE Std 1588-2019 also allows the use of end-to-end delay measurement.

f) For FDE and HDE links, gPTP requires the use of two-step processing (use of Follow_Up and Pdelay_Resp_Follow_Up messages to communicate timestamps), with an optional one-step processing mode for FDE that embeds timestamps in the Sync "on the fly" as they are being transmitted (gPTP does not specify one-step processing for peer delay messages). IEEE Std 1588-

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2019 allows either two-step or one-step processing to be required (for both Sync and peer delay messages) depending on a specific profile.

g) All PTP Instances in a gPTP domain are logically syntonized; in other words, they all measure time intervals using the same frequency. This is done by the process described in 7.4.3 and is mandatory. Syntonization in IEEE Std 1588-2019 is optional. The syntonization method used by gPTP is supported as an option in IEEE Std 1588-2019, but uses a TLV standardized as part of IEEE Std 1588-2019 (this feature is new for IEEE Std 1588-2019), while gPTP uses the ORGANIZATION_EXTENSION TLV specified in 11.4.4.3.

h) Finally, this standard includes formal interface definitions, including primitives, for the time-aware applications (see Clause 9). IEEE Std 1588-2019 describes external interfaces without describing specific interface primitives.

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8. IEEE 802.1AS concepts

8.1 gPTP domain

A gPTP domain consists of one or more PTP Instances and links that meet the requirements of this standard and communicate with each other as defined by the IEEE 802.1AS protocol. A gPTP domain defines the scope of gPTP message communication, state, operations, data sets, and timescale.

A gPTP domain is identified by two attributes: domainNumber and sdoId. The sdoId of a gPTP domain is a 12-bit unsigned integer. The sdoId is structured as a two-part attribute as follows:

— The most significant 4 bits are named the majorSdoId, and
— The least significant 8 bits are named the minorSdoId.

A time-aware system shall support one or more gPTP domains, each with a distinct domainNumber in the range 0 through 127. Unless otherwise specified in this standard, the operation of gPTP and the timescale in any given gPTP domain is independent of operation in any other domain.

The value of majorSdoId for a gPTP domain shall be 0x1. The value of minorSdoId for a gPTP domain shall be 0x00 (see 16.5, 7.1.3.1, and E.4 of IEEE Std 1588-2019).

Since the IEEE 802.1 Working Group has only one single unique sdoId value, the PTP profile specified in the present standard is isolated from other PTP profiles (see 16.5.2 of IEEE Std 1588-2019).

Both the domainNumber and the sdoId are carried in the common header of all PTP messages (see 10.6.2.2).

NOTE 1—The above requirements for majorSdoId and minorSdoId are for gPTP domains. The requirements for the CMLDS are given in 11.2.16.

NOTE 2—In the 2011 edition of this standard, the attribute majorSdoId was named transportSpecific, and its value was specified as 0x1 in 10.5.2.2.1 of Corrigendum 1. The attribute minorSdoId did not exist in the 2011 edition, but its location in the common header was a reserved field, which was specified to be transmitted as 0 and ignored on receipt.

Unless otherwise stated, information in the remainder of this document is per gPTP domain.

NOTE 3—In steady state, all PTP Instances in a gPTP domain are traceable to a single Grandmaster PTP Instance.

8.2 Timescale

8.2.1 Introduction

The timescale for a gPTP domain is established by the Grandmaster Clock. There are two types of timescales supported by gPTP:

— The timescale PTP: The epoch is the PTP epoch (see 8.2.2), and the timescale is continuous. The unit of measure of time is the second defined by International Atomic Time (TAI) (for the definition of TAI, see Service de la Rotation Terrestre [B38], with further amplification in IAU [B37]; and for more information on TAI, see Jekeli [B32], international system of units (SI) brochure [B25], and Petit and Luzum [B36]). The timescale of domain 0 shall be PTP. See IEEE Std 1588-2019 for more details.
— The timescale ARB (arbitrary): The epoch is the domain startup time and can be set by an administrative procedure. Between invocations of the administrative procedure, the timescale is continuous. Additional invocations of the administrative procedure can introduce discontinuities in

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the overall timescale. The unit of measure of time is determined by the Grandmaster Clock. The second used in the operation of the protocol can differ from the SI second.

8.2.2 Epoch

The epoch is the origin of the timescale of a gPTP domain.

The PTP epoch (epoch of the timescale PTP) is 1 January 1970 00:00:00 TAI.

NOTE—The common portable operating system interface (POSIX) algorithms can be used for converting elapsed seconds since the PTP epoch to the ISO 8601:2004 [B26] printed representation of time of day on the TAI timescale (see also ISO/IEC 9945:2003 [B27]).

See Annex C for information on converting between common timescales.

8.2.3 UTC offset

When the timescale is PTP, it is possible to calculate Coordinated Universal Time (UTC) time using the value of currentUtcOffset. The value of currentUtcOffset is given by:

currentUtcOffset = TAI – UTC

where the difference TAI – UTC is derived from UTC – TAI (the negative of currentUtcOffset) specified in IERS Bulletin C.

The value of currentUtcOffset for the current Grandmaster PTP Instance is maintained in the currentUtcOffset member of the time properties data set (see 14.5.2).

NOTE 1—As of 0 hours 1 January 2017 UTC, UTC was behind TAI by 37 s, i.e., TAI – UTC = +37 s. At that moment, the IEEE-802.1AS-defined value of currentUtcOffset was +37 s, as designated in the applicable IERS Bulletin C (see Clause 2; see also Service de la Rotation Terrestre [B38] and U.S. Naval Observatory [B39]).

NOTE 2—Leap second events and the value of UTC – TAI are posted well in advance in IERS Bulletin C. A list of all leap second events is maintained by the U.S. Naval Observatory [B39], which also offers an extensive discussion of timescales, leap seconds, and related time issues.

NOTE 3—The value of currentUtcOffset represents the difference between TAI and UTC. Since 1972, only integral values are permitted for this difference. Due to leap seconds, UTC cannot be correctly represented as a single integer but can be expressed in the ISO 8601:2004 [B26] print form. See also C.2 for an example of the use of the POSIX algorithm to compute the correct print form.

When the timescale is PTP, it is possible to calculate local time from the time provided by a gPTP domain using the currentUtcOffset, leap59, and leap61 member values of the time properties data set and knowledge of the local time zone and whether and when daylight savings time is observed.

When the timescale is ARB, the values of currentUtcOffset, leap59, and leap61 cannot be used to compute UTC.

The mechanism for computing UTC or any other time does not change the synchronized time (see 3.36), i.e., the PTP Instance time, of a PTP Instance.

8.2.4 Measurement of time within a gPTP domain

Time in a gPTP domain shall be measured as elapsed time since the epoch of the timescale of that domain.

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8.3 Link asymmetry

This standard requires the measurement of the mean propagation time (also known as the *mean propagation delay*) between the time-aware systems that comprise the two endpoints of a link. The measurement is performed when one of the time-aware systems (the initiator time-aware system) sends a message to the other time-aware system (the responder time-aware system). The responder then sends a message back to the initiator at a later time. The departure of the message sent by the initiator time-aware system is timestamped, and the timestamp value is retained by that system. The arrival of this message at the responder time-aware system is timestamped, and the timestamp value is conveyed to the initiator time-aware system in a subsequent message. The departure of the response message sent by the responder time-aware system (in response to the message it receives from the initiator time-aware system) is timestamped, and the timestamp value is conveyed to the initiator time-aware system in a subsequent message. The arrival of this response message at the initiator time-aware system is timestamped, and the timestamp value is retained by that system. The mean propagation time is computed by the initiator time-aware system after receiving the response message, from the four timestamp values it has at this point.

Typically, the propagation time is not exactly the same in both directions, and the degree to which it differs in the two directions is characterized by the delay asymmetry. The relation between the individual propagation times in the two directions, the mean propagation time, and the delay asymmetry is as follows. Let t_{ir} be the propagation time from the initiator to the responder, t_{ri} be the propagation time from the responder to the initiator, meanLinkDelay be the mean propagation time (see 10.2.5.8), and delayAsymmetry be the delay asymmetry. The propagation times in the two directions are illustrated in Figure 8-1.

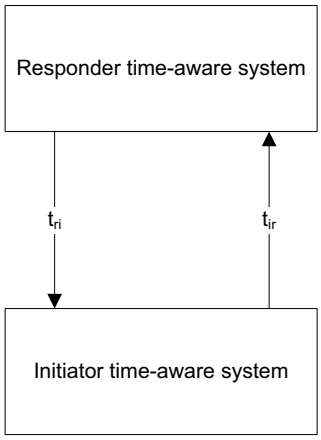


Figure 8-1—Propagation asymmetry

The meanLinkDelay is the mean value of t_{ir} and t_{ri}, i.e., meanLinkDelay = (t_{ir} + t_{ri}) / 2. The delayAsymmetry is defined as:

t_{ir} = meanLinkDelay – delayAsymmetry

t_{ri} = meanLinkDelay + delayAsymmetry

In other words, delayAsymmetry is defined to be positive when the responder to initiator propagation time is longer than the initiator to responder propagation time.

This standard does not explicitly require the measurement of delayAsymmetry; however, if delayAsymmetry is modeled, it shall be modeled as specified in this clause.

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NOTE 1—A time-aware system can change the value of delayAsymmetry during operation (see 14.10.11, 14.18.9, and Annex G).

NOTE 2—A time-aware system PTP Port cannot measure the value of delayAsymmetry during live operation of the system (i.e., asymmetryMeasurementMode is FALSE; see 10.2.5.2); therefore, the value of delayAsymmetry must be defined separately using information from the supplier or additional testing before running the live system. The methods for measuring asymmetry are not specified in this standard. These values can be added to the system configuration to improve the accuracy of time synchronization. The inaccuracy caused by asymmetry is half the value of the difference between t_{ri} and t_{ir}, and these inaccuracies can either accumulate over successive hops or, if the successive asymmetries have different signs, cancel each other over the successive hops.

8.4 Messages

8.4.1 General

All communications occur via PTP messages and/or media-dependent messages.

8.4.2 Message attributes

8.4.2.1 General

All messages used in this standard have the following attributes:

a) Message class
b) Message type

The message class attribute is defined in this clause. The message type attribute is defined in 3.22. Some messages have additional attributes; these are defined in the subclauses where the respective messages are defined.

8.4.2.2 Message class

There are two message classes, the event message class and the general message class. Event messages are timestamped on egress from a PTP Instance and ingress to a PTP Instance. General messages are not timestamped. Every message is either an event message or a general message.

8.4.3 Generation of event message timestamps

All event messages are timestamped on egress and ingress. The timestamp shall be the time, relative to the LocalClock entity (see 10.1) at which the message timestamp point passes the reference plane marking the boundary between the PTP Instance and the network media.

The definition of the timestamp measurement plane (see 3.40), along with the corrections defined as follows, allows transmission delays to be measured in such a way (at such a low layer) that they appear fixed and symmetrical to gPTP even though the MAC client might otherwise observe substantial asymmetry and transmission variation. For example, the timestamp measurement plane is located below any retransmission and queuing performed by the MAC.

NOTE 1—If an implementation generates event message timestamps using a point other than the message timestamp point, then the generated timestamps should be appropriately corrected by the time interval (fixed or otherwise) between the actual time of detection and the time the message timestamp point passed the reference plane. Failure to make these corrections results in a time offset between PTP Instances.

NOTE 2—In general, the timestamps can be generated at a timestamp measurement plane that is removed from the reference plane. Furthermore, the timestamp measurement plane, and therefore the time offset of this plane from the

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reference plane, is likely to be different for inbound and outbound event messages. To meet the requirement of this clause, the generated timestamps should be corrected for these offsets. Figure 8-2 illustrates these offsets. Based on this model the appropriate corrections are as follows:

egressTimestamp = egressMeasuredTimestamp + egressLatency

ingressTimestamp = ingressMeasuredTimestamp – ingressLatency

where the timestamps relative to the reference plane, egressTimestamp and ingressTimestamp, are computed from the timestamps relative to the timestamp measurement plane, egressMeasuredTimestamp and ingressMeasuredTimestamp, respectively, using their respective latencies, egressLatency and ingressLatency. Failure to make these corrections results in a time offset between the timeReceiver and timeTransmitter clocks.

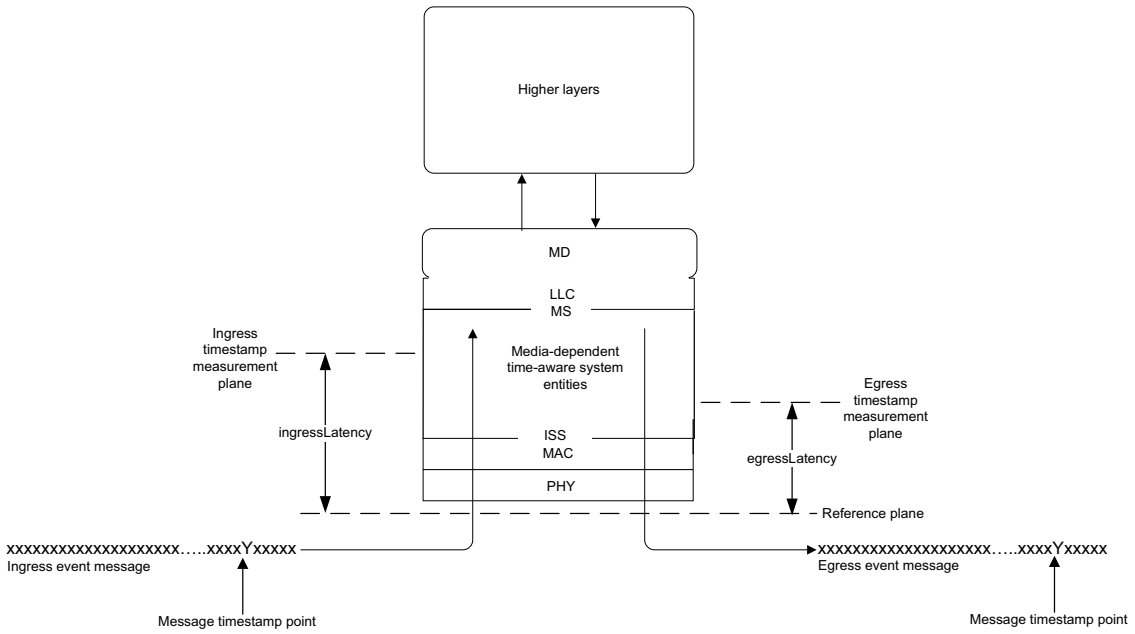


Figure 8-2—Definition of message timestamp point, reference plane, timestamp measurement plane, and latency constants

It is advised to use the ingress and egress timestamping measurement planes recommended by the specification for the medium used. For example, for IEEE 802.3 networks, the recommendation is to follow Clause 90 of IEEE Std 802.3-2022, as amended by IEEE Std 802.3cx-2023, IEEE Std 802.3de-2022, and IEEE Std 802.3df-2024.

8.4.4 Priorities

IEEE Std 802.1AS messages can be delayed in internal queues depending on their transmission priority relative to other frames. IEEE Std 802.1AS messages are sensitive to delays, and long bursts of other traffic can cause loss of synchronization due to gPTP timeouts. Delays caused by queues on the IEEE Std 802.1AS messages effectively increases residence time, which degrades achievable time accuracy.

8.5 Ports

8.5.1 General

The PTP Instances in a gPTP domain interface with the network media via physical ports. gPTP defines a PTP Port in such a way that communication between PTP Instances is point-to-point or point-to-multipoint. The choice of point-to-point (e.g., see Clause 16) or point-to-multipoint (e.g. see Clause 19) communication

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is specified in each media dependent specification in this standard. A PTP port, consists of one PortSync entity and one media-dependent (MD) entity.

8.5.2 Port identity

8.5.2.1 General

A PTP Port is identified by a port identity of type PortIdentity (see 6.4.3.7). The value is maintained in portDS.portIdentity (see 14.10.2). A port identity consists of the following two attributes:

a) portIdentity.clockIdentity
b) portIdentity.portNumber

8.5.2.2 clockIdentity

The clockIdentity attribute shall be as specified in 7.5.2.2 of IEEE Std 1588-2019.

8.5.2.3 Port number

The portNumber values for the PTP Ports on a time-aware system shall be distinct in the range 0x0001 through 0xFFFE.

The portNumber value 0 is assigned to the interface between the ClockTimeTransmitter and ClockSource entities (see 10.1 and Figure 10-1). The assignment of portNumber 0 to this interface helps to simplify the SiteSync and PortStateSelection state machines.

The portNumber value 0xFFFF is reserved.

8.5.2.4 Ordering of clockIdentity and portIdentity values

Two clockIdentity values X and Y are compared as follows. Let x be the unsigned integer formed by concatenating octets 0 through 7 of X such that octet j+1 follows octet j (i.e., is less significant than octet j) in x (j = 0, 1, ..., 6). Let y be the unsigned integer formed by concatenating octets 0 through 7 of Y such that octet j+1 follows octet j (i.e., is less significant than octet j) in y (j = 0, 1, ..., 6). Then:

a) X = Y if and only if $x = y$,
b) X > Y if and only if $x > y$, and
c) X < Y if and only if $x < y$.

Two portIdentity values A and B with members clockIdentity and portNumber are compared as follows. Let a be the unsigned integer formed by concatenating octets 0 through 7 of A.clockIdentity, such that octet j+1 follows octet j (i.e., is less significant than octet j) in a (j = 0, 1, ..., 6), followed by octet 0 of A.portNumber, followed by octet 1 of A.portNumber. Let b be the unsigned integer formed by concatenating octets 0 through 7 of B.clockIdentity, such that octet j+1 follows octet j (i.e., is less significant than octet j) in b (j = 0, 1, ..., 6), followed by octet 0 of B.portNumber, followed by octet 1 of B.portNumber. Then:

d) A = B if and only if $a = b$,
e) A > B if and only if $a > b$, and
f) A < B if and only if $a < b$.

A portIdentity A with members clockIdentity and portNumber and a clockIdentity B are compared as follows. The unsigned integer a is formed from portIdentity A as described above. The unsigned integer b is

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formed by first forming a portIdentity B' whose clockIdentity is B and portNumber is 0. *b* is then formed from B' as described above. A and B are then compared as described in items d) through f) in this subclause.

8.6 PTP Instance characterization

8.6.1 PTP Instance type

There are two types of PTP Instances used in a gPTP domain, as follows:

a) PTP End Instance
b) PTP Relay Instance

All PTP Instances are identified by clockIdentity.

In addition, PTP Instances are characterized by the following attributes:

c) priority1
d) clockClass
e) clockAccuracy
f) offsetScaledLogVariance
g) priority2
h) clockIdentity
i) timeSource
j) numberPorts

NOTE—Attributes c) through i) can be considered to be associated with the ClockTimeTransmitter entity of the PTP Instance.

8.6.2 PTP Instance attributes

8.6.2.1 priority1

priority1 is used in the execution of the BTCA (see 10.3). The value of priority1 is an integer selected from the range 0 through 255. The ordering of priority1 in the operation of the BTCA (see 10.3.4 and 10.3.5) is specified as follows. A ClockTimeTransmitter A shall be deemed better than a ClockTimeTransmitter B if the value of priority1 of A is numerically less than that of B.

The value of priority1 shall be 255 for a PTP Instance that is not grandmaster-capable. The value of priority1 shall be less than 255 for a PTP Instance that is grandmaster-capable. The value 0 shall be reserved for management use, i.e., the value of priority1 shall be set to 0 only via management action. The default value shall be set to one of the values listed in Table 8-1, and the choice of value from Table 8-1 is left to the implementer of the PTP Instance.

NOTE 1—Care must be applied to multi-function device design, specifically for an end station that also contains a Bridge. The Bridge function inside multi-function devices must not be powered down when the end station function is powered down, or else the data and controls (like gPTP) passing through the Bridge function to other network devices will cease. Example devices of this are a television and/or amplifier/receiver, each of which contains a Bridge.

NOTE 2—The BTCA (see 10.3) considers priority1 before other attributes; the priority1 attribute can therefore be used to force a desired ordering of PTP Instances for best timeTransmitter selection.

NOTE 3—The settings for priority1 in Table 8-1 guarantee that a PTP Instance that is grandmaster-capable is always preferred by the BTCA over a PTP Instance that is not grandmaster-capable.

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Table 8-1—Default values for priority1, for the respective media

PTP Instance type	Default value for priority1
PTP Instances that are a central/critical part of the network and are not expected to ever be turned off during normal network operation (e.g., Bridges, wireless access points, and grandmaster-capable end nodes that are intended by the manufacturer to be Grandmaster PTP Instances). (End node Grandmaster PTP Instances are not a "central" part of the network, but they are "critical" to this gPTP operation.)	246
PTP Instances that (A) can be turned off at any time and therefore cannot be considered a central/critical part of the network; or (B) are a central/critical part of the network, are not expected to ever be turned off during normal network operation, and are grandmaster-capable, but are not intended by the manufacturer to be Grandmaster PTP Instances. PTP Instances that can be turned off and are turned off affect only the function(s) they support and do not affect any other functions of the network (e.g., desktop computers, fixed (heavy or otherwise) end nodes, speakers, receivers, amplifiers, and televisions).	248
PTP Instances that can go away (physically or otherwise) at any time (e.g., PTP Instances that are designed to be transient to the network such as laptop computers, cell phones, and battery-powered speakers).	250
PTP Instance that is not grandmaster-capable.	255

NOTE 4—These values are assigned so that PTP Instances with priority1 value of 246 are selected as Grandmaster PTP Instances over PTP Instances with priority1 values of 248 or 250. (PTP Instances with priority1 value of 255 are never selected as Grandmaster PTP Instances.)

NOTE 5—These default values are suitable for applications in which the availability of the Grandmaster PTP Instance is the most important criterion for Grandmaster PTP Instance selection. A PTP Instance built for a specific application for which this is not the case can be capable of having priority1 changed via management.

8.6.2.2 clockClass

The clockClass attribute denotes the traceability of the synchronized time distributed by a ClockTimeTransmitter when it is the Grandmaster PTP Instance.

The value shall be selected as follows:

a) If defaultDS.gmCapable is TRUE (see 14.2.7), then
   1) clockClass is set to the value that reflects the combination of the LocalClock and ClockSource entities; else
   2) If the value that reflects the LocalClock and ClockSource entities is not specified or not known, clockClass is set to 248;
b) If defaultDS.gmCapable is FALSE, clockClass is set to 255 (see 8.6.2.1).

The ordering of clockClass in the operation of the best timeTransmitter clock algorithm (see 10.3.4 and 10.3.5) is specified as follows. When comparing clockClass values, PTP Instance A shall be deemed better than PTP Instance B if the value of the clockClass of A is lower than that of B.

See 7.6.2.5 of IEEE Std 1588-2019 for a more detailed description of clockClass.

NOTE—The PTP Instance has a LocalClock entity, which can be the free-running quartz crystal that just meets the IEEE 802.3 requirements, but could also be better. There can be a ClockSource entity, e.g., timing taken from a GNSS, available in the local system that provides timing to the ClockSource entity. The time provided by the PTP Instance, if it is the Grandmaster PTP Instance, is reflected by the combination of these two entities, and the clockClass reflects this

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combination as specified in 7.6.2.5 of IEEE Std 1588-2019. For example, when the LocalClock entity uses a quartz oscillator that meets the requirements of IEEE Std 802.3-2022 and B.1 of this standard, clockClass is set to 248. But, if a GNSS receiver is present and synchronizes the PTP Instance, then the clockClass is set to the value 6, indicating traceability to a primary reference time source (see 7.6.2.5 of IEEE Std 1588-2019).

8.6.2.3 clockAccuracy

The clockAccuracy attribute indicates the expected time accuracy of a ClockTimeTransmitter.

The value shall be selected as follows:

a) clockAccuracy is set to the value that reflects the combination of the LocalClock and ClockSource entities; else

b) If the value that reflects the LocalClock and ClockSource entities is not specified or unknown, clockAccuracy is set to 254 (0xFE).

The ordering of clockAccuracy in the operation of the best timeTransmitter clock algorithm (see 10.3.4 and 10.3.5) is specified as follows. When comparing clockAccuracy values, PTP Instance A shall be deemed better than PTP Instance B if the value of the clockAccuracy of A is lower than that of B.

See 7.6.2.6 of IEEE Std 1588-2019 for more detailed description of clockAccuracy.

8.6.2.4 offsetScaledLogVariance

The offsetScaledLogVariance is a scaled, offset representation of an estimate of the PTP variance. The PTP variance characterizes the precision and frequency stability of the ClockTimeTransmitter. The PTP variance is the square of PTP Deviation (PTPDEV) (see B.1.3.2).

The value shall be selected as follows:

a) offsetScaledLogVariance is set to the value that reflects the combination of the LocalClock and ClockSource entities; else

b) If the value that reflects these entities is not specified or not known, offsetScaledLogVariance is set to 17258 (0x436A). This value corresponds to the value of PTPDEV for observation interval equal to the default Sync message transmission interval (i.e., observation interval of 0.125 s; see 11.5.2.3 and B.1.3.2).

The ordering of offsetScaledLogVariance in the operation of the best timeTransmitter clock algorithm (see 10.3.4 and 10.3.5) is specified as follows. When comparing offsetScaledLogVariance values, PTP Instance A shall be deemed better than PTP Instance B if the value of the offsetScaledLogVariance of A is lower than that of B.

See 7.6.3 of IEEE Std 1588-2019 for more detailed description of PTP variance and offsetScaledLogVariance. (Subclause 7.6.3.3 of IEEE Std 1588-2019 provides a detailed description of the computation of offsetScaledLogVariance from PTP variance, along with an example.)

8.6.2.5 priority2

priority2 is used in the execution of the BTCA (see 10.3). The value of priority2 shall be an integer selected from the range 0 through 255. The ordering of priority2 in the operation of the BTCA is the same as the ordering of priority1 (see 8.6.2.1).

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The default value of priority2 shall be 247 or 248. The default value for a PTP Relay Instance should be 247. The default value for a PTP End Instance should be 248. See 7.6.2.4 of IEEE Std 1588-2019 for a more detailed description of priority2.

NOTE—IEEE 802.1AS performance is improved when the number of hops between the Grandmaster PTP Instance and a timeReceiver PTP End Instance is reduced. When BTCA attributes are equal in a network, the preceding recommendations for priority2 select a PTP Relay Instance in order to reduce the number of hops (rather than use clockIdentity alone).

8.6.2.6 clockIdentity

The clockIdentity value for a PTP Instance shall be as specified in 8.5.2.2.

8.6.2.7 timeSource

The timeSource is an information only attribute indicating the type of source of time used by a ClockTimeTransmitter. The value is not used in the selection of the Grandmaster PTP Instance. The data type of timeSource shall be TimeSource, which is an Enumeration8. The values of TimeSource are specified in Table 8-2. These represent categories. For example, the global positioning system (GPS) entry includes not only the GPS system of the U.S. Department of Defense but the European Galileo system and other present and future GNSSs.

Table 8-2—TimeSource enumeration

Value	Time source	Description
0x10	ATOMIC_CLOCK	Any PTP Instance, or PTP Instance directly connected to such a device, that is based on atomic resonance for frequency and that has been calibrated against international standards for frequency and time.
0x20	GPS (see NOTE 1)	Any PTP Instance synchronized to any of the satellite systems that distribute time and frequency tied to international standards.
0x30	TERRESTRIAL_RADIO	Any PTP Instance synchronized via any of the radio distribution systems that distribute time and frequency tied to international standards.
0x40	PTP	Any PTP Instance synchronized to an IEEE 1588 PTP-based source of time external to the gPTP domain (see NOTE 2).
0x50	NTP	Any PTP Instance synchronized via the network time protocol (NTP) to servers that distribute time and frequency tied to international standards.
0x60	HAND_SET	Used in all cases for any PTP Instance whose time has been set by means of a human interface based on observation of an international standards source of time to within the claimed clock accuracy.

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Table 8-2—TimeSource enumeration *(continued)*

Value	Time source	Description
0x90	OTHER	Any source of time and/or frequency not covered by other values, or for which the source is not known.
0xA0	INTERNAL_OSCILLATOR	Any PTP Instance whose frequency is not based on atomic resonance nor calibrated against international standards for frequency, and whose time is based on a free-running oscillator with epoch determined in an arbitrary or unknown manner.
NOTE 1—In this standard, this value refers to any GNSS or Regional Navigation Satellite System (i.e., not only GPS). NOTE 2—For example, a clock that implements both a gPTP domain and a separate IEEE 1588 (i.e., PTP) domain, and is synchronized by the separate IEEE 1588 domain, would have time source of PTP in the gPTP domain.		

All unused values are reserved.

See 7.6.2.8 of IEEE Std 1588-2019 for a more detailed description of timeSource.

The initialization value is selected as follows:

a) If the timeSource (8.6.2.7 and Table 8-2) is known at the time of initialization, the value is derived from the table, else

b) The value is set to 0xA0 (INTERNAL_OSCILLATOR).

8.6.2.8 numberPorts

The numberPorts indicates the number of PTP Ports of the PTP Instance.

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9. Application interfaces

9.1 Overview of the interfaces

Unless otherwise stated, information in this clause is per domain.

The following subclauses define one application interface between the ClockSource entity and ClockTimeTransmitter entity (see 10.1.2) and four application interfaces between the ClockTarget entity and ClockTimeReceiver entity (see 10.1.2). The ClockSource is an entity that can be used as an external timing source for the gPTP domain. The ClockSource entity either contains or has access to a clock (see 3.3). The ClockTarget entity represents any application that uses information provided by the ClockTimeReceiver entity via any of the application interfaces.

NOTE 1—The manner in which the ClockSource entity obtains time from a clock is outside the scope of this standard. The manner in which the ClockTarget uses the information provided by application interfaces is outside the scope of this standard.

The five interfaces are illustrated in Figure 9-1:

a) ClockSourceTime interface, which provides external timing to a PTP Instance,
b) ClockTargetEventCapture interface, which returns the synchronized time of an event signaled by a ClockTarget entity,
c) ClockTargetTriggerGenerate interface, which causes an event to be signaled at a synchronized time specified by a ClockTarget entity,
d) ClockTargetClockGenerator interface, which causes a periodic sequence of results to be generated, with a phase and rate specified by a ClockTarget entity, and
e) ClockTargetPhaseDiscontinuity interface, which supplies information that an application can use to determine if a discontinuity in Grandmaster Clock phase or frequency has occurred.

NOTE 2—The application interfaces described in this clause are models for behavior and not application program interfaces. Other application interfaces besides items a) through e) in this subclause are possible, but are not described here. In addition, there can be multiple instances of a particular interface.

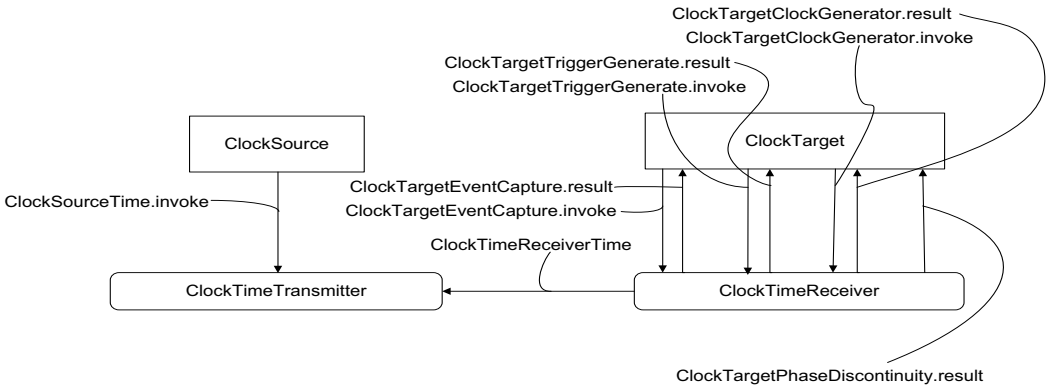


Figure 9-1—Application interfaces

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9.2 ClockSourceTime interface

9.2.1 General

This interface is used by the ClockSource entity to provide time to the ClockTimeTransmitter entity of a PTP Instance. The ClockSource entity invokes the ClockSourceTime.invoke function. The function provides the time, relative to the ClockSource, at which the function was invoked.

9.2.2 ClockSourceTime.invoke function parameters

```
ClockSourceTime.invoke {
        domainNumber,
        sourceTime,
        timeBaseIndicator,
        lastGmPhaseChange,
        lastGmFreqChange
}
```

9.2.2.1 domainNumber (UInteger8)

This parameter is the domainNumber of the gPTP domain to which this ClockSource entity is providing time.

9.2.2.2 sourceTime (ExtendedTimestamp)

The value of sourceTime is the time this function is invoked by the ClockSource entity.

9.2.2.3 timeBaseIndicator (UInteger16)

The timeBaseIndicator is a value that is set by the ClockSource entity. The ClockSource entity changes the value whenever its time base changes. The ClockSource entity shall change the value of timeBaseIndicator if and only if there is a phase or frequency change.

NOTE—While the clock that supplies time to the ClockSource entity can be lost, the ClockSource entity itself is not lost, the ClockSource entity ensures that timeBaseIndicator changes if the source of time is lost.

9.2.2.4 lastGmPhaseChange (ScaledNs)

The value of lastGmPhaseChange is the phase change (i.e., change in sourceTime) that occurred on the most recent change in timeBaseIndicator. The value is initialized to 0.

9.2.2.5 lastGmFreqChange (Float64)

The value of lastGmFreqChange is the fractional frequency change (i.e., frequency change expressed as a pure fraction) that occurred on the most recent change in timeBaseIndicator. The value is initialized to 0.

9.3 ClockTargetEventCapture interface

9.3.1 General

This interface is used by the ClockTarget entity to request the synchronized time of an event that it signals to the ClockTimeReceiver entity of a PTP Instance. The ClockTarget entity invokes the ClockTargetEventCapture.invoke function to signal an event to the ClockTimeReceiver entity. The

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ClockTimeReceiver entity invokes the ClockTargetEventCapture.result function to return the time of the event relative to the current Grandmaster Clock or, if no PTP Instance is grandmaster-capable, the LocalClock. The ClockTargetEventCapture.result function also returns gmPresent, to indicate to the ClockTarget whether a Grandmaster PTP Instance is present.

9.3.2 ClockTargetEventCapture.invoke parameters

```
ClockTargetEventCapture.invoke {
        domainNumber
}
```

9.3.2.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to provide the synchronized time of the signaled event.

9.3.3 ClockTargetEventCapture.result parameters

```
ClockTargetEventCapture.result {
        domainNumber,
        timeReceiverTimeCallback,
        gmPresent,
        isSynced,
        grandmasterIdentity
}
```

9.3.3.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is providing the synchronized time of the signaled event.

9.3.3.2 timeReceiverTimeCallback (ExtendedTimestamp)

The value of timeReceiverTimeCallback is the time, relative to the Grandmaster Clock, that the corresponding ClockTargetEventCapture.invoke function is invoked.

NOTE—The invocation of the ClockTargetEventCapture.invoke function and the detection of this invocation by the ClockTimeReceiver entity are simultaneous in this abstract interface.

9.3.3.3 gmPresent (Boolean)

The value of gmPresent is set equal to the value of the global variable gmPresent (see 10.2.4.13). This parameter indicates to the ClockTarget whether a Grandmaster PTP Instance is present.

9.3.3.4 isSynced (Boolean)

The value of the variable isSynced (see 18.4.1.1). This parameter shall be present if the optional hot standby feature is implemented (see Clause 18).

9.3.3.5 grandmasterIdentity (ClockIdentity)

If driftTrackingTlvSupport is TRUE and the value of the global variable syncGrandmasterIdentity is not 0xFFFFFFFFFFFFFFFF, grandmasterIdentity is the value of the global variable syncGrandmasterIdentity. Else, the value of grandmasterIdentity is the clockIdentity of the Grandmaster PTP Instance.

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NOTE—If the transfer of valid grandmasterIdentity information in the Drift_Tracking TLV provides a value that identifies the grandmaster upon which the API call is triggered or the time information conveyed, this value is used. In all other cases, the grandmasterIdentity as defined in 10.6.3.2.5 is used.

9.4 ClockTargetTriggerGenerate interface

9.4.1 General

This interface is used by the ClockTarget entity to request that the ClockTimeReceiver entity send a result at a specified time relative to the Grandmaster Clock. The ClockTarget entity invokes the ClockTargetTriggerGenerate.invoke function to indicate the synchronized time of the event. The ClockTimeReceiver entity invokes the ClockTargetTriggerGenerate.result function to either signal the event at the requested synchronized time or indicate an error condition.

9.4.2 ClockTargetTriggerGenerate.invoke parameters

```
ClockTargetTriggerGenerate.invoke {
        domainNumber,
        timeReceiverTimeCallback
}
```

9.4.2.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to signal an event at the specified time.

9.4.2.2 timeReceiverTimeCallback (ExtendedTimestamp)

If timeReceiverTimeCallback is nonzero, its value is the synchronized time the corresponding ClockTargetTriggerGenerate.result function, i.e., the trigger, is to be invoked. If timeReceiverTimeCallback is zero, any previous ClockTargetTriggerGenerate.invoke function for which a ClockTargetTriggerGenerate.result function has not yet been issued is canceled.

9.4.3 ClockTargetTriggerGenerate.result parameters

```
ClockTargetTriggerGenerate.result {
        domainNumber,
        errorCondition,
        gmPresent,
        isSynced,
        grandmasterIdentity
}
```

9.4.3.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is triggering an event at the specified time.

9.4.3.2 errorCondition (Boolean)

A value of FALSE indicates that the ClockTargetTriggerGenerate.result function was invoked at the time, relative to the Grandmaster Clock, contained in the corresponding ClockTargetTriggerGenerate.invoke function. A value of TRUE indicates that the ClockTargetTriggerGenerate.result function could not be

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invoked at the synchronized time contained in the corresponding ClockTargetTriggerGenerate.invoke function.

NOTE—For example, the ClockTargetTriggerGenerate.result function is invoked with errorCondition = TRUE if the requested timeReceiverTimeCallback is a time prior to the synchronized time when the corresponding ClockTargetTriggerGenerate.invoke function is invoked. As another example, the ClockTargetTriggerGenerate.result function is invoked with errorCondition = TRUE if a discontinuity in the synchronized time causes the requested timeReceiverTimeCallback to be skipped over.

9.4.3.3 gmPresent (Boolean)

As specified in 9.3.3.3.

9.4.3.4 isSynced (Boolean)

As specified in 9.3.3.4.

9.4.3.5 grandmasterIdentity (ClockIdentity)

As specified in 9.3.3.5.

9.4.4 ClockTargetTriggerGenerate interface definition

The invocation of the ClockTargetTriggerGenerate.invoke function causes the ClockTimeReceiver entity to store the value of the timeReceiverTimeCallback parameter in an internal variable (replacing any previous value of that variable) until the synchronized time, or LocalClock time if gmPresent is FALSE, equals the value of that variable, at which time the ClockTargetTriggerGenerate.result function is invoked with errorCondition = FALSE. If it is not possible to invoke the ClockTargetTriggerGenerate.result function at timeReceiverTimeCallback, e.g., if timeReceiverTimeCallback is earlier than the synchronized time (or LocalClock time if gmPresent is FALSE) when the ClockTargetTriggerGenerate.invoke function is invoked, the ClockTargetTriggerGenerate.result function is invoked with errorCondition = TRUE. Invocation of the ClockTargetTriggerGenerate.invoke function with timeReceiverTimeCallback = 0 (which is earlier than any synchronized time) is used to cancel a pending request.

9.5 ClockTargetClockGenerator interface

9.5.1 General

This interface is used by the ClockTarget entity to request that the ClockTimeReceiver entity deliver a periodic clock signal of specified period and phase. The ClockTarget entity invokes the ClockTargetClockGenerator.invoke function to request that the ClockTimeReceiver entity generate the periodic clock signal. The ClockTimeReceiver entity invokes the ClockTargetClockGenerator.result function at significant instants of the desired clock signal.

9.5.2 ClockTargetClockGenerator.invoke parameters

```
ClockTargetClockGenerator.invoke {
        domainNumber,
        clockPeriod,
        timeReceiverTimeCallbackPhase
}
```


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9.5.2.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is requested to deliver a periodic clock signal.

9.5.2.2 clockPeriod (TimeInterval)

The value of clockPeriod is the period between successive invocations of the ClockTargetClockGenerator.result function. A value that is zero or negative causes any existing periodic clock signal generated via this application interface to be terminated.

9.5.2.3 timeReceiverTimeCallbackPhase (ExtendedTimestamp)

The value of timeReceiverTimeCallbackPhase describes the phase of the generated clock signal by specifying a point on the timescale in use such that ClockTargetClockGenerator.result invocations occur at synchronized times that differ from timeReceiverTimeCallbackPhase by $n \times$ clockPeriod, where n is an integer.

NOTE—The value of timeReceiverTimeCallbackPhase can be earlier or later than the synchronized time the ClockTargetClockGenerator.invoke function is invoked; use of a timeReceiverTimeCallbackPhase value in the future does not imply that the initiation of the periodic clock signal is suppressed until that synchronized time.

9.5.3 ClockTargetClockGenerator.result parameters

```
ClockTargetClockGenerator.result {
        domainNumber,
        gmPresent,
        timeReceiverTimeCallback,
        isSynced,
        grandmasterIdentity
}
```

9.5.3.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is delivering a periodic clock signal.

9.5.3.2 gmPresent

As specified in 9.3.3.3.

9.5.3.3 timeReceiverTimeCallback (ExtendedTimestamp)

The value of timeReceiverTimeCallback is the synchronized time of this event.

9.5.3.4 isSynced (Boolean)

As specified in 9.3.3.4.

9.5.3.5 grandmasterIdentity (ClockIdentity)

As specified in 9.3.3.5.

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9.6 ClockTargetPhaseDiscontinuity interface

9.6.1 General

This interface provides discontinuity information, sent from the Grandmaster PTP Instance, to an application within an end station. It is used by the ClockTimeReceiver entity to supply sufficient information to the ClockTarget entity to enable the ClockTarget entity to determine whether a phase or frequency discontinuity has occurred. The ClockTimeReceiver invokes the ClockTargetPhaseDiscontinuity.result function in the SEND_SYNC_INDICATION block of the ClockTimeReceiverSync state machine (see 10.2.13 and Figure 10-9). The invocation occurs when a PortSyncSync structure is received, after the needed information has been computed by the ClockTimeReceiverSync state machine.

9.6.2 ClockTargetPhaseDiscontinuity.result parameters

```
ClockTargetPhaseDiscontinuity.result {
        domainNumber,
        grandmasterIdentity,
        gmTimeBaseIndicator,
        lastGmPhaseChange,
        lastGmFreqChange,
        isSynced
}
```

9.6.2.1 domainNumber (UInteger8)

This parameter is the domainNumber of the ClockTimeReceiver entity that is providing discontinuity information.

9.6.2.2 grandmasterIdentity (ClockIdentity)

If gmPresent (see 10.2.4.13) is TRUE, the value of grandmasterIdentity is the ClockIdentity of the current Grandmaster PTP Instance. If gmPresent is FALSE, the value of grandmasterIdentity is 0x0.

9.6.2.3 gmTimeBaseIndicator (UInteger16)

The value of gmTimeBaseIndicator is the timeBaseIndicator of the current Grandmaster PTP Instance.

9.6.2.4 lastGmPhaseChange (ScaledNs)

The value of the global lastGmPhaseChange parameter (see 10.2.4.16) received from the Grandmaster PTP Instance.

9.6.2.5 lastGmFreqChange (Float64)

The value of lastGmFreqChange parameter (see 10.2.4.17) received from the Grandmaster PTP Instance.

9.6.2.6 isSynced (Boolean)

As specified in 9.3.3.4.

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10. Media-independent layer specification

10.1 Overview

10.1.1 General

Unless otherwise stated, information in this clause is per domain.

10.1.2 Model of operation

A PTP Instance contains a best timeTransmitter selection function and a synchronization function. These functions include PTP Port-specific aspects and aspects associated with the PTP Instance as a whole. The functions are distributed among a number of entities, which together describe the behavior of a compliant implementation. The functions are specified by a number of state machines.

The model for the media-independent layer of a PTP Instance is shown in Figure 10-1. It includes a single SiteSync entity, ClockTimeTransmitter entity, and ClockTimeReceiver entity for the PTP Instance as a whole, plus one PortSync for each PTP Port. The PTP Instance also includes one MD entity for each PTP Port, which is part of the media-dependent layer. The media-dependent functions performed by the MD entity are described in the clauses for the respective media. In addition to the entities, Figure 10-1 shows the information that flows between the entities via the PortSyncSync, MDSyncSend, and MDSyncReceive structures (see 10.2.2.3, 10.2.2.1, and 10.2.2.2, respectively).

The SiteSync, ClockTimeTransmitter, ClockTimeReceiver, and PortSync entities each contain a number of cooperating state machines, which are described later in this clause (the MD entity state machines are described in the respective media-dependent clauses). The ClockTimeTransmitter entity receives information from an external time source, known as a *ClockSource entity* (see 9.2), via an application interface, and provides the information to the SiteSync entity. The ClockTimeReceiver entity receives Grandmaster Clock time-synchronization and current Grandmaster PTP Instance information from the SiteSync entity, and makes the information available to an external application, known as a *clockTarget entity* (see 9.3 through 9.6), via one or more application service interfaces. If the PTP Instance uses the BTCA, then the SiteSync entity executes the portion of best timeTransmitter clock selection associated with the PTP Instance as a whole, i.e., it uses the best timeTransmitter information received on each PTP Port to determine which PTP Port has received the best information, and updates the states of all the ports (see 10.3.1.1 for a discussion of PTP Port states). If the PTP Instance uses external port configuration, port roles are defined by engineering and the best timeTransmitter information received is not used for this purpose. It also distributes synchronization information received on the TimeReceiverPort to all the ports whose state is TimeTransmitterPort (see 10.3.1.1). The PortSync entity for a TimeReceiverPort receives best timeTransmitter selection information from the PTP Instance at the other end of the associated link, compares this to the current best timeTransmitter information that it has, and forwards the result of the comparison to the Site Sync entity. The PortSync entity for a TimeReceiverPort also receives time-synchronization information from the MD entity associated with the PTP Port, and forwards it to the SiteSync entity. The PortSync entity for a TimeTransmitterPort sends best timeTransmitter selection and time-synchronization information to the MD entity for the PTP Port, which in turn sends the respective messages.

NOTE—This clause does not require a one-to-one correspondence between the PortSync entities of PTP Instances attached to the same gPTP communication path (see 3.14), i.e., more than two PTP Instances can be attached to a gPTP communication path that uses a shared medium and meet the requirements of this clause. However, it is possible for a media-dependent clause to have additional requirements that limit the gPTP communication paths to point-to-point links for that medium; in this case, each link has exactly two PortSync entities, which can be considered to be in one-to-one correspondence. One example of this is the FDE media-dependent layer specified in Clause 11. In addition, one or more gPTP communication paths can be logically point-to-point but traverse the same shared medium.

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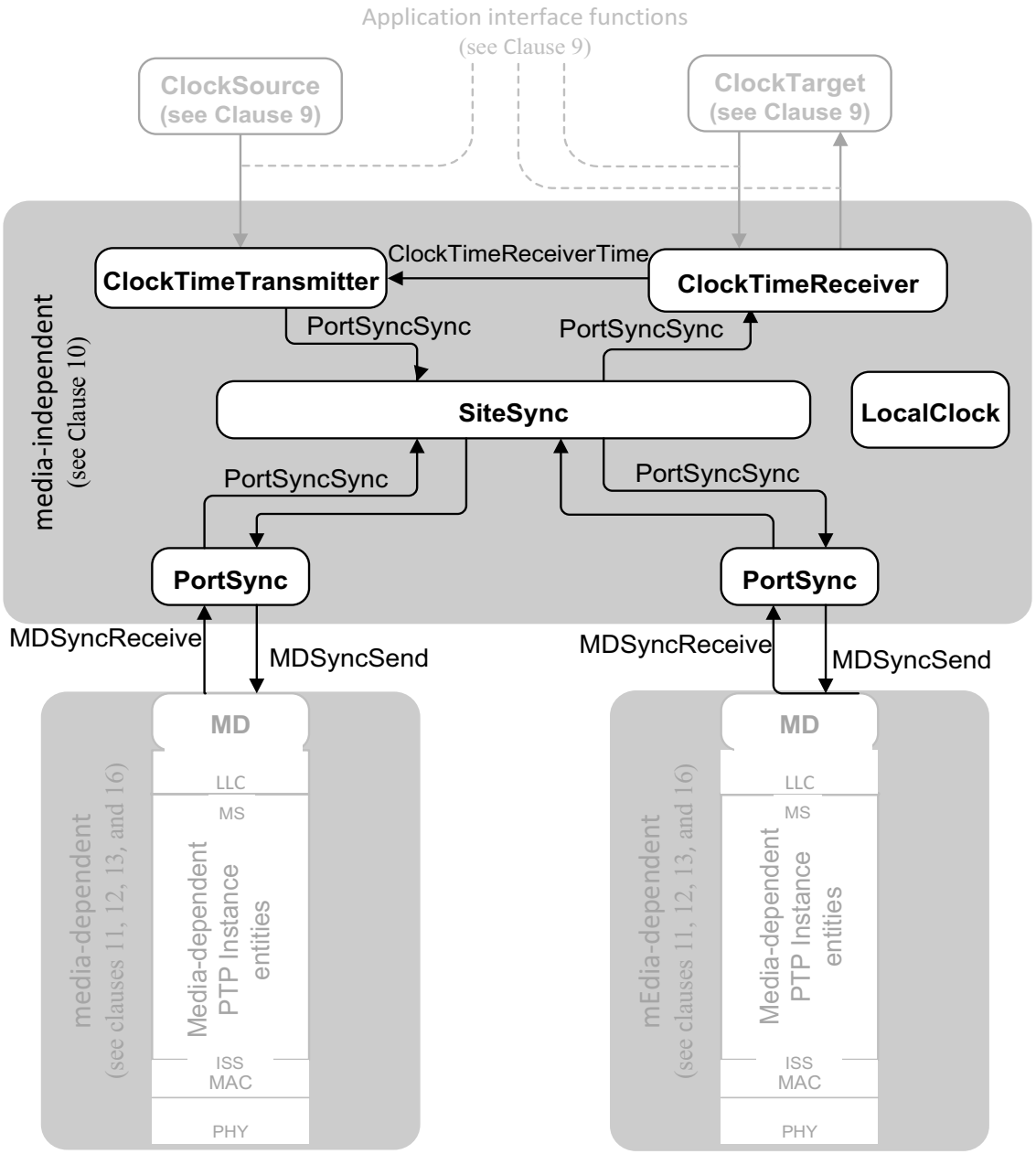


Figure 10-1—Model for media-independent layer of PTP Instance

The time-synchronization state machines are described in 10.2. The best timeTransmitter clock selection state machines are described in 10.3. The attributes and format of the Announce message are described in 10.5 and 10.6. The timing characterization of the protocol is described in 10.7.

10.1.2.1 LocalClock entity

The LocalClock entity is a free-running local clock (see 3.20) that provides a common time to the PTP Instance, relative to an arbitrary epoch. A PTP Instance contains a LocalClock entity. The requirements for the LocalClock entity are specified in B.1. All timestamps are taken relative to the LocalClock entity

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(see 8.4.3). The LocalClock entity also provides the value of currentTime (see 10.2.4.12), which is used in the state machines to specify the various timers.

NOTE—The epoch for the LocalClock entity can be the time that the PTP Instance is powered on.

10.1.3 Grandmaster-capable PTP Instance

A PTP Instance may be grandmaster-capable. An implementation may provide the ability to configure a PTP Instance as grandmaster-capable via a management interface.

NOTE 1—The managed object gmCapable is read only (see Table 14-1). gmCapable is configured by setting the value of the managed object priority1, which is read/write (see Table 14-1). If the value of priority1 is 255, then:

a) gmCapable is set to FALSE (see 8.6.2.1), and
b) The value of the managed object clockClass, which is read only, is set to 255 (see 8.6.2.2).

NOTE 2—While a PTP Instance that is not grandmaster-capable can never be the Grandmaster PTP Instance of the gPTP domain, such a PTP Instance contains a best timeTransmitter selection function, invokes the best timeTransmitter selection algorithm, and conveys synchronization information received from the current Grandmaster PTP Instance.

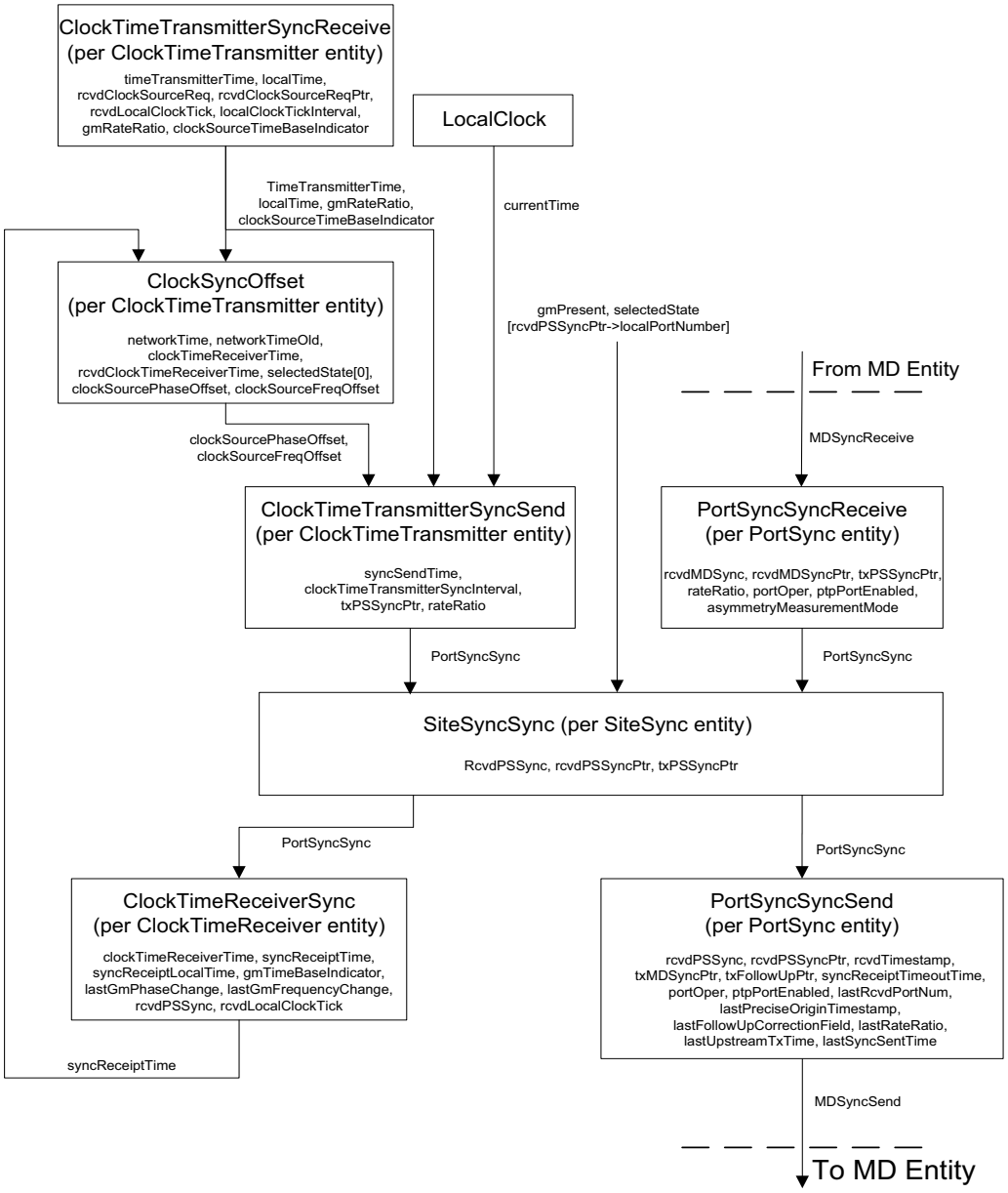
10.2 Time-synchronization state machines

10.2.1 Overview

The time-synchronization function in a PTP Instance is specified by a number of cooperating state machines. Figure 10-2 illustrates these state machines, their local variables, their interrelationships, and the global variables and structures used to communicate between them. The figure indicates the interaction between the state machines and the media-dependent layer and LocalClock entity.

The ClockTimeTransmitterSyncReceive, ClockTimeTransmitterSyncOffset, and ClockTimeTransmitterSyncSend state machines are optional for PTP Instances that are not grandmaster-capable (see 8.6.2.1 and 10.1.3). These state machines may be present in a PTP Instance that is not grandmaster-capable; however, any information supplied by them, via the ClockTimeTransmitterSyncSend state machine, to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

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Notes:
a) selectedState for each port and gmPresent are set by Port State Selection state machine (see 10.3.12)
b) currentTime is a global variable that is always equal to the current time relative to the local oscillator
c) application interfaces to higher layers are not shown
d) the ClockTimeTransmitterSyncReceive, ClockTimeTransmitterSyncSend, and ClockTimeTransmitterSyncOffset state machines are optional for PTP Instances that are not grandmaster-capable.

Figure 10-2—Time-synchronization state machines—overview and interrelationships

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The media-independent layer state machines in Figure 10-2 are as follows:

a) ClockTimeTransmitterSyncReceive (one instance per PTP Instance): receives ClockSourceTime.invoke functions from the ClockSource entity and notifications of LocalClock entity ticks (see 10.2.4.18), updates timeTransmitterTime, and provides timeTransmitterTime to ClockTimeTransmitterSyncOffset and ClockTimeTransmitterSyncSend state machines.

b) ClockTimeTransmitterSyncOffset (one instance per PTP Instance): receives syncReceiptTime from the ClockTimeReceiver entity and timeTransmitterTime from the ClockTimeTransmitterSyncReceive state machine, computes phase offset and frequency offset between timeTransmitterTime and syncReceiptTime if the PTP Instance is not the Grandmaster PTP Instance, and provides the frequency and phase offsets to the ClockTimeTransmitterSyncSend state machine.

c) ClockTimeTransmitterSyncSend (one instance per PTP Instance): receives timeTransmitterTime from the ClockTimeTransmitterSyncReceive state machine, receives phase and frequency offset between timeTransmitterTime and syncReceiptTime from the ClockTimeTransmitterSyncOffset state machine, and provides timeTransmitterTime (i.e., synchronized time) and the phase and frequency offset to the SiteSync entity using a PortSyncSync structure.

d) PortSyncSyncReceive (one instance per PTP Instance, per PTP Port): receives time-synchronization information from the MD entity of the corresponding PTP Port, computes accumulated rateRatio, computes syncReceiptTimeoutTime, and sends the information to the SiteSync entity.

e) SiteSyncSync (one instance per PTP Instance): receives time-synchronization information, accumulated rateRatio, and syncReceiptTimeoutTime from the PortSync entity of the current timeReceiver port or from the ClockTimeTransmitter entity; and sends the information to the PortSync entities of all the ports and to the ClockTimeReceiver entity.

f) PortSyncSyncSend (one instance per PTP Instance, per PTP Port): receives time-synchronization information from the SiteSync entity, requests that the MD entity of the corresponding PTP Port send a time-synchronization event message, receives the syncEventEgressTimestamp for this event message from the MD entity, uses the most recent time-synchronization information received from the SiteSync entity and the timestamp to compute time-synchronization information that will be sent by the MD entity in a general message (e.g., for FDE IEEE 802.3 media) or a subsequent event message (e.g., for IEEE 802.11 media), and sends this latter information to the MD entity.

g) ClockTimeReceiverSync (one instance per PTP Instance): receives time-synchronization information from the SiteSync entity; computes clockTimeReceiverTime and syncReceiptTime; sets syncReceiptLocalTime, GmTimeBaseIndicator, lastGmPhaseChange, and lastGmFreqChange; sends clockTimeReceiverTime to the ClockTimeTransmitter entity; and provides information to the ClockTarget entity (via the ClockTargetPhaseDiscontinuity interface; see 9.6) to enable that entity to determine if a phase or frequency discontinuity has occurred.

10.2.2 Data structures communicated between state machines

Subclauses 10.2.2.1 through 10.2.2.3 describe the data structures communicated between the time-synchronization state machines.

10.2.2.1 MDSyncSend

10.2.2.1.1 General

This structure contains information that is sent by the PortSync entity of a PTP Port to the MD entity of that PTP Port when requesting that the MD entity cause time-synchronization information to be sent. The structure contains information that reflects the most recent time-synchronization information received by this PTP Instance and is used to determine the contents of the time-synchronization event message and possibly separate general message that will be sent by this PTP Port.

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```
MDSyncSend      {
        domainNumber,
        followUpCorrectionField,
        sourcePortIdentity,
        logMessageInterval,
        preciseOriginTimestamp,
        upstreamTxTime,
        rateRatio,
        gmTimeBaseIndicator,
        lastGmPhaseChange,
        lastGmFreqChange
}
```

The members of the structure are defined in the following subclauses.

10.2.2.1.2 domainNumber (UInteger8)

This parameter is the domainNumber of the gPTP domain in which this structure is sent.

NOTE—The domainNumber member is not essential because the state machines that send and receive this structure are per domain, and each state machine implicitly knows the number of the domain in which it operates.

10.2.2.1.3 followUpCorrectionField (ScaledNs)

The followUpCorrectionField contains the accumulated time since the preciseOriginTimestamp was captured by the Grandmaster PTP Instance. This is equal to the elapsed time, relative to the Grandmaster Clock, between the time the Grandmaster PTP Instance sent the received time-synchronization event message, truncated to the nearest nanosecond, and the time at which that event message was sent by the upstream PTP Instance. The followUpCorrectionField is equal to the value of the followUpCorrectionField member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.5).

10.2.2.1.4 sourcePortIdentity (PortIdentity)

The sourcePortIdentity is the portIdentity of this PTP Port (see 8.5.2).

10.2.2.1.5 logMessageInterval (Integer8)

The logMessageInterval is the value of currentLogSyncInterval for this PTP Port (see 10.7.2.3).

10.2.2.1.6 preciseOriginTimestamp (Timestamp)

The preciseOriginTimestamp is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP Instance, with any fractional nanoseconds truncated, when the received time-synchronization information was sent by the Grandmaster PTP Instance. The preciseOriginTimestamp is the value of the preciseOriginTimestamp member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.8).

10.2.2.1.7 upstreamTxTime (UScaledNs)

The upstreamTxTime is given by the following equation:

$$\text{upstreamTxTime} = \text{syncEventIngressTimestamp} - \frac{\text{meanLinkDelay}}{\text{neighborRateRatio}}$$

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where

syncEventIngressTimestamp	corresponds to the receipt of the time-synchronization information at the timeReceiver port of this PTP Instance
meanLinkDelay	is defined in 10.2.5.8
neighborRateRatio	is defined in 10.2.5.7
upstreamTxTime	is the value of the upstreamTxTime member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.9)

NOTE 1—When using Clause 11 MD entity, nrrPdelay is used instead of neighborRateRatio (see 11.2.12.14).

NOTE 2—Depending on the medium, delayAsymmetry also needs to be considered.

10.2.2.1.8 rateRatio (Float64)

The rateRatio is the value of the rateRatio member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.10). It is equal to the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of this PTP Instance (see 10.2.8.1.4).

10.2.2.1.9 gmTimeBaseIndicator (UInteger16)

The gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity of the current Grandmaster PTP Instance. It is set equal to the gmTimeBaseIndicator of the received time-synchronization information. The gmTimeBaseIndicator is the value of the gmTimeBaseIndicator member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.11).

10.2.2.1.10 lastGmPhaseChange (ScaledNs)

The lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or the step change in the time of the current Grandmaster Clock at the time of the most recent gmTimeBaseIndicator change. It is set equal to the lastGmPhaseChange of the received time-synchronization information. The lastGmPhaseChange is the value of the lastGmPhaseChange member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.12).

10.2.2.1.11 lastGmFreqChange (Float64)

The lastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator. It is set equal to the lastGmFreqChange of the received time-synchronization information. The lastGmFreqChange is the value of the lastGmFreqChange member of the most recently received PortSyncSync structure from the PortSync entity of this PTP Port (see 10.2.2.3.13).

10.2.2.2 MDSyncReceive

10.2.2.2.1 General

This structure contains information that is sent by the MD entity of a PTP Port to the PortSync entity of that PTP Port. It provides the PortSync entity with timeTransmitter clock timing information and timestamp of receipt of a time-synchronization event message compensated for propagation time on the upstream link. The information is sent to the PortSync entity upon receipt of time-synchronization information by the MD entity of the PTP Port. The information is in turn provided by the PortSync entity to the SiteSync entity. The

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information is used by the PortSyncSyncReceive state machine of the PortSync entity to compute the rate ratio of the Grandmaster Clock relative to the local clock and is communicated to the SiteSync entity, and then the SiteSync entity communicates it to the other PortSync entities for use in computing timeTransmitter clock timing information.

```
MDSyncReceive {
        domainNumber,
        followUpCorrectionField,
        sourcePortIdentity,
        logMessageInterval,
        preciseOriginTimestamp,
        upstreamTxTime,
        rateRatio,
        gmTimeBaseIndicator,
        lastGmPhaseChange,
        lastGmFreqChange
}
```

The members of the structure are defined in the following subclauses.

10.2.2.2.2 domainNumber (UInteger8)

This parameter is the domainNumber of the gPTP domain in which this structure is sent.

NOTE—The domainNumber member is not essential because the state machines that send and receive this structure are per domain, and each state machine implicitly knows the number of the domain in which it operates.

10.2.2.2.3 followUpCorrectionField (ScaledNs)

The followUpCorrectionField contains the elapsed time, relative to the Grandmaster Clock, between the time the Grandmaster PTP Instance sent the received time-synchronization information, truncated to the nearest nanosecond, and the time at which this information was sent by the upstream PTP Instance.

NOTE 1—The sum of followUpCorrectionField and preciseOriginTimestamp is the synchronized time that corresponds to the time the most recently received time-synchronization event message was sent by the upstream PTP Instance.

NOTE 2—For a medium that uses separate event and general messages (for example, FDE media described in Clause 11), the event message corresponding to the most recently received network synchronization information is the event message that corresponds to the most recently received general message. For a medium that places synchronization information based on the event message timestamp in the next event message (for example, IEEE 802.11 media described in Clause 12), the event message corresponding to the most recently received network synchronization information is the previous event message; in this case, the time-synchronization information in the current event message refers to the previous event message.

10.2.2.2.4 sourcePortIdentity (PortIdentity)

The sourcePortIdentity is the value of the sourcePortIdentity of the time-synchronization event message received by this PTP Port. It is the portIdentity of the upstream TimeTransmitterPort that sent the event message.

10.2.2.2.5 logMessageInterval (Integer8)

The logMessageInterval is the value of the logMessageInterval of the time-synchronization event message received by this PTP Port. It is the currentLogSyncInterval (see 10.7.2.3) of the upstream TimeTransmitterPort that sent the event message.

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10.2.2.2.6 preciseOriginTimestamp (Timestamp)

The preciseOriginTimestamp is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP Instance, with any fractional nanoseconds truncated, when the time-synchronization event message was sent by the Grandmaster PTP Instance.

10.2.2.2.7 upstreamTxTime (UScaledNs)

The upstreamTxTime is given by the following equation:

$$\text{upstreamTxTime} = \text{syncEventIngressTimestamp} - \frac{\text{meanLinkDelay}}{\text{neighborRateRatio}}$$

where

syncEventIngressTimestamp	corresponds to the receipt of the time-synchronization information at the timeReceiver port of this PTP Instance (i.e., at this PTP Port)
meanLinkDelay	is defined in 10.2.5.8
neighborRateRatio	is defined in 10.2.5.7

NOTE—When using Clause 11 MD entity, nrrPdelay is used instead of neighborRateRatio (see 11.2.12.14).

10.2.2.2.8 rateRatio (Float64)

The rateRatio is the value of rateRatio of the received time-synchronization information. It is equal to the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of the PTP Instance at the other end of the link attached to this PTP Port, i.e., the PTP Instance that sent the most recently received time-synchronization event message (see 10.2.8.1.4).

10.2.2.2.9 gmTimeBaseIndicator (UInteger16)

The gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity of the current Grandmaster PTP Instance. It is set equal to the gmTimeBaseIndicator of the received time-synchronization information.

10.2.2.2.10 lastGmPhaseChange (ScaledNs)

The lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or the step change in the time of the current Grandmaster Clock at the time of the most recent gmTimeBaseIndicator change. It is set equal to the lastGmPhaseChange of the received time-synchronization information.

10.2.2.2.11 lastGmFreqChange (Float64)

The lastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator. It is set equal to the lastGmFreqChange of the received time-synchronization information.

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10.2.2.3 PortSyncSync

10.2.2.3.1 General

This structure is sent by the PortSync and ClockTimeTransmitter entities to the SiteSync entity and also from the SiteSync entity to the PortSync and ClockTimeReceiver entities.

When sent from the PortSync or ClockTimeTransmitter entity, it provides the SiteSync entity with timeTransmitter clock timing information, timestamp of receipt of a time-synchronization event message compensated for propagation time on the upstream link, and the time at which sync receipt timeout occurs if a subsequent Sync message is not received by then. The information is used by the SiteSync entity to compute the rate ratio of the Grandmaster Clock relative to the local clock and is communicated to the other PortSync entities for use in computing timeTransmitter clock timing information.

When sent from the SiteSync entity to the PortSync or ClockTimeReceiver entity, the structure contains information needed to compute the synchronization information that will be included in respective fields of the time-synchronization event and general messages that will be sent and also to compute the synchronized time that the ClockTimeReceiver entity will supply to the ClockTarget entity.

```
PortSyncSync    {
        domainNumber,
        localPortNumber,
        syncReceiptTimeoutTime,
        followUpCorrectionField,
        sourcePortIdentity,
        logMessageInterval,
        preciseOriginTimestamp,
        upstreamTxTime,
        rateRatio,
        gmTimeBaseIndicator,
        lastGmPhaseChange,
        lastGmFreqChange
}
```

The parameters of the PortSyncSync structure are defined in the following subclauses for when the structure is sent from the PortSync or ClockTimeTransmitter entity to the SiteSync entity. If the structure is sent from the SiteSync entity to the PortSync or ClockTimeReceiver entity, the member values are copied from the most recently received PortSyncSync structure where the PTP Port that received this structure has PTP Port state of TimeReceiverPort.

10.2.2.3.2 domainNumber (UInteger8)

This parameter is the domainNumber of the gPTP domain in which this structure is sent.

NOTE—The domainNumber member is not essential because the state machines that send and receive this structure are per domain, and each state machine implicitly knows the number of the domain in which it operates.

10.2.2.3.3 localPortNumber (UInteger16)

If the structure is sent by a PortSync entity, the localPortNumber is the port number of the PTP Port whose PortSync entity sent this structure. If the structure is sent by a ClockTimeTransmitter entity, the localPortNumber is zero.

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10.2.2.3.4 syncReceiptTimeoutTime (UScaledNs)

If the structure is sent by a PortSync entity, the syncReceiptTimeoutTime is the value of the local time (i.e., the free-running, local clock time) at which sync receipt timeout occurs if a subsequent time-synchronization event message is not received by that time. If the structure is sent by a ClockTimeTransmitter entity, the syncReceiptTimeoutTime is 0xFFFFFFFFFFFFFFFF [see item h) in 10.2.9.2.1].

10.2.2.3.5 followUpCorrectionField (ScaledNs)

If the structure is sent by a PortSync entity, the followUpCorrectionField is the value of the followUpcorrectionField member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.2). If the structure is sent by a ClockTimeTransmitter entity, the followUpCorrectionField is the sub-nanosecond portion of the ClockTimeTransmitter time.

10.2.2.3.6 sourcePortIdentity (PortIdentity)

If the structure is sent by a PortSync entity, the sourcePortIdentity is the value of the sourcePortIdentity member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.4). If the structure is sent by a ClockTimeTransmitter entity, the clockIdentity member of the sourcePortIdentity is the clockIdentity of this PTP Instance, and the portNumber member of the sourcePortIdentity is 0.

10.2.2.3.7 logMessageInterval (Integer8)

If the structure is sent by a PortSync entity, the logMessageInterval is the value of the logMessageInterval member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.5). If the structure is sent by a ClockTimeTransmitter entity, the logMessageInterval is the value of clockTimeTransmitterLogSyncInterval (see 10.7.2.4).

10.2.2.3.8 preciseOriginTimestamp (Timestamp)

If the structure is sent by a PortSync entity, the preciseOriginTimestamp is the value of the preciseOriginTimestamp member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.6). If the structure is sent by a ClockTimeTransmitter entity, the preciseOriginTimestamp is the ClockTimeTransmitter time truncated to the next lower nanosecond.

10.2.2.3.9 upstreamTxTime (UScaledNs)

If the structure is sent by a PortSync entity, the upstreamTxTime is the value of the upstreamTxTime member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.7). If the structure is sent by a ClockTimeTransmitter entity, the upstreamTxTime is the local clock time corresponding to the ClockTimeTransmitter time.

10.2.2.3.10 rateRatio (Float64)

If the structure is sent by a PortSync entity, the rateRatio is the value of the rateRatio member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.8). It is equal to the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of the PTP Instance at the other end of the link attached to this PTP Port, i.e., the PTP Instance that sent the most recently-received time-synchronization event message (see 10.2.8.1.4). If the structure is sent by a ClockTimeTransmitter entity, the rateRatio is equal to gmRateRatio (see 10.2.4.14).

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10.2.2.3.11 gmTimeBaseIndicator (UInteger16)

If the structure is sent by a PortSync entity, the gmTimeBaseIndicator is the value of the gmTimeBaseIndicator member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.9). If the structure is sent by a ClockTimeTransmitter entity, the gmTimeBaseIndicator is equal to clockSourceTimeBaseIndicator (see 10.2.4.8).

10.2.2.3.12 lastGmPhaseChange (ScaledNs)

If the structure is sent by a PortSync entity, the lastGmPhaseChange is the value of the lastGmPhaseChange member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.9). If the structure is sent by a ClockTimeTransmitter entity, the lastGmPhaseChange is equal to clockSourcePhaseOffset (see 10.2.4.7).

10.2.2.3.13 lastGmFreqChange (Float64)

If the structure is sent by a PortSync entity, the lastGmFreqChange is the value of the lastGmFreqChange member of the MDSyncReceive structure whose receipt caused the sending of this structure (see 10.2.2.2.9). If the structure is sent by a ClockTimeTransmitter entity, the lastGmFreqChange is equal to clockSourceFreqOffset (see 10.2.4.6).

10.2.3 Overview of global variables used by time synchronization state machines

Subclauses 10.2.4 and 10.2.5 define global variables used by time synchronization state machines whose scopes are as follows:

— Per PTP Instance (i.e., per domain)
— Per PTP Instance, per PTP Port
— Instance used by the CMLDS (see 11.2.16) (i.e., variable is common across all Link Ports)
— Instance used by CMLDS, per Link Port

Table 10-1 summarizes the scope of each global variable of 10.2.4 and 10.2.5.

Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5)

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
BEGIN	10.2.4.1	Boolean	Per PTP Instance	Yes	No
clockTimeTransmitterSyncInterval	10.2.4.2	UScaledNs	Per PTP Instance	No	No
clockTimeReceiverTime	10.2.4.3	ExtendedTimestamp	Per PTP Instance	No	No
syncReceiptTime	10.2.4.4	ExtendedTimestamp	Per PTP Instance	No	No

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Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5) *(continued)*

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
syncReceiptLocalTime	10.2.4.5	UScaledNs	Per PTP Instance	No	No
clockSourceFreqOffset	10.2.4.6	Float64	Per PTP Instance	No	No
clockSourcePhaseOffset	10.2.4.7	ScaledNs	Per PTP Instance	No	No
clockSourceTimeBaseIndicator	10.2.4.8	UInteger16	Per PTP Instance	No	No
clockSourceTimeBaseIndicatorOld	10.2.4.9	UInteger16	Per PTP Instance	No	No
clockSourceLastGmPhaseChange	10.2.4.10	ScaledNs	Per PTP Instance	No	No
clockSourceLastGmFreqChange	10.2.4.11	Float64	Per PTP Instance	No	No
currentTime	10.2.4.12	UScaledNs	Per PTP Instance	No	No
gmPresent	10.2.4.13	Boolean	Per PTP Instance	No	No
gmRateRatio	10.2.4.14	Float64	Per PTP Instance	No	No
gmTimeBaseIndicator	10.2.4.15	UInteger16	Per PTP Instance	No	No
lastGmPhaseChange	10.2.4.16	ScaledNs	Per PTP Instance	No	No
lastGmFreqChange	10.2.4.17	Float64	Per PTP Instance	No	No
localClockTickInterval	10.2.4.18	TimeInterval	Per PTP Instance	No	No
localTime	10.2.4.19	UScaledNs	Per PTP Instance	No	No
selectedState	10.2.4.20	Enumeration2 array	Per PTP Instance	No	No
timeTransmitterTime	10.2.4.21	ExtendedTimestamp	Per PTP Instance	No	No
thisClock	10.2.4.22	ClockIdentity	Per PTP Instance	Yes	No
parentLogSyncInterval	10.2.4.23	Integer8	Per PTP Instance	No	No
instanceEnable	10.2.4.24	Boolean	Per PTP Instance	No	No

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Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5) *(continued)*

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
syncReceiptTimeoutTime	10.2.4.25	UScaledNs	Per PTP Instance	No	No
asCapable	10.2.5.1	Boolean	Per PTP Port	No	No
asymmetryMeasurementMode[a]	10.2.5.2	Boolean	Per PTP Port	No	Yes
syncReceiptTimeoutTimeInterval	10.2.5.3	UScaledNs	Per PTP Port	No	No
currentLogSyncInterval	10.2.5.4	Integer8	Per PTP Port	No	No
initialLogSyncInterval	10.2.5.5	Integer8	Per PTP Port	No	No
syncInterval	10.2.5.6	UScaledNs	Per PTP Port	No	No
neighborRateRatio	10.2.5.7	Float64	Per PTP Port	No	No
meanLinkDelay	10.2.5.8	ScaledNs	Per PTP Port	No	Yes
delayAsymmetry	10.2.5.9	ScaledNs	Per PTP Port	No	Yes
computeNeighborRateRatio	10.2.5.10	Boolean	Per PTP Port	No	Yes
computeMeanLinkDelay	10.2.5.11	Boolean	Per PTP Port	No	Yes
portOper[a]	10.2.5.12	Boolean	Per PTP Port	No	Yes
ptpPortEnabled	10.2.5.13	Boolean	Per PTP Port	No	No
thisPort	10.2.5.14	UInteger16	Per PTP Port	No	Yes
syncLocked	10.2.5.15	Boolean	Per PTP Port	No	No
neighborGptpCapable	10.2.5.16	Boolean	Per PTP Port	No	No
syncSlowdown	10.2.5.17	Boolean	Per PTP Port	No	No
oldSyncInterval	10.2.5.18	UScaledNs	Per PTP Port	No	No
gPtpCapableMessageSlowdown	10.2.5.19	Boolean	Per PTP Port	No	No
gPtpCapableMessageInterval	10.2.5.20	UScaledNs	Per PTP Port	No	No
oldGptpCapableMessageInterval	10.2.5.21	UScaledNs	Per PTP Port	No	No
currentLogGptpCapableMessageInterval	10.2.5.22	Integer8	Per PTP Port	No	No
initialLogGptpCapableMessageInterval	10.2.5.23	Integer8	Per PTP Port	No	No
syncGrandmasterIdentity	10.2.4.26	ClockIdentity	Per PTP Instance	No	No
syncStepsRemoved	10.2.4.27	UInteger16	Per PTP Instance	No	No
driftTrackingTlvSupport	10.2.4.28	Boolean	Per PTP Instance	No	No

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Table 10-1—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5) *(continued)*

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
rcvdPSSyncCSS	10.2.4.29	Boolean	Per PTP Instance	No	No
rcvdLocalClockTickCSS	10.2.4.30	Boolean	Per PTP Instance	No	No
rateRatioDrift	10.2.4.31	Integer32	Per PTP Instance	No	No

[a] There is one instance of this variable per physical port, which is accessible by all PTP Ports and Link Ports associated with the physical port.

10.2.4 Per PTP Instance global variables

10.2.4.1 BEGIN: A Boolean controlled by the system initialization. If BEGIN is true, all state machines, including per-PTP Port state machines, continuously execute their initial state. See Annex E of IEEE Std 802.1Q-2022.

10.2.4.2 clockTimeTransmitterSyncInterval: A variable containing the mean time interval between successive messages providing time-synchronization information by the ClockTimeTransmitter entity to the SiteSync entity. This value is given by $1\,000\,000\,000 \times 2^{\text{clockTimeTransmitterLogSyncInterval}}$ ns, where clockTimeTransmitterLogSyncInterval is the logarithm to base 2 of the mean time between the successive providing of time-synchronization information by the ClockTimeTransmitter entity (see 10.7.2.4).

10.2.4.3 ClockTimeReceiverTime: The synchronized time maintained, at the timeReceiver, at the granularity of the LocalClock entity [i.e., a new value is computed every localClockTickInterval (see 10.2.4.18) by the ClockTimeReceiver entity].

10.2.4.4 syncReceiptTime: The synchronized time computed by the ClockTimeReceiver entity at the instant time-synchronization information, contained in a PortSyncSync structure, is received.

10.2.4.5 syncReceiptLocalTime: The value of currentTime (i.e., the time relative to the LocalClock entity) corresponding to syncReceiptTime.

10.2.4.6 clockSourceFreqOffset: The fractional frequency offset of the ClockSource entity frequency relative to the current Grandmaster Clock frequency.

10.2.4.7 clockSourcePhaseOffset: The time provided by the ClockSource entity, minus the synchronized time.

10.2.4.8 clockSourceTimeBaseIndicator: A global variable that is set equal to the timeBaseIndicator parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.3), by the ClockTimeTransmitter entity. The parameter timeBaseIndicator of ClockSourceTime.invoke is set by the ClockSource entity and is changed by that entity whenever the time base changes.

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10.2.4.9 clockSourceTimeBaseIndicatorOld: A global variable that is set equal to the previous value of clockSourceTimeBaseIndicator.

10.2.4.10 clockSourceLastGmPhaseChange: A global variable that is set equal to the lastGmPhaseChange parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.4). That parameter is set by the ClockSource entity and is changed by that entity whenever the time base changes.

10.2.4.11 clockSourceLastGmFreqChange: A global variable that is set equal to the lastGmFreqChange parameter of the ClockSourceTime.invoke application interface function (see 9.2.2.5). That parameter is set by the ClockSource entity and is changed by that entity whenever the time base changes.

10.2.4.12 currentTime: The current value of time relative to the LocalClock entity clock.

10.2.4.13 gmPresent: A Boolean that indicates whether a grandmaster-capable PTP Instance is present in the domain. If TRUE, a grandmaster-capable PTP Instance is present; if FALSE, a grandmaster-capable PTP Instance is not present.

10.2.4.14 gmRateRatio: The measured ratio of the frequency of the ClockSource entity to the frequency of the LocalClock entity.

10.2.4.15 gmTimeBaseIndicator: The most recent value of gmTimeBaseIndicator provided to the ClockTimeReceiverSync state machine via a PortSyncSync structure. The data type for gmTimeBaseIndicator is UInteger16.

10.2.4.16 lastGmPhaseChange: The most recent value of lastGmPhaseChange provided to the ClockTimeReceiverSync state machine via a PortSyncSync structure.

10.2.4.17 lastGmFreqChange: The most recent value of lastGmFreqChange provided to the ClockTimeReceiverSync state machine via a PortSyncSync structure.

10.2.4.18 localClockTickInterval: The time interval between two successive significant instants (i.e., “ticks”) of the LocalClock entity.

10.2.4.19 localTime: The value of currentTime when the most recent ClockSourceTime.invoke function (see 9.2) was received from the ClockSource entity, or when the LocalClock entity most recently updated its time.

10.2.4.20 selectedState: An Enumeration2 array of length numberPorts + 1 (see 8.6.2.8). selectedState[*j*] is set equal to the PTP Port State (see Table 10-2) of the PTP Port whose portList index is *j*.

10.2.4.21 timeTransmitterTime: The time maintained by the ClockTimeTransmitter entity, based on information received from the ClockSource and LocalClock entities.

10.2.4.22 thisClock: The clockIdentity of the current PTP Instance.

10.2.4.23 parentLogSyncInterval: The most recent logMessageInterval value received on the timeReceiver port. If this PTP Instance is the Grandmaster PTP Instance, then this is the clockTimeTransmitterLogSyncInterval (see 10.7.2.4).

10.2.4.24 instanceEnable: A per-domain Boolean used to enable gPTP on all ports that are enabled for that domain (i.e., ports for which portOper and ptpPortEnabled are both TRUE). Setting instanceEnable to FALSE causes all per-domain state machines to go to the initial state.

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NOTE—instanceEnable has no effect on the operation of the MDPdelayReq (see 11.2.18) and MDPdelayResp (see 11.2.19) state machines because those state machines are not per domain (i.e., there is a single instance of each of those state machines, per link, for all domains).

10.2.4.25 syncReceiptTimeoutTime: The value of the syncReceiptTimeoutTime member of the most recently received PortSyncSync structure.

10.2.4.26 syncGrandmasterIdentity: The clockIdentity carried in the syncGrandmassterIdentity field of the Drift_Tracking TLV carried by the most recently received Sync message (twoStep flag FALSE) or Follow_Up message (twoStep flag TRUE). If the received Sync or Follow_Up message does not carry a Drift_Tracking TLV, syncGrandmasterIdentity is set to the null value 0xFFFF FFFF FFFF FFFF (see the section "Unassigned and NULL EUI values" of the IEEE Registration Authority tutorial "Guidelines for Use of Extended Unique Identifier (EUI), Organizationally Unique Identifier (OUI), and Company ID (CID)" [B7]).

10.2.4.27 syncStepsRemoved: The value of the syncStepsRemoved field of the Drift_Tracking TLV carried by the most recently received Sync message (twoStep flag FALSE) or Follow_Up message (twoStep flag TRUE). If the received Sync or Follow_Up message does not carry a Drift_Tracking TLV, syncstepsRemoved is set to 0xFFFF.

10.2.4.28 driftTrackingTlvSupport: An indicator of whether the PTP Instance supports the Drift_Tracking TLV and the feature is enabled. The value is TRUE if the Drift_Tracking TLV is supported and the managed object driftTrackingTlvSupportEnabled (see 14.9) is TRUE. The value is FALSE if the Drift_Tracking TLV is not supported, or if the TLV is supported and the managed object driftTrackingTlvSupport is FALSE.

NOTE—The Drift_Tracking TLV is transported only on FDE links (as specified in Clause 11) and HDE links (as specified in Clause 19). Even if driftTrackingTlvSupport is TRUE, the Drift_Tracking TLV is not transported on links other than FDE and HDE links, and is not received by the PTP Ports at the other end of these links.

10.2.4.29 rcvdPSSyncCSS: A Boolean variable that is set to TRUE when a PortSyncSync structure is received from the SiteSyncSync state machine of the SiteSync entity by the ClockTimeReceiverSync state machine. This variable is reset by the ClockTimeReceiverSync state machine.

10.2.4.30 rcvdLocalClockTickCSS: A Boolean variable that is set to TRUE when the LocalClock entity updates its time. This variable is reset by the ClockTimeReceiverSync state machine.

10.2.4.31 rateRatioDrift: The value of the rateRatioDrift field of the Drift_Tracking TLV carried by the most recently received Sync message (twoStep flag FALSE) or Follow_Up message (twoStep flag TRUE). If the received Sync or Follow_Up message does not carry a Drift_Tracking TLV, rateRatioDrift is set to 0xFFFFFFFF.

10.2.5 Per-port global variables

10.2.5.1 asCapable: A Boolean that is TRUE if and only if it is determined that this PTP Instance and the PTP Instance at the other end of the link attached to this PTP Port can interoperate with each other via the IEEE 802.1AS protocol. As a result,

a) This PTP Instance is capable of executing the IEEE 802.1AS protocol,

b) The PTP Instance at the other end of the link is capable of executing the IEEE 802.1AS protocol, and

c) There are no non-IEEE-802.1AS systems in between this PTP Instance and the PTP Instance at the other end of the link that introduce sufficient impairments that the end-to-end time-synchronization performance of B.3 cannot be met.

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The determination of asCapable is different for each medium and is described in the respective media-dependent clauses.

NOTE—For FDE links, asCapableAcrossDomains (see 11.2.12.13) is common across, and accessible by, all the domains. For HDE links, asCapableAcrossDomains is a per-PTP Instance (i.e., per domain). asCapableAcrossDomains is computed by the MDPdelayReq state machine (see 11.2.18). For FDE links (see Clause 11) and HDE links (see Clause 19), asCapableAcrossDomains is used when setting the instance of asCapable for each domain (for the link in question).

10.2.5.2 asymmetryMeasurementMode: A Boolean that contains the value of the managed object asymmetryMeasurementMode (see 14.10.46). For FDE IEEE 802.3 media, the value is TRUE if an asymmetry measurement is being performed for the link attached to this port and FALSE otherwise. For all other media, the value is FALSE. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

10.2.5.3 syncReceiptTimeoutTimeInterval: The time interval after which sync receipt timeout occurs if time-synchronization information has not been received during the interval. The value of syncReceiptTimeoutTimeInterval is equal to syncReceiptTimeout (see 10.7.3.1) multiplied by the syncInterval (see 10.2.5.6) for the PTP Port at the other end of the link to which this PTP Port is attached. The value of syncInterval for the PTP Port at the other end of the link is computed from logMessageInterval of the received Sync message (see 10.6.2.2.14).

10.2.5.4 currentLogSyncInterval: The current value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive time-synchronization event messages (see 10.7.2.3). This value is set in the SyncIntervalSetting state machine (see 10.2.14).

10.2.5.5 initialLogSyncInterval: The initial value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive time-synchronization event messages (see 10.7.2.3).

10.2.5.6 syncInterval: A variable containing the mean time-synchronization event message transmission interval for the PTP Port. This value is set in the SyncIntervalSetting state machine (see 10.2.14).

10.2.5.7 neighborRateRatio: The measured ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the link attached to this port, to the frequency of the LocalClock entity of this time-aware system. There is one instance of this variable for each domain, i.e., each PTP Instance (per port).

10.2.5.8 meanLinkDelay: The measured mean propagation delay (see 8.3) on the link attached to this port, relative to the LocalClock entity of the time-aware system at the other end of the link (i.e., expressed in the time base of the time-aware system at the other end of the link). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

NOTE—The variable meanLinkDelay was named neighborPropDelay in the 2011 edition of this standard.

10.2.5.9 delayAsymmetry: The asymmetry in the propagation delay on the link attached to this port. If propagation delay asymmetry is not modeled, then delayAsymmetry is zero. There is one instance of this variable for CMLDS (see 11.2.16), and there is also one instance of this variable for each domain. The instance of this variable for CMLDS is relative to the local clock. The instance of this variable for each domain is relative to the grandmaster time base for that domain. The instance of delayAsymmetry for CMLDS is used where needed in all computations done by the CMLDS, and also by computations using the syncEgressTimestamp field of the Drift_Tracking TLV (see 11.4.4.2.3) if CMLDS is present. The instance of delayAsymmetry for a domain is used where needed in all computations done for that domain, and also by computations using the Drift_Tracking TLV if CMLDS is not present.

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10.2.5.10 computeNeighborRateRatio: A Boolean, set by the LinkDelayIntervalSetting state machine (see 11.2.20), that indicates whether nrrPdelay is computed for this port. There is one instance of this variable for each domain, i.e., each PTP Instance (per port), and one instance of this variable for CMLDS. If the instance of this variable for a domain is TRUE, nrrPdelay for that domain is computed.

10.2.5.11 computeMeanLinkDelay: A Boolean, set by the LinkDelayIntervalSetting state machine (see 11.2.20), that indicates whether meanLinkDelay is to be computed by this port. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

10.2.5.12 portOper: A Boolean that is TRUE if and only if the port is up and able to send and receive messages. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port). The variable is accessible by all the domains. The term *port* in this definition is a physical port.

NOTE 1—portOper is an indicator, and not a control, and reflects the operational status of the underlying medium. It is not administratively set by gPTP.

NOTE 2—The variable portOper corresponds to the variable portEnabled in the 2011 edition of this standard. The change is reflected in many state machines.

NOTE 3—portOper is the same as MAC_Operational (see IEEE Std 802.1AC-2016).

NOTE 4—When portOper is referenced for a logical port, the reference is to portOper for the physical port that corresponds to the logical port.

10.2.5.13 ptpPortEnabled: A Boolean that is administratively set to TRUE if time-synchronization is to be enabled on this PTP Port.

NOTE 1—It is expected that the value of ptpPortEnabled is set via the management interface (see 14.10.4). A physical port can be enabled for data transport but not for synchronization transport.

NOTE 2—The variable ptpPortEnabled was named pttPortEnabled in the 2011 edition of this standard. Only the name of this variable has changed; the definition and function of this variable are the same as in the 2011 edition. The name change is reflected in many state machines.

10.2.5.14 thisPort: The portNumber of the current port.

10.2.5.15 syncLocked: A Boolean, set by the PortSyncSyncSend state machine (see 10.2.12.3), that indicates that this PTP Port, when operating as a timeTransmitter port, shall transmit a Sync as soon as possible after the timeReceiver port received a Sync (ignoring syncInterval). If FALSE, the PTP Port shall use the timing set by syncInterval.

10.2.5.16 neighborGptpCapable: A Boolean, set by the GptpCapableReceive state machine (see 10.4.3), that indicates that the neighbor of this PTP Port (i.e, the PTP Port at the other end of the link attached to this PTP Port) is capable of invoking gPTP.

10.2.5.17 syncSlowdown: A Boolean that is set to TRUE if the SyncIntervalSetting state machine (see Figure 10-10 in 10.2.14.3) receives a TLV that requests a larger sync interval (see 10.7.2.3) and FALSE otherwise. When syncSlowdown is set to TRUE, the PortSyncSyncSend state machine (see Figure 10-8) continues to send time synchronization event messages (see 11.4.3, 12.1, 12.2, and 13.3.1) at the old (i.e., faster) rate until the number of time synchronization event messages equal to syncReceiptTimeout (see 10.7.3.1) have been sent, but with the respective time synchronization event message transmission interval field (see 11.4.2.9, 12.7, Figure 12-8, and 13.3.1.2.10) of the time synchronization event message set equal to the new sync interval (i.e., corresponding to the slower rate). After syncReceiptTimeout Sync messages

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have been sent, subsequent time synchronization event messages are sent at the new (i.e., slower) rate and with the respective time synchronization event message transmission interval field of the time synchronization event message set to the new sync interval. When syncSlowdown is set to FALSE, the PortSyncSyncSend state machine immediately sends time synchronization event messages at the new (i.e., faster or the same) rate.

NOTE—If a receiver of time synchronization event messages (see 11.4.3, 12.1, 12.2, and 13.3.1) requests a slower rate, the receiver will continue to use the upstream sync interval value, which it obtains from the respective time synchronization event message transmission interval field (see 11.4.2.9, 12.7, Figure 12-8, and 13.3.1.2.10) of the received time synchronization event message, until it receives a time synchronization event message where that value has changed. If, immediately after requesting a slower time synchronization event message rate, up to syncReceiptTimeout consecutive time synchronization event messages sent to the receiver are lost, sync receipt timeout could occur if the sender had changed to the slower rate immediately. Delaying the slowing down of the sending rate of time synchronization event messages for syncReceiptTimeout messages prevents this timeout from happening.

10.2.5.18 oldSyncInterval: The saved value of the previous sync interval, when a new time-synchronization event message transmission interval is requested via a Signaling message that contains a message interval request TLV.

10.2.5.19 gPtpCapableMessageSlowdown: A Boolean that is set to TRUE if the GptpCapableIntervalSetting state machine (see Figure 10-20 in 10.3.17.3) receives a TLV that requests a larger gPTP-capable message interval (see 10.7.2.5) and FALSE otherwise. When gPtpCapableMessageSlowdown is set to TRUE, the GptpCapableTransmit state machine (see Figure 10-21 in 10.4.2.3) continues to send Signaling messages containing the gPTP-capable TLV at the old (i.e., faster) rate until a number of Signaling messages containing the gPTP-capable TLV, equal to gPtpCapableReceiptTimeout (see 10.7.3.3), have been sent, but with the logGptpCapableMessageInterval field of the gPTP-capable TLV (see 10.6.4.5.6) set equal to the new gPTP-capable message interval (i.e., corresponding to the slower rate). After gPtpCapableReceiptTimeout Signaling messages containing the gPTP-capable TLV have been sent, subsequent such Signaling messages are sent at the new (i.e., slower) rate and with the logGptpCapableMessageInterval field of the gPTP-capable TLV set to the new gPTP-capable message interval. When gPtpCapableSlowdown is set to FALSE, the GptpCapableTransmit state machine immediately sends Signaling messages containing the gPTP-capable TLV at the new (i.e., faster or the same) rate.

NOTE—If a receiver of Signaling messages containing the gPTP-capable TLV requests a slower rate, the receiver will continue to use the old gPTP-capable message interval value in determining, via the GptpCapableReceive state machine (see 10.4.3), if its neighbor is no longer capable of invoking gPTP, until it has received gPtpCapableReceiptTimeout such Signaling messages. If, immediately after requesting a slower rate, up to gPtpCapableReceiptTimeout consecutive Signaling messages, containing the gPTP-capable TLV, sent to the receiver are lost, a declaration that the sender is no longer capable of invoking gPTP could occur if the sender had changed to the slower rate immediately. Delaying the slowing down of the sending rate of Signaling messages containing the gPTP-capable TLV for gPtpCapableReceiptTimeout messages prevents this timeout from happening.

10.2.5.20 gPtpCapableMessageInterval: A variable whose value is the mean time, in seconds, between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.7.2.5 and 10.6.4.4).

10.2.5.21 oldGptpCapableMessageInterval: The saved value of the previous interval between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.2.5.20), when a new such interval is requested via a Signaling message that contains a gPTP-capable message interval request TLV.

10.2.5.22 currentLogGptpCapableMessageInterval: The current value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.6.4.4). This value is set in the GptpCapableIntervalSetting state machine (see 10.4.4).

10.2.5.23 initialLogGptpCapableMessageInterval: The initial value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Signaling messages that carry the gPTP-capable TLV (see 10.6.4.4).

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10.2.6 Function used by multiple state machines

10.2.6.1 random(): Returns a uniformly-distributed pseudo-random number whose data type is UInteger16 (i.e., the function returns a uniformly distributed, pseudo-random integer in the range [0, $2^{16} - 1$]).

10.2.7 SiteSyncSync state machine

10.2.7.1 State machine variables

The following variables are used in the state diagram in Figure 10-3 (in 10.2.7.3):

10.2.7.1.1 rcvdPSSyncSSS: A Boolean variable that notifies the current state machine when a PortSyncSync structure (see 10.2.2.3) is received from the PortSyncSyncReceive state machine of a PortSync entity or from the ClockTimeTransmitterSyncSend state machine of the ClockTimeTransmitter entity. This variable is reset by this state machine.

10.2.7.1.2 rcvdPSSyncPtrSSS: A pointer to the received PortSyncSync structure indicated by rcvdPSSyncSSS.

10.2.7.1.3 txPSSyncPtrSSS: A pointer to the PortSyncSync structure transmitted by the state machine.

10.2.7.2 State machine functions

10.2.7.2.1 setPSSyncSend (rcvdPSSyncPtrSSS): Creates a PortSyncSync structure to be transmitted, and returns a pointer to this structure. The members are copied from the received PortSyncSync structure pointed to by rcvdPSSyncPtrSSS.

10.2.7.2.2 txPSSync (txPSSyncPtrSSS): Transmits a copy of the PortSyncSync structure pointed to by txPSSyncPtrSSS to the PortSyncSyncSend state machine of each PortSync entity and the ClockTimeReceiverSync state machine of the ClockTimeReceiver entity of this PTP Instance.

10.2.7.3 State diagram

The SiteSyncSync state machine shall implement the function specified by the state diagram in Figure 10-3, the local variables specified in 10.2.7.1, the functions specified in 10.2.7.2, the structure specified in 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives time-synchronization information, accumulated rateRatio, and syncReceiptTimeoutTime from the PortSync entity (PortSyncSyncReceive state machine) of the current timeReceiver port or from the ClockTimeTransmitter entity (ClockTimeTransmitterSyncSend state machine). If the information was sent by a PortSync entity, the state machine also receives the portIdentity of the PTP Port on the upstream PTP Instance that sent the information to this PTP Instance (if the information was sent by the ClockTimeTransmitter entity, the portIdentity is that of the ClockTimeTransmitter entity, i.e., it has clockIdentity equal to the clockIdentity of this PTP Instance and portNumber 0). The state machine sends a PortSyncSync structure to the PortSync entities of all the ports and to the ClockTimeReceiver entity.

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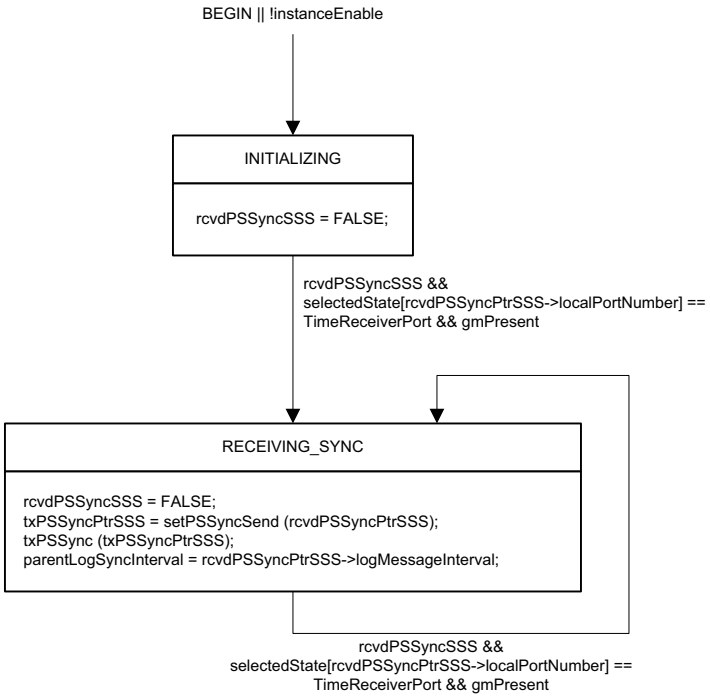


Figure 10-3—SiteSyncSync state machine

10.2.8 PortSyncSyncReceive state machine

10.2.8.1 State machine variables

The following variables are used in the state diagram in Figure 10-4 (in 10.2.8.3):

10.2.8.1.1 rcvdMDSyncPSSR: A Boolean variable that notifies the current state machine when an MDSyncReceive structure is received from an MD entity of the same PTP Port (see 10.2.2.1). This variable is reset by this state machine.

10.2.8.1.2 rcvdMDSyncPtrPSSR: A pointer to the received MDSyncReceive structure indicated by rcvdMDSyncPSSR.

10.2.8.1.3 txPSSyncPtrPSSR: A pointer to the PortSyncSync structure transmitted by the state machine.

10.2.8.1.4 rateRatio: A Float64 variable that holds the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity. This frequency ratio is computed by:

a) Measuring the ratio of the Grandmaster Clock frequency to the LocalClock frequency at the Grandmaster PTP Instance and initializing rateRatio to this value in the ClockTimeTransmitterSyncSend state machine of the Grandmaster PTP Instance and

b) Accumulating, in the PortSyncSyncReceive state machine of each PTP Instance, the frequency offset of the LocalClock entity of the PTP Instance at the remote end of the link attached to that PTP Port to the frequency of the LocalClock entity of this PTP Instance.

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10.2.8.1.5 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-4). The data type for TEMP is Integer16.

10.2.8.2 State machine functions

10.2.8.2.1 setPSSyncPSSR (rcvdMDSyncPtrPSSR syncReceiptTimeoutTimeInterval, rateRatio): Creates a PortSyncSync structure to be transmitted, and returns a pointer to this structure. The members are set as follows:

a) localPortNumber is set equal to thisPort.

b) domainNumber, followUpCorrectionField, sourcePortIdentity, logMessageInterval, preciseOriginTimestamp, and upstreamTxTime are copied from the received MDSyncReceive structure pointed to by rcvdMDSyncPtrPSSR.

c) syncReceiptTimeoutTime is set equal to currentTime plus syncReceiptTimeoutTimeInterval (see 10.2.5.3).

d) The function argument rateRatio is set equal to the local variable rateRatio (computed just prior to invoking setPSSyncPSSR (see Figure 10-4)). The rateRatio member of the PortSyncSync structure is then set equal to the function argument rateRatio.

10.2.8.2.2 txPSSyncPSSR (txPSSyncPtrPSSR): Transmits a copy of the PortSyncSync structure pointed to by txPSSyncPtrPSSR to the SiteSyncSync state machine of this PTP Instance.

10.2.8.3 State diagram

The PortSyncSyncReceive state machine shall implement the function specified by the state diagram in Figure 10-4, the local variables specified in 10.2.8.1, the functions specified in 10.2.8.2, the structures specified in 10.2.2.1 and 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives time-synchronization information, accumulated rateRatio, and syncReceiptTimeoutTime from the MD entity of the same PTP Port. The state machine adds, to rateRatio, the fractional frequency offset of the LocalClock entity relative to the LocalClock entity of the upstream PTP Instance at the remote end of the link attached to this PTP Port. The state machine computes syncReceiptTimeoutTime. The state machine sends this information to the SiteSync entity (SiteSyncSync state machine).

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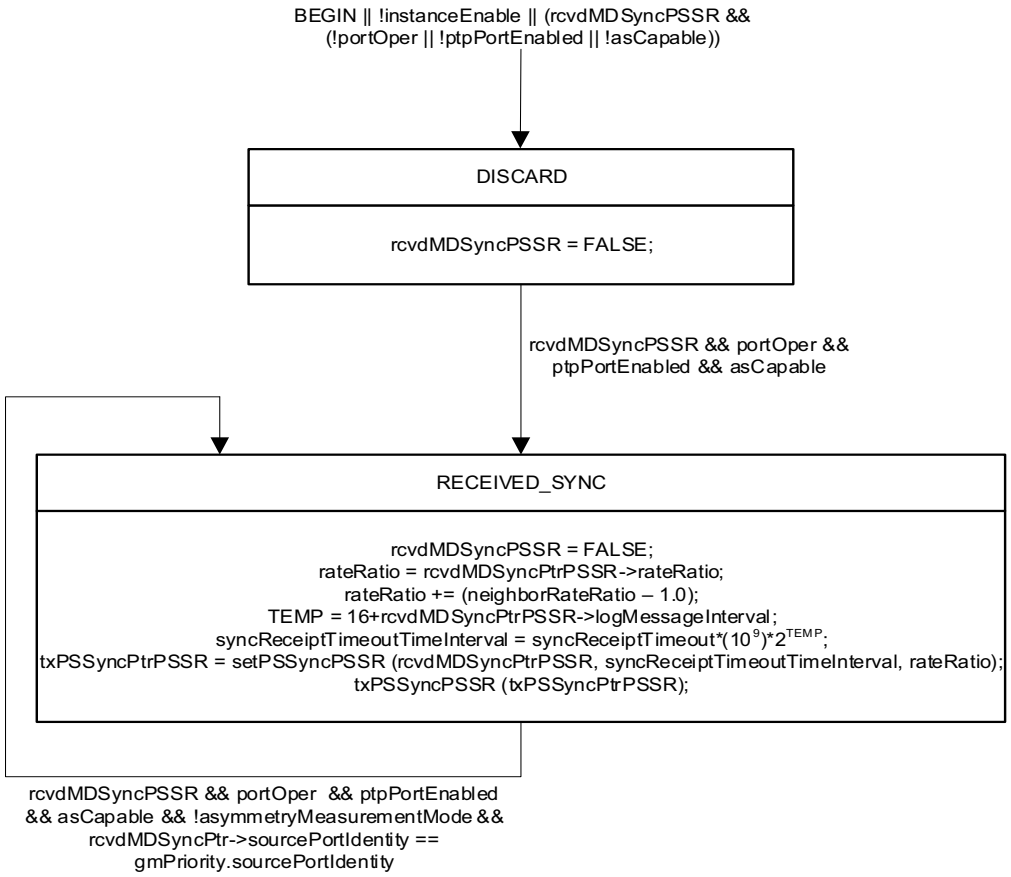


Figure 10-4—PortSyncSyncReceive state machine

10.2.9 ClockTimeTransmitterSyncSend state machine

10.2.9.1 State machine variables

The following variables are used in the state diagram in Figure 10-5 (in 10.2.9.3):

10.2.9.1.1 syncSendTime: The time in seconds, relative to the LocalClock entity, when synchronization information will next be sent to the SiteSync entity, via a PortSyncSync structure. The data type for syncSendTime is UScaledNs.

10.2.9.1.2 txPSSyncPtrCMSS: A pointer to the PortSyncSync structure transmitted by the state machine.

10.2.9.2 State machine functions

10.2.9.2.1 setPSSyncCMSS (gmRateRatio): Creates a PortSyncSync structure to be transmitted, and returns a pointer to this structure. The members are set as follows:

a) localPortNumber is set to 0.

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b) preciseOriginTimestamp is set equal to the timeTransmitterTime, with any fractional nanoseconds truncated.

c) followUpCorrectionField is set equal to the sum of:

1) The fractional nanoseconds portion of timeTransmitterTime.fractionalNanoseconds and

2) The quantity gmRateRatio × (currentTime – localTime).

NOTE 1—Both localTime (10.2.4.19) and timeTransmitterTime (10.2.4.21) are updated by the ClockTimeTransmitterSyncReceive state machine (10.2.11). The updates occur both when sourceTime (9.2.2.2) is received from the ClockSource (9.2)—indicated by the variable rcvdClockSourcReq (10.2.11.1.1) being TRUE—and when the LocalClock entity (10.1.2.1) itself updates by one tick—indicated by the variable rcvdLocalClockTickCMSR (10.2.11.1.3) being TRUE. timeTransmitterTime is updated by updateTimeTransmitterTime() (see 10.2.11.2.2). localTime is updated by setting it equal to currentTime (Figure 10-7). The result is that localTime is equal to the value of currentTime when timeTransmitterTime was most recently updated. localTime, currentTime, and timeTransmitterTime are per PTP Instance global variables. When they are updated, their values are known to all state machines.

NOTE 2—It is possible that currentTime (10.2.4.12) and localTime (10.2.4.19) are not equal, despite the statement localTime = currentTime in the RECEIVE_SOURCE_TIME state of the ClockTimeTransmitterSyncReceive state machine (10.2.11). For example, if the ClockTimeTransmitterSyncReceive state machine is implemented as a different module than the ClockTimeTransmitterSyncSend state machine (10.2.9), with timeTransmitterTime (10.2.4.21) and localTime being sent from the ClockTimeTransmitterSyncReceive state machine to the ClockTimeTransmitterSyncSend state machine, then currentTime and localTime will not be equal. In this case, currentTime in c) 2) is the value of the LocalClock entity (10.1.2.1) when the ClockTimeTransmitterSyncSend state machine receives timeTransmitterTime and localTime from the ClockTimeTransmitterSyncReceive state machine.

d) The clockIdentity member of sourcePortIdentity is set equal to the clockIdentity of this PTP Instance.

e) The portNumber member of the sourcePortIdentity is set to 0.

NOTE 3—This quantity and localPortNumber are redundant; both are retained so that the SiteSync entity can process PortSyncSync structures received from a PortSync entity or the ClockTimeTransmitter entity in the same manner.

f) logMessageInterval is set to clockTimeTransmitterLogSyncInterval.

g) upstreamTxTime is set equal to localTime.

h) syncReceiptTimeoutTime is set equal to 0xFFFFFFFFFFFFFFFF, which indicates that there is no sync receipt timeout.

NOTE 4—A ClockTimeTransmitter entity does not receive Sync messages, and there is no notion of sync receipt timeout.

i) rateRatio is set equal to gmRateRatio.

j) gmTimeBaseIndicator is set equal to clockSourceTimeBaseIndicator.

k) lastGmPhaseChange is set equal to clockSourcePhaseOffset.

l) lastGmFreqChange is set equal to clockSourceFreqOffset.

m) domainNumber is set equal to the domainNumber of this gPTP domain.

10.2.9.2.2 txPSSyncCMSS (txPSSyncPtrCMSS): Transmits a copy of the PortSyncSync structure pointed to by txPSSyncPtrCMSS to the SiteSync state machine.

10.2.9.2.3 computeClockTimeTransmitterSyncInterval(): Computes the value of clockTimeTransmitterSyncInterval (see 10.2.4.2) as $1\,000\,000\,000 \times 2^{\text{clockTimeTransmitterLogSyncInterval}}$ ns, where clockTimeTransmitterLogSyncInterval is the minimum currentLogSyncInterval value, taken over all the PTP Ports of the PTP Instance (see 10.7.2.4).

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10.2.9.3 State diagram

The ClockTimeTransmitterSyncSend state machine shall implement the function specified by the state diagram in Figure 10-5, the local variables specified in 10.2.9.1, the functions specified in 10.2.9.2, the structure specified in 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives timeTransmitterTime and clockSourceTimeBaseIndicator from the ClockTimeTransmitterSyncReceive state machine, and phase and frequency offset between timeTransmitterTime and syncReceiptTime from the ClockTimeTransmitterSyncOffset state machine. It provides timeTransmitterTime (i.e., synchronized time) and the phase and frequency offset to the SiteSync entity via a PortSyncSync structure.

The ClockTimeTransmitterSyncSend state machine is optional for PTP Instances that are not grandmaster-capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance that is not grandmaster-capable; however, any information supplied by it to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

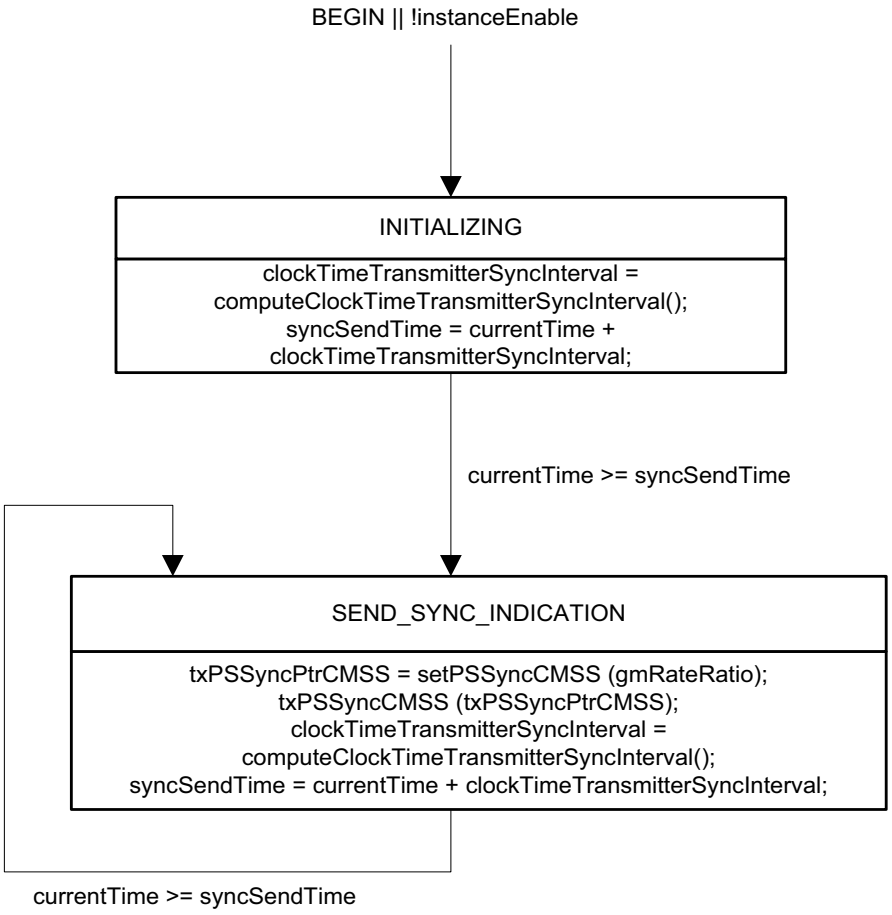


Figure 10-5—ClockTimeTransmitterSyncSend state machine

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10.2.10 ClockTimeTransmitterSyncOffset state machine

10.2.10.1 State machine variables

The following variable is used in the state diagram in Figure 10-6 (in 10.2.10.3):

10.2.10.1.1 rcvdSyncReceiptTime: A Boolean variable that notifies the current state machine that syncReceiptTime has been updated by the ClockTimeReceiver entity. This variable is reset by this state machine.

10.2.10.2 State machine functions

10.2.10.2.1 computeClockSourceFreqOffset(): Computes and returns clockSourceFreqOffset (see 10.2.4.6), using successive values of timeTransmitterTime computed by the ClockTimeTransmitterSyncReceive state machine (see 10.2.11) and successive values of syncReceiptTime computed by the ClockTimeReceiverSync state machine (see 10.2.13). The data type for the returned value is Float64. Any scheme that uses this information to compute clockSourceFreqOffset is acceptable as long as the performance requirements specified in B.2.4 are met.

NOTE—As one example, clockSourceFreqOffset can be estimated as the ratio of the duration of a time interval measured by the ClockSource entity to the duration of the same time interval computed from ClockTimeReceiverTime values, minus 1.

10.2.10.3 State diagram

The ClockTimeTransmitterSyncOffset state machine shall implement the function specified by the state diagram in Figure 10-6, the local variable specified in 10.2.10.1, the function specified in 10.2.10.2, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives syncReceiptTime from the ClockTimeReceiverSync state machine and timeTransmitterTime from the ClockTimeTransmitterSyncReceive state machine. It computes clockSourcePhaseOffset and clockSourceFrequency offset if this PTP Instance is not currently the Grandmaster PTP Instance, i.e., if selectedState[0] is equal to PassivePort.

The ClockTimeTransmitterSyncOffset state machine is optional for PTP Instances that are not grandmaster-capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance that is not grandmaster-capable; however, any information supplied by it, via the ClockTimeTransmitterSyncSend state machine, to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

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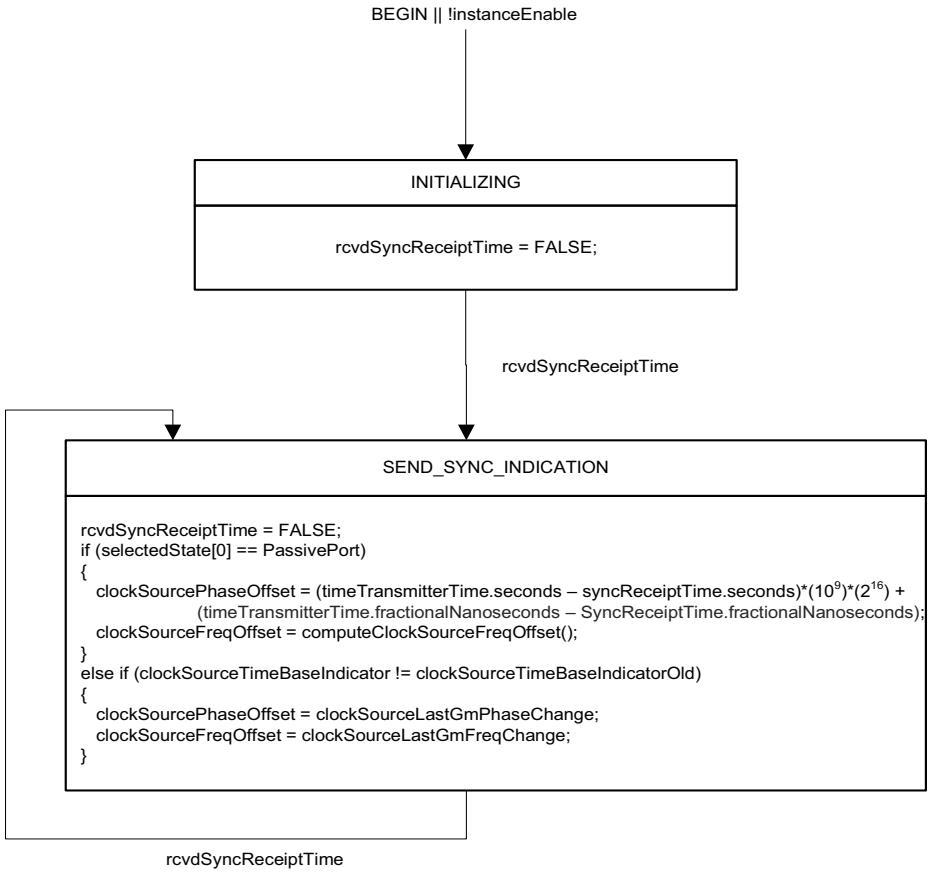


Figure 10-6—ClockTimeTransmitterSyncOffset state machine

10.2.11 ClockTimeTransmitterSyncReceive state machine

10.2.11.1 State machine variables

The following variables are used in the state diagram in Figure 10-7 (in 10.2.11.3):

10.2.11.1.1 rcvdClockSourceReq: A Boolean variable that notifies the current state machine when sourceTime is received from the ClockSource entity, due to the ClockSourceTime.invoke primitive having been invoked at that entity. This variable is reset by this state machine.

10.2.11.1.2 rcvdClockSourceReqPtr: A pointer to the received ClockSourceTime.invoke function parameters.

10.2.11.1.3 rcvdLocalClockTickCMSR: A Boolean variable that notifies the current state machine when the LocalClock entity updates its time. This variable is reset by this state machine.

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10.2.11.2 State machine functions

10.2.11.2.1 computeGmRateRatio(): Computes gmRateRatio (see 10.2.4.14), using values of sourceTime conveyed by successive ClockSourceTime.invoke functions (see 9.2.2.1), and corresponding values of localTime (see 10.2.4.19). Any scheme that uses this information, along with any other information conveyed by the successive ClockSourceTime.invoke functions and corresponding values of localTime, to compute gmRateRatio is acceptable as long as the performance requirements specified in B.2.4 are met.

NOTE—As one example, gmRateRatio can be estimated as the ratio of the elapsed time of the ClockSource entity that supplies time to this PTP Instance, to the elapsed time of the LocalClock entity of this PTP Instance. This ratio can be computed for the time interval between a received ClockSourceTime.invoke function and a second received ClockSourceTime.invoke function some number of ClockSourceTime.invoke functions later, i.e.,

$$\frac{\text{ClockSource.invoke.sourceTime}_N - \text{ClockSource.invoke.sourceTime}_0}{\text{localTime}_N - \text{localTime}_0}$$

where the successive received ClockSourceTime.invoke functions are indexed from 0 to *N*, with the first such function indexed as 0, and localTime_j is the value of localTime when the ClockSourceTime.invoke function whose index is *j* is received. Another example is to use a sliding window for the calculation of gmRateRatio.

10.2.11.2.2 updateTimeTransmitterTime(): Updates the global variable timeTransmitterTime (see 10.2.4.21), based on information received from the ClockSource and LocalClock entities. It is the responsibility of the application to filter timeTransmitter times appropriately. As one example, timeTransmitterTime can be set equal to the sourceTime member of the ClockSourceTime.invoke function when this function is invoked at the ClockSource entity and can be incremented by localClockTickInterval (see 10.2.4.18) multiplied by gmRateRatio (see 10.2.4.14) when rcvdLocalClockTickCMSR is TRUE.

10.2.11.3 State diagram

The ClockTimeTransmitterSyncReceive state machine shall implement the function specified by the state diagram in Figure 10-7, the local variables specified in 10.2.11.1, the functions specified in 10.2.11.2, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine updates the global variable timeTransmitterTime with information received from the ClockSource entity via the ClockSourceTime.invoke function and information received from the LocalClock entity. It also computes gmRateRatio, i.e., the ratio of the ClockSource entity frequency and the LocalClock entity frequency.

The ClockTimeTransmitterSyncReceive state machine is optional for PTP Instances that are not grandmaster-capable (see 8.6.2.1, 10.1.3, and 10.2.1). This state machine may be present in a PTP Instance that is not grandmaster-capable; however, any information supplied by it, via the ClockTimeTransmitterSyncSend state machine, to the SiteSyncSync state machine is not used by the SiteSyncSync state machine if the PTP Instance is not grandmaster-capable.

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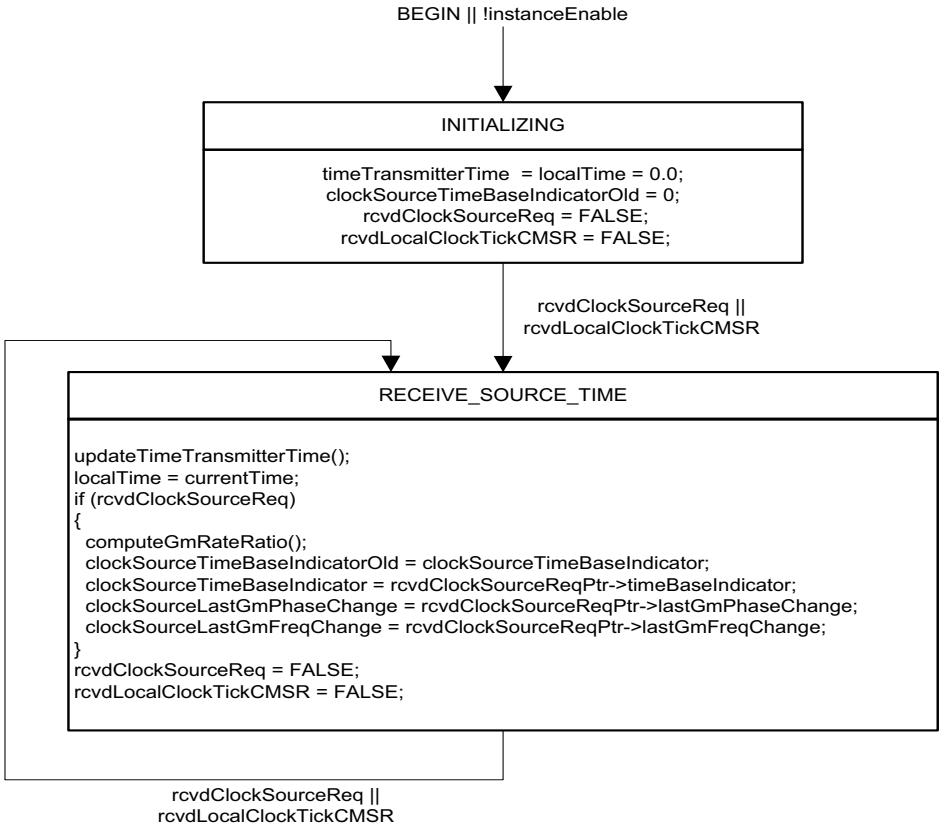


Figure 10-7—ClockTimeTransmitterSyncReceive state machine

10.2.12 PortSyncSyncSend state machine

10.2.12.1 State machine variables

The following variables are used in the state diagram in Figure 10-8 (in 10.2.12.3):

10.2.12.1.1 rcvdPSSyncPSSS: A Boolean variable that notifies the current state machine when a PortSyncSync structure is received from the SiteSyncSync state machine of the SiteSync entity of the PTP Instance (see 10.2.2.3). This variable is reset by this state machine.

10.2.12.1.2 rcvdPSSyncPtrPSSS: A pointer to the received PortSyncSync structure indicated by rcvdPSSyncPSSS.

10.2.12.1.3 lastPreciseOriginTimestamp: The preciseOriginTimestamp member of the most recently received PortSyncSync structure. The data type for lastPreciseOriginTimestamp is Timestamp.

10.2.12.1.4 lastFollowUpCorrectionField: The followUpCorrectionField member of the most recently received PortSyncSync structure. The data type for lastFollowUpCorrectionField is ScaledNs.

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10.2.12.1.5 lastRateRatio: The rateRatio member of the most recently received PortSyncSync structure. The data type for lastRateRatio is Float64.

10.2.12.1.6 lastUpstreamTxTime: The upstreamTxTime member of the most recently received PortSyncSync structure. The data type for lastUpstreamTxTime is UScaledNs.

10.2.12.1.7 lastSyncSentTime: The value of currentTime (i.e., the time relative to the LocalClock entity) when the most recent MDSyncSend structure was sent. The data type for lastSyncSentTime is UScaledNs.

NOTE—lastSyncSentTime is the time the abstract MDSyncSend structure was sent, NOT the time the corresponding Sync message (or equivalent) was sent on a physical link.

10.2.12.1.8 lastRcvdPortNum: The portNumber of the PTP Port on which time-synchronization information was most recently received. The data type for lastRcvdPortNum is UInteger16.

10.2.12.1.9 lastGmTimeBaseIndicator: The gmTimeBaseIndicator of the most recently received PortSyncSync structure. The data type for lastGmTimeBaseIndicator is UInteger16.

10.2.12.1.10 lastGmPhaseChangePSSS: The lastGmPhaseChange of the most recently received PortSyncSync structure. The data type for lastGmPhaseChange is ScaledNs.

10.2.12.1.11 lastGmFreqChangePSSS: The lastGmFreqChange of the most recently received PortSyncSync structure. The data type for lastGmPhaseChange is Float64.

10.2.12.1.12 txMDSyncPtr: A pointer to the MDSyncSend structure sent to the MD entity of this PTP Port.

10.2.12.1.13 numberSyncTransmissions: A count of the number of consecutive Sync message transmissions after the SyncIntervalSetting state machine (see Figure 10-10) has set syncSlowdown (see 10.2.5.17) to TRUE. The data type for numberSyncTransmissions is UInteger8.

10.2.12.1.14 interval1: A local variable that holds either syncInterval or oldSyncInterval. The data type for interval1 is UScaledNs.

10.2.12.2 State machine functions

10.2.12.2.1 setMDSync(): Creates an MDSyncSend structure, and returns a pointer to this structure. The members are set as follows:

a) sourcePortIdentity is set to the portIdentity of this PTP Port (see 8.5.2).
b) logMessageInterval is set equal to the value of currentLogSyncInterval for this PTP Port (see 10.7.2.3).
c) preciseOriginTimestamp is set equal to lastPreciseOriginTimestamp (see 10.2.12.1.3).
d) rateRatio is set equal to lastRateRatio (see 10.2.12.1.5).
e) followUpCorrectionField is set equal to lastFollowUpCorrectionField (see 10.2.12.1.4).
f) upstreamTxTime is set equal to lastUpstreamTxTime (see 10.2.12.1.6).
g) gmTimeBaseIndicator is set to lastGmTimeBaseIndicator (see 10.2.12.1.9).
h) lastGmPhaseChange (structure member) is set to lastGmPhaseChangePSSS (see 10.2.12.1.10).
i) lastGmFreqChange (structure member) is set to lastGmFreqChangePSSS (see 10.2.12.1.11).
j) domainNumber is set equal to the domainNumber of this gPTP domain (see 8.1).

10.2.12.2.2 txMDSync(txMDSyncPtr): Transmits the MDSyncSend structure pointed to by txMDSyncPtr, to the MD entity of this PTP Port.

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10.2.12.3 State diagram

The PortSyncSyncSend state machine shall implement the function specified by the state diagram in Figure 10-8, the local variables specified in 10.2.12.1, the functions specified in 10.2.12.2, the structures specified in 10.2.2.1 through 10.2.2.3, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives time-synchronization information from the SiteSyncSync state machine, corresponding to the receipt of the most recent synchronization information on either the timeReceiver port, if this PTP Instance is not the Grandmaster PTP Instance, or from the ClockTimeTransmitterSyncSend state machine, if this PTP Instance is the Grandmaster PTP Instance. The state machine causes time-synchronization information to be sent to the MD entity if this PTP Port is a TimeTransmitterPort.

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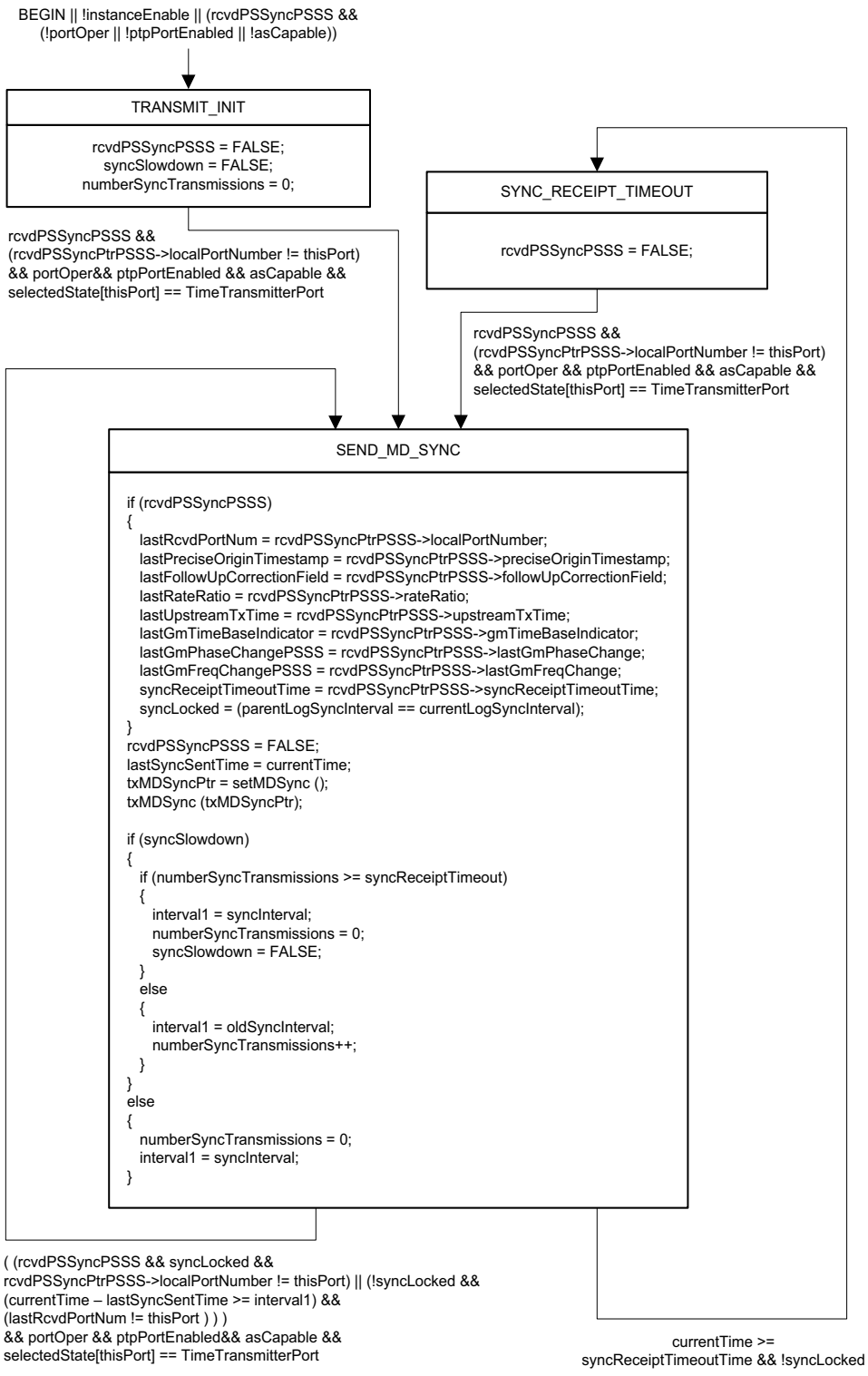


Figure 10-8—PortSyncSyncSend state machine

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10.2.13 ClockTimeReceiverSync state machine

10.2.13.1 State machine variable

The following variable is used in the state diagram in Figure 10-9 (in 10.2.13.3):

10.2.13.1.1 rcvdPSSyncPtrCSS: A pointer to the received PortSyncSync structure.

10.2.13.2 State machine functions

10.2.13.2.1 updateTimeReceiverTime(): Updates the global variable clockTimeReceiverTime (see 10.2.4.3), based on information received from the SiteSync and LocalClock entities. It is the responsibility of the application to filter timeReceiver times appropriately (see B.3 and B.4 for examples). As one example, clockTimeReceiverTime can be:

a) Set to syncReceiptTime at every LocalClock update immediately after a PortSyncSync structure is received, and

b) Incremented by localClockTickInterval (see 10.2.4.18) multiplied by the rateRatio member of the previously received PortSyncSync structure during all other LocalClock updates.

If no PTP Instance is grandmaster-capable, i.e., gmPresent is FALSE, then clockTimeReceiverTime is set to the time provided by the LocalClock. This function is invoked when rcvdLocalClockTickCSS is TRUE.

10.2.13.2.2 invokeApplicationInterfaceFunction (functionName): Invokes the application interface function whose name is functionName. For the ClockTimeReceiverSync state machine, functionName is clockTargetPhaseDiscontinuity.result (see 9.6.2).

10.2.13.3 State diagram

The ClockTimeReceiverSync state machine shall implement the function specified by the state diagram in Figure 10-9, the local variables specified in 10.2.13.1, the functions specified in 10.2.13.2, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives a PortSyncSync structure from the SiteSyncSync state machine. It computes syncReceiptTime and clockTimeReceiverTime, and sets syncReceiptLocalTime (i.e., the time relative to the LocalClock entity corresponding to syncReceiptTime), GmTimeBaseIndicator, lastGmPhaseChange, and lastGmFreqChange. It provides clockTimeReceiverTime to the ClockTimeTransmitterSyncOffset state machine, and provides information to the ClockTarget entity (via the ClockTargetPhaseDiscontinuity interface; see 9.6) to enable that entity to determine if a phase or frequency discontinuity has occurred.

The per-PTP Port global variables used in the ClockTimeReceiverSync state machine are determined based on rcvdPSSyncPtrCSS->localPortNumber, as follows:

a) If rcvdPSSyncPtrCSS->localPortNumber > 0, the per-PTP Port global variables of PTP Port number rcvdPSSyncPtrCSS->localPortNumber are used.

b) If rcvdPSSyncPtrCSS->localPortNumber == 0, the values of the used per-PTP Port global variables are fixed as follows:

    1) meanLinkDelay = 0

    2) delayAsymmetry = 0

    3) neighborRateRatio = 1.0

For Clause 11, nrrPdelay is used instead neighborRateRatio.

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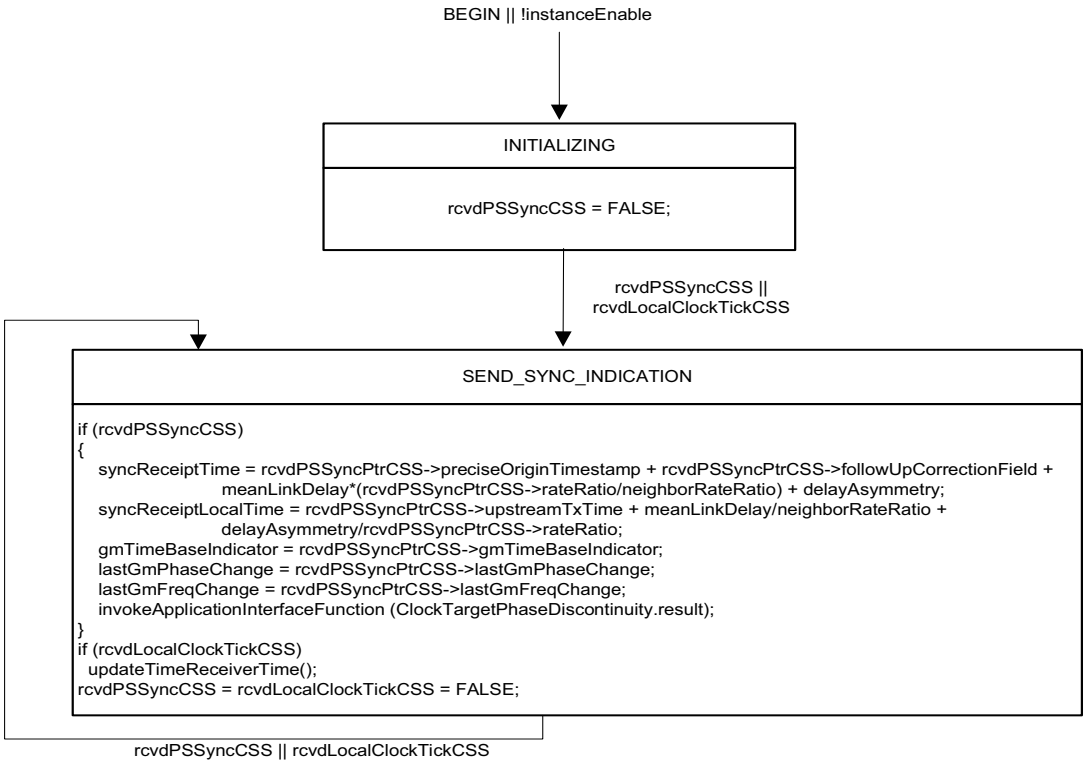


Figure 10-9—ClockTimeReceiverSync state machine

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10.2.14 SyncIntervalSetting state machine

10.2.14.1 State machine variables

The following variables are used in the state diagram in Figure 10-10 (in 10.2.14.3):

10.2.14.1.1 rcvdSignalingMsg3: A Boolean variable that notifies the current state machine when a Signaling message that contains a message interval request TLV (see 10.6.4.3) is received. This variable is reset by the current state machine.

10.2.14.1.2 rcvdSignalingPtrSIS: A pointer to a structure whose members contain the values of the fields of the received Signaling message that contains a message interval request TLV (see 10.6.4.3).

10.2.14.1.3 logSupportedSyncIntervalMax: The maximum supported logarithm to base 2 of the sync interval. The data type for logSupportedSyncIntervalMax is Integer8.

10.2.14.1.4 logSupportedClosestLongerSyncInterval: The logarithm to base 2 of the sync interval, such that logSupportedClosestLongerSyncInterval > logRequestedSyncInterval, that is numerically closest to logRequestedSyncInterval, where logRequestedSyncInterval is the argument of the function computeLogSyncInterval() (see 10.2.14.2.2). The data type for logSupportedClosestLongerSyncInterval is Integer8.

10.2.14.1.5 computedLogSyncInterval: A variable used to hold the result of the function computeLogSyncInterval(). The data type for computedLogSyncInterval is Integer8.

10.2.14.1.6 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-10). The data type for TEMP is Integer16.

10.2.14.2 State machine functions

10.2.14.2.1 isSupportedLogSyncInterval (logSyncInterval): A Boolean function that returns TRUE if the sync interval given by the argument logSyncInterval is supported by the PTP Port and FALSE if the sync interval is not supported by the PTP Port. The argument logSyncInterval has the same data type and format as the field syncInterval of the message interval request TLV (see 10.6.4.3.7).

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10.2.14.2.2 computeLogSyncInterval (logRequestedSyncInterval): An Integer8 function that computes and returns the logSyncInterval, based on the logRequestedSyncInterval. This function is defined as indicated below. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
Integer8 computeLogSyncInterval (logRequestedSyncInterval)
Integer8 logRequestedSyncInterval;
{
        Integer8 logSupportedSyncIntervalMax, logSupportedClosestLongerSyncInterval;
        if (isSupportedLogSyncInterval (logRequestedSyncInterval))
                // The requested Sync Interval is supported and returned
                return (logRequestedSyncInterval)
        else
        {
                if (logRequestedSyncInterval > logSupportedSyncIntervalMax)
                        // Return the largest supported logSyncInterval, even if smaller than the
requested interval
                        return (logSupportedSyncIntervalMax);
                else
                        // Return the smallest supported logSyncInterval that is still larger than
                        // the requested interval.
                        return (logSupportedClosestLongerSyncInterval);
        }
}
```

10.2.14.3 State diagram

The SyncIntervalSetting state machine shall implement the function specified by the state diagram in Figure 10-10, the local variables specified in 10.2.14.1, the functions specified in 10.2.14.2, the messages specified in 10.6, the relevant global variables specified in 10.2.5, the relevant managed objects specified in 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting the global variables that give the duration of the mean intervals between successive Sync messages, both at initialization and in response to the receipt of a Signaling message that contains a message interval request TLV (see 10.6.4.3).

10.3 Best timeTransmitter clock selection, external port configuration, and announce interval setting state machines

10.3.1 Best timeTransmitter clock selection and external port configuration overview

10.3.1.1 General

There are two methods for setting the Grandmaster PTP Instance and synchronization tree for a gPTP domain:

a) The BTCA is used to determine the Grandmaster PTP Instance for a gPTP domain and construct the synchronization tree with that Grandmaster PTP Instance as the root. In this case, the network is configured automatically, i.e., the PTP Port states are set, using the results of the BTCA.

b) The PTP Port states are configured to force a desired Grandmaster PTP Instance and to construct a desired synchronization tree with the Grandmaster PTP Instance as the root.

The PTP Port state definitions are given in Table 10-2.

The per PTP Instance global variable externalPortConfigurationEnabled indicates whether method a) or b) is used; a value of TRUE indicates method b), and a value of FALSE indicates method a) (see 10.3.9.24).

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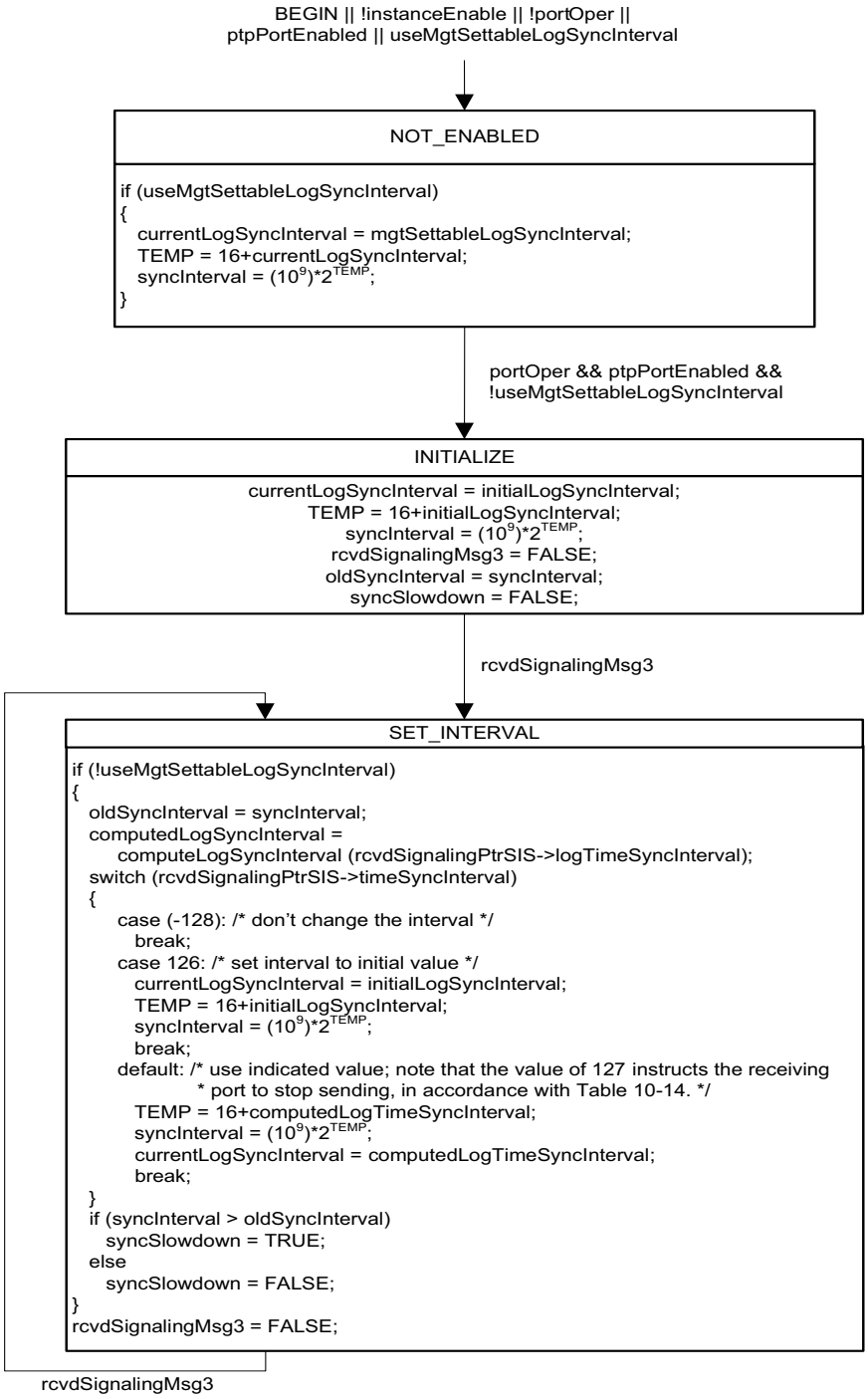


Figure 10-10—SyncIntervalSetting state machine

For domain 0, the following statements apply:

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c) Method a), i.e., BTCA is implemented.

d) Both methods a) and b) can be implemented.

e) The default value of externalPortConfigurationEnabled is FALSE, unless Hot standby is activated.

For domains other than domain 0, the following statements apply:

f) At least one of the possibilities [method a) or b)] is implemented.

g) Both methods a) and b) can be implemented.

h) If both methods a) and b) are implemented, the default value of externalPortConfigurationEnabled is FALSE, unless Hot standby is activated.

Once a GM is indicated in an Announce message transmitted by a PTP Port, subsequent timing information (see 7.5) transmitted by that PTP Port is derived from the Grandmaster PTP Instance indicated in that Announce message.

Table 10-2—PTP Port state definitions

PTP Port state	Description
TimeTransmitterPort	Any PTP Port, X, of the PTP Instance that is closer to the root than any other PTP Port of the gPTP communication path connected to X.
TimeReceiverPort	The one PTP Port of the PTP Instance that is closest to the root PTP Instance. If the root is grandmaster-capable, the TimeReceiverPort is also closest to the Grandmaster PTP Instance. The PTP Instance does not transmit Sync or Announce messages on the TimeReceiverPort.
PassivePort	Any PTP Port of the PTP Instance whose PTP Port state is not TimeTransmitterPort, TimeReceiverPort, or DisabledPort.
DisabledPort	Any PTP Port of the PTP Instance for which the variables portOper, ptpPortEnabled, and asCapable are not all TRUE.
NOTE—PTP Port states are per PTP Port and per domain (i.e., per PTP Instance; see 8.1).	

NOTE—Information contained in Sync and associated Follow_Up messages received on PTP Ports whose PTP Port state is PassivePort is discarded; the SiteSyncSync state machine (see 10.2.7) uses only information received from a PTP Port whose PTP Port state is TimeReceiverPort.

An example timeTransmitter/timeReceiver hierarchy of PTP Instances is shown in Figure 10-11. The Grandmaster PTP Instance ports all have a PTP Port state of TimeTransmitterPort. All the other PTP Instances have exactly one timeReceiver port. The synchronization tree is composed of the PTP Instances and the links that do not have an endpoint PTP Port whose PTP Port state is PassivePort.

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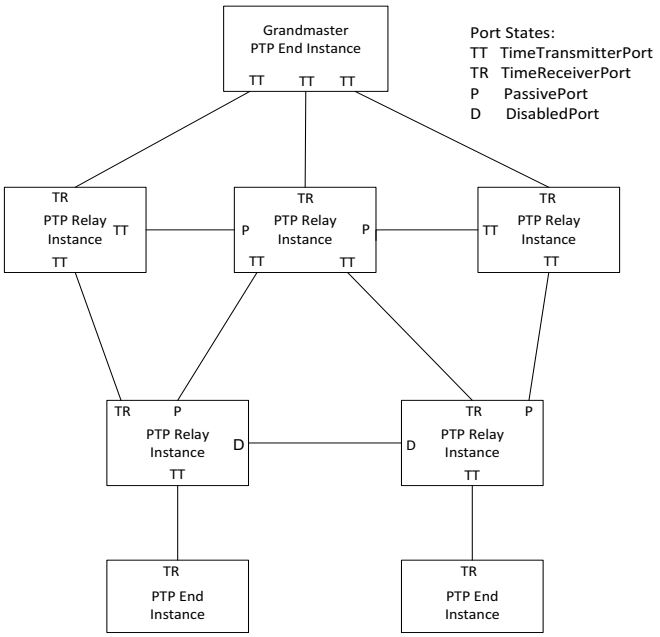


Figure 10-11—Example timeTransmitter/timeReceiver hierarchy of PTP Instances

10.3.1.2 Best timeTransmitter clock algorithm overview

In the BTCA (i.e., method a) of 10.3.1.1), best timeTransmitter selection information is exchanged between PTP Instances of time-aware systems via Announce messages (see 10.5 and 10.6). Each Announce message contains synchronization tree vector information that identifies one PTP Instance as the root of the synchronization tree and, if the PTP Instance is grandmaster-capable, the Grandmaster PTP Instance. Each PTP Instance in turn uses the information contained in the Announce messages it receives, along with its knowledge of itself, to compute which of the PTP Instances that it has knowledge of ought to be the root of the synchronization tree and, if grandmaster-capable, the Grandmaster PTP Instance. As part of constructing the synchronization tree, each PTP Port of each PTP Instance is assigned a PTP Port state from Table 10-2 by state machines associated with the ports and with the PTP Instance as a whole.

NOTE—The BTCA described in this standard is the default BTCA according to the specifications of 9.3 of IEEE Std 1588-2019. It is also equivalent to a subset of the Rapid Spanning Tree Protocol (RSTP) described in IEEE Std 802.1Q-2022 (though the full RSTP described in IEEE Std 802.1Q-2022 is not equivalent to the full BTCA described in IEEE Std 1588-2019). The BTCA description here uses the formalism of the RSTP description in IEEE Std 802.1Q-2022.

10.3.1.3 External port configuration overview

In external port configuration [i.e., method b) of 10.3.1.1], an external entity determines the synchronization tree and sets the PTP Port states accordingly. The method used by the external entity to determine the synchronization tree is outside the scope of this standard. However, as with the BTCA, Announce messages are used to transport information on the synchronization tree and Grandmaster PTP Instance time properties information from one PTP Instance to the next in the synchronization tree. The external entity sets the state of a PTP Port by setting the value of externalPortConfigurationPortDS.desiredState to the desired state.

In the case of external port configuration, the synchronization tree and the desired PTP Port states are controlled by an external entity, but the Grandmaster PTP Instance might change. If the timeReceiver port of

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a PTP Instance that is gmCapable (priority1 < 255, see 8.6.2.1) is no longer asCapable, the state of this PTP Port changes from TimeReceiverPort to DisabledPort (see 10.3.6.2) and the PTP Instance becomes grandmaster for the synchronization (sub-)tree where it is the root [10.3.15.2.2 f) 3)]. The original synchronization tree can be split into disjunct subtrees with different, mutually unsynchronized, Grandmaster PTP Instances. If the port becomes asCapable again, its PTP Port state is again set to the desired state. If the timeReceiver port of a PTP Instance that is not gmCapable is no longer asCapable, the state of this PTP Port changes from TimeReceiverPort to DisabledPort (see 10.3.6.2). However, in this case gmPresent is set to FALSE [see 10.3.15.2.2 f) 3)], and the PTP Instance does not send Sync messages on any of its ports.

In external port configuration, there is no supervision of announce receipt timeout (see 10.7.3.2).

10.3.2 systemIdentity

The systemIdentity attribute of a PTP Instance is a UInteger112 (i.e., a 14-byte, unsigned integer) formed by concatenating the following attributes, in the following order, from most significant to least significant octet:

a) priority1 (1 octet; see 8.6.2.1)
b) clockClass (1 octet; see 8.6.2.2 and 6.4.3.8)
c) clockAccuracy (1 octet; see 8.6.2.3 and 6.4.3.8)
d) offsetScaledLogVariance (2 octets; see 8.6.2.4 and 6.4.3.8)
e) priority2 (1 octet; see 8.6.2.5)
f) clockIdentity (8 octets; see 8.5.2.2 and 6.4.3.6)

The systemIdentity attribute is defined for convenience when comparing two PTP Instances to determine, when using the BTCA (i.e., method a) of 10.3.1.1), which is a better candidate for root and if the PTP Instance is grandmaster-capable (i.e., the value of priority1 is less than 255; see 8.6.2.1). Two PTP Instances are compared as follows. Let the systemIdentity of PTP Instance A be S_A and the systemIdentity of PTP Instance B be S_B. Let the clockIdentity of A be C_A and the clockIdentity of B be C_B. Then, if $C_A \neq C_B$, i.e., A and B represent different PTP Instances:

g) A is better than B if and only if $S_A < S_B$, and
h) B is better than A if and only if $S_B < S_A$.

If $C_A = C_B$, i.e., A and B represent the same PTP Instance:

i) $S_A < S_B$ means that A represents an upgrading of the PTP Instance compared to B or, equivalently, B represents a downgrading of the PTP Instance compared to A,
j) $S_B < S_A$ means that B represents an upgrading of the PTP Instance compared to A or, equivalently, A represents a downgrading of the PTP Instance compared to B, and
k) $S_A = S_B$ means that A and B represent the same PTP Instance that has not changed.

Comparisons g) and h) in this subclause imply that, with the ordering of attributes in the systemIdentity, the clockIdentity is a tie-breaker when two different PTP Instances that have identical attributes a) through e) are compared.

Comparisons g) and h) also imply that a PTP Instance that is grandmaster-capable is always better than another PTP Instance that is not grandmaster-capable because the priority1 is less than 255 if the PTP Instance is grandmaster-capable and is equal to 255 if it is not grandmaster-capable (see 8.6.2.1).

The cases where A and B represent different PTP Instances and where they represent the same PTP Instance are handled separately in the BTCA. When comparing two different PTP Instances, the better PTP Instance

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is selected as the Grandmaster PTP Instance candidate. However, if A and B represent the same PTP Instance with attributes that have changed, the PTP Instance is considered as having the most recent attributes when doing subsequent comparisons with other PTP Instances.

10.3.3 stepsRemoved

Every PTP Instance has a stepsRemoved associated with it. For the root PTP Instance, and therefore the Grandmaster PTP Instance when the root is grandmaster-capable, it is zero. For all other PTP Instances, it is the number of gPTP communication paths in the path from the root to the respective PTP Instance.

NOTE—For example, stepsRemoved for a timeReceiver port on the same gPTP communication path as the Grandmaster PTP Instance will have a value of 1, indicating that a single path was traversed.

The stepsRemoved attributes of different ports of a PTP Instance are compared after comparisons of other attributes that take precedence (i.e., priority1, clockClass, clockAccuracy, offsetScaledLogVariance, priority2) do not result in one PTP Port being declared better than the other. Among the ports whose stepsRemoved attributes are compared, the PTP Port on the PTP Instance with the lowest stepsRemoved is assigned the state of TimeReceiverPort for that PTP Instance (the root PTP Instance does not have a TimeReceiverPort). This lowest stepsRemoved is also considered the stepsRemoved for the PTP Instance. If a PTP Instance has two or more ports with the same stepsRemoved, then the PTP Port with the smallest sourePortIdentity is selected as the TimeReceiverPort. If a PTP Instance has two or more ports with the same stepsRemoved and the same sourcePortIdentity, then the PTP Port with the smallest portNumber is selected as the TimeReceiverPort (see 10.3.4).

10.3.4 Synchronization tree priority vectors

PTP Instances send best timeTransmitter selection information to each other in Announce messages. The information is structured in a synchronization tree priority vector. Synchronization tree priority vectors provide the basis for a concise specification of the BTCA's determination of the synchronization tree and Grandmaster PTP Instance. A priority vector is formed by concatenating the following attributes, in the following order, from most significant to least significant octet:

a) rootSystemIdentity (14 octets; see 10.3.2)
b) stepsRemoved (2 octets; see 10.3.3)
c) sourcePortIdentity (i.e., portIdentity of the transmitting PTP Instance; 10 octets; see 8.5.2 and 10.6.2)
d) portNumber of the receiving PTP Port (2 octets; see 8.5.2.3)

The first two components of a priority vector are significant throughout the gPTP domain; they are propagated via Announce messages and updated through invocation of BTCA state machines. The next component is assigned hop-by-hop for each gPTP communication path or PTP Instance and thus is of local significance only. It is used as a tie-breaker in decisions between synchronization tree priority vectors that are otherwise equal. The fourth component is not conveyed in Announce messages, but is used as a tie-breaker within a PTP Instance.

The set of all synchronization tree priority vectors is totally ordered. For all components, a lesser numerical value is better, and earlier components in the preceding list are more significant. In addition, as mentioned earlier, a priority vector that reflects a root PTP Instance that is grandmaster-capable is always better than a priority vector that reflects a root PTP Instance that is not grandmaster-capable. As each PTP Port receives a priority vector, via an Announce message, from ports closer to the root, additions are made to one or more components to yield a worse priority vector. This process of receiving information, adding to it, and passing it on, can be described in terms of the message priority vector received and a set of priority vectors used to facilitate the computation of a priority vector for each PTP Port, to be transmitted in further Announce Messages to PTP Instances further from the root.

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10.3.5 Priority vector calculations

The portPriorityVector is the synchronization tree priority vector held for the PTP Port when the reception of Announce messages and any pending update of information has been completed:

portPriorityVector = {rootSystemIdentity : stepsRemoved : sourcePortIdentity : portNumber}

A messagePriorityVector is the synchronization tree priority vector conveyed in a received Announce Message. For a PTP Instance S receiving an Announce Message on PTP Port P_S with portNumber PN_S, from a TimeTransmitterPort with portIdentity P_M on PTP Instance M claiming a rootSystemIdentity of R_M and a stepsRemoved of SR_M:

messagePriorityVector = {R_M : SR_M : P_M : PN_S}

This messagePriorityVector is superior to the portPriorityVector and will replace it if, and only if, the messagePriorityVector is better than the portPriorityVector, or the Announce message has been transmitted from the same timeTransmitter PTP Instance and TimeTransmitterPort as the portPriorityVector, i.e., if the following is true:

((R_M < rootSystemIdentity)) ||

((R_M == rootSystemIdentity) && (SR_M < stepsRemoved)) ||

((R_M == rootSystemIdentity) && (SR_M == stepsRemoved) && (P_M < sourcePortIdentity (of current timeTransmitter PTP Instance))) ||

((R_M == rootSystemIdentity) && (SR_M == stepsRemoved) && (P_M == sourcePortIdentity (of current timeTransmitter PTP Instance)) && (PN_S < portNumber)) ||

((P_M.clockIdentity == sourcePortIdentity.clockIdentity (of current timeTransmitter PTP Instance)) && (P_M.portNumber == sourcePortIdentity.PortNumber (of the current timeTransmitter PTP Instance)))

A gmPathPriorityVector can be calculated from a portPriorityVector (that contains information from a received messagePriorityVector) by adding one to the stepsRemoved component:

gmPathPriorityVector = {R_M : SR_M + 1 : P_M : PN_S}

The systemPriorityVector for a PTP Instance S with systemIdentity S_S and clockIdentity C_S is the priority vector that would, with the portIdentity of the TimeReceiverPort set equal to the portIdentity of the transmitting PTP Port, be used as the message priority vector in Announce Messages transmitted on S's ports whose state is TimeTransmitterPort if S was selected as the root:

systemPriorityVector = {S_S : 0 : {C_S : 0} : 0}

The gmPriorityVector for S is the best of the set comprising the systemPriorityVector vector plus every gmPathPriorityVector for which the clockIdentity of the timeTransmitter PTP Instance portIdentity is not the

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clockIdentity of S. If the systemPriorityVector is best, S has been selected as the root. When the best gmPathPriorityVector is that of PTP Port PN_S above, then:

gmPriorityVector = $\{S_S : 0 : \{C_S : 0\} : 0\}$ if S is better than R_M, or

gmPriorityVector = $\{R_M : SR_M + 1 : P_M : PN_S\}$ if S is worse than R_M.

The timeTransmitterPriorityVector for a PTP Port Q on PTP Instance S is the gmPriorityVector with S's clockIdentity C_S substituted for the clockIdentity of the timeTransmitter portIdentity, and Q's portNumber PN_Q substituted for the portNumber of the timeTransmitter portIdentity and for the portNumber of the receiving PTP Port:

timeTransmitterPriorityVector = $\{S_S : 0 : \{C_S : PN_Q\} : PN_Q\}$ if S is better than R_M, or
timeTransmitterPriorityVector = $\{R_M : SR_M + 1 : \{C_S : PN_Q\} : PN_Q\}$ if S is worse than R_M.

If the timeTransmitterPriorityVector is better than the portPriorityVector, the PTP Port will be the TimeTransmitterPort for the attached gPTP communication path and the portPriorityVector will be updated. The messagePriorityVector information in Announce messages transmitted by a PTP Port always includes the first three components of the timeTransmitterPriorityVector of the PTP Port.

NOTE—The consistent use of lower numerical values to indicate better information is deliberate as the TimeTransmitterPort that is closest to the root, i.e., has a numerically lowest path cost component, is selected from among potential alternatives for any given gPTP communication path. Adopting the conventions that lower numerical values indicate better information, that where possible more significant priority components are encoded earlier in the octet sequence of an Announce message, and that earlier octets in the encoding of individual components are more significant allows concatenated octets that compose a priority vector to be compared as if they were a multiple octet encoding of a single number, without regard to the boundaries between the encoded components. To reduce the confusion that naturally arises from having the lesser of two numerical values represent the better of the two, i.e., the one to be chosen all other factors being equal, this clause uses the following consistent terminology. Relative numeric values are described as "least," "lesser," "equal," and "greater," and their comparisons as "less than," "equal to," or "greater than," while relative synchronization tree priorities are described as "best," "better," "the same," "different," and "worse" and their comparisons as "better than," "the same as," "different from," and "worse than." The operators "<" and "==" represent less than and equal to, respectively. The terms "superior" and "inferior" are used for comparisons that are not simply based on priority, but can include the fact that the priority vector of a TimeTransmitterPort can replace an earlier vector transmitted in an Announce message by the same PTP Port.

10.3.6 PTP Port state assignments

10.3.6.1 PTP Port state assignments when the BTCA is used

The BTCA assigns one of the following PTP Port states to each PTP Port: TimeTransmitterPort, TimeReceiverPort, PassivePort, or DisabledPort.

The DisabledPort state is assigned if portOper is FALSE (see 10.2.5.12), ptpPortEnabled is FALSE (see 10.2.5.13), or asCapable is FALSE (see 10.2.5.1).

A PTP Port for which portOper, ptpPortEnabled, and asCapable are all TRUE has its PTP Port state assigned according to the source and relative priority of the synchronization tree portPriorityVector (see 10.3.4 and 10.3.5) as follows:

a) If the PTP Instance is not the root, the source of the gmPriorityVector is the TimeReceiverPort.
b) Each PTP Port whose portPriorityVector is its timeTransmitterPriorityVector is a TimeTransmitterPort.

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c) Each PTP Port, other than the TimeReceiverPort, whose portPriorityVector has been received from another PTP Instance or another PTP Port on this PTP Instance is a PassivePort.

10.3.6.2 PTP Port state assignments when external port configuration is used

If external port configuration is used, one of the states TimeTransmitterPort, TimeReceiverPort, PassivePort, or DisabledPort is assigned to each PTP Port by an external entity, as described in this subclause.

The DisabledPort state is assigned if portOper is FALSE (see 10.2.5.12), ptpPortEnabled is FALSE (see 10.2.5.13), or asCapable is FALSE (see 10.2.5.1).

The member externalPortConfigurationPortDS.desiredState (see 14.14.2) is used by an external entity to set the state of the respective PTP Port to TimeTransmitterPort, TimeReceiverPort, or PassivePort. When this member is set, its value is copied to the per PTP Port local variable portStateInd (see 10.3.15.1.5). If portOper, ptpPortEnabled, and asCapable are all TRUE for this PTP Port, the PTP Port state is set equal to the value of externalPortConfigurationPortDS.desiredState by copying the value of this member to the element of the selectedState array (see 10.2.4.20) for this PTP Port.

10.3.7 Overview of best timeTransmitter clock selection, external port configuration, and announce interval setting state machines

10.3.7.1 Best timeTransmitter clock selection state machines overview

The best timeTransmitter clock selection function in a PTP Instance is specified by a number of cooperating state machines. Figure 10-12 is not itself a state machine, but illustrates the machines, their local variables, their interrelationships, their performance parameters, and the global variables and structures used to communicate between them.

NOTE—The BTCA state machines are all invoked by the media-independent layer, i.e., by the SiteSync and PortSync entities. The media-dependent layer, i.e., the MD entity, simply takes an Announce message received from the PortSync entity of the same PTP Port and gives it to the next lower layer (e.g., IEEE 802.3, IEEE 802.11). It is the PortSync entity that generates and consumes Announce messages.

The following media-independent layer state machines are in Figure 10-12:

a) PortAnnounceReceive (one instance per PTP Instance, per PTP Port): receives Announce information from the MD entity of the same PTP Port, determines if the Announce message is qualified and, if so, sets the rcvdMsg variable. This state machine is invoked by the PortSync entity of the PTP Port.

b) PortAnnounceInformation (one instance per PTP Instance, per PTP Port): receives new qualified Announce information from the PortAnnounceReceive state machine, determines if the Announce information is better than the current best timeTransmitter information it knows about, and updates the current best timeTransmitter information when it receives updated PTP Port state information from the PortStateSelection state machine and when announce receipt timeout or, when gmPresent is TRUE, sync receipt timeout occurs. This state machine is invoked by the PortSync entity of the PTP Port.

c) PortStateSelection (one instance per PTP Instance): updates the gmPathPriority vector for each PTP Port of the PTP Instance, the gmPriorityVector for the PTP Instance, and the timeTransmitterPriorityVector for each PTP Port of the PTP Instance; determines the PTP Port state for each PTP Port; and updates gmPresent. This state machine is invoked by the SiteSync entity of the PTP Instance.

d) PortAnnounceTransmit (one instance per PTP Instance, per PTP Port): If the PTP Port state is TimeTransmitterPort, transmits Announce information to the MD entity when an announce interval has elapsed; or PTP Port states have been updated; or portPriority and portStepsRemoved

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information has been updated with newly determined timeTransmitterPriority and timeTransmitterStepsRemoved information. This state machine is invoked by the PortSync entity of the PTP Port and is also used when external port configuration is used.

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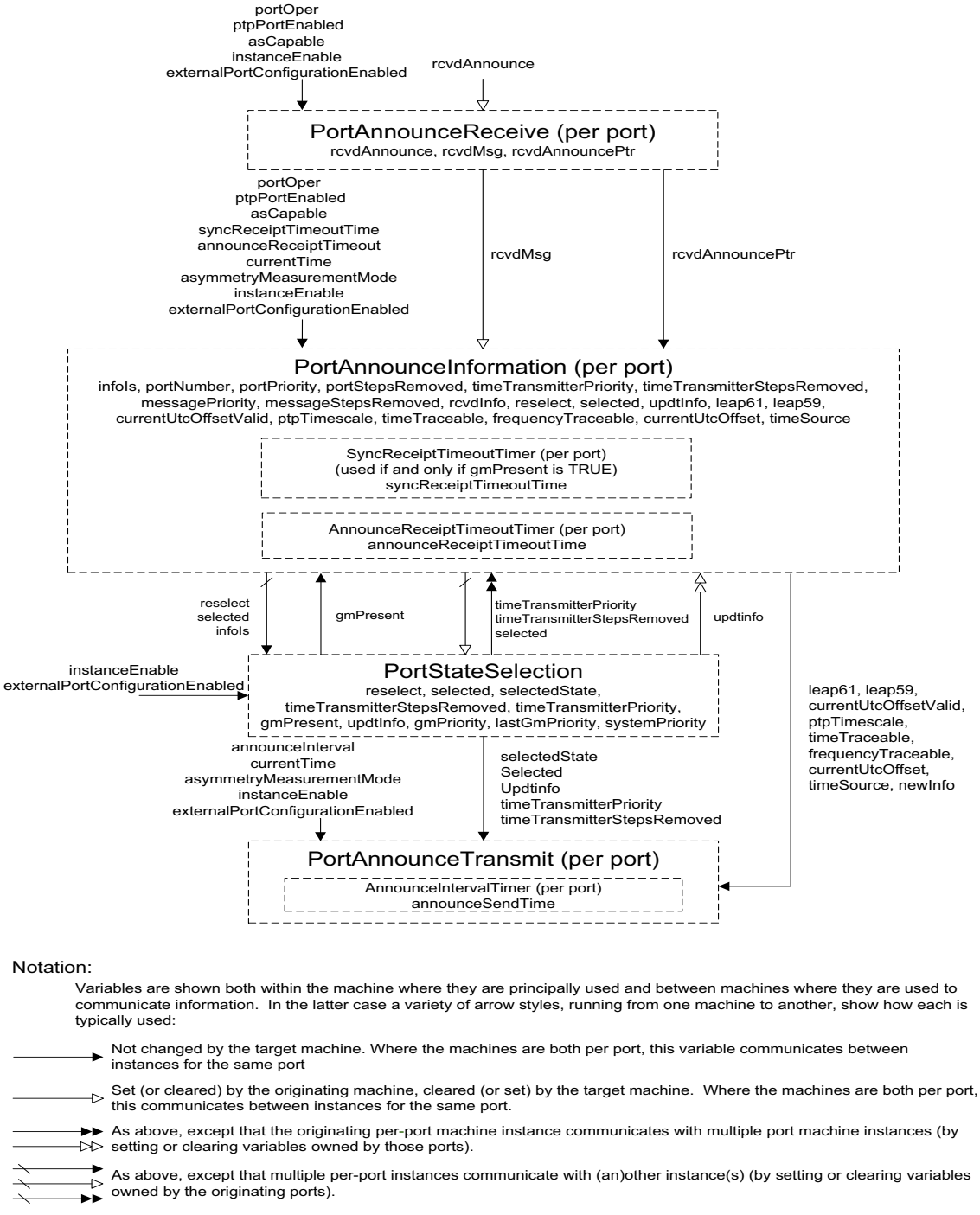


Figure 10-12—Best timeTransmitter clock selection state machines—overview and interrelationships

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10.3.7.2 External port configuration state machines overview

The external port configuration function in a PTP Instance is specified by a number of cooperating state machines. Figure 10-13 is not itself a state machine, but illustrates the machines, their local variables, their interrelationships, their performance parameters, and the global variables and structures used to communicate between them.

NOTE—The external port configuration state machines are all invoked by the media-independent layer and are per PTP Port, i.e., they are invoked by the PortSync entity for the respective PTP Port. The media-dependent layer, i.e., the MD entity, simply takes an Announce message received from the PortSync entity of the same PTP Port and gives it to the next lower layer (e.g., IEEE 802.3, IEEE 802.11). It is the PortSync entity that generates and consumes Announce messages.

The following media-independent layer state machines are in Figure 10-13:

a) PortAnnounceInformationExt (one instance per PTP Instance, per PTP Port): Receives and stores new Announce information received in Announce messages.

b) PortStateSettingExt (one instance per PTP Instance): Copies the desired PTP Port state for the PTP Port to the respective selectedState array element, updates gmPresent, computes timeTransmitterStepsRemoved, stores the time properties information in the respective global variables, and computes the gmPriorityVector and timeTransmitterPriorityVector.

c) PortAnnounceTransmit (one instance per PTP Instance, per PTP Port): If the PTP Port state is TimeTransmitterPort, transmits Announce information to the MD entity when an announce interval has elapsed, or PTP Port states have been updated, or portPriority and portStepsRemoved information has been updated with newly determined TimeTransmitterPriority and timeTransmitterStepsRemoved information. This state machine is invoked by the PortSync entity of the PTP Port and is also used when the BTCA is used.

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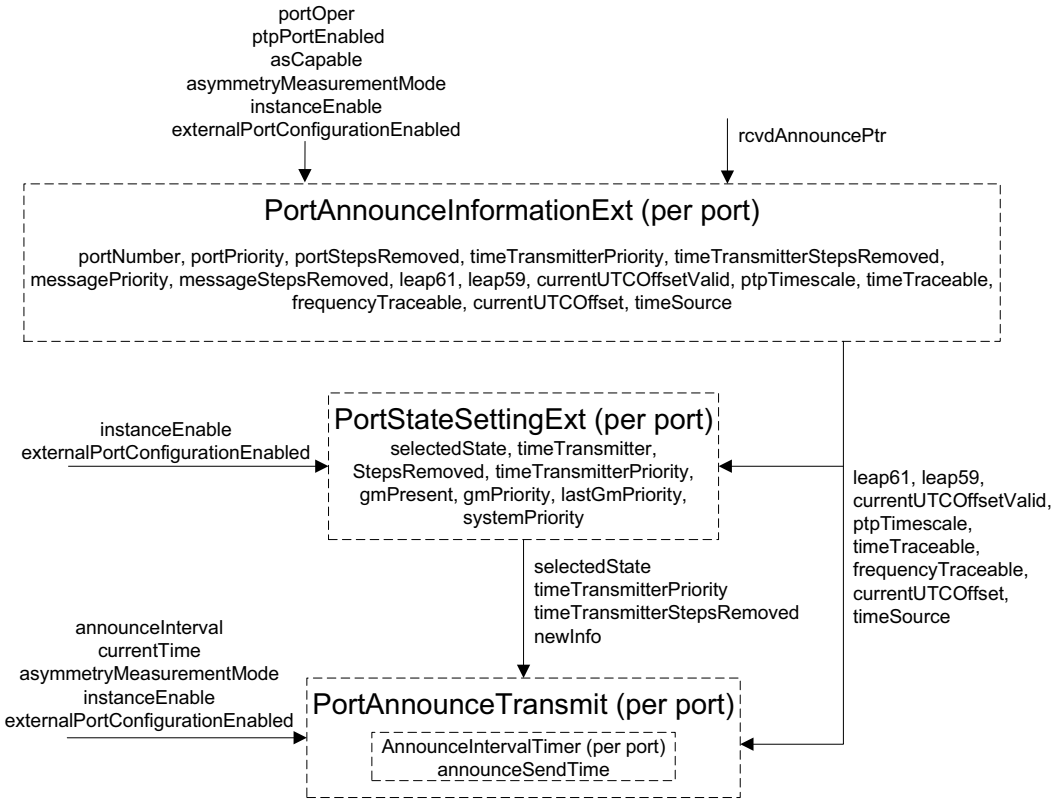


Figure 10-13—External port configuration state machines—overview and interrelationships

10.3.7.3 AnnounceIntervalSetting state machine overview

An additional state machine, the AnnounceIntervalSetting state machine, receives a Signaling message that contains a message interval request TLV (see 10.6.4.3) and sets the global variables that give the duration of the mean interval between successive Announce messages.

10.3.8 Overview of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines

10.3.9 and 10.3.10 define global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines whose scopes are as follows:

— Per PTP Instance (i.e., per domain)
— Per PTP Instance, per PTP Port
— Instance used by CMLDS (see 11.2.16) (i.e., variable is common across all Link Ports)
— Instance used by CMLDS, per Link Port

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Table 10-3 summarizes the scope of each global variable of 10.3.9 and 10.3.10.

Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10)

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
reselect	10.3.9.1	Boolean array	Per PTP Instance	No	No
selected	10.3.9.2	Boolean array	Per PTP Instance	No	No
timeTransmitterStepsRemoved	10.3.9.3	UInteger16	Per PTP Instance	No	No
leap61	10.3.9.4	Boolean	Per PTP Instance	No	No
leap59	10.3.9.5	Boolean	Per PTP Instance	No	No
currentUtcOffsetValid	10.3.9.6	Boolean	Per PTP Instance	No	No
ptpTimescale	10.3.9.7	Boolean	Per PTP Instance	No	No
timeTraceable	10.3.9.8	Boolean	Per PTP Instance	No	No
frequencyTraceable	10.3.9.9	Boolean	Per PTP Instance	No	No
currentUtcOffset	10.3.9.10	UInteger16	Per PTP Instance	No	No
timeSource	10.3.9.11	TimeSource	Per PTP Instance	No	No
sysLeap61	10.3.9.12	Boolean	Per PTP Instance	No	No
sysLeap59	10.3.9.13	Boolean	Per PTP Instance	No	No
sysCurrentUtcOffsetValid	10.3.9.14	Boolean	Per PTP Instance	No	No
sysPtpTimescale	10.3.9.15	Boolean	Per PTP Instance	No	No
sysTimeTraceable	10.3.9.16	Boolean	Per PTP Instance	No	No
sysFrequencyTraceable	10.3.9.17	Boolean	Per PTP Instance	No	No
sysCurrentUtcOffset	10.3.9.18	Integer16	Per PTP Instance	No	No

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Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10) *(continued)*

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
sysTimeSource	10.3.9.19	TimeSource	Per PTP Instance	No	No
systemPriority	10.3.9.20	UInteger224	Per PTP Instance	No	No
gmPriority	10.3.9.21	UInteger224	Per PTP Instance	No	No
lastGmPriority	10.3.9.22	UInteger224	Per PTP Instance	No	No
pathTrace	10.3.9.23	ClockIdentity[N]	Per PTP Instance	No	No
externalPortConfigurationEnabled	10.3.9.24	Boolean	Per PTP Instance	No	No
lastAnnouncePort	10.3.9.25	UInteger16	Per PTP Instance	No	No
announceReceiptTimeoutTimeInterval	10.3.10.1	UScaledNs	Per PTP Port	No	No
announceSlowdown	10.3.10.2	Boolean	Per PTP Port	No	No
oldAnnounceInterval	10.3.10.3	UScaledNs	Per PTP Port	No	No
infoIs	10.3.10.4	Enumeration2	Per PTP Port	No	No
timeTransmitterPriority	10.3.10.5	UInteger224	Per PTP Port	No	No
currentLogAnnounceInterval	10.3.10.6	Integer8	Per PTP Port	No	No
initialLogAnnounceInterval	10.3.10.7	Integer8	Per PTP Port	No	No
announceInterval	10.3.10.8	UScaledNs	Per PTP Port	No	No
messageStepsRemoved	10.3.10.9	UInteger16	Per PTP Port	No	No
newInfo	10.3.10.10	Boolean	Per PTP Port	No	No
portPriority	10.3.10.11	UInteger224	Per PTP Port	No	No
portStepsRemoved	10.3.10.12	UInteger16	Per PTP Port	No	No

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Table 10-3—Summary of scope of global variables used by best timeTransmitter clock selection, external port configuration, and announce interval setting state machines (see 10.3.9 and 10.3.10) *(continued)*

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
rcvdAnnouncePtr	10.3.10.13	Pointer to a structure	Per PTP Port	No	No
rcvdMsg	10.3.10.14	Boolean	Per PTP Port	No	No
updtInfo	10.3.10.15	Boolean	Per PTP Port	No	No
annLeap61	10.3.10.16	Boolean	Per PTP Port	No	No
annLeap59	10.3.10.17	Boolean	Per PTP Port	No	No
annCurrentUtcOffsetValid	10.3.10.18	Boolean	Per PTP Port	No	No
annPtpTimescale	10.3.10.19	Boolean	Per PTP Port	No	No
annTimeTraceable	10.3.10.20	Boolean	Per PTP Port	No	No
annFrequencyTraceable	10.3.10.21	Boolean	Per PTP Port	No	No
annCurrentUtcOffset	10.3.10.22	Integer16	Per PTP Port	No	No
annTimeSource	10.3.10.23	TimeSource	Per PTP Port	No	No
receivedPathTrace	10.3.10.24	ClockIdentity[N]	Per PTP Port	No	No

10.3.9 Per PTP Instance global variables

10.3.9.1 reselect: A Boolean array of length numberPorts + 1 (see 8.6.2.8). Setting reselect[j], where $0 \le j \le$ numberPorts, to TRUE causes the STATE_SELECTION block of the PortStateSelection state machine (see 10.3.13) to be re-entered, which in turn causes the PTP Port state of each PTP Port of the PTP Instance to be updated (via the function updtStatesTree(); see 10.3.13.2.4). This variable is used only by the BTCA, i.e., not by the external port configuration option.

10.3.9.2 selected: A Boolean array of length numberPorts + 1 (see 8.6.2.8). selected[j], where $0 \le j \le$ numberPorts, is set to TRUE immediately after the PTP Port states of all the ports are updated. This value indicates to the PortAnnounceInformation state machine (see 10.3.12) that it can update the portPriorityVector and other variables for each PTP Port. This variable is used by both the BTCA and the external port configuration option; however, its value does not impact the external port configuration option (see the NOTE in 10.3.16.3).

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NOTE—Array elements 0 of the reselect and selected arrays are not used, except that the function clearReselectTree() sets reselect[0] to FALSE when it sets the entire array to zero and the function setSelectedTree() sets selected[0] to TRUE when it sets the entire array to TRUE. This action is taken only for convenience, so that array element *j* can correspond to PTP Port *j*. Note also that, in contrast, selectedState[0] is used (see 10.2.4.20).

10.3.9.3 timeTransmitterStepsRemoved: The value of stepsRemoved for the PTP Instance, after the PTP Port states of all the ports have been updated (see 10.3.13.2.4 for details on the computation of timeTransmitterStepsRemoved). This variable is used by both the BTCA and the external port configuration option.

10.3.9.4 leap61: A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative to the current Grandmaster Clock, contains 61 s and FALSE if the last minute of the current UTC day does not contain 61 s. This variable is used by both the BTCA and the external port configuration option.

10.3.9.5 leap59: A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative to the current Grandmaster Clock, contains 59 s and FALSE if the last minute of the current UTC day does not contain 59 s. This variable is used by both the BTCA and the external port configuration option.

10.3.9.6 currentUtcOffsetValid: A Boolean variable whose value is TRUE if currentUtcOffset (see 10.3.9.10), relative to the current Grandmaster Clock, is known to be correct and FALSE if currentUtcOffset is not known to be correct. This variable is used by both the BTCA and the external port configuration option.

10.3.9.7 ptpTimescale: A Boolean variable whose value is TRUE if the timescale of the current Grandmaster Clock is PTP (see 8.2.1) and FALSE if the timescale is ARB. This variable is used by both the BTCA and the external port configuration option.

10.3.9.8 timeTraceable: A Boolean variable whose value is TRUE if both ClockTimeReceiverTime [i.e., the synchronized time maintained at the timeReceiver (see 10.2.4.3)] and currentUtcOffset (see 10.3.9.10), relative to the current Grandmaster Clock, are traceable to a primary reference and FALSE if one or both are not traceable to a primary reference. This variable is used by both the BTCA and the external port configuration option.

10.3.9.9 frequencyTraceable: A Boolean variable whose value is TRUE if the frequency that determines clockTimeReceiverTime, i.e., the frequency of the LocalClockEntity multiplied by the most recently computed rateRatio by the PortSyncSyncReceive state machine (see 10.2.8.1.4), is traceable to a primary reference and FALSE if this frequency is not traceable to a primary reference. This variable is used by both the BTCA and the external port configuration option.

10.3.9.10 currentUtcOffset: The difference between TAI time and UTC time, i.e., TAI time minus UTC time, in seconds, and relative to the current Grandmaster Clock, when known. Otherwise, the value has no meaning (see 10.3.9.6). This variable is used by both the BTCA and the external port configuration option.

NOTE—For example, 2006-01-01 00:00:00 UTC and 2006-01-01 00:00:33 TAI represent the same instant of time. At this time, currentUtcOffset was equal to 33 s.[19]

10.3.9.11 timeSource: The value of the timeSource attribute of the current Grandmaster PTP Instance. The data type for timeSource is TimeSource (see 8.6.2.7). This variable is used by both the BTCA and the external port configuration option.

10.3.9.12 sysLeap61: A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative to the ClockTimeTransmitter entity of this PTP Instance, contains 61 s and FALSE if the last minute

[19] Note also that a leap second was not added at the end of the last UTC minute of 2005-12-31.

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of the current UTC day does not contain 61 s. This variable is used by both the BTCA and the external port configuration option.

10.3.9.13 sysLeap59: A Boolean variable whose value is TRUE if the last minute of the current UTC day, relative to the ClockTimeTransmitter entity of this PTP Instance, contains 59 s and FALSE if the last minute of the current UTC day does not contain 59 s. This variable is used by both the BTCA and the external port configuration option.

10.3.9.14 sysCurrentUtcOffsetValid: A Boolean variable whose value is TRUE if currentUtcOffset (see 10.3.9.10), relative to the ClockTimeTransmitter entity of this PTP Instance, is known to be correct and FALSE if currentUtcOffset is not known to be correct. This variable is used by both the BTCA and the external port configuration option.

10.3.9.15 sysPtpTimescale: A Boolean variable whose value is TRUE if the timescale of the ClockTimeTransmitter entity of this PTP Instance is PTP (see 8.2.1) and FALSE if the timescale of the ClockTimeTransmitter entity of this PTP Instance is ARB. This variable is used by both the BTCA and the external port configuration option.

10.3.9.16 sysTimeTraceable: A Boolean variable whose value is TRUE if both timeTransmitterTime [i.e., the time maintained by the ClockTimeTransmitter entity of this PTP Instance (see 10.2.4.21)] and currentUtcOffset (see 10.3.9.10), relative to the ClockTimeTransmitter entity of this PTP Instance, are traceable to a primary reference and FALSE if one or both are not traceable to a primary reference. This variable is used by both the BTCA and the external port configuration option.

10.3.9.17 sysFrequencyTraceable: A Boolean variable whose value is TRUE if the frequency that determines timeTransmitterTime of the ClockTimeTransmitter entity of this PTP Instance, i.e., the frequency of the LocalClockEntity multiplied by the most recently computed gmRateRatio by the ClockTimeTransmitterSyncReceive state machine (see 10.2.4.14 and 10.2.11), is traceable to a primary reference and FALSE if this frequency is not traceable to a primary reference. This variable is used by both the BTCA and the external port configuration option.

10.3.9.18 sysCurrentUtcOffset: The difference between TAI time and UTC time, i.e., TAI time minus UTC time, in seconds, and relative to the ClockTimeTransmitter entity of this PTP Instance, when known. Otherwise, the value has no meaning (see 10.3.9.14). This variable is used by both the BTCA and the external port configuration option.

NOTE—See the NOTE in 10.3.9.10 for more detail on the sign convention.

10.3.9.19 sysTimeSource: The value of the timeSource attribute of the ClockTimeTransmitter entity of this PTP Instance (see 8.6.2.7).

10.3.9.20 systemPriority: The systemPriority vector for this PTP Instance. The data type for systemPriority is UInteger224 (see 10.3.5).

10.3.9.21 gmPriority: The current gmPriorityVector for the PTP Instance.

10.3.9.22 lastGmPriority: The previous gmPriorityVector for the PTP Instance, prior to the most recent invocation of the PortStateSelection state machine. lastGmPriority is used only by the BTCA, i.e., not by the external port configuration option.

10.3.9.23 pathTrace: An array that contains the clockIdentities of the successive PTP Instances that receive, process, and send Announce messages. The data type for pathTrace is ClockIdentity[N], where N is the number of PTP Instances, including the Grandmaster PTP Instance, that the Announce information has traversed. This variable is used by both the BTCA and the external port configuration option.

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NOTE 1—N is equal to stepsRemoved + 1 (see 10.6.3.2.6). The size of the pathTrace array can change after each reception of an Announce message, up to the maximum size for the respective medium. For example, the maximum value of N for a FDE IEEE 802.3 medium is 179. This is obtained from the fact that the number of PTP octets in an Announce message is 68 + 8N, where N is the number of entries in the pathTrace array (see 10.6.3.1 and Table 10-11), and the maximum payload size for FDE IEEE 802.3 media is 1500 octets. Setting 68 + 8N = 1500, and solving for N gives N = 179.

NOTE 2—The current behavior for the path trace feature is documented in 10.3.11.2.1 and 10.3.16.2.1 and is as follows:

- Item c) of 10.3.11.2.1, the description of the qualifyAnnounce() function of the PortAnnounceReceive state machine, indicates that if a path trace TLV is present and one of the elements of the pathSequence array field is equal to the clockIdentity of the clock where the TLV is being processed, the Announce message is not qualified.
- Item d) of 10.3.11.2.1 (qualifyAnnounce() function) indicates that if the Announce message is qualified and a path trace TLV is present, the pathSequence array of the TLV is copied to the pathTrace array (described in this subclause) and the clockIdentity of the PTP Instance that processes the Announce message is appended to the array. However, if a path trace TLV is not present, the path trace array is empty.
- Item f) of 10.3.16.2.1, the description of the txAnnounce() function of the PortAnnounceTransmit state machine, indicates that a path trace TLV is constructed and appended to an Announce message just before the Announce message is transmitted only if the pathTrace array is not empty and appending the TLV does not cause the media-dependent layer frame to exceed any respective maximum size. If appending the TLV does cause a respective maximum frame size to be exceeded or if the pathTrace array is empty, the TLV is not appended.
- As a result of the behaviors of the qualifyAnnounce() and txAnnounce() functions described in this note, the path trace feature is no longer used, i.e., a path trace TLV is not appended to an Announce message and the pathTrace array is empty, when appending a clockIdentity to the TLV would cause the frame carrying the Announce message to exceed its maximum size.

NOTE 3—Once the value of stepsRemoved of an Announce message reaches 255, the Announce message is not qualified [see item b) of 10.3.11.2.1].

10.3.9.24 externalPortConfigurationEnabled: A variable whose value indicates whether PTP Port states are externally configured or determined by the BTCA. The value TRUE indicates that the PTP Port states are externally configured; the value FALSE indicates that the PTP Port states are determined by the BTCA. This variable is used by both the BTCA and the external port configuration option.

10.3.9.25 lastAnnouncePort: The PTP Port number of the PTP Port on which the most recent Announce message was received. This variable is used by the PortAnnounceInformationExt and PortStateSettingExt state machines for the external port configuration option. This variable is not used by the BTCA.

10.3.10 Per-port global variables

10.3.10.1 announceReceiptTimeoutTimeInterval: The time interval after which announce receipt timeout occurs if an Announce message has not been received during the interval. The value of announceReceiptTimeoutTimeInterval is equal to announceReceiptTimeout (see 10.7.3.2) multiplied by the announceInterval (see 10.3.10.8) for the PTP Port at the other end of the link to which this PTP Port is attached. The value of announceInterval for the PTP Port at the other end of the link is computed from logMessageInterval of the received Announce message (see 10.6.2.2.14). This variable is used only by the BTCA, i.e., not by the external port configuration option.

10.3.10.2 announceSlowdown: A Boolean that is set to TRUE if the AnnounceIntervalSetting state machine (see Figure 10-20 in 10.3.17.3) receives a TLV that requests a larger Announce message transmission interval (see 10.7.2.2) and FALSE otherwise. When announceSlowdown is set to TRUE, the PortAnnounceTransmit state machine (see Figure 10-19) continues to send Announce messages at the old (i.e., faster) rate until a number of Announce messages equal to announceReceiptTimeout (see 10.7.3.2) have been sent, but with the logMessageInterval field of the PTP common header set equal to the new announce interval (i.e., corresponding to the slower rate). After announceReceiptTimeout Announce

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messages have been sent, subsequent Announce messages are sent at the new (i.e., slower) rate and with the logMessageInterval field of the PTP common header set to the new announce interval. This variable is used by both the BTCA and the external port configuration option. When announceSlowdown is set to FALSE, the PortAnnounceTransmit state machine immediately sends Announce messages at the new (i.e., faster or the same) rate.

NOTE—If a receiver of Announce messages requests a slower rate, the receiver continues to use the upstream announceInterval value, which it obtains from the logMessageInterval field of received Announce messages, until it receives an Announce message where that value has changed. If, immediately after requesting a slower Announce message rate, up to announceReceiptTimeout minus one consecutive Announce messages sent to the receiver are lost, announce receipt timeout could occur if the sender had changed to the slower rate immediately. Delaying the slowing down of the sending rate of Announce messages for announceReceiptTimeout messages prevents announce receipt timeout from occurring until at least announceReceiptTimeout Announce messages have been lost. Note that networks with high packet loss can still experience announce receipt timeout under high-packet-loss conditions; however, the announce receipt timeout condition occurs only after at least announceReceiptTimeout Announce messages have been lost.

10.3.10.3 oldAnnounceInterval: The saved value of the previous announce interval, when a new announce interval is requested via a Signaling message that contains a message interval request TLV. This variable is used by both the BTCA and the external port configuration option.

10.3.10.4 infoIs: An Enumeration2 that takes the values Received, Mine, Aged, or Disabled to indicate the origin and state of the PTP Port's synchronization tree information:

a) If infoIs is Received, the PTP Port has received current information (i.e., announce receipt timeout has not occurred and, if gmPresent is TRUE, sync receipt timeout also has not occurred) from the timeTransmitter PTP Instance for the attached gPTP communication path.

b) If infoIs is Mine, information for the PTP Port has been derived from the TimeReceiverPort for the PTP Instance (with the addition of TimeReceiverPort stepsRemoved). This includes the possibility that the TimeReceiverPort is the PTP Port whose portNumber is 0, i.e., the PTP Instance is the root of the gPTP domain.

c) If infoIs is Aged, announce receipt timeout or, when gmPresent is TRUE, sync receipt timeout has occurred.

d) If portOper, ptpPortEnabled, and asCapable are not all TRUE, infoIs is Disabled.

The variable infoIs is used only by the BTCA, i.e., not by the external port configuration option.

10.3.10.5 timeTransmitterPriority: The timeTransmitterPriorityVector for the PTP Port. This variable is used by both the BTCA and the external port configuration option.

10.3.10.6 currentLogAnnounceInterval: The current value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Announce messages (see 10.7.2.2). This value is set in the AnnounceIntervalSetting state machine (see 10.3.17). This variable is used by both the BTCA and the external port configuration option.

10.3.10.7 initialLogAnnounceInterval: The initial value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Announce messages (see 10.7.2.2). This variable is used by both the BTCA and the external port configuration option.

10.3.10.8 announceInterval: A variable containing the mean Announce message transmission interval for the PTP Port. This value is set in the AnnounceIntervalSetting state machine (see 10.3.17). This variable is used by both the BTCA and the external port configuration option.

10.3.10.9 messageStepsRemoved: The value of stepsRemoved contained in the received Announce information. This variable is used by both the BTCA and the external port configuration option.

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10.3.10.10 newInfo: A Boolean variable that is set to cause a PTP Port to transmit Announce information; specifically, it is set when an announce interval has elapsed (see Figure 10-19), or PTP Port states have been updated, or portPriority and portStepsRemoved information has been updated with newly determined timeTransmitterPriority and timeTransmitterStepsRemoved information. This variable is used by both the BTCA and the external port configuration option.

10.3.10.11 portPriority: The portPriorityVector for the PTP Port. This variable is used only by the BTCA, i.e., not by the external port configuration option.

10.3.10.12 portStepsRemoved: The value of stepsRemoved for the PTP Port. portStepsRemoved is set equal to timeTransmitterStepsRemoved (see 10.3.9.3) after timeTransmitterStepsRemoved is updated. This variable is used by both the BTCA and the external port configuration option.

10.3.10.13 rcvdAnnouncePtr: A pointer to a structure that contains the fields of a received Announce message. This variable is used by both the BTCA and the external port configuration option.

10.3.10.14 rcvdMsg: A Boolean variable that is TRUE if a received Announce message is qualified and FALSE if it is not qualified. This variable is used only by the BTCA, i.e., not by the external port configuration option.

10.3.10.15 updtInfo: A Boolean variable that is set to TRUE to indicate that the PortAnnounceInformation state machine (see 10.3.12) should copy the newly determined timeTransmitterPriority and timeTransmitterStepsRemoved to portPriority and portStepsRemoved, respectively. This variable is used by both the BTCA and the external port configuration option; however, its value does not impact the external port configuration option (see the NOTE in 10.3.16.3).

10.3.10.16 annLeap61: A global variable in which the leap61 flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.17 annLeap59: A global variable in which the leap59 flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.18 annCurrentUtcOffsetValid: A global variable in which the currentUtcOffsetValid flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.19 annPtpTimescale: A global variable in which the ptpTimescale flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.20 annTimeTraceable: A global variable in which the timeTraceable flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.21 annFrequencyTraceable: A global variable in which the frequencyTraceable flag (see 10.6.2.2.8) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.22 annCurrentUtcOffset: A global variable in which the currentUtcOffset field (see 10.6.3.2.1) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

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10.3.10.23 annTimeSource: A global variable in which the timeSource field (see 10.6.3.2.1) of a received Announce message is saved. This variable is used by both the BTCA and the external port configuration option.

10.3.10.24 receivedPathTrace: An array in which the pathSequence array field of the path trace TLV of the most recently received Announce message is saved. The data type for receivedPathTrace is ClockIdentity[N], where N is the number of entries in the pathSequence array field.

10.3.11 PortAnnounceReceive state machine

10.3.11.1 State machine variables

The following variable is used in the state diagram in Figure 10-14 (in 10.3.11.3):

10.3.11.1.1 rcvdAnnouncePAR: A Boolean variable that notifies the current state machine when Announce message information is received from the MD entity of the same PTP Port. This variable is reset by this state machine.

10.3.11.2 State machine functions

10.3.11.2.1 qualifyAnnounce (rcvdAnnouncePtr): Qualifies the received Announce message pointed to by rcvdAnnouncePtr as follows:

a) If the Announce message was sent by the current PTP Instance, i.e., if sourcePortIdentity.clockIdentity (see 10.6.2.2.11 and 8.5.2) is equal to thisClock (see 10.2.4.22), the Announce message is not qualified, and FALSE is returned;

b) If the stepsRemoved field is greater than or equal to 255, the Announce message is not qualified, and FALSE is returned;

c) If a path trace TLV is present and one of the elements of the pathSequence array field of the path trace TLV is equal to thisClock (i.e., the clockIdentity of the current PTP Instance; see 10.2.4.22), the Announce message is not qualified, and FALSE is returned;

d) Otherwise, the Announce message is qualified, and TRUE is returned. If a path trace TLV is present, it is saved in the per port global variable receivedPathTrace. If a path trace TLV is not present, the per port global variable receivedPathTrace is set to the empty array.

10.3.11.3 State diagram

The PortAnnounceReceive state machine shall implement the function specified by the state diagram in Figure 10-14, the local variable specified in 10.3.11.1, the function specified in 10.3.11.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. The state machine is not used if externalPortConfigurationEnabled is TRUE. The machine receives Announce information from the MD entity of the same PTP Port, determines if the Announce message is qualified, and if so, sets the rcvdMsg variable.

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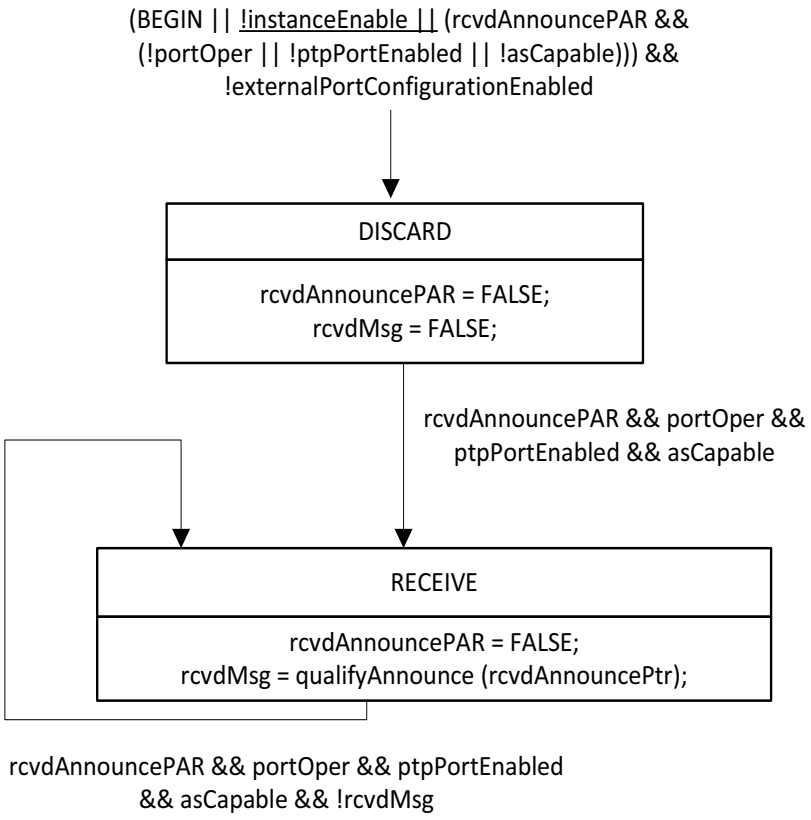


Figure 10-14—PortAnnounceReceive state machine

10.3.12 PortAnnounceInformation state machine

10.3.12.1 State machine variables

The following variables are used in the state diagram in Figure 10-15 (in 10.3.12.3):

10.3.12.1.1 announceReceiptTimeoutTime: A variable used to save the time at which announce receipt timeout occurs. The data type for announceReceiptTimeoutTime is UScaledNs.

10.3.12.1.2 messagePriorityPAI: The messagePriorityVector corresponding to the received Announce information. The data type for messagePriorityPAI is UInteger224 (see 10.3.4).

10.3.12.1.3 rcvdInfo: An Enumeration2 that holds the value returned by rcvInfo() (see 10.3.12.2.1).

10.3.12.1.4 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-15). The data type for TEMP is Integer16.

10.3.12.2 State machine functions

10.3.12.2.1 rcvInfo (rcvdAnnouncePtr): Decodes the messagePriorityVector (see 10.3.4 and 10.3.5) and stepsRemoved 10.6.3.2.6) field from the Announce information pointed to by rcvdAnnouncePtr (see 10.3.10.13), and then:

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a) Stores the messagePriorityVector and stepsRemoved field value in messagePriorityPAI and messageStepsRemoved, respectively, and then:
   1) If the received message conveys the PTP Port state TimeTransmitterPort and the messagePriorityVector is the same as the portPriorityVector of the PTP Port, returns RepeatedTimeTransmitterInfo; else
   2) If the received message conveys the PTP Port state TimeTransmitterPort and the messagePriorityVector is superior to the portPriorityVector of the PTP Port, returns SuperiorTimeTransmitterInfo; else
   3) If the received message conveys the PTP Port state TimeTransmitterPort, and the messagePriorityVector is worse than the portPriorityVector of the PTP Port, returns InferiorTimeTransmitterInfo; else
   4) Returns OtherInfo.

NOTE—In accordance with 10.3.5, the messagePriorityVector is superior to the portPriorityVector of the PTP Port if, and only if, the messagePriorityVector is better than the portPriorityVector, or the Announce message has been transmitted from the same timeTransmitter PTP Instance and TimeTransmitterPort as the portPriorityVector. In steps a) 1) to a) 4) in this subclause, rcvInfo() first checks whether the messagePriorityVector and portPriorityVector are the same (and the received message conveys the PTP Port state TimeTransmitterPort), before checking whether the messagePriorityVector is superior to the portPriorityVector. The reason for this sequence is that RepeatedTimeTransmitterInfo needs to be returned if the messagePriorityVector and portPriorityVector are the same, while SuperiorTimeTransmitterInfo needs to be returned in other instances where the Announce message has been transmitted from the same timeTransmitter PTP Instance and TimeTransmitterPort as the portPriorityVector (if the test for SuperiorTimeTransmitterInfo were done before the test for RepeatedTimeTransmitterInfo, SuperiorTimeTransmitterInfo would be returned when RepeatedTimeTransmitterInfo is desired).

10.3.12.2.2 recordOtherAnnounceInfo(): Saves the flags leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource, of the received Announce message for this PTP Port. The values are saved in the per-PTP Port global variables annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable, annCurrentUtcOffset, and annTimeSource (see 10.3.10.16 through 10.3.10.23).

10.3.12.3 State diagram

The PortAnnounceInformation state machine shall implement the function specified by the state diagram in Figure 10-15, the local variables specified in 10.3.12.1, the functions specified in 10.3.12.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if externalPortConfigurationEnabled is FALSE (if this variable is TRUE, the PortAnnounceInformationExt state machine of 10.3.14.3 is used instead). The state machine receives new qualified Announce information from the PortAnnounceReceive state machine (see 10.3.11) of the same PTP Port and determines if the Announce information is better than the current best timeTransmitter information it knows. The state machine also updates the current best timeTransmitter information when it receives updated PTP Port state information from the PortStateSelection state machine (see 10.3.13) and when announce receipt timeout or, when gmPresent is TRUE, sync receipt timeout occurs.

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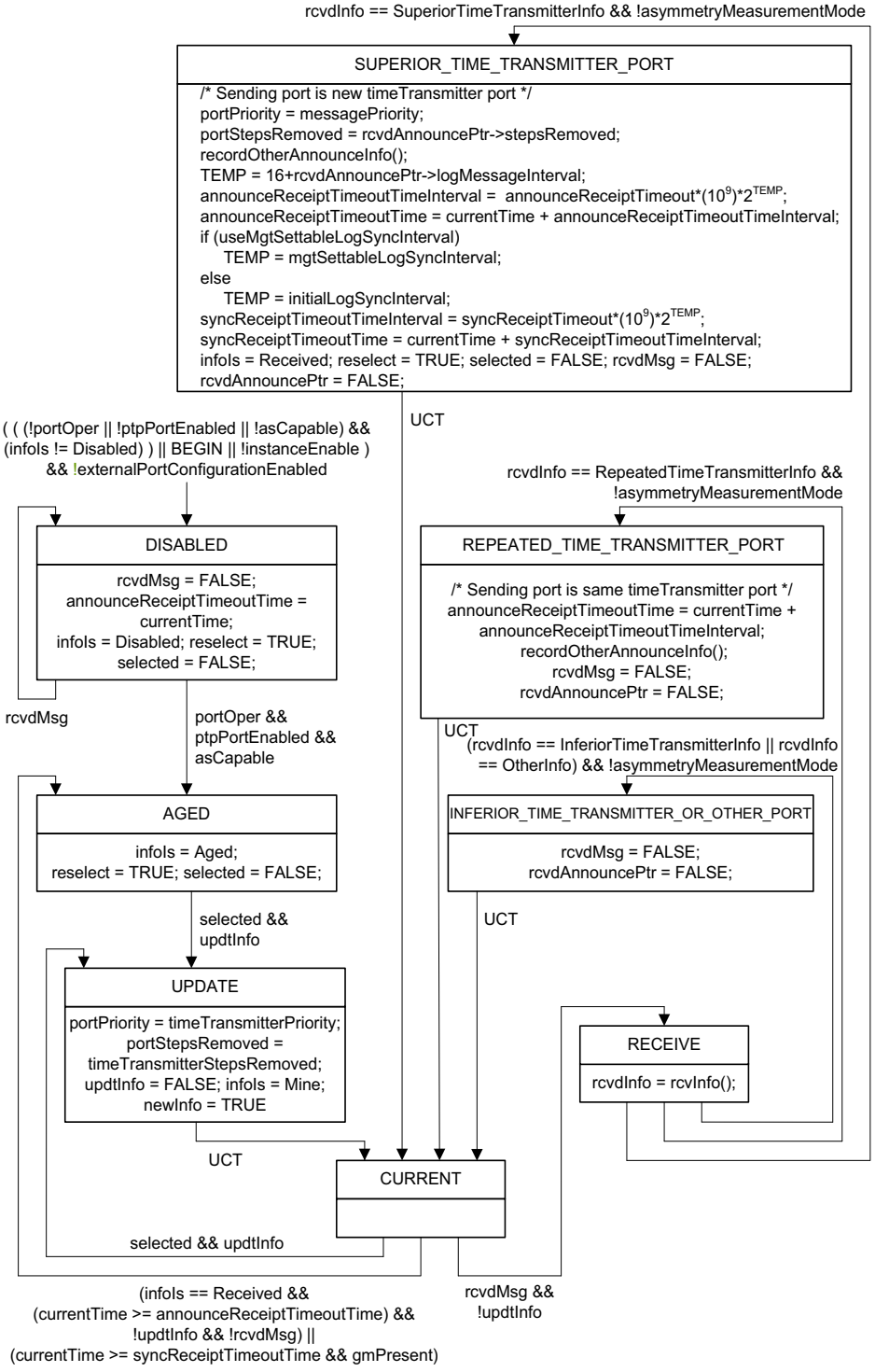


Figure 10-15—PortAnnounceInformation state machine

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10.3.13 PortStateSelection state machine

10.3.13.1 State machine variables

The following variables are used in the state diagram in Figure 10-16 (in 10.3.13.3):

10.3.13.1.1 systemIdentityChange: A Boolean variable that notifies the current state machine when the systemIdentity (see 10.3.2) of this PTP Instance has changed. This variable is reset by this state machine.

The systemIdentity changes when at least one of the attributes priority1, clockClass, clockAccuracy, offsetScaledLogVariance, and priority2 changes (e.g., due to management action, degradation or loss of the ClockSource). The systemIdentity also includes the attribute clockIdentity, but this attribute does not change.

10.3.13.1.2 asymmetryMeasurementModeChange: A Boolean variable that notifies the current state machine when the per-port variable asymmetryMeasurementMode (see 10.2.5.2) changes on any port. This variable is reset by this state machine. There is one instance of asymmetryMeasurementModeChange for all the domains (per port). The variable is accessible by all the domains.

10.3.13.2 State machine functions

10.3.13.2.1 updtStateDisabledTree(): Sets all the elements of the selectedState array (see 10.2.4.20) to DisabledPort. Sets lastGmPriority to all ones. Sets the pathTrace array (see 10.3.9.23) to contain the single element thisClock (see 10.2.4.22).

10.3.13.2.2 clearReselectTree(): Sets all the elements of the reselect array (see 10.3.9.1) to FALSE.

10.3.13.2.3 setSelectedTree(): Sets all the elements of the selected array (see 10.3.9.2) to TRUE.

10.3.13.2.4 updtStatesTree(): Performs the following operations (see 10.3.4 and 10.3.5 for details on the priority vectors):

a) Computes the gmPathPriorityVector for each PTP Port that has a portPriorityVector and for which neither announce receipt timeout nor, if gmPresent is TRUE, sync receipt timeout have occurred.

b) Saves gmPriority (see 10.3.9.21) in lastGmPriority (see 10.3.9.22), computes the gmPriorityVector for the PTP Instance and saves it in gmPriority, chosen as the best of the set consisting of the systemPriorityVector (for this PTP Instance) and the gmPathPriorityVector for each PTP Port for which the clockIdentity of the timeTransmitter port is not equal to thisClock (see 10.2.4.22).

c) Sets the per PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource as follows:

    1) If the gmPriorityVector was set to the gmPathPriorityVector of one of the ports, then leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource are set to annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable, annCurrentUtcOffset, and annTimeSource, respectively, for that PTP Port.

    2) If the gmPriorityVector was set to the systemPriorityVector, then leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource are set to sysLeap61, sysLeap59, sysCurrentUtcOffsetValid, sysPtpTimescale, sysTimeTraceable, sysFrequencyTraceable, sysCurrentUtcOffset, and sysTimeSource, respectively.

d) Updates the timePropertiesDS members leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource with the values of the corresponding global variables computed in item c) above.

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e) Updates the parentDS members grandmasterIdentity, grandmasterClockQuality.clockClass, grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance, grandmasterPriority1, grandmasterPriority2 with the clockIdentity, clockClass, clockAccuracy, offsetScaledLogVariance, priority1, and priority2 attributes, respectively, of the systemIdentity component of the gmPrioirity computed in item b) above.

f) Updates the parentDS member parentPortIdentity with the sourcePortIdentity component of the gmPriority computed in item b) above.

g) Computes the timeTransmitterPriorityVector for each PTP Port.

h) Computes timeTransmitterStepsRemoved, which is equal to one of the following:

1) messageStepsRemoved (see 10.3.10.9) for the PTP Port associated with the gmPriorityVector, incremented by 1, if the gmPriorityVector is not the systemPriorityVector, or

2) 0 if the gmPriorityVector is the systemPriorityVector.

i) Sets currentDS.stepsRemoved equal to timeTransmitterStepsRemoved.

j) Assigns the PTP Port state for PTP Port *j*, and sets selectedState[*j*] equal to this PTP Port state, as follows, for *j* = 1, 2, ..., numberPorts:

1) If the PTP Port is disabled (infoIs == Disabled), then selectedState[*j*] is set to DisabledPort.

2) If asymmetryMeasurementMode is TRUE, then selectedState[*j*] is set to PassivePort, and updtInfo is set to FALSE.

3) If announce receipt timeout, or sync receipt timeout with gmPresent set to TRUE, has occurred (infoIs = Aged), then selectedState[*j*] is set to TimeTransmitterPort, and updtInfo is set to TRUE.

4) If the portPriorityVector was derived from another PTP Port on the PTP Instance or from the PTP Instance itself as the root (infoIs == Mine), then selectedState[*j*] is set to TimeTransmitterPort. In addition, updtInfo is set to TRUE if the portPriorityVector differs from the timeTransmitterPriorityVector or portStepsRemoved differs from timeTransmitterStepsRemoved.

5) If the portPriorityVector was received in an Announce message, neither announce receipt timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received), and the gmPriorityVector is now derived from the portPriorityVector, then selectedState[*j*] is set to TimeReceiverPort, and updtInfo is set to FALSE. The per port global variable receivedPathTrace, for this port, is copied to the per PTP Instance global array pathTrace, and, if it is not empty, thisClock is appended to pathTrace.

6) If the portPriorityVector was received in an Announce message, neither announce receipt timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received), the gmPriorityVector is not now derived from the portPriorityVector, the timeTransmitterPriorityVector is not better than the portPriorityVector, and the sourcePortIdentity component of the portPriorityVector *does not* reflect another PTP Port on the PTP Instance, then selectedState[*j*] is set to PassivePort, and updtInfo is set to FALSE.

7) If the portPriorityVector was received in an Announce message, neither announce receipt timeout nor sync receipt timeout with gmPresent TRUE has not occurred (infoIs == Received), the gmPriorityVector is not now derived from the portPriorityVector, the timeTransmitterPriorityVector is not better than the portPriorityVector, and the sourcePortIdentity component of the portPriorityVector *does* reflect another PTP Port on the PTP Instance, then selectedState[*j*] set to PassivePort, and updtInfo is set to FALSE.

8) If the portPriorityVector was received in an Announce message, neither announce receipt timeout nor sync receipt timeout with gmPresent TRUE has occurred (infoIs == Received), the gmPriorityVector is not now derived from the portPriorityVector, and the timeTransmitterPriorityVector is better than the portPriorityVector, then selectedState[*j*]is set to TimeTransmitterPort, and updtInfo is set to TRUE.

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k) Updates gmPresent as follows:
   1) gmPresent is set to TRUE if the priority1 field of the rootSystemIdentity of the gmPriorityVector is less than 255.
   2) gmPresent is set to FALSE if the priority1 field of the rootSystemIdentity of the gmPriorityVector is equal to 255.

l) Assigns the PTP Port state for PTP Port 0 (see 8.5.2.3), and sets selectedState[0] as follows:
   1) if selectedState[*j*] is set to TimeReceiverPort for any PTP Port with portNumber *j*, *j* = 1, 2, ..., numberPorts, selectedState[0] is set to PassivePort.
   2) if selectedState[*j*] is *not* set to TimeReceiverPort for any PTP Port with portNumber *j*, *j* = 1, 2, ..., numberPorts, selectedState[0] is set to TimeReceiverPort.

m) If the clockIdentity member of the systemIdentity (see 10.3.2) member of gmPriority (see 10.3.9.21) is equal to thisClock (see 10.2.4.22), i.e., if the current PTP Instance is the Grandmaster PTP Instance, the pathTrace array is set to contain the single element thisClock (see 10.2.4.22).

10.3.13.3 State diagram

The PortStateSelection state machine shall implement the function specified by the state diagram in Figure 10-16, the functions specified in 10.3.13.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if externalPortConfigurationEnabled is FALSE (if this variable is TRUE, the PortStateSettingExt state machine is used instead). The state machine updates the gmPathPriority vector for each PTP Port of the PTP Instance, the gmPriorityVector for the PTP Instance, and the timeTransmitterPriorityVector for each PTP Port of the PTP Instance. The state machine determines the PTP Port state for each PTP Port and updates gmPresent.

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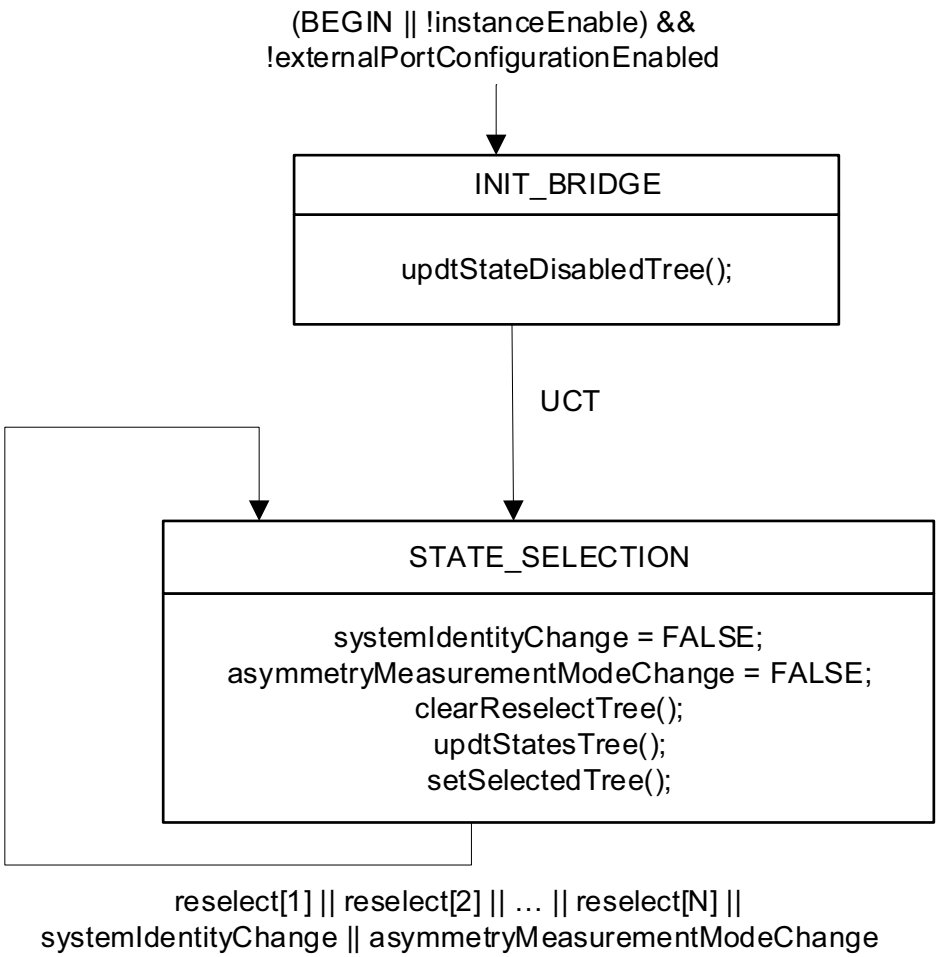


Figure 10-16—PortStateSelection state machine

10.3.14 PortAnnounceInformationExt state machine

10.3.14.1 State machine variables

The following variables are used in the state diagram in Figure 10-17 (in 10.3.14.3):

10.3.14.1.1 rcvdAnnouncePAIE: A Boolean variable that notifies the current state machine when Announce message information is received from the MD entity of the same PTP Port. This variable is reset by this state machine.

10.3.14.1.2 messagePriorityPAIE: The messagePriorityVector corresponding to the received Announce information. The data type for messagePriorityPAIE is UInteger224 (see 10.3.4).

10.3.14.2 State machine functions

10.3.14.2.1 rcvInfoExt (rcvdAnnouncePtr): Decodes the messagePriorityVector (see 10.3.4 and 10.3.5) and stepsRemoved (see 10.6.3.2.6) field from the Announce information pointed to by rcvdAnnouncePtr (see 10.3.10.13), and then stores the messagePriorityVector and stepsRemoved field value in messagePriorityPAIE and messageStepsRemoved, respectively. If a path trace TLV is present in the

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Announce message and the portState of the PTP Port is TimeReceiverPort, the pathSequence array field of the TLV is copied to the global array pathTrace, and thisClock is appended to pathTrace (i.e., is added to the end of the array).

10.3.14.2.2 recordOtherAnnounceInfo(): Saves the flags leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource, of the received Announce message for this PTP Port. The values are saved in the per-PTP Port global variables annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable, annCurrentUtcOffset, and annTimeSource (see 10.3.10.16 through 10.3.10.23).

10.3.14.3 State diagram

The PortAnnounceInformationExt state machine shall implement the function specified by the state diagram in Figure 10-17, the local variables specified in 10.3.14.1, the functions specified in 10.3.14.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if externalPortConfigurationEnabled is TRUE (if this variable is FALSE, the PortAnnounceInformation state machine of 10.3.12.3 is used instead). The state machine receives Announce information from the MD entity of the same PTP Port and saves the information.

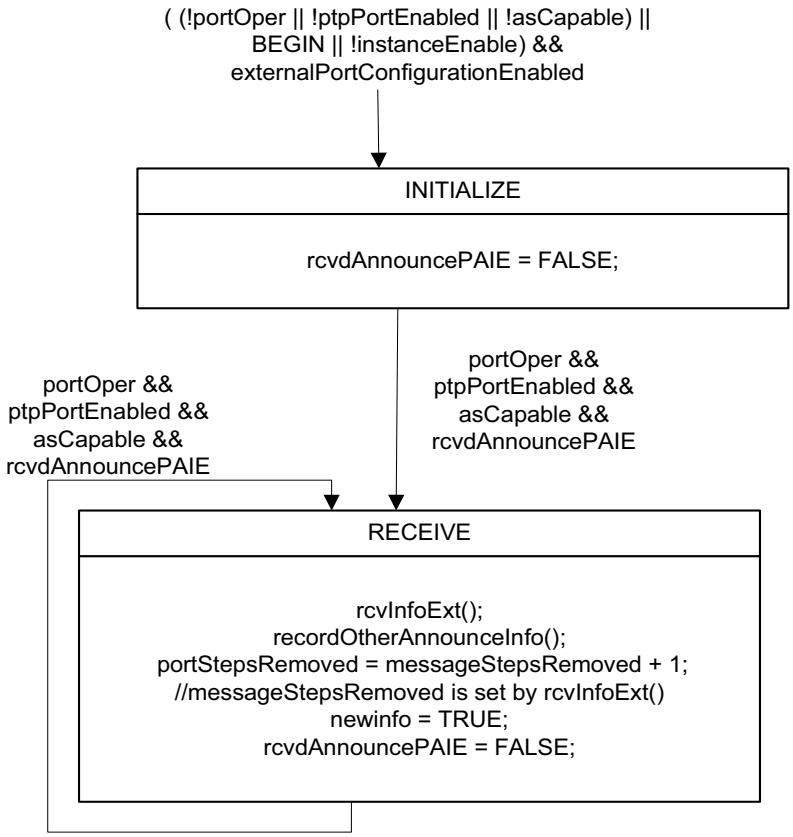


Figure 10-17—PortAnnounceInformationExt state machine

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10.3.15 PortStateSettingExt state machine

10.3.15.1 State machine variables

The following variables are used in the state diagram in Figure 10-18 (in 10.3.15.3):

10.3.15.1.1 disabledExt: A Boolean variable that notifies the current state machine (when it is set to TRUE) when at least one of the variables portOper, ptpPortEnabled, or asCapable, for this PTP Port, has changed from TRUE to FALSE. This variable is reset by this state machine.

10.3.15.1.2 reenabledExt: A Boolean variable that notifies the current state machine (when it is set to TRUE) when all of the variables portOper, ptpPortEnabled, and asCapable, for this PTP Port, that are FALSE have changed to TRUE. This variable is reset by this state machine.

10.3.15.1.3 asymmetryMeasurementModeChangeThisPort: A Boolean variable that notifies the current state machine when the per-port variable asymmetryMeasurementMode (see 10.2.5.2) changes on this port. This variable is reset by this state machine. There is one instance of asymmetryMeasurementModeChangeThisPort for all the domains (per port). The variable is accessible by all the domains.

10.3.15.1.4 rcvdPortStateInd: A Boolean variable that notifies the current state machine (when it is set to TRUE) when the PTP Port state of this PTP Port has been externally set. This variable is reset by this state machine.

10.3.15.1.5 portStateInd: An Enumeration2 that indicates the PTP Port state that has been set. The values are TimeTransmitterPort, TimeReceiverPort, and PassivePort (see 14.14.2).

NOTE—The PTP Port state can be externally set to DisabledPort by setting portOper or ptpPortEnabled to FALSE. The PTP Port state is set to DisabledPort when asCapable becomes FALSE.

10.3.15.2 State machine functions

10.3.15.2.1 resetStateTree(j): Sets selectedState[*j*] (see 10.2.4.20) to externalPortConfigurationPortDS.desiredState (see 14.14.2). Sets the pathTrace array (see 10.3.9.23) to contain the single element thisClock (see 10.2.4.22) if no PTP Port of the PTP Instance has the PTP Port state TimeReceiverPort.

10.3.15.2.2 updtPortState(j): Performs the following operations for PTP Port *j* (see 10.3.4 and 10.3.5 for details on the priority vectors):

a) Sets the per PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource as follows:
   1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort, then leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource are set to annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, annFrequencyTraceable, annCurrentUtcOffset, and annTimeSource, respectively, for that PTP Port.
   2) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort, then leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource are set to sysLeap61, sysLeap59, sysCurrentUtcOffsetValid, sysPtpTimescale, sysTimeTraceable, sysFrequencyTraceable, sysCurrentUtcOffset, and sysTimeSource, respectively.

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b) Update the timePropertiesDS members leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource with the values of the corresponding global variables computed in item a) above.

c) Computes timeTransmitterStepsRemoved as follows:

1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort, then timeTransmitterStepsRemoved is set equal to portStepsRemoved for that PTP Port.

2) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort, then timeTransmitterStepsRemoved is set equal to 0.

d) Sets currentDS.stepsRemoved equal to timeTransmitterStepsRemoved.

e) Assigns the PTP Port state for PTP Port *j*, and sets selectedState[*j*] equal to this PTP Port state, as follows:

1) If disabledExt is TRUE, selectedState[*j*] is set to DisabledPort, else

2) If asymmetryMeasurementMode is TRUE, selectedState[*j*] is set to PassivePort, else

3) selectedState[*j*] is set to portStateInd. If portStateInd is equal to TimeTransmitterPort, newInfo is set to TRUE.

f) Updates gmPresent as follows:

1) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort and the priority1 field of the rootSystemIdentity of the messagePriorityPAIE of the timeReceiver port is less than 255, gmPresent is set to TRUE, else

2) If the PTP Port state of any PTP Port of this PTP Instance (see 8.5.2.3) is TimeReceiverPort and the priority1 field of the rootSystemIdentity of the messagePriorityPAIE of the timeReceiver PTP Port is equal to 255, gmPresent is set to FALSE, else

3) If no PTP Port of this PTP Instance (see 8.5.2.3) has the PTP Port state TimeReceiverPort, gmPresent is set to TRUE if priority1 for this PTP Instance is less than 255 and FALSE if priority1 for this PTP Instance is equal to 255.

g) Assigns the PTP Port state for PTP Port 0, and sets selectedState[0] as follows:

1) If selectedState[*j*] is set to TimeReceiverPort, selectedState[0] is set to PassivePort.

2) If selectedState[*j*] is *not* set to TimeReceiverPort and selectedState[*k*] is not equal to TimeReceiverPort for every *k* not equal to 0 or *j*, selectedState[0] is set to TimeReceiverPort.

h) Computes the gmPriorityVector as follows:

1) If selectedState[*j*] is set to TimeReceiverPort, the gmPriorityVector is set equal to messagePriorityPAIE for PTP Port *j*.

2) If selectedState[*j*] is *not* set to TimeReceiverPort and selectedState[*k*] is not equal to TimeReceiverPort for every *k* not equal to 0 or *j*, the gmPriorityVector is set equal to the systemPriorityVector.

i) Update the parentDS members grandmasterIdentity, grandmasterClockQuality.clockClass, grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance, grandmasterPriority1, grandmasterPriority2 with the clockIdentity, clockClass, clockAccuracy, offsetScaledLogVariance, priority1, and priority2 attributes, respectively, of the systemIdentity component of the gmPrioirityVector computed in item h) above.

j) Update the parentDS member parentPortIdentity with the sourcePortIdentity component of the gmPriorityVector computed in item h) above.

k) Computes the timeTransmitterPriorityVector for PTP Port *j*.

l) If no PTP Port of this PTP Instance has the PTP Port state TimeReceiverPort, the pathTrace array is set to contain the single element thisClock (see 10.2.4.22).

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10.3.15.3 State diagram

The PortStateSettingExt state machine shall implement the function specified by the state diagram in Figure 10-18, the local variables specified in 10.3.15.1, the functions specified in 10.3.15.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. This state machine is used only if externalPortConfigurationEnabled is TRUE (if this variable is FALSE, the PortStateSelection state machine of 10.3.13.3 is used instead). A separate instance of this state machine runs on each PTP Port (unlike the PortStateSelection state machine, for which a single instance runs in the PTP Instance and performs operations on all the ports).

The state machine updates the gmPriorityVector for the PTP Instance and the timeTransmitterPriorityVector for each PTP Port of the PTP Instance. The state machine determines the PTP Port state for each PTP Port and updates gmPresent.

NOTE—It is possible to use the external port configuration mechanism to mis-configure the network, e.g., to produce a configuration where one or more PTP Instances have more than one timeReceiver port. Detecting and correcting mis-configurations is outside the scope of this standard.

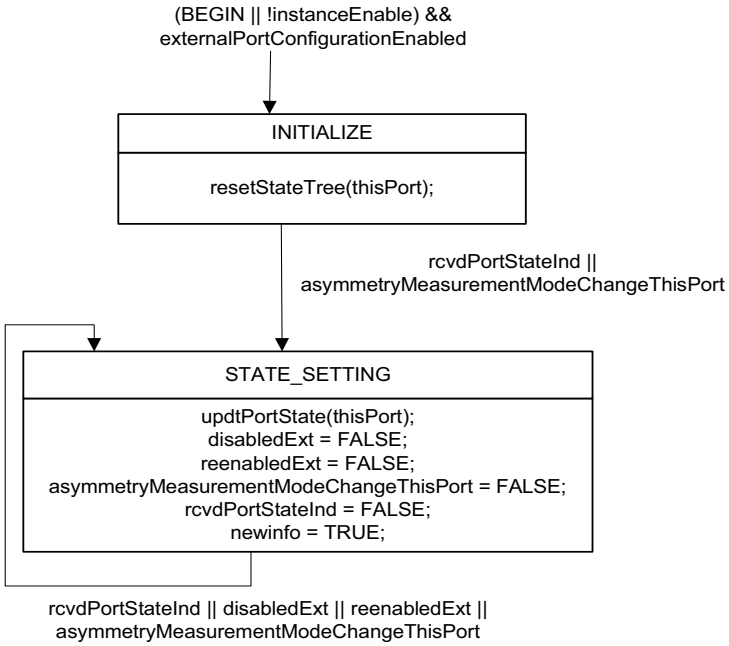


Figure 10-18—PortStateSettingExt state machine

10.3.16 PortAnnounceTransmit state machine

10.3.16.1 State machine variables

The following variables are used in the state diagram in Figure 10-19 (in 10.3.16.3):

10.3.16.1.1 announceSendTime: The time, relative to the LocalClock, at which the next transmission of Announce information is to occur. The data type for announceSendTime is UScaledNs.

10.3.16.1.2 numberAnnounceTransmissions: A count of the number of consecutive Announce message transmissions after the AnnounceIntervalSetting state machine (see Figure 10-20 in 10.3.17.3) has set

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announceSlowdown (see 10.3.10.2) to TRUE. The data type for numberAnnounceTransmissions is UInteger8.

10.3.16.1.3 interval2: A local variable that holds either announceInterval or oldAnnounceInterval. The data type for interval2 is UScaledNs.

10.3.16.2 State machine functions

10.3.16.2.1 txAnnounce(): Transmits Announce information to the MD entity of this PTP Port. The Announce information is set as follows:

a) The components of the messagePriorityVector are set to the values of the respective components of the timeTransmitterPriorityVector of this PTP Port.

b) The grandmasterIdentity, grandmasterClockQuality, grandmasterPriority1, and grandmasterPriority2 fields of the Announce message are set equal to the corresponding components of the messagePriorityVector.

c) The value of the stepsRemoved field of the Announce message is set equal to timeTransmitterStepsRemoved.

d) The Announce message flags leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the Announce message fields currentUtcOffset and timeSource, are set equal to the values of the global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, frequencyTraceable, currentUtcOffset, and timeSource, respectively (see 10.3.9.4 through 10.3.9.11).

e) The sequenceId field of the Announce message is set in accordance with 10.5.7.

f) A path trace TLV (see 10.6.3.3) is constructed, with its pathSequence field (see 10.6.3.3.4) set equal to the pathTrace array (see 10.3.9.23). If appending the path trace TLV to the Announce message does not cause the media-dependent layer frame to exceed any respective maximum size, the path trace TLV is appended to the Announce message; otherwise, it is not appended. If the pathTrace array is empty, the path trace TLV is not appended. See 10.3.9.23 for a description of the path trace feature.

10.3.16.3 State diagram

The PortAnnounceTransmit state machine shall implement the function specified by the state diagram in Figure 10-19, the local variables specified in 10.3.16.1, the functions specified in 10.3.16.2, and the relevant global variables specified in 10.2.4, 10.2.5, 10.3.9, and 10.3.10. The state machine transmits Announce information to the MD entity when an announce interval has elapsed, PTP Port states have been updated, and portPriority and portStepsRemoved information has been updated with newly determined timeTransmitterPriority and timeTransmitterStepsRemoved information.

NOTE—When the external port configuration option is used (i.e., externalPortConfigurationEnabled is TRUE; see 10.3.9.24) the values of the variables updtInfo and selected do not affect the operation of the PortAnnounceTransmit state machine because the term of the conditions in which they appear, i.e., (selected && !updtInfo) || externalPortConfiguartionEnabled, evaluates to TRUE when externalPortConfigurationEnabled is TRUE.

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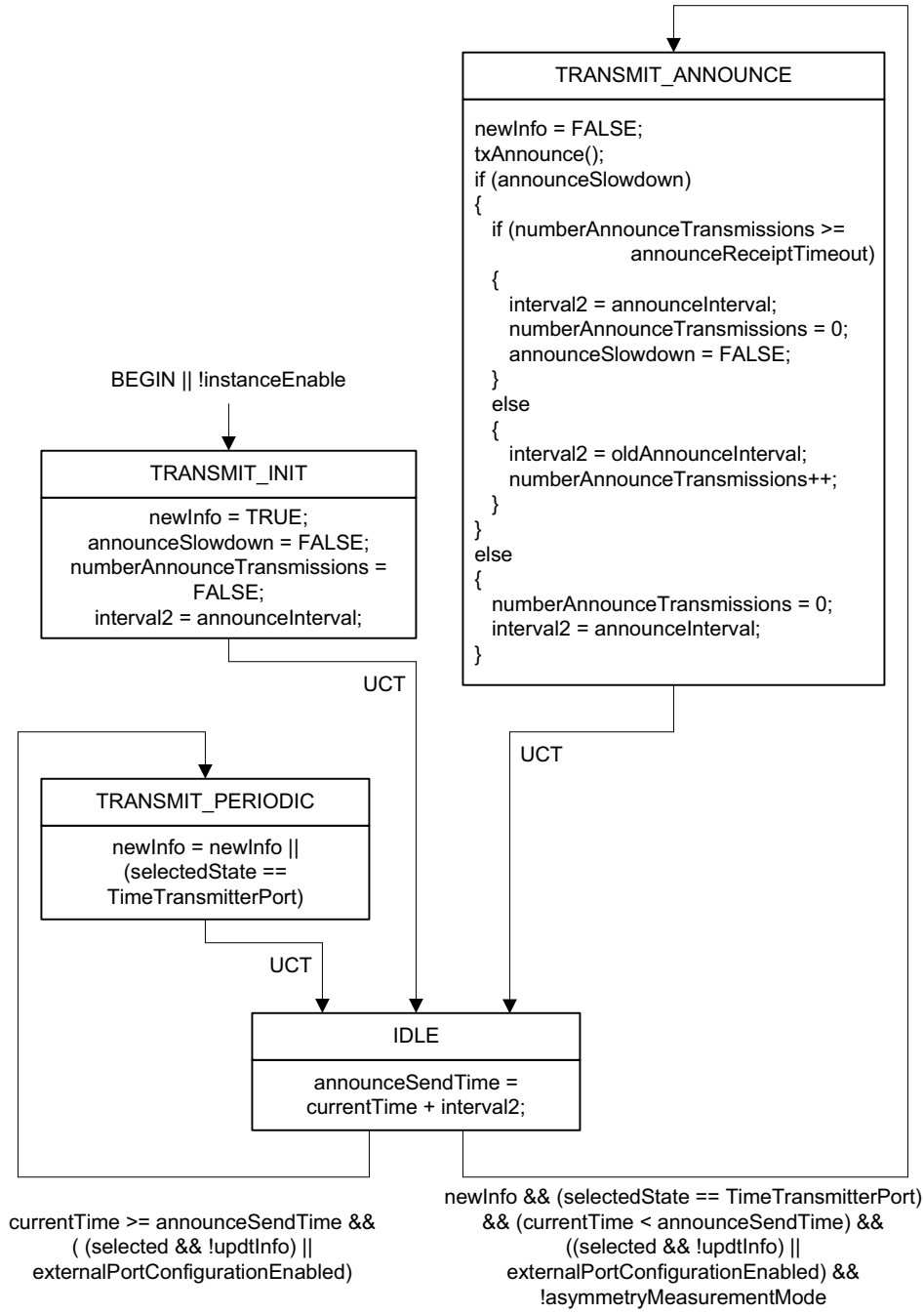


Figure 10-19—PortAnnounceTransmit state machine

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10.3.17 AnnounceIntervalSetting state machine

10.3.17.1 State machine variables

The following variables are used in the state diagram in Figure 10-20 (in 10.3.17.3):

10.3.17.1.1 rcvdSignalingMsg2: A Boolean variable that notifies the current state machine when a Signaling message that contains a message interval request TLV (see 10.6.4.3) is received. This variable is reset by the current state machine.

10.3.17.1.2 rcvdSignalingPtrAIS: A pointer to a structure whose members contain the values of the fields of the received Signaling message that contains a message interval request TLV (see 10.6.4.3).

10.3.17.1.3 logSupportedAnnounceIntervalMax: The maximum supported logarithm to base 2 of the announce interval. The data type for logSupportedAnnounceIntervalMax is Integer8.

10.3.17.1.4 logSupportedClosestLongerAnnounceInterval: The logarithm to base 2 of the announce interval, such that logSupportedClosestLongerAnnounceInterval > logRequestedAnnounceInterval, that is numerically closest to logRequestedAnnounceInterval, where logRequestedAnnounceInterval is the argument of the function computeLogAnnounceInterval() (see 10.3.17.2.2). The data type for logSupportedClosestLongerAnnounceInterval is Integer8.

10.3.17.1.5 computedLogAnnounceInterval: A variable used to hold the result of the function computeLogAnnounceInterval(). The data type for computedLogAnnounceInterval is Integer8.

10.3.17.1.6 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-20). The data type for TEMP is Integer16.

10.3.17.2 State machine functions

10.3.17.2.1 isSupportedLogAnnounceInterval (logAnnounceInterval): A Boolean function that returns TRUE if the announce interval given by the argument logAnnounceInterval is supported by the PTP Port and FALSE if the announce interval is not supported by the PTP Port. The argument logAnnounceInterval has the same data type and format as the field logAnnounceInterval of the message interval request TLV (see 10.6.4.3.8).

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10.3.17.2.2 computeLogAnnounceInterval (logRequestedAnnounceInterval): An Integer8 function that computes and returns the logAnnounceInterval, based on the logRequestedAnnounceInterval. This function is defined as indicated below. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
Integer8 computeLogAnnounceInterval (logRequestedAnnounceInterval)
Integer8 logRequestedAnnounceInterval;
{
        Integer8 logSupportedAnnounceIntervalMax,
                                logSupportedClosestLongerAnnounceInterval;
        if (isSupportedLogAnnounceInterval (logRequestedAnnounceInterval))
                // The requested Announce Interval is supported and returned
                return (logRequestedAnnounceInterval)
        else
        {
                if (logRequestedAnnounceInterval > logSupportedAnnounceIntervalMax)
                        // Return the largest supported logAnnounceInterval, even if smaller than
the requested interval
                        return (logSupportedAnnounceIntervalMax);
                else
                        // Return the smallest supported logAnnounceInterval that is still larger
than
                        // the requested interval.
                        return (logSupportedClosestLongerAnnounceInterval);
        }
}
```

10.3.17.3 State diagram

The AnnounceIntervalSetting state machine shall implement the function specified by the state diagram in Figure 10-20, the local variables specified in 10.3.17.1, the functions specified in 10.3.17.2, the messages specified in 10.6, the relevant global variables specified in 10.2.5 and 10.3.10, the relevant managed objects specified in 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting the global variables that give the duration of the mean interval between successive Announce messages, both at initialization and in response to the receipt of a Signaling message that contains a message interval request TLV (see 10.6.4.3).

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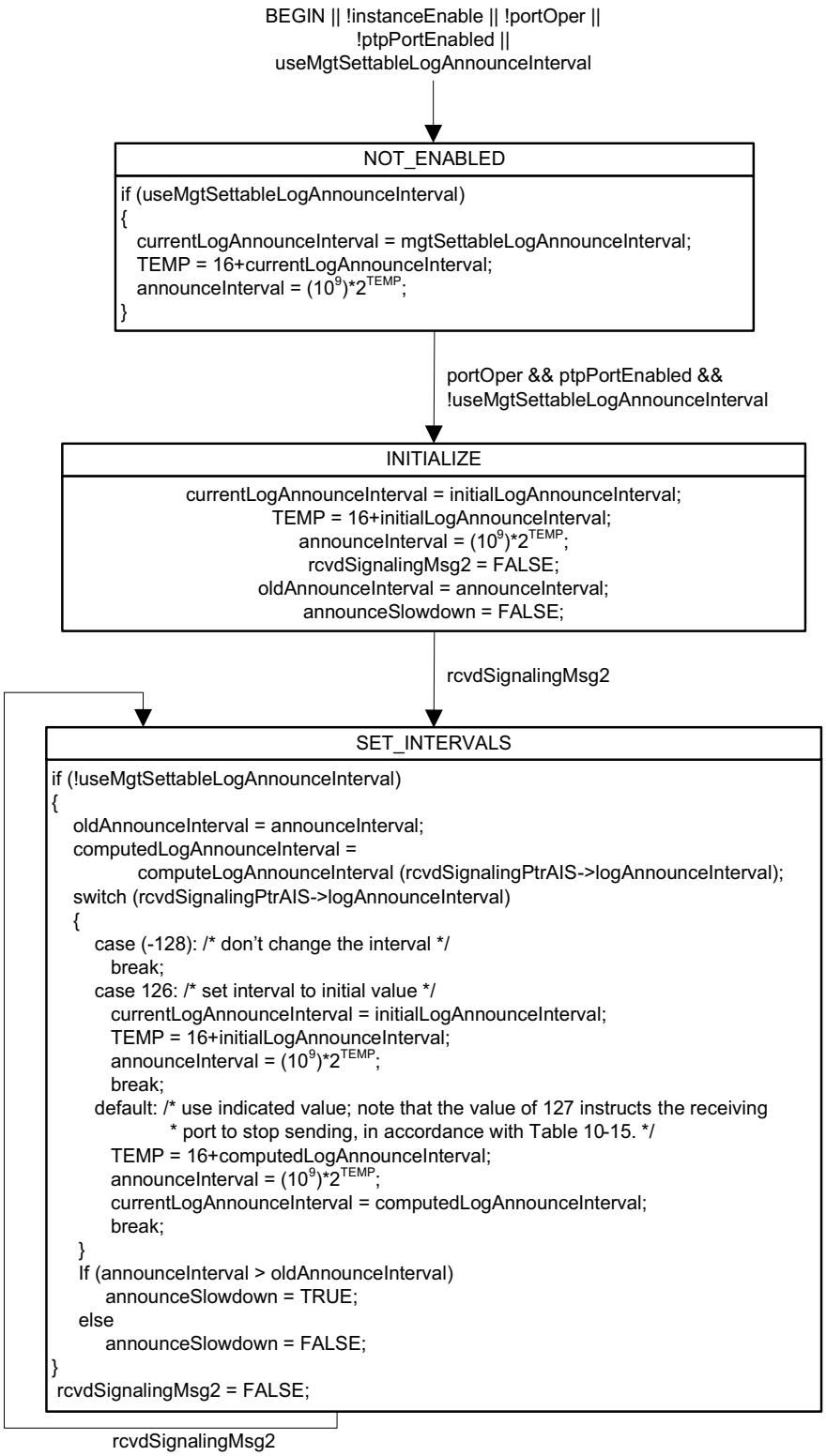


Figure 10-20—AnnounceIntervalSetting state machine

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10.4 State machines related to signaling gPTP capability

10.4.1 Enabling and disabling the state machines

If the managed object gptpCapableStateMachinesEnabled (see 14.10.56) is TRUE, the GptpCapableTransmit and GptpCapableReceive state machines shall be enabled and shall function as specified in 10.4.2 and 10.4.3, respectively. The GptpCapableIntervalSetting state machine is enabled if it is implemented.

If the managed object gptpCapableStateMachinesEnabled is FALSE, the GptpCapableTransmit and GptpCapableReceive state machines shall be disabled. The GptpCapableIntervalSetting state machine shall be disabled if it is implemented.

If the managed object gptpCapableStateMachinesEnabled is FALSE, the global variable neighborGptpCapable for the port (see 10.2.5.16) shall be set to TRUE.

NOTE 1—The global variable neighborGptpCapable indicates to a PTP Port whether the PTP Port at the other end of the attached PTP Link is capable of invoking gPTP, and is used in the determination of asCapable (see 10.2.5.1, 11.2.2, 12.4, and 13.4). If gptpCapableStateMachinesEnabled is TRUE, the variable is set to TRUE or FALSE by the GptpCapableReceive state machine. If gptpCapableStateMachinesEnabled is FALSE, the variable is automatically set to TRUE, that is, it is assumed that the network is engineered such that the PTP Port at the other end of the attached link is capable of invoking gPTP. In this case, it is essential that the network be engineered to fulfill this condition; if neighborGptpCapable is TRUE and the PTP Port at the other end of the link is not capable of invoking gPTP, undesirable behavior can occur.

NOTE 2—If the GptpCapableTransmit and GptpCapableReceive state machines are disabled, the exchange of peer delay messages can occur immediately, that is, it is not necessary to wait until gPTP capability is determined by the GptpCapableTransmit and GptpCapableReceive state machines.

10.4.2 GptpCapableTransmit state machine

10.4.2.1 State machine variables

The following variables are used in the state diagram in Figure 10-21 (in 10.4.2.3):

10.4.2.1.1 gPtpCapableSendTime: The time, relative to the LocalClock entity, when a Signaling message containing a gPTP-capable TLV is next sent. The data type for gPtpCapableSendTime is UScaledNs.

10.4.2.1.2 txSignalingMsgPtr: A pointer to a structure whose members contain the values of the fields of a Signaling message to be transmitted, which contains a gPTP-capable TLV (see 10.6.4.4).

10.4.2.1.3 interval3: A local variable that holds either gPtpCapableMessageInterval or oldGptpCapableMessageInterval. The data type for interval3 is UScaledNs.

10.4.2.1.4 numberGptpCapableMessageTransmissions: A count of the number of consecutive transmissions of Signaling messages that contain a gPTP-capable TLV, after the GptpCapableIntervalSetting state machine (see Figure 10-23 in 10.4.4.3) has set gPtpCapableMessageSlowdown (see 10.2.5.19) to TRUE. The data type for numberGptpCapableMessageTransmissions is UInteger8.

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10.4.2.2 State machine functions

10.4.2.2.1 setGptpCapableTlv(): Creates a structure containing the parameters of a Signaling message that contains a gPTP-capable TLV, to be transmitted (see 10.6.4), and returns a pointer to this structure. The parameters are set as follows:

a) logGptpCapableMessageInterval is set to the value of the managed object currentLogGptpCapableMessageInterval (see 14.10.29).

b) The remaining parameters are set as specified in 10.6.4 and 10.6.4.4.

10.4.2.3 State diagram

The GptpCapableTransmit state machine shall implement the function specified by the state diagram in Figure 10-21, the local variables specified in 10.4.2.1, the functions specified in 10.4.2.2, the relevant parameters specified in 10.6.4 and 10.6.4.4, the relevant timing attributes specified in 10.7, and the relevant global variables defined in 10.2.5. This state machine is responsible for setting the parameters of each Signaling message that contains the gPTP-capable TLV, and causing these Signaling messages to be transmitted at a regular rate.

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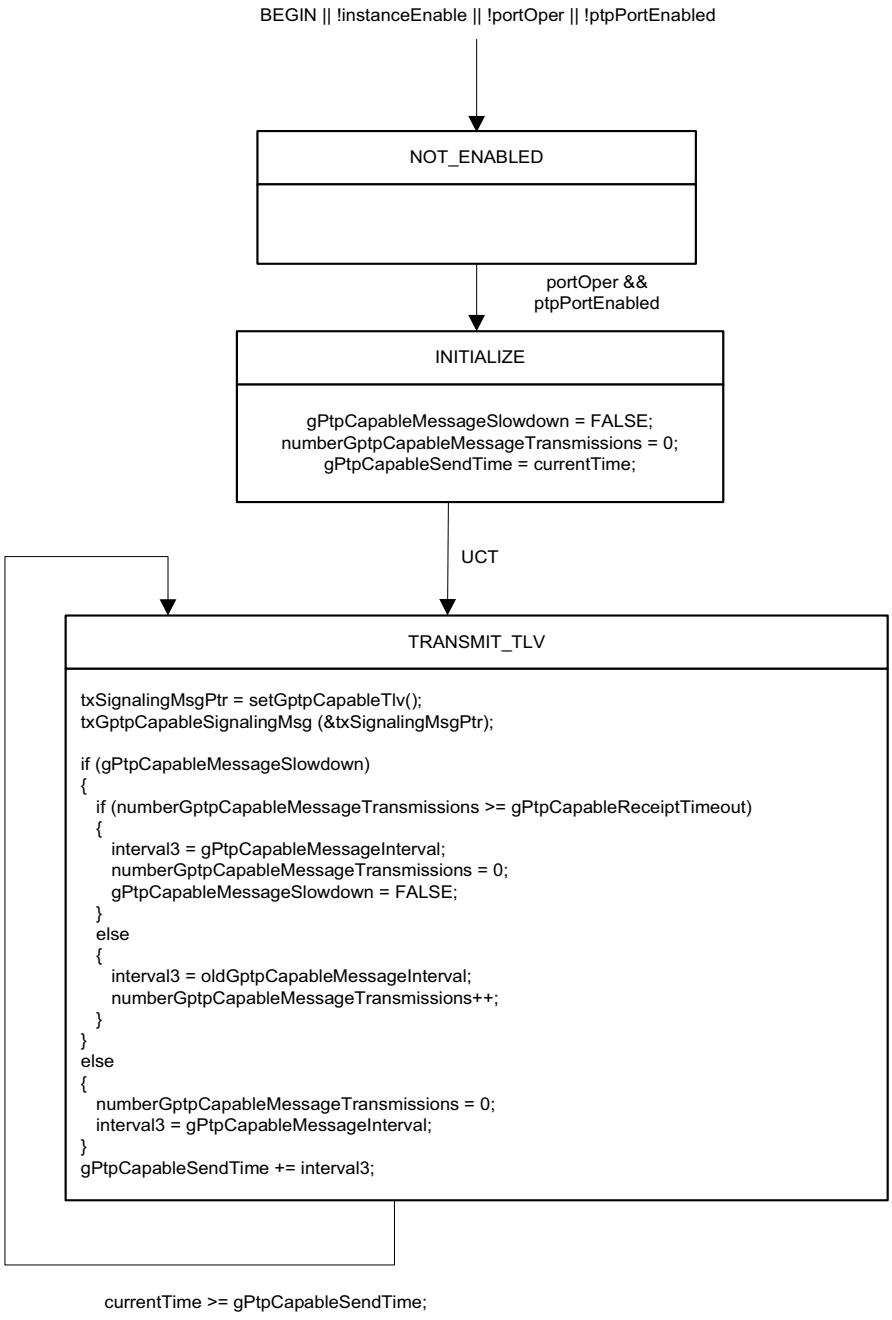


Figure 10-21—GptpCapableTransmit state machine

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10.4.3 GptpCapableReceive state machine

10.4.3.1 State machine variables

The following variables are used in the state diagram in Figure 10-22 (in 10.4.3.2):

10.4.3.1.1 rcvdGptpCapableTlv: A Boolean variable that notifies the current state machine when a Signaling message containing a gPTP-capable TLV is received. This variable is reset by the current state machine.

10.4.3.1.2 rcvdSignalingMsgPtr: A pointer to a structure whose members contain the values of the fields of a Signaling message whose receipt is indicated by rcvdGptpCapableTlv (see 10.4.3.1.1).

10.4.3.1.3 gPtpCapableReceiptTimeoutTimeInterval: The time interval after which, if a Signaling message containing a gPTP-capable TLV is not received, the neighbor of this PTP Port is considered to no longer be invoking gPTP. The data type for gPtpCapableReceiptTimeoutTimeInterval is UScaledNs.

10.4.3.1.4 timeoutTime: A variable used to save the time at which the neighbor of this PTP Port is considered to no longer be invoking gPTP if a Signaling message containing a gPTP-capable TLV is not received. The data type for timeoutTime is UScaledNs.

10.4.3.1.5 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-22). The data type for TEMP is Integer16.

10.4.3.2 State diagram

The GptpCapableReceive state machine shall implement the function specified by the state diagram in Figure 10-22, the local variables specified in 10.4.3.1, the relevant parameters specified in 10.6.4 and 10.6.4.4, the relevant timing attributes specified in 10.7, and the relevant global variables defined in 10.2.5. This state machine is responsible for setting neighborGptpCapable to TRUE on receipt of a Signaling message containing the gPTP-capable TLV, and setting the timeout time after which neighborGptpCapable is set to FALSE.

10.4.4 GptpCapableIntervalSetting state machine

10.4.4.1 State machine variables

The following variables are used in the state diagram in Figure 10-23 (in 10.4.4.3):

10.4.4.1.1 rcvdSignalingMsg5: A Boolean variable that notifies the current state machine when a Signaling message that contains a gPTP-capable message interval request TLV (see 10.6.4.5) is received. This variable is reset by the current state machine.

10.4.4.1.2 rcvdSignalingPtrGIS: A pointer to a structure whose members contain the values of the fields of the received Signaling message that contains a gPTP-capable message interval request TLV (see 10.6.4.5).

10.4.4.1.3 logSupportedGptpCapableMessageIntervalMax: The maximum supported logarithm to base 2 of the gPTP-capable message interval. The data type for logSupportedGptpCapableMessageIntervalMax is Integer8.

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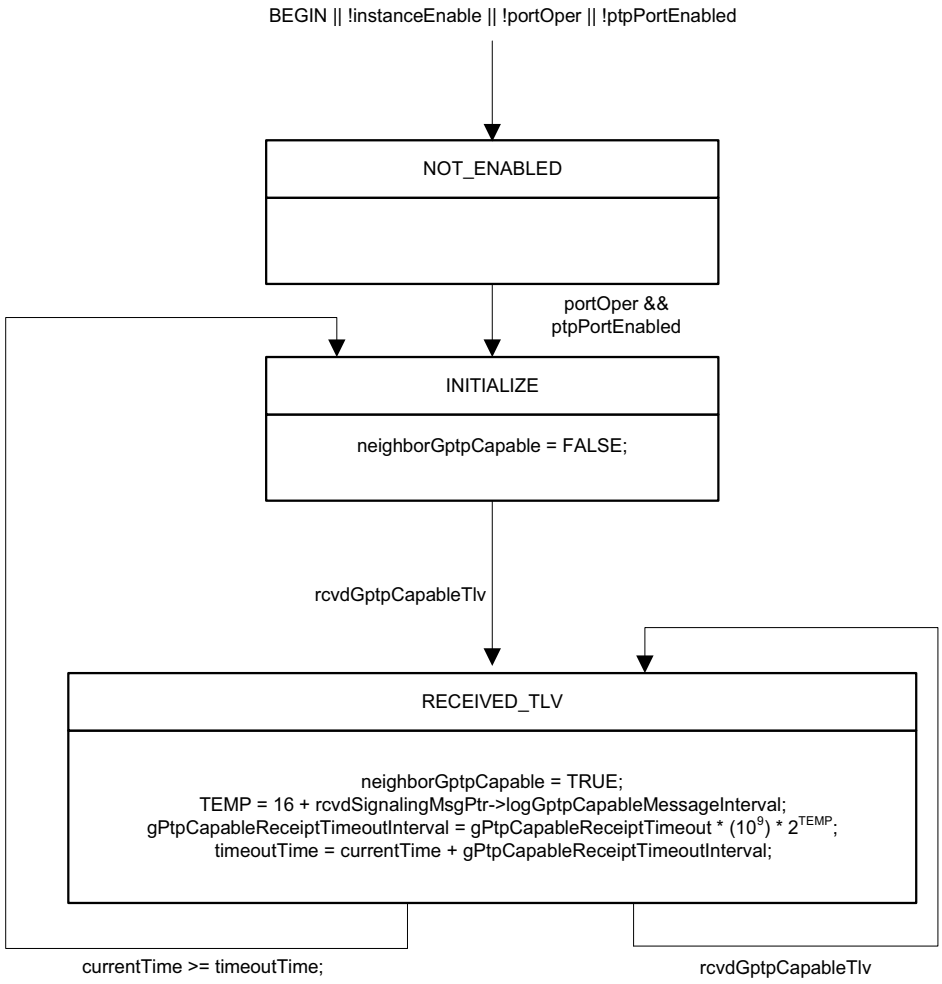


Figure 10-22—GptpCapableReceive state machine

10.4.4.1.4 logSupportedClosestLongerGptpCapableMessageInterval: The logarithm to base 2 of the gPTP-capable message interval, such that logSupportedClosestLongerGptpCapableMessageInterval > logRequestedGptpCapableMessageInterval, that is numerically closest to logRequestedGptpCapableMessageInterval, where logRequestedGptpCapableMessageInterval is the argument of the function computeLogGptpCapableMessageInterval() (see 10.4.4.2.2). The data type for logSupportedClosestLongerGptpCapableMessageInterval is Integer8.

10.4.4.1.5 computedLogGptpCapableMessageInterval: A variable used to hold the result of the function computeLogGptpCapableMessageInterval(). The data type for computedLogGptpCapableMessageInterval is Integer8.

10.4.4.1.6 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 10-23). The data type for TEMP is Integer16.

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10.4.4.2 State machine functions

10.4.4.2.1 isSupportedLogGptpCapableMessageInterval (logGptpCapableMessageInterval): A Boolean function that returns TRUE if the gPTP-capable message interval given by the argument logGptpCapableMessageInterval is supported by the PTP Port and FALSE if the gPTP-capable message interval is not supported by the PTP Port. The argument logGptpCapableMessageInterval has the same data type and format as the field logGptpCapableMessageInterval of the gPTP-capable message interval request TLV (see 10.6.4.5.6).

10.4.4.2.2 computeLogGptpCapableMessageInterval (logRequestedGptpCapableMessageInterval): An Integer8 function that computes and returns the logGptpCapableMessageInterval, based on the logRequestedGptpCapableMessageInterval. This function is defined as indicated below. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
Integer8 computeLogGptpCapableMessageInterval (logRequestedGptpCapableMessageInterval)
Integer8 logRequestedGptpCapableMessageInterval;
{
        Integer8 logSupportedGptpCapableMessageIntervalMax,
                        logSupportedClosestLongerGptpCapableMessageInterval;
        if (isSupportedLogGptpCapableMessageInterval
                                        (logRequestedGptpCapableMessageInterval))
                // The requested gPTP-capable Message Interval is supported and returned
                return (logRequestedGptpCapableMessageInterval)
        else
        {
                if (logRequestedGptpCapableMessageInterval >
                                        logSupportedGptpCapableMessageIntervalMax)
                        // Return the largest supported logGptpCapableMessageInterval, even if
smaller than the requested interval
                        return (logSupportedGptpCapableMessageIntervalMax);
                else
                        // Return the smallest supported logGptpCapableMessageInterval that is
still larger than
                        // the requested interval.
                        return (logSupportedClosestLongerGptpCapableMessageInterval);
        }
}
```

10.4.4.3 State diagram

The GptpCapableIntervalSetting state machine shall implement the function specified by the state diagram in Figure 10-23, the local variables specified in 10.4.4.1, the functions specified in 10.4.4.2, the messages specified in 10.6, the relevant global variables specified in 10.2.5, the relevant managed objects specified in 14.10, and the relevant timing attributes specified in 10.7. This state machine is responsible for setting the global variables that give the duration of the mean intervals between successive Signaling messages containing the gPTP-capable TLV, both at initialization and in response to the receipt of a Signaling message that contains a gPTP-capable message interval request TLV (see 10.6.4.5).

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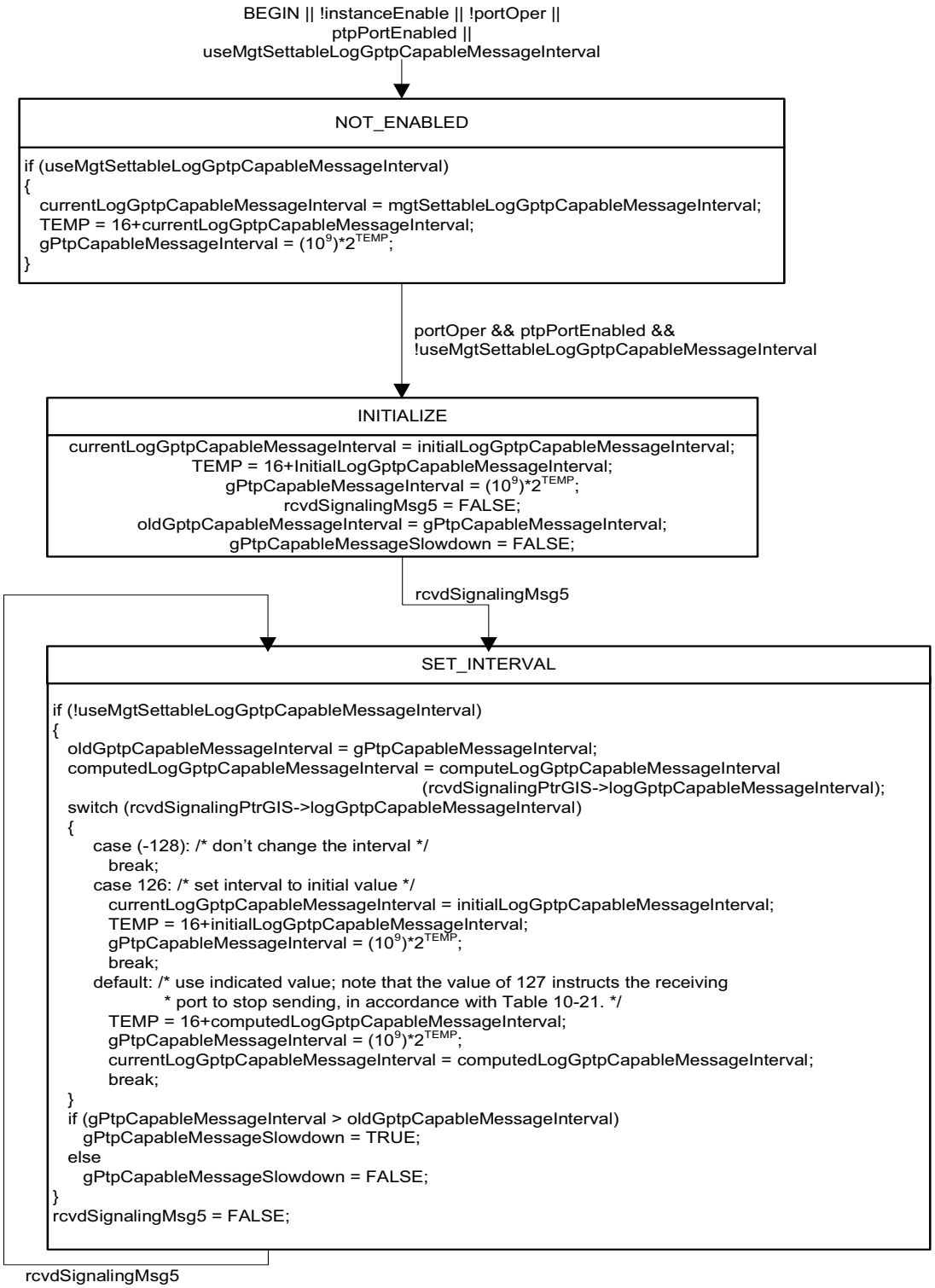


Figure 10-23—GptpCapableIntervalSetting state machine

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10.5 Message attributes

10.5.1 General

This subclause describes media-independent attributes of the Announce message and the Signaling message that are not described in 8.4.2 and whose descriptions are not generic to all messages used in this standard. This subclause also describes media-independent attributes of all time-synchronization event messages.

10.5.2 Message class

The Announce message is a general message, i.e., it is not timestamped. An Announce message provides status and characterization information of the PTP Instance that transmitted the message and the Grandmaster PTP Instance. This information is used by the receiving PTP Instance when executing the BTCA.

The Signaling message is a general message, i.e., it is not timestamped. A Signaling message carries information, requests, and/or commands between PTP Instances, via one or more TLVs.

NOTE—In this standard, the Signaling message is used by a port of a PTP Instance to request that the port at the other end of the link send time-synchronization event messages, link delay measurement messages, or Announce messages at desired intervals; to indicate whether the port at the other end of the link should compute neighborRateRatio and/or meanLinkDelay; and to indicate whether a PTP Port can receive and correctly process one-step Syncs. The message interval request TLV is defined to carry this information (see 10.6.4.3). One usage of this functionality is to allow a time-aware system in power-saving mode to remain connected to a gPTP domain via the port on which the Signaling message is sent.

10.5.3 Addresses

The destination address of the Announce and Signaling messages shall be the reserved multicast address given in Table 10-4 unless otherwise specified in a media-dependent clause (see 12.2 and 16.2).

Table 10-4—Destination address for Announce and Signaling messages

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2022.

NOTE—Frames whose destination address is the address of Table 10-4 are never forwarded, according to IEEE 802.1Q protocol. Use of this address is shared by IEEE 802.1AS and other IEEE 802.1 protocols.

If the transport is FDE or half-duplex IEEE 802.3, all Announce and Signaling messages transmitted shall use the MAC address of the respective egress physical port as the source address.

10.5.4 EtherType

The EtherType of the Announce and Signaling messages shall be the EtherType given in Table 10-5.

NOTE—This EtherType is used for all PTP messages.

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Table 10-5—EtherType for Announce and Signaling messages

EtherType
88-F7

10.5.5 Subtype

The subtype for the Announce and Signaling messages is indicated by the majorSdoId field (see 10.6.2.2.1).

10.5.6 Source port identity

The Announce message, Signaling message, and all time-synchronization messages contain a sourcePortIdentity field (see 10.6.2.2.11), which identifies the egress port (see 8.5) on which the respective message is sent.

10.5.7 Sequence number

Each PortSync entity of a PTP Instance maintains a separate sequenceId pool for each of the message types Announce and Signaling transmitted by the MD entity of the PTP Port.

Each Announce and Signaling message contains a sequenceId field (see 10.6.2.2.12), which carries the message sequence number. The sequenceId of an Announce message shall be one greater than the sequenceId of the previous Announce message sent by the transmitting PTP Port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field. The sequenceId of a Signaling message shall be one greater than the sequenceId of the previous Signaling message sent by the transmitting PTP Port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field.

10.6 Message formats

10.6.1 General

The PTP messages Announce and Signaling each have a header, body, and, if present, a suffix that contains one or more TLVs (see 10.6.2, 10.6.3, and 10.6.4 of this standard and Clause 14 of IEEE Std 1588-2019). Reserved fields shall be transmitted with all bits of the field set to 0 and ignored by the receiver, unless otherwise specified. The data type of the field shall be the type indicated in brackets in the title of each subclause.

Subclause 10.6 defines the path trace TLV, which is carried by the Announce message (see 10.6.3.2.8), and the message interval request TLV, which is carried by the Signaling message (see 10.6.4.3).

PTP Management Messages are not used in this standard. They are specified in IEEE Std 1588-2019.

IEEE Std 1588-2019 specifies various optional features that have associated TLVs. These optional features, including the associated TLVs, may be supported by an implementation of this standard. IEEE Std 1588-2019 also specifies that certain TLVs are propagated by a Boundary Clock if they are attached to an Announce message and are not supported (see 14.2.2.2 and Table 52 of IEEE Std 1588-2019). These TLVs are listed in Table 10-6. The TLV Propagate requirement in IEEE Std 1588-2019 means that a Propagate TLV is propagated through a PTP Relay Instance (e.g., from an ingress PTP Port in the TimeReceiver state to an egress PTP Port in the TimeTransmitter state, even when the TLV is unsupported by the PTP Relay Instance). If the corresponding optional feature is not supported by the PTP Relay Instance, the PTP Relay Instance shall propagate the TLV unchanged.

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Table 10-6—Propagate TLVs of IEEE Std 1588-2019

tlvType values	Value (hex)	TLV defined in
PATH_TRACE	0008	16.2 of IEEE Std 1588-2019, and required by 10.6.3.3 of this standard
ALTERNATE_TIME_OFFSET_INDICATOR	0009	16.3 of IEEE Std 1588-2019
ORGANIZATION_EXTENSION_PROPAGATE	4000	14.3 of IEEE Std 1588-2019
ENHANCED_ACCURACY_METRICS	4001	16.12 of IEEE Std 1588-2019
Reserved for assignment by the IEEE 1588 Working Group for TLVs that propagate	4002–7EFF	
Experimental values (see 4.2.9 of IEEE Std 1588-2019)	7FF0–7FFF	

If a PTP Instance cannot parse a non-forwarding TLV, it shall ignore it and attempt to parse the next TLV (see 14.1 of IEEE Std 1588-2019).

NOTE—Any overhead specific to the respective medium is added to each message.

10.6.2 Header

10.6.2.1 General header specifications

The common header for all PTP messages shall be as specified in Table 10-7 and 10.6.2.2.

Table 10-7—PTP message header

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
majorSdoId				messageType				1	0
minorVersionPTP				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
minorSdoId								1	5
flags								2	6
correctionField								8	8
messageTypeSpecific								4	16
sourcePortIdentity								10	20
sequenceId								2	30
controlField								1	32
logMessageInterval								1	33

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10.6.2.2 Header field specifications

10.6.2.2.1 majorSdoId (Nibble)

The value is specified in 8.1 for all transmitted PTP messages of a gPTP domain. The value is specified in 11.2.16 for all transmitted PTP messages of the CMLDS. Any PTP message received for which the value is not one of the values specified in those subclauses shall be ignored.

10.6.2.2.2 messageType (Enumeration4)

The value indicates the type of the message, as defined in Table 10-8.

The most significant bit of the messageType field divides this field in half between event and general messages, i.e., it is 0 for event messages and 1 for general messages.

Table 10-8—Values for messageType field

Message type	Message class	Value
Announce	General	0xB
Signaling	General	0xC
NOTE—Values for the messageType field for other PTP messages that are used only for specific media are defined in the respective media-dependent clause(s).		

10.6.2.2.3 minorVersionPTP (UInteger4)

For transmitted messages, the value shall be 1 (see 7.5.4 and 13.3.2.5 of IEEE Std 1588-2019). For received messages, the value is ignored.

NOTE—minorVersionPTP indicates the minor version number of IEEE 1588 PTP used in the PTP profile contained in this standard for information only.

10.6.2.2.4 versionPTP (UInteger4)

For transmitted messages, the value shall be 2 (see 7.5.4 and 13.3.2.4 of IEEE Std 1588-2019). For received messages, if the value is not 2, the entire message shall be ignored.

NOTE—versionPTP indicates the version number of IEEE 1588 PTP used in the PTP profile contained in this standard.

10.6.2.2.5 messageLength (UInteger16)

The value is the total number of octets that form the PTP message. The counted octets start with, and include, the first octet of the header and terminate with, and include, the last octet of the last TLV or, if there are no TLVs, the last octet of the message as defined in this clause.

NOTE—For example, the Follow_Up message (see 11.4.4) contains a PTP header (34 octets), preciseOriginTimestamp (10 octets), and Follow_Up information TLV (32 octets). The value of the messageLength field is 34+10+32 = 76.

10.6.2.2.6 domainNumber (UInteger8)

The value is the gPTP domainNumber specified in 8.1.

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10.6.2.2.7 minorSdoId (UInteger8)

The value is specified in 8.1 for all transmitted PTP messages of a gPTP domain. The value is specified in 11.2.16 for all transmitted PTP messages of the CMLDS. Any PTP message received for which the value is not one of the values specified in those subclauses shall be ignored.

10.6.2.2.8 flags (Octet2)

The value of the bits of the array are defined in Table 10-9. For message types where the bit is not defined in Table 10-9, the value of the bit is set to FALSE.

Table 10-9—Values of flag bits

Octet	Bit	Message types	Name	Value
0	0	All	alternateTimeTransmitterFlag in Announce, Sync, Follow_Up, and Delay_Resp messages	Not used in this standard; transmitted as FALSE and ignored on reception.
0	1	Sync, Pdelay_Resp	twoStepFlag	*For Sync messages:* a) For a one-step transmitting PTP Port (see 11.1.3 and 11.2.12.10), the value is FALSE. b) For a two-step transmitting PTP Port, the value is TRUE. *For Pdelay_Resp messages:* The value is transmitted as TRUE and ignored on reception.
0	2	All	unicastFlag	Not used in this standard; transmitted as FALSE and ignored on reception.
0	3	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception.
0	4	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception.
0	5	All	PTP profileSpecific 1	Not used in this standard; transmitted as FALSE and ignored on reception.
0	6	All	PTP profileSpecific 2	Not used in this standard; transmitted as FALSE and ignored on reception.
0	7	All	Reserved	Not used in this standard; transmitted as FALSE and ignored on reception.
1	0	Announce	leap61	The value of the global variable leap61 (see 10.3.9.4).
1	1	Announce	leap59	The value of the global variable leap59 (see 10.3.9.5).
1	2	Announce	currentUtcOffsetValid	The value of the global variable currentUtcOffsetValid (see 10.3.9.6).

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Table 10-9—Values of flag bits *(continued)*

Octet	Bit	Message types	Name	Value
1	3	Announce	ptpTimescale	For domain 0, transmitted as TRUE and ignored on receipt. For domains other than domain 0, the value of the global variable ptpTimescale (see 10.3.9.7).
1	4	Announce	timeTraceable	The value of the global variable timeTraceable (see 10.3.9.8).
1	5	Announce	frequencyTraceable	The value of the global variable frequencyTraceable (see 10.3.9.9).
1	6	All	Reserved	Not used by IEEE Std 1588-2019; reserved as FALSE and ignored on reception.
1	7	All	Reserved	Not used in this standard; reserved as FALSE and ignored on reception.

10.6.2.2.9 correctionField (Integer64)

The value is 0.

10.6.2.2.10 messageTypeSpecific (Octet4)

The value of the messageTypeSpecific field varies, based on the value of the messageType field, as described in Table 10-10.

For Event messages only, the four octets of the messageTypeSpecific field may be used for internal implementation of a PTP Instance and its ports. For example, if the clock consists of multiple hardware components that are not synchronized, messageTypeSpecific can be used to transfer an internal timestamp between components (e.g., a physical layer chip and the clock's processor).

The messageTypeSpecific field is not used for features of this standard, and it has no meaning from one clock to another. In the on-the-wire format at each PTP Port, for all messageType values, the messageTypeSpecific field is transmitted with all bits of the field 0 and ignored on receive.

Table 10-10—messageTypeSpecific semantics

Value of messageType	Description
Follow_Up, Pdelay_Resp_Follow_Up, Announce, Signaling, Management	For the General message class, this field is reserved; it is transmitted as 0 and ignored on reception.
Sync, Pdelay_Req, Pdelay_Resp	For the Event message class, this field may be used for internal implementation as specified in this subclause.

10.6.2.2.11 sourcePortIdentity (PortIdentity)

The value is the PTP Port identity attribute (see 8.5.2) of the PTP Port that transmits the PTP message.

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10.6.2.2.12 sequenceId (UInteger16)

The sequenceId field is assigned as specified in 10.5.7.

10.6.2.2.13 controlField (UInteger8)

The value is 0.

10.6.2.2.14 logMessageInterval (Integer8)

For an Announce message, the value is the value of currentLogAnnounceInterval (see 10.3.10.6) for the PTP Port that transmits the Announce message. For a Signaling message, the value is transmitted as 0x7F and ignored on reception.

10.6.3 Announce message

10.6.3.1 General Announce message specifications

The fields of the body of the Announce message shall be as specified in Table 10-11 and 10.6.3.2.

Table 10-11—Announce message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 10.6.2)								34	0
reserved								10	34
currentUtcOffset								2	44
reserved								1	46
grandmasterPriority1								1	47
grandmasterClockQuality								4	48
grandmasterPriority2								1	52
grandmasterIdentity								8	53
stepsRemoved								2	61
timeSource								1	63
path trace TLV								4+8N	64

10.6.3.2 Announce message field specifications

10.6.3.2.1 currentUtcOffset (Integer16)

The value is the value of currentUtcOffset (see 10.3.9.10) for the PTP Instance that transmits the Announce message.

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10.6.3.2.2 grandmasterPriority1 (UInteger8)

The value is the value of the priority1 component of the rootSystemIdentity of the gmPriorityVector (see 10.3.5) of the PTP Instance that transmits the Announce message.

10.6.3.2.3 grandmasterClockQuality (ClockQuality)

The value is the clockQuality formed by the clockClass, clockAccuracy, and offsetScaledLogVariance of the rootSystemIdentity of the gmPriorityVector (see 10.3.5) of the PTP Instance that transmits the Announce message.

10.6.3.2.4 grandmasterPriority2 (UInteger8)

The value is the value of the priority2 component of the rootSystemIdentity of the gmPriorityVector (see 10.3.5) of the PTP Instance that transmits the Announce message.

10.6.3.2.5 grandmasterIdentity (ClockIdentity)

The value is the value of the clockIdentity component of the rootSystemIdentity of the gmPriorityVector (see 10.3.5) of the PTP Instance that transmits the Announce message.

10.6.3.2.6 stepsRemoved (UInteger16)

The value is the value of timeTransmitterStepsRemoved (see 10.3.9.3) for the PTP Instance that transmits the Announce message.

10.6.3.2.7 timeSource (TimeSource)

The value is the value of timeSource (see 8.6.2.7 and 10.3.9.11) for the PTP Instance that transmits the Announce message.

10.6.3.2.8 Path trace TLV

The Announce message carries the path trace TLV, defined in 10.6.3.3.

10.6.3.3 Path trace TLV definition

10.6.3.3.1 General

The fields of the path-trace TLV shall be as specified in Table 10-12 and in 10.6.4.3.2 through 10.6.4.3.9. This TLV and its use are defined in IEEE Std 1588-2019 (see 16.2 and Table 52 of IEEE Std 1588-2019).

10.6.3.3.2 tlvType (Enumeration16)

The value of the tlvType field is 0x8.

NOTE—This value indicates the TLV is a path trace TLV, as specified in 16.2.5.1 and Table 52 of IEEE Std 1588-2019. The value 0x8 is specified in that standard as PATH_TRACE.

10.6.3.3.3 lengthField (UInteger16)

The value of the lengthField is 8N.

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Table 10-12—Path trace TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
pathSequence								8N	4

10.6.3.3.4 pathSequence (ClockIdentity[N])

The value of pathSequence is a ClockIdentity array. The array elements are the clockIdentities of the successive PTP Instances that receive and send an Announce message. The quantity N is the number of PTP Instances, including the Grandmaster PTP Instance, that the Announce information has traversed.

NOTE—N is equal to stepsRemoved + 1 (see 10.6.3.2.6). The size of the pathSequence array increases by 1 for each PTP Instance that the Announce information traverses.

10.6.4 Signaling message

10.6.4.1 General Signaling message specifications

The fields of the body of the Signaling message shall be as specified in Table 10-13 and 10.6.4.2.

Table 10-13—Signaling message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 10.6.2)								34	0
targetPortIdentity								10	34
one or more TLVs								N	44

10.6.4.2 Signaling message field specifications

10.6.4.2.1 targetPortIdentity (PortIdentity)

The value of targetPortIdentity.clock identity is all ones, i.e., 0xFFFFFFFFFFFFFFFF. The value of targetPortIdentity.portNumber is all ones, i.e., 0xFFFF.

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10.6.4.2.2 TLVs carried in one Signaling message

The Signaling message carries either:

a) One or both interval request TLVs, defined in 10.6.4.3 and 10.6.4.5, or
b) The gPTP-capable TLV, defined in 10.6.4.4.

10.6.4.3 Message interval request TLV definition

10.6.4.3.1 General

The fields of the message interval request TLV shall be as specified in Table 10-14 and in 10.6.4.3.2 through 10.6.4.3.9. This TLV is a standard organization extension TLV for the Signaling message, as specified in 14.3 of IEEE Std 1588-2019.

Table 10-14—Message interval request TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logLinkDelayInterval								1	10
logTimeSyncInterval								1	11
logAnnounceInterval								1	12
flags								1	13
reserved								2	14

10.6.4.3.2 tlvType (Enumeration16)

The value of the tlvType field is 0x3.

NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a value of 0x3.

10.6.4.3.3 lengthField (UInteger16)

The value of the lengthField is 12.

10.6.4.3.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

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10.6.4.3.5 organizationSubType (Enumeration24)

The value of organizationSubType is 2.

10.6.4.3.6 logLinkDelayInterval (Integer8)

The value is the logarithm to base 2 of the mean time interval, desired by the port that sends this TLV, between successive Pdelay_Req messages sent by the port at the other end of the link. The format and allowed values of logLinkDelayInterval are the same as the format and allowed values of initialLogPdelayReqInterval (see 11.5.2.2).

The values 127, 126, and –128 are interpreted as defined in Table 10-15.

Table 10-15—Interpretation of special values of logLinkDelayInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the port that receives this TLV to stop sending link delay measurement messages.
126	Instructs the port that receives this TLV to set currentLogPdelayReqInterval to the value of initialLogPdelayReqInterval (see 11.5.2.2).
–128	Instructs the port that receives this TLV not to change the mean time interval between successive Pdelay_Req messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

10.6.4.3.7 logTimeSyncInterval (Integer8)

The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV, between successive time-synchronization event messages sent by the PTP Port at the other end of the link. The format and allowed values of logTimeSyncInterval are the same as the format and allowed values of initialLogSyncInterval (see 10.7.2.3, 11.5.2.3, 12.8, and 13.9.2).

The values 127, 126, and –128 are interpreted as defined in Table 10-16.

Table 10-16—Interpretation of special values of logTimeSyncInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending time-synchronization event messages.
126	Instructs the PTP Port that receives this TLV to set currentLogSyncInterval to the value of initialLogSyncInterval (see 10.7.2.3, 11.5.2.3, 12.8, and 13.9.2).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive time-synchronization event messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

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When a Signaling message that contains this TLV is sent by a PTP Port, the value of syncReceiptTimeoutTimeInterval for that PTP Port (see 10.2.5.3) shall be set equal to syncReceiptTimeout (see 10.7.3.1) multiplied by the value of the interval, in seconds, reflected by logTimeSyncInterval.

10.6.4.3.8 logAnnounceInterval (Integer8)

The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV, between successive Announce messages sent by the PTP Port at the other end of the link. The format and allowed values of logAnnounceInterval are the same as the format and allowed values of initialLogAnnounceInterval (see 10.7.2.2).

The values 127, 126, and –128 are interpreted as defined in Table 10-17.

Table 10-17—Interpretation of special values of logAnnounceInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending Announce messages.
126	Instructs the PTP Port that receives this TLV to set currentLogAnnounceInterval to the value of initialLogAnnounceInterval (see 10.7.2.2).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive Announce messages.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

When a Signaling message that contains this TLV is sent by a PTP Port, the value of announceReceiptTimeoutTimeInterval for that PTP Port (see 10.3.10.1) shall be set equal to announceReceiptTimeout (see 10.7.3.2) multiplied by the value of the interval, in seconds, reflected by logAnnounceInterval.

10.6.4.3.9 flags (Octet)

Bits 0 through 2 of the octet are defined in Table 10-18 and take on the values TRUE and FALSE. Bits not defined in Table 10-18 are set to FALSE and ignored on receipt.

Table 10-18—Definitions of bits of flags field of message interval request TLV

Bit	Name
0	computeNeighborRateRatio
1	computeMeanLinkDelay
2	oneStepReceiveCapable

NOTE—For FDE links (see Clause 11) and HDE links (see Clause 19), it is expected that the PTP Port sending this TLV will set bits 0 and/or 1 to FALSE if this PTP Port will not provide valid timing information in its subsequent responses (Pdelay_Resp and Pdelay_Resp_Follow_Up) to Pdelay_Req messages. Similarly, it is expected that the PTP Port sending this TLV will set bit 2 to TRUE if it is capable of receiving and correctly processing one-step Sync messages.

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10.6.4.4 gPTP-capable TLV definition

10.6.4.4.1 General

The fields of the gPTP-capable TLV shall be as specified in Table 10-19 and in 10.6.4.4.2 through 10.6.4.4.7. This TLV is a standard organization extension TLV for the Signaling message, as specified in 14.3 of IEEE Std 1588-2019.

Table 10-19—gPTP-capable TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logGptpCapableMessageInterval								1	10
flags								1	11
reserved								4	12

10.6.4.4.2 tlvType (Enumeration16)

The value of the tlvType field is 0x8000.

NOTE—This value indicates that the TLV is a vendor and standard organization extension TLV that is not propagated, as specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION_DO_NOT_PROPAGATE with a value of 0x8000.

10.6.4.4.3 lengthField (UInteger16)

The value of the lengthField is 12.

10.6.4.4.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

10.6.4.4.5 organizationSubType (Enumeration24)

The value of organizationSubType is 4.

10.6.4.4.6 logGptpCapableMessageInterval (Integer8)

The value of logGptpCapableMessageInterval is the logarithm to base 2 of the mean gPTP-capable message interval in seconds (see 10.7.2.1 and 10.7.2.5).

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10.6.4.4.7 flags (Octet)

The flag bits shall be transmitted as FALSE and ignored on receipt.

10.6.4.5 gPTP-capable message interval request TLV definition

10.6.4.5.1 General

The fields of the gPTP-capable message interval request TLV shall be as specified in Table 10-20 and in 10.6.4.5.2 through 10.6.4.5.6. This TLV is a standard organization extension TLV for the Signaling message, as specified in 14.3 of IEEE Std 1588-2019.

Table 10-20—gPTP-capable message interval request TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
logGptpCapableMessageInterval								1	10
reserved								3	11

10.6.4.5.2 tlvType (Enumeration16)

The value of the tlvType field is 0x8000.

NOTE—This value indicates that the TLV is a vendor and standard organization extension TLV that is not propagated, as specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION_DO_NOT_PROPAGATE with a value of 0x8000.

10.6.4.5.3 lengthField (UInteger16)

The value of the lengthField is 10.

10.6.4.5.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

10.6.4.5.5 organizationSubType (Enumeration24)

The value of organizationSubType is 5.

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10.6.4.5.6 logGptpCapableMessageInterval (Integer8)

The value is the logarithm to base 2 of the mean time interval, desired by the PTP Port that sends this TLV, between successive Signaling messages that contain the gPTP-capable TLV (see 10.6.4.4), sent by the PTP Port at the other end of the link. The format and allowed values of logGptpCapableMessageInterval are the same as the format and allowed values of initialLogGptpCapableMessageInterval (see 10.7.2.5).

The values 127, 126, and –128 are interpreted as defined in Table 10-21.

Table 10-21—Interpretation of special values of logGptpCapableMessageInterval

Value	Instruction to PTP Instance that receives this TLV
127	Instructs the PTP Port that receives this TLV to stop sending Signaling messages that contain the gPTP-capable TLV.
126	Instructs the PTP Port that receives this TLV to set currentlogGptpCapableMessageInterval to the value of initialLogGptpCapableMessageInterval (see 10.7.2.4).
–128	Instructs the PTP Port that receives this TLV not to change the mean time interval between successive Signaling messages that contain the gPTP-capable TLV.
All values in the ranges [–127, –25] and [25, 125] are reserved.	

When a Signaling message that contains this TLV is sent by a PTP Port, the value of gPtpCapableReceiptTimeoutTimeInterval for that PTP Port (see 10.3.10.1) shall be set equal to gPtpCapableReceiptTimeout (see 10.7.3.3) multiplied by the value of the interval, in seconds, reflected by logGptpCapableMessageInterval.

10.7 Protocol timing characterization

10.7.1 General

This subclause specifies timing and timeout attributes for the media-independent sublayer state machines.

10.7.2 Message transmission intervals

10.7.2.1 General interval specification

The mean time interval between the sending of successive Announce messages, known as the *announce interval*, is specified in 10.7.2.2.

The mean time interval between the sending of successive time-synchronization event messages, known as the *sync interval*. The sync interval is specified in 10.7.2.3.

The mean time interval between the sending of successive Signaling messages that contain the gPTP-capable TLV, known as the *gPTP-capable message interval*, is specified in 10.7.2.5.

10.7.2.2 Announce message transmission interval

The logarithm to the base 2 of the announce interval (in seconds) is carried in the logMessageInterval field of the Announce message.

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When useMgtSettableLogAnnounceInterval (see 14.10.15) is FALSE, the initialLogAnnounceInterval specifies the announce interval when the PTP Port is initialized and the value to which the announce interval is set when a message interval request TLV is received with the logAnnounceInterval field set to 126 (see the AnnounceIntervalSetting state machine in 10.3.17). The currentLogAnnounceInterval specifies the current value of the announce interval. The default value of initialLogAnnounceInterval is 0. Every PTP Port supports the value 127; the PTP Port does not send Announce messages when currentLogAnnounceInterval has this value (see 10.3.17). A PTP Port may support other values, except for the reserved values indicated in Table 10-17. A PTP Port ignores requests for unsupported values (see 10.3.17). The initialLogAnnounceInterval and currentLogAnnounceInterval are per-PTP Port attributes.

When useMgtSettableLogAnnounceInterval is TRUE, currentLogAnnounceInterval is set equal to mgtSettableLogAnnounceInterval (see 14.10.16), and initialLogAnnounceInterval is ignored.

Announce messages shall be transmitted such that the value of the arithmetic mean of the intervals, in seconds, between message transmissions is within ±30% of $2^{\text{currentLogAnnounceInterval}}$. In addition, a PTP Port shall transmit Announce messages such that at least 90% of the inter-message intervals are within ±30% of the value of $2^{\text{currentLogAnnounceInterval}}$. The interval between successive Announce messages should not exceed twice the value of $2^{\text{currentLogAnnounceInterval}}$ in order to prevent causing an announceReceiptTimeout event. The PortAnnounceTransmit state machine (see 10.3.16) is consistent with these requirements, i.e., the requirements here and the requirements of the PortAnnounceTransmit state machine can be met simultaneously.

NOTE 1—In transient situations (i.e., during a synchronization tree reconfiguration), Announce messages can be transmitted with a shorter interval then described above.

NOTE 2—A minimum number of inter-message intervals is necessary in order to verify that a PTP Port meets these requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples. For more detailed discussion of statistical analyses, see Papoulis [B35].

NOTE 3—If useMgtSettableLogAnnounceInterval is FALSE, the value of initialLogAnnounceInterval is the value of the mean time interval between successive Announce messages when the PTP Port is initialized. The value of the mean time interval between successive Announce messages can be changed, e.g., if the PTP Port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in currentLogAnnounceInterval. The value of the mean time interval between successive Announce messages can be reset to the initial value, e.g., by a message interval request TLV for which the value of the field logAnnounceInterval is 126 (see 10.6.4.3.8).

NOTE 4—A PTP Port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and 10.3.17) that the PTP Port at the other end of the attached link set its currentLogAnnounceInterval to a specific value can determine if the request was honored by examining the logMessageInterval field of subsequent received Announce messages.

10.7.2.3 Time-synchronization event message transmission interval

The logarithm to the base 2 of the sync interval (in seconds) is carried in the logMessageInterval field of the time-synchronization messages.

When useMgtSettableLogSyncInterval (see 14.10.20) is FALSE, the initialLogSyncInterval specifies the sync interval when the PTP Port is initialized and the value to which the sync interval is set when a message interval request TLV is received with the logTimeSyncInterval field set to 126 (see the SyncIntervalSetting state machine in 10.2.14). The default value is media-dependent; the value is specified in the respective media-dependent clauses. The initialLogSyncInterval is a per-PTP Port attribute.

The currentLogSyncInterval specifies the current value of the sync interval and is a per-PTP Port attribute.

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When useMgtSettableLogSyncInterval is TRUE, currentLogSyncInterval is set equal to mgtSettableLogSyncInterval (see 14.10.21), and initialLogSyncInterval is ignored.

When the value of syncLocked is FALSE, time-synchronization messages shall be transmitted such that the value of the arithmetic mean of the intervals, in seconds, between message transmissions is within ±30% of $2^{\text{currentLogSyncInterval}}$. In addition, a PTP Port shall transmit time-synchronization messages such that at least 90% of the inter-message intervals are within ±30% of the value of $2^{\text{currentLogSyncInterval}}$. The interval between successive time-synchronization messages should not exceed twice the value of $2^{\text{currentLogSyncInterval}}$ in order to prevent causing a syncReceiptTimeout event. The PortSyncSyncSend state machine (see 10.2.12) is consistent with these requirements, i.e., the requirements here and the requirements of the PortSyncSyncSend state machine can be met simultaneously.

NOTE 1—A minimum number of inter-message intervals is necessary in order to verify that a PTP Port meets these requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples. For more detailed discussion of statistical analyses, see Papoulis [B35].

NOTE 2—If useMgtSettableLogSyncInterval is FALSE the value of initialLogSyncInterval is the value of the sync interval when the PTP Port is initialized. The value of the sync interval can be changed, e.g., if the PTP Port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in currentLogSyncInterval. The value of the sync interval can be reset to the initial value, e.g., by a message interval request TLV for which the value of the field logTimeSyncInterval is 126 (see 10.6.4.3.7).

10.7.2.4 Interval for providing synchronization information by ClockTimeTransmitter entity

The clockTimeTransmitterLogSyncInterval specifies the mean time interval between successive instants at which the ClockTimeTransmitter entity provides time-synchronization information to the SiteSync entity. The value is less than or equal to the smallest currentLogSyncInterval (see 10.7.2.3) value for all the ports of the PTP Instance. The clockTimeTransmitterLogSyncInterval is an internal, per PTP Instance variable.

10.7.2.5 Interval for sending the gPTP-capable TLV Signaling message

The logarithm to the base 2 of the gPTP-capable message interval (in seconds) is carried in the logGptpCapableMessageInterval field of the gPTP-capable TLV. The default value shall be 0. The range shall be –24 through 24. Other values in the range [–128, 127] shall be reserved.

When useMgtSettableLogGptpCapableMessageInterval (see 14.10.15) is FALSE, the initialLogGptpCapableMessageInterval specifies the following:

a) The mean gPTP-capable message interval when the PTP Port is initialized, and
b) The value to which the gPTP-capable message interval is set when a gPTP-capable TLV is received with the logGptpCapableMessageInterval field set to 126 (see the GptpCapableMessageIntervalSetting state machine in 10.4.4).

The currentLogGptpCapableMessageInterval specifies the current value of the gPTP-capable message interval. Every PTP Port supports the value 127; the PTP Port does not send Signaling messages containing a gPTP-capable TLV when currentLogGptpCapableMessageInterval has this value (see 10.4.4). A PTP Port may support other values, except for the reserved values indicated in Table 10-21. A PTP Port shall ignore requests for unsupported values (see 10.4.4). The initialLogGptpCapableMessageInterval and currentLogGptpCapableMessageInterval are per-PTP Port attributes.

When useMgtSettableLogGptpCapableMessageInterval is TRUE, currentLogGptpCapableMessageInterval is set equal to mgtSettableLogGptpCapableMessageInterval (see 14.10.16), and initialLogGptpCapableMessageInterval is ignored.

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NOTE 1—It is recommended that the interval between successive Signaling messages containing the gPTP-capable TLV not exceed twice the value of $2^{currentLogGptpCapableMessageInterval}$ in order to prevent causing an gPtpCapableReceiptTimeout event at the neighbor.

NOTE 2—If useMgtSettableLogGptpCapableMessageInterval is FALSE, the value of initialLogGptpCapableMessageInterval is the value of the mean time interval between successive Signaling messages containing the gPTP-capable TLV when the PTP Port is initialized. The value of the mean time interval between successive Signaling messages containing the gPTP-capable TLV can be changed, e.g., if the PTP Port receives a Signaling message that carries a gPTP-capable TLV (see 10.6.4.4) and the current value is stored in currentLogGptpCapableMessageInterval. The value of the mean time interval between successive Signaling messages containing the gPTP-capable TLV can be reset to the initial value, e.g., by a Signaling message containing a gPTP-capable message interval request TLV for which the value of the field logGptpCapableMessageInterval is 126 (see 10.6.4.5.6).

NOTE 3—A PTP Port that requests (using a Signaling message that contains a gPTP-capable message interval request TLV; see 10.6.4 and 10.6.4.5) that the PTP Port at the other end of the attached link set its currentLogGptpCapableMessageInterval to a specific value can determine if the request was honored by examining the logGptpCapableMessageInterval field of subsequent received Signaling messages containing the gPTP-capable TLV.

NOTE 4—The GptpCapableTransmit state machine ensures that the times between transmission of successive Signaling messages, containing the gPTP-capable TLV, in seconds, are not smaller than $2^{currentLogGptpCapableMessaggeInterval}$. This is consistent with the manner in which Pdelay_Req messages are transmitted (see NOTE 3 of 11.5.2.2).

10.7.3 Timeouts

10.7.3.1 syncReceiptTimeout

The value of this attribute tells a timeReceiver port the number of sync intervals to wait without receiving synchronization information, before assuming that the timeTransmitter is no longer transmitting synchronization information and that the BTCA needs to be run, if appropriate. The condition of the timeReceiver port not receiving synchronization information for syncReceiptTimeout sync intervals is known as *sync receipt timeout.*

The default value shall be 3. The range shall be 2 through 255. The syncReceiptTimeout is a per-PTP Port attribute.

10.7.3.2 announceReceiptTimeout

The value of this attribute tells a timeReceiver port the number of announce intervals to wait without receiving an Announce message, before assuming that the timeTransmitter is no longer transmitting Announce messages, and that the BTCA needs to be run, if appropriate. The condition of the timeReceiver port not receiving an Announce message for announceReceiptTimeout announce intervals is known as *announce receipt timeout.*

The default value shall be 3. The range shall be 2 through 255. The announceReceiptTimeout is a per-PTP Port attribute.

10.7.3.3 gPtpCapableReceiptTimeout

The value of this attribute tells a PTP Port the number of gPTP-capable message intervals to wait without receiving from its neighbor a Signaling message containing a gPTP-capable TLV, before determining that its neighbor is no longer invoking gPTP.

NOTE—A determination that its neighbor is no longer invoking gPTP causes the PTP Port to set asCapable to FALSE.

The default value shall be 9. The range shall be 2 through 255.

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11. Media-dependent layer specification for FDE links

11.1 Overview

11.1.1 General

A port attached to a HDE PTP Link uses the PTP peer delay protocol to measure propagation delay on the PTP Link. An overview of the propagation delay measurement is given in 11.1.2. Synchronization information is transported using the PTP messages Sync and Follow_Up. An overview of the transport of synchronization information is given in 11.1.3. An overview of the MD entity model for a FDE medium is given in 11.1.4.

PTP links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode are specified in Clause 19. Since Clause 19 references some of the state machines defined in this clause, Clause 11 also includes the state machine variables and conditions relevant to links operating in half-duplex mode.

11.1.2 Propagation delay measurement

The measurement of propagation delay on a FDE PTP Link using the peer-to-peer delay mechanism is illustrated in Figure 11-1. The instance of the peer-to-peer delay mechanism that runs only on domain 0 is referred to as the transport-specific peer-to-peer delay mechanism. The instance of the peer-to-peer delay mechanism that runs as part of the Common Mean Link Delay Service (CMLDS, see 11.2.16) is referred to as CMLDS. The mechanism is the same as the peer-to-peer delay mechanism described in IEEE Std 1588-2019, specialized to a two-step PTP Port[20] and sending the requestReceiptTimestamp and the responseOriginTimestamp separately [see item c) 8) of 11.4.2 in IEEE Std 1588-2019]. The transport-specific peer-to-peer delay mechanism is mathematically equivalent to CMLDS, though its computations are organized differently. The description in this subclause focuses on the transport-specific peer-to-peer delay mechanism. In addition, when it is not important in this standard to distinguish the transport-specific peer-to-peer delay mechanism from CMLDS, the mechanism is simply referred to as the "peer-to-peer delay mechanism," or more briefly as the "peer delay mechanism."

The measurement is made by each port at the end of every FDE PTP Link. Thus, both ports sharing a PTP Link will independently make the measurement, and both ports will know the propagation delay as a result. This allows the time-synchronization information described in 11.1.3 to be transported irrespective of the direction taken by a Sync message. The propagation delay measurement is made on ports otherwise blocked by non-PTP algorithms (e.g., Rapid Spanning Tree Protocol) used to eliminate cyclic topologies. This enables either no loss of synchronization or faster resynchronization, after a reconfiguration, because propagation delays are already known and do not have to be initially measured when the reconfiguration occurs.

Since the propagation delay measurement is made using timestamps relative to the LocalClock entities at each port at the ends of the PTP Link and the resulting mean link delay is expressed in the responder timebase (see 11.2.18.3.4), there is no need to measure the mean link delay for the PTP Link in each domain because the mean link delay is the same in each domain. In addition, the quantity nrrPdelay (see 11.2.12.14) is the ratio of the responder to requester LocalClock frequency and is also the same in all domains. Therefore, the propagation delay measurement and nrrPdelay measurement, made using the peer-to-peer delay mechanism (i.e., nrrPdelay, see 11.2.12.14), are domain-independent. Single instances of the respective state machines that cause these measurements to be made are invoked, rather than one instance per domain, and the results are available to all domains. The PTP messages used for the measurements (i.e., Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up; see 11.4.5 through 11.4.7) carry 0 in the domainNumber field, but this value is not used.

[20] See 3.1.86 of IEEE Std 1588-2019 for the definition of a *two-step PTP Port*.

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In Figure 11-1, the propagation delay measurement is initiated by a time-aware system at one end of a PTP Link; this time-aware system is known as the *peer delay initiator*. For purposes of the measurement, the other time-aware system is the *peer delay responder*. A similar measurement occurs in the opposite direction, with the initiator and responder interchanged and the directions of the messages in Figure 11-1 reversed.

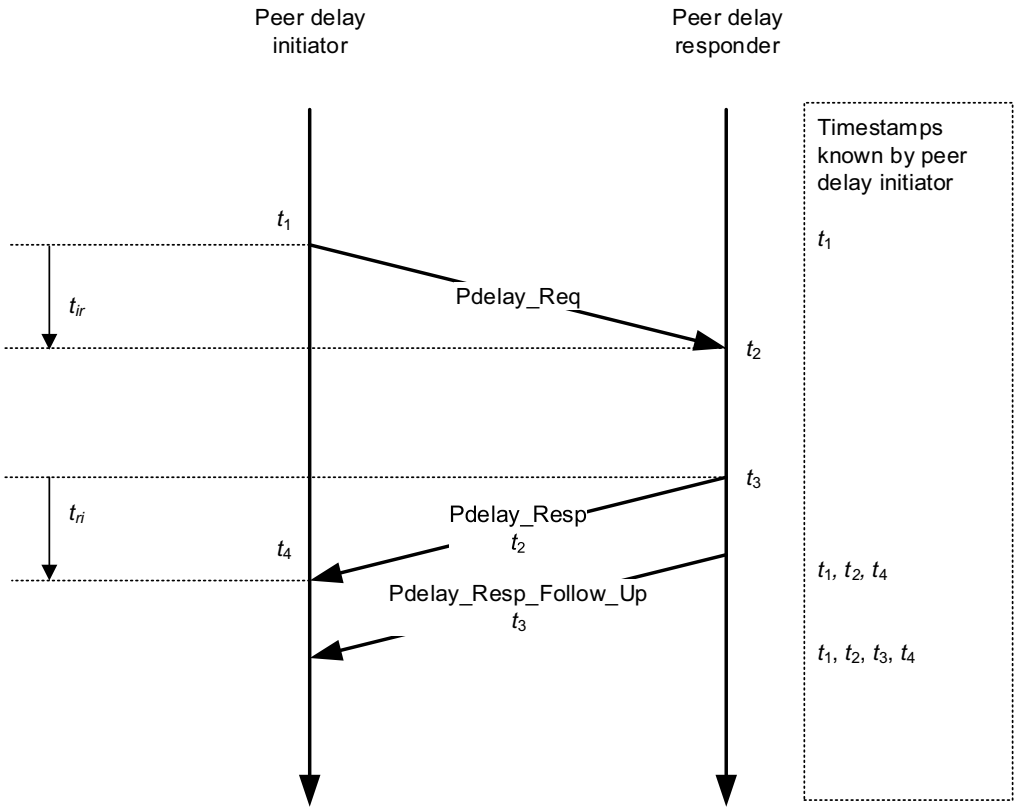


Figure 11-1—Propagation delay measurement using peer-to-peer delay mechanism

The propagation delay measurement starts with the initiator issuing a Pdelay_Req message and generating a timestamp, t_1. The responder receives this message and timestamps it with time t_2. The responder returns a Pdelay_Resp message and timestamps it with time t_3. The responder returns the time t_2 in the Pdelay_Resp message and the time t_3 in a Pdelay_Resp_Follow_Up message. The initiator generates a timestamp, t_4, upon receiving the Pdelay_Resp message. The initiator then uses these four timestamps to compute the mean propagation delay (i.e., meanLinkDelay; see 8.3) as shown in Equation (11-1).

$$\begin{aligned} t_{ir} &= t_2 - t_1 \\ t_{ri} &= t_4 - t_3 \\ D &= \frac{t_{ir} + t_{ri}}{2} = \frac{(t_4 - t_1) - (t_3 - t_2)}{2} \end{aligned} \tag{11-1}$$

where D is the measured mean propagation delay and the other quantities are defined in Figure 11-1.

NOTE—Equation (11-1) does not take into consideration neighbor rate ratio.

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Note that it is the mean propagation delay that is measured here. Any PTP Link asymmetry is modeled as described in 8.3. Any asymmetry that is not corrected for introduces an error in the transported synchronized time value.

The accuracy of the mean propagation delay measurement depends on how accurately the times t_1, t_2, t_3, and t_4 are measured. In addition, Equation (11-1) assumes that the initiator and responder timestamps are taken relative to clocks that have the same frequency. In practice, t_1 and t_4 are measured relative to the LocalClock entity of the initiator time-aware system, and t_2 and t_3 are measured relative to the LocalClock entity of the responder time-aware system. If the propagation delay measurement is desired relative to the responder time base, the term $(t_4 - t_1)$ in Equation (11-1) must be multiplied by the rate ratio of the responder relative to the initiator, otherwise there will be an error equal to $0.5y(t_4 - t_1)$, where y is the frequency offset of the responder relative to the initiator. Likewise, if the propagation delay measurement is desired relative to the initiator time base, the term $(t_3 - t_2)$ in Equation (11-1) must be multiplied by the rate ratio of the initiator relative to the responder, otherwise there will be an error equal to $0.5y(t_3 - t_2)$, where y is the frequency offset of the initiator relative to the responder. Finally, if the propagation delay measurement is desired relative to the Grandmaster Clock time base, each term must be multiplied by the rate ratio of the Grandmaster Clock relative to the time base in which the term is expressed.

The calculation of meanLinkDelay may result in negative values if very short cables are used and link delay compensation is not accurate, or because of random time stamping errors.

There can also be an error in measured propagation delay due to time measurement granularity (see B.1.2). For example, if the time measurement granularity is 40 ns (as specified in B.1.2), the timestamps t_1, t_2, t_3, and/or t_4 can undergo 40 ns step changes. When this occurs, the measured propagation delay, D, will change by 20 ns (or by a multiple of 20 ns if more than one of the timestamps has undergone a 40 ns step change). The actual propagation delay has not changed by 20 ns; the effect is due to time measurement granularity. The effect can be reduced, and the accuracy improved, by averaging successive measured propagation delay values. For example, an exponential averaging filter can be used, i.e., as shown in Equation (11-2).

$$D_{avg,k} = aD_{avg,k-1} + (1-a)D_{k-1} \tag{11-2}$$

where

D_k is the kth propagation delay measurement
$D_{avg,k}$ is the kth computed average propagation delay
k is an index for the propagation delay measurements (i.e., peer delay message exchange)

The quantity a is the exponential weighting factor; it can be set so that the weight of a past propagation delay measurement is 1/e after M measurements, i.e., as shown in Equation (11-3).

$$a = \mathrm{e}^{-\frac{1}{M}} \tag{11-3}$$

The above averager must be initialized. One method is to use a simple average (i.e., the sum of the sample values divided by the number of samples) of the measurements made up to the current measurement until a window of M measurements has been accumulated. In this case, Equation (11-2) is used only for $k > M$. For $k \leq M$, the averaged propagation delay is given by Equation (11-4).

$$D_{avg,k} = \frac{(k-1)D_{avg,k-1} + D_{k-1}}{k} \tag{11-4}$$

The rate ratio of the responder relative to the initiator is the quantity nrrPdelay (see 11.2.12.14). When computed using peer-to-peer delay messages, it is computed by the function computePdelayRateRatio()

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(see 11.2.18.3.3) of the MDPdelayReq state machine (see 11.2.18) using successive values of t_3 and t_4. As indicated in the description of computePdelayRateRatio(), any scheme that uses this information is acceptable as long as the performance requirements of B.2.4 are met. One example scheme is given in NOTE 1 of 11.2.18.3.3.

11.1.3 Transport of time-synchronization information

The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up (or just Sync) messages, is illustrated in Figure 11-2. The mechanism is mathematically equivalent to the mechanism described in IEEE Std 1588-2019 for a peer-to-peer Transparent Clock that is syntonized (see 10.1, 10.3, 11.1, and 11.4 of IEEE Std 1588-2019). However, the processes of transporting synchronization by a peer-to-peer Transparent Clock that is syntonized and by a Boundary Clock are mathematically and functionally equivalent. The main functional difference between the two types of clocks is that the Boundary Clock participates in best timeTransmitter selection and invokes the BTCA, while the peer-to-peer Transparent Clock does not participate in best timeTransmitter selection and does not invoke the BTCA (and implementations of the two types of clocks can be different).

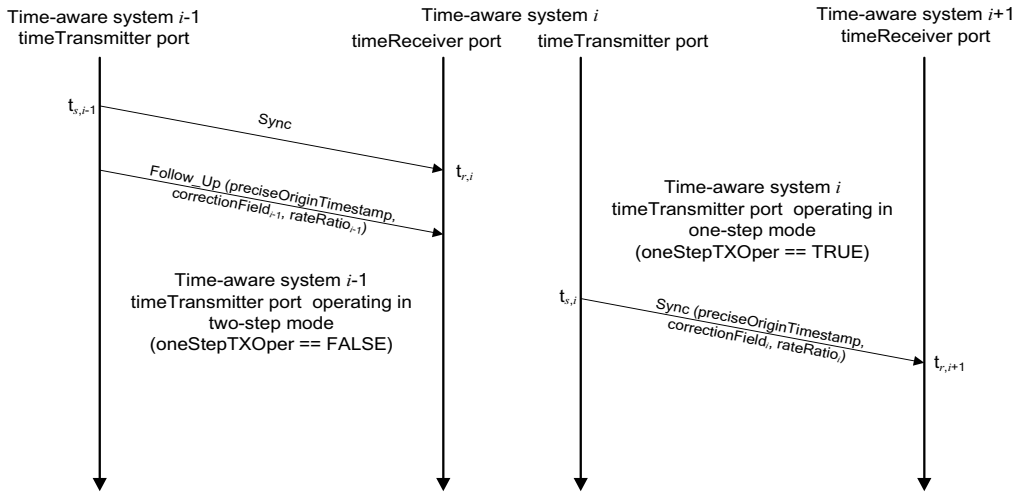


Figure 11-2—Transport of time-synchronization information

Figure 11-2 shows three adjacent PTP Instances, indexed $i - 1$, i, and $i + 1$. Synchronization is transported from PTP Instance $i - 1$ to PTP Instance i using two-step time transport, and then to PTP Instance $i + 1$ using one-step time transport. PTP Instance $i - 1$ sends a Sync message to PTP Instance i at time $t_{s,i-1}$, relative to the LocalClock entity of PTP Instance $i - 1$. At a later time (since it is using two-step time transport), PTP Instance $i - 1$ sends an associated Follow_Up message to PTP Instance i, which contains a preciseOriginTimestamp, correctionField$_{i-1}$, and rateRatio$_{i-1}$. The preciseOriginTimestamp contains the time of the Grandmaster Clock when it originally sent this synchronization information. It is not indexed here because it normally does not change as the Sync and Follow_Up messages traverse the network. The quantity correctionField$_{i-1}$ contains the difference between the synchronized time when the Sync message is sent (i.e., the synchronized time that corresponds to the local time $t_{s,i-1}$) and the preciseOriginTimestamp. The sum of preciseOriginTimestamp and correctionField$_{i-1}$ gives the synchronized time that corresponds to $t_{s,i-1}$. The quantity rateRatio$_{i-1}$ is the ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity of PTP Instance $i - 1$.

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PTP Instance i receives the Sync message from PTP Instance $i-1$ at time $t_{r,i}$, relative to its LocalClock entity. It timestamps the receipt of the Sync message, and the timestamp value is $t_{r,i}$. It receives the associated Follow_Up message some time later.

PTP Instance i will eventually send a new Sync message at time $t_{s,i}$, relative to its LocalClock entity. The time $t_{s,i}$ occurs after the Follow_Up message is received from PTP Instance $i-1$. It will have to compute correctionField$_i$, i.e., the difference between the synchronized time that corresponds to $t_{s,i}$ and the preciseOriginTimestamp. To do this calculation, it must compute the value of the time interval between $t_{s,i-1}$ and $t_{s,i}$, expressed in the Grandmaster Clock time base. This interval is equal to the sum of the following quantities:

a) The propagation delay on the PTP Link between PTP Instances $i-1$ and i, expressed in the Grandmaster Clock time base, and

b) The difference between $t_{s,i}$ and $t_{r,i}$ (i.e., the residence time), expressed in the Grandmaster Clock time base.

The mean propagation delay on the PTP Link between PTP Instances $i-1$ and i, relative to the LocalClock entity of PTP Instance $i-1$, is equal to meanLinkDelay (see 10.2.5.8). This must be multiplied by rateRatio$_{i-1}$ to express it in the Grandmaster Clock time base. The total propagation delay is equal to the mean propagation delay plus the quantity delayAsymmetry (see 8.3 and 10.2.5.9); delayAsymmetry is already expressed in the Grandmaster Clock time base. The residence time, $t_{s,i}-t_{r,i}$, must be multiplied by rateRatio$_i$ to express it in the Grandmaster Clock time base.

The preceding computation is organized slightly differently in the state machines of 11.2.13 and 11.2.14. Rather than explicitly expressing the PTP Link propagation delay in the Grandmaster Clock time base, the local time at PTP Instance i that corresponds to $t_{s,i-1}$ is computed; this is the upstreamTxTime member of the MDSyncReceive structure (see 10.2.2.2.7; recall that $t_{s,i-1}$ is relative to the LocalClock entity of PTP Instance $i-1$). upstreamTxTime is equal to the quantity $t_{r,i}$ minus the PTP Link propagation delay expressed relative to the LocalClock entity of PTP Instance i. The PTP Link propagation delay expressed relative to the LocalClock entity of PTP Instance i is equal to the sum of the following:

c) The quantity meanLinkDelay (see 10.2.5.8) divided by nrrPdelay (see 11.2.12.14), and

d) The quantity delayAsymmetry (see 10.2.5.9) divided by rateRatio$_i$.

The division of delayAsymmetry by rateRatio$_i$ is performed after rateRatio$_i$ has been updated. The computation of upstreamTxTime is done by the MDSyncReceiveSM state machine in the function setMDSyncReceiveMDSR() (see 11.2.13.2.1). When PTP Instance i sends a Sync message to PTP Instance $i+1$, it computes the sum of the PTP Link propagation delay and residence time, expressed in the Grandmaster Clock time base, as shown in item e).

e) The quantity ($t_{s,i}$ – upstreamTxTime)(rateRatio$_i$).

As in item d) in this subclause, this computation is performed after rateRatio$_i$ has been updated. The quantity of item e) is added to correctionField$_{i-1}$ to obtain correctionField$_i$. The computation of item e) and correctionField$_i$ is done by the MDSyncSendSM state machine in the function setFollowUp() (see 11.2.14.2.3). The quantity correctionField$_i$ is inserted in the Sync message sent by PTP Instance i.

The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to the time bases of the PTP Instance at the other end of the attached PTP Link or of the current PTP Instance is usually negligible. The former can be obtained from the latter by multiplying the latter by the ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity of the PTP Instance at the other end of the PTP Link attached to this PTP Port. This ratio differs from 1 by 200 ppm or less. For example, for a worst-case frequency offset of the LocalClock entity of the PTP Instance at the other end of the PTP Link,

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relative to the Grandmaster Clock, of 200 ppm, and a measured propagation time of 100 ns, the difference in *D* relative to the two time bases is 20 ps. The corresponding difference for PTP Link delay asymmetry in this example is also negligible because the magnitude of the PTP Link delay asymmetry is of the same order of magnitude as the mean propagation time, or less. However, the difference is usually not negligible for residence time because residence time can be much larger (see B.2.2).

It was indicated in the first paragraph of this subclause that the processes of transporting synchronization by a peer-to-peer Transparent Clock that is syntonized and by a Boundary Clock are mathematically and functionally equivalent. The reason for this equivalency is that the computations described so far in this subclause compute the synchronized time when the Sync message is sent by the PTP Instance. The same computations are done if PTP Instance *i* sends a Sync message without having received a new Sync message, i.e., if Sync receipt timeout occurs (see 10.7.3.1). In this case, PTP Instance *i* uses the most recently received time-synchronization information from PTP Instance $i - 1$, which would be prior to when PTP Instance *i* sent its most recent Sync message. The synchronized time corresponding to the sending of a Sync message is equal to the sum of the preciseOriginTimestamp and correctionField. Normally a Boundary Clock places this entire value, except for any sub-nanosecond portion, in the preciseOriginTimestamp, while a Transparent Clock retains the preciseOriginTimestamp and updates the correctionField. However, the sum of the two fields is equal to the synchronized time when the Sync message is sent in both cases.

The ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity at PTP Instance *i*, rateRatio_i, is equal to the same quantity at PTP Instance $i - 1$, rateRatio_{i-1}, multiplied by the ratio of the frequency of the LocalClock entity at PTP Instance $i - 1$ to the frequency of the LocalClock entity at PTP Instance *i*, nrrPdelay (see 11.2.12.14). If nrrPdelay is sufficiently small, this is approximately equal to the sum of rateRatio_{i-1} and the quantity nrrPdelay – 1, which is the frequency offset of PTP Instance $i - 1$ relative to PTP Instance *i*. This computation is done by the PortSyncSyncReceive state machine (see 10.2.8).

NOTE—The sending of time-synchronization information by the timeTransmitter ports of a PTP Instance might or might not be tightly synchronized with the receipt of time-synchronization information by the timeReceiver port. If a timeTransmitter port has the same logMessageInterval as the timeReceiver port, it will transmit timing event messages as soon as possible after the timeReceiver port has received the corresponding timing event messages and the timeTransmitter port is operating in “syncLocked” mode (see 10.2.5.15). If a timeTransmitter port and timeReceiver port have different logMessageInterval values, then the timeTransmitter port can send timing event messages without any synchronization with the timeReceiver port.

11.1.4 Model of operation

A PTP Instance contains one MD entity per PTP Instance, per PTP Port. This entity contains functions generic to all media, which are described in Clause 10, and functions specific to the respective medium for the PTP Link. Functions specific to FDE links are described in the current clause.

NOTE—FDE links are in the category of links specified in this clause.

The model for a PTP Instance of a time-aware system with FDE links is shown in Figure 11-3. It assumes the presence of one FDE MD entity per PTP Port. The media-independent entities shown in Figure 11-3 are described in 10.1.2.

A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the LocalClock entity when the message timestamp point (see 3.21) crosses the timestamp measurement plane (see 3.40). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a timestamp relative to the reference plane (see 3.31). The corrected timestamp value is provided to the MD entity.

The MD entity behavior and detailed state machines specific to FDE links are described in 11.2. The behavior of the MD entity that is generic to all media is described in Clause 10.

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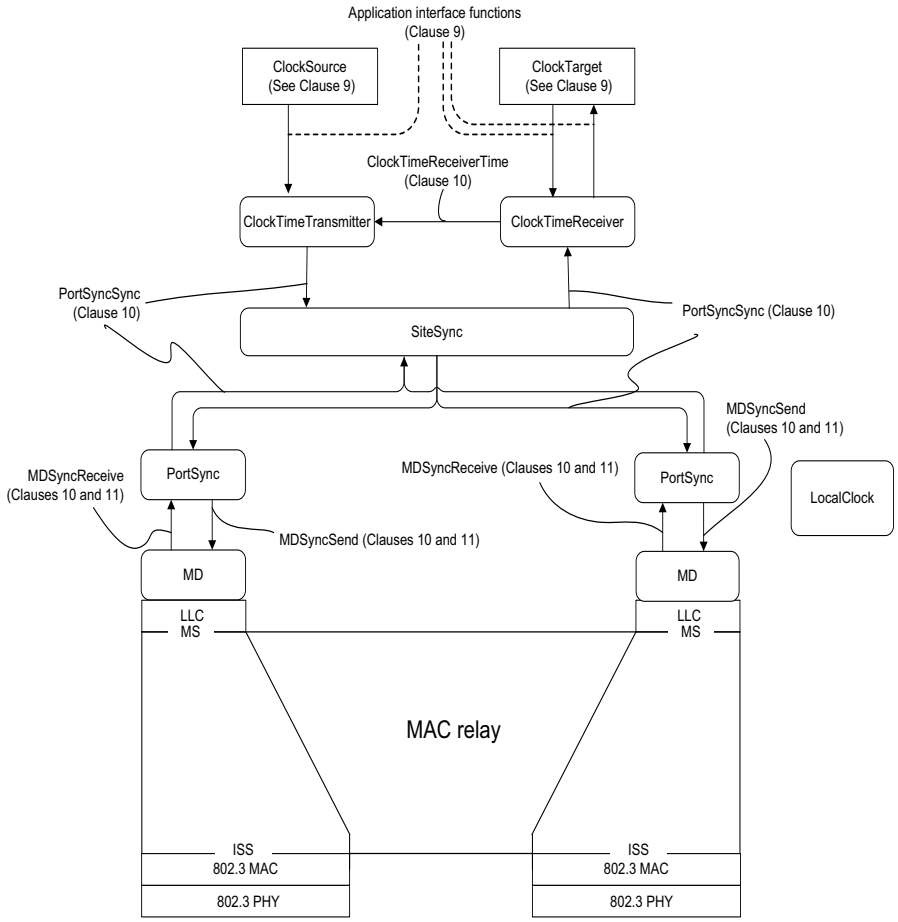


Figure 11-3—Model for a PTP Instance of a time-aware system with FDE links

11.2 State machines for MD entity specific to FDE links

11.2.1 General

This subclause describes the media-dependent state machines for an MD entity, for FDE links. The state machines are all per port because an instance of each is associated with an MD entity. The state machines are as follows:

a) MDSyncReceiveSM (shown in Figure 10-2, and in more detail in Figure 11-4): receives Sync and, if the received information is two-step, Follow_Up messages, and sends the time-synchronization information carried in these messages to the PortSync entity of the same PTP Port. In addition, if the global variable driftTrackingTlvSupport is TRUE, this state machine computes the ratio of the frequency of the LocalClock entity in the PTP Instance at the other end of the attached PTP Link to the frequency of the LocalClock entity in this PTP Instance using the information contained in successive Drift_Tracking TLVs, and places the result in the global variable nrrSync (see 11.2.12.15). The global variable neighborRateRatio (see 10.2.5.7) is set equal to nrrSync or nrrPdelay (see 11.2.12.14) depending on whether the value of the global variable nrrCompMethod

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(see 11.2.12.16) is equal to Sync or Pdelay, respectively. There is one instance of this state machine per PTP Port, per domain.

b) MDSyncSendSM (shown in Figure 10-2, and in more detail in Figure 11-4): receives an MDSyncSend structure from the PortSync entity of the same PTP Port; transmits a Sync message; uses the syncEventEgressTimestamp and information contained in the MDSyncSend structure to compute information needed for the Sync message if the PTP Port is currently operating as a one-step PTP Port and for the corresponding Follow_Up message if the PTP Port is currently operating as a two-step PTP Port; and transmits the Follow_Up message if the PTP Port is two-step. There is one instance of this state machine per PTP Port, per domain.

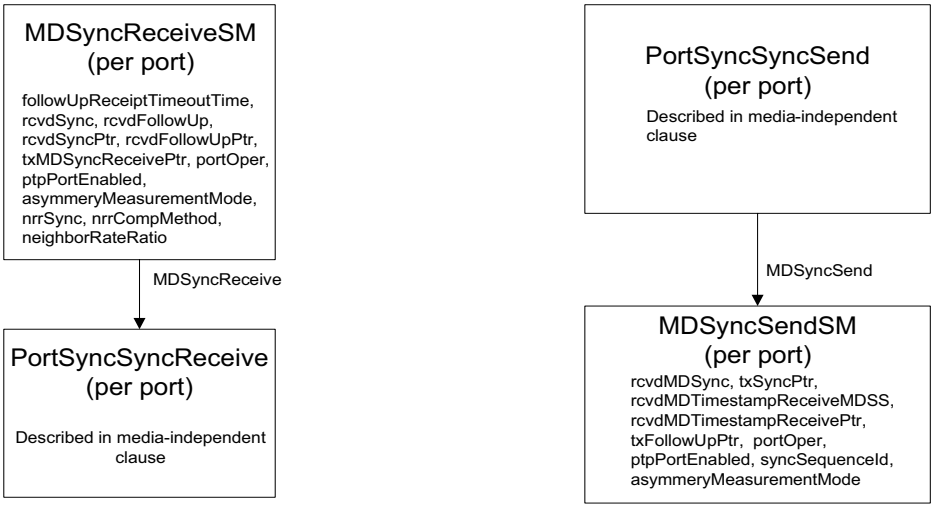


Figure 11-4—Detail of MD entity time-synchronization state machines for FDE links

c) MDPdelayReq (shown in Figure 11-5): transmits a Pdelay_Req message, receives Pdelay_Resp and Pdelay_Resp_Follow_Up messages corresponding to the transmitted Pdelay_Req message, uses the information contained in successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages to compute the ratio of the frequency of the LocalClock entity in the PTP Instance at the other end of the attached PTP Link to the frequency of the LocalClock entity in this PTP Instance, places the result in the global variable nrrPdelay (see 11.2.12.14), and uses the information obtained from the message exchange and the computed frequency ratio to compute propagation delay on the attached PTP Link. There is one instance of this state machine for all the domains, per port.

d) MDPdelayResp (shown in Figure 11-5): receives a Pdelay_Req message from the MD entity at the other end of the attached PTP Link, and responds with Pdelay_Resp and Pdelay_Resp_Follow_Up messages. There is one instance of this state machine for all the domains, per port.

e) SyncIntervalSetting state machine (not shown): receives a Signaling message that contains a message interval request TLV (see 10.6.4.3), and sets the global variables that give the duration of the mean intervals between successive Sync messages. There is one instance of this state machine per PTP Port, per domain.

f) LinkDelayIntervalSetting state machine (not shown): receives a Signaling message that contains a message interval request TLV (see 10.6.4.3), and sets the global variables that give the duration of the mean intervals between successive Pdelay_Req messages. There is one instance of this state machine for all the domains, per port.

Figure 10-2, Figure 11-4, and Figure 11-5 are not themselves state machines, but illustrate the machines, their interrelationships, the principle variables and messages used to communicate between them, their local

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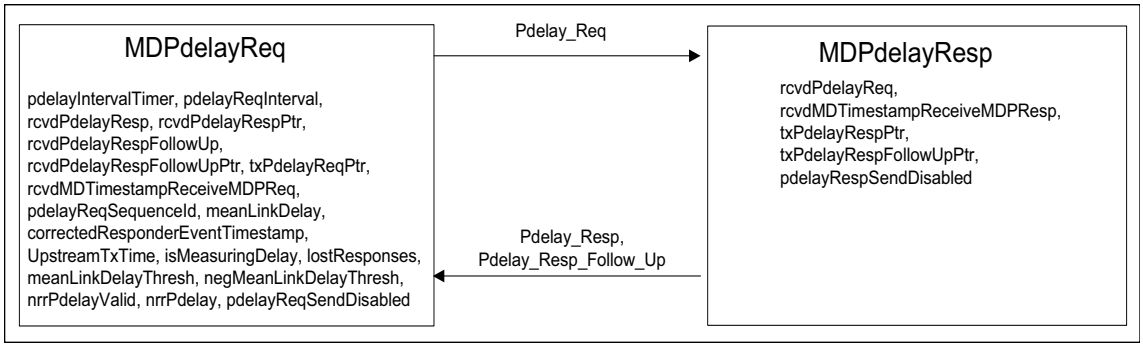


Figure 11-5—Peer-to-peer delay mechanism state machines—overview and interrelationships

variables, and performance parameters. The figures do not show the service interface primitives between the media-dependent layer and the logical link control (LLC). Figure 11-5 is analogous to Figure 10-2; while Figure 10-2 applies to the general time-synchronization protocol, Figure 11-5 is limited to the peer-to-peer delay mechanism for measurement of propagation delay in FDE links. Figure 11-4 shows greater detail of the MDSyncReceiveSM and MDSyncSendSM state machines for FDE links than Figure 10-2.

The state machines described in 11.2 use some of the global per PTP Instance system variables specified in 10.2.4, the global per-port variables specified in 10.2.5, and the functions specified in 10.2.6.

11.2.2 Determination of asCapable and asCapableAcrossDomains

There is one instance of the global variable asCapable (see 10.2.5.1) per PTP Port, per domain. For FDE links, there is one instance of the global variable asCapableAcrossDomains (see 11.2.12.13), per port, that is common across, and accessible by, all the domains. For HDE links, there is one instance of asCapableAcrossDomains per PTP Instance (i.e., per domain) (see 11.2.12.13).

The global variable asCapable (see 10.2.5.1) indicates whether the IEEE 802.1AS protocol is operating, in this domain, on the PTP Link attached to this PTP Port, and can provide the required time-synchronization performance described. asCapable is used by the PortSync entity, which is media-independent; however, the determination of asCapable is media-dependent.

The per-port global variable asCapableAcrossDomains is set by the MDPdelayReq state machine (see 11.2.18 and Figure 11-9). For a port attached to a FDE PTP Link or to an HDE link, asCapableAcrossDomains shall be set to TRUE if and only if either:

a) It is determined, via the transport-specific peer-to-peer delay mechanism or CMLDS, that the following conditions hold for the port:
   1) The port is exchanging peer delay messages with its neighbor,
   2) The measured delay neither exceeds meanLinkDelayThresh nor is below negMeanLinkDelayThresh,
   3) For FDE links, the port does not receive multiple Pdelay_Resp or Pdelay_Resp_Follow_Up messages in response to a single Pdelay_Req message,
   4) For HDE links, the port does not receive multiple Pdelay_Resp or multiple Pdelay_Resp_Follow_Up messages with the requestingPortIdentity.Clock Identity equal to the PTP Instance's clock identity in response to a single Pdelay_Req message, and
   5) The port does not receive a response from itself or another PTP Port of the same PTP Instance.

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or:

b) pdelayReqSendDisabled is set to TRUE.

NOTE 1—If a PTP Instance implements only domain 0 and the MDPdelayReq and MDPdelayResp state machines are invoked on domain 0 (see 11.2.18), asCapableAcrossDomains is still set by the MDPdelayReq state machine.

The default value of meanLinkDelayThresh shall be set as specified in Table 11-1.

Table 11-1—Value of meanLinkDelayThresh for various links

Link	Value of meanLinkDelayThresh (ns) (see NOTE)
100BASE-TX, 1000BASE-T	800_{10}
100BASE-FX, 1000BASE-X	0x7FFFFFFF
NOTE—The actual propagation delay for 100BASE-TX and 1000BASE-T links is expected to be smaller than the above respective threshold. If the measured mean propagation delay (i.e., meanLinkDelay; see 10.2.5.8) exceeds this threshold, it is assumed that this is due to the presence of equipment that does not implement gPTP. For 100BASE-FX and 1000BASE-X links, the actual propagation delay can be on the order of, or larger than, the delay produced by equipment that does not implement gPTP; therefore, such equipment cannot be detected by comparing measured propagation delay with a threshold. In this case, meanLinkDelayThresh is set to the largest possible positive value (i.e., in binary notation a zero followed by all 1s). The default value for negMeanLinkDelayThresh is not defined and it is implementation specific. This table does not list propagation delay threshold values for all types of PHYs because the values are implementation specific and depend on the application.	

The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if and only if the following conditions hold:

c) The value of asCapableAcrossDomains is TRUE, and

d) One of the following conditions holds:

1) The value of neighborGptpCapable for this PTP Port is TRUE, or

2) The value of domainNumber is zero, and the value of sdoId for peer delay messages received on this PTP Port is 0x100.

NOTE 2—Condition d) 2) ensures backward compatibility with the 2011 edition of this standard. A PTP Instance compliant with the current edition of this standard that is attached, via a FDE PTP Link, to a PTP Instance compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP-capable TLV and will not set neighborGptpCapable to TRUE. However, condition d) 2) ensures that asCapable for this PTP Port and domain (i.e., domain 0) will still be set to TRUE if condition c) holds because the peer delay messages received from the time-aware system compliant with the 2011 edition of this standard will have sdoId set to 0x100.

11.2.3 Use of MAC Control PAUSE operation

A PTP Instance shall not use the MAC Control PAUSE operation.

11.2.4 Use of priority-based flow control

A PTP Instance that implements priority-based flow control shall neither transmit nor honor upon receipt priority-based flow control messages that act on the IEEE 802.1AS message priority code point (see 8.4.4).

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11.2.5 Use of link aggregation

gPTP PDUs, when used with physical ports that are attached to an IEEE 802.1AX Link Aggregation Group, shall be transmitted and received over the physical ports (Aggregation Ports), not the Aggregator Port. Using the Parser/Multiplexer functions described in 6.1.3 and 6.2.7 of IEEE Std 802.1AX-2020, received gPTP PDUs are recognized as control frames, separated from the incoming data stream, and directed to the gPTP layers. gPTP PDUs output from the gPTP layers are integrated into the port's data stream by the Parser/Multiplexer.

Assuming that ptpPortEnabled is true for all the physical links (Aggregation Links) in a Link Aggregation Group and that all the physical links are connected to the same two systems, gPTP will measure the delay on each physical link, and the IEEE 802.1AS protocol will choose one of the physical links for transmitting time from the timeTransmitter clock.

11.2.6 M-UNITDATA.request

This service primitive is used by an MD entity to request to the associated LLC the transmission of a Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up message. The primitive is described in IEEE Std 802.1AC.

11.2.7 M-UNITDATA.indication

This service primitive is used by the LLC to indicate to the associated MD entity the receipt of a Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up message. The primitive is described in IEEE Std 802.1AC.

11.2.8 MDTimestampReceive

11.2.8.1 General

This structure provides the timestamp, relative to the timestamp measurement plane, of the event message that was just sent or just received. The structure is received by the MD entity of the PTP Port. The MD entity corrects this timestamp for any ingressLatency or egressLatency (see 8.4.3) before using it and/or passing it to higher layer entities.

```
MDTimestampReceive {
        timestamp
}
```

The member of the structure is defined in the following subclause.

11.2.8.2 timestamp (UScaledNs)

The timestamp is the value of the timestamp of the event message (i.e., Sync, Pdelay_Req, Pdelay_Resp) that was just transmitted or received. This timestamp is taken relative to the timestamp measurement plane.

11.2.9 MDSyncReceive

This structure is specified in 10.2.2.2.

11.2.10 MDSyncSend

This structure is specified in 10.2.2.1.

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11.2.11 Overview of MD entity global variables

Subclause 11.2.12 defines global variables used by the MD entity state machines whose scopes are as follows:

— Per PTP Instance (i.e., per domain)
— Per PTP Instance, per PTP Port
— Instances used by CMLDS (see 11.2.16) (i.e., variable is common across all Link Ports)
— Instances used by CMLDS, per Link Port

Table 11-2 summarizes the scope of each global variable of 11.2.12.

Table 11-2—Summary of scope of global variables used by time synchronization state machines (see 10.2.4 and 10.2.5)

Variable name	Subclause of definition	Data type	Scope	Instance used by CMLDS (i.e., variable is common across all Link Ports)	Instance used by CMLDS, per Link Port
currentLogPdelayReqInterval	11.2.12.1	Integer8	Per PTP Port	No	Yes
initialLogPdelayReqInterval	11.2.12.2	Integer8	Per PTP Port	No	Yes
pdelayReqInterval	11.2.12.3	UScaledNs	Per PTP Port	No	Yes
allowedLostResponses	11.2.12.4	UInteger8	Per PTP Port	No	Yes
allowedFaults	11.2.12.5	UInteger8	Per PTP Port	No	Yes
isMeasuringDelay	11.2.12.6	Boolean	Per PTP Port	No	Yes
meanLinkDelayThresh	11.2.12.7	ScaledNs	Per PTP Port	No	Yes
negMeanLinkDelayThresh	11.2.12.8	ScaledNs	Per PTP Port	No	Yes
syncSequenceId	11.2.12.9	UInteger16	Per PTP Port	No	No
oneStepReceive	11.2.12.10	Boolean	Per PTP Port	No	No
oneStepTransmit	11.2.12.11	Boolean	Per PTP Port	No	No
oneStepTxOper	11.2.12.12	Boolean	Per PTP Port	No	No
asCapableAcrossDomains	11.2.12.13	Boolean	Per PTP Port	No	Yes
nrrPdelay	11.2.12.14	Float64	Per PTP Port	No	Yes
nrrSync	11.2.12.15	Float64	Per PTP Port	No	No
nrrCompMethod	11.2.12.16	Enumeration1	Per PTP Port	No	No
NOTE—asCapableAcrossDomains is per PTP Instance for HDE links.					

11.2.12 MD entity global variables

11.2.12.1 currentLogPdelayReqInterval: The current value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Pdelay_Req messages (see 11.5.2.2). This value is set in the LinkDelayIntervalSetting state machine (see 11.2.20). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

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11.2.12.2 initialLogPdelayReqInterval: The initial value of the logarithm to base 2 of the mean time interval, in seconds, between the sending of successive Pdelay_Req messages (see 11.5.2.2). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.3 pdelayReqInterval: A variable containing the mean Pdelay_Req message transmission interval for the port corresponding to this MD entity. The value is set in the LinkDelayIntervalSetting state machine (see 11.2.20). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.4 allowedLostResponses: The number of Pdelay_Req messages without valid responses above which a port is considered to be not exchanging peer delay messages with its neighbor. The default value and range of allowedLostResponses is given in 11.5.3. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.5 allowedFaults: The number of faults [i.e., instances where meanLinkDelay (see 10.2.5.8) exceeds meanLinkDelayThresh (see 11.2.12.7), or is below the negMeanLinkDelayThresh (see 11.2.12.8), and/or the computation of nrrPdelay (see 11.2.12.14) is invalid (see 11.2.18.2.10)] above which asCapableAcrossDomains is set to FALSE, i.e., the port is considered not capable of interoperating with its neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). The default value and range of allowedFaults is given in 11.5.4. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.6 isMeasuringDelay: A Boolean that is TRUE if the port is receiving Pdelay_Resp and Pdelay_Resp_Follow_Up messages from the port at the other end of the PTP Link. For a FDE PTP Link, the port is measuring PTP Link propagation delay if it is receiving Pdelay_Resp and Pdelay_Resp_Follow_Up messages from the port at the other end of the PTP Link (i.e., it performs the measurement using the peer-to-peer delay mechanism). There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.7 meanLinkDelayThresh: The propagation time threshold above which a port is considered not capable of participating in the IEEE 802.1AS protocol. If meanLinkDelay (see 10.2.5.8) exceeds meanLinkDelayThresh, then asCapableAcrossDomains (see 11.2.12.13) is set to FALSE. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

11.2.12.8 negMeanLinkDelayThresh: The negative propagation time threshold determines the point below which a port is considered not capable of participating in the IEEE 802.1AS protocol. If meanLinkDelay (see 10.2.5.8) falls below the negMeanLinkDelayThresh, then asCapableAcrossDomains (see 11.2.12.13) is set to FALSE. There is one instance of this variable for all the domains, i.e., all the PTP Instances (per port), and also one instance of this variable for domain 0 if domain 0 is implemented. The variable is accessible by all the domains.

NOTE—The variable meanLinkDelayThresh was named neighborPropDelayThresh in the 2011 edition of this standard.

11.2.12.9 syncSequenceId: The sequenceId (see 11.4.2.8) for the next Sync message to be sent by this MD entity.

11.2.12.10 oneStepReceive: A Boolean variable that is TRUE if the PTP Port is capable of receiving one-step Sync messages.

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11.2.12.11 oneStepTransmit: A Boolean variable that is TRUE if the PTP Port is capable of transmitting one-step Sync messages.

11.2.12.12 oneStepTxOper: A Boolean variable that is TRUE if the PTP Port will be transmitting one-step Sync messages (see 11.1.3).

NOTE—Since a one-step port modifies Sync message fields (e.g., the correctionField) "on the fly" as the Sync message is being transmitted, the modifying of the correctionField of the Sync message is often implemented in hardware. Each distinct PTP Instance of a time-aware system runs in a distinct domain and tracks a distinct time. If a time-aware system supports more than one PTP Instance, support for one-step often requires hardware for each PTP Instance (e.g., four domains requires four hardware components), but support for two-step can share hardware among all PTP Instances.

11.2.12.13 asCapableAcrossDomains: A Boolean that is TRUE if and only if either: 11.2.2 item a) or 11.2.2 item b) is satisfied. asCapableAcrossDomains is set by the MDPdelayReq state machine and is used in determining asCapable for a port (see 11.2.2). There is one instance of this variable for all the domains (per port) for FDE links, and the variable is accessible by all the domains. There is one instance of this variable per PTP Instance (i.e., per domain) for HDE links (see 19.2.2). When only one domain is active, asCapableAcrossDomains is equivalent to the variable asCapable (see 10.2.5.1).

11.2.12.14 nrrPdelay: The neighborRateRatio (see 10.2.5.7) computed using the transport-specific peer-to-peer delay mechanism or CMLDS (see 11.2.18).

11.2.12.15 nrrSync: The neighborRateRatio (see 10.2.5.7) computed using the Drift_Tracking TLV carried by successive Sync or, if two-step, Follow_Up messages (see 11.2.13).

11.2.12.16 nrrCompMethod: An Enumeration1 that takes on the values Sync and Pdelay to indicate the source of the value of neighborRateRatio (see 10.2.5.7):

a) If the value is Sync and the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE, neighborRateRatio is populated with the value of nrrSync whenever a new value of nrrSync is computed.

b) If the value is Pdelay or if the global variable driftTrackingTlvSupport (10.2.4.28) is FALSE, neighborRateRatio is populated with the value of nrrPdelay whenever a new value of nrrPdelay is computed.

11.2.13 MDSyncReceiveSM state machine

11.2.13.1 State machine variables

The following variables are used in the state diagram in Figure 11-6 (in 11.2.13.3):

11.2.13.1.1 followUpReceiptTimeoutTime: A variable used to save the time at which the information conveyed by a received Sync message will be discarded if the associated Follow_Up message is not received by then. The data type for followUpReceiptTimeoutTime is UScaledNs.

11.2.13.1.2 rcvdSync: A Boolean variable that is set to TRUE when a Sync message is received. This variable is reset to FALSE by the current state machine.

11.2.13.1.3 rcvdFollowUp: A Boolean variable that notifies the current state machine when a Follow_Up message is received. This variable is reset by the current state machine.

11.2.13.1.4 rcvdSyncPtr: A pointer to a structure whose members contain the values of the fields of the Sync message whose receipt is indicated by rcvdSync (see 11.2.13.1.2).

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11.2.13.1.5 rcvdFollowUpPtr: A pointer to a structure whose members contain the values of the fields of the Follow_Up message whose receipt is indicated by rcvdFollowUp (see 11.2.13.1.3).

11.2.13.1.6 txMDSyncReceivePtrMDSR: A pointer to a structure whose members contain the values of the parameters of an MDSyncReceive structure to be transmitted.

11.2.13.1.7 upstreamSyncInterval: The sync interval (see 10.7.2.1) for the upstream PTP Port that sent the received Sync message. The data type for upstreamSyncInterval is UScaledNs.

11.2.13.1.8 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 11-6). The data type for TEMP is Integer16.

11.2.13.2 State machine functions

11.2.13.2.1 setMDSyncReceiveMDSR(): Creates an MDSyncReceive structure, and returns a pointer to this structure. The members of this structure are set as follows:

a) If twoStepFlag of the most recently received Sync message is TRUE, followUpCorrectionField is set equal to the sum of the correctionField (see 11.4.2.6) of the most recently received Sync message, plus the correctionField of the most recently received Follow_Up message. Otherwise, followUpCorrectionField is set equal to the correctionField of the most recently received Sync message.

b) sourcePortIdentity is set equal to the sourcePortIdentity (see 11.4.2.7) of the most recently received Sync message (see 11.4.3).

c) logMessageInterval is set equal to the logMessageInterval (see 11.4.2.9) of the most recently received Sync message (see 11.4.3).

d) If twoStepFlag of the most recently received Sync message is TRUE, preciseOriginTimestamp is set equal to the preciseOriginTimestamp (see 11.4.4.2.1) of the most recently received Follow_Up message. Otherwise, preciseOriginTimestamp is set equal to the originTimestamp (see 11.4.3.2.1) of the most recently received Sync message.

e) rateRatio is set equal to the quantity $(\text{cumulativeScaledRateOffset} \times 2^{-41}) + 1.0$, where the cumulativeScaledRateOffset field is for the most recently received Follow_Up information TLV (see 11.4.4.3.6).

f) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the most recently received Sync message (if twoStepFlag is FALSE) or Follow_Up message (if twoStepFlag is TRUE) contains a Drift_Tracking TLV:

   1) syncGrandmasterIdentity (see 10.2.4.26) is set equal to the syncGrandmasterIdentity of the Drift_Tracking TLV.

   2) If the value of syncStepsRemoved of the Drift_Tracking TLV is not equal to 0xFFFF, syncStepsRemoved (see 10.2.4.27) is set equal to the value of syncStepsRemoved of the Drift_Tracking TLV plus one. If the value of syncStepsRemoved of the Drift_Tracking TLV is equal to 0xFFFF, syncStepsRemoved (see 10.2.4.27) is set equal to 0xFFFF.

   3) nrrSync (see 11.2.12.15) is computed using the following information conveyed by successive Sync and, if twoStepFlag is TRUE, Follow_Up messages; the specific algorithms are beyond the scope of this document:

      i) The syncEventIngressTimestamp values for the respective Sync messages.

      ii) The successive correctedSyncEgressTimestamp values, whose data type is UScaledNs, obtained by adding (A) the seconds field of the syncEgressTimestamp of the Drift_Tracking TLV, multiplied by 10^9, (B) the nanoseconds field of the syncEgressTimestamp of the Drift_Tracking TLV, and (C) CMLDS delayAsymmetry (i.e., cmldsLinkPort.delayAsymmetry, see 10.2.5.9) if CMLDS is provided, otherwise

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instance-specific delayAsymmetry (i.e., portDS.delayAsymmetry, see 14.10.11) divided by rateRatio (see 10.2.8.1.4).

iii) The rateRatioDrift field of the Drift_Tracking TLV.

NOTE 1—If delayAsymmetry does not change during the time interval over which nrrSync is computed, it is not necessary to add it in f) 3) ii) (C) because in that case it will be canceled when computing the difference between earlier and later correctedSyncEgressTimestamps.

NOTE 2—The methodology in f) 3) for computing nrrSync follows the methodology described in 11.2.18.3.3 for computing nrrPdelay using information contained in peer delay messages. See the NOTEs and description in 11.2.18.3.3 for more detail. The variable correctedSyncEgressTimestamp is computed in a manner analogous to the computation of the variable correctedResponderEventTimestamp in 11.2.18.3.3.

NOTE 3—The use of the rateRatioDrift, e.g., in compensating for frequency drift when computing rateRatio (see the RECEIVED_SYNC state of the PortSyncSyncReceive state machine of Figure 10-4), is implementation specific or profile standard specific.

g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the most recently received Sync message (if twoStepFlag is FALSE) or Follow_Up message (if twoStepFlag is TRUE) does not contain a Drift_Tracking TLV:

1) syncGrandmasterIdentity (see 10.2.4.26) is set equal to 0xFFFF FFFF FFFF FFFF.

2) syncStepsRemoved (see 10.2.4.27) is set equal to 0xFFFF.

h) The value of neighborRateRatio (see 10.2.5.7) is set equal to:

1) The value of nrrSync (see 11.2.12.15) if nrrCompMethod (see 11.2.12.16) is Sync and driftTrackingTlvSupport (see 10.2.4.28) is TRUE, or

2) The value of nrrPdelay (see 11.2.12.14) based on the value of nrrCompMethod (see 11.2.12.16) is Pdelay or driftTrackingTlvSupport (see 10.2.4.28) is FALSE

i) upstreamTxTime is set equal to the syncEventIngressTimestamp for the most recently received Sync message (see 11.4.3), minus the mean propagation time on the PTP Link attached to this PTP Port (meanLinkDelay; see 10.2.5.8) divided by nrrPdelay (see 11.2.12.14), minus delayAsymmetry (see 10.2.5.9) for this PTP Port divided by the sum of rateRatio [see item e) in this subclause] and the quantity neighborRateRatio – 1. The syncEventIngressTimestamp is equal to the timestamp value measured relative to the timestamp measurement plane, minus any ingressLatency (see 8.4.3). The upstreamTxTime can be written as follows:

$$\text{upstreamTxTime} = \text{syncEventIngressTimestamp} - (\text{meanLinkDelay}/\text{nrrPdelay}) - (\text{delayAsymmetry}/(\text{rateRatio} \times \text{neighborRateRatio}))$$

NOTE 4—The syncEventIngressTimestamp is measured relative to the time base of the current PTP Instance. The mean propagation delay is divided by neighborRateRatio to convert it from the time base of the PTP Instance at the other end of the attached PTP Link to the time base of the current PTP Instance. delayAsymmetry (i.e., portDS.delayAsymmetry, see 14.10.11) is divided by the new rateRatio to convert it from the time base of the Grandmaster Clock to the time base of the current PTP Instance.

NOTE 5—meanLinkDelay is divided by nrrPdelay because the meanLinkDelay calculation is done using the peer-to-peer delay message timestamps. However, neighborRateRatio appears in the denominator of the delayAsymmetry term because delayAsymmetry, which is supplied via the managed object portDS.delayAsymmetry or cmldsLinkPort.delayAsymmetry, is obtained via measurements outside of gPTP and is relative to the grandmaster time base. delayAsymmetry is converted to the local clock time base based on the respective method used to measure neighbor rate ratio.

NOTE 6—This represents a media-dependent modification to the calculation given in 10.2.2.2.7.

NOTE 7—The difference between the mean propagation time in the Grandmaster Clock time base, the time base of the PTP Instance at the other end of the PTP Link, and the time base of the current PTP Instance is usually negligible. The same is true of any delayAsymmetry (see NOTE 2 of 11.2.18.3.4).

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j) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator of the most recently received Follow_Up information TLV (see 11.4.4.3.7).

k) lastGmPhaseChange is set equal to the lastGmPhaseChange of the most recently received Follow_Up information TLV (see 11.4.4.3.8).

l) lastGmFreqChange is set equal to the scaledLastGmFreqChange of the most recently received Follow_Up information TLV (see 11.4.4.3.9), multiplied by 2^{-41}.

m) domainNumber is set equal to the domainNumber of the most recently received Sync message (see 11.4.3).

11.2.13.2.2 txMDSyncReceive (txMDSyncReceivePtrMDSR): Transmits an MDSyncReceive structure to the PortSyncSyncReceive state machine of the PortSync entity of this PTP Port.

11.2.13.3 State diagram

The MDSyncReceiveSM state machine shall implement the function specified by the state diagram in Figure 11-6, the local variables specified in 11.2.13.1, the functions specified in 11.2.13.2, the structure specified in 10.2.2.2, the messages specified in 11.4, the MD entity global variables specified in 11.2.12, and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives Sync and, if the received information is two-step, Follow_Up messages, places the time-synchronization information in an MDSyncReceive structure, and sends the structure to the PortSyncSyncReceive state machine of the PortSync entity of this PTP Port.

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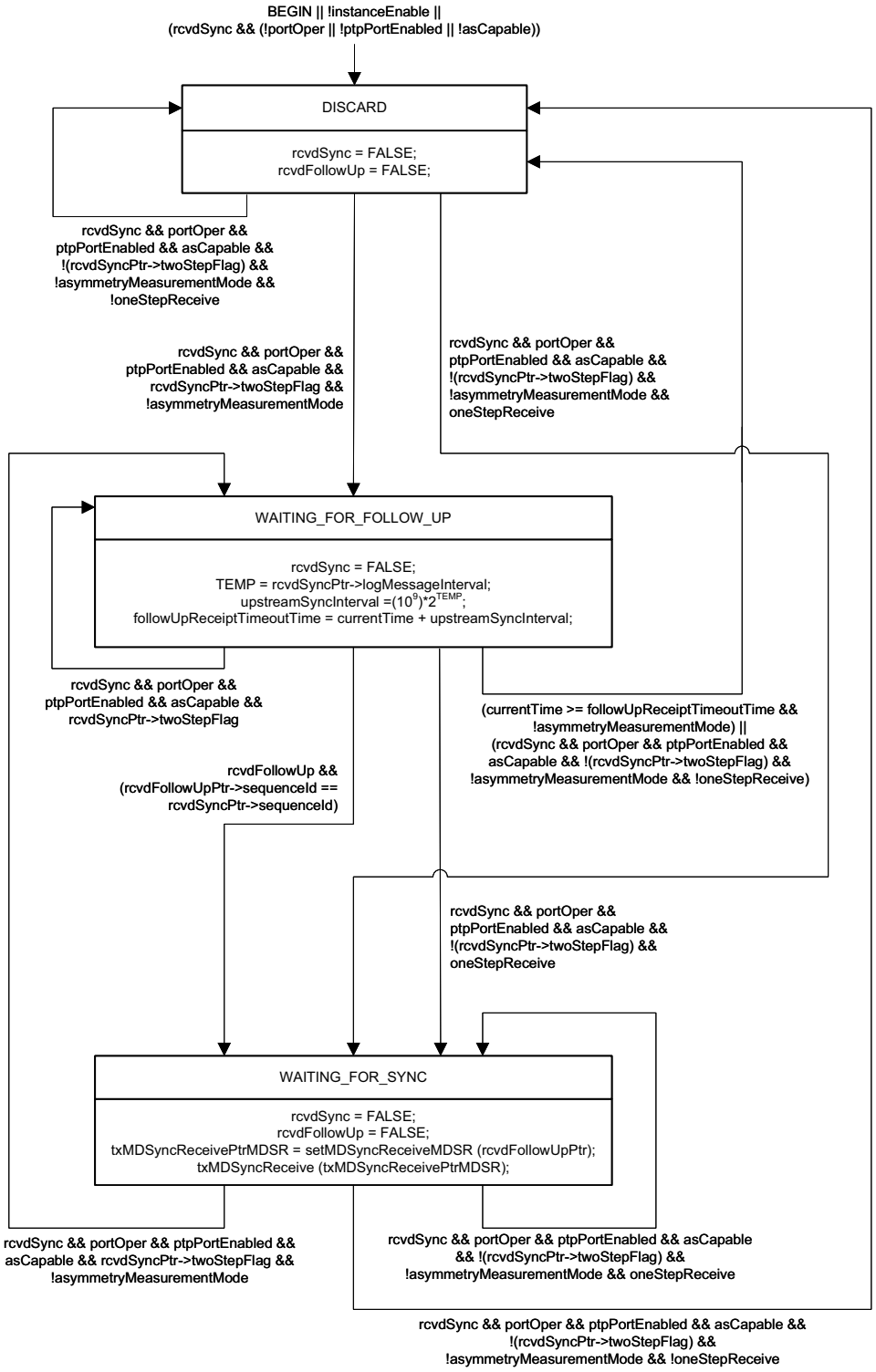


Figure 11-6—MDSyncReceiveSM state machine

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11.2.14 MDSyncSendSM state machine

11.2.14.1 State machine variables

The following variables are used in the state diagram in Figure 11-7 (in 11.2.14.3):

11.2.14.1.1 rcvdMDSyncMDSS: A Boolean variable that notifies the current state machine when an MDSyncSend structure is received. This variable is reset by the current state machine.

11.2.14.1.2 txSyncPtr: A pointer to a structure whose members contain the values of the fields of a Sync message to be transmitted.

11.2.14.1.3 rcvdMDTimestampReceiveMDSS: A Boolean variable that notifies the current state machine when the syncEventEgressTimestamp (see 11.3.2.1) for a transmitted Sync message is received. This variable is reset by the current state machine.

11.2.14.1.4 rcvdMDTimestampReceivePtr: A pointer to the received MDTimestampReceive structure (see 11.2.8).

11.2.14.1.5 txFollowUpPtr: A pointer to a structure whose members contain the values of the fields of a Follow_Up message to be transmitted.

11.2.14.2 State machine functions

11.2.14.2.1 setSyncTwoStep(): Creates a structure whose parameters contain the fields (see 11.4) of a Sync message to be transmitted, and returns a pointer to this structure. The parameters are set as follows:

a) correctionField is set equal to 0.
b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).
c) sequenceId is set equal to syncSequenceId (see 11.2.12.9).
d) logMessageInterval is set equal to the logMessageInterval member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).
e) The remaining fields are set as specified in 11.4.2 and 11.4.3 when twoStepFlag is TRUE.

11.2.14.2.2 txSync (txSyncPtr): Transmits a Sync message from this MD entity, whose fields contain the parameters in the structure pointed to by txSyncPtr (see 11.2.14.1.2).

11.2.14.2.3 setFollowUp(): Creates a structure whose parameters contain the fields (see 11.4) of a Follow_Up message to be transmitted, and returns a pointer to this structure. The parameters are set as follows:

a) followUpCorrectionField is set equal to the sum of the following:
   1) The followUpCorrectionField member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10)
   2) The quantity:

$$\text{rateRatio} \times (\text{syncEventEgressTimestamp} - \text{upstreamTxTime})$$

   where

rateRatio	is the rateRatio member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10)
upstreamTxTime	is the upstreamTxTime member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10
syncEventEgressTimestamp	is the timestamp pointed to by rcvdMDTimestampReceivePtr, corrected for egressLatency (see 8.4.3)

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NOTE—If the PTP Instance that contains this PortSync entity is a PTP Relay Instance, the quantity:

syncEventEgressTimestamp − upstreamTxTime

is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the LocalClock entity, and the quantity:

rateRatio × (syncEventEgressTimestamp − upstreamTxTime)

is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the Grandmaster Clock (see 11.2.13.2.1 for details on the setting of upstreamTxTime).

b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

c) sequenceId is set equal to syncSequenceId (see 11.2.12.9).

d) logMessageInterval is set equal to the logMessageInterval member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

e) preciseOriginTimestamp is set equal to the preciseOriginTimestamp member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

f) rateRatio is set equal to the rateRatio member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP Instance is the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted Follow_Up message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and rateRatioDrift fields set as follows:

1) syncGrandmasterIdentity is set equal to the defaultDS.clockIdentity.

2) syncStepsRemoved is set equal to 0.

3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync message.

4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the frequency of the Grandmaster Clock relative to the frequency of the local clock at the Grandmaster PTP Instance. If the Grandmaster Clock and the local clock at the Grandmaster PTP Instance are the same, rateRatioDrift is zero.

h) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP Instance is not the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted Follow_Up message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and rateRatioDrift fields set as follows:

1) syncGrandmasterIdentity is set equal to the value of the global variable syncGrandmasterIdentity (see 10.2.4.26).

2) syncStepsRemoved is set equal to the value of the global variable syncStepsRemoved (see 10.2.4.27).

3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync message.

4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the frequency of the Grandmaster Clock relative to the frequency of the local clock of the PTP Instance that sends the message.

i) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

j) lastGmPhaseChange is set equal to the lastGmPhaseChange member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

k) scaledLastGmFreqChange is set equal to the lastGmFreqChange member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10), multiplied by 2^{41}.

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l) domainNumber is set equal to the domainNumber of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

m) The remaining fields are set as specified in 11.4.2 and 11.4.4.

11.2.14.2.4 txFollowUp (txFollowUpPtr): Transmits a Follow_Up message from this MD entity, whose fields contain the parameters in the structure pointed to by txFollowUpPtr (see 11.2.14.1.5).

11.2.14.2.5 setSyncOneStep(): Creates a structure whose parameters contain the fields (see 11.4) of a Sync message to be transmitted, and returns a pointer to this structure. The parameters are set as follows:

a) correctionField is set equal to the followUpCorrectionField member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

b) sourcePortIdentity is set equal to the sourcePortIdentity member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

c) sequenceId is set equal to syncSequenceId (see 11.2.12.9).

d) logMessageInterval is set equal to the logMessageInterval member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

e) originTimestamp is set equal to the preciseOriginTimestamp member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

f) rateRatio is set equal to the rateRatio member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

g) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP Instance is the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted Sync message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and rateRatioDrift fields set as follows:

    1) syncGrandmasterIdentity is set equal to the defaultDS.clockIdentity.

    2) syncStepsRemoved is set equal to 0.

    3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync message.

    4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the frequency of the Grandmaster Clock relative to the frequency of the local clock at the Grandmaster PTP Instance. If the Grandmaster Clock and the local clock at the Grandmaster PTP Instance are the same, rateRatioDrift is zero.

h) If the value of the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and the PTP Instance is not the Grandmaster PTP Instance, a Drift_Tracking TLV is attached to the transmitted Sync message with the syncGrandmasterIdentity, syncStepsRemoved, syncEgressTimestamp, and rateRatioDrift fields set as follows:

    1) syncGrandmasterIdentity is set equal to the value of the global variable syncGrandmasterIdentity (see 10.2.4.26).

    2) syncStepsRemoved is set equal to the value of the global variable syncStepsRemoved (see 10.2.4.27).

    3) syncEgressTimestamp is set equal to the syncEventEgressTimestamp of the associated Sync message.

    4) rateRatioDrift (see 11.4.4.4.9) is the measured estimate of the rate of change per second of the frequency of the Grandmaster Clock relative to the frequency of the local clock of the PTP Instance that sends the message.

i) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

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j) lastGmPhaseChange is set equal to the lastGmPhaseChange member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

k) scaledLastGmFreqChange is set equal to the lastGmFreqChange member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10), multiplied by 2^{41}.

l) domainNumber is set equal to the domainNumber of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10).

m) The remaining fields are set as specified in 11.4.2 and 11.4.3 when twoStepFlag is FALSE.

11.2.14.2.6 modifySync(): Adds to the correctionField of the transmitted Sync message, after the syncEventEgressTimestamp is taken and known, the quantity:

rateRatio × (syncEventEgressTimestamp − upstreamTxTime)

where

rateRatio	is the rateRatio member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10)
upstreamTxTime	is the upstreamTxTime member of the most recently received MDSyncSend structure (see 10.2.2.1 and 11.2.10)
syncEventEgressTimestamp	is the timestamp pointed to by rcvdMDTimestampReceivePtr, corrected for egressLatency (see 8.4.3)

NOTE—If the PTP Instance that contains this PortSync entity is a PTP Relay Instance, the quantity:

syncEventEgressTimestamp − upstreamTxTime

is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the LocalClock entity, and the quantity:

rateRatio × (syncEventEgressTimestamp − upstreamTxTime)

is the sum of the residence time and PTP Link propagation delay on the upstream PTP Link, relative to the Grandmaster Clock (see 11.2.13.2.1 for details on the setting of upstreamTxTime).

11.2.14.3 State diagram

The MDSyncSendSM state machine shall implement the function specified by the state diagram in Figure 11-7; the local variables specified in 11.2.14.1; the functions specified in 11.2.14.2; the structures specified in 11.2.8 and 10.2.2.1; the messages specified in 11.4; the MD entity global variables specified in 11.2.12; and the relevant global variables and functions specified in 10.2.4 through 10.2.6. The state machine receives an MDSyncSend structure from the PortSyncSyncSend state machine of the PortSync entity of this PTP Port and transmits a Sync and corresponding Follow_Up message.

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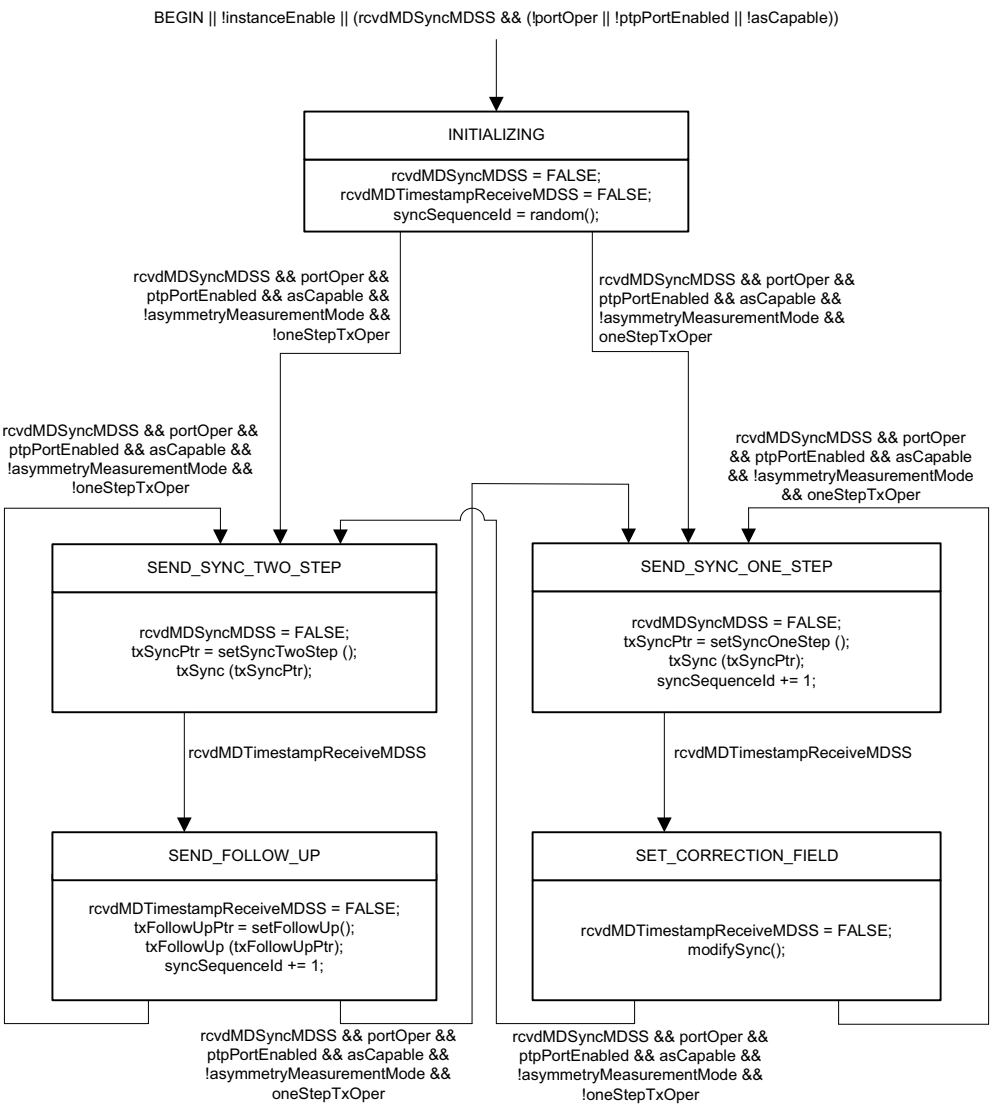


Figure 11-7—MDSyncSendSM state machine

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11.2.15 OneStepTxOperSetting state machine

11.2.15.1 State machine variables

The following variables are used in the state diagram in Figure 11-8 (in 11.2.15.2):

11.2.15.1.1 rcvdSignalingMsg4: A Boolean variable that notifies the current state machine when a Signaling message that contains a message interval request TLV (see 10.6.4.3) is received. This variable is reset by the current state machine.

11.2.15.1.2 rcvdSignalingPtrOSTOS: A pointer to a structure whose members contain the values of the fields of the received Signaling message that contains a message interval request TLV (see 10.6.4.3).

11.2.15.2 State diagram

The OneStepTxOperSetting state machine shall implement the function specified by the state diagram in Figure 11-8, the local variables specified in 11.2.15.1, the messages specified in 10.6 and 11.4, the relevant global variables specified in 10.2.5 and 11.2.12, and the relevant managed objects specified in 14.10. This state machine is responsible for setting the relevant global variables and managed objects pertaining to one-step/two-step operation.

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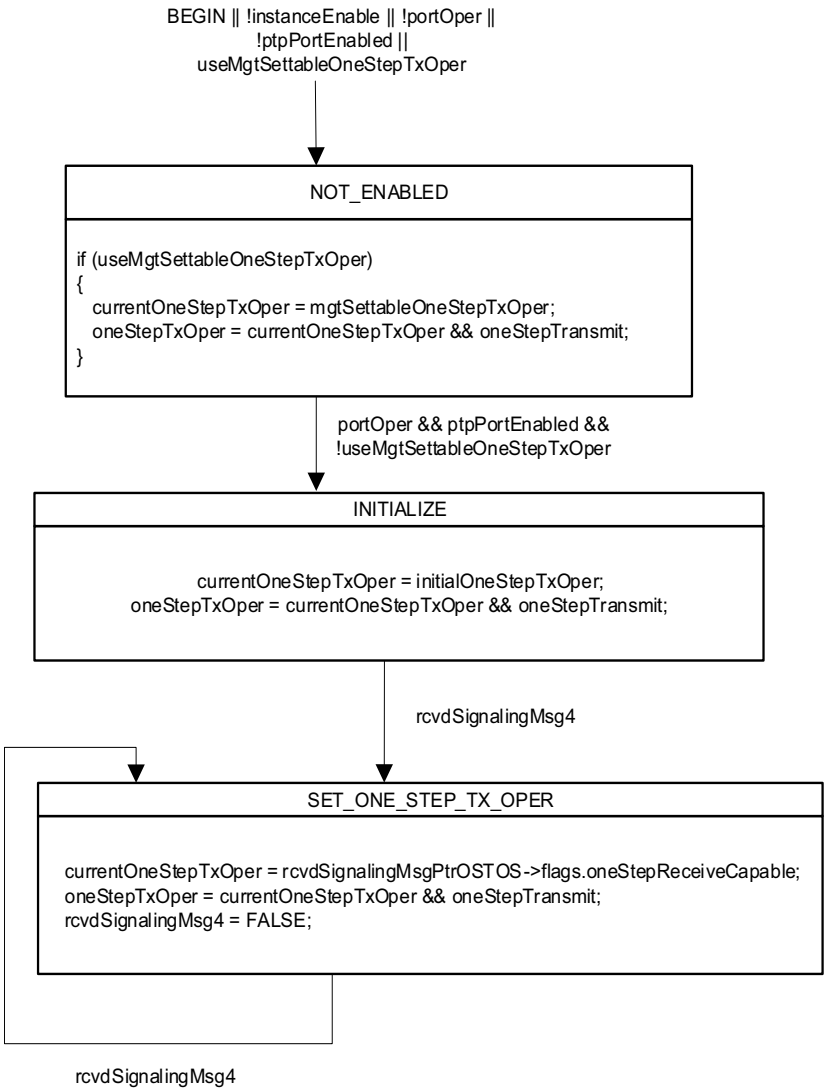


Figure 11-8—OneStepTxOperSetting state machine

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11.2.16 Common Mean Link Delay Service (CMLDS)

11.2.16.1 General

Each PTP Port or Link Port of a time-aware system invokes a single instance of the MDPdelayReq state machine (see 11.2.18) and the MDPdelayResp state machine (see 11.2.19). If the time-aware system implements more than one domain or if domainNumber 0 is not present, these two state machines shall provide a Common Mean Link Delay Service (CMLDS), as described in this subclause, that measures mean propagation delay on the PTP Link attached to the port and the neighbor rate ratio for the port (i.e., the ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the PTP Link attached to this port, to the frequency of the LocalClock entity of this time-aware system). The CMLDS makes the mean propagation delay and neighbor rate ratio available to all active domains. If the time-aware system implements one domain, and if the domainNumber of this domain is 0 (see 8.1), these two state machines may provide the CMLDS; however, if the state machines do not provide the CMLDS (i.e., if only the transport-specific peer delay mechanism is provided), they shall be invoked on domain 0. In other words, if the domainNumber is not 0, portDS.delayMechanism (see Table 14-11 in 14.10.5) must not be P2P. If CMLDS is used, the LocalClock entity for CMLDS and the LocalClock entity for each PTP Instance shall be the same LocalClock.

NOTE 1—In the above, the case where the time-aware system implements only one domain implicitly assumes that IEEE 802.1AS is the only PTP profile present on the respective port of the time-aware system, i.e., no other PTP profiles are implemented on that port. If another PTP profile (see IEEE Std 1588) besides IEEE Std 802.1AS is active on the port, the CMLDS provides the mean propagation delay and neighbor rate ratio to all of them.

In accordance with IEEE Std 1588-2019, the term *Link Port* refers to a port of the CMLDS. A PTP Port for which portDS.delayMechanism is COMMON_P2P uses the CMLDS provided by the Link Port whose cmldsLinkPortDS.portIdentity.portNumber (see 14.18.2) is equal to the commonServicesPortDS.cmldsLinkPortPortNumber (see 14.16.2) for this PTP Port.

The value of majorSdoId for the CMLDS shall be 0x2. The value of minorSdoId for the CMLDS shall be 0x00. As a result, the value of sdoId for the CMLDS is 0x200 (see 16.5, 7.1.3.1, and E.4 of IEEE Std 1588-2019).

NOTE 2—The above requirements for majorSdoId and minorSdoId are for the CMLDS. The requirements for gPTP domains, including transport-specific peer delay messages, are given in 8.1.

NOTE 3—The requirements of this subclause for CMLDS are consistent with the requirements of IEEE Std 1588-2019.

If a PTP Port on a physical interface that also supports CMLDS and a Link Port receives a Pdelay_Req message with majorSdoId value of 0x1, minorSdoId value of 0x00, and domainNumber value of 0, the PTP Port shall respond with PTP transport-specific peer delay messages (i.e., the Pdelay_Resp and Pdelay_Resp_Follow_Up corresponding to this Pdelay_Req) using the transport-specific peer-to-peer delay mechanism if domain 0 is enabled. These transport-specific messages have majorSdoId value of 0x1, minorSdoId value of 0x00, and domainNumber value of 0.

The above requirement provides:

a) Backward compatibility with time-aware systems that comply with the 2011 version of this standard provided that delayMechanism is set to 0x02 (P2P, see 14.10.5), and

b) Compatibility with time-aware systems that implement only one domain and invoke the MDPdelayReq and MDPdelayResp state machines on domain 0, provided that delayMechanism is set to 0x02 (P2P, see 14.10.5).

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In general, a port can receive:

c) Peer delay messages of the CMLDS (with sdoId of 0x200),

d) Transport-specific peer delay messages of domain 0 (with sdoId of 0x100), and

e) If there are other PTP profiles on the neighbor port that use transport-specific peer delay, peer delay messages of those profiles (i.e., with an sdoId of neither 0x100 nor 0x200).

The port responds to the messages of type c) if it invokes CMLDS, the messages of type d) if it invokes gPTP domain 0, and the messages of type e) if it invokes the respective other PTP profiles.

The CMLDS shall be enabled on a Link Port if the value of portDS.delayMechanism (see 14.10.5) is COMMON_P2P for at least one PTP Port that is enabled (i.e., for which portOper and ptpPortEnabled are both TRUE) and corresponds to the same physical port as the Link Port (see 14.1). The value of cmldsLinkPortEnabled is TRUE if the CMLDS is enabled on the Link Port and FALSE if the CMLDS is not enabled on the Link Port.

11.2.16.2 Differences between transport-specific peer-to-peer delay mechanism and CMLDS in computations of meanLinkDelay and nrrPdelay

The MDPdelayReq state machine (see 11.2.18), and the MDPdelay_Resp state machine (see 11.2.19), are invoked either by the transport-specific peer-to-peer delay mechanism or by CMLDS. The resulting values of meanLinkDelay and nrrPdelay are the same for both transport-specific peer delay and CMLDS; however, the organization of the computations for transport-specific peer delay and CMLDS are different. Some of the computations are done at the Initiator and some of the computations are done at the Responder. It is necessary for the Initiator and the Responder to both perform the transport-specific peer delay computations or both perform the CMLDS computations in order to obtain the correct results for meanLinkDelay and nrrPdelay. The differences are:

a) Both transport-specific peer delay and CMLDS compute meanLinkDelay averaged over the two directions, and add delayAsymmetry separately when computing upstreamTxTime by the setMDSyncReceiveMDSR() function of the MDSyncReceive state machine. However, in the computation of meanLinkDelay, CMLDS subtracts delayAsymmetry from the correctionField when sending the pdelayReq message at the Initiator (see Figure 11-1) and adds delayAsymmetry back when computing the quantity t_3 in the function computePropTime() (at the Initiator), while transport-specific peer delay does not subtract and add back delayAsymmetry.

b) Transport-specific peer delay sets the correctionField of a transmitted Pdelay_Req message to 0, while CMLDS sets it to –delayAsymmetry (see 11.2.18.3.1).

c) Transport-specific peer delay sets the correctionField of Pdelay_Resp equal to the fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req, while CMLDS sets the correctionField of Pdelay_Resp equal to minus the fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req (see 11.2.19.3.1).

d) Transport-specific peer delay sets the correctionField of Pdelay_Resp_Follow_Up equal to the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp, while CMLDS sets the correctionField of Pdelay_Resp_Follow_Up equal to the sum of the correctionField of the corresponding Pdelay_Req and the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp (see 11.2.19.3.3).

e) When computing the quantity t_2 for the meanLinkDelay computation [i.e., the quantity D in Equation (11-5), see 11.2.18.3.4], the correctionField of the Pdelay_Resp message, divided by 2^{16}, is added if transport-specific peer delay is used, while it is subtracted if CMLDS is used (see 11.2.18.3.4), and the quantity –delayAsymmetry, divided by 2^{16}, is added if CMLDS is used (i.e., delayAsymmetry is subtracted in the case of CMLDS).

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f) When computing nrrPdelay, the computation of the correctedResponderEventTimestamp is corrected for delayAsymmetry if, and only if, CMLDS is used. The reason for this correction is that, with CMLDS, delayAsymmetry was subtracted from the Pdelay_Req correctionField, and then the Pdelay_Resp_Follow_Up correctionField was set equal to the sum of the Pdelay_Req correctionField and the fractional nanoseconds portion of the PdelayRespEventEgressTimestamp, while with transport-specific peer delay, the correctionField of Pdelay_Req was set equal to 0 (see 11.2.18.3.1, 11.2.18.3.3, and 11.2.19.3.3).

The computations in this standard for the transport-specific peer-to-peer delay mechanism are the same as in IEEE Std 802.1AS-2011, for backward compatibility. However, the computations in this standard for CMLDS need to be consistent with IEEE Std 1588-2019 because CMLDS can be used by other PTP profiles, in addition to the PTP profile included in IEEE Std 802.1AS, that might be present in a gPTP node. Therefore, the computations for the transport-specific peer-to-peer delay mechanism and CMLDS are different (i.e., are organized differently), even though they produce the same results.

11.2.17 Common Mean Link Delay Service (CMLDS) global variable

11.2.17.1 cmldsLinkPortEnabled: A per-Link-Port Boolean that is TRUE if both the value of portDS.delayMechanism is Common_P2P and the value of portDS.ptpPortEnabled is TRUE, for at least one PTP Port that uses the CMLDS that is invoked on the Link Port; otherwise, the value is FALSE.

11.2.18 MDPdelayReq state machine

11.2.18.1 General

This state machine is either invoked as part of the CMLDS or by the transport-specific peer-to-peer delay mechanism. There is one instance of this state machine for all the domains (per port). As a result, there also is one instance of each of the state machine variables of 11.2.18.2, state machine functions of 11.2.18.3, and relevant global variables of 10.2.5, 11.2.12, and 11.2.17 for all the domains (per port). None of the variables used or functions invoked in this state machine are specific to a single domain. However, the single instances of all of these objects or entities are accessible to all the domains.

NOTE—This state machine uses the variable asCapableAcrossDomains (see 11.2.12.13). When only one domain is active, asCapableAcrossDomains is equivalent to the variable asCapable.

11.2.18.2 State machine variables

The following variables are used in the state diagram in Figure 11-9 (in 11.2.18.4):

11.2.18.2.1 pdelayIntervalTimer: A variable used to save the time at which the Pdelay_Req interval timer is started. A Pdelay_Req message is sent when this timer expires. The data type for pdelayIntervalTimer is UScaledNs.

11.2.18.2.2 rcvdPdelayResp:

a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp message is received. This variable is reset by the current state machine.

b) For half-duplex mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp message is received AND its requestingPortIdentity.clockIdentity is equal to the clockIdentity of the current PTP Instance. This variable is reset by the current state machine.

11.2.18.2.3 rcvdPdelayRespPtr: A pointer to a structure whose members contain the values of the fields of the Pdelay_Resp message whose receipt is indicated by rcvdPdelayResp (see 11.2.18.2.2).

11.2.18.2.4 rcvdPdelayRespFollowUp:

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a) For full-duplex point-to-point mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp_Follow_Up message is received. This variable is reset by the current state machine.

b) For half-duplex mode: A Boolean variable that notifies the current state machine when a Pdelay_Resp_Follow_Up message is received AND its requestingPortIdentity.clockIdentity is equal to the clockIdentity of the current PTP instance. This variable is reset by the current state machine.

11.2.18.2.5 rcvdPdelayRespFollowUpPtr:

A pointer to a structure whose members contain the values of the fields of the Pdelay_Resp_Follow_Up message whose receipt is indicated by rcvdPdelayRespFollowUp (see 11.2.18.2.4).

11.2.18.2.6 txPdelayReqPtr: A pointer to a structure whose members contain the values of the fields of a Pdelay_Req message to be transmitted.

11.2.18.2.7 rcvdMDTimestampReceiveMDPReq: A Boolean variable that notifies the current state machine when the pdelayReqEventEgressTimestamp (see 11.3.2.1) for a transmitted Pdelay_Req message is received. This variable is reset by the current state machine.

11.2.18.2.8 pdelayReqSequenceId: A variable that holds the sequenceId for the next Pdelay_Req message to be transmitted by this MD entity. The data type for pdelayReqSequenceId is UInteger16.

11.2.18.2.9 lostResponses: A count of the number of consecutive Pdelay_Req messages sent by the port, for which Pdelay_Resp and/or Pdelay_Resp_Follow_Up messages are not received. The data type for lostResponses is UInteger16.

11.2.18.2.10 nrrPdelayValid: A Boolean variable that indicates whether the function computePdelayRateRatio() (see 11.2.18.3.3) successfully computed nrrPdelay (see 11.2.12.14). The default value of nrrPdelayValid is TRUE.

11.2.18.2.11 detectedFaults: A count of the number of consecutive faults (see 11.5.4 for the definition of a fault).

11.2.18.2.12 portEnabled0: A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to cmldsLinkPortEnabled (see 11.2.17.1) if this state machine is invoked by CMLDS.

11.2.18.2.13 s: A variable whose value is +1 if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and –1 if this state machine is invoked by CMLDS. The data type for s is Integer8.

11.2.18.2.14 pdelayReqSendDisabled: A Boolean that is administratively set to TRUE if Pdelay_Req messages are not transmitted by this port. The default value for this variable shall be FALSE.

11.2.18.3 State machine functions

11.2.18.3.1 setPdelayReq(): Creates a structure containing the parameters (see 11.4) of a Pdelay_Req message to be transmitted, and returns a pointer, txPdelayReqPtr (see 11.2.18.2.6), to this structure. The parameters are set as follows:

a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity (see 8.5.2).

b) sequenceId is set equal to pdelayReqSequenceId (see 11.2.18.2.8).

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c) correctionField is set to
   1) 0 if this state machine is invoked by the transport-specific peer-to-peer delay mechanism, and
   2) –delayAsymmetry (i.e., the negative of delayAsymmetry) if this state machine is invoked by the CMLDS.

d) The remaining parameters are set as specified in 11.4.2 and 11.4.5.

11.2.18.3.2 txPdelayReq(txPdelayReqPtr): Transmits a Pdelay_Req message from the MD entity, containing the parameters in the structure pointed to by txPdelayReqPtr (see 11.2.18.2.6).

11.2.18.3.3 computePdelayRateRatio(): Computes nrrPdelay (see 11.2.12.14) using the following information conveyed by successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages:

a) The pdelayRespEventIngressTimestamp (see 11.3.2.1) values for the respective Pdelay_Resp messages.

b) The correctedResponderEventTimestamp values, whose data type is UScaledNs, obtained by adding the following fields of the received Pdelay_Resp_Follow_Up message:
   1) The seconds field of the responseOriginTimestamp field, multiplied by 10^9.
   2) The nanoseconds field of the responseOriginTimestamp field.
   3) The correctionField, divided by 2^{16}.
   4) delayAsymmetry (see 10.2.5.9), if and only if this state machine is invoked by CMLDS.

NOTE 1—If delayAsymmetry does not change during the time interval over which nrrPdelay is computed, it is not necessary to add it if this state machine is invoked by CMLDS because in that case it will be canceled when computing the difference between earlier and later correctedResponderEventTimestamps.

Any scheme that uses the preceding information, along with any other information conveyed by the successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages, to compute nrrPdelay is acceptable as long as the performance requirements specified in B.2.4 are met. If nrrPdelay is successfully computed, the Boolean nrrPdelayValid (see 11.2.18.2.10) is set to TRUE. If nrrPdelay is not successfully computed (e.g., if the MD entity has not yet exchanged a sufficient number of peer delay messages with its peer), the Boolean nrrPdelayValid is set to FALSE.

NOTE 2—As one example, nrrPdelay can be estimated as the ratio of the elapsed time of the LocalClock entity of the time-aware system at the other end of the PTP Link attached to this port, to the elapsed time of the LocalClock entity of this time-aware system. This ratio can be computed for the time interval between a set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages and a second set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages some number of Pdelay_Req message transmission intervals later, i.e.,

$$\frac{\text{correctedResponderEventTimestamp}_N - \text{correctedResponderEventTimestamp}_0}{\text{pdelayRespEventIngressTimestamp}_N - \text{pdelayRespEventIngressTimestamp}_0}$$

where N is the number of Pdelay_Req message transmission intervals separating the first set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages and the second set, and the successive sets of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages are indexed from 0 to N with the first set indexed 0.

NOTE 3—This function must account for non-receipt of Pdelay_Resp and/or Pdelay_Resp_Follow_Up for a Pdelay_Req message and also for receipt of multiple Pdelay_Resp messages within one Pdelay_Req message transmission interval.

11.2.18.3.4 computePropTime(): Computes the mean propagation delay on the PTP Link attached to this MD entity, D, and returns this value. D is given by Equation (11-5).

$$D = \frac{r \times (t_4 - t_1) - (t_3 - t_2)}{2} \tag{11-5}$$

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where

t_4 is pdelayRespEventIngressTimestamp (see 11.3.2.1) for the Pdelay_Resp message received in response to the Pdelay_Req message sent by the MD entity, in nanoseconds; the pdelayRespEventIngressTimestamp is equal to the timestamp value measured relative to the timestamp measurement plane, minus any ingressLatency (see 8.4.3)

t_1 is pdelayReqEventEgressTimestamp (see 11.3.2.1) for the Pdelay_Req message sent by the peer-to-peer Port entity, in nanoseconds

t_2 is the sum of (1) the ns field of the requestReceiptTimestamp, (2) the seconds field of the requestReceiptTimestamp multiplied by 10^9, (3) the correctionField multiplied by s (see 11.2.18.2.13) and then divided by 2^{16} (i.e., the correctionField is expressed in nanoseconds plus fractional nanoseconds), of the Pdelay_Resp message received in response to the Pdelay_Req message sent by the MD entity, and (4) –delayAsymmetry divided by 2^{16} if this state machine is invoked by CMLDS (i.e., delayAsymmetry is subtracted in the case of CMLDS)

t_3 is the sum of (1) the ns field of the responseOriginTimestamp, (2) the seconds field of the responseOriginTimestamp multiplied by 10^9, and (3) the correctionField divided by 2^{16} (i.e., the correctionField is expressed in nanoseconds plus fractional nanoseconds), of the Pdelay_Resp_Follow_Up message received in response to the Pdelay_Req message sent by the MD entity

r is the current value of nrrPdelay for this MD entity (see 11.2.12.14)

When CMLDS is used, Equation (11-5) can be rewritten as shown in Equation (11-6) (see NOTE 3 below) using the definitions of t_2 and t_3 above for the CMLDS case.

Propagation delay averaging may be performed, as described in 11.1.2 by Equation (11-2), Equation (11-3), and Equation (11-4). In this case, the successive values of propagation delay computed using Equation (11-5) are input to either Equation (11-2) or Equation (11-4), and the computed average propagation delay is returned by this function.

NOTE 1—Equation (11-5) defines D as the mean propagation delay relative to the time base of the time-aware system at the other end of the attached PTP Link. It is divided by nrrPdelay (see 11.2.12.14) to convert it to the time base of the current time-aware system when subtracting from syncEventIngressTimestamp to compute upstreamTxTime [see item i) in 11.2.13.2.1].

NOTE 2—The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to the time base of the time-aware system at the other end of the attached PTP Link is usually negligible. To see this, note that the former can be obtained from the latter by multiplying the latter by the ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity of the time-aware system at the other end of the PTP Link attached to this port. This ratio differs from 1 by 200 ppm or less. For example, for a worst-case frequency offset of the LocalClock entity of the time-aware system at the other end of the PTP Link, relative to the Grandmaster Clock, of 200 ppm, and a measured propagation time of 100 ns, the difference in D relative to the two time bases is 20 ps.

NOTE 3—In IEEE Std 1588-2019, the computation of meanLinkDelay is organized differently from the organization used for transport-specific peer delay in the present standard. Using the definitions of t_2 and t_3 above, Equation (11-5) can be rewritten as shown in Equation (11-6).

$$D = [r\times(t_4 - t_1) - (\text{responseOriginTimestamp} - \text{requestReceiptTimestamp}) + s\times(\text{correctionField of Pdelay_Resp}) - (\text{correctionField of Pdelay_Resp_Follow_Up}) - \text{delayAsymmetry}] / 2 \quad (11\text{-}6)$$

where each term is expressed in units of nanoseconds as described in the definitions of t_1, t_2, t_3, and t_4 above. In IEEE Std 1588-2019, the fractional nanoseconds portion of t_2 is subtracted from the correctionField of Pdelay_Resp, rather than added as in the present standard for transport-specific peer delay [see 11.2.19.3.1 d) 1)]; however, the correctionField of Pdelay_Resp is then subtracted in Equation (11-6) rather than added [in Equation (11-6) for transport-specific peer delay, where s=1], and the two minus signs result in the same sign for the fractional nanoseconds portion of t_2 and the seconds and nanoseconds portion of t_2 (requestReceiptTimestamp). The computations of D in this standard and IEEE Std 1588-2019 are mathematically equivalent. The organization of the computation with CMLDS needs to be used in the present standard for interoperability with IEEE Std 1588-2019 (see 11.2.16.2). This organization in the present standard, for the case where CMLDS is used, is consistent with the organization in IEEE Std 1588-2019.

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11.2.18.4 State diagram

The MDPdelayReq state machine shall implement the function specified by the state diagram in Figure 11-9, the local variables specified in 11.2.18.2, the functions specified in 11.2.18.3, the messages specified in 11.4, and the relevant global variables and functions specified in 11.2.12 and 11.2.17 and 10.2.4 through 10.2.6. This state machine is responsible for the following:

a) Sending Pdelay_Req messages and restarting the pdelayIntervalTimer.
b) Detecting that the peer mechanism is running.
c) Detecting if Pdelay_Resp and/or Pdelay_Resp_Follow_Up messages corresponding to a Pdelay_Req message sent are not received.
d) Detecting whether more than one Pdelay_Resp is received within one Pdelay_Req message transmission interval (see 11.5.2.2).
e) Computing propagation time on the attached PTP Link when Pdelay_Resp and Pdelay_Resp_Follow_Up messages are received.
f) Computing the ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the attached PTP Link to the frequency of the LocalClock entity of the current time-aware system.

NOTE 1—The ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the attached PTP Link to the frequency of the LocalClock entity of the current time-aware system, nrrPdelay, retains its most recent value when a Pdelay_Resp and/or Pdelay_Resp_Follow_Up message is lost.

NOTE 2—Normally, Pdelay_Resp should be received within a time interval after sending Pdelay_Req that is less than or equal to the Pdelay turnaround time plus twice the mean propagation delay. However, while receiving Pdelay_Resp after a time interval that is longer than this can result in worse time-synchronization performance (see 11.1.2 and B.2.3), the peer delay protocol will still operate. It is expected that a peer delay initiator can receive and process Pdelay_Resp and Pdelay_Resp_Follow_Up messages within a time interval after sending Pdelay_Req that is as large as the current Pdelay Req message transmission interval (see 11.5.2.2).

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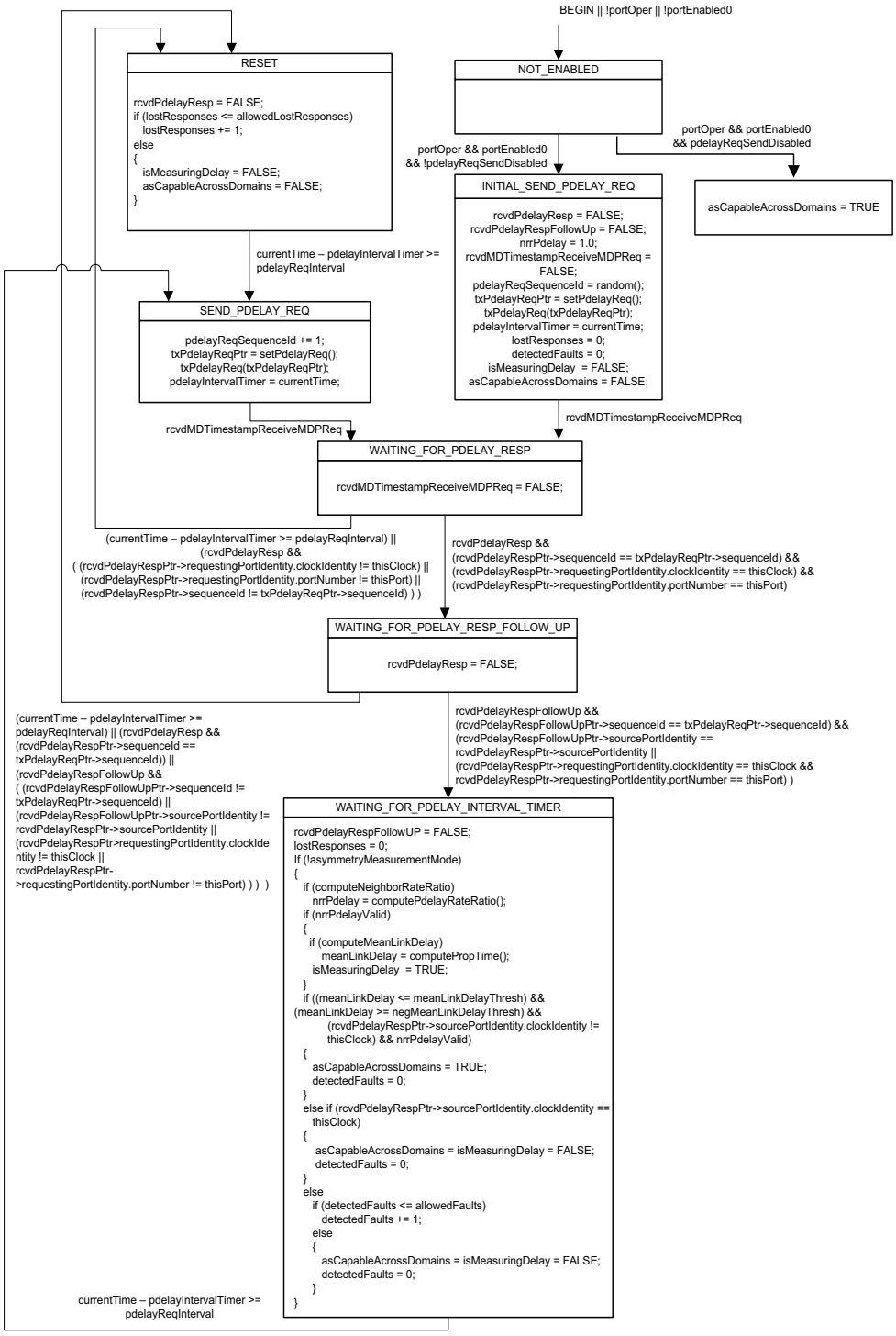


Figure 11-9—MDPdelayReq state machine

NOTE 3—A change in the value of the variable pdelayReqSendDisabled takes effect only when portEnabled0 (see 11.2.19.2.12) is FALSE.

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11.2.19 MDPdelayResp state machine

11.2.19.1 General

This state machine is either invoked as part of the CMLDS or by the transport-specific peer-to-peer delay mechanism. There is one instance of this state machine for all the domains (per port). As a result, there also is one instance of each of the state machine variables of 11.2.19.2, state machine functions of 11.2.19.3, and relevant global variables of 10.2.5, 11.2.12, and 11.2.17 for all the domains (per port). None of the variables used or functions invoked in this state machine are specific to a single domain. However, the single instances of all of these objects or entities are accessible to all the domains.

11.2.19.2 State machine variables

The following variables are used in the state diagram in Figure 11-10 (in 11.2.19.4):

11.2.19.2.1 rcvdPdelayReq: A Boolean variable that notifies the current state machine when a Pdelay_Req message is received. This variable is reset by the current state machine.

11.2.19.2.2 rcvdMDTimestampReceiveMDPResp: A Boolean variable that notifies the current state machine when the pdelayRespEventEgressTimestamp (see 11.3.2.1) for a transmitted Pdelay_Resp message is received. This variable is reset by the current state machine.

11.2.19.2.3 txPdelayRespPtr: A pointer to a structure whose members contain the values of the fields of a Pdelay_Resp message to be transmitted.

11.2.19.2.4 txPdelayRespFollowUpPtr: A pointer to a structure whose members contain the values of the fields of a Pdelay_Resp_Follow_Up message to be transmitted.

11.2.19.2.5 portEnabled1: A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to cmldsLinkPortEnabled (see 11.2.17.1) if this state machine is invoked by the CMLDS.

11.2.19.2.6 pdelayRespSendDisabled: A Boolean that is administratively set to TRUE to indicate that corresponding Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by this port in response to a Pdelay_Req message. The default value for this variable shall be FALSE.

11.2.19.3 State machine functions

11.2.19.3.1 setPdelayResp(): Creates a structure containing the parameters (see 11.4) of a Pdelay_Resp message to be transmitted, and returns a pointer, txPdelayRespPtr (see 11.2.19.2.3), to this structure. The parameters are set as follows:

a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity (see 8.5.2).

b) sequenceId is set equal to the sequenceId field of the corresponding Pdelay_Req message.

c) requestReceiptTimestamp is set equal to the pdelayReqEventIngressTimestamp (see 11.3.2) of the corresponding Pdelay_Req message, with any fractional nanoseconds portion truncated.

d) correctionField is set equal to the following:

   1) The fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req message if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and

   2) Minus the fractional nanoseconds portion of the pdelayReqEventIngressTimestamp of the corresponding Pdelay_Req message if this state machine is invoked by CMLDS.

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e) requestingPortIdentity is set equal to the sourcePortIdentity field of the corresponding Pdelay_Req message.

f) The remaining parameters are set as specified in 11.4.2 and 11.4.6.

11.2.19.3.2 txPdelayResp(txPdelayRespPtr): Transmits a Pdelay_Resp message from the MD entity, containing the parameters in the structure pointed to by txPdelayRespPtr (see 11.2.19.2.3).

11.2.19.3.3 setPdelayRespFollowUp(): Creates a structure containing the parameters (see 11.4) of a Pdelay_Resp_Follow_Up message to be transmitted, and returns a pointer, txPdelayRespFollowUpPtr (see 11.2.19.2.4), to this structure. The parameters are set as follows:

a) sourcePortIdentity is set equal to the port identity of the port corresponding to this MD entity (see 8.5.2).

b) sequenceId is set equal to the sequenceId field of the corresponding Pdelay_Req message.

c) responseOriginTimestamp is set equal to the pdelayRespEventEgressTimestamp (see 11.3.2) of the corresponding Pdelay_Resp message, with any fractional nanoseconds truncated.

d) correctionField is set equal to the following:

1) The fractional nanoseconds portion of the pdelayRespEventEgressTimestamp of the corresponding Pdelay_Resp message if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and

2) The sum of the correctionField of the corresponding Pdelay_Req message and the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp of the corresponding Pdelay_Resp message if this state machine is invoked by CMLDS.

e) requestingPortIdentity is set equal to the sourcePortIdentity field of the corresponding Pdelay_Req message.

f) The remaining parameters are set as specified in 11.4.2 and 11.4.6.

11.2.19.3.4 txPdelayRespFollowUp(txPdelayRespFollowUpPtr): Transmits a Pdelay_Resp_Follow_Up message from the peer-to-peer Port entity containing the parameters in the structure pointed to by txPdelayRespFollowUpPtr (see 11.2.19.2.4).

11.2.19.4 State diagram

The MDPdelayResp state machine shall implement the function specified by the state diagram in Figure 11-10, the local variables specified in 11.2.19.2, the functions specified in 11.2.19.3, the messages specified in 11.4, and the relevant global variables and functions specified in 10.2.4 through 10.2.6, 11.2.12, and 11.2.17. This state machine is responsible for responding to Pdelay_Req messages, received from the MD entity at the other end of the attached PTP Link, with Pdelay_Resp and Pdelay_Resp_Follow_Up messages.

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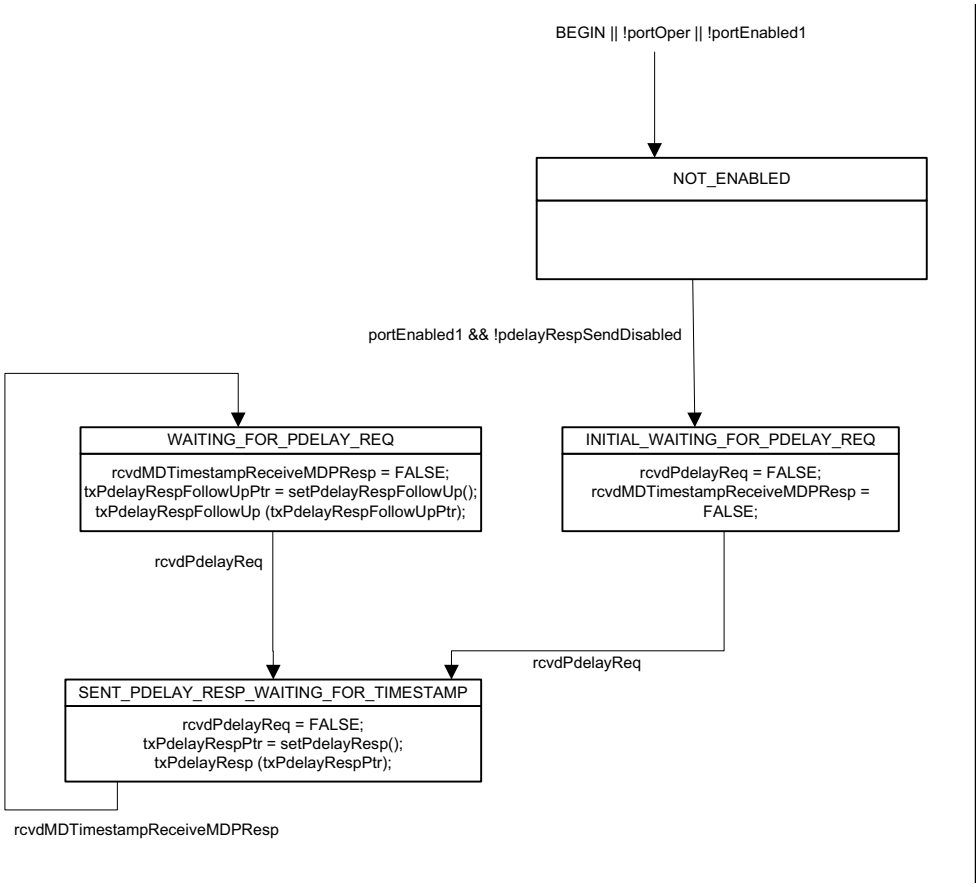


Figure 11-10—MDPdelayResp state machine

NOTE—A change in the value of the variable pdelayRespSendDisabled takes effect only when portEnabled1 (see 11.2.20.2.5) is FALSE.

11.2.20 LinkDelayIntervalSetting state machine

11.2.20.1 General

This state machine is either invoked as part of the CMLDS or by the transport-specific peer-to-peer delay mechanism. There is one instance of this state machine per port, for the common service of the time-aware system.

11.2.20.2 State machine variables

The following variables are used in the state diagram in Figure 11-11 (in 11.2.20.4):

11.2.20.2.1 rcvdSignalingMsg1: A Boolean variable that notifies the current state machine when a Signaling message that contains a message interval request TLV (see 10.6.4.3) is received. This variable is reset by the current state machine.

11.2.20.2.2 rcvdSignalingPtrLDIS: A pointer to a structure whose members contain the values of the fields of the received Signaling message that contains a message interval request TLV (see 10.6.4.3).

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11.2.20.2.3 portEnabled3: A Boolean variable whose value is equal to ptpPortEnabled (see 10.2.5.13) if this state machine is invoked by the transport-specific peer-to-peer delay mechanism and is equal to cmldsLinkPortEnabled (see 11.2.17.1) if this state machine is invoked by the CMLDS.

11.2.20.2.4 logSupportedPdelayReqIntervalMax: The maximum supported logarithm to base 2 of the Pdelay_Req interval. The data type for logSupportedPdelayReqIntervalMax is Integer8.

11.2.20.2.5 logSupportedClosestLongerPdelayReqInterval: The logarithm to base 2 of the Pdelay_Req interval, such that logSupportedClosestLongerPdelayReqInterval > logRequestedPdelayReqInterval, that is numerically closest to logRequestedPdelayReqInterval, where logRequestedPdelayReqInterval is the argument of the function computeLogPdelayReqInterval() (see 11.2.20.3.2). The data type for logSupportedClosestLongerPdelayReqInterval is Integer8.

11.2.20.2.6 computedLogPdelayReqInterval: A variable used to hold the result of the function computeLogPdelayReqInterval(). The data type for computedLogPdelayReqInterval is Integer8.

11.2.20.2.7 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 11-11). The data type for TEMP is Integer16.

11.2.20.3 State machine functions

11.2.20.3.1 isSupportedLogPdelayReqInterval (logPdelayReqInterval): A Boolean function that returns TRUE if the Pdelay_Req interval given by the argument logPdelaytReqInterval is supported by the PTP Port and FALSE if the Pdelay_Req interval is not supported by the PTP Port. The argument logPdelayReqInterval has the same data type and format as the field logLinkDelayInterval of the message interval request TLV (see 10.6.4.3.6).

11.2.20.3.2 computeLogPdelayReqInterval (logRequestedPdelayReqInterval): An Integer8 function that computes and returns the logPdelayReqInterval, based on the logRequestedPdelayReqInterval. This function is defined as indicated below. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
Integer8 computeLogPdelayReqInterval (logRequestedPdelayReqInterval)
Integer8 logRequestedPdelayReqInterval;
{
        Integer8 logSupportedPdelayReqIntervalMax,
                                logSupportedClosestLongerPdelayReqInterval;
        if (isSupportedLogPdelayReqInterval (logRequestedPdelayReqInterval))
                // The requested Pdelay_Req Interval is supported and returned
                return (logRequestedPdelayReqInterval)
        else
        {
                if (logRequestedPdelayReqInterval > logSupportedPdelayReqIntervalMax)
                        // Return the largest supported logPdelayReqInterval, even if smaller than
the requested interval
                        return (logSupportedPdelayReqIntervalMax);
                else
                        // Return the smallest supported logPdelayReqInterval that is still larger
than
                        // the requested interval.
                        return (logSupportedClosestLongerPdelayReqInterval);
        }
}
```


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11.2.20.4 State diagram

The LinkDelayIntervalSetting state machine shall implement the function specified by the state diagram in Figure 11-11, the local variables specified in 11.2.20.2, the functions specified in 11.2.20.3, the messages specified in 10.6 and 11.4, the relevant global variables specified in 10.2.5, 11.2.12, and 11.2.17, the relevant managed objects specified in 14.10 and 14.14, and the relevant timing attributes specified in 10.7 and 11.5. This state machine is responsible for setting the global variables that give the duration of the mean interval between successive Pdelay_Req messages and also the global variables that control whether meanLinkDelay and nrrPdelay are computed, both at initialization and in response to the receipt of a Signaling message that contains a message interval request TLV (see 10.6.4.3).

NOTE—A Signaling message received by this state machine that carries the message interval request TLV (see 10.6.4.3) is ignored if multiple profiles are present and the Signaling message is directed to the CMLDS (i.e., if the value of sdoId of the Signaling message is 0x200).

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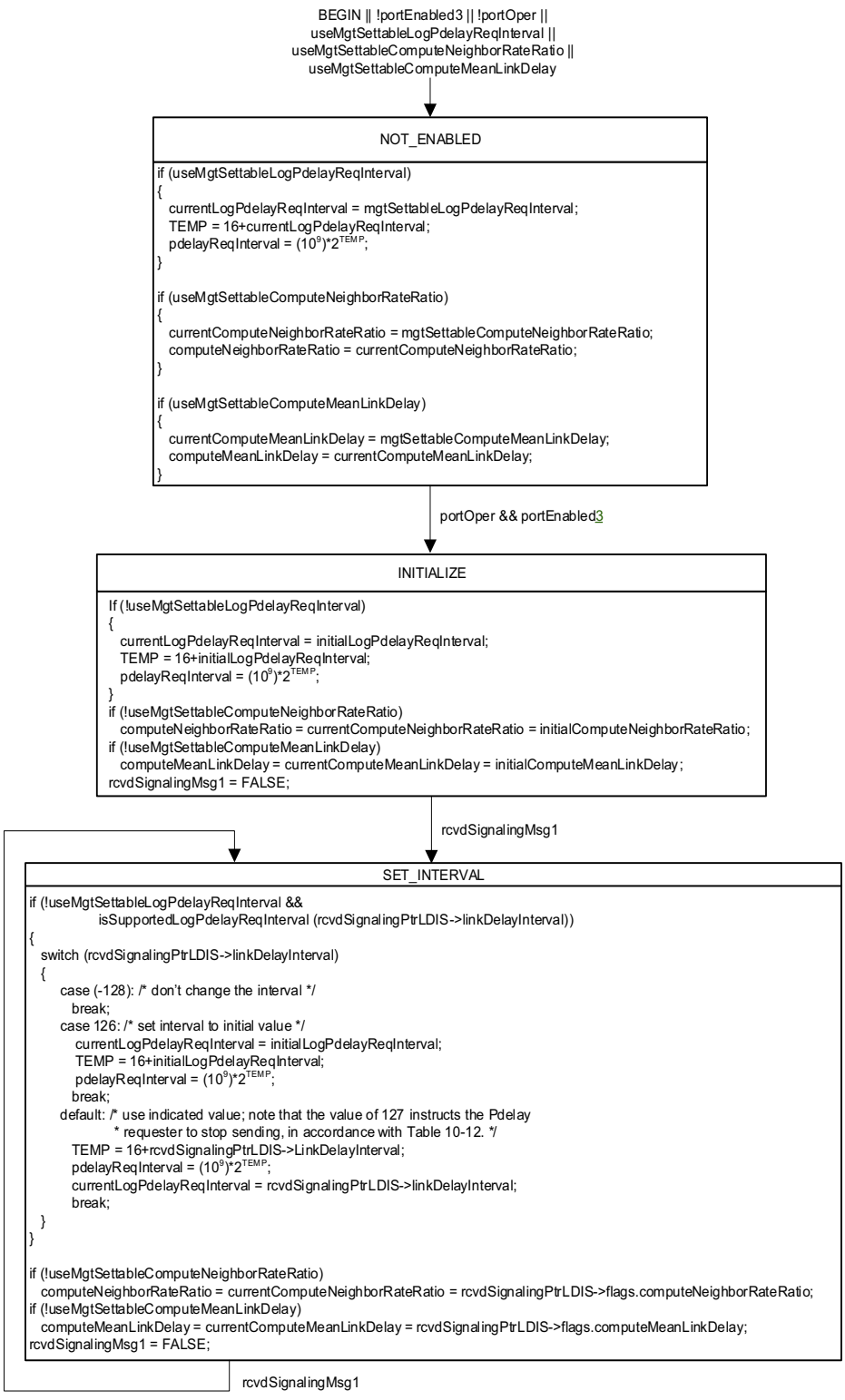


Figure 11-11—LinkDelayIntervalSetting state machine

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11.3 Message attributes

11.3.1 General

This subclause describes attributes of the Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages that are not described in 8.4.2.

11.3.2 Message types contained in each message class

11.3.2.1 Event message class

The event message class contains the following message types:

a) Sync: A Sync message contains time-synchronization information that originates at a ClockTimeTransmitter entity. The appearance of a Sync message at the reference plane of the PTP Port corresponding to an MD entity is an event to which the LocalClock assigns a timestamp, the syncEventIngressTimestamp or syncEventEgressTimestamp, based on the time of the LocalClock. The syncEventIngressTimestamp and syncEventEgressTimestamp are calculated based on the measured timestamp at the timestamp measurement plane and the ingress and egress latencies, respectively (see 8.4.3). The Sync message is followed by a Follow_Up message containing synchronization information that is based in part on the sum of the syncEventEgressTimestamp (see 8.4.3).

b) Pdelay_Req: A Pdelay_Req message is transmitted by an MD entity to another MD entity as part of the peer-to-peer delay mechanism (see 11.2.18 and Figure 11-9) to determine the delay on the PTP Link between them. The appearance of a Pdelay_Req message at the reference plane of the port of an MD entity is an event to which the LocalClock assigns a timestamp, the pdelayReqEventIngressTimestamp or pdelayReqEventEgressTimestamp, based on the time of the LocalClock. The pdelayReqEventIngressTimestamp and pdelayReqEventEgressTimestamp are measured relative to the timestamp measurement plane; the MD entity corrects them for ingress and egress latencies, respectively (see 8.4.3).

c) Pdelay_Resp: A Pdelay_Resp message is transmitted by an MD entity to another MD entity in response to the receipt of a Pdelay_Req message. The Pdelay_Resp message contains the pdelayReqEventIngressTimestamp of the Pdelay_Req message to which it is transmitted in response. The appearance of a Pdelay_Resp message at the reference plane of the port of an MD entity is an event to which the LocalClock assigns a timestamp, the pdelayRespEventIngressTimestamp or pdelayRespEventEgressTimestamp, based on the time of the LocalClock. The pdelayRespEventIngressTimestamp and pdelayRespEventEgressTimestamp are measured relative to the timestamp measurement plane; the MD entity corrects them for ingress and egress latencies, respectively (see 8.4.3). The Pdelay_Resp message is followed by a Pdelay_Resp_Follow_Up message containing the sum of the pdelayRespEventEgressTimestamp and any egressLatency (see 8.4.3).

Event messages shall be assigned the timestamps previously defined, in accordance with 8.4.3.

11.3.2.2 General message class

The general message class contains the following message types:

a) Follow_Up: A Follow_Up message communicates the value of the syncEventEgressTimestamp for the associated Sync message.

b) Pdelay_Resp_Follow_Up: A Pdelay_Resp_Follow_Up message communicates the value of the PdelayRespEventEgressTimestamp for the associated Pdelay_Resp message.

General messages are not required to be timestamped.

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11.3.3 VLAN tag

A frame that carries an IEEE 802.1AS message shall not have a VLAN tag nor a priority tag (see 3.190 of IEEE Std 802.1Q-2022).

11.3.4 Addresses

The destination address of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages transmitted shall be the reserved multicast address given in Table 11-3.

Table 11-3—Destination address for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages

Destination address
01-80-C2-00-00-0E
NOTE—This address is taken from Table 8-1, Table 8-2, and Table 8-3 of IEEE Std 802.1Q-2020.

All Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages transmitted shall use the MAC address of the respective egress physical port as the source address.

11.3.5 EtherType

The EtherType of Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages transmitted shall be the EtherType given in Table 11-4.

Table 11-4—EtherType for Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages

EtherType
88-F7

11.3.6 Subtype

The subtype for the Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages is indicated by the majorSdoId field (see 10.6.2.2.1).

11.3.7 Source port identity

The Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages each contain a sourcePortIdentity field (see 11.4.2.7) that identifies the egress port (see 8.5) on which the respective message is sent.

11.3.8 Sequence number

Each MD entity shall maintain a separate sequenceId pool for each of the message types Sync and Pdelay_Req.

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Each Sync and Pdelay_Req message contains a sequenceId field (see 11.4.2.8), which carries the message sequence number. The sequenceId of a Sync message shall be one greater than the sequenceId of the previous Sync message sent by the transmitting port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field. The sequenceId of a Pdelay_Req message shall be one greater than the sequenceId of the previous Pdelay_Req message sent by the transmitting port, subject to the constraints of the rollover of the UInteger16 data type used for the sequenceId field.

Separate pools of sequenceId are not maintained for the following message types:

a) Pdelay_Resp
b) Follow_Up
c) Pdelay_Resp_Follow_Up

For these exceptions, the sequenceId value is specified in Table 11-7 (in 11.4.2.8).

11.3.9 Event message timestamp point

The message timestamp point for a PTP event message shall be the beginning of the first symbol following the start of frame delimiter.

11.4 Message formats

11.4.1 General

The PTP messages Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up shall each have a header, body, and if present, a suffix that contains one or more TLVs (see 10.6.2, 11.4.3, 11.4.4, 11.4.5, 11.4.6, and 11.4.7 of this standard and Clause 14 of IEEE Std 1588-2019). Reserved fields shall be transmitted with all bits of the field 0 and ignored by the receiver, unless otherwise specified. The data type of the field shall be the type indicated in brackets in the title of each subclause.

Subclause 11.4.4.3 defines the Follow_Up information TLV, which is carried by the Follow_Up message if the corresponding Sync message is two-step (i.e., twoStepFlag of the Sync message is TRUE; see 10.6.2.2.8) and by the Sync message if the message is one-step (i.e., twoStepFlag is FALSE). The Follow_Up information TLV is the first TLV of the Follow_Up message or Sync message. The requirements for the parsing and forwarding of TLVs are given in 10.6.1.

NOTE—The standard Ethernet header and FCS (18 bytes total) must be added to each message.

11.4.2 Header

11.4.2.1 General

The common header for the PTP messages Sync, Follow_Up, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up shall be as specified in 10.6.2, except as noted in the following subclauses.

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11.4.2.2 messageType (Enumeration4)

The value indicates the type of the message, as defined in Table 11-5.

Table 11-5—Values for messageType field

Message type	Message class	Value
Sync	Event	0x0
Pdelay_Req	Event	0x2
Pdelay_Resp	Event	0x3
Follow_Up	General	0x8
Pdelay_Resp_Follow_Up	General	0xA
NOTE—Other values for the messageType field, except for 0xB that is used for the Announce message and 0xC that is used for the Signaling message (see 10.6.2.2.2), are not used in this standard.		

The most significant bit of the messageType field divides this field in half between event and general messages, i.e., it is 0 for event messages and 1 for general messages.

NOTE—The reserved nibble immediately following messageType is reserved for future expansion of the messageType field.

11.4.2.3 messageLength (UInteger16)

The value is the total number of octets that form the PTP message. The counted octets start with and include the first octet of the header and terminate with and include the last octet of the last TLV or, if there are no TLVs, with the last octet of the message as defined in this subclause.

NOTE—See 10.6.2.2.5 for an example.

11.4.2.4 domainNumber (UInteger8)

For FDE, Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted with domainNumber set to 0. Any received Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up message with a domainNumber other than 0 shall be ignored. For HDE, see 19.2.18 and 19.2.19. The domainNumber for all other PTP messages is as specified in 10.6.2.2.6.

11.4.2.5 flags (Octet2)

The value of the bits of the array are defined in Table 10-9.

11.4.2.6 correctionField (Integer64)

The correctionField is the value of the correction as specified in Table 11-6, measured in nanoseconds and multiplied by 2^{16}. For example, 2.5 ns is represented as 0x0000000000028000.

A value of one in all bits of the field, except the most significant, indicates that the correction is too big to be represented.

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Table 11-6—Value of correctionField

Message type	Value
Follow_Up Sync (sent by a one-step PTP Port; see 11.1.3 and 11.2.12.10)	Corrections for fractional nanoseconds (see 10.2.9 and Figure 10-5), difference between preciseOriginTimestamp field (if sent by a two-step PTP Port) or originTimestamp field (if sent by a one-step PTP Port) and current synchronized time (see 11.2.14.2.3 and Figure 11-7), and asymmetry (see 8.3, 11.2.13.2.1, and 11.2.14.2.3. The quantity delayAsymmetry is used in the computation of upstreamTxTime in 11.2.13.2.1, and upstreamTxTime is used in computing an addition to the correctionField in 11.2.14.2.3).
Pdelay_Resp, Pdelay_Resp_Follow_Up	Corrections for fractional nanoseconds (see Figure 11-9 and Figure 11-10) if the message is sent by the transport-specific peer-to-peer delay mechanism; or For Pdelay_Resp, minus the corrections for fractional nanoseconds (see 11.2.19.3.1, Figure 11-9, and Figure 11-10), and for Pdelay_Resp_Follow_Up, the sum of the correctionField of the corresponding Pdelay_Req message and the fractional nanoseconds portion of the pdelayRespEventEgressTimestamp of the corresponding Pdelay_Resp message, if this state machine is invoked by CMLDS.
Sync (sent by a two-step PTP Port), Pdelay_Req, Announce, Signaling	The value is 0 (see 10.6.2.2.9) if the message is sent by the transport-specific peer-to-peer delay mechanism, or The value is 0 for Sync (sent by a two-step PTP Port), Announce, and Signaling, and –delayAsymmetry for Pdelay_Req (see 11.2.18.3.1), if the message is sent by CMLDS.
NOTE—IEEE Std 1588-2019 describes asymmetry corrections for the Pdelay_Req and Pdelay_Resp messages. However, the peer-to-peer delay mechanism computes the mean propagation delay. Here, where the gPTP communication path is a FDE PTP Link, these corrections cancel in the mean propagation delay computation and therefore are not needed.	

11.4.2.7 sourcePortIdentity (PortIdentity)

The value is the portIdentity of the egress port (see 8.5.2) on which the respective message is sent.

11.4.2.8 sequenceId (UInteger16)

The value is assigned by the originator of the message in conformance with 11.3.8, except for Follow_Up, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages. The sequenceId field values for these exceptions are defined in the state diagrams given in the figures referenced in Table 11-7.

Table 11-7—References for sequenceId value exceptions

Message type	Reference
Follow_Up	See 11.2.14 and Figure 11-7
Pdelay_Resp	See 11.2.19 and Figure 11-10
Pdelay_Resp_Follow_Up	See 11.2.19 and Figure 11-10

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11.4.2.9 logMessageInterval (Integer8)

For Sync and Follow_Up messages, the value is the value of currentLogSyncInterval (see 10.2.5.4 and 10.7.2.3). For Pdelay_Req messages, the value is the value of currentLogPdelayReqInterval. For Pdelay_Resp and Pdelay_Resp_Follow_Up messages, the value is transmitted as 0x7F and ignored on reception.

11.4.3 Sync message

11.4.3.1 General Sync message specifications

If the twoStep flag of the PTP common header (see Table 10-9) of the Sync message is TRUE, the fields of the Sync message shall be as specified in Table 11-8. If the twoStep flag of the PTP common header of the Sync message is FALSE, the fields of the Sync message shall be as specified in Table 11-9 and 11.4.3.2. Carrying the Drift_Tracking TLV is optional.

Table 11-8—Sync message fields if twoStep flag is TRUE

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
reserved								10	34

Table 11-9—Sync message fields if twoStep flag is FALSE

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
originTimestamp								10	34
Follow_Up information TLV								32	44
Drift_Tracking TLV (optional)								36	76

11.4.3.2 Sync message field specifications if twoStep flag is FALSE

11.4.3.2.1 originTimestamp (Timestamp)

The value of the originTimestamp field is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP Instance, when the Sync message was sent by that Grandmaster PTP Instance, with any fractional nanoseconds truncated (see 10.2.9).

The sum of the correctionField and the originTimestamp field of the Sync message is the value of the synchronized time corresponding to the syncEventEgressTimestamp at the PTP Instance that sent the Sync message, including any fractional nanoseconds.

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11.4.3.2.2 Follow_Up information TLV

The Sync message carries the Follow_Up information TLV, defined in 11.4.4.3. This TLV shall be the first TLV after the fixed fields.

11.4.3.2.3 Drift_Tracking TLV

The Sync message may carry the Drift_Tracking TLV, defined in 11.4.4.4.

11.4.4 Follow_Up message

11.4.4.1 General Follow_Up message specifications

The fields of the Follow_Up message shall be as specified in Table 11-10 and 11.4.4.2. Carrying the Drift_Tracking TLV is optional.

Table 11-10—Follow_Up message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
preciseOriginTimestamp								10	34
Follow_Up information TLV								32	44
Drift_Tracking TLV (optional)								36	76

11.4.4.2 Follow_Up message field specifications

11.4.4.2.1 preciseOriginTimestamp (Timestamp)

The value of the preciseOriginTimestamp field is the sourceTime of the ClockTimeTransmitter entity of the Grandmaster PTP Instance, when the associated Sync message was sent by that Grandmaster PTP Instance, with any fractional nanoseconds truncated (see 10.2.9).

The sum of the correctionFields in the Follow_Up and associated Sync messages, added to the preciseOriginTimestamp field of the Follow_Up message, is the value of the synchronized time corresponding to the syncEventEgressTimestamp at the PTP Instance that sent the associated Sync message, including any fractional nanoseconds.

11.4.4.2.2 Follow_Up information TLV

The Follow_Up message carries the Follow_Up information TLV, defined in 11.4.4.3. This TLV shall be the first TLV after the fixed fields.

11.4.4.2.3 Drift_Tracking TLV

The Follow_Up message may carry the Drift_Tracking TLV, defined in 11.4.4.4.

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11.4.4.3 Follow_Up information TLV definition

11.4.4.3.1 General

The fields of the Follow_Up information TLV shall be as specified in Table 11-11 and in 11.4.4.3.2 through 11.4.4.3.9. This TLV is a standard organization extension TLV for the Follow_Up message, as specified in 14.3 of IEEE Std 1588-2019.

Table 11-11—Follow_Up information TLV

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
cumulativeScaledRateOffset								4	10
gmTimeBaseIndicator								2	14
lastGmPhaseChange								12	16
scaledLastGmFreqChange								4	28

NOTE—The Follow_Up information TLV is different from the CUMULATIVE_RATE_RATIO TLV of IEEE Std 1588-2019 (see 16.10 and Table 52 of IEEE Std 1588-2019). While both TLVs carry cumulative rate offset information, the Follow_Up information TLV also carries information on the Grandmaster Clock time base, most recent phase change, and most recent frequency change. The CUMULATIVE_RATE_RATIO TLV is not used by gPTP.

11.4.4.3.2 tlvType (Enumeration16)

The value of the tlvType field is 0x3.

NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a value of 0x3.

11.4.4.3.3 lengthField (UInteger16)

The value of the lengthField is 28.

11.4.4.3.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

11.4.4.3.5 organizationSubType (Enumeration24)

The value of organizationSubType is 1.

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11.4.4.3.6 cumulativeScaledRateOffset (Integer32)

The value of cumulativeScaledRateOffset is equal to (rateRatio – 1.0) × (2^{41}), truncated to the next smaller signed integer, where rateRatio is the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity in the PTP Instance that sends the message.

NOTE—The above scaling allows the representation of fractional frequency offsets in the range [–(2^{-10} – 2^{-41}), 2^{-10} –2^{-41}], with granularity of 2^{-41}. This range is approximately [–9.766 × 10^{-4}, 9.766 × 10^{-4}].

11.4.4.3.7 gmTimeBaseIndicator (UInteger16)

The value of gmTimeBaseIndicator is the timeBaseIndicator of the ClockSource entity for the current Grandmaster PTP Instance (see 9.2.2.3).

NOTE—The timeBaseIndicator is supplied by the ClockSource entity to the ClockTimeTransmitter entity via the ClockSourceTime.invoke function (see 9.2.2.3).

11.4.4.3.8 lastGmPhaseChange (ScaledNs)

The value of lastGmPhaseChange is the time of the current Grandmaster Clock minus the time of the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance. The value is copied from the lastGmPhaseChange member of the MDSyncSend structure whose receipt causes the MD entity to send the Follow_Up message (see 11.2.10).

11.4.4.3.9 scaledLastGmFreqChange (Integer32)

The value of scaledLastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator, multiplied by 2^{41} and truncated to the next smaller signed integer. The value is obtained by multiplying the lastGmFreqChange member of MDSyncSend whose receipt causes the MD entity to send the Follow_Up message (see 11.2.10) by 2^{41}, and truncating to the next smaller signed integer.

NOTE—The above scaling allows the representation of fractional frequency offsets in the range [–(2^{-10} – 2^{-41}), 2^{-10} – 2^{-41}], with granularity of 2^{-41}. This range is approximately [–9.766 × 10^{-4}, 9.766 × 10^{-4}].

11.4.4.4 Drift_Tracking TLV definition

11.4.4.4.1 General

The fields of the Drift_Tracking TLV shall be as specified in Table 11-12 and in 11.4.4.4.2 through 11.4.4.4.9. This TLV is a standard organization extension TLV for the Sync or Follow_Up message, as specified in 14.3 of IEEE Std 1588-2019.

11.4.4.4.2 tlvType (Enumeration16)

The value of the tlvType is 0x3.

NOTE—This value indicates the TLV is a vendor and standard-organization extension TLV, as specified in 14.3.2.1 and Table 52 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a value of 0x3.

11.4.4.4.3 lengthField (UInteger16)

The value of lengthField is 32.

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Table 11-12—Drift_Tracking TLV

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
syncEgressTimestamp								12	10
syncGrandmasterIdentity								8	22
syncStepsRemoved								2	30
rateRatioDrift								4	32

11.4.4.4.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

11.4.4.4.5 organizationSubType (Enumeration24)

The value of organizationSubType is 6.

11.4.4.4.6 syncEgressTimestamp (ExtendedTimestamp)

The value of the syncEgressTimestamp field is the timestamp, based on the local clock, when the Sync message was sent by that PTP Instance (see 11.4.3.2).

11.4.4.4.7 syncGrandmasterIdentity (ClockIdentity)

The value of syncGrandmasterIdentity as determined by the MDSyncSend state machine (see 11.2.14.2.3 and 11.2.14.2.5).

11.4.4.4.8 syncStepsRemoved (UInteger16)

The value of syncStepsRemoved as determined by the MDSyncSend state machine (see 11.2.14.2.3 and 11.2.14.2.5).

11.4.4.4.9 rateRatioDrift (Integer32)

The value of rateRatioDrift is equal to (RRdrift – 1.0) × (2^{41}), truncated to the next smaller signed integer, where RRdrift is the measured estimate of the rate of change per second of the ratio of the frequency of the Grandmaster Clock to the frequency of the local clock entity in the PTP Instance that sends the message.

NOTE—The above scaling allows the representation of rates of change of fractional frequency offset in the range [–($2^{-10} – 2^{-41}$), $2^{-10} – 2^{-41}$] s^{-1}, with granularity of 2^{-41}. This range is approximately [$–9.766 \times 10^{-4}$, 9.766×10^{-4}] s^{-1}.

11.4.5 Pdelay_Req message

The fields of the Pdelay_Req message shall be as specified in Table 11-13.

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Table 11-13—Pdelay_Req message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
reserved								10	34
reserved								10	44

11.4.6 Pdelay_Resp message

11.4.6.1 General Pdelay_Resp message specifications

The fields of the Pdelay_Resp message shall be as specified in Table 11-14 and 11.4.6.2.

Table 11-14—Pdelay_Resp message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
requestReceiptTimestamp								10	34
requestingPortIdentity								10	44

11.4.6.2 Pdelay_Resp message field specifications

11.4.6.2.1 requestReceiptTimestamp (Timestamp)

The value is the seconds and nanoseconds portion of the pdelayReqEventIngressTimestamp of the associated Pdelay_Req message (see 11.2.18).

11.4.6.2.2 requestingPortIdentity (PortIdentity)

The value is the value of the sourcePortIdentity field of the associated Pdelay_Req message (see 11.4.5).

11.4.7 Pdelay_Resp_Follow_Up message

11.4.7.1 General Pdelay_Resp_Follow_Up message specifications

The fields of the Pdelay_Resp_Follow_Up message shall be as specified in Table 11-15 and 11.4.7.2.

11.4.7.2 Pdelay_Resp_Follow_Up message field specifications

11.4.7.2.1 responseOriginTimestamp (Timestamp)

The value is the seconds and nanoseconds portion of the pdelayRespEventEgressTimestamp of the associated Pdelay_Resp message (see 11.4.6.2).

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Table 11-15—Pdelay_Resp_Follow_Up message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (see 11.4.2)								34	0
responseOriginTimestamp								10	34
requestingPortIdentity								10	44

11.4.7.2.2 requestingPortIdentity (PortIdentity)

The value is the value of the sourcePortIdentity field of the associated Pdelay_Req message (see 11.4.5).

11.5 Protocol timing characterization

11.5.1 General

This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

11.5.2 Message transmission intervals

11.5.2.1 General interval specification

The mean time interval between successive Pdelay_Req messages is represented as the logarithm to the base 2 of this time interval measured in seconds. The value of this logarithmic attribute shall be as specified in 11.5.2.2.

The mean time interval between successive Sync messages shall be as specified in 10.7.2.1, 10.7.2.3, and 11.5.2.3.

11.5.2.2 Pdelay_Req message transmission interval

When useMgtSettableLogPdelayReqInterval (see 14.18.13) is FALSE, the initialLogPdelayReqInterval specifies the following:

a) The mean time interval between successive Pdelay_Req messages sent over a PTP Link when the port is initialized, and

b) The value to which the mean time interval between successive Pdelay_Req messages is set when a message interval request TLV is received with the logLinkDelayIntervalField set to 126 (see 11.2.20).

The currentLogPdelayReqInterval specifies the current value of the mean time interval between successive Pdelay_Req messages. The default value of initialLogPdelayReqInterval is 0. Every port supports the value 127; the port does not send Pdelay_Req messages when currentLogPdelayReqInterval has this value (see 11.2.20). A port may support other values, except for the reserved values indicated in Table 10-15. A port shall ignore requests for unsupported values (see 11.2.20). The initialLogPdelayReqInterval and currentLogPdelayReqInterval are per-port attributes.

When useMgtSettableLogPdelayReqInterval is TRUE, currentLogPdelayReqInterval is set equal to mgtSettableLogPdelayReqInterval (see 14.18.14), and initialLogPdelayReqInterval is ignored.

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Pdelay_Req messages shall be transmitted such that the value of the arithmetic mean of the intervals, in seconds, between Pdelay_Req message transmissions is not less than the value of $0.9 \times 2^{\text{currentLogPdelayReqInterval}}$.

NOTE 1—A minimum number of inter-message intervals is necessary to verify that a PTP Port meets these requirements. The arithmetic mean is the sum of the inter-message interval samples divided by the number of samples. For more detailed discussion of statistical analyses, see Papoulis [B35].

NOTE 2—If useMgtSettableLogPdelayReqInterval is FALSE, the value of initialLogPdelayReqInterval is the value of the mean time interval between successive Pdelay_Req messages when the port is initialized. The value of the mean time interval between successive Pdelay_Req messages can be changed, e.g., if the port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in currentLogPdelayReqInterval. The value of the mean time interval between successive Pdelay_Req messages can be reset to the initial value, e.g., by a message interval request TLV for which the value of the field logLinkDelayInterval is 126 (see 10.6.4.3.6).

NOTE 3—A port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and 11.2.20) that the port at the other end of the attached PTP Link set its currentLogPdelayReqInterval to a specific value can determine if the request was honored by examining the logMessageInterval field of subsequent received Pdelay_Req messages.

NOTE 4—The MDPdelayReq state machine ensures that the times between transmission of successive Pdelay_Req messages, in seconds, are not smaller than $2^{\text{currentLogPdelayReqInterval}}$. This is consistent with the requirements of the current subclause and IEEE Std 1588-2019, which require that the value of the arithmetic mean of the intervals, in seconds, between Pdelay_Req message transmissions is not less than the value of $0.9 \times 2^{\text{currentLogPdelayReqInterval}}$.The sending of Pdelay_Req messages is governed by the LocalClock and not the synchronized time (i.e., the estimate of the Grandmaster Clock time). Since the LocalClock frequency can be slightly larger than the Grandmaster Clock frequency (e.g., by 100 ppm, which is the specified frequency accuracy of the LocalClock; see B.1.1), it is possible for the time intervals between successive Pdelay_Req messages to be slightly less than $2^{\text{currentLogPdelayReqInterval}}$ when measured relative to the synchronized time. The factor 0.9 allows a margin of 10% for the arithmetic mean of the successive intervals between Pdelay_Req messages.

11.5.2.3 Sync message transmission interval default value

The default value of initialLogSyncInterval (see 10.7.2.3) is –3. Every PTP Port supports the value 127; the PTP Port does not send Sync messages when currentLogSyncInterval has this value (see 10.2.14). A PTP Port may support other values, except for the reserved values indicated in Table 10-16. A PTP Port ignores requests for unsupported values (see 10.2.14).

NOTE—A PTP Port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and 10.2.14) that the PTP Port at the other end of the attached PTP Link set its currentLogSyncInterval to a specific value can determine if the request was honored by examining the logMessageInterval field of subsequent received Sync messages.

11.5.3 allowedLostResponses

The variable allowedLostResponses (see 11.2.12.4) is the number of Pdelay_Req messages without valid responses above which a port is considered to be not exchanging peer delay messages with its neighbor. The default value of allowedLostResponses shall be 9. The range shall be 1 through 255.

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11.5.4 allowedFaults

The variable allowedFaults (see 11.2.12.5) is the number of faults above which asCapableAcrossDomains is set to FALSE, i.e., the port is considered not capable of interoperating with its neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). In this context, the term *faults* refers to instances where:

a) The computed mean propagation delay, i.e., meanLinkDelay (see 10.2.5.8), exceeds the threshold, meanLinkDelayThresh (see 11.2.12.7), or is below the negMeanLinkDelayThresh (see 11.2.12.8), and/or

b) The computation of nrrPdelay is invalid (see 11.2.18.2.10).

The default value of allowedFaults shall be 9. The range shall be 1 through 255.

NOTE—The above description of allowedFaults uses the variable asCapableAcrossDomains (see 11.2.12.13). When only one domain is active, asCapableAcrossDomains is equivalent to the variable asCapable.

11.6 Control of computation of neighborRateRatio

If the global variable driftTrackingTlvSupport (see 10.2.4.28) is TRUE and nrrCompMethod (see 11.2.12.16) is equal to Sync, the value of neighborRateRatio (see 10.2.5.7) is set equal to nrrSync (see 11.2.12.15).

If the global variable driftTrackingTlvSupport (see 10.2.4.28) is FALSE or nrrCompMethod (see 11.2.12.16) is equal to Pdelay, the value of neighborRateRatio (see 10.2.5.7) is set equal to nrrPdelay (see 11.2.12.14). The computation of nrrPdelay is controlled as described below. The description below applies only to the computation of nrrPdelay, and not to the computation of nrrSync.

The variable computeNeighborRateRatio (see 10.2.5.10) indicates whether nrrPdelay is to be computed by this port when the peer-to-peer delay mechanism is invoked.

When useMgtSettableComputeNeighborRateRatio (see 14.18.17) is FALSE, computeNeighborRateRatio is initialized to the value of initialComputeNeighborRateRatio.

The currentComputeNeighborRateRatio specifies the current value of computeNeighborRateRatio. The default value of initialComputeNeighborRateRatio is TRUE. The initialComputeNeighborRateRatio and currentComputeNeighborRateRatio are per-port attributes.

When useMgtSettableComputeNeighborRateRatio is TRUE, currentComputeNeighborRateRatio is set equal to mgtSettableComputeNeighborRateRatio (see 14.18.18), and initialComputeNeighborRateRatio is ignored.

NOTE—If useMgtSettableComputeNeighborRateRatio is FALSE, the value of initialComputeNeighborRateRatio determines whether nrrPdelay is computed by the peer delay mechanism when the port is initialized. The value of computeNeighborRateRatio can be changed, e.g., if the port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in currentComputeNeighborRateRatio.

11.7 Control of computation of meanLinkDelay

The variable computeMeanLinkDelay (see 10.2.5.10) indicates whether meanLinkDelay is to be computed by this port when the peer-to-peer delay mechanism is invoked.

When useMgtSettableComputeMeanLinkDelay (see 14.18.21) is FALSE, computeMeanLinkDelay is initialized to the value of initialComputeMeanLinkDelay.

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The currentComputeMeanLinkDelay specifies the current value of computeMeanLinkDelay. The default value of initialComputeMeanLinkDelay is TRUE. The initialComputeMeanLinkDelay and currentComputeMeanLinkDelay are per-port attributes.

When useMgtSettableComputeMeanLinkDelay is TRUE, currentComputeMeanLinkDelay is set equal to mgtSettableComputeMeanLinkDelay (see 14.18.22), and initialComputeMeanLinkDelay is ignored.

NOTE—If useMgtSettableComputeMeanLinkDelay is FALSE, the value of initialComputeMeanLinkDelay determines whether meanLinkDelay is computed by the peer delay mechanism when the port is initialized. The value of computeMeanLinkDelay can be changed, e.g., if the port receives a Signaling message that carries a message interval request TLV (see 10.6.4.3) and the current value is stored in currentComputeMeanLinkDelay.

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12. Media-dependent layer specification for IEEE 802.11 links

12.1 Overview

12.1.1 General

Accurate synchronized time is distributed across a domain through time measurements between adjacent PTP Instances in a packet network. Time is communicated from the root of the synchronization tree (i.e., the Grandmaster PTP Instance) to the leaves of the synchronization tree, by recursively propagating time from a leaf-facing "timeTransmitter" port to some number of root-facing "timeReceiver" ports in PTP Instances at the next level of the synchronization tree through measurements made across the links connecting the PTP Instances. While the time semantics are consistent across the time-aware packet network, the method for communicating synchronized time from a timeTransmitter port to the immediate downstream link partner varies depending on the type of link interconnecting the two systems.

This clause specifies the interface primitives and state machines that provide accurate synchronized time across wireless IEEE 802.11 links as part of a packet network. This clause builds upon time measurement features defined in IEEE Std 802.11-2016 and makes no distinction between stations with an access point function and stations without an access point function.

12.1.2 IEEE 802.11 timing measurement and fine timing measurement procedures

12.1.2.1 General

IEEE Std 802.11-2016 defines a family of wireless measurements, including both "timing measurement" (TM) and "fine timing measurement" (FTM), which captures timestamps of the transmit time and receive time of a round-trip message exchange between associated wireless local area network (WLAN) stations.

In contrast to the protocol defined for FDE links, this clause does not define any new frames nor the transmission of any frames. Rather, it makes use of a MAC layer management entity (MLME) interface, which causes the IEEE 802.11 layer to not only take timestamps of measurement frames as they are transmitted and received, but to also *generate* and *consume* the measurement frames, all within the IEEE 802.11 MLME layer, and then to provide timestamp information from the MLME to this media-dependent layer through a set of well-defined service primitives. However, as an aid to the reader, the protocol and frames used by the IEEE 802.11 MLME for both timing measurement and fine timing measurement are described briefly as follows and illustrated in Figure 12-1 and Figure 12-2, respectively.

Both timing measurement and fine timing measurement are accomplished through a round-trip frame exchange. For timing measurement, the first frame of the round-trip measurement is generated by the timeTransmitter within the IEEE 802.11 MLME when the MLME-TIMINGMSMT.request primitive is invoked. For fine timing measurement, an initial Fine Timing Measurement Request frame is generated by the timeReceiver within the IEEE 802.11 MLME when the MLME-FINETIMINGMSMTRQ.request primitive is invoked. After this frame is successfully received by the timeTransmitter, the first frame of the round-trip measurement is generated by the TimeTransmitter within the IEEE 802.11 MLME when the MLME-FINETIMINGMSMT.request primitive is invoked. As defined by IEEE Std 802.11-2016, upon receipt of the resulting Timing Measurement or Fine Timing Measurement frame, the timeReceiver station transmits an IEEE 802.11 Ack control frame to the timeTransmitter station. Four timestamps are captured during this two-frame exchange, as follows:

a) t1 is when (in the timeTransmitter station's time base) the request frame is transmitted
b) t2 is when (in the timeReceiver station's time base) the request frame is received
c) t3 is when (in the timeReceiver station's time base) the Ack control frame is transmitted
d) t4 is when (in the timeTransmitter station's time base) the Ack control frame is received

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When the timeTransmitter sends either a Fine Timing Measurement or a Timing Measurement frame, it passes the t1 and t4 timestamps (and other end-to-end synchronization information) and FollowUpInformation, from the previous measurement to the timeReceiver. A pair of tokens is passed in each Timing or Fine Timing Measurement frame, one to identify the current measurement and the other to allow the timeReceiver to associate the timestamp information with the previous measurement.

NOTE 1—TM also can include a Timing Measurement Request frame; however, this frame type is not used by this standard.

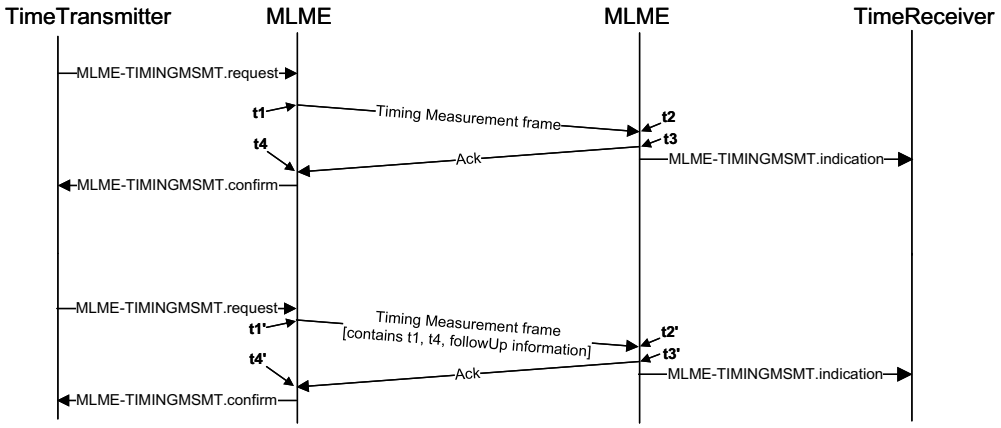


Figure 12-1—Timing measurement procedure for IEEE 802.11 links

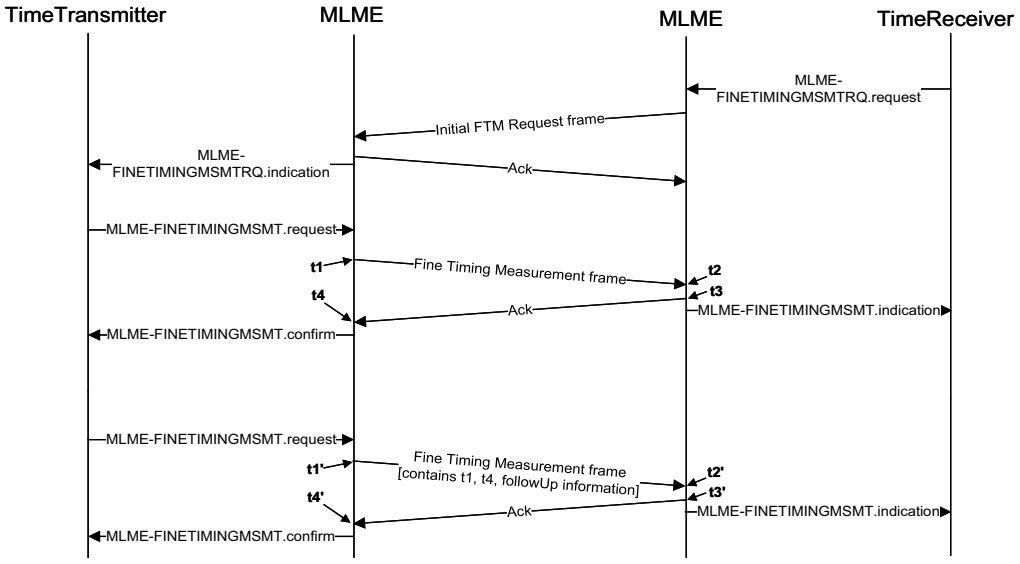


Figure 12-2—Fine timing measurement procedure for IEEE 802.11 links

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Note that, unlike full-duplex point-to-point ports, IEEE 802.11 ports do not compute the link delay measurements in both directions since only a PTP Port in the TimeReceiver state makes use of that information. A PTP Port that transitions from the TimeTransmitter state to the TimeReceiver state (e.g., due to selection of a new Grandmaster PTP Instance) can collect a number of link delay measurements and perform averaging or other filtering to achieve the desired accuracy.

NOTE 2—Fine timing measurement can be used for time synchronization as described in this standard; however, it can also be used for location services as defined in IEEE Std 802.11-2016. Since FTM supports only one configuration at a time, it is possible that setting that single configuration for time synchronization might disable its previous configuration for use by another application for location services or, conversely, an application that configures FTM for location services could disable this standard's use for time synchronization. Implementations that use FTM for time synchronization and other applications need to coordinate usage of the FTM protocol.

The timeTransmitter generates MLME-TIMINGMSMT.request primitives for timing measurement, as described in this standard, in a manner such that the requirements of 10.7.2.3 and 12.8 for the time synchronization message interval are satisfied. Timing Measurement frames are then sent from the timeTransmitter to the timeReceiver continuously and at a rate that satisfies the requirements of those two subclauses. It is not necessary for the timeReceiver to continually request timing information from the timeTransmitter. In contrast, the timeReceiver must request timing information from the timeTransmitter for fine timing measurement. A Fine Timing Measurement frame carries timestamp information for a previous measurement. The timeReceiver requests a burst of Fine Timing Measurement frames from the timeTransmitter. Figure 12-2 shows an example of a burst of Fine Timing Measurement frames (there are two frames in that example). The fine timing measurement process is described in more detail in 12.1.2.2 and is illustrated in Figure 12-3. In that discussion, the focus is on the transmission of the frames. In the simplified example, service primitives are omitted from Figure 12-3.

12.1.2.2 Detailed description of fine timing measurement (FTM)

Figure 12-3 is adapted from Figure 11-37 of IEEE Std 802.11-2016. Additional details on the FTM procedure are given in 11.24.6 of IEEE Std 802.11-2016. The example of this figure is for when the initiating station (STA), i.e., the timeReceiver, requests a single burst of three FTM frames from the responding STA, i.e., the timeTransmitter, as soon as possible. The timeReceiver makes this request by sending an initial FTM Request to the timeTransmitter with respective parameters set to appropriate values. The FTM parameters that are relevant to time synchronization in this standard are described in 12.6, and all the FTM parameters are described in more detail in 9.4.2.168 of IEEE Std 802.11-2016. However, in the example here, the parameter ASAP is set to 1 to indicate to the timeTransmitter that the FTM frames are desired as soon as possible, and the Number of Bursts Exponent parameter is set to 0 to indicate a single burst. Figure 12-3 is a simplified view; the timeReceiver causes the frame to be sent by invoking the MLME-FINETIMINGMSMTRQ.request primitive, which includes the FTM parameter values. The timeTransmitter sends an acknowledgment (Ack) frame to the timeReceiver to indicate it received the initial FTM request. The timeTransmitter then sends an initial FTM frame at a time that is recommended to be no more than 10 ms later than the receipt of the initial FTM request. The initial FTM frame indicates to the timeReceiver whether the timeTransmitter was able to grant the values of the FTM parameters that the timeReceiver requested. If the requested parameters are granted, the procedure continues (Figure 12-3 illustrates this case). If the requested parameters are not granted, the timeReceiver sends a new initial FTM request for a burst of two FTM frames. If the new request is granted, the procedure continues. If the new request is not granted, the timeReceiver and timeTransmitter use TM if they both support TM. If, at this point, the timeReceiver or the timeTransmitter, or both, do not support TM, the procedure terminates and asCapable is set to FALSE (see 12.4).

NOTE—IEEE Std 802.11-2016 allows various options in case the timeTransmitter does not grant the request. The above procedure is used in this subclause.

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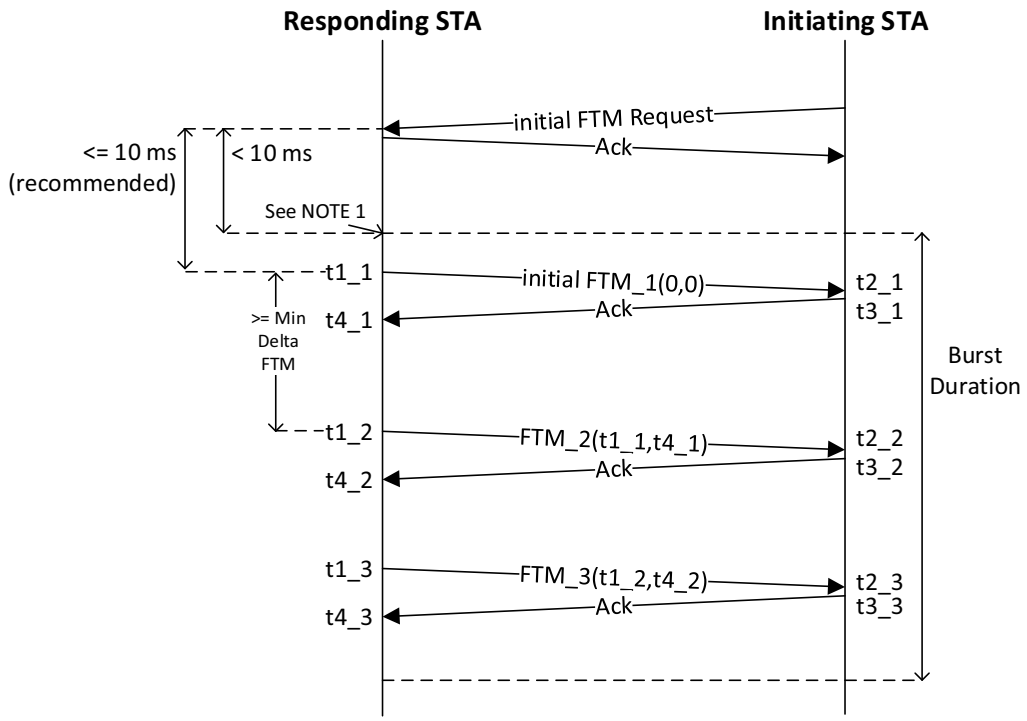


Figure 12-3—Illustration of fine timing measurement burst

The initial FTM frame (initial FTM_1 in Figure 12-3) sent by the timeTransmitter is timestamped with the value t1_1 on transmission from the timeTransmitter and timestamped with the value t2_1 on receipt by the timeReceiver. The initial FTM frame has fields that carry the t1 and t4 timestamps of the previous FTM frame and corresponding Ack; however, since this is the first FTM frame of the burst, these fields are set to zero. The timeReceiver responds to the timeTransmitter with an Ack, which is timestamped with the value t3_1 on transmission from the timeReceiver and with the value t4_1 on receipt by the timeTransmitter.

The timeTransmitter sends the second FTM frame (FTM_2 in Figure 12-3) after a time interval since sending the initial FTM frame that is greater than or equal to the Min Delta FTM parameter that was requested by the timeReceiver. This FTM frame is timestamped with t1_2 on transmission and t2_2 on reception. This FTM frame also carries the values of the timestamps t1_1 and t4_1 of the initial FTM frame and corresponding Ack. On receipt of the second FTM frame, the timeReceiver sends an Ack frame to the timeTransmitter; this frame is timestamped with t3_2 on transmission and t4_2 on reception. Finally, the timeTransmitter sends the third FTM frame (FTM_3 in Figure 12-3) to the timeReceiver, also at a time interval since sending FTM_2 that is greater than or equal to the Min Delta FTM parameter that was requested by the timeReceiver. As with FTM_2, this frame is timestamped with t1_3 on transmission and t2_3 on reception. This FTM frame also carries the values of the timestamps t1_2 and t4_2 of the second FTM frame and corresponding Ack. On receipt of the third FTM frame, the timeReceiver sends an Ack frame to the timeTransmitter; this frame is timestamped with t3_3 on transmission and t4_3 on reception.

On completion of the above exchanges of FTM frames and corresponding acknowledgments, the timeReceiver knows the transmission and reception times for the initial FTM frame (t1_1, t2_1, t3_1, and t4_1) and second FTM frame (t1_2, t2_2, t3_2, and t4_2) and the reception time for the third FTM frame (t2_3) and transmission time for the corresponding Ack (t3_3). The timeReceiver can use this information (along with FollowUpInformation contained in the VendorSpecific information element; see 12.7) to synchronize to the timeTransmitter. In this standard, timestamps for the minimum delay FTM frames are

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used. Specifically, the timeReceiver computes the quantities D1 = t2_1 – t1_1, and D2 = t2_2 – t1_2, and uses the timestamps t1_*i* and t2_*i*, where *i* = 1 if D1 < D2 and *i* = 2 if D1 ≥ D2, to compute the respective members of the MDSyncReceive structure. The timestamps of FTM_3 and its corresponding Ack are not used; FTM_3 is used only to convey the timestamps of FTM_2 and its corresponding Ack.

If the timeTransmitter does not grant the parameters requested initially by the timeReceiver, i.e., for a burst of three FTM frames, but it does grant the subsequent request for a burst of two FTM frames, the timeReceiver has a full set of timestamps for only the initial FTM_1 frame. In this case, the timeReceiver uses the timestamps t1_1, t2_1, t3_1, and t4_1 to compute the respective members of the MDSyncReceive structure.

With the above procedure for FTM, the timeReceiver controls the rate at which time synchronization information is sent from the timeTransmitter. Therefore, the timeReceiver requests time synchronization information from the timeTransmitter at an average rate equal to the inverse of the current synchronization message interval currentLogSyncInterval (see 12.8 and 14.10.19). In addition, the actual intervals between successive requests by the timeReceiver for time synchronization information meet the requirements of 10.7.2.3. Also, the value of syncLocked at the timeTransmitter port will not affect the sending of time synchronization information from the timeTransmitter to the timeReceiver; the requests for time synchronization information from the timeReceiver are asynchronous to the receipt of time synchronization information from upstream at the node that contains the timeTransmitter port.

NOTE—The rate at which time synchronization information is sent from the timeTransmitter can be controlled by the timeTransmitter instead of the timeReceiver for non-FTM cases (e.g., TM).

12.1.3 Layering for IEEE 802.11 links

The *media-dependent* (MD) entity is tailored to the link technology and is responsible for translating the PortSync entity's media-independent actions to media-dependent PDUs or primitives as necessary for communicating synchronized time from the timeTransmitter port over the link to a single timeReceiver port. For an IEEE 802.11 link, this one-to-one relationship between the MD entities of the timeTransmitter and timeReceiver implies that if the one physical IEEE 802.11 port is associated with multiple stations, each association requires its own instantiation of the IEEE 802.1AS PortSync entity and MD entity. The MLME-TIMINGMSMT and MLME-FINETIMINGMSMT service primitives defined in IEEE Std 802.11-2016 are used to perform timing measurements and fine timing measurements, respectively, between a timeTransmitter IEEE 802.11 station and associated IEEE 802.11 timeReceiver stations. Figure 12-4 illustrates how the MD entity interacts with the higher and lower layers.

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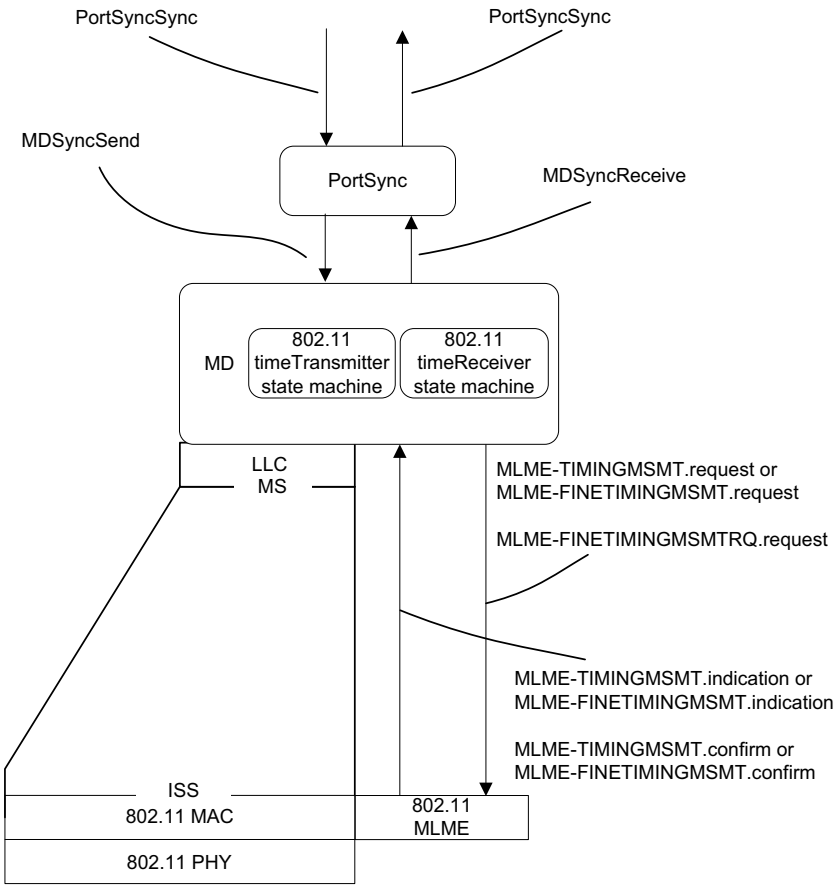


Figure 12-4—Media-dependent and lower entities in stations with IEEE 802.11 links

12.2 Messages

All media-dependent frames are generated and consumed by the lower-layer IEEE 802.11 MLME and thus none are defined here. Also, since the IEEE 802.11 event messages are timestamped by the MAC/PHY, the timestamp point is defined in IEEE Std 802.11-2016 as well. Media-independent messages, i.e., Announce and Signaling messages, are transmitted using the unicast address of the WLAN station instead of the group address defined in 10.5.3.

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12.3 Determination of timing measurement and fine timing measurement capability

The bits of the per-PTP Port global variable tmFtmSupport (see 12.5.1.5) shall be set as indicated in Table 12-1.

Table 12-1—Values of bits of tmFtmSupport

Bit	Value
0	TRUE if: a) The port supports timing measurement and b) The Timing Measurement bit in the Extended Capabilities information element defined in Table 9-135 of IEEE Std 802.11-2016 indicates that the peer IEEE 802.11 station is capable of participating in the timing measurement protocol. FALSE otherwise.
1	TRUE if: a) The port supports fine timing measurement and b) The Fine Timing Measurement Responder and Initiator bits in the Extended Capabilities information element defined in Table 9-135 of IEEE Std 802.11-2016 indicate that the peer IEEE 802.11 station is capable of participating in the fine timing measurement protocol. FALSE otherwise.
2–7	Reserved as FALSE.

12.4 Determination of asCapable

The per-PTP Port, per-domain instance of the global variable asCapable (see 10.2.5.1) is set to TRUE if the following conditions hold (see 12.5.1 and 12.5.2):

a) The value of tmFtmSupport is not zero.
b) neighborGptpCapable is TRUE.
c) At least one of the following conditions hold:
    1) Bit 0 of tmFtmSupport is TRUE.
    2) Bit 1 of tmFtmSupport is TRUE and, if the PTP Port is a timeTransmitter port, it can support (i.e., grant) the parameters requested by the timeReceiver with either FTMs per burst equal to 3 or FTMs per burst equal to 2.
    3) Bit 1 of tmFtmSupport is TRUE and, if the PTP Port is a timeReceiver port, the timeTransmitter port at the other end of the link can support (i.e., grant) the parameters requested by the timeReceiver with either FTMs per burst equal to 3 or FTMs per burst equal to 2.

If the value of domainNumber is zero (which is required for support of the 2011 edition of this standard) and bit 0 of tmFtmSupport is TRUE, asCapable can be set to TRUE. In all other instances, asCapable shall be set to FALSE.

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NOTE—The above conditions ensure backward compatibility with the 2011 edition of this standard. A time-aware system that is compliant with the 2011 edition of this standard will not process the gPTP-capable TLV, and asCapable will be determined as specified in the 2011 edition. A PTP Instance of a time-aware system compliant with the current edition of this standard that is attached, via an IEEE 802.11 link, to a node compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP-capable TLV and will not set neighborGptpCapable to TRUE. However, item c) 1) in this subclause ensures that asCapable for this PTP Port and domain (i.e., domain 0) will still be set in a manner consistent with the 2011 edition of this standard.

12.5 State machines

12.5.1 Media-dependent timeTransmitter state machines

12.5.1.1 Overview

The MD entity of an IEEE 802.11 port whose port state is TimeTransmitterPort (see Table 10-2) shall behave in a manner that is indistinguishable, relative to an observer external to a system, from a strict implementation of the timeTransmitter state machines in Figure 12-5 and Figure 12-6 (denoted as timeTransmitter state machine A and timeTransmitter state machine B, respectively, in 12.5.1.2), the local variables specified in 12.5.1.3, the functions specified in 12.5.1.4, the shared variables specified in 12.5.1.5, and the primitives defined in 12.5.1.6.

For timing measurement, timeTransmitter state machine A is responsible for initiating a time measurement whenever the PortSync entity requests it do so, as indicated by the rcvdMDSyncDot11TimeTransmitterA Boolean (see 12.5.1.3.8). TimeTransmitter state machine A invokes the IEEE 802.11 MLME-TIMINGMSMT.request primitive and waits for the subsequent MLME-TIMINGMSMT.confirm primitive. It collects local timestamp information from the measurement (t1 and t4, provided by the confirm primitive) and includes the information in the subsequent request. See 8.4.3 for more information on timestamps. TimeTransmitter state machine B is not used for timing measurement.

For fine timing measurement, timeTransmitter state machine A receives and stores information from the PortSync entity. TimeTransmitter state machine B receives the MLME-FINETIMINGMSMTRQ.indication caused by the initial FTM request from the timeReceiver. It sets asCapable as specified in 12.4. It then generates successive MLME-FINETIMINGMSMT.request primitives to indicate to the timeReceiver whether it can grant the requested parameters and also to cause information saved by timeTransmitter state machine A to be sent to the timeReceiver. It receives MLME-FINETIMINGMSMT.confirm primitives caused by Acks received from the timeReceiver. It collects local timestamp information from the current measurement (t1 and t4, provided by the confirm primitive) and includes the information in the subsequent MLME-FINETIMINGMSMT.request.

12.5.1.2 State diagrams

NOTE—In the computation of the burstDuration in timeTransmitter state machine B, the burst duration parameter from IEEE Std 802.11-2016 is converted to UScaledNs (i.e., units of 2^{-16} ns; see 6.4.3.2). The burst duration in UScaledNs (see 6.4.3.2) is related to the quantity A = initReqParamsDot11TimeTransmitterB.burstDuration – 2 by:

$$\text{burst duration in UScaledNs} = 1000 \times 2^{16} \times 250 \times 2^{A}$$

i.e., A is the logarithm to base 2 of the burst duration, in microseconds, divided by 250. Also, it is assumed that the burst duration starts when the initial FTM request is received. In actuality, the timer begins by the partial Timing Synchronization Function timer (see 9.4.2.168 of IEEE Std 802.11-2016) value indicated in the initial FTM frame, which is slightly after the initial FTM request is received.

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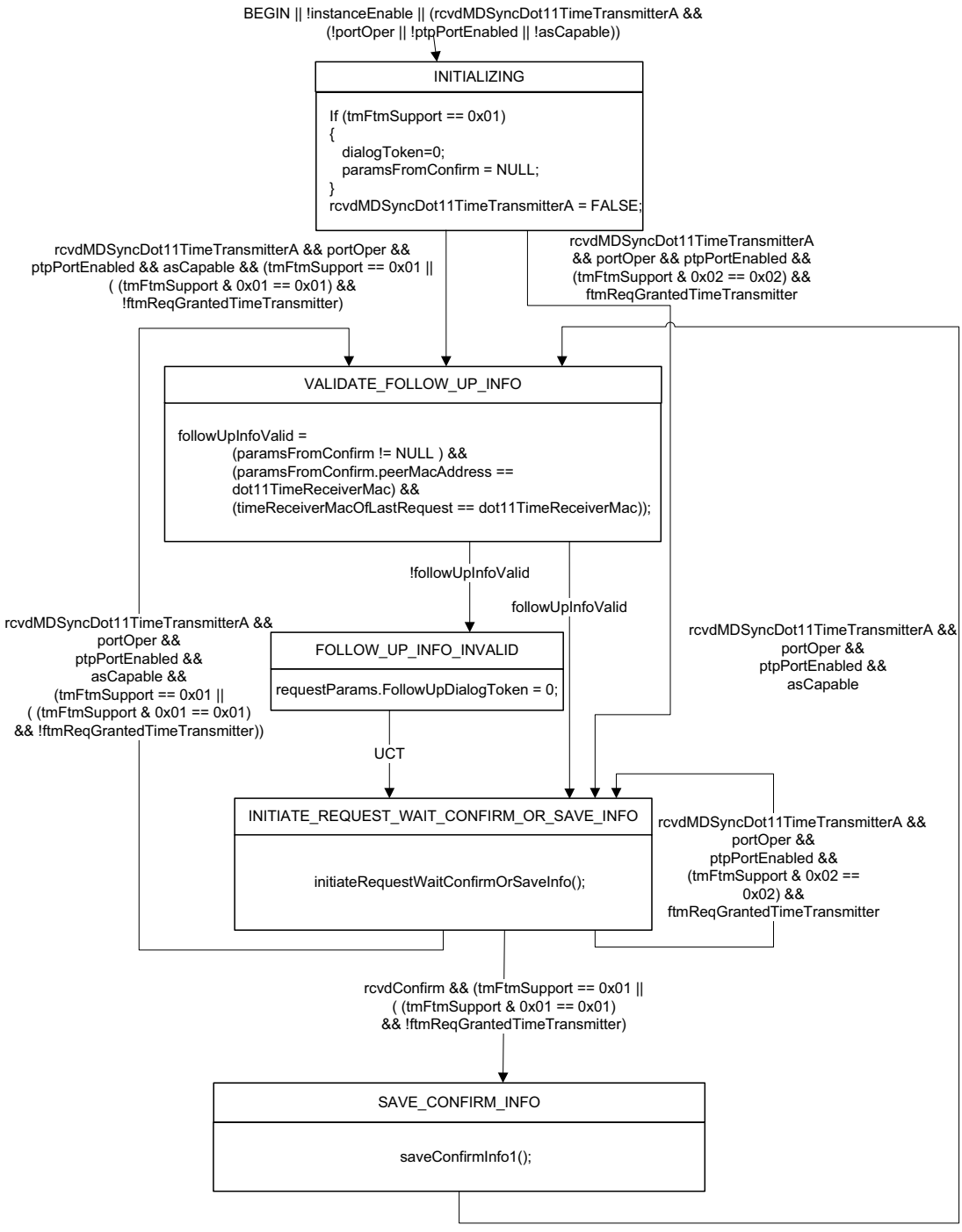


Figure 12-5—TimeTransmitter state machine A
(a) For TM, receives information from the PortSync entity and sends to timeReceiver, and
(b) For FTM, receives and stores information from the PortSync entity

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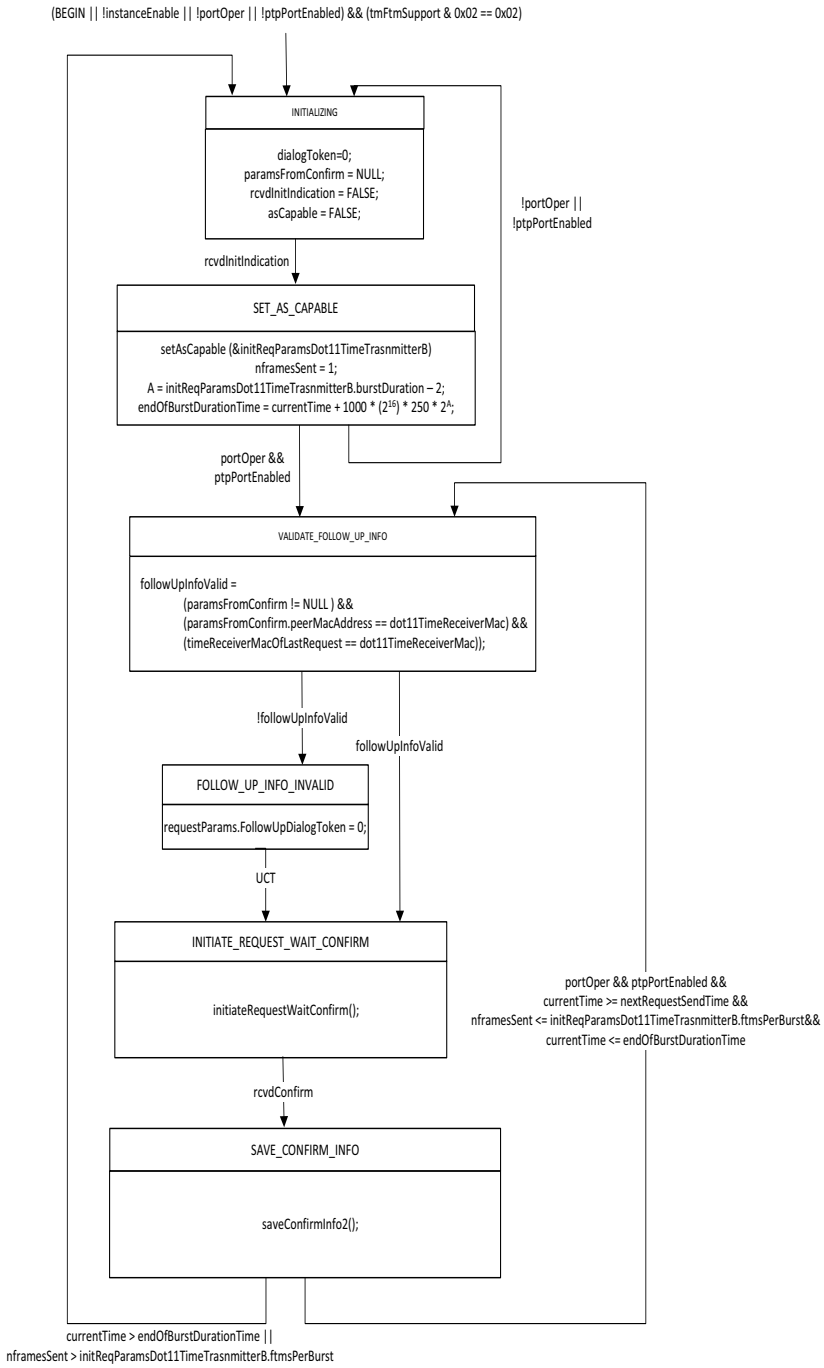


Figure 12-6—TimeTransmitter state machine B
(a) For TM, not invoked and
(b) For FTM, receives initial FTM request from timeReceiver and sends information received from upstream to timeReceiver in successive FTM frames

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12.5.1.3 State machine local variables

12.5.1.3.1 dialogToken: An unsigned 8-bit integer used to identify a measurement from among those preceding and following it.

12.5.1.3.2 followUpInfoValid: A Boolean variable indicating whether the FollowUpInformation (e.g., timestamps and rateRatio; see 12.7) and the link partner are unchanged since the last timing measurement or fine timing measurement.

12.5.1.3.3 requestParams: A structure whose members contain the values of the fields of either the MLME-TIMINGMSMT.request or MLME-FINETIMINGMSMT.request primitive.

12.5.1.3.4 paramsFromConfirm: A structure whose members contain the values of the fields of either the MLME-TIMINGMSMT.confirm or MLME-FINETIMINGMSMT.confirm primitive.

12.5.1.3.5 dot11TimeReceiverMac: The MAC address of the station associated with the current port.

12.5.1.3.6 timeReceiverMacOfLastRequest: The MAC address of the station of the previous request, used to validate FollowUpInformation.

12.5.1.3.7 residenceTime: A temporary variable that holds the computation of the time between receipt of the last synchronization information and transmission of synchronization information.

12.5.1.3.8 rcvdMDSyncDot11TimeTransmitterA: A Boolean variable that is set to TRUE when an MDSyncSend structure is provided by the PortSync entity.

12.5.1.3.9 rcvdConfirm: A Boolean variable that is set to TRUE when either the MLME-TIMINGMSMT.confirm or MLME-FINETIMINGMSMT.confirm primitive is received.

12.5.1.3.10 nframesSent: An unsigned 8-bit integer used to count the number of frames sent by the timeReceiver (the number of indications received is counted).

12.5.1.3.11 rcvdInitIndication: A Boolean variable that is set to TRUE when the initial MLME-FINETIMINGMSMTRQ.indication primitive is received.

12.5.1.3.12 initReqParamsDot11TimeTransmitterB: A structure whose members contain the values of the fields of an MLME-FINETIMINGMSMTRQ.indication primitive.

12.5.1.3.13 endOfBurstDurationTime: A UScaledNs variable whose value is the time at which the current burst ends.

12.5.1.3.14 nextRequestSendTime: A UScaledNs variable whose value is the expected time that the next MLME-FINETIMINGMSMTRQ.indication primitive, for the next request from the timeReceiver, will be received.

12.5.1.4 State machine functions

12.5.1.4.1 setRequestParams(&requestParams, MDSyncSend): Assigns values to the parameters of the request primitive of either MLME-TIMINGMSMT or MLME-FINETIMINGMSMT (see 12.5.1.6) as follows:

a) Members of the FollowUpInformation member of the VendorSpecific information element, as defined in 12.7, are assigned as defined in 11.4.4 with the exception of the correctionField, which is assigned, for TM, by the function saveConfirmInfo1() in TimeTransmitter state machine A

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(see Figure 12-5) and, for FTM, by the function saveConfirmInfo2() in TimeTransmitter state machine B (see Figure 12-6).

b) The other fields of the VendorSpecific information element are assigned as follows:

1) ElementID is assigned the value 221 as defined in Table 9-77 (Element IDs) of IEEE Std 802.11-2016, indicating that the information element is of type Vendor Specific.

2) The Length field is set to 80.

NOTE—This is equal to the length of the Follow_Up payload defined in 11.4.4 (including the common header) plus the length of the OUI or CID field and the Type field (see Figure 12-8).

3) The OUI or CID field is set to 00-80-C2.

4) The Type field is set to 0.

c) Max t1 Error, Max t4 Error are set to zero.

d) For Fine Timing Measurement frames, the location configuration information (LCI) Report and Location Civic Report are not present. All other members are left unchanged.

12.5.1.4.2 setAsCapable (&initReqParamsDot11TimeTransmitterB): Determines the value of asCapable consistent with 12.4 and whether the timeTransmitter is able to grant the parameters requested by the timeReceiver. This function is used only for FTM.

12.5.1.4.3 initiateRequestWaitConfirmOrSaveInfo(): This function is defined as indicated below. It is used in TimeTransmitter state machine A. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
initiateRequestWaitConfirmOrSaveInfo()
{
        rcvdMDSyncDot11TimeTransmitterA = FALSE;

        If (tmFtmSupport == 0x01)
        {
                if ((++dialogToken % 256) == 0) dialogToken++;
                requestParams.DialogToken=dialogToken;
                requestParams.PeerMACAddress = dot11TimeReceiverMac;
                setRequestParams(&requestParams, MDSyncSend);
                MLME-TIMINGMSMT.request(requestParams);
                requestParams.FollowUpDialogToken = 0;
                                        //In case no confirm is received
                timeReceiverMacOfLastRequest = dot11TimeReceiverMac;
        }
}
```

12.5.1.4.4 saveConfirmInfo1(): This function is defined as indicated below. It is used in TimeTransmitter state machine A. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
saveConfirmInfo1()
{
        MLME-TIMINGMSMT.confirm(&paramsFromConfirm);

        requestParams.FollowUpDialogToken = paramsFromConfirm.DialogToken;
        requestParams.T1 = paramsFromConfirm.T1;
        requestParams.T4 = paramsFromConfirm.T4;

        // NOTE: In timing measurement, T1 is in units of 10 ns.
        // upstreamTxTime is units of 2^-16 ns.
```


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```
                K = 1;
                // K is 1 for timing measurement.
                residenceTime = MDSyncSend.rateRatio *
                        (paramsFromConfirm.T1 * 10^K*(2^16) - MDSyncSend.upstreamTxTime);

                requestParams.VendorSpecific.correctionField =
                        residenceTime + MDSyncSend.followUpCorrectionField;
                // NOTE: T1 and T4 are timestamps from a single
                // local clock source. The roll-over of the 32-bit timestamps returned by
                // MLME-TIMINGMSMT.request and MLME-TIMINGMSMT.indication
                //      must be accounted for.
        }
```

12.5.1.4.5 initiateRequestWaitConfirm(): This function is defined as indicated below. It is used in TimeTransmitter state machine B. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
        initiateRequestWaitConfirm()
        {
                If ((++dialogToken % 256) == 0) dialogToken++;
                If (nframesSent == initReqParamsDot11TimeTransmitterB.ftmsPerBurst)
                        dialogToken = 0;

                requestParams.DialogToken=dialogToken;
                requestParams.PeerMACAddress = dot11TimeReceiverMac;
                setRequestParams(&requestParams, MDSyncSend);
                // In the following statement, MinDeltaFTM, which is in units of 100
                //     microseconds, is converted to UScaledNs (i.e., units of 2^-16 ns; see 6.3.3.2)
                nextRequestSendTime = currentTime +
                        initReqParamsDot11TimeTransmitterB.MinDeltaFTM * (65536 x 10^5);
                MLME-FINETIMINGMSMT.request(requestParams);
                requestParams.FollowUpDialogToken = 0;    //In case no confirm is received
                timeReceiverMacOfLastRequest = dot11TimeReceiverMac;
        }
```

12.5.1.4.6 saveConfirmInfo2(): This function is defined as indicated below. It is used in TimeTransmitter state machine B. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
        saveConfirmInfo2()
        {
                MLME-FINETIMINGMSMT.confirm(&paramsFromConfirm);

                requestParams.FollowUpDialogToken = paramsFromConfirm.DialogToken;
                requestParams.T1 = paramsFromConfirm.T1;
                requestParams.T4 = paramsFromConfirm.T4;

                // NOTE: In fine timing measurement, T1 is in units of 0.1 ns.
                // upstreamTxTime is units of 2^-16 ns.

                K = -3;
                // K is 1 for timing measurement and -3 for fine timing measurement.
                residenceTime = MDSyncSend.rateRatio *
                        (paramsFromConfirm.T1 * 10^K*(2^16) - MDSyncSend.upstreamTxTime);

                requestParams.VendorSpecific.correctionField =
                residenceTime + MDSyncSend.followUpCorrectionField;
```


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```
                // NOTE: T1 and T4 are timestamps from a single
                // local clock source. The roll-over of the 48-bit timestamps returned by
                // MLME-FINETIMINGMSMT.request and MLME-FINETIMINGMSMT.indication
                // must be accounted for.

                // A frame is only counted as being sent, for purposes of number of frames in a
                // burst, if a confirm is received. It is up to IEEE Std 802.11-2016 to handle the
                // case where a confirm is not received.
                nframesSent += 1;
        }
```

12.5.1.5 Shared variables

12.5.1.5.1 MDSyncSend: A structure as defined in 10.2.2.1.

12.5.1.5.2 portOper: A Boolean as defined in 10.2.5.12.

12.5.1.5.3 ptpPortEnabled: A Boolean as defined in 10.2.5.13.

12.5.1.5.4 asCapable: A Boolean whose value is specified in 12.4 and 10.2.5.1.

12.5.1.5.5 tmFtmSupport: An Octet whose bits are interpreted as Booleans and whose values are specified in Table 12-1.

12.5.1.5.6 ftmReqGrantedTimeTransmitter: A Boolean whose value is TRUE if the timeTransmitter grants the current initial FTM request and FALSE otherwise.

12.5.1.6 TimeTransmitter primitives

12.5.1.6.1 MLME-TIMINGMSMT.request

The MLME-TIMINGMSMT.request primitive is used by a timeTransmitter station to initiate a timing measurement and also communicates timestamps t1 and t4 captured by the timeTransmitter during a previous measurement. The primitive and its parameters are specified in 6.3.57.2 of IEEE Std 802.11-2016.

12.5.1.6.2 MLME-TIMINGMSMT.confirm

The MLME-TIMINGMSMT.confirm primitive indicates that a timing measurement request has completed. The primitive and its parameters are specified in 6.3.57.3 of IEEE Std 802.11-2016.

12.5.1.6.3 MLME-FINETIMINGMSMT.request

The MLME-FINETIMINGMSMT request primitive is used by a timeTransmitter station to initiate a fine timing measurement and also communicates timestamps t1 and t4 captured by the timeTransmitter during a previous measurement. The primitive and its parameters are specified in 6.3.58.2 of IEEE Std 802.11-2016.

12.5.1.6.4 MLME-FINETIMINGMSMT.confirm

The MLME-FINETIMINGMSMT request primitive indicates that a fine timing measurement request has completed. The primitive and its parameters are specified in 6.3.58.3 of IEEE Std 802.11-2016.

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12.5.1.6.5 MLME-FINETIMINGMSMTRQ.indication

The MLME-FINETIMINGMSMTRQ.indication primitive indicates to a timeTransmitter station that the timeReceiver is requesting a burst of FTM frames with the indicated parameters. The primitive and its parameters are specified in 6.3.70.3 of IEEE Std 802.11-2016.

12.5.2 Media-dependent timeReceiver state machine

12.5.2.1 Overview

The MD entity of an IEEE 802.11 port whose PTP Port state is TimeReceiverPort or PassivePort (see 10.3.6) shall behave in a manner that is indistinguishable, relative to an observer external to a system, from a strict implementation of the timeReceiver state machine in 12.5.2.2, the local variables specified in 12.5.2.3, the functions specified in 12.5.2.4, the shared variables specified in 12.5.2.5, and the primitives defined in 12.5.2.6.

The timeReceiver state machine is responsible for collecting information from the timing measurement or fine timing measurement indications, constructing an MDSyncReceive structure with the relevant information, and passing the structure to the PortSync entity for further processing. In order to do this, the state machine saves locally captured timestamps (i.e., t2 and t3) received in the indication and associates them with the timestamps sent from the timeTransmitter port in a future indication (i.e., t1 and t4). In addition, for fine timing measurement, the timeReceiver state machine is responsible for generating the MLME-FINETIMINGMSMTRQ.request primitive, which causes the initial FTM request frame to be sent to the timeTransmitter.

12.5.2.2 State diagram

Figure 12-7 presents the timeReceiver state machine. While quantities are shown to be computed from information in consecutive indications, an implementation can choose to compute over longer intervals as long as the clock performance requirements of Annex B are met.

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1

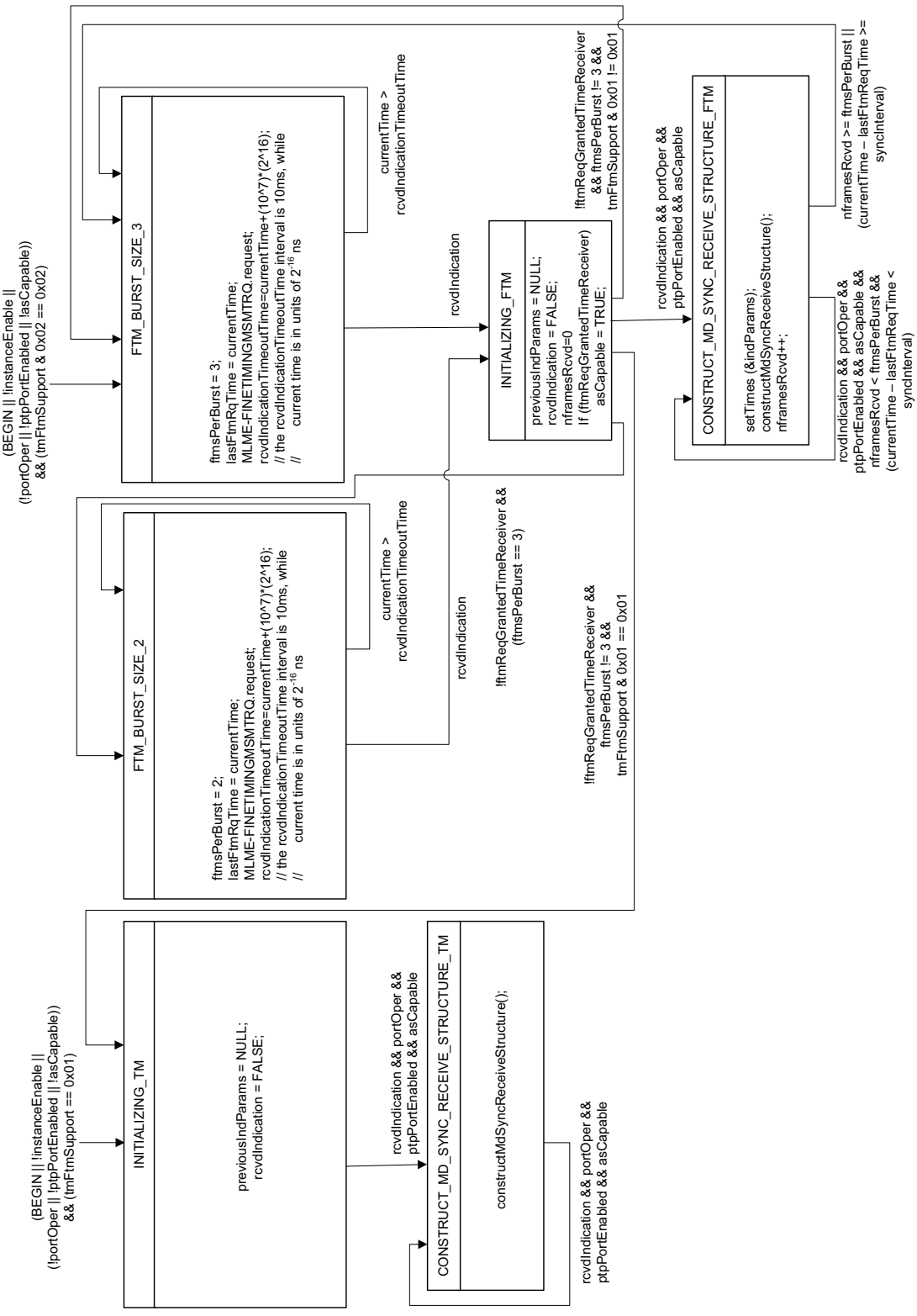


Figure 12-7—TimeReceiver state machine

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12.5.2.3 State machine local variables

12.5.2.3.1 indParams: A structure whose members contain the values of the fields of the MLME-TIMINGMSMT.indication primitive or MLME-FINETIMINGMSMT.indication primitive, as defined in 12.5.2.6, depending on whether TM or FTM, respectively, is used.

12.5.2.3.2 previousIndParams: A structure with members identical to those of indParams, used to save parameters from the previous indication.

12.5.2.3.3 rcvdIndication: A Boolean that is set to TRUE when either the MLME-TIMINGMSMT.indication or MLME-FINETIMINGMSMT.indication primitive is received.

12.5.2.3.4 rcvdIndicationTimeoutTime: A UScaledNs variable whose value is the time after which the state machine will not wait any longer for the next MLME-FINETIMINGMSMT.indication, and a new burst is initiated.

12.5.2.3.5 nframesRcvd: An unsigned 8-bit integer used to count the number of frames received by the timeTransmitter in the burst (the number of indications received from the timeTransmitter are counted).

12.5.2.3.6 initReqParamsDot11TimeReceiver: A structure whose members contain the values of the fields of an MLME-FINETIMINGMSMTRQ.request primitive.

12.5.2.3.7 ftmsPerBurst: The value of the FTM parameter ‘FTMs per burst’ (see 12.6), i.e., the number of FTM frames in the burst granted by the timeTransmitter.

12.5.2.3.8 ftmReqGrantedTimeReceiver: A Boolean that is TRUE if the timeTransmitter has granted the respective request for a burst and FALSE otherwise.

12.5.2.3.9 t1_1, t1_2, t2_1, t2_2, t2_3, t3_1, t3_2, t3_3, t4_1, t4_2: Temporary local variables used to hold the values of the timestamps t1, t2, t3, and t4 returned by the MLME-FINETIMINGMSMT.indication primitive in the structure indParams.

12.5.2.3.10 D1, D2: Temporary local variables used to hold the values of delay computed for the first two FTM frames and corresponding Acks when the timeTransmitter grants the request for three FTM frames.

12.5.2.4 State machine functions

12.5.2.4.1 setMDSyncReceiveDot11TimeReceiver(indParams): Creates an MDSyncReceive structure and returns the structure. All fields are assigned from FollowUpInformation (contained in the VendorSpecific information element) of indParams as in 11.2.13.2.1.

12.5.2.4.2 passMDSyncReceiveToPortSync(): Passes an MDSyncReceive structure to the PortSync entity of this PTP Port.

12.5.2.4.3 setTimes (&indParams): extracts the timestamp values from the successive indParams structures returned by the multiple FTM frame exchanges of a burst, and places the correct times (corresponding to minimum delay frames and Acks) in the final indParams structure. This final indParams structure is then used in the function constructMdSyncReceiveStructure(). This procedure is needed when the timeTransmitter grants three FTM frames for the burst. If the timeTransmitter grants only two FTM frames for the burst, the timestamp values returned in the indication primitive of the second frame are used. The function setTimes is used in the TimeReceiver state machine and is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

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```
setTimes (&indParams)
{
        if (nframesRcvd == 1)
        {
                t2_1 = indParams.T2;
                t3_1 = indParams.T3;
        }
        if (nframesRcvd == 2)
        {
                t1_1 = indParams.T1;
                t2_2 = indParams.T2;
                t3_2 = indParams.T3;
                t4_1 = indParams.T4
        }
        if (nframesRcvd == 3)
        {
                t1_2 = indParams.T1;
                t2_3 = indParams.T2;
                t3_3 = indParams.T3;
                t4_2 = indParams.T4;
                if (ftmsPerBurst == 3)
                {
                        D1 = t2_1 - t1_1;
                        D2 = t2_2 - t1_2
                        if (D2 <= D1)
                        {
                                indParams.T2 = t2_2;
                                indParams.T1 = t1_2;
                        }
                        else
                        {
                                indParams.T2 = t2_1;
                                indParams.T1 = t1_1;
                        }
                        D1 = t4_1 - t3_1;
                        D2 = t4_2 - t3_2
                        if (D2 <= D1)
                        {
                                indParams.T4 = t4_2;
                                indParams.T3 = t3_2;
                        }
                        else
                        {
                                indParams.T4 = t4_1;
                                indParams.T3 = t3_1;
                        }
                }
        }
}
```

12.5.2.4.4 constructMdSyncReceiveStructure(): This function constructs the MDSyncReceive structure and is defined as indicated below. It is used in the TimeReceiver state machine. It is defined here so that the detailed code that it invokes does not need to be placed into the state machine diagram.

```
constructMdSyncReceiveStructure()
{
        if (tmFtmSupport == 0x01)
                MLME-TIMINGMSMT.indication(&indParams);
```


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```
        else if (tmFtmSupport & 0x02 == 0x02)
        {
                MLME-FINETIMINGMSMT.indication(&indParams);
        }

        if ((previousIndParams != NULL) &&
                (previousIndParams.PeerMacAddress == dot11TimeReceiverMac) &&
                (indParams.FollowUpDialogToken != 0))
        {

                neighborRateRatio =
                        (indParams.T1-previousIndParams.T1) /
                        (indParams.T2-previousIndParams.T2);
                //NOTE: Other methods of computing neighborRateRatio
                                can be used.

                if (tmFtmSupport == 0x01)
                        K = 1;
                else if (tmFtmSupport & 0x02 == 0x02)
                        K = -3;
                //K = 1 for timing measurement and K = -3 for fine timing measurement
                meanLinkDelay =
                        (((indParams.T4 - indParams.T1) -
                        neighborRateRatio * (indParams.T3 - indParams.T2)) /
                                        (2.0)) * (10^K);

                //NOTE: Other methods of computing meanLinkDelay
                                can be used.

                MDSyncReceive = setMDSyncReceiveDot11TimeReceiver(indParams);
                MDSyncReceive.VendorSpecific.rateRatio +=
                        (neighborRateRatio – 1);
                MDSyncReceive.VendorSpecific.upstreamTxTime =
                                indParams.T2* (2^16) *(10^K) -
                                meanLinkDelay*(2^16)/neighborRateRatio;
                //NOTE: Actions performed with the timestampError
                        parameters of indParams are implementation independent.

                passMDSyncReceiveToPortSync(&MDSyncReceive);
        }
        previousIndParams = indParams;
        rcvdIndication = FALSE;
}
```

12.5.2.5 State machine shared variables

12.5.2.5.1 MDSyncReceive: A structure used for passing information between MD and PortSync, as defined in 10.2.2.2.

12.5.2.5.2 portOper: A Boolean as defined in 10.2.5.12.

12.5.2.5.3 ptpPortEnabled: A Boolean as defined in 10.2.5.

12.5.2.5.4 asCapable: A Boolean as defined in 12.4 and 10.2.5.1.

12.5.2.5.5 meanLinkDelay: The delay over the link to the associated WLAN station as defined in 10.2.5.8.

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12.5.2.5.6 neighborRateRatio: The measured ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the link attached to this PTP Port, to the frequency of the LocalClock entity of this time-aware system, as defined in 10.2.5.7.

12.5.2.5.7 tmFtmSupport: An Octet whose value is specified in 12.3.

12.5.2.6 TimeReceiver primitives

12.5.2.6.1 MLME-TIMINGMSMT.indication

The MLME-TIMINGMSMT.indication primitive is received by a timeReceiver station as the natural result of the peer timeTransmitter station issuing the corresponding request primitive and carries the same parameters plus local timestamp information. The primitive and its parameters are specified in 6.3.57.4 of IEEE Std 802.11-2016.

12.5.2.6.2 MLME-FINETIMINGMSMT.indication

The MLME-FINETIMINGMSMT.indication primitive is received by a timeReceiver station as the natural result of the peer timeTransmitter station issuing the corresponding request primitive and carries the same parameters plus local timestamp information. The primitive and its parameters are specified in 6.3.58.4 of IEEE Std 802.11-2016.

12.5.2.6.3 MLME-FINETIMINGMSMTRQ.request

The MLME-FINETIMINGMSMTRQ.request primitive is used by the timeReceiver to request a burst of FTM frames from the timeTransmitter, with respective FTM parameters. The primitive and its parameters are specified in 6.3.70.2 of IEEE Std 802.11-2016.

12.6 FTM parameters

The values of the FTM parameters that are relevant to time synchronization transport in this standard are specified in Table 12-2, along with a brief description of each parameter. These parameter values are carried in the initial FTM Request. More detailed descriptions of these and other FTM parameters are given in 9.4.2.168 of IEEE Std 802.11-2016. The parameters Burst Duration and Min Delta FTM depend on the time synchronization message interval, i.e., on currentLogSyncInterval (see 10.7.2.3 and 12.8). The values for these parameters are specified in Table 12-3. In this table, the Burst Duration encoding (i.e., value and corresponding duration in ms) is taken from Table 9-257 of IEEE Std 802.11-2016. The Min Delta FTM encoding values are in multiples of 100 μs (see 9.4.2.168 of IEEE Std 802.11-2016).

The values of the FTM parameters given in Table 12-2 shall be used in the MLME-FINETIMINGMSMTRQ.request invoked by the timeReceiver STA (i.e., in the initial FTM Request). The values for Burst Duration and Min Delta FTM given in Table 12-3 shall be used in the MLME-FINETIMINGMSMTRQ.request invoked by the timeReceiver STA.

Background on the derivation of the FTM parameters of Table 12-2 and Table 12-3 is given in Garner [B5].

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Table 12-2—FTM parameters relevant to time-synchronization transport

Parameter	Value	Description
Number of Bursts Exponent	0	Log to base 2 of the number of bursts requested by the timeReceiver (value of 0 indicates that one burst is requested)
Burst Duration	See Table 12-3	Duration of the burst of FTM frames and their corresponding Acks
Min Delta FTM	See Table 12-3	Minimum time between consecutive FTM frames
Partial Timing Synchronization Function Timer	1	See 9.4.2.168 of IEEE Std 802.11-2016
Partial Timing Synchronization Function Timer No Preference	Reserved in the initial FTM request	See 9.4.2.168 of IEEE Std 802.11-2016
ASAP	1	ASAP = 1 indicates that the timeReceiver would like the timeTransmitter to respond as soon as possible
ASAP Capable	Reserved in the initial FTM request	See 9.4.2.168 of IEEE Std 802.11-2016
FTMs per burst	3 in the first initial FTM Request 2 in the first retry, if the first initial FTM Request is not granted	Desired number of FTM frames and corresponding Acks in the requested burst
Burst Period	Reserved when Number of Bursts Exponent is zero	See 9.4.2.168 of IEEE Std 802.11-2016

Table 12-3—Values of Burst Duration and Min Delta FTM, for each value of currentLogSyncInterval

currentLogSyncInterval	Nominal message rate (messages/s)	Burst duration	Min delta FTM
–24 through –6, inclusive	64 through 16 777 216	6 (4 ms)	6 (0.6 ms)
–5	32	8 (16 ms)	25 (2.5 ms)
–4	16	9 (32 ms)	50 (5 ms)
–3	8	10 (64 ms)	100 (10 ms)
–2 through 24, inclusive	4	11 (128 ms)	200 (20 ms)

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12.7 Format of VendorSpecific information element

The IEEE 802.11 MLME request and indication primitives for timing measurement and fine timing measurement support an ability to carry data transparently between stations using the VendorSpecific information element. The Type field within the VendorSpecific Content identifies the type of information that follows the Type field. See Figure 12-8 and Table 12-4.

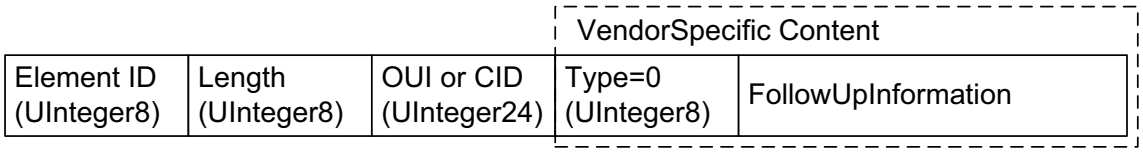


Figure 12-8—Format of VendorSpecific information element when Type = 0

Table 12-4—Values of the Type field in the VendorSpecific information element

Value	Description
0	The Type field is followed by FollowUpInformation
1–255	Reserved

This mechanism shall be used to carry end-to-end link-independent timing information from the timeTransmitter port to the associated timeReceiver port, including preciseOriginTimestamp, rateRatio, correctionField, and other fields of the Follow_Up message, as described in 12.5.1.4. For consistency, all of these fields are packed into the FollowUpInformation field using exactly the same format as used for FDE links. In other words, the timeTransmitter state machine communicates an entire Follow_Up message [i.e., including all the fields of the common header (see 11.4.2 and 10.6.2), the preciseOriginTimestamp, and all the fields of the Follow_Up information TLV (see 11.4.4)] using this mechanism. The Type field, illustrated in Figure 12-8, identifies this use of the OUI or CID within the VendorSpecific information element. Table 12-4 lists values for the Type field.

12.8 Synchronization message interval

12.8.1 General synchronization message interval specification

The mean time interval between successive synchronization messages shall be as specified in 10.7.2.1, 10.7.2.3, and 12.8.2.

12.8.2 Synchronization message interval default value

The default value of initialLogSyncInterval (see 10.7.2.3) is –3. Every PTP Port supports the value 127; the PTP Port does not send Sync messages when currentLogSyncInterval has this value (see 10.2.14). A PTP Port may support other values, except for the reserved values indicated in Table 10-16. A PTP Port ignores requests (see 10.2.14) for unsupported values.

Processing of the message interval request TLV carried in a Signaling message (see 10.6.4) shall be supported, as specified by the SyncIntervalSetting state machine of 10.2.14 (see Figure 10-10), except that

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the logLinkDelayInterval (which is not relevant to IEEE 802.11 ports) is set to –128 by the sender of the Signaling message, the logLinkDelayInterval is ignored by the receiver, and unsupported values of logTimeSyncInterval are ignored by the receiver.

NOTE 1—For TM, a timeReceiver port that requests (using a Signaling message that contains a message interval request TLV; see 10.6.4 and 10.2.14) that the PTP Port at the other end of the attached link set its currentLogSyncInterval to a specific value can determine if the request was honored by examining the logMessageInterval field of a FollowUpInformation field contained in the VendorSpecific information element of a subsequent MLME indication primitive.

NOTE 2—The time interval between every pair of adjacent timing or fine timing measurements is not guaranteed to be precisely the same. Some variation is expected, due to factors such as the MAC protocol (e.g., delay in accessing the medium and/or packet retries) and the power state of the associated station. However, timestamp t1 is not captured when either TIMINGMSMT.request or FINETIMINGMSMT.request is invoked, but only after the frame resulting from the request is actually transmitted.

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13. Media-dependent layer specification for interface to IEEE 802.3 Ethernet passive optical network link

13.1 Overview

13.1.1 General

This clause specifies the service interface primitives, state machines, and message formats that provide accurate synchronized time across IEEE 802.3 Ethernet passive optical network (EPON) links, through the use of the timing process and measurements specified in 64.2.1.1, 64.3.2.4, 77.2.1.1, and 76.1.2 of IEEE Std 802.3-2022. For purposes of this clause, an EPON link is an EPON that contains one optical line terminal (OLT) and associated optical network units (ONUs).

A time-aware system may contain more than one OLT and/or ONU. Each PTP Instance of a time-aware system uses at most one ONU port, but may serve, i.e., provide timing to, more than one OLT port (i.e., each PTP Instance of a time-aware system is a clock timeReceiver to at most one EPON link, but can be clock timeTransmitter to more than one EPON link). Two different PTP Instances of a time-aware system may use different ONU ports.

13.1.2 Description of the EPON timing process

The timing process in EPON relies on the 32-bit counters (see 64.2.2.2 and 77.2.2.2 of IEEE Std 802.3-2022) at both the OLT and the ONU. The 32-bit counter used by EPON is the LocalClock entity of the PTP Instance that uses the respective OLT or ONU. These counters increment every time_quantum, which is equal to 16 ns (see 64.2.2.1 and 77.2.2.1 of IEEE Std 802.3-2022). IEEE Std 802.3-2022 defines multipoint control protocol (MPCP), which is one of the protocols that enable MAC clients to communicate over a point-to-multipoint optical network. When either the clock timeTransmitter (OLT) or the clock timeReceiver (ONU) transmits an MPCP data unit (MPCPDU), its counter value is mapped into the timestamp field. Clause 64 and Clause 77 of IEEE Std 802.3-2022 specify the EPON timing mechanism.

13.1.3 Best timeTransmitter selection

13.1.3.1 General

An EPON link contains one OLT and the associated ONUs. The OLT is the clock timeTransmitter and the associated ONUs are clock timeReceivers. The OLT initiates the time synchronization as a requester. The ONUs are the responders of the time synchronization. In other words, the invocation of the BTCA results in the OLT having the PTP Port state TimeTransmitterPort and the ONU having the PTP Port state TimeReceiverPort (see 10.3.1.1 and Table 10-2), for all PTP Instances using these PTP Ports, regardless of the attributes of PTP Instances downstream from the ONU. This behavior is achieved using the acceptable timeTransmitter table feature defined in 17.5 of IEEE Std 1588-2019.

A PTP Instance that contains an ONU port shall maintain a configured table, the acceptableTimeTransmitterTable, and a per-PTP Port Boolean variable acceptableTimeTransmitterTableEnabled. The data type of acceptableTimeTransmitterTable is AcceptableTimeTransmitterTable (see 13.1.3.2).

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13.1.3.2 AcceptableTimeTransmitterTable

The AcceptableTimeTransmitterTable type represents a table of AcceptableTimeTransmitter entries.

```
struct AcceptableTimeTransmitterTable {
        UInteger16 maxTableSize;
        UInteger16 actualTableSize;
        AcceptableTimeTransmitter[actualTableSize] acceptableTimeTransmitter;
}
```

The maxTableSize member is the maximum size of the AcceptableTimeTransmitterTable. The actualTableSizeMember is the actual size of the AcceptableTimeTransmitterTable. The AcceptableTimeTransmitter array contains a list of AcceptableTimeTransmitter PTP Ports. The value of maxTableSize is implementation specific. actualTableSize shall be less than or equal to maxTableSize.

An AcceptableTimeTransmitterTable is configurable and may contain a number of AcceptableTimeTransmitter entries up to maxTableSize.

13.1.3.3 AcceptableTimeTransmitter

The AcceptableTimeTransmitter type represents a PTP Port that can be considered, in the execution of the BTCA, as a candidate for timeTransmitter.

```
struct AcceptableTimeTransmitter {
        PortIdentity acceptablePortIdentity;
        UInteger8 alternatePriority1;
}
```

The acceptablePortIdentity member is the PortIdentity of an acceptable timeTransmitter port. The alternatePriority1 member contains an alternate value for the priority1 attribute of the acceptable timeTransmitter port (see 13.1.3.4).

13.1.3.4 Acceptable timeTransmitter table feature

The acceptable timeTransmitter table feature shall modify the operation of the BTCA (see 10.3) as follows:

a) If acceptableTimeTransmitterTableEnabled for a PTP Port is FALSE, the BTCA operates as described in 10.3.

b) If acceptableTimeTransmitterTableEnabled for a PTP Port is TRUE, then the following apply:

   1) The function qualifyAnnounce() of the PortAnnounceReceive state machine (see 10.3.11.2.1) is replaced by the following:

      qualifyAnnounce (rcvdAnnouncePtr): qualifies the received Announce message pointed to by rcvdAnnouncePtr as follows:

      i) If the Announce message was sent by the current PTP Instance, i.e., if sourcePortIdentity.clockIdentity (see 10.6.2.2.11 and 8.5.2) is equal to thisClock (see 10.2.4.22), the Announce message is not qualified, and FALSE is returned;

      ii) If the stepsRemoved field is greater than or equal to 255, the Announce message is not qualified, and FALSE is returned;

      iii) If the sourcePortIdentity of the Announce message is not equal to the sourcePortIdentity of one of the entries of the acceptableTimeTransmitterTable, FALSE is returned;

      iv) If a path trace TLV is present and one of the elements of the pathSequence array field of the path trace TLV is equal to thisClock (i.e., the clockIdentity of the current PTP Instance; see 10.2.4.22), the Announce message is not qualified, and FALSE is returned;

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v) Otherwise, the Announce message is qualified, and TRUE is returned. If a path trace TLV is present, it is saved in the per port global variable receivedPathTrace. If a path trace TLV is not present, the per port global variable receivedPathTrace is set to the empty array.

2) If the alternatePriority1 member of the AcceptableTimeTransmitter array element that corresponds to the sourcePortIdentity of a received Announce message is 0, the alternatePriority1 member has no effect on the operation of the BTCA.

3) If the alternatePriority1 member of the AcceptableTimeTransmitter array element that corresponds to the sourcePortIdentity of a received Announce message is greater than 0, the value of the grandmasterPriority1 field of the Announce message is replaced by the value of alternatePriority1 of this AcceptableTimeTransmitter array element for use in the invocation of the BTCA.

13.1.3.5 Default configuration of acceptable timeTransmitter table feature

The default configuration of the acceptable timeTransmitter table feature for a PTP Instance that is attached to an IEEE 802.3 EPON link shall be as follows:

a) If the PTP Instance does not contain an ONU port, the default acceptableTimeTransmitterTable is empty, i.e., the member actualTableSize is 0 and there are no AcceptableTimeTransmitter array entries. The variable acceptableTimeTransmitterTableEnabled for each PTP Port is set to FALSE.

b) If the PTP Instance contains an ONU port, the default acceptableTimeTransmitterTable contains one element in the AcceptableTimeTransmitter array. The member actualTableSize is 1. The acceptablePortIdentity of that element is set equal to the portIdentity of the OLT port that the ONU port is attached to, and alternatePriority1 set equal to 244. The variable acceptableTimeTransmitterTableEnabled for each PTP Port is set to TRUE.

NOTE—These default settings ensure that, with the default priority1 values of 8.6.2.1, Table 8-1, used for all PTP Instances, the PTP Instance that contains the ONU port will consider Announce messages only from the OLT that the ONU port is attached to when invoking the BTCA. The alternatePriority1 value of 244 ensures that the OLT will be considered better than the ONU in the sense of the BTCA, which will cause the OLT port state to be set to TimeTransmitterPort and the ONU port state to be set to TimeReceiverPort. All other PTP Ports of this PTP Instance that are not disabled and for which asCapable is TRUE will have PTP Port states of either TimeTransmitterPort or PassivePort. If all PTP Instances downstream from the ONU have priority1 greater than 244, then the PTP Port at the other end of each link attached to each non-ONU port that is not disabled and for which asCapable is TRUE will have PTP Port states of either TimeReceiverPort or PassivePort; in this case, the downstream network portions will get their timing through the EPON. However, if a downstream PTP Instance has priority1 less than 244, or priority1 equal to 244 and is better than the Grandmaster PTP Instance information contained in the Announce message received by the ONU based on other attributes, then the portion of the network that is downstream of the ONU and includes that better PTP Instance will get its timing from that better downstream PTP Instance. In this case, the endpoints of the link of that network portion attached to the PTP Instance that contains the ONU will both have PTP Port states of TimeTransmitterPort, and the PTP Ports at each end of the link will send Announce messages. However, the Announce messages sent by the downstream PTP Instance will be ignored by the PTP Instance that contains the ONU because the sourcePortIdentity of those Announce messages will not be contained in the acceptableTimeTransmitterTable. The Announce messages sent by the PTP Instance that contains the ONU will be used in the invocation of the BTCA at the downstream PTP Instance; however, those Announce messages will not reflect the best timeTransmitter because one of the downstream PTP Instances is better.

13.1.4 Time synchronization in EPON

Transmission in the EPON downstream direction (from OLT to ONUs) utilizes time division multiplexing (TDM). In the upstream direction (from ONUs to OLT), time division multiple access (TDMA) is employed. Due to the frame queuing in TDMA, the downstream delay is different from the upstream delay. Asymmetric delay also occurs in the EPON physical layer due to upstream and downstream transmission using different wavelengths. The index of refraction is frequency dependent, which results in the upstream and downstream delays being asymmetric. The accurate time synchronization across the EPON links is

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operated as follows. It is assumed that the clock timeTransmitter (the OLT) has an accurate synchronized time. The clock timeTransmitter informs the clock timeReceiver (the ONU) what the accurate synchronized time will be when the counter of the clock timeReceiver reaches a certain value. The information transfer can be accomplished using the organization-specific slow protocol (OSSP) message (see Clause 57 of IEEE Std 802.3-2022).

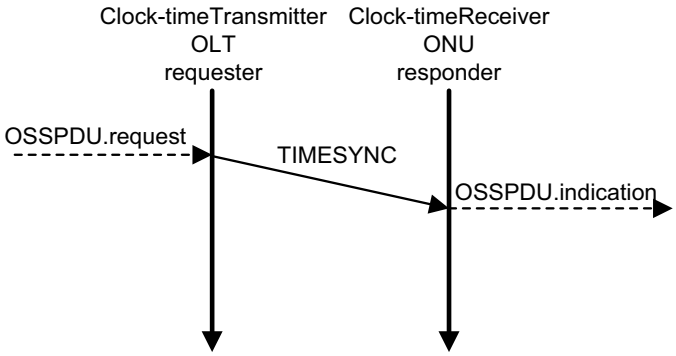


Figure 13-1—IEEE 802.3 EPON time-synchronization interfaces

The following reference process, illustrated schematically in Figure 13-1, will result in the clock timeReceiver of an ONU being synchronized to the clock timeTransmitter of the OLT:

a) The clock timeTransmitter selects a value *X* of the local MPCP counter that is used as the timing reference. Any value can be chosen, provided it is relative to the current epoch of the MPCP counter.

b) The clock timeTransmitter calculates the $ToD_{X,i}$ based on $ToD_{X,o}$ using Equation (13-1).

$$ToD_{X,i} = ToD_{X,o} + RTT_i \times \frac{\text{ndown}}{(\text{nup} + \text{ndown})} \times \text{rateRatio} \tag{13-1}$$

where $ToD_{X,i}$ is the synchronized time when the MPCP counter at the clock timeReceiver *i* reaches a value equal to the timestamp X minus the *onuLatencyFactor*; $ToD_{X,o}$ is the synchronized time when the MPCP counter at the clock timeTransmitter reaches a value equal to the timestamp X plus the *oltLatencyFactor*; RTT_i is the round-trip time measured by the clock timeTransmitter for clock timeReceiver *i*, i.e., ONU *i*; nup is the effective refraction index of the light propagating in the upstream channel; ndown is the effective refraction index of the light propagating in the downstream channel; and rateRatio is the rateRatio member of the most recently received MDSyncSend structure. The *onuLatencyFactor* and *oltLatencyFactor* are given in Equation (13-2) and Equation (13-3), respectively. The impact of the worst-case variation in the transmission wavelength for the clock timeTransmitter and clock timeReceiver transmitters is examined in appendix VII of ITU-T G.984.3, Amendment 2 (11/2009).

$$onuLatencyFactor = onuIngressLatency - (onuIngressLatency + onuEgressLatency) \times \frac{\text{ndown}}{(\text{nup} + \text{ndown})} \times \text{rateRatio} \tag{13-2}$$

$$oltLatencyFactor = oltEgressLatency - (oltIngressLatency + oltEgressLatency) \times \frac{\text{ndown}}{(\text{nup} + \text{ndown})} \times \text{rateRatio} \tag{13-3}$$

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c) The clock timeTransmitter sends the pair of values (X, $ToD_{X,i}$) to clock timeReceiver i via the downstream TIMESYNC message.

NOTE—After the clock timeReceiver receives the downstream TIMESYNC message, it can compute the synchronized time, *ToD*, when the value of the local MPCP counter is equal to *S*; *ToD* is given by the following equation:

$$ToD = ToD_{X,i} + [(S - X) \bmod (2^{32})](16\text{ ns})(\text{rateRatio})$$

where (*A*) mod (*B*) is *A* modulo *B*.

The OSSP message is a general message (see 3.13), analogous to Follow_Up. Note that the preceding synchronized time values correspond to timestamps that are referenced to the MAC control sublayer. Both the clock timeTransmitter and clock timeReceiver are responsible for compensating their processing delays (e.g., the ingressLatency and egressLatency, as described in 8.4.3). RTT_i is measured using MPCPDU timestamps, inserted into the frame structure as specified by 64.2.1.1 and 77.2.1.1 of IEEE Std 802.3-2022.

13.2 Message attributes

13.2.1 Message class

The TIMESYNC message is a general message (see 3.13 and 8.4.2.2). It is transmitted in the downstream direction, from OLT to ONU.

13.3 Message format

13.3.1 TIMESYNC message

13.3.1.1 General TIMESYNC message specifications

The fields of the body of the TIMESYNC message shall be as specified in Table 13-1 and 13.3.1.2.

13.3.1.2 TIMESYNC message field specifications

13.3.1.2.1 Destination Address (Octet6)

The Destination Address field is equal to 01-80-C2-00-00-02 (see 57A.3 of IEEE Std 802.3-2022).

13.3.1.2.2 Source Address (Octet6)

The Source Address field is the individual MAC address associated with the port through which the TIMESYNC message is transmitted (see 57B.1.1 of IEEE Std 802.3-2022).

13.3.1.2.3 Length/Type (Octet2)

The value of this field is equal to 0x8809 (see 57A.4 of IEEE Std 802.3-2022).

13.3.1.2.4 Subtype (Octet)

The value of this field is equal to 0x0A (see 57A.4 of IEEE Std 802.3-2022).

13.3.1.2.5 OUI or CID (Octet3)

This field contains the OUI or CID that identifies the Organization-Specific Data. The value is 00-80-C2, i.e., the OUI assigned to IEEE 802.1.

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Table 13-1—TIMESYNC message fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
Destination Address								6	0
Source Address								6	6
Length/Type								2	12
Subtype								1	14
OUI or CID								3	15
Message Identifier								2	18
X								4	20
$ToD_{X,i}$								10	24
sourcePortIdentity								10	34
logMessageInterval								1	44
rateRatio								8	45
gmTimeBaseIndicator								2	53
lastGmPhaseChange								12	55
scaledLastGmFreqChange								4	67
domainNumber								1	71
majorSdoId				reserved				1	72
minorSdoId								1	73
reserved								0	74
FCS								4	

13.3.1.2.6 Message Identifier (Octet2)

This field is the TIMESYNC message identifier. The value of this field is 1.

13.3.1.2.7 *X* (UInteger32)

The *X* field is the selected timestamp that will be used as the timing reference as specified in 13.1.4.

13.3.1.2.8 $ToD_{X,i}$ (Timestamp)

$ToD_{X,i}$ is the synchronized time when the MPCP counter at the clock timeReceiver *i* reaches a value equal to *X* minus the *onuLatencyFactor* (see 13.1.4). *X* is carried in the respective TIMESYNC message. Synchronization of the MPCP clock is described in detail in 64.2.1.1 and 77.2.1.1 in IEEE Std 802.3-2022, for 1G-EPON and 10G-EPON, respectively.

NOTE—Any subnanosecond portion of synchronized time (in this case, time of day), normally transported in a correction field (see 10.2.2.1.2, 10.2.2.2.2, and 10.2.2.3.5), is not transported over EPON.

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13.3.1.2.9 sourcePortIdentity (PortIdentity)

This field is specified as the sourcePortIdentity member of the MDSyncSend structure most recently received from the PortSync entity of the OLT (see 10.2.2.1.4).

13.3.1.2.10 logMessageInterval (Integer8)

This field is specified as the logMessageInterval member of the MDSyncSend structure most recently received from the PortSync entity of the OLT (see 10.2.2.1.5). It is the value of the currentLogSyncInterval for this PTP Port (see 10.7.2.3).

13.3.1.2.11 rateRatio (Float64)

This field is specified as the rateRatio member of the MDSyncSend structure most recently received from the PortSync entity of the OLT (see 10.2.2.1.8).

13.3.1.2.12 gmTimeBaseIndicator (UInteger16)

This field is specified as the gmTimeBaseIndicator member of the MDSyncSend structure most recently received from the PortSync entity of the OLT (see 10.2.2.1.9).

13.3.1.2.13 lastGmPhaseChange (ScaledNs)

This field is specified as the lastGmPhaseChange member of the MDSyncSend structure most recently received from the PortSync entity of the OLT (see 10.2.2.1.10).

13.3.1.2.14 scaledLastGmFreqChange (Integer32)

The value of scaledLastGmFreqChange is the fractional frequency offset of the current Grandmaster Clock relative to the previous Grandmaster Clock, at the time that the current Grandmaster PTP Instance became the Grandmaster PTP Instance, or relative to itself prior to the last change in gmTimeBaseIndicator, multiplied by 2^{41} and truncated to the next smaller signed integer. The value is obtained by multiplying the lastGmFreqChange member of MDSyncSend (see 10.2.2.1) whose receipt causes the MD entity to send the TIMESYNC message by 2^{41}, and truncating to the next smaller signed integer.

NOTE—The above scaling allows the representation of fractional frequency offsets in the range $[-(2^{-10} - 2^{-41}), 2^{-10} - 2^{-41}]$, with granularity of 2^{-41}. This range is approximately $[-9.766 \times 10^{-4}, 9.766 \times 10^{-4}]$.

13.3.1.2.15 domainNumber (UInteger8)

This field is specified as the gPTP domainNumber (see 8.1).

13.3.1.2.16 majorSdoId (Nibble)

The value is the same as the value specified in 8.1 for all transmitted PTP messages of a gPTP domain. Any TIMESYNC message received for which the value is not one of the values specified in 8.1 shall be ignored.

NOTE—The nibble that immediately follows majorSdoId is reserved (see 10.6.1).

13.3.1.2.17 minorSdoId (UInteger8)

The value is the same as the value specified in 8.1 for all transmitted PTP messages of a gPTP domain. Any TIMESYNC message received for which the value is not one of the values specified in 8.1 shall be ignored.

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13.3.1.2.18 reserved

The reserved field that follows minorSdoId has variable length. The field shall have zero length on transmission. Any bytes between the minorSdoId and the FCS field shall be ignored on reception.

13.3.1.2.19 FCS (Octet4)

This field is the frame check sequence (see 57B.1.1 of IEEE Std 802.3-2022).

13.4 Determination of asCapable

The default value of the per-PTP Port, per-domain global variable asCapable shall be TRUE.

The per-PTP Port, per-domain global variable asCapable shall be set to TRUE if the value of neighborGptpCapable for this PTP Port is TRUE.

NOTE—The above conditions ensure backward compatibility with the 2011 edition of this standard. A time-aware system that is compliant with the 2011 edition of this standard will not process the gPTP-capable TLV, and asCapable will be determined as specified in the 2011 edition. A PTP Instance of a time-aware system compliant with the current edition of this standard that is attached, via an EPON link, to a node compliant with the 2011 edition of this standard will not receive Signaling messages that contain the gPTP-capable TLV and will not set neighborGptpCapable to TRUE. However, the above ensures that asCapable for this PTP Port and domain (i.e., domain 0) will still be set in a manner consistent with that of the 2011 edition of this standard because the default value of asCapable is TRUE in that edition.

13.5 Layering for IEEE 802.3 EPON links

The MD entity is media-dependent and is responsible for translating the media-independent layer to media-dependent PDUs or primitives as necessary for communicating synchronized time over the EPON link from the OLT to a single ONU. This implies that if one OLT port is associated with multiple ONUs, it will require one IEEE 802.1AS PortSync entity and one MD entity per associated ONU. The OSSPDU primitives are used to communicate synchronized time information. Figure 13-2 illustrates how the MD entity interacts with the OSSP sublayer.

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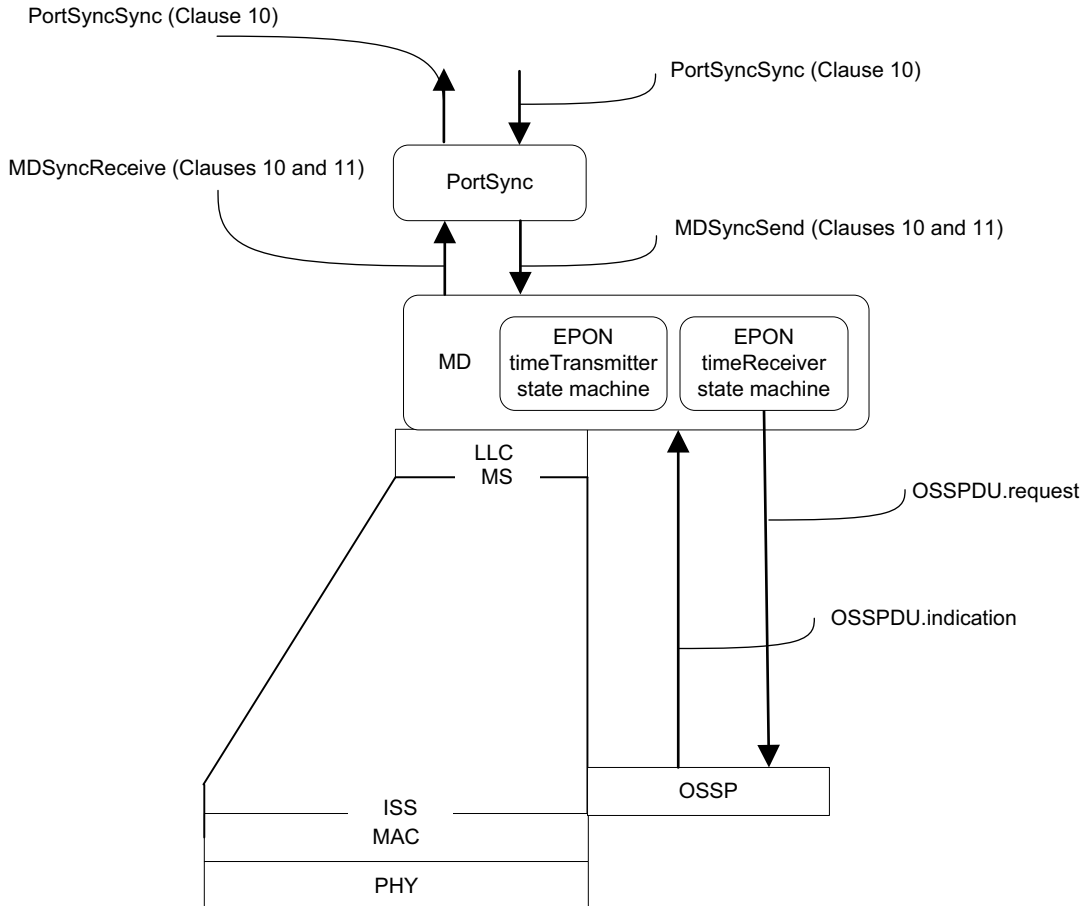


Figure 13-2—IEEE 802.3 EPON interface model

13.6 Service interface definitions

13.6.1 OSSPDU.request

13.6.1.1 General

This service interface primitive is generated periodically by the MD entity of the clock timeTransmitter every sync interval (see 10.7.2.1). It triggers transmission of a TIMESYNC message from the clock timeTransmitter to the clock timeReceiver. The values of the parameters of the primitive are sent to the clock timeReceiver via the TIMESYNC message.

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13.6.1.2 OSSPDU.request parameters

13.6.1.2.1 Syntax of the primitive

```
OSSPDU.request {
        X
        ToD_X,i
        sourcePortIdentity
        logMessageInterval
        rateRatio
        gmTimeBaseIndicator
        lastGmPhaseChange
        scaledLastGmFreqChange
        domainNumber
        sdold
}
```

13.6.1.2.2 *X* (Integer32)

The *X* field is the selected timestamp that will be used as the timing reference as specified in 13.1.4.

13.6.1.2.3 $ToD_{X,i}$ (Timestamp)

$ToD_{X,i}$ is the synchronized time when the MPCP counter at the clock timeReceiver *i* reaches a value equal to *X* minus the *onuLatencyFactor* (see 13.1.4). *X* is carried in the respective TIMESYNC message. Synchronization of the MPCP clock is described in detail in 64.2.1.1 and 77.2.1.1 in IEEE Std 802.3-2022, for 1G-EPON and 10G-EPON, respectively.

13.6.1.2.4 sourcePortIdentity (PortIdentity)

This parameter identifies the sourcePortIdentity value for this PTP Port (see 13.3.1.2.9).

13.6.1.2.5 logMessageInterval (Integer8)

This parameter identifies the currentLogSyncInterval value for this PTP Port (see 13.3.1.2.10).

13.6.1.2.6 rateRatio (Float64)

This parameter identifies the rateRatio value for this PTP Port (see 13.3.1.2.11).

13.6.1.2.7 gmTimeBaseIndicator (UInteger16)

This parameter identifies the gmTimeBaseIndicator value for this PTP Port (see 13.3.1.2.12).

13.6.1.2.8 lastGmPhaseChange (ScaledNs)

This parameter identifies the lastGmPhaseChange value for this PTP Port (see 13.3.1.2.13).

13.6.1.2.9 scaledLastGmFreqChange (Integer32)

This parameter identifies the scaledLastGmFreqChange value for this PTP Port (see 13.3.1.2.14).

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13.6.1.2.10 domainNumber

This parameter identifies the domainNumber for this instance of gPTP (see 13.3.1.2.15).

13.6.1.2.11 sdoId

This parameter identifies the sdoId for this instance of gPTP (see 13.3.1.2.16 and 13.3.1.2.17).

13.6.1.3 When generated

This primitive is generated by the clock timeTransmitter every $2^{\text{currentLogSyncInterval}}$ seconds when it is in the TIME_TRANSMITTER state, as the first phase of synchronized time information transfer.

13.6.1.4 Effect of receipt

Upon receipt of this primitive, a TIMESYNC message is enqueued for transmission.

NOTE—Arrival of the TIMESYNC message at the ONU after the selected time *X* does not impede proper operation of the synchronization mechanism defined in this clause.

13.6.2 OSSPDU.indication

13.6.2.1 General

This service interface primitive is generated on receipt of a TIMESYNC message by the responder, and provides the values contained in the corresponding OSSPDU.request primitive to the clock timeReceiver.

13.6.2.2 OSSPDU.indication parameters

OSSPDU.indication {
    X
    $ToD_{X,i}$
    sourcePortIdentity
    logMessageInterval
    rateRatio
    gmTimeBaseIndicator
    lastGmPhaseChange
    scaledLastGmFreqChange
    domainNumber
    sdoId
}

The parameters of the OSSPDU.indication are set equal to the corresponding fields of the most recently received TIMESYNC message. Their definitions are given in 13.3.1.2.7 through 13.3.1.2.17, respectively.

13.6.2.3 When generated

This primitive is generated by the receipt of a TIMESYNC message during the phase of synchronized time information transfer.

13.6.2.4 Effect of receipt

Upon receipt, the OSSPDU.indication parameters are used by the MD entity to compute the parameters of the MDSyncReceive structure that will be transmitted to the PortSync entity of this PTP Port.

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13.7 MD entity global variables

13.7.1 RTT_i: Is used only by the OLT MD entity. RTT_i is the RTT between the clock timeTransmitter and clock timeReceiver. The data type for RTT_i is UInteger32.

NOTE—RTT is measured and updated by the MPCP using the mechanism specified in IEEE Std 802.3-2022 and stored in RTT_i when measured and updated. RTT_i is not used by the ONU and is set to zero in an ONU MD entity.

13.8 State machines

13.8.1 Requester state machine

13.8.1.1 Function

This state machine generates and consumes primitives, at the requester, used to provide accurate synchronized time across EPON links to the responder.

13.8.1.2 State machine variables

The following variables are used in the state diagram in Figure 13-3 (in 13.8.1.4):

13.8.1.2.1 ndown: The effective index of the light propagating in the downstream channel. The data type for ndown is Float64.

13.8.1.2.2 nup: The effective index of the light propagating in the upstream channel. The data type for ndown is Float64.

13.8.1.2.3 rcvdMDSyncEponReq: A Boolean variable that notifies the current state machine when an MDSyncSend structure is received. This variable is reset by the current state machine.

13.8.1.2.4 rcvdMDSyncPtrEponReq: A pointer to the received MDSyncSend structure.

13.8.1.2.5 registered: A Boolean variable that indicates an ONU has registered to EPON.

13.8.1.2.6 $ToD_{X,i}$: The synchronized time when the MPCP counter at the clock timeReceiver *i* reaches a value equal to *X* (see 13.8.1.2.8) minus the *onuLatencyFactor* (see 13.1.4). The data type for $ToD_{X,i}$ is Timestamp.

13.8.1.2.7 $ToD_{X,o}$: The synchronized time when the MPCP counter at the clock timeTransmitter reaches a value equal to *X* (see 13.8.1.2.8) plus the *oltLatencyFactor* (see 13.1.4). The data type for $ToD_{X,o}$ is Timestamp.

13.8.1.2.8 *X*: The value of the timestamp [see item a) of 13.1.4] that is selected as the reference time. The data type for *X* is UInteger32.

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13.8.1.3 State machine functions

13.8.1.3.1 setToDXo(): Computes the state machine variable $ToD_{X,o}$ (see 13.8.1.2.7) as the sum of the following:

a) The preciseOriginTimestamp member of the most recently received MDSyncSend structure
b) The followUpCorrectionField of the most recently received MDSyncSend structure
c) The quantity in Equation (13-4):

$$\text{rateRatio} \times (X \times (16\ \text{ns}) - \text{upstreamTxTime}) \quad (13\text{-}4)$$

where

rateRatio	is the rateRatio member of the most recently received MDSyncSend structure
upstreamTxTime	is the upstreamTxTime member of the most recently received MDSyncSend structure
X	is defined in 13.8.1.2.8

13.8.1.4 State diagram

The requester state machine shall implement the function specified by the state diagram in Figure 13-3, the local variables specified in 13.8.1.2, the function specified in 13.8.1.3, the service interface primitives specified in 13.6, the structure specified in 10.2.2.1, the message specified in 13.3, and the relevant global variables specified in 10.2.5 and 13.7. The state machine receives an MDSyncSend structure from the PortSyncSyncSend state machine of the PortSync entity of this PTP Port and transmits an OSSPDU.request primitive to cause a TIMESYNC message to be sent to the responder (ONU).

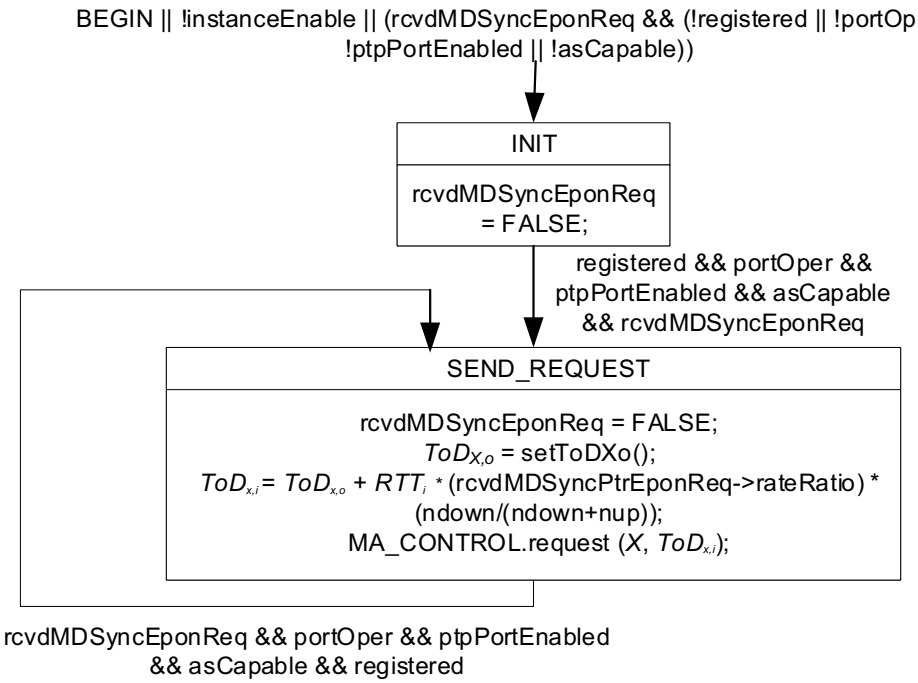


Figure 13-3—State machine for IEEE 802.3 EPON requester

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13.8.2 Responder state machine

13.8.2.1 Function

This state machine responds to EPON-specific primitives generated by receipt of a TIMESYNC message from the requester.

13.8.2.2 State machine variables

The following variables are used in the state diagram in Figure 13-4 (in 13.8.2.4):

13.8.2.2.1 rcvdOSSPDUind: a Boolean variable that notifies the responder state machine when a TIMESYNC message is received and the OSSPDU.indication primitive is generated.

13.8.2.2.2 txMDSyncReceivePtrEponResp: a pointer to a structure whose members contain the values of the parameters of an MDSyncReceive structure to be transmitted.

13.8.2.2.3 rcvdOSSPDUptr: a pointer to a structure whose members contain the values of the parameters of the OSSPDU.indication primitive whose receipt is indicated by rcvdOSSPDUind (see 13.8.2.2.1).

13.8.2.3 State machine functions

13.8.2.3.1 setMDSyncReceiveEponResp(): creates an MDSyncReceive structure (see 10.2.2.2) using members of the structure pointed to by rcvdOSSPDUptr (see 13.8.2.2.3), and returns a pointer to this structure. The members of this structure are set as follows:

a) followUpCorrectionField is set equal to 0.

b) sourcePortIdentity is set equal to the sourcePortIdentity field of the most recently received TIMESYNC message (see 13.3.1.2.9).

c) logMessageInterval is set equal to the logMessageInterval of the most recently received TIMESYNC message (see 13.3.1.2.10).

d) preciseOriginTimestamp is set equal to the $ToD_{X,i}$ field of the most recently received TIMESYNC message (see 13.3.1.2.8).

e) rateRatio is set to the rateRatio of the most recently received TIMESYNC message (see 13.3.1.2.11).

f) upstreamTxTime is set equal to *X* multiplied by 16 ns, where *X* is the value of the *X* field of the most recently received TIMESYNC message (see 13.3.1.2.7).

g) gmTimeBaseIndicator is set equal to the gmTimeBaseIndicator of the most recently received TIMESYNC message (see 13.3.1.2.12).

h) lastGmPhaseChange is set equal to the lastGmPhaseChange of the most recently received TIMESYNC message (see 13.3.1.2.13).

i) lastGmFreqChange is set equal to the scaledLastGmFreqChange of the most recently received TIMESYNC message (see 13.3.1.2.14), divided by 2^{41}.

j) domainNumber is set equal to the domainNumber of the most recently received TIMESYNC message (see 13.3.1.2.15).

13.8.2.3.2 txMDSyncReceive (txMDSyncReceivePtrEponResp): transmits an MDSyncReceive structure to the PortSyncSyncReceive state machine of the PortSync entity of this PTP Port.

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13.8.2.4 State diagram

The responder state machine shall implement the function specified by the state diagram in Figure 13-4, the local variables specified in 13.8.2.2, the functions specified in 13.8.2.3, the service interface primitives specified in 13.6, the structure specified in 10.2.2.2, the message specified in 13.3, and the relevant global variables specified in 10.2.5 and 13.7. The state machine receives an OSSPDU.indication primitive in response to its having received a TIMESYNC message from the requester (OLT), and transmits an MDSyncReceive structure to the PortSync entity of this PTP Port.

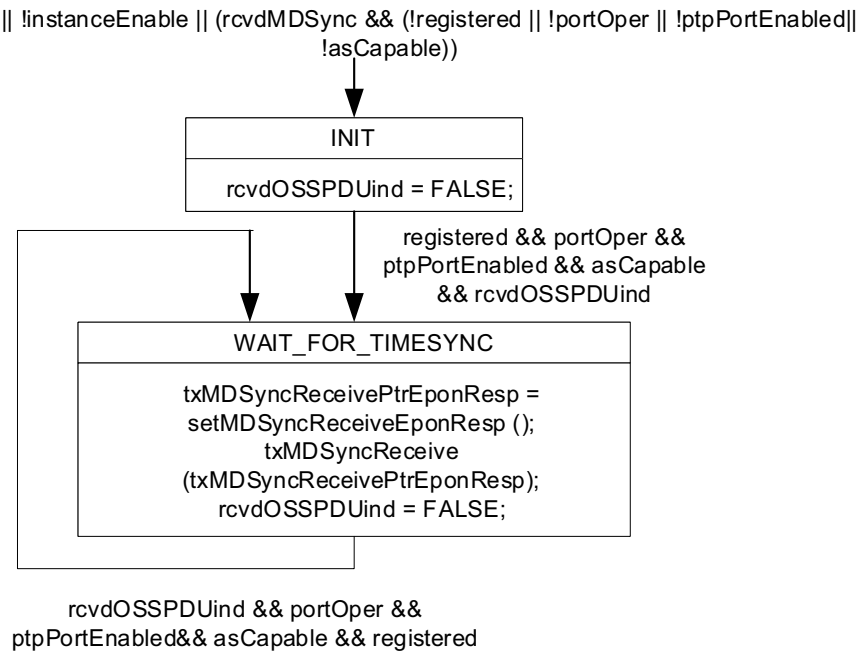


Figure 13-4—State machine for IEEE 802.3 EPON responder

13.9 Message transmission intervals

13.9.1 General interval specification

The mean time interval between successive TIMESYNC messages shall be as specified in 10.7.2.1, 10.7.2.3, and 13.9.2.

13.9.2 TIMESYNC message transmission interval default value

The default value of initialLogSyncInterval (see 10.7.2.4) is –3. Every PTP Port supports the value 127; the PTP Port does not send TIMESYNC messages when currentLogSyncInterval has this value. A PTP Port may support other values, except for the reserved values indicated in Table 10-16.

Processing of the message interval request TLV carried in a Signaling message (see 10.6.4) shall be supported, as specified by the SyncIntervalSetting state machine of 10.2.14 (see Figure 10-10), except that the logLinkDelayInterval (which is not relevant to IEEE 802.3 EPON ports) is set to 128 by the sender of the Signaling message, the logLinkDelayInterval is ignored by the receiver, and unsupported values of logTimeSyncInterval are ignored by the receiver.

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14. Timing and synchronization management

14.1 General

14.1.1 Data set hierarchy

This clause defines the set of managed objects, and their functionality, that allow administrative configuration of clock parameters and timing and synchronization protocols.

Management data models typically represent data for the physical device (i.e., time-aware system). The specifications for discovery, management address, and security for the physical device are typically covered by standards of the management mechanism, which are outside the scope of this standard. For the management information model of this standard, the scope of work is the data contained within a time-aware system. From a management perspective, the time-aware system contains a list of one or more PTP Instances. Each entry in the list is a set of managed data sets for the respective PTP Instance.

Conformance for each managed object is optional. This standard operates correctly using default values; therefore, management is not essential. Since the management mechanism is not limited to remote protocols (e.g., SNMP, NETCONF), management can use a local mechanism with a simple interface (e.g., DIP switches). Therefore, each product can determine the support of managed objects as appropriate for its management mechanism.

The following hierarchy summarizes the managed data sets within a gPTP Node:

a) instanceList[]
    1) defaultDS (see 14.2)
    2) currentDS (see 14.3)
    3) parentDS (see 14.4)
    4) timePropertiesDS (see 14.5)
    5) pathTraceDS (see 14.6)
    6) acceptableTimeTransmitterTableDS (see 14.7)
    7) ptpInstanceSyncDS (see 14.8)
    8) driftTrackingDS (see 14.9)
    9) portList[]
        i) portDS (see 14.10)
        ii) descriptionPortDS (see 14.11)
        iii) portStatisticsDS (see 14.12)
        iv) acceptableTimeTransmitterPortDS (see 14.13)
        v) externalPortConfigurationPortDS (see 14.14)
        vi) asymmetryMeasurementModeDS (see 14.15)
        vii) commonServicesPortDS (see 14.16)
b) commonServices
    1) commonMeanLinkDelayService
        i) cmldsDefaultDS (see 14.17)
        ii) cmldsLinkPortList[]
            — cmldsLinkPortDS (see 14.18)
            — cmldsLinkPortStatisticsDS (see 14.19)
            — cmldsAsymmetryMeasurementModeDS (see 14.20)

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2) hotStandbyService
   i) hotStandbySystemList[]
      — hotStandbySystemDS (see 14.21)
      — hotStandbySystemDescriptionDS (see 14.22)
3) fttService
   i) fttEntityList[]
      — fttEntityDS (see 14.23)
      — fttEntityDescriptionDS (see 14.24)

The instanceList is a structure for the gPTP datasets, and it is indexed using a number that is unique per PTP Instance within the time-aware system, applicable to the management context only (i.e., not used in PTP messages). The domainNumber of the PTP Instance must not be used as the index to instanceList since it is possible for a time-aware system to contain multiple PTP Instances using the same domainNumber. The portList is indexed using a number that is unique per logical port (i.e., PTP Port) in the PTP Instance (see 8.5.1). Since the portNumber of a logical port can have any value in the range 1, 2, 3, ..., 0xFFFE (see 8.5.2.3), the portList index and portNumber values for a logical port are not necessarily the same. PTP Instances and logical ports may be created or deleted dynamically in implementations that support dynamic create/delete of devices. Unless otherwise indicated, the data sets and managed objects under the instanceList[] are maintained separately for each PTP Instance supported by the time-aware system.

The commonServices is an overall structure containing the commonMeanLinkDelayService structure for the data sets used for CMLDS, the hotStandbyService structure for the data sets used for Hot Standby Service, and the fttService structure for the data sets used for fault-tolerant timing with time integrity service.

The commonMeanLinkDelayService structure contains the cmldsLinkPortList, which is a list of CMLDS Link Ports (see 11.2.16), of the time-aware system that runs the common service. The CMLDS must be implemented on every physical port for which there is a PTP Port of a PTP Instance that can use the CMLDS. Therefore, the cmldsLinkPortList[] must include Link Ports that correspond to all such physical ports. As is the case for the portList of a PTP Instance, the cmldsLinkPortList is indexed using a number that is unique per Link Port that invokes the CMLDS (see 8.5.1). Since the portNumber of a logical port (i.e., PTP Port or CMLDS Link Port) can have any value in the range 1, 2, 3, ..., 0xFFFE (see 8.5.2.3), the cmldsLinkPortList index and cmldsLinkPortDS.portIdentity.portNumber values for a logical port of the CMLDS are not necessarily the same. CMLDS Link Ports may be created or deleted dynamically in implementations that support dynamic create/delete of devices.

The CMLD Data Sets are not maintained separately for each PTP Instance. Rather, a single copy of the commonServices.cmldsDefaultDS is maintained for the time-aware system, and a single copy of each data set under the cmldsLinkPortList[] is maintained per Link Port of the time-aware system.

A PTP Instance can use the commonServicesPortDS to determine which Link Port it must use when it obtains information provided by the CMLD (see 14.16).

The hotStandbyService structure contains the hotStandbySystemList, which is a list of instances of the Hot Standby Service. Since a Hot Standby Service Instance is associated with two PTP Instances, but a time-aware system can have more than two PTP Instances, there can, in general, be multiple instances of the Hot Standby Service (i.e., one Hot Standby Service Instance associated with each set of two distinct PTP Instances). A hotStandbySystemDS and a hotStandbySystemDescriptionDS are maintained for each instance of the Hot Standby Service.

Each instance of the Hot Standby Service (i.e., each hotStandbySystemList member) is associated with two PTP Instances, i.e., the primary PTP Instance and the secondary PTP Instance (see 18.1 and 18.2). The indices of these PTP Instances in the instanceList are contained in the members primaryPtpInstanceIndex

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and secondaryPtpInstanceIndex, respectively, of the hotStandbySystemDS for this instance of the Hot Standby Service.

The fttService structure contains the fttEntityList, which is a list of instances of FTT entities. An fftEntityDS and an fttEntityDescriptionDS are maintained for each FTT entity.

Each FTT entity is associated with a set of PTP Instances. The indices of these PTP Instances are mapped to inputs of the FTT entity instance via the member fttPtpInstancetoInput of the fttEntityDS for this instance of the FTT entity.

14.2 Default Parameter Data Set (defaultDS)

14.2.1 General

The defaultDS represents the native capabilities of a PTP Instance, i.e., a PTP Relay Instance or a PTP End Instance.

The parameters of defaultDS are summarized in Table 14-1 and subsequently specified.

Table 14-1—defaultDS

Name	Data type	Operations supported[a]	References
clockIdentity	ClockIdentity	R	14.2.2
numberPorts	UInteger16	R	14.2.3
clockQualtiy.clockClass	UInteger8	R	14.2.4.2; 7.6.2.5 of IEEE Std 1588-2019
clockQuality.clockAccuracy	Enumeration8	R	14.2.4.3; 7.6.2.6 of IEEE Std 1588-2019
clockQuality.offsetScaledLogVariance	UInteger16	R	14.2.4.4
priority1	UInteger8	RW	14.2.5
priority2	UInteger8	RW	14.2.6
gmCapable	Boolean	R	14.2.7
currentUtcOffset	Integer16	R	14.2.8
currentUtcOffsetValid	Boolean	R	14.2.9
leap59	Boolean	R	14.2.10
leap61	Boolean	R	14.2.11
timeTraceable	Boolean	R	14.2.12
frequencyTraceable	Boolean	R	14.2.13
ptpTimescale	Boolean	R	14.2.14
timeSource	TimeSource	R	14.2.15 and Table 8-2
domainNumber	UInteger8	RW	14.2.16

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Table 14-1—defaultDS

Name	Data type	Operations supported[a]	References
sdoId	UInteger16	R	14.2.17
externalPortConfigurationEnabled	Boolean	RW	14.2.18
instanceEnable	Boolean	RW	14.2.19

[a] R = Read only access; RW = Read/write access.

14.2.2 clockIdentity

The value is the clockIdentity (see 8.5.2.2) of the PTP Instance.

14.2.3 numberPorts

The value is the number of PTP Ports of the PTP Instance (see 8.6.2.8). For an end station the value is 1.

14.2.4 clockQuality

14.2.4.1 General

This is a structure whose data type is ClockQuality (see 6.4.3.8).

14.2.4.2 clockQuality.clockClass

The value is the clockClass of the PTP Instance, which implements the clockClass specifications of 8.6.2.2.

14.2.4.3 clockQuality.clockAccuracy

The value is the clockAccuracy of the PTP Instance, which implements the clockAccuracy specifications of 8.6.2.3.

14.2.4.4 clockQuality.offsetScaledLogVariance

The value is the offsetScaledLogVariance of the PTP Instance, which implements the offsetScaledLogVariance specifications of 8.6.2.4.

14.2.5 priority1

The value is the priority1 attribute of the PTP Instance (see 8.6.2.1).

14.2.6 priority2

The value is the priority2 attribute of the PTP Instance (see 8.6.2.5).

14.2.7 gmCapable

The value is TRUE if the PTP Instance is capable of being a Grandmaster PTP Instance and FALSE if the PTP Instance is not capable of being a Grandmaster PTP Instance.

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14.2.8 currentUtcOffset

The value is the offset between TAI and UTC, relative to the ClockTimeTransmitter entity of this PTP Instance. It is equal to the global variable sysCurrentUtcOffset (see 10.3.9.18). The value is in units of seconds.

The default value is selected as follows:

a) The value is the value obtained from a primary reference if the value is known at the time of initialization, else
b) The value is the current IERS defined value of TAI – UTC (see IERS Bulletin C) when the PTP Instance is designed.

14.2.9 currentUtcOffsetValid

The value is TRUE if the currentUtcOffset, relative to the ClockTimeTransmitter entity of this PTP Instance, is known to be correct. It is equal to the global variable sysCurrentUtcOffsetValid (see 10.3.9.14).

The default value is TRUE if the value of currentUtcOffset is known to be correct; otherwise, it is set to FALSE.

14.2.10 leap59

A TRUE value indicates that the last minute of the current UTC day, relative to the ClockTimeTransmitter entity of this PTP Instance, will contain 59 s. It is equal to the global variable sysLeap59 (see 10.3.9.13).

The value is selected as follows:

a) The value is obtained from a primary reference if known at the time of initialization, else
b) The value is set to FALSE.

14.2.11 leap61

A TRUE value indicates that the last minute of the current UTC day, relative to the ClockTimeTransmitter entity of this PTP Instance, will contain 61 s. It is equal to the global variable sysLeap61 (see 10.3.9.12).

The value is selected as follows:

a) The value is obtained from a primary reference if known at the time of initialization, else
b) The value is set to FALSE.

14.2.12 timeTraceable

The value is set to TRUE if the timescale and the value of currentUtcOffset, relative to the ClockTimeTransmitter entity of this PTP Instance, are traceable to a primary reference standard; otherwise the value is set to FALSE. It is equal to the global variable sysTimeTraceable (see 10.3.9.16).

The value is selected as follows:

a) If the time and the value of currentUtcOffset are traceable to a primary reference standard at the time of initialization, the value is set to TRUE, else
b) The value is set to FALSE.

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14.2.13 frequencyTraceable

The value is set to TRUE if the frequency determining the timescale of the ClockTimeTransmitter Entity of this PTP Instance is traceable to a primary standard; otherwise the value is set to FALSE. It is equal to the global variable sysFrequencyTraceable (see 10.3.9.17).

The value is selected as follows:

a) If the frequency is traceable to a primary reference standard at the time of initialization the value is set to TRUE, else
b) The value is set to FALSE.

14.2.14 ptpTimescale

The value is set to TRUE if the clock timescale of the ClockTimeTransmitter Entity of this PTP Instance is PTP (see 8.2) and FALSE otherwise.

14.2.15 timeSource

The value is the source of time used by the Grandmaster Clock (see 8.6.2.7).

14.2.16 domainNumber

The value is the domainNumber of the gPTP domain for this instance of gPTP supported by the time-aware system (see 8.1).

NOTE—The PTP Instance for which domainNumber is 0 has constraints applied to it, e.g., timescale (see 8.2.1).

14.2.17 sdoId

The value is the sdoId of the gPTP domain for this instance of gPTP supported by the time-aware system (see 8.1).

NOTE—The attribute sdoId is specified as a 12-bit unsigned integer in 8.1. The data type for the managed object sdoId is UInteger16 in Table 14-1, for compatibility with IEEE Std 1588-2019. The range of the managed object is limited to 12 bits; in addition, only the single value 0x100 is specified in this standard for the gPTP domain of a PTP Instance.

14.2.18 externalPortConfigurationEnabled

The value is the externalPortConfigurationEnabled attribute of the PTP Instance (see 10.3.9.24).

14.2.19 instanceEnable

The value is the instanceEnable attribute of the PTP Instance (see 10.2.4.24).

14.3 Current Parameter Data Set (currentDS)

14.3.1 General

The currentDS represents the position of a local system and other information, relative to the Grandmaster PTP Instance.

The parameters of currentDS are summarized in Table 14-2 and subsequently specified.

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Table 14-2—currentDS

Name	Data type	Operations supported[a]	References
stepsRemoved	UInteger16	R	14.3.2
offsetFromTimeTransmitter	TimeInterval	R	14.3.3
lastGmPhaseChange	ScaledNs	R	14.3.4
lastGmFreqChange	Float64	R	14.3.5
gmTimebaseIndicator	UInteger16	R	14.3.6
gmChangeCount	UInteger32	R	14.3.7
timeOfLastGmChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.8
timeOfLastGmPhaseChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.9
timeOfLastGmFreqChangeEvent	UInteger32 (sysUp Time, IETF RFC 3418)	R	14.3.10

[a] R = Read only access; RW = Read/write access.

14.3.2 stepsRemoved

The value is the number of gPTP communication paths traversed between this PTP Instance and the Grandmaster PTP Instance, as specified in 10.3.3.

14.3.3 offsetFromTimeTransmitter

The value is an implementation-specific or profile standard-specific representation of the current value of the time difference between a timeReceiver and the Grandmaster Clock, as computed by the timeReceiver, and as specified in 10.2.10. The value is computed by an algorithm that is implementation-specific or profile standard-specific. The data type shall be TimeInterval. The default value is implementation-specific.

NOTE—For example, the inputs to this implementation-specific algorithm could be the successive values of clockSourcePhaseOffset (see 10.2.4.7) of the ClockSyncOffset state machine (see 10.2.10 and Figure 10-6).

14.3.4 lastGmPhaseChange

The value (see 10.2.4.16) is the phase change that occurred on the most recent change in either Grandmaster PTP Instance or gmTimeBaseIndicator (see 9.2.2.3).

14.3.5 lastGmFreqChange

The value (see 10.2.4.17) is the frequency change that occurred on the most recent change in either Grandmaster PTP Instance or gmTimeBaseIndicator (see 9.2.2.3).

14.3.6 gmTimebaseIndicator

The value is the value of timeBaseIndicator of the current Grandmaster PTP Instance (see 9.2.2.3 and 9.6.2.3).

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14.3.7 gmChangeCount

This statistics counter tracks the number of times the Grandmaster PTP Instance has changed in a gPTP domain. This counter increments when the PortAnnounceInformation state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-15).

14.3.8 timeOfLastGmChangeEvent

This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent Grandmaster PTP Instance change occurred in a gPTP domain. This timestamp is updated when the PortAnnounceInformation state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-15).

14.3.9 timeOfLastGmPhaseChangeEvent

This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent change in Grandmaster Clock phase occurred due to a change of either the Grandmaster PTP Instance or the Grandmaster Clock time base. This timestamp is updated when one of the following occurs:

a) The PortAnnounceInformation state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-15), or

b) The gmTimebaseIndicator managed object (see 14.3.6) changes and the lastGmPhaseChange field of the most recently received Follow_Up information TLV is nonzero.

14.3.10 timeOfLastGmFreqChangeEvent

This timestamp takes the value of sysUpTime (see IETF RFC 3418) when the most recent change in Grandmaster Clock frequency occurred due to a change of either the Grandmaster PTP Instance or the Grandmaster Clock time base. This timestamp is updated when one of the following occurs:

a) The PortAnnounceInformation state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state (see 10.3.12 and Figure 10-15), or

b) The gmTimebaseIndicator managed object (see 14.3.6) changes, and the lastGmFreqChange field of the most recently received Follow_Up information TLV is nonzero.

14.4 Parent Parameter Data Set (parentDS)

14.4.1 General

The parentDS represents capabilities of the upstream system, toward the Grandmaster PTP Instance, as measured at a local system.

The parameters of parentDS are summarized in Table 14-3 and subsequently specified.

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Table 14-3—parentDS

Name	Data type	Operations supported[a]	References
parentPortIdentity	PortIdentity (see 6.4.3.7)	R	14.4.2
cumulativeRateRatio	Integer32	R	14.4.3
grandmasterIdentity	ClockIdentity	R	14.4.4
grandmasterClockQuality.clockClass	UInteger8	R	14.4.5.2; 7.6.2.5 of IEEE Std 1588-2019
grandmasterClockQuality.clockAccuracy	Enumeration8	R	14.4.5.3; 7.6.2.6 of IEEE Std 1588-2019
grandmasterClockQuality.offsetScaledLogVariance	UInteger16	R	14.4.5.4
grandmasterPriority1	UInteger8	R	14.4.6
grandmasterPriority2	UInteger8	R	14.4.7
gmPresent	Boolean	R	14.4.8

[a] R = Read only access; RW = Read/write access.

14.4.2 parentPortIdentity

If this PTP Instance is the Grandmaster PTP Instance, the value is a portIdentity whose clockIdentity is the clockIdentity of this PTP Instance, and whose portNumber is 0.

If this PTP Instance is not the Grandmaster PTP Instance, the value is the portIdentity of the TimeTransmitterPort (see Table 10-7) of the gPTP communication path attached to the single timeReceiver port of this PTP Instance.

The default value is a portIdentity for which the following apply:

a) The clockIdentity member is the value of the clockIdentity member of the default data set.
b) The portNumber member is 0.

14.4.3 cumulativeRateRatio

The value is an estimate of the ratio of the frequency of the Grandmaster Clock to the frequency of the LocalClock entity of this PTP Instance. cumulativeRateRatio is expressed as the fractional frequency offset multiplied by 2^{41}, i.e., the quantity $(\text{rateRatio} - 1.0)(2^{41})$, where rateRatio is computed by the PortSyncSyncReceive state machine (see 10.2.8.1.4).

14.4.4 grandmasterIdentity

The value is the clockIdentity attribute (see 8.5.2.2) of the Grandmaster PTP Instance.

The default value is the value of defaultDS.clockIdentity (14.2.2).

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14.4.5 grandmasterClockQuality

14.4.5.1 General

This is a structure whose data type is ClockQuality (see 6.4.3.8).

14.4.5.2 grandmasterClockQuality.clockClass

The value is the clockClass (see 8.6.2.2) of the Grandmaster PTP Instance.

The default value is the clockClass member of the default data set.

14.4.5.3 grandmasterClockQuality.clockAccuracy

The value is the clockAccuracy (see 8.6.2.3) of the Grandmaster PTP Instance.

The default value is the clock accuracy member of the default data set.

14.4.5.4 grandmasterClockQuality.offsetScaledLogVariance

The value is the offsetScaledLogVariance (see 8.6.2.4) of the Grandmaster PTP Instance.

The default value is the offsetScaledLogVariance member of the default data set.

14.4.6 grandmasterPriority1

The value is the priority1 attribute (see 8.6.2.1) of the Grandmaster PTP Instance.

The default value is the priority1 value of the default data set.

14.4.7 grandmasterPriority2

The value is the priority2 attribute (see 8.6.2.5) of the Grandmaster PTP Instance.

The default value is the priority2 value of the default data set.

14.4.8 gmPresent

The value is the value of the per PTP Instance global variable gmPresent (see 10.2.4.13).

14.5 Time Properties Parameter Data Set (timePropertiesDS)

14.5.1 General

The timePropertiesDS represents capabilities of the Grandmaster PTP Instance, as measured at a local system.

The parameters of timePropertiesDS are summarized in Table 14-4 and subsequently specified.

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Table 14-4—timePropertiesDS

Name	Data type	Operations supported[a]	References
currentUtcOffset	Integer16	R	14.5.2
currentUtcOffsetValid	Boolean	R	14.5.3
leap59	Boolean	R	14.5.4
leap61	Boolean	R	14.5.5
timeTraceable	Boolean	R	14.5.6
frequencyTraceable	Boolean	R	14.5.7
ptpTimescale	Boolean	R	14.5.8
timeSource	TimeSource	R	14.5.9 and Table 8-2

[a] R = Read only access; RW = Read/write access.

14.5.2 currentUtcOffset

The value is currentUtcOffset for the current Grandmaster PTP Instance (see 14.2.8). It is equal to the value of the global variable currentUtcOffset (see 10.3.9.10). The value is in units of seconds.

14.5.3 currentUtcOffsetValid

The value is currentUtcOffsetValid for the current Grandmaster PTP Instance (see 14.2.9). It is equal to the global variable currentUtcOffsetValid (see 10.3.9.6).

14.5.4 leap59

The value is leap59 for the current Grandmaster PTP Instance (see 14.2.10). It is equal to the global variable leap59 (see 10.3.9.5).

14.5.5 leap61

The value is leap61 for the current Grandmaster PTP Instance (see 14.2.11). It is equal to the global variable leap61 (see 10.3.9.4).

14.5.6 timeTraceable

The value is timeTraceable for the current Grandmaster PTP Instance (see 14.2.12). It is equal to the global variable timeTraceable (see 10.3.9.8).

14.5.7 frequencyTraceable

The value is frequencyTraceable for the current Grandmaster PTP Instance (see 14.2.13). It is equal to the global variable frequencyTraceable (see 10.3.9.9).

14.5.8 ptpTimescale

The value is ptpTimescale for the current Grandmaster PTP Instance (see 14.2.14).

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14.5.9 timeSource

The value is timeSource for the current Grandmaster PTP Instance (see 14.2.15). It is equal to the global variable timeSource (see 10.3.9.11).

14.6 Path Trace Parameter Data Set (pathTraceDS)

14.6.1 General

The pathTraceDS represents the current path trace information available at the PTP Instance.

The parameters of pathTraceDS are summarized in Table 14-5 and subsequently specified.

Table 14-5—pathTraceDS

Name	Data type	Operations supported[a]	References
list	ClockIdentity[N], where N is defined in 10.3.9.23	R	14.6.2
enable	Boolean	R	14.6.3

[a] R = Read only access; RW = Read/write access.

14.6.2 list

The value is the array of ClockIdentity values contained in the pathTrace array (see 10.3.9.23), which represents the current path trace information and that is carried in the path trace TLV (see 10.6.3.3).

The initialization value shall be the empty list (i.e., an array of length 0).

14.6.3 enable

The value is TRUE.

NOTE—This member is included for compatibility with IEEE Std 1588-2019. In IEEE Std 1588-2019, the path trace mechanism is optional, and the pathTraceDS.enable member is configurable (its value in IEEE Std 1588-2019 is TRUE or FALSE, depending on whether the path trace mechanism is operational or not operational, respectively. However, the pathTrace mechanism is mandatory in this standard, and the value of enable is always TRUE.

14.7 Acceptable TimeTransmitter Table Parameter Data Set (acceptableTimeTransmitterTableDS)

14.7.1 General

The acceptableTimeTransmitterTableDS represents the acceptable timeTransmitter table used when an EPON port is used by a PTP Instance of a time-aware system.

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The parameters of acceptableTimeTransmitterTableDS are summarized in Table 14-6 and subsequently specified.

Table 14-6—acceptableTimeTransmitterTableDS

Name	Data type	Operations supported[a]	References
maxTableSize	UInteger16	R	14.7.2
actualTableSize	UInteger16	RW	14.7.3
acceptableTimeTransmitterArray	AcceptableTimeTransmitter[actualTableSize] (see 13.1.3.3)	RW	14.7.4

[a] R = Read only access; RW = Read/write access.

14.7.2 maxTableSize

The value is the maximum size of the AcceptableTimeTransmitterTable. It is equal to the maxTableSize member of the AcceptableTimeTransmitterTable structure (see 13.1.3.2).

14.7.3 actualTableSize

The value is the actual size of the AcceptableTimeTransmitterTable. It is equal to the actualTableSize member of the AcceptableTimeTransmitterTable structure (see 13.1.3.2 and 13.1.3.5), i.e., the current number of elements in the acceptable timeTransmitter array. The actual table size is less than or equal to the maxTableSize.

14.7.4 acceptableTimeTransmitterArray

Each element of this array is an AcceptableTimeTransmitter structure (see 13.1.3.3 and 13.1.3.5).

14.8 PTP Instance Synchronization Parameter Data Set (ptpInstanceSyncDS)

14.8.1 General

The ptpInstanceSyncDS describes the synchronization status of the PTP Instance. The ptpInstanceSyncDS shall be implemented if the optional hot standby feature (see Clause 18) is implemented and may be implemented otherwise.

The parameters of ptpInstanceSyncDS are summarized in Table 14-7. and subsequently specified.

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Table 14-7—ptpInstanceSyncDS

Name	Data type	Operations supported[a]	References
isSynced	Boolean	R	14.8.2
offsetFromTimeTransmitterMax	TimeInterval	RW	14.8.3
rxSyncCountTimeReceiverPThresh	UInteger32	RW	14.8.4
offsetMaxExceededCountThresh	UInteger32	RW	14.8.5
offsetMaxMetCountThresh	UInteger32	RW	14.8.6

[a] R = Read only access; RW = Read/write access.

14.8.2 isSynced

The value of the global variable isSynced (see 18.4.1.1).

14.8.3 offsetFromTimeTransmitterMax

The value is the threshold for offsetFromTimeTransmitter (see 18.4.1.2), below which the PTP Instance is considered to be synchronized. For values less than or equal to zero, the PTP Instance is considered synchronized if and only if offsetFromTimeTransmitter is zero. For values greater than zero, the PTP Instance is considered synchronized if and only if:

$$-\text{offsetFromTimeTransmitterThreshold} \leq \text{offsetFromTimeTransmitter} \leq \text{offsetFromTimeTransmitterThreshold}$$

is satisfied.

14.8.4 rxSyncCountTimeReceiverPThresh

The value of rxSyncCountTimeReceiverPThresh is the threshold for rxSyncCountTimeReceiverP (see 18.4.1.4), above which the PTP Instance is considered to be synchronized.

14.8.5 offsetMaxExceededCountThresh

The value of offsetMaxExceededCountThresh (see 18.4.1.7) is the threshold for the number of consecutive exceedances of offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter (see 18.4.1.2), at which isSynced (see 18.4.1.1) is no longer TRUE.

14.8.6 offsetMaxMetCountThresh

The value of offsetMaxMetCountThresh (18.4.1.9) is the threshold for the number of consecutive occurrences of offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax (see 18.4.1.3), at which isSynced (see 18.4.1.1) is changed to TRUE if it currently is FALSE.

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14.9 Drift Tracking Parameter Data Set (driftTrackingDS)

14.9.1 General

The driftTrackingDS contains a managed object that is used to enable or disable the optional Drift_Tracking TLV.

The parameters of driftTrackingDS are summarized in Table 14-8 and subsequently specified.

Table 14-8—driftTrackingDS

Name	Data type	Operations supported[a]	References
driftTrackingTlvSupport	Boolean	RW	14.9.2
syncGrandmasterIdentity	ClockIdentity	R	14.9.3
syncStepsRemoved	UInteger16	R	14.9.4
rateRatioDrift	Integer32	R	14.9.5

[a] R = Read only access; RW = Read/write access.

14.9.2 driftTrackingTlvSupport

The value of driftTrackingTlvSupport indicates whether the Drift_Tracking TLV is enabled or disabled. If the value is TRUE, the TLV is enabled, i.e., the global variable driftTrackingTlvSupport (see 10.2.4.28) is set to TRUE. If the value is FALSE, the TLV is disabled, i.e., the global variable driftTrackingTlvSupport is set to FALSE.

NOTE—Setting the value of the managed object to TRUE only has an effect if the instance supports the Drift_Tracking TLV. If the instance does not support the Drift_Tracking TLV, reading the managed object will always return FALSE.

14.9.3 syncGrandmasterIdentity

The value of syncGrandmasterIdentity as determined by the MDSyncSend state machine (see 11.2.14.2.3 and 11.2.14.2.5).

14.9.4 syncStepsRemoved

The value of syncStepsRemoved as determined by the MDSyncSend state machine (see 11.2.14.2.3 and 11.2.14.2.5).

14.9.5 rateRatioDrift

The value of rateRatioDrift is equal to (RRdrift – 1.0) × (2^{41}), truncated to the next smaller signed integer, where RRdrift is the measured estimate of the rate of change per second of the ratio of the frequency of the Grandmaster Clock to the frequency of the local clock entity in the PTP Instance that sends the message.

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14.10 Port Parameter Data Set (portDS)

14.10.1 General

The portDS represents PTP Port time-aware capabilities for a PTP Instance of a time-aware system.

For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the portDS is maintained as the basis for making protocol decisions and providing values for message fields. The number of such data sets is the same as the value of defaultDS.numberPorts.

There is one set of portDS parameters per PTP Port, per PTP Instance of a time-aware system. Each portDS contains a set of parameters for each PTP Port that supports the time-synchronization capability, as detailed in Table 14-9. Each set of portDS can be created or removed dynamically in implementations that support dynamic configuration of PTP Ports and components.

The parameters of portDS are summarized in Table 14-9 and subsequently specified.

Table 14-9—portDS

Name	Data type	Operations supported[a]	References
portIdentity	PortIdentity (see 6.4.3.7)	R	14.10.2
portState	Enumeration8	R	14.10.3, Table 14-10
ptpPortEnabled	Boolean	RW	14.10.4
delayMechanism	Enumeration8	RW	14.10.5
isMeasuringDelay	Boolean	R	14.10.6
asCapable	Boolean	R	14.10.7
meanLinkDelay	TimeInterval	R	14.10.8
meanLinkDelayThresh	TimeInterval	RW	14.10.9
negMeanLinkDelayThresh	TimeInterval	RW	14.10.10
delayAsymmetry	TimeInterval	RW	14.10.11
neighborRateRatio	Integer32	R	14.10.12
initialLogAnnounceInterval	Integer8	RW	14.10.13
currentLogAnnounceInterval	Integer8	R	14.10.14
useMgtSettableLogAnnounceInterval	Boolean	RW	14.10.15
mgtSettableLogAnnounceInterval	Integer8	RW	14.10.16
announceReceiptTimeout	UInteger8	RW	14.10.17
initialLogSyncInterval	Integer8	RW	14.10.18
currentLogSyncInterval	Integer8	R	14.10.19
useMgtSettableLogSyncInterval	Boolean	RW	14.10.20
mgtSettableLogSyncInterval	Integer8	RW	14.10.21
syncReceiptTimeout	UInteger8	RW	14.10.22

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Table 14-9—portDS *(continued)*

Name	Data type	Operations supported[a]	References
syncReceiptTimeoutTimeInterval	UScaledNs	R	14.10.23
initialLogPdelayReqInterval	Integer8	RW	14.10.24
currentLogPdelayReqInterval	Integer8	R	14.10.25
useMgtSettableLogPdelayReqInterval	Boolean	RW	14.10.26
mgtSettableLogPdelayReqInterval	Integer8	RW	14.10.27
initialLogGptpCapableMessageInterval	Integer8	RW	14.10.28
currentLogGptpCapableMessageInterval	Integer8	R	14.10.29
useMgtSettableLogGptpCapableMessageInterval	Boolean	RW	14.10.30
mgtSettableLogGptpCapableMessageInterval	Integer8	RW	14.10.31
initialComputeNeighborRateRatio	Boolean	RW	14.10.32
currentComputeNeighborRateRatio	Boolean	R	14.10.33
useMgtSettableComputeNeighborRateRatio	Boolean	RW	14.10.34
mgtSettableComputeNeighborRateRatio	Boolean	RW	14.10.35
initialComputeMeanLinkDelay	Boolean	RW	14.10.36
currentComputeMeanLinkDelay	Boolean	R	14.10.37
useMgtSettableComputeMeanLinkDelay	Boolean	RW	14.10.38
mgtSettableComputeMeanLinkDelay	Boolean	RW	14.10.39
allowedLostResponses	UInteger8	RW	14.10.40
allowedFaults	UInteger8	RW	14.10.41
gPtpCapableReceiptTimeout	UInteger8	RW	14.10.42
versionNumber	UInteger4	R	14.10.43
nup (NOTE)	Float64	RW	14.10.44
ndown (NOTE)	Float64	RW	14.10.45
oneStepTxOper	Boolean	R	14.10.46
oneStepReceive	Boolean	R	14.10.47
oneStepTransmit	Boolean	R	14.10.48
initialOneStepTxOper	Boolean	RW	14.10.49
currentOneStepTxOper	Boolean	R	14.10.50
useMgtSettableOneStepTxOper	Boolean	RW	14.10.51
mgtSettableOneStepTxOper	Boolean	RW	14.10.52
syncLocked	Boolean	R	14.10.53
pdelayTruncatedTimestampsArray	UInteger48[4]	R	14.10.54

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Table 14-9—portDS *(continued)*

Name	Data type	Operations supported[a]	References
minorVersionNumber	UInteger4	R	14.10.55
gptpCapableStateMachinesEnabled	Boolean	RW	14.10.56
nrrPdelay	Integer32	R	14.10.57
nrrSync	Integer32	R	14.10.58
nrrCompMethod	Enumeration1	RW	14.10.59
asCapableAcrossDomains	Boolean	R	14.10.60
pdelayReqSendDisabled	Boolean	RW	14.10.61
pdelayRespSendDisabled	Boolean	RW	14.10.62
NOTE—The values of nup and ndown in Table 14-9 depend on the particular PHY used.			

[a] R = Read only access; RW = Read/write access.

14.10.2 portIdentity

The value is the portIdentity attribute of the local PTP Port (see 8.5.2).

14.10.3 portState

The value is the value of the PTP Port state of this PTP Port (see Table 10-2) and is taken from the enumeration in Table 14-10. It is equal to the value of the global variable selectedState (see 10.2.4.20) for this PTP Port.

Table 14-10—portState enumeration

State	Value
DisabledPort	3
TimeTransmitterPort	6
PassivePort	7
TimeReceiverPort	9
	All other values reserved
NOTE—The enumeration values are consistent with Table 20 in IEEE Std 1588-2019.	

The default value is 3 (DisabledPort).

14.10.4 ptpPortEnabled

The value is equal to the value of the Boolean ptpPortEnabled (see 10.2.5.13). Setting the managed object ptpPortEnabled causes the Boolean ptpPortEnabled to have the same value.

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14.10.5 delayMechanism

The value of delayMechanism indicates the mechanism for measuring mean propagation delay and neighbor rate ratio on the link attached to this PTP Port and is taken from the enumeration in Table 14-11. portDS.delayMechanism may be P2P (see 11.2.16) only if the only domainNumber active on the physical port is 0.

Table 14-11—delayMechanism enumeration

Delay mechanism	Value	Specification
P2P	02	The PTP Port uses the transport-specific peer-to-peer delay mechanism
COMMON_P2P	03	The PTP Port uses the CMLDS
SPECIAL	04	The PTP Port uses a transport that has a native time transfer mechanism and, therefore, does not use the peer-to-peer delay mechanism (e.g., IEEE 802.11, IEEE 802.3 EPON)
	All other values reserved	
NOTE—The enumeration values are consistent with Table 21 in IEEE Std 1588-2019.		

14.10.6 isMeasuringDelay

The value is equal to the value of the per-port, per PTP Instance global variable isMeasuringDelay (see 11.2.12.6 and 16.4.3.3).

14.10.7 asCapable

The value is equal to the value of the Boolean asCapable (see 10.2.5.1).

14.10.8 meanLinkDelay

The value is equal to the value of the per-PTP Port global variable meanLinkDelay (see 10.2.5.8). It is an estimate of the current one-way propagation time on the link attached to this PTP Port, measured as specified for the respective medium (see 11.2.16, 12.5.2, and 16.4). The value is zero for PTP Ports attached to IEEE 802.3 EPON links and for the timeTransmitter port of an IEEE 802.11 link, because one-way propagation delay is not measured on the latter and not directly measured on the former. The data type shall be TimeInterval. The default value is zero.

14.10.9 meanLinkDelayThresh

The value is equal to the value of the per-PTP Port global variable meanLinkDelayThresh (see 11.2.12.7). It is the propagation time threshold above which a PTP Port is considered not capable of participating in the IEEE 802.1AS protocol. Setting the managed object meanLinkDelayThresh causes the per PTP Port global variable meanLinkDelayThresh to have the same value.

14.10.10 negMeanLinkDelayThresh

The value is equal to the value of the per-PTP Port global variable negMeanLinkDelayThresh (see 11.2.12.8). It is the negative propagation time threshold below which a PTP Port is considered not capable of participating in the IEEE 802.1AS protocol. Setting the managed object

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negMeanLinkDelayThresh causes the per PTP Port global variable negMeanLinkDelayThresh to have the same value.

14.10.11 delayAsymmetry

The value is the asymmetry in the propagation delay on the link attached to this PTP Port relative to the Grandmaster Clock time base, as defined in 10.2.5.9 and 8.3. If propagation delay asymmetry is not modeled, then delayAsymmetry is 0.

NOTE—The underlying per-port global variable delayAsymmetry is of type ScaledNS, which is a 96-bit value (see 6.4.3.1). delayAsymmetry values that are larger than the maximum value that can be represented by the TimeInterval data type, i.e., 0x7FFF FFFF FFFF FFFF (where the units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this largest value. delayAsymmetry values that are less than the minimum value that can be represented by the TimeInterval data type, i.e., 0x8000 0000 0000 0001 written in twos complement form (where the units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this smallest value.

14.10.12 neighborRateRatio

The value is an estimate of the ratio of the frequency of the LocalClock entity of the PTP Instance at the other end of the link attached to this PTP Port, to the frequency of the LocalClock entity of this PTP Instance (see 10.2.5.7). neighborRateRatio is expressed as the fractional frequency offset multiplied by 2^{41}, i.e., the quantity (neighborRateRatio – 1.0)(2^{41}).

14.10.13 initialLogAnnounceInterval

If useMgtSettableLogAnnounceInterval is FALSE, the value is the logarithm to base 2 of the announce interval used when:

a) The PTP Port is initialized or

b) A message interval request TLV is received with the logAnnounceInterval field set to 126 (see 10.7.2.2 and the AnnounceIntervalSetting state machine in 10.3.17).

14.10.14 currentLogAnnounceInterval

The value is the logarithm to the base 2 of the current announce interval (see 10.7.2.2).

14.10.15 useMgtSettableLogAnnounceInterval

The managed object is a Boolean that determines the source of the announce interval. If the value is TRUE, the value of currentLogAnnounceInterval is set equal to the value of mgtSettableLogAnnounceInterval (see 14.10.16). If the value is FALSE, the value of currentLogAnnounceInterval is determined by the AnnounceIntervalSetting state machine (see 10.3.17). The default value of useMgtSettableLogAnnounceInterval is FALSE for domain 0 and TRUE for domains other than domain 0.

14.10.16 mgtSettableLogAnnounceInterval

The value is the logarithm to base 2 of the announce interval used if useMgtSettableLogAnnounceInterval is TRUE. The value is not used if useMgtSettableLogAnnounceInterval is FALSE.

14.10.17 announceReceiptTimeout

The value is the number of Announce message transmission intervals that a timeReceiver port waits without receiving an Announce message before assuming that the timeTransmitter is no longer transmitting Announce messages and the BTCA needs to be run, if appropriate (see 10.7.3.2).

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14.10.18 initialLogSyncInterval

If useMgtSettableLogSyncInterval is FALSE, the value is the logarithm to base 2 of the sync interval used when:

a) The PTP Port is initialized or

b) A message interval request TLV is received with the logTimeSyncInterval field set to 126 (see 10.7.2.3, 11.5.2.3, 12.8.2, 13.9.2, and the SyncIntervalSetting state machine in 10.2.14).

14.10.19 currentLogSyncInterval

The value is the logarithm to the base 2 of the current time-synchronization transmission interval (see 10.7.2.3).

14.10.20 useMgtSettableLogSyncInterval

The managed object is a Boolean that determines the source of the sync interval. If the value is TRUE, the value of currentLogSyncInterval is set equal to the value of mgtSettableLogSyncInterval (see 14.10.21). If the value of the managed object is FALSE, the value of currentLogSyncInterval is determined by the SyncIntervalSetting state machine (see 10.2.14). The default value of useMgtSettableLogSyncInterval is FALSE for domain 0 and TRUE for domains other than domain 0.

14.10.21 mgtSettableLogSyncInterval

The value is the logarithm to base 2 of the sync interval if useMgtSettableLogSyncInterval is TRUE. The value is not used if useMgtSettableLogSyncInterval is FALSE.

14.10.22 syncReceiptTimeout

The value is the number of time-synchronization transmission intervals that a timeReceiver port waits without receiving synchronization information before assuming that the timeTransmitter is no longer transmitting synchronization information and that the BTCA needs to be run, if appropriate (see 10.7.3.1).

14.10.23 syncReceiptTimeoutTimeInterval

The value is equal to the value of the per-PTP Port global variable syncReceiptTimeoutTimeInterval (see 10.2.5.3). It is the time interval after which sync receipt timeout occurs if time-synchronization information has not been received during the interval.

14.10.24 initialLogPdelayReqInterval

For full-duplex and half-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req message transmission interval used when:

a) The PTP Port is initialized or

b) A message interval request TLV is received with the logLinkDelayInterval field set to 126 (see 11.5.2.2 and the LinkDelayIntervalSetting state machine in 11.2.20).

For all other media, the value is 127.

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14.10.25 currentLogPdelayReqInterval

For full-duplex and half-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req message transmission interval (see 11.5.2.2).

For all other media, the value is 127.

14.10.26 useMgtSettableLogPdelayReqInterval

The managed object is a Boolean that determines the source of the mean time interval between successive Pdelay_Req messages. If the value is TRUE, the value of currentLogPdelayReqInterval is set equal to the value of mgtSettableLogPdelayReqInterval (see 14.10.27). If the value of the managed object is FALSE, the value of currentLogPdelayReqInterval is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableLogPdelayReqInterval is FALSE.

14.10.27 mgtSettableLogPdelayReqInterval

The value is the logarithm to base 2 of the mean time interval between successive Pdelay_Req messages if useMgtSettableLogPdelayReqInterval is TRUE. The value is not used if useMgtSettableLogPdelayReqInterval is FALSE.

14.10.28 initialLogGptpCapableMessageInterval

The value is the logarithm to base 2 of the gPTP capable message interval used when:

a) The PTP Port is initialized; or

b) A gPtpCapableMessage interval request TLV is received with the logGptpCapableMessageInterval field set to 126 (see 10.6.4.5 and the GptpCapableIntervalSetting state machine in 10.4.4).

14.10.29 currentLogGptpCapableMessageInterval

The value is the logarithm to the base 2 of the current gPTP capable message interval (see 10.7.2.5).

14.10.30 useMgtSettableLogGptpCapableMessageInterval

The managed object is a Boolean that determines the source of the gPTP capable message interval. If the value is TRUE, the value of currentLogGptpCapableMessageInterval is set equal to the value of mgtSettableLogGptpCapableMessageInterval (see 14.10.31). If the value of the managed object is FALSE, the value of currentLogGptpCapableMessageInterval is determined by the GptpCapableMessageIntervalSetting state machine (see 10.4.4). The default value of useMgtSettableLogGptpCapableMessageInterval is FALSE.

14.10.31 mgtSettableLogGptpCapableMessageInterval

The value is the logarithm to base 2 of the gPtpCapableMessageInterval if useMgtSettableLogGptpCapableMessageInterval is TRUE. The value is not used if useMgtSettableLogGptpCapableMessageInterval is FALSE.

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14.10.32 initialComputeNeighborRateRatio

If useMgtSettableComputeNeighborRateRatio is FALSE, then for full-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial value of computeNeighborRateRatio (see 10.2.5.10).

For all other media, the value is TRUE.

14.10.33 currentComputeNeighborRateRatio

For FDE and half-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

For all other media, the value is TRUE.

14.10.34 useMgtSettableComputeNeighborRateRatio

The managed object is a Boolean that determines the source of the value of computeNeighborRateRatio. If the value is TRUE, the value of computeNeighborRateRatio is set equal to the value of mgtSettableComputeNeighborRateRatio (see 14.18.18). If the value of the managed object is FALSE, the value of currentComputeNeighborRateRatio is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableComputeNeighborRateRatio is FALSE.

14.10.35 mgtSettableComputeNeighborRateRatio

computeNeighborRateRatio is configured to this value if useMgtSettableComputeNeighborRateRatio is TRUE. The value is not used if useMgtSettableComputeNeighborRateRatio is FALSE.

14.10.36 initialComputeMeanLinkDelay

If useMgtSettableComputeMeanLinkDelay is FALSE, then for FDE and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial value of computeMeanLinkDelay (see 10.2.5.10).

For all other media, the value is TRUE.

14.10.37 currentComputeMeanLinkDelay

For FDE and half-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

For all other media, the value is TRUE.

14.10.38 useMgtSettableComputeMeanLinkDelay

The managed object is a Boolean that determines the source of the value of computeMeanLinkDelay. If the value is TRUE, the value of computeMeanLinkDelay is set equal to the value of mgtSettableComputeMeanLinkDelay (see 14.10.39). If the value of the managed object is FALSE, the value of currentComputeMeanLinkDelay is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableComputeMeanLinkDelay is FALSE.

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14.10.39 mgtSettableComputeMeanLinkDelay

computeMeanLinkDelay is configured to this value if useMgtSettableComputeMeanLinkDelay is TRUE. The value is not used if useMgtSettableComputeMeanLinkDelay is FALSE.

14.10.40 allowedLostResponses

The value is equal to the value of the per-PTP Port global variable allowedLostResponses (see 11.5.3 and 11.2.12.4). It is the number of Pdelay_Req messages without valid responses above which a PTP Port is considered to be not exchanging peer delay messages with its neighbor. Setting the managed object allowedLostResponses causes the per PTP Port global variable allowedLostResponses to have the same value.

14.10.41 allowedFaults

The value is equal to the value of the per-PTP Port global variable allowedFaults (see 11.5.4 and 11.2.12.5). It is the number of faults (see 11.5.4) above which asCapable is set to FALSE, i.e., a PTP Port is considered not capable of interoperating with its neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). Setting the managed object allowedLostResponses causes the per PTP Port global variable allowedFaults to have the same value.

14.10.42 gPtpCapableReceiptTimeout

The value is the number of transmission intervals that a PTP Port waits without receiving the gPTP-capable TLV before assuming that the neighbor PTP Port is no longer invoking gPTP (see 10.7.3.3).

14.10.43 versionNumber

This value is set to versionPTP as specified in 10.6.2.2.4.

14.10.44 nup

For an OLT port of an IEEE 802.3 EPON link, the value is the effective index of refraction for the EPON upstream wavelength light of the optical path (see 13.1.4 and 13.8.1.2.2). The default value is 1.46770 for 1 Gb/s upstream links and 1.46773 for 10 Gb/s upstream links.

For all other PTP Ports, the value is 0.

14.10.45 ndown

For an OLT port of an IEEE 802.3 EPON link, the value is the effective index of refraction for the EPON downstream wavelength light of the optical path (see 13.1.4 and 13.8.1.2.1). The default value is 1.46805 for 1 Gb/s downstream links and 1.46851 for 10 Gb/s downstream links.

For all other PTP Ports, the value is 0.

14.10.46 oneStepTxOper

The value is equal to the value of the per-PTP Port global variable oneStepTxOper (see 11.2.12.12). Its value is TRUE if the PTP Port is sending one-step Sync messages and FALSE if the PTP Port is sending two-step Sync and Follow_Up messages.

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14.10.47 oneStepReceive

The value is equal to the value of the per-PTP Port global variable oneStepReceive (see 11.2.12.10). Its value is TRUE if the PTP Port is capable of receiving and processing one-step Sync messages.

14.10.48 oneStepTransmit

The value is equal to the value of the per-PTP Port global variable oneStepTransmit (see 11.2.12.11). Its value is TRUE if the PTP Port is capable of transmitting one-step Sync messages.

14.10.49 initialOneStepTxOper

If useMgtSettableOneStepTxOper is FALSE, the value is used to initialize currentOneStepTxOper when the PTP Port is initialized. If useMgtSettableOneStepTxOper is TRUE, the value of initialOneStepTxOper is not used. The default value of initialOneStepTxOper shall be FALSE.

14.10.50 currentOneStepTxOper

The value is TRUE if it is desired, either via management or via a received Signaling message, that the PTP Port transmit one-step Sync messages. The value is FALSE if it is not desired, either via management or via a received Signaling message, that the PTP Port transmit one-step Sync messages.

NOTE—The PTP Port will send one-step Sync messages only if currentOneStepTxOper and oneStepTransmit (see 14.10.48) are both TRUE (see 11.2.15 and Figure 11-8).

14.10.51 useMgtSettableOneStepTxOper

The managed object is a Boolean that determines the source of currentOneStepTxOper. If the value is TRUE, the value of currentOneStepTxOper is set equal to the value of mgtSettableOneStepTxOper (see 14.10.52). If the value is FALSE, the value of currentOneStepTxOper is determined by the OneStepTxOperSetting state machine (see 11.2.15 and Figure 11-8). The default value of useMgtSettableOneStepTxOper is TRUE.

14.10.52 mgtSettableOneStepTxOper

If useMgtSettableOneStepTxOper is TRUE, currentOneStepTxOper is set equal to the value of mgtSettableOneStepTxOper. The value of mgtSettableOneStepTxOper is not used if useMgtSettableOneStepTxOper is FALSE. The default value of mgtSettableOneStepTxOper is FALSE for domains other than domain 0.

14.10.53 syncLocked

The value is equal to the value of the per-PTP Port global variable syncLocked (see 10.2.5.15). Its value is TRUE if the PTP Port will transmit a Sync as soon as possible after the timeReceiver PTP Port receives a Sync.

14.10.54 pdelayTruncatedTimestampsArray

For FDE and half-duplex IEEE 802.3 media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table 14-12. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1, t2, t3, and t4, respectively, in Figure 11-1 and are expressed in units of 2^{-16} ns (i.e., the value of each array element is equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2^{-16} ns, by 2^{48}). At any given time, the timestamp values stored in the array are for the same, and most recently completed, peer delay message exchange.

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Table 14-12—Description of pdelayTruncatedTimestampsArray

Array element	Timestamp description	Corresponding timestamp of Figure 11-1	Units
0	pdelayReqEventEgressTimestamp for Pdelay_Req message, of most recently completed peer delay message exchange, transmitted by this PTP Instance (NOTE 1)	t1	2^{-16} ns
1	pdelayReqEventIngressTimestamp for Pdelay_Req message received at peer delay responder to which this Link Port sends Pdelay_Req, of most recently completed peer delay message exchange; it is equal to the sum of the following: a) The ns field of the requestReceiptTimestamp (see Table 11-14), multiplied by 2^{16} b) The correctionField (see Table 11-6) (NOTE 2)	t2	2^{-16} ns
2	pdelayRespEventEgressTimestamp for Pdelay_Resp message, of most recently completed peer delay message exchange, transmitted by peer delay responder to which this Link Port sends Pdelay_Req; it is equal to the sum of the following: a) The ns field of the responseOriginTimestamp (see Table 11-15), multiplied by 2^{16} b) The correctionField (see Table 11-6) (NOTE 3)	t3	2^{-16} ns
3	pdelayRespEventIngressTimestamp for Pdelay_Resp message, of most recently completed peer delay message exchange, received by this PTP Instance (NOTE 4)	t4	2^{-16} ns

NOTE 1—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the pdelayReqEventEgressTimestamp. Rather, it is equal to [pdelayReqEventEgressTimestamp.seconds × $(10^9)(2^{16})$ + pdelayReqEventEgressTimestamp.fractionalNanoseconds] mod 2^{48}, where pdelayReqEventEgressTimestamp is expressed as an ExtendedTimestamp (see 6.4.3.5). Its units are 2^{-16} ns.

NOTE 2—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the pdelayReqEventIngressTimestamp. Rather, it is equal to [pdelayReqEventIngressTimestamp>.seconds × $(10^9)(2^{16})$ + pdelayReqEventIngressTimestamp.fractionalNanoseconds] mod 2^{48}, where pdelayReqEventIngressTimestamp is expressed as an ExtendedTimestamp (see 6.4.3.5). Its units are 2^{-16} ns.

NOTE 3—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the pdelayRespEventEgressTimestamp. Rather, it is equal to [pdelayRespEventEgressTimestamp.seconds × $(10^9)(2^{16})$ + pdelayRespEventEgressTimestamp.fractionalNanoseconds] mod 2^{48}, where pdelayRespEventEgressTimestamp is expressed as an ExtendedTimestamp (see 6.4.3.5). Its units are 2^{-16} ns.

NOTE 4—This quantity is not simply the nanoseconds plus fractional nanoseconds portion of the pdelayRespEventIngressTimestamp. Rather, it is equal to [pdelayRespEventIngressTimestamp.seconds × $(10^9)(2^{16})$ + pdelayRespEventIngressTimestamp.fractionalNanoseconds] mod 2^{48}, where pdelayRespEventIngressTimestamp is expressed as an ExtendedTimestamp (see 6.4.3.5). Its units are 2^{-16} ns.

14.10.55 minorVersionNumber

This value is set to minorVersionPTP as specified in 10.6.2.2.3.

14.10.56 gptpCapableStateMachinesEnabled

A Boolean that is used to enable or disable the GptpCapableTransmit, GptpCapableReceive, and GptpCapableIntervalSetting state machines. If the value is TRUE, the GptpCapableTransmit and

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GptpCapableReceive state machines are enabled, and the GptpCapableIntervalSetting state machine is enabled if it is implemented. If the value is FALSE, the GptpCapableTransmit and GptpCapableReceive state machines are disabled, and the GptpCapableIntervalSetting state machine is disabled if it is implemented. The default value is TRUE.

14.10.57 nrrPdelay

The value of the global variable nrrPdelay (see 11.2.12.14), expressed as Integer32.

The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-aware system (see 10.2.5.7). The managed object nrrPdelay is expressed as the fractional frequency offset stored in the global variable nrrPdelay (see 11.2.12.14) multiplied by 2^{41}, i.e., the quantity (nrrPdelay(global variable) – 1.0)(2^{41}). The default value of the managed object nrrPdelay is 0.0, corresponding to a rate ratio of 1.0.

14.10.58 nrrSync

The value of the global variable nrrSync (see 11.2.12.15), expressed as Integer32.

The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-aware system (see 10.2.5.7). The managed object nrrSync is expressed as the fractional frequency offset stored in the global variable nrrSync (see 11.2.12.15) multiplied by 2^{41}, i.e., the quantity (nrrSync(global variable) – 1.0)(2^{41}).

14.10.59 nrrCompMethod

The value of the global variable nrrCompMethod (see 11.2.12.16). If its value is Pdelay or driftTrackingTlvSupport is FALSE, neighborRateRatio is populated with the value of nrrPdelay. If the value is Sync and driftTrackingTlvSupport is TRUE, neighborRateRatio is populated with the value of nrrSync. The default value for nrrCompMethod shall be Pdelay

14.10.60 asCapableAcrossDomains

The value is equal to the value of the Boolean asCapableAcrossDomains (see 11.2.2 and 11.2.12.13).

14.10.61 pdelayReqSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayReqSendDisabled (see 11.2.18.2.14). If its value is TRUE, Pdelay_Req messages are not transmitted by the PTP Port. The default value for this variable shall be FALSE.

14.10.62 pdelayRespSendDisabled

The value is equal to the value of the per-PTP Port variable pdelayRespSendDisabled (see 11.2.19.2.6). If its value is TRUE, Pdelay_Resp and Pdelay_Resp_Follow_Up messages are not transmitted by the PTP Port in response to Pdelay_Req message. The default value for this variable shall be FALSE.

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14.11 Description Port Parameter Data Set (descriptionPortDS)

14.11.1 General

The descriptionPortDS contains the profileIdentifier for this PTP profile, as specified in F.2.

The parameter of descriptionPortDS is summarized in Table 14-13 and subsequently specified.

Table 14-13—descriptionPortDS

Name	Data type	Operations supported[a]	References
profileIdentifier	Octet6	R	14.11.2

[a] R= Read only access.

14.11.2 profileIdentifier

The value is the profileIdentifier for this PTP profile (see F.2).

14.12 Port Parameter Statistics Data Set (portStatisticsDS)

14.12.1 General

For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the portStatisticsDS provides counters associated with PTP Port capabilities at a given PTP Instance. The number of such statistics sets is the same as the value of defaultDS.numberPorts.

There is one set of portStatisticsDS parameters per PTP Port, per PTP Instance of a time-aware system. Each portStatisticsDS contains a set of counters for each PTP Port that supports the time-synchronization capability, as detailed in Table 14-14. Each set of portStatisticsDS can be created or removed dynamically in implementations that support dynamic configuration of PTP Ports and components.

The parameters of portStatisticsDS are summarized in Table 14-14 and subsequently specified.

Table 14-14—portStatisticsDS

Name	Data type	Operations supported[a]	References
rxSyncCount	UInteger32	R	14.12.2
rxOneStepSyncCount	UInteger32	R	14.12.3
rxFollowUpCount	UInteger32	R	14.12.4
rxPdelayRequestCount	UInteger32	R	14.12.5
rxPdelayResponseCount	UInteger32	R	14.12.6
rxPdelayResponseFollowUpCount	UInteger32	R	14.12.7
rxAnnounceCount	UInteger32	R	14.12.8

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Table 14-14—portStatisticsDS

Name	Data type	Operations supported[a]	References
rxPtpPacketDiscardCount	UInteger32	R	14.12.9
syncReceiptTimeoutCount	UInteger32	R	14.12.10
announceReceiptTimeoutCount	UInteger32	R	14.12.11
pdelayAllowedLostResponsesExceededCount	UInteger32	R	14.12.12
txSyncCount	UInteger32	R	14.12.13
txOneStepSyncCount	UInteger32	R	14.12.14
txFollowUpCount	UInteger32	R	14.12.15
txPdelayRequestCount	UInteger32	R	14.12.16
txPdelayResponseCount	UInteger32	R	14.12.17
txPdelayResponseFollowUpCount	UInteger32	R	14.12.18
txAnnounceCount	UInteger32	R	14.12.19
rxSyncCountTimeReceiverP	UInteger32	R	14.12.20

[a] R= Read only access.

14.12.2 rxSyncCount

This counter increments every time synchronization information is received, denoted by one of the following events:

— A transition to TRUE from FALSE of the rcvdSync variable of the MDSyncReceiveSM state machine (see 11.2.13.1.2 and Figure 11-6) when in the DISCARD, WAITING_FOR_SYNC, or WAITING_FOR_FOLLOW_UP states; or
— rcvdIndication transitions to TRUE (see Figure 12-7).

14.12.3 rxOneStepSyncCount

This counter increments every time a one-step Sync message is received, denoted by a transition to TRUE from FALSE of the rcvdSync variable of the MDSyncReceiveSM state machine (see 11.2.13.1.2 and Figure 11-6) with the variable rcvdSyncPtr->twoStepFlag FALSE when in the DISCARD or WAITING_FOR_SYNC states.

14.12.4 rxFollowUpCount

This counter increments every time a Follow_Up message is received, denoted by a transition to TRUE from FALSE of the rcvdFollowUp variable of the MDSyncReceiveSM state machine (see 11.2.13.1.3 and Figure 11-6) when in the WAITING_FOR_FOLLOW_UP state.

14.12.5 rxPdelayRequestCount

This counter increments every time a Pdelay_Req message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.19.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ states.

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14.12.6 rxPdelayResponseCount

This counter increments every time a Pdelay_Resp message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayResp variable of the MDPdelayReq state machine (see 11.2.18.2.2 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP state.

14.12.7 rxPdelayResponseFollowUpCount

This counter increments every time a Pdelay_Resp_Follow_Up message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayRespFollowUp variable of the MDPdelayReq state machine (see 11.2.18.2.4 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state.

14.12.8 rxAnnounceCount

This counter increments every time an Announce message is received, denoted by a transition to TRUE from FALSE of the rcvdAnnounce variable of the PortAnnounceReceive state machine (see 10.3.11 and Figure 10-14) when in the DISCARD or RECEIVE states.

14.12.9 rxPtpPacketDiscardCount

This counter increments every time a PTP message of the respective PTP Instance is discarded, caused by the occurrence of any of the following conditions:

a) A received Announce message is not qualified, denoted by the function qualifyAnnounce (see 10.3.11.2.1 and 13.1.3.4) of the PortAnnounceReceive state machine (see 10.3.11 and Figure 10-14) returning FALSE.

b) A Follow_Up message corresponding to a received Sync message is not received, denoted by a transition of the condition (currentTime ≥ followUpReceiptTimeoutTime) to TRUE from FALSE when in the WAITING_FOR_FOLLOW_UP state of the MDSyncReceiveSM state machine (see 11.2.13 and Figure 11-6).

c) A Pdelay_Resp message corresponding to a transmitted Pdelay_Req message is not received, denoted by a transition from the WAITING_FOR_PDELAY_RESP state to the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

d) A Pdelay_Resp_Follow_Up message corresponding to a transmitted Pdelay_Req message is not received, denoted by a transition from the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state to the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

14.12.10 syncReceiptTimeoutCount

This counter increments every time sync receipt timeout occurs, denoted by entering the AGED state of the PortAnnounceInformation state machine (see 10.3.12 and Figure 10-15) with the condition (currentTime ≥ syncReceiptTimeoutTime && gmPresent) evaluating to TRUE.

14.12.11 announceReceiptTimeoutCount

This counter increments every time announce receipt timeout occurs, denoted by entering the AGED state of the PortAnnounceInformation state machine from the CURRENT state of the PortAnnounceInformation state machine (see 10.3.12 and Figure 10-15) with the condition (currentTime ≥ announceReceiptTimeoutTime) evaluating to TRUE.

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14.12.12 pdelayAllowedLostResponsesExceededCount

This counter increments every time the value of the variable lostResponses (see 11.2.18.2.9) exceeds the value of the variable allowedLostResponses (see 11.2.12.4), in the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

14.12.13 txSyncCount

This counter increments every time synchronization information is transmitted, denoted by one of the following events:

— A transition to TRUE from FALSE of the rcvdMDSyncMDSS variable of the MDSyncSendSM state machine (see 11.2.14.1.1 and Figure 11-7) when in the INITIALIZING, SEND_FOLLOW_UP, or SET_CORRECTION_FIELD states; or
— The INITIATE_REQUEST_WAIT_CONFIRM_OR_SAVE_INFO state is entered in Figure 12-5 and TM is being used [i.e., (tmFtmSupport == 0x01 || ((tmFtmSupport & 0x01 == 0x01) && ftmReqGrantedTimeTransmitter)) in timeTransmitter state machine A of Figure 12-5]; or
— The INITIATE_REQUEST_WAIT_CONFIRM state is entered in Figure 12-6 (in this case FTM is being used).

14.12.14 txOneStepSyncCount

This counter increments every time a one-step Sync message is transmitted, denoted by a transition to TRUE from FALSE of the rcvdMDSyncMDSS variable of the MDSyncSendSM state machine (see 11.2.14.1.1 and Figure 11-7) with the variable oneStepTxOper TRUE when in the INITIALIZING, SEND_SYNC_ONE_STEP, or SET_CORRECTION_FIELD states.

14.12.15 txFollowUpCount

This counter increments every time a Follow_Up message is transmitted, denoted by a transition to TRUE from FALSE of the rcvdMDTimestampReceive variable of the MDSyncSendSM state machine (see 11.2.14.1.3 and Figure 11-7) when in the SEND_SYNC state.

14.12.16 txPdelayRequestCount

This counter increments every time a Pdelay_Req message is transmitted, denoted by entering the INITIAL_SEND_PDELAY_REQ or SEND_PDELAY_REQ states of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

14.12.17 txPdelayResponseCount

This counter increments every time a Pdelay_Resp message is transmitted, denoted by the following events:

— A transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.19.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ state and
— The resulting entry to the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state.

14.12.18 txPdelayResponseFollowUpCount

This counter increments every time a Pdelay_Resp_Follow_Up message is transmitted, denoted by the following events:

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— A transition to TRUE from FALSE of the rcvdMDTimestampReceive variable of the MDPdelayResp state machine (see 11.2.19.2.2 and Figure 11-10) when in the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state and
— The resulting entry to the WAITING_FOR_PDELAY_REQ state.

14.12.19 txAnnounceCount

This counter increments every time an Announce message is transmitted, denoted by entering the TRANSMIT_ANNOUNCE state of the PortAnnounceReceive state machine (see 10.3.16 and Figure 10-19).

14.12.20 rxSyncCountTimeReceiverP

This counter increments whenever time synchronization information is received on a PTP Port when its port state is TimeReceiverPort. The receipt of time synchronization information is denoted by one of the following events:

— A transition to TRUE from FALSE of the rcvdSync variable of the MDSyncReceiveSM state machine (see 11.2.13.1.2 and Figure 11-6) when in the DISCARD, WAITING_FOR_SYNC, or WAITING_FOR_FOLLOW_UP states; or
— rcvdIndication transitions to TRUE (see Figure 12-7).

This counter is initialized to zero, and resets to zero when the port state is not TimeReceiverPort.

14.13 Acceptable TimeTransmitter Port Parameter Data Set (acceptableTimeTransmitterPortDS)

14.13.1 General

The acceptableTimeTransmitterPortDS represents the capability to enable/disable the acceptable timeTransmitter table feature on a PTP Port. For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, this data set contains the single member acceptableTimeTransmitterTableEnabled, which is used to enable/disable the Acceptable TimeTransmitter Table Feature. The number of such data sets is the same as the value of defaultDS.numberPorts.

The parameter of acceptableTimeTransmitterPortDS is summarized in Table 14-15 and subsequently specified.

Table 14-15—acceptableTimeTransmitterPortDS

Name	Data type	Operations supported[a]	References
acceptableTimeTransmitterTableEnabled	Boolean	RW	14.13.2

[a] R = Read only access; RW = Read/write access.

14.13.2 acceptableTimeTransmitterTableEnabled

The value is equal to the value of the Boolean acceptableTimeTransmitterTableEnabled (see 13.1.3.2 and 13.1.3.5).

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14.14 External Port Configuration Port Parameter Data Set (externalPortConfigurationPortDS)

14.14.1 General

The externalPortConfigurationPortDS is used with the external port configuration option to indicate the desired state for the PTP Port. This data set contains the single member desiredState, which indicates the desired state for the PTP Port. The number of such data sets is the same as the value of defaultDS.numberPorts.

The parameter of externalPortConfigurationPortDS is summarized in Table 14-16 and subsequently specified.

Table 14-16—externalPortConfigurationPortDS

Name	Data type	Operations supported[a]	References
desiredState	Enumeration8 (see 6.4.2)	RW	14.14.2, Table 14-10

[a] R = Read only access; RW = Read/write access.

14.14.2 desiredState

When the value of defaultDS.externalPortConfigurationEnabled is TRUE, the value of externalPortConfigurationPortDS.desiredState is the desired state of the PTP Port. This member sets the value of the variable portStateInd (see 10.3.15.1.5). When a new value is written to the member by management, the variable rcvdPortStateInd (see 10.3.15.1.4) is set to TRUE.

14.15 Asymmetry Measurement Mode Parameter Data Set (asymmetryMeasurementModeDS)

14.15.1 General

The asymmetryMeasurementModeDS represents the capability to enable/disable the Asymmetry Compensation Measurement Procedure on a PTP Port (see Annex G). This data set is used instead of the cmldsAsymmetryMeasurementModeDS when only domain 0 is present and CMLDS is not used.

For the single PTP Port of a PTP End Instance and for each PTP Port of a PTP Relay Instance, the asymmetryMeasurementModeDS contains the single member asymmetryMeasurementMode, which is used to enable/disable the Asymmetry Compensation Measurement Procedure. The number of such data sets is the same as the value of defaultDS.numberPorts.

There is one asymmetryMeasurementModeDS for the single PTP Instance whose domainNumber is 0, per PTP Port, as detailed in Table 14-17. This data set is used only when there is a single gPTP domain and CMLDS is not used.

The parameter of asymmetryMeasurementModeDS is summarized in Table 14-17 and subsequently specified.

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Table 14-17—asymmetryMeasurementModeDS

Name	Data type	Operations supported[a]	References
asymmetryMeasurementMode	Boolean	RW	14.15.2

[a] R = Read only access; RW = Read/write access.

14.15.2 asymmetryMeasurementMode

The value is equal to the value of the Boolean asymmetryMeasurementMode (see G.3). For FDE media, the value is TRUE if an asymmetry measurement is being performed for the link attached to this PTP Port and FALSE otherwise. For all other media, the value shall be FALSE (see 10.2.5.2). Setting the managed object asymmetryMeasurementMode causes the Boolean asymmetryMeasurementMode to have the same value.

NOTE—If an asymmetry measurement is being performed for a link, asymmetryMeasurementMode must be TRUE for the PTP Ports at each end of the link.

14.16 Common Services Port Parameter Data Set (commonServicesPortDS)

14.16.1 General

The commonServicesPortDS enables a PTP Port of a PTP Instance to determine which port of the respective common service corresponds to that PTP Port.

This member contains the port number of the CMLDS Link Port that corresponds to this PTP Port.

The parameter of commonServicesPortDS is summarized in Table 14-18 and subsequently specified.

Table 14-18—commonServicesPortDS

Name	Data type	Operations supported[a]	References
cmldsLinkPortPortNumber	UInteger16	R	14.16.2

[a] R = Read only access.

14.16.2 cmldsLinkPortPortNumber

The value is the portNumber attribute of the cmldsLinkPortDS.portIdentity (see 14.18.2) of the Link Port that corresponds to this PTP Port.

14.17 Common Mean Link Delay Service Default Parameter Data Set (cmldsDefaultDS)

14.17.1 General

The cmldsDefaultDS describes the per-time-aware-system attributes of the CMLD.

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NOTE—The value of sdoId for the CMLD is fixed as 0x200 (see 11.2.16) and cannot be changed. Therefore, a corresponding data set member for sdoId is not needed.

The parameters of cmldsDefaultDS are summarized in Table 14-19 and subsequently specified.

Table 14-19—cmldsDefaultDS

Name	Data type	Operations supported[a]	References
clockIdentity	ClockIdentity	R	14.17.2
numberLinkPorts	UInteger16	R	14.17.3

[a] R = Read only access; RW = Read/write access.

14.17.2 clockIdentity

The value is the clockIdentity (see 8.5.2.2) that will be used to identify the CMLD.

14.17.3 numberLinkPorts

The value is the number of Link Ports of the time-aware system on which the CMLD is implemented. For an end station the value is 1.

14.18 Common Mean Link Delay Service Link Port Parameter Data Set (cmldsLinkPortDS)

14.18.1 General

The cmldsLinkPortDS represents time-aware Link Port capabilities for the CMLD of a Link Port of a time-aware system.

For every Link Port of the CMLD of a time-aware system, the cmldsLinkPortDS is maintained as the basis for making protocol decisions and providing values for message fields. The number of such data sets is the same as the value of cmldsDefaultDS.numberLinkPorts.

There is one set of cmldsLinkPortDS parameters per Link Port, for the PTP Instance of a time-aware system. Each cmldsLinkPortDS contains a set of parameters for each Link Port that supports the time-synchronization capability, as detailed in Table 14-20. Each set of cmldsLinkPortDS can be created or removed dynamically in implementations that support dynamic configuration of Link Ports and components.

The parameters of cmldsLinkPortDS are summarized in Table 14-20 and subsequently specified.

14.18.2 portIdentity

The value is the portIdentity attribute of the local port (see 8.5.2).

14.18.3 cmldsLinkPortEnabled

The value is equal to the value of the Boolean cmldsLinkPortEnabled (see 11.2.17.1).

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Table 14-20—cmldsLinkPortDS

Name	Data type	Operations supported[a]	References
portIdentity	PortIdentity (see 6.4.3.7)	R	14.18.2
cmldsLinkPortEnabled	Boolean	R	14.18.3
isMeasuringDelay	Boolean	R	14.18.4
asCapableAcrossDomains	Boolean	R	14.18.5
meanLinkDelay	TimeInterval	R	14.18.6
meanLinkDelayThresh	TimeInterval	RW	14.18.7
negMeanLinkDelayThresh	TimeInterval	RW	14.18.8
delayAsymmetry	TimeInterval	RW	14.18.9
nrrPdelay	Integer32	R	14.18.10
initialLogPdelayReqInterval	Integer8	RW	14.18.11
currentLogPdelayReqInterval	Integer8	R	14.18.12
useMgtSettableLogPdelayReqInterval	Boolean	RW	14.18.13
mgtSettableLogPdelayReqInterval	Integer8	RW	14.18.14
initialComputeNeighborRateRatio	Boolean	RW	14.18.15
currentComputeNeighborRateRatio	Boolean	R	14.18.16
useMgtSettableComputeNeighborRateRatio	Boolean	RW	14.18.17
mgtSettableComputeNeighborRateRatio	Boolean	RW	14.18.18
initialComputeMeanLinkDelay	Boolean	RW	14.18.19
currentComputeMeanLinkDelay	Boolean	R	14.18.20
useMgtSettableComputeMeanLinkDelay	Boolean	RW	14.18.21
mgtSettableComputeMeanLinkDelay	Boolean	RW	14.18.22
allowedLostResponses	UInteger8	RW	14.18.23
allowedFaults	UInteger8	RW	14.18.24
versionNumber	UInteger4	R	14.18.25
pdelayTruncatedTimestampsArray	UInteger48[4]	R	14.18.26
minorVersionNumber	UInteger4	R	14.18.27

[a] R = Read only access; RW = Read/write access.

14.18.4 isMeasuringDelay

The value is equal to the value of the instance of the Boolean isMeasuringDelay (see 11.2.12.6 and 16.4.3.3) that is per Link Port and across all domains.

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14.18.5 asCapableAcrossDomains

The value is equal to the value of the Boolean asCapableAcrossDomains (see 11.2.2 and 11.2.12.13).

14.18.6 meanLinkDelay

The value is equal to the value of the per-port global variable meanLinkDelay (see 10.2.5.8). It is an estimate of the current one-way propagation time on the link attached to this Link Port, measured as specified for the respective medium (see 11.2.16, 12.5.2, and 16.4). The value is zero for Link Ports attached to IEEE 802.3 EPON links and for the timeTransmitter port of an IEEE 802.11 link because one-way propagation delay is not measured on the latter and not directly measured on the former. The data type shall be TimeInterval. The default value is zero.

14.18.7 meanLinkDelayThresh

The value is equal to the value of the per-Link-Port global variable meanLinkDelayThresh (see 11.2.12.7). It is the propagation time threshold above which a Link Port (and therefore any PTP Ports that use the CMLDS on this Link Port) is considered not capable of participating in the IEEE 802.1AS protocol. Setting the managed object meanLinkDelayThresh causes the per-Link-Port global variable meanLinkDelayThresh to have the same value.

14.18.8 negMeanLinkDelayThresh

The value is equal to the value of the per-Link-Port global variable negMeanLinkDelayThresh (see 11.2.12.7). It is the negative propagation time threshold below which a Link Port (and therefore any PTP Ports that use the CMLDS on this Link Port) is considered not capable of participating in the IEEE 802.1AS protocol. Setting the managed object negMeanLinkDelayThresh causes the per-Link-Port global variable negMeanLinkDelayThresh to have the same value.

14.18.9 delayAsymmetry

The value is the asymmetry in the propagation delay on the link attached to this Link Port relative to the local clock, as defined in 10.2.5.9 and 8.3. If propagation delay asymmetry is not modeled, then delayAsymmetry is 0.

NOTE—The underlying per-port global variable delayAsymmetry is of type ScaledNS, which is a 96-bit value (see 6.4.3.1). delayAsymmetry values that are larger than the maximum value that can be represented by the TimeInterval data type, i.e., 0x7FFF FFFF FFFF FFFF (where the units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this largest value. delayAsymmetry values that are less than the minimum value that can be represented by the TimeInterval data type, i.e., 0x8000 0000 0000 0001 written in twos complement form (where the units are 2^{-16} ns; see 6.4.3.3), used for this managed object are set to this smallest value.

14.18.10 nrrPdelay

The value is an estimate of the ratio of the frequency of the LocalClock entity of the time-aware system at the other end of the link attached to this Link Port, to the frequency of the LocalClock entity of this time-aware system (see 10.2.5.7). The managed object nrrPdelay is expressed as the fractional frequency offset stored in the global variable nrrPdelay (see 11.2.12.14) multiplied by 2^{41}, i.e., the quantity (nrrPdelay(global variable) – 1.0)(2^{41}). The default value of the managed object nrrPdelay is 0.0, corresponding to a rate ratio of 1.0.

NOTE—This data set member corresponds to the scaledNeighborRateRatio member of the CommonMeanLinkDelayInformation Structure in 16.6.3.2 of IEEE Std 1588-2019.

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14.18.11 initialLogPdelayReqInterval

If useMgtSettableLogPdelayReqInterval is FALSE, then for FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to base 2 of the Pdelay_Req message transmission interval used when:

a) The Link Port is initialized, or
b) A message interval request TLV is received with the logLinkDelayInterval field set to 126 (see 11.5.2.2 and the LinkDelayIntervalSetting state machine in 11.2.20).

For all other media, the value is 127.

14.18.12 currentLogPdelayReqInterval

For FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the logarithm to the base 2 of the current Pdelay_Req message transmission interval (see 11.5.2.2).

For all other media, the value is 127.

14.18.13 useMgtSettableLogPdelayReqInterval

The managed object is a Boolean that determines the source of the sync interval and mean time interval between successive Pdelay_Req messages. If the value is TRUE, the value of currentLogPdelayReqInterval is set equal to the value of mgtSettableLogPdelayReqInterval (see 14.18.14). If the value of the managed object is FALSE, the value of currentLogPdelayReqInterval is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableLogPdelayReqInterval is FALSE.

14.18.14 mgtSettableLogPdelayReqInterval

The value is the logarithm to base 2 of the mean time interval between successive Pdelay_Req messages if useMgtSettableLogPdelayReqInterval is TRUE. The value is not used if useMgtSettableLogPdelayReqInterval is FALSE.

14.18.15 initialComputeNeighborRateRatio

If useMgtSettableComputeNeighborRateRatio is FALSE, then for FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial value of computeNeighborRateRatio (see 10.2.5.10).

For all other media, the value is TRUE.

14.18.16 currentComputeNeighborRateRatio

For FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeNeighborRateRatio.

For all other media, the value is TRUE.

14.18.17 useMgtSettableComputeNeighborRateRatio

The managed object is a Boolean that determines the source of the value of computeNeighborRateRatio. If the value is TRUE, the value of computeNeighborRateRatio is set equal to the value of

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mgtSettablecomputeNeighborRateRatio (see 14.18.18). If the value of the managed object is FALSE, the value of currentComputeNeighborRateRatio is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableComputeNeighborRateRatio is FALSE.

14.18.18 mgtSettableComputeNeighborRateRatio

computeNeighborRateRatio is configured to this value if useMgtSettableComputeNeighborRateRatio is TRUE. The value is not used if useMgtSettableComputeNeighborRateRatio is FALSE.

14.18.19 initialComputeMeanLinkDelay

If useMgtSettableComputeMeanLinkDelay is FALSE, then for FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the initial value of computeMeanLinkDelay (see 10.2.5.10).

For all other media, the value is TRUE.

14.18.20 currentComputeMeanLinkDelay

For FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the value is the current value of computeMeanLinkDelay.

For all other media, the value is TRUE.

14.18.21 useMgtSettableComputeMeanLinkDelay

The managed object is a Boolean that determines the source of the value of computeMeanLinkDelay. If the value is TRUE, the value of computeMeanLinkDelay is set equal to the value of mgtSettableComputeMeanLinkDelay (see 14.18.18). If the value of the managed object is FALSE, the value of currentComputeMeanLinkDelay is determined by the LinkDelayIntervalSetting state machine (see 11.2.20). The default value of useMgtSettableComputeMeanLinkDelay is FALSE.

14.18.22 mgtSettableComputeMeanLinkDelay

computeMeanLinkDelay is configured to this value if useMgtSettableComputeMeanLinkDelay is TRUE. The value is not used if useMgtSettableComputeMeanLinkDelay is FALSE.

14.18.23 allowedLostResponses

The value is equal to the value of the per-Link-Port global variable allowedLostResponses (see 11.5.3 and 11.2.12.4). It is the number of Pdelay_Req messages without valid responses above which a Link Port is considered to be not exchanging peer delay messages with its neighbor. Setting the managed object allowedLostResponses causes the per-Link-Port global variable allowedLostResponses to have the same value.

14.18.24 allowedFaults

The value is equal to the value of the per-Link-Port global variable allowedFaults (see 11.5.4 and 11.2.12.5). It is the number of faults (see 11.5.4) above which asCapableAcrossDomains is set to FALSE, i.e., a Link Port is considered not capable of interoperating with its neighbor via the IEEE 802.1AS protocol (see 10.2.5.1). Setting the managed object allowedFaults causes the per-Link-Port global variable allowedFaults to have the same value.

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14.18.25 versionNumber

This value is set to versionPTP as specified in 10.6.2.2.4.

14.18.26 pdelayTruncatedTimestampsArray

For FDE media and for CSN media that use the peer-to-peer delay mechanism to measure path delay (see 16.4.3.2), the values of the four elements of this array are as described in Table 14-12. For all other media, the values are zero. Array elements 0, 1, 2, and 3 correspond to the timestamps t1, t2, t3, and t4, modulo 2^{32}, respectively, in Figure 11-1 and are expressed in units of 2^{-16} ns (i.e., the value of each array element is equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2^{-16} ns, by 2^{48}). At any given time, the timestamp values stored in the array are for the same, and most recently completed, peer delay message exchange.

NOTE—This managed object is used with the Asymmetry Compensation Measurement Procedure, which is based on line-swapping.

14.18.27 minorVersionNumber

This value is set to minorVersionPTP as specified in 10.6.2.2.3.

14.19 Common Mean Link Delay Service Link Port Parameter Statistics Data Set (cmldsLinkPortStatisticsDS)

14.19.1 General

For every Link Port of the CMLD of a time-aware system, the cmldsLinkPortStatisticsDS provides counters associated with Link Port capabilities at a given time-aware system. The number of such statistics sets is the same as the value of cmldsDefaultDS.numberLinkPorts.

There is one set of cmldsLinkPortStatisticsDS parameters per Link Port of a time-aware system. The cmldsLinkPortStatisticsDS contains a set of counters for each Link Port that supports the time-synchronization capability, as detailed in Table 14-21. Each set of parameters can be created or removed dynamically in implementations that support dynamic configuration of Link Ports and components.

The parameters of cmldsLinkPortStatisticsDS are summarized in Table 14-21 and subsequently specified.

Table 14-21—cmldsLinkPortStatisticsDS

Name	Data type	Operations supported[a]	References
rxPdelayRequestCount	UInteger32	R	14.19.2
rxPdelayResponseCount	UInteger32	R	14.19.3
rxPdelayResponseFollowUpCount	UInteger32	R	14.19.4
rxPtpPacketDiscardCount	UInteger32	R	14.19.5
pdelayAllowedLostResponsesExceededCount	UInteger32	R	14.19.6
txPdelayRequestCount	UInteger32	R	14.19.7

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Table 14-21—cmldsLinkPortStatisticsDS

Name	Data type	Operations supported[a]	References
txPdelayResponseCount	UInteger32	R	14.19.8
txPdelayResponseFollowUpCount	UInteger32	R	14.19.9

[a] R= Read only access.

14.19.2 rxPdelayRequestCount

This counter increments every time a Pdelay_Req message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.19.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ states.

14.19.3 rxPdelayResponseCount

This counter increments every time a Pdelay_Resp message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayResp variable of the MDPdelayReq state machine (see 11.2.18.2.2 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP state.

14.19.4 rxPdelayResponseFollowUpCount

This counter increments every time a Pdelay_Resp_Follow_Up message is received, denoted by a transition to TRUE from FALSE of the rcvdPdelayRespFollowUp variable of the MDPdelayReq state machine (see 11.2.18.2.4 and Figure 11-9) when in the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state.

14.19.5 rxPtpPacketDiscardCount

This counter increments every time a PTP message of the CMLD is discarded, caused by the occurrence of any of the following conditions:

— A Pdelay_Resp message corresponding to a transmitted Pdelay_Req message is not received, denoted by a transition from the WAITING_FOR_PDELAY_RESP state to the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).
— A Pdelay_Resp_Follow_Up message corresponding to a transmitted Pdelay_Req message is not received, denoted by a transition from the WAITING_FOR_PDELAY_RESP_FOLLOW_UP state to the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

14.19.6 pdelayAllowedLostResponsesExceededCount

This counter increments every time the value of the variable lostResponses (see 11.2.18.2.9) exceeds the value of the variable allowedLostResponses (see 11.2.12.4) in the RESET state of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

14.19.7 txPdelayRequestCount

This counter increments every time a Pdelay_Req message is transmitted, denoted by entering the INITIAL_SEND_PDELAY_REQ or SEND_PDELAY_REQ states of the MDPdelayReq state machine (see 11.2.18 and Figure 11-9).

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14.19.8 txPdelayResponseCount

This counter increments every time a Pdelay_Resp message is transmitted, denoted by the following events:

— A transition to TRUE from FALSE of the rcvdPdelayReq variable of the MDPdelayResp state machine (see 11.2.19.2.1 and Figure 11-10) when in the WAITING_FOR_PDELAY_REQ or INITIAL_WAITING_FOR_PDELAY_REQ states; and
— The resulting entry to the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state.

14.19.9 txPdelayResponseFollowUpCount

This counter increments every time a Pdelay_Resp_Follow_Up message is transmitted, denoted by the following events:

— A transition to TRUE from FALSE of the rcvdMDTimestampReceiveMDPResp variable of the MDPdelayResp state machine (see 11.2.19.2.2 and Figure 11-10) when in the SENT_PDELAY_RESP_WAITING_FOR_TIMESTAMP state; and
— The resulting entry to the WAITING_FOR_PDELAY_REQ state.

14.20 Common Mean Link Delay Service Asymmetry Measurement Mode Parameter Data Set (cmldsAsymmetryMeasurementModeDS)

14.20.1 General

The cmldsAsymmetryMeasurementModeDS represents the capability to enable/disable the Asymmetry Compensation Measurement Procedure on a Link Port (see Annex G).

For every Link Port of the CMLD of a time-aware system, the cmldsAsymmetryMeasurementModeDS contains the single member asymmetryMeasurementMode, which is used to enable/disable the Asymmetry Compensation Measurement Procedure. The number of such data sets is the same as the numberLinkPorts value of the cmldsDefaultDS.

The parameter of cmldsAsymmetryMeasurementModeDS is summarized in Table 14-22 and subsequently specified.

Table 14-22—cmldsAsymmetryMeasurementModeDS

Name	Data type	Operations supported[a]	References
asymmetryMeasurementMode	Boolean	RW	14.20.2

[a] R = Read only access; RW = Read/write access.

14.20.2 asymmetryMeasurementMode

The value is equal to the value of the Boolean asymmetryMeasurementMode (see G.3). For HDE media, the value is TRUE if an asymmetry measurement is being performed for the link attached to this Link Port and FALSE otherwise. For all other media, the value shall be FALSE (see 10.2.5.2). Setting the managed object asymmetryMeasurementMode causes the Boolean asymmetryMeasurementMode to have the same value.

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NOTE—If an asymmetry measurement is being performed for a link, asymmetryMeasurementMode must be TRUE for the Link Ports at each end of the link.

14.21 Hot Standby System Parameter Data Set (hotStandbySystemDS)

14.21.1 General

The hotStandbySystemDS describes the attributes of the respective instance of the Hot Standby Service.

The parameters of hotStandbyDS are summarized in Table 14-19 and subsequently specified.

Table 14-23—hotStandbySystemDS

Name	Data type	Operations supported[a]	References
primaryPtpInstanceIndex	UInteger32	RW	14.21.2
secondaryPtpInstanceIndex	UInteger32	RW	14.21.3
hotStandbySystemEnable	Boolean	RW	14.21.4
hotStandbySystemState	Enumeration8	R	14.21.5
hotStandbySystemSplitFunctionality	Boolean	RW	14.21.6
primarySecondaryOffset	ScaledNs	R	14.21.7
primarySecondaryOffsetThresh	ScaledNs	RW	14.21.8
hotStandbySystemLogSyncTimeThresh	Integer8	RW	14.21.9

[a] R = Read only access; RW = Read/write access.

14.21.2 primaryPtpInstanceIndex

The value of primaryPtpInstanceIndex is the index (see 14.1.1) of the primary PTP Instance associated with this hotStandbySystem instance.

14.21.3 secondaryPtpInstanceIndex

The value of secondaryPtpInstanceIndex is the index (see 14.1.1) of the secondaryPTP Instance associated with this hotStandbySystem instance.

14.21.4 hotStandbySystemEnable

The value is the hotStandbySystemEnable attribute of the HotStandbySystem entity (see 18.5.1.2).

14.21.5 hotStandbySystemState

The value of hotStandbySystemState is the state of the hotStandbySystem, i.e., the value of the global variable hotStandbySystemState (see 18.5.1.1).

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14.21.6 hotStandbySystemSplitFunctionality

If the value is TRUE, the optional split functionality (see 18.6.4) is used. If the value is FALSE, the optional split functionality is not used.

14.21.7 primarySecondaryOffset

The absolute value of the difference between the clockTimeReceiverTimes (see 10.2.4.3) of the primary and secondary PTP Instances.

14.21.8 primarySecondaryOffsetThresh

The threshold for hotStandbySystemDS.primarySecondaryOffset (see 14.21.7), above which the hotStandbySytemState transitions from REDUNDANT to NOT_REDUNDANT, or does not transition from NOT_REDUNDANT or OUT_OF_SYNC to REDUNDANT even if other conditions for these transitions are satisfied.

14.21.9 hotStandbySystemLogSyncTimeThresh

The value of hotStandbySystemLogSyncTimeThresh is the logarithm to base 2 of the time interval, in seconds, after which the hotStandbySystem transitions from the OUT_OF_SYNC state to either the NOT-REDUNDANT or REDUNDANT state, or from the NOT_REDUNDANT to the REDUNDANT state, if all other conditions for the respective transition are met. The value –128 means that the transition time is zero, i.e., the transition occurs immediately.

14.22 Hot Standby System Description Parameter Data Set (hotStandbySystemDescriptionDS)

14.22.1 General

The hotStandbySystemDescriptionDS contains descriptive information for the respective instance of the Hot Standby Service.

The parameter of hotStandbySystemDescriptionDS is summarized in Table 14-24 and subsequently specified.

Table 14-24—hotStandbySystemDescriptionDS

Name	Data type	Operations supported[a]	References
userDescription	Octet128	RW	14.22.2

[a] RW= Read/write access.

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14.22.2 userDescription

The user description is a character string whose maximum length is 128.

14.23 Fault-Tolerant Timing with Time Integrity Entity Parameter Data Set (fttEntityDS)

14.23.1 General

The fttEntityDS describes the attributes of the respective instance of the FTT entity.

The parameters of fttEntityDS are summarized in Table 14-25 and subsequently specified.

Table 14-25—fttEntityDS

Name	Data type	Operations supported[a]	References
fttMaxInputs	UInteger8	R	14.23.2
fttNumInputs	UInteger8	RW	14.23.3
fttMaxDtsfs	UInteger8	R	14.23.4
fttNumDtsfs	UInteger8	RW	14.23.5
fttDtsfMaxInputs	Uinteger8	R	14.23.6
fttInputsToTsf	Uinteger16[fttNumDtsfs + 1]	RW	14.23.7
fttPtpInstanceToInput	UInteger32[fttNumInputs]	RW	14.23.8
fttHyst	TimeInterval[fttNumInputs] [fttNumInputs]	RW	14.23.9
fttMaxAs	TimeInterval[fttNumInputs] [fttNumInputs]	RW	14.23.10
fttSelChangeThresh	TimeInterval[fttNumDtsfs + 1]	RW	14.23.11
fttRRThresh	UInteger32	RW	14.23.12
fttRRThreshStdDev	UInteger32	RW	14.23.13
fttUseNQ	Boolean[fttNumDtsfs + 1]	RW	14.23.14
fttUseFTTClk	Boolean	RW	14.23.15
fttTsfSelInput	UInteger8[fttNumDtsfs + 1]	R	14.23.16
fttSelInstance	UInteger32	R	14.23.17
fttTsfSelChangeCnt	Uinteger16[fttNumDtsfs + 1]	R	14.23.18
fttInputTrustState	Enumeration2[fttNumInputs]	R	14.23.19
fttTrustState	Enumeration2	R	14.23.20
fttStateChangeCnt	Uinteger16	R	14.23.21

[a] RW = Read/Write access; R = Read only access.

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14.23.2 fttMaxInputs

The fttMaxInputs object specifies the maximum number of input ClockTarget interfaces available on the FTT entity. The value ranges from 1 to 16. The default value is implementation specific.

14.23.3 fttNumInputs

The fttNumInputs object specifies the number of enabled inputs to the FTT entity, where an enabled input is one that has been mapped to either a DTSF instance or to the ITSF. The value ranges from 1 to fttMaxInputs.

14.23.4 fttMaxDtsfs

The fttMaxDtsfs object specifies the maximum number of DTSF instances available in the FTT entity. The default value is implementation specific.

14.23.5 fttNumDtsfs

The fttNumDtsfs object specifies the number of DTSF instances enabled in the FTT entity, with the value ranging from 0 to fttMaxDtsfs.

14.23.6 fttDtsfMaxInputs

The fttDtsfMaxInputs object gives the maximum number of inputs available on each of the DTSF instances in the FTT entity. The value ranges from 1 to 16. The default value is implementation specific.

NOTE—All DTSFs of one FTT entity share the same limit on the maximum number of inputs allowed.

14.23.7 fttInputsToTsf

The fttInputsToTsf object provides a list of the FTT input index numbers mapped to each time selection function (TSF, see 20.2.3). fttInputsToTsf[*j*], where *j* is a value from 0 to fttNumDtsfs, provides the bit map of FTT inputs that are assigned to TSF instance *j*. The most-significant bit corresponds to FTT input index 16, the least-significant bit corresponds to FTT input index 1. The number of inputs assigned to any DTSF instance cannot be greater than fttDtsfMaxInputs and the total number of assigned inputs to all TSF instances cannot exceed fttMaxInputs.

14.23.8 fttPtpInstanceToInput

The fttPtpInstanceToInput object provides a mapping between FTT input index number and the PTP Instance index number (see 14.1.1). fttPtpInstanceToInput[*j*] is the index number of the PTP Instance that is connected to the FTT input with input index *j*.

14.23.9 fttHyst

This fttHyst object is a two-dimensional array of TimeInterval (see 6.4.3.3) values. The array has a size of fttNumInputs in each dimension. The object fttHyst[*x*][*y*] holds the hysteresis to be added to fttMaxAs[*x*][*y*] (see 14.23.10) for the time values of the two FTT inputs with index numbers *x* and *y*.

14.23.10 fttMaxAs

The fttMaxAs object is a two-dimensional array of TimeInterval (see 6.4.3.3) values. The array has a size of fttNumInputs in each dimension. The fttMaxAs[*x*][*y*] object defines the maximum expected skew between time values provided by the FTT inputs with index *x* and *y*, when those time values are not faulty.

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14.23.11 fttSelChangeThresh

The fttSelChangeThresh object provides the TimeInterval thresholds used by each TSF instance to determine the time difference required to switch between FTT inputs that meet the selection criteria (see 20.3.2.4). fttSelChangeThresh[*j*] is the threshold value used by TSF instance with instance number *j*.

14.23.12 fttRRThresh

The fttRRThresh object specifies the maximum rate-ratio offset magnitude, between the FTT entity's selected time clock rate and the FTT entity's local clock (FTT_CLK) rate, that is deemed to be acceptable to go to or remain in the FREQ_TRUSTED state (see 20.2.4). The rateRatio offset magnitude is expressed as a fractional frequency offset multiplied by 2^{41}. The default value for fttRRThresh is 0.

14.23.13 fttRRThreshStdDev

The fttRRThreshStdDev object specifies the maximum standard deviation of the rate-ratio offset, between the FTT entity's selected time clock rate and the FTT entity's local clock (FTT_CLK) rate, that is deemed to be acceptable to go to or remain in the FREQ_TRUSTED state (see 20.3.4). The standard deviation of the rateRatio offset is expressed as a fractional frequency offset multiplied by 2^{41}, and has a default value of 0.

14.23.14 fttUseNQ

The fttUseNQ object enables or disables the use of not qualified (NQ) input (see 20.2.4) by a TSF instance when no trusted time is found. fttUseNQ[*j*], where *j* is a value from 0 to fttNumDtsfs, is the boolean value of fttUseNQ for TSF instance *j*.

If fttUseNQ[*j*] is TRUE, the use of the NQ input of TSF instance *j* is enabled.

If fttUseNQ[*j*] is FALSE, the use of the NQ input of TSF instance *j* is not enabled.

The default values are implementation-specific.

14.23.15 fttUseFTTClk

The fttUseFTTClk object defines whether the FTT_CLK frequency is used as a reference for checking time integrity (see 20.2.2).

If fttUseFTTClk is TRUE, the FTT_CLK frequency is used as a reference for checking time integrity.

If fttUseFTTClk is FALSE, the FTT_CLK frequency is not used as a reference for checking time integrity.

The default value is implementation-specific.

14.23.16 fttTsfSelInput

The fttTsfSelInput object specifies the FTT input index that is selected by each TSF instance. It is the array value of the global variable fttTsfSelInput (see 20.3.2.3).

14.23.17 fttSelInstance

The fttSelInstance object specifies the index of the PTP Instance corresponding to the FTT input that was selected by the FTT entity as its output.

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14.23.18 fttTsfSelChangeCnt

The fttTsfSelChangeCnt object counts the number of times each TSF has changed its selection, i.e., selected input that is determined to be trusted. fttTsfSelChangeCnt[*j*], where *j* is a value from 0 to fttNumDtsfs, is the number of times TSF instance *j* has changed its selected input. The default value is 0.

14.23.19 fttInputTrustState

The trust state of all FTT inputs, i.e., the array value of the global variable fttInputTrustState (see 20.3.2.2).

14.23.20 fttTrustState

The value of fttTrustState is the trust state of the FTT entity, i.e., the value of the global variable fttTrustState (see 20.3.2.1).

14.23.21 fttStateChangeCnt

The fttStateChangeCnt object counts the number of times the FTT entity has changed its trust state (see 14.23.20). The default value is 0.

14.24 Fault-Tolerant Timing with Time Integrity Entity Description Parameter Data Set (fttEntityDescriptionDS)

14.24.1 General

The fttEntityDescriptionDS contains descriptive information for the respective instance of the FTT entity. There is one fttEntityDescriptionDS table per FTT entity.

The parameter of fttEntityDescriptionDS is summarized in Table 14-26 and subsequently specified.

Table 14-26—fttEntityDescriptionDS

Name	Data type	Operations supported[a]	References
userDescription	Octet128	RW	14.24.2

[a] RW = Read/Write access.

14.24.2 userDescription

The user description is a character string whose maximum length is 128.

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15. Management Information Base (MIB)

15.1 Internet Standard Management Framework

For a detailed overview of the documents that describe the current Internet Standard Management Framework, refer to section 7 of IETF RFC 3410 (Dec. 2002).

Managed objects are accessed via a virtual information store, termed the *Management Information Base* (MIB). MIB objects are generally accessed through the Simple Network Management Protocol (SNMP). Objects in the MIB are defined using the mechanisms defined in the Structure of Management Information (SMI). This clause specifies a MIB module that is compliant to the SMIv2, which is described in IETF STD 58, comprising IETF RFC 2578 [B14], IETF RFC 2579 [B15], and IETF RFC 2580 [B16].

This clause contains a complete SMIv2 MIB set for all features of this standard, except for hot standby (see Clause 18), the drift tracking feature (see 11.4.4.4), and new managed objects included in this revision of the standard.

15.2 Structure of the MIB

The IEEE 802.1AS MIB provides objects to configure and manage the IEEE 802.1AS timing and synchronization for time-sensitive applications.

The MIB contains a set of textual conventions and is additionally subdivided into the following subtrees, each of which is organized as a set of related objects:

a) The Default Parameter Data Set (defaultDS) represents the native capabilities of a PTP Instance.

b) The Current Parameter Data Set (currentDS) represents topological position of a local PTP Instance relative to the Grandmaster PTP Instance.

c) The Parent Parameter Data Set (parentDS) represents capabilities of the upstream PTP Instance toward the Grandmaster PTP Instance, as measured at a local PTP Instance.

d) The Time Properties Parameter Data Set (timePropertiesDS) represents capabilities of the Grandmaster PTP Instance, as measured at a local PTP Instance.

e) The Path Trace Parameter Data Set (pathTraceDS) represents the current path trace information (see 10.3.9.23) available at the PTP Instance.

f) The Acceptable TimeTransmitter Table Parameter Data Set (acceptableTimeTransmitterTableDS) represents the acceptable timeTransmitter table used when the media-dependent PTP Port type of EPON is present in a PTP Instance.

g) The Port Parameter Data Set (portDS) represents time-aware capabilities at a given PTP Port, as a set of augmentation to the interface table entry (ifEntry).

h) The Description Port Parameter Data Set (descriptionPortDS) contains the profileIdentifier for this PTP profile as specified in F.2.

i) The Port Parameter Statistics Data Set (portStatisticsDS) represents statistics and counters associated with time-aware capabilities at a given PTP Relay Instance or PTP End Instance port.

j) The Acceptable TimeTransmitter Port Parameter Data Set (acceptableTimeTransmitterPortDS) represents the capability to enable/disable the acceptable timeTransmitter table feature on a PTP Port.

k) The External Port Configuration Port Parameter Data Set (externalPortConfigurationPortDS) is used with the external port configuration option to indicate the desired state of a PTP Port.

l) The Asymmetry Measurement Mode Parameter Data Set (asymmetryMeasurementModeDS) represents the capability to enable/disable the Asymmetry Compensation Measurement Procedure

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on a port (see Annex G) and is used instead of the cmldsAsymmetryMeasurementModeDS when CMLDS is not used and there is a single gPTP domain.

m) The Common Services Port Parameter Data Set (commonServicesPortDS) enables a PTP Port of a PTP Instance to determine which port of the respective common service corresponds to that PTP Port.

n) The CMLD Default Parameter Data Set (cmldsDefaultDS) describes the per-time-aware-system attributes of the CMLD.

o) The CMLD Link Port Parameter Data Set (cmldsLinkPortDS) represents time-aware Link Port capabilities for the CMLD of a time-aware system.

p) The CMLD Link Port Parameter Statistics Data Set (cmldsLinkPortStatisticsDS) represents statistics and counters associated with Link Port capabilities at a given time-aware system.

q) The CMLD Asymmetry Measurement Mode Parameter Data Set (cmldsAsymmetryMeasurementModeDS) represents the capability to enable/disable the Asymmetry Compensation Measurement Procedure on a Link Port (see Annex G).

Table 15-1 shows the structure of the MIB and the relationship of the MIB objects to the above data sets.

Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference

MIB table	MIB object	Reference
ieee8021AsV3DefaultDS		defaultDS (Table 14-1)
	ieee8021AsV3DefaultDSClockIdentity	14.2.2
	ieee8021AsV3DefaultDSNumberPorts	14.2.3
	ieee8021AsV3DefaultDSClockQualityClockClass	14.2.4.2
	ieee8021AsV3DefaultDSClockQualityClockAccuracy	14.2.4.3
	ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance	14.2.4.4
	ieee8021AsV3DefaultDSPriority1	14.2.5
	ieee8021AsV3DefaultDSPriority2	14.2.6
	ieee8021AsV3DefaultDSGmCapable	14.2.7
	ieee8021AsV3DefaultDSCurrentUtcOffset	14.2.8
	ieee8021AsV3DefaultDSCurrentUtcOffsetValid	14.2.9
	ieee8021AsV3DefaultDSLeap59	14.2.10
	ieee8021AsV3DefaultDSLeap61	14.2.11
	ieee8021AsV3DefaultDSTimeTraceable	14.2.12
	ieee8021AsV3DefaultDSFrequencyTraceable	14.2.13
	ieee8021AsV3DefaultDSPtpTimescale	14.2.14
	ieee8021AsV3DefaultDSTimeSource	14.2.15
	ieee8021AsV3DefaultDSDomainNumber	14.2.16
	ieee8021AsV3DefaultDSSdoId	14.2.17
	ieee8021AsV3DefaultDSExternalPortConfigurationEnabled	14.2.18
	ieee8021AsV3DefaultDSInstanceEnable	14.2.19

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
ieee8021AsV3CurrentDS		currentDS (Table 14-2)
	ieee8021AsV3CurrentDSStepsRemoved	14.3.2
	ieee8021AsV3CurrentDSOffsetFromTimeTransmitter	14.3.3
	ieee8021AsV3CurrentDSLastGmPhaseChange	14.3.4
	ieee8021AsV3CurrentDSLastGmFreqChange	14.3.5
	ieee8021AsV3CurrentDSGmTimebaseIndicator	14.3.6
	ieee8021AsV3CurrentDSGmChangeCount	14.3.7
	ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent	14.3.8
	ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent	14.3.9
	ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent	14.3.10
ieee8021AsV3ParentDS		parentDS (Table 14-3)
	ieee8021AsV3ParentDSParentClockIdentity	14.4.2
	ieee8021AsV3ParentDSParentPortNumber	14.4.2
	ieee8021AsV3ParentDSCumulativeRateRatio	14.4.3
	ieee8021AsV3ParentDSGrandmasterIdentity	14.4.4
	ieee8021AsV3ParentDSGrandmasterClockQualityclockClass	14.4.5.2
	ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy	14.4.5.3
	ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar	14.4.5.4
	ieee8021AsV3ParentDSGrandmasterPriority1	14.4.6
	ieee8021AsV3ParentDSGrandmasterPriority2	14.4.7
ieee8021AsV3TimePropertiesDS		timePropertiesDS (Table 14-4)
	ieee8021AsV3TimePropertiesDSCurrentUtcOffset	14.5.2
	ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid	14.5.3
	ieee8021AsV3TimePropertiesDSLeap59	14.5.4
	ieee8021AsV3TimePropertiesDSLeap61	14.5.5
	ieee8021AsV3TimePropertiesDSTimeTraceable	14.5.6
	ieee8021AsV3TimePropertiesDSFrequencyTraceable	14.5.7
	ieee8021AsV3TimePropertiesDSPtpTimescale	14.5.8
	ieee8021AsV3TimePropertiesDSTimeSource	14.5.9
ieee8021AsV3PathTraceDS		pathTraceDS (Table 14-5)
	ieee8021AsV3PathTraceDSEnable	14.6.3

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
ieee8021AsV3PathTraceDSArray		pathTraceDS (Table 14-5)
	ieee8021AsV3PathTraceDSArrayList	14.6.2
ieee8021AsV3AcceptableTimeTransmitterTableDS		acceptableTimeTransmitterTable DS (Table 14-6)
	ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize	14.7.2
	ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize	14.7.3
ieee8021AsV3AcceptableTimeTransmitterTableDSArray		acceptableTimeTransmitterTable DS (Table 14-6)
	ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity	14.7.4
	ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1	14.7.4
ieee8021AsV3PortDS		portDS (Table 14-9)
	ieee8021AsV3PortDSClockIdentity	14.10.2
	ieee8021AsV3PortDSPortNumber	14.10.2
	ieee8021AsV3PortDSPortState	14.10.3
	ieee8021AsV3PortDSPtpPortEnabled	14.10.4
	ieee8021AsV3PortDSdelayMechanism	14.10.5
	ieee8021AsV3PortDSIsMeasuringDelay	14.10.6
	ieee8021AsV3PortDSAsCapable	14.10.7
	ieee8021AsV3PortDSMeanLinkDelay	14.10.8
	ieee8021AsV3PortDSMeanLinkDelayThresh	14.10.9
	ieee8021AsV3PortDSDelayAsym	14.10.11
	ieee8021AsV3PortDSNbrRateRatio	14.10.12
	ieee8021AsV3PortDSInitialLogAnnounceInterval	14.10.13
	ieee8021AsV3PortDSCurrentLogAnnounceInterval	14.10.14
	ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval	14.10.15
	ieee8021AsV3PortDSMgtSettableLogAnnounceInterval	14.10.16
	ieee8021AsV3PortDSAnnounceReceiptTimeout	14.10.17
	ieee8021AsV3PortDSInitialLogSyncInterval	14.10.18
	ieee8021AsV3PortDSCurrentLogSyncInterval	14.10.19
	ieee8021AsV3PortDSUseMgtSettableLogSyncInterval	14.10.20
	ieee8021AsV3PortDSMgtSettableLogSyncInterval	14.10.21
	ieee8021AsV3PortDSSyncReceiptTimeout	14.10.22
	ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval	14.10.23
	ieee8021AsV3PortDSInitialLogPdelayReqInterval	14.10.24

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
	ieee8021AsV3PortDSCurrentLogPdelayReqInterval	14.10.25
	ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval	14.10.26
	ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval	14.10.27
	ieee8021AsV3PortDSInitialLogGptpCapableMessageInterval	14.10.28
	ieee8021AsV3PortDSCurrentLogGptpCapableMessageInterval	14.10.29
	ieee8021AsV3PortDSUseMgtSettableLogGptpCapableMessageInterval	14.10.30
	ieee8021AsV3PortDSMgtSettableLogGptpCapableMessageInterval	14.10.31
	ieee8021AsV3PortDSInitialComputeNbrRateRatio	14.10.32
	ieee8021AsV3PortDSCurrentComputeNbrRateRatio	14.10.33
	ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio	14.10.34
	ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio	14.10.35
	ieee8021AsV3PortDSInitialComputeMeanLinkDelay	14.10.36
	ieee8021AsV3PortDSCurrentComputeMeanLinkDelay	14.10.37
	ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay	14.10.38
	ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay	14.10.39
	ieee8021AsV3PortDSAllowedLostRsp	14.10.40
	ieee8021AsV3PortDSAllowedFaults	14.10.41
	ieee8021AsV3PortDSGPtpCapableReceiptTimeout	14.10.42
	ieee8021AsV3PortDSVersionNumber	14.10.43
	ieee8021AsV3PortDSNup	14.10.44
	ieee8021AsV3PortDSNdown	14.10.45
	ieee8021AsV3PortDSOneStepTxOper	14.10.46
	ieee8021AsV3PortDSOneStepReceive	14.10.47
	ieee8021AsV3PortDSOneStepTransmit	14.10.48
	ieee8021AsV3PortDSInitialOneStepTxOper	14.10.49
	ieee8021AsV3PortDSCurrentOneStepTxOper	14.10.50
	ieee8021AsV3PortDSUseMgtSettableOneStepTxOper	14.10.51
	ieee8021AsV3PortDSMgtSettableOneStepTxOper	14.10.52
	ieee8021AsV3PortDSSyncLocked	14.10.53
	ieee8021AsV3PortDSPdelayTruncTST1	14.10.54
	ieee8021AsV3PortDSPdelayTruncTST2	14.10.54
	ieee8021AsV3PortDSPdelayTruncTST3	14.10.54

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
	ieee8021AsV3PortDSPdelayTruncTST4	14.10.54
	ieee8021AsV3PortDSMinorVersionNumber	14.10.55
ieee8021AsV3DescriptionPortDS		descriptionPortDS (Table 14-13)
	ieee8021AsV3DescriptionPortDSProfileIdentifier	14.11.2
ieee8021AsV3PortStatDS		portStatisticsDS (Table 14-14)
	ieee8021AsV3PortStatRxSyncCount	14.12.2
	ieee8021AsV3PortStatRxOneStepSyncCount	14.12.3
	ieee8021AsV3PortStatRxFollowUpCount	14.12.4
	ieee8021AsV3PortStatRxPdelayRequestCount	14.12.5
	ieee8021AsV3PortStatRxPdelayRspCount	14.12.6
	ieee8021AsV3PortStatRxPdelayRspFollowUpCount	14.12.7
	ieee8021AsV3PortStatRxAnnounceCount	14.12.8
	ieee8021AsV3PortStatRxPtpPacketDiscardCount	14.12.9
	ieee8021AsV3PortStatSyncReceiptTimeoutCount	14.12.10
	ieee8021AsV3PortStatAnnounceReceiptTimeoutCount	14.12.11
	ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount	14.12.12
	ieee8021AsV3PortStatTxSyncCount	14.12.13
	ieee8021AsV3PortStatTxOneStepSyncCount	14.12.14
	ieee8021AsV3PortStatTxFollowUpCount	14.12.15
	ieee8021AsV3PortStatTxPdelayRequestCount	14.12.16
	ieee8021AsV3PortStatTxPdelayRspCount	14.12.17
	ieee8021AsV3PortStatTxPdelayRspFollowUpCount	14.12.18
	ieee8021AsV3PortStatTxAnnounceCount	14.12.19
ieee8021AsV3AcceptableTimeTransmitterPortDS		acceptableTimeTransmitterTableDS (Table 14-15)
	ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled	14.13.2
ieee8021AsV3ExternalPortConfigurationPortDS		externalPortConfigurationPortDS (Table 14-16)
	ieee8021AsV3ExternalPortConfigurationPortDSDesiredState	14.14.2
ieee8021AsV3AsymMeasurementModeDS		asymmetryMeasurementModeDS (Table 14-17)
	ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode	14.15.2

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
ieee8021AsV3CommonServicesPortDS		commonServicesPortDS (Table 14-18)
	ieee8021AsV3CommonServicesPortDSCmldsLinkPortPortNumber	14.16.2
ieee8021AsV3CommonMeanLinkDelayServiceDefaultDS		cmldsDefaultDS (Table 14-19)
	ieee8021AsV3CmldsDefaultDSClockIdentity	14.17.2
	ieee8021AsV3CmldsDefaultDSNumberLinkPorts	14.17.3
ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDS		cmldsLinkPortDS (Table 14-20)
	ieee8021AsV3CmldsLinkPortDSClockIdentity	14.18.2
	ieee8021AsV3CmldsLinkPortDSPortNumber	14.18.2
	ieee8021AsV3CmldsLinkPortDSCmldsLinkPortEnabled	14.18.3
	ieee8021AsV3CmldsLinkPortDSIsMeasuringDelay	14.18.4
	ieee8021AsV3CmldsLinkPortDSAsCapableAcrossDomains	14.18.5
	ieee8021AsV3CmldsLinkPortDSMeanLinkDelay	14.18.6
	ieee8021AsV3CmldsLinkPortDSMeanLinkDelayThresh	14.18.7
	ieee8021AsV3CmldsLinkPortDSDelayAsym	14.18.9
	ieee8021AsV3CmldsLinkPortDSNbrRateRatio	14.18.10
	ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval	14.18.11
	ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval	14.18.12
	ieee8021AsV3CmldsLinkPortDSUseMgtSettableLogPdelayReqInterval	14.18.13
	ieee8021AsV3CmldsLinkPortDSMgtSettableLogPdelayReqInterval	14.18.14
	ieee8021AsV3CmldsLinkPortDSInitialComputeNbrRateRatio	14.18.15
	ieee8021AsV3CmldsLinkPortDSCurrentComputeNbrRateRatio	14.18.16
	ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio	14.18.17
	ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio	14.18.18
	ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay	14.18.19
	ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay	14.18.20
	ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay	14.18.21
	ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay	14.18.22
	ieee8021AsV3CmldsLinkPortDSAllowedLostRsp	14.18.23
	ieee8021AsV3CmldsLinkPortDSAllowedFaults	14.18.24
	ieee8021AsV3CmldsLinkPortDSVersionNumber	14.18.25
	ieee8021AsV3CmldsLinkPortDSPdelayTruncTST1	14.18.26

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Table 15-1—IEEE8021-AS-V3 MIB structure and object cross reference *(continued)*

MIB table	MIB object	Reference
	ieee8021AsV3CmldsLinkPortDSPdelayTruncTST2	14.18.26
	ieee8021AsV3CmldsLinkPortDSPdelayTruncTST3	14.18.26
	ieee8021AsV3CmldsLinkPortDSPdelayTruncTST4	14.18.26
	ieee8021AsV3CmldsLinkPortDSMinorVersionNumber	14.18.27
ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDS		cmldsLinkPortStatisticsDS (Table 14-21)
	ieee8021AsV3CmldsLinkPortStatDSRxPdelayRequestCount	14.19.2
	ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspCount	14.19.3
	ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspFollowUpCount	14.19.4
	ieee8021AsV3CmldsLinkPortStatDSRxPtpPacketDiscardCount	14.19.5
	ieee8021AsV3CmldsLinkPortStatDSPdelayAllowedLostRspExceedCount	14.19.6
	ieee8021AsV3CmldsLinkPortStatDSTxPdelayRequestCount	14.19.7
	ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspCount	14.19.8
	ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspFollowUpCount	14.19.9
ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDS		cmldsAsymmetryMeasurementModeDS (Table 14-22)
	ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode	14.20.2

15.3 Relationship to MIB in IEEE Std 802.1AS-2011

The version 1 MIB module (IEEE8021-AS MIB) that was published in IEEE Std 802.1AS-2011 has been superseded by the version 2 MIB module (IEEE8021-AS-V2 MIB) specified in 15.6 of IEEE Std 802.1AS-2020. The version 3 MIB module (IEEE8021-AS-V3 MIB) specified in the current standard remains unchanged from the IEEE8021-AS-V2 MIB except updating terminology. Support of the version 3 module is a requirement for conformance to the required or optional capabilities (Clause 5) in the current standard.

For an implementation that supports a single PTP Instance, version 1, version 2, and version 3 implementations can successfully co-exist and interoperate.

15.4 Security considerations

SNMP versions prior to SNMPv3 did not include adequate security. Even if the network itself is secure (for example by using IPsec), there is no control as to who on the secure network is allowed to access and GET/SET (read/change/create/delete) the objects in these MIB module.

It is recommended that implementers consider the security features as provided by the SNMPv3 framework [see section 8 in IETF RFC 3410 (Dec. 2002)], including full support for the SNMPv3 cryptographic mechanisms (for authentication and privacy).

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Further, deployment of SNMP versions prior to SNMPv3 is not recommended. Instead, it is recommended to deploy SNMPv3 and to enable cryptographic security. It is then a customer/operator responsibility to ensure that the SNMP entity giving access to an instance of these MIB modules is properly configured to give access to the objects only to the principals (users) that have legitimate rights to indeed GET or SET (change/create/delete) them.

A number of management objects defined in the IEEE8021-AS-V3 MIB module have a MAX-ACCESS clause of read-write and/or read-create. Such objects might be considered sensitive or vulnerable in some network environments. The support for SET operations in a non-secure environment without proper protection can have a negative effect on network operations.

Some of the readable objects in this MIB module (i.e., objects with a MAX-ACCESS other than "not-accessible") might be considered sensitive or vulnerable in some network environments. It is thus important to control all types of access (including GET and/or NOTIFY) to these objects and possibly to even encrypt the values of these objects when sending them over the network via SNMP.

The following objects in the IEEE8021-AS-V3 MIB can be manipulated to interfere with the operation of timing synchronization. This could, for example, be used to force a reinitialization of state machines to cause timing synchronization and network instability. Another possibility would be for an attacker to override Grandmaster PTP Instance status to give a user (or an attacker) unauthorized control over the network time.

Improper manipulation of the following writable objects could result in an unintended Grandmaster PTP Instance to be elected when a system is grandmaster-capable in a gPTP domain. It could also be used maliciously to cause frequent Grandmaster PTP Instance changes that could affect network stability.

ieee8021AsV3DefaultDSPriority1
ieee8021AsV3DefaultDSPriority2

Improper manipulation of the following writable objects could result in a segmented time-aware network, could compromise the expected accuracy, and could interrupt paths of the gPTP domain.

ieee8021AsV3PortDSPtpPortEnabled
ieee8021AsV3PortDSDelayAsymmetry

Unintended access to any of the readable tables or variables in the IEEE8021-AS-V3 MIB alerts the reader that timing synchronization in gPTP domain is configured, and on which values timing parameters are configured, and which system is the current Grandmaster PTP Instance. This information can suggest to an attacker what applications are being run, and thus suggest application-specific attacks, or can enable the attacker to detect whether their attacks are being successful. It is thus important to control even GET access to these objects and possibly to even encrypt the values of these objects when sending them over the network via SNMP.

15.5 Textual conventions defined in this MIB

The following textual conventions are defined in this MIB:

a) Ieee8021AsV3ClockIdentity. IEEE 802 MAC address represented in "canonical" order defined by IEEE Std 802-2024, 64-bit Network Unique Identifier (NUI-64) as described in IEEE Std 802-2024.
b) Ieee8021AsV3GPtpProfileIdentifier. Profile identifier (see 14.11.2).
c) Ieee8021AsV3ClockClassValue. Clock class value (see 8.6.2.2).
d) Ieee8021AsV3ClockAccuracyValue. Clock accuracy value (see 8.6.2.3).

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e) Ieee8021AsV3TimeSourceValue. Source of time used by Grandmaster PTP Instance (see 8.6.2.7).

f) Ieee8021ASV3PtpTimeInterval. Time intervals in units of 2^{-16} ns (see 6.4.3.3).

g) Ieee8021ASV3PtpPortIdentity. Identifies a port of a PTP Instance (see 6.4.3.7).

h) Ieee8021ASV3ScaledNs. Represents signed values of time and time interval in units of 2^{-16} ns (see 6.4.3.1).

i) Ieee8021ASV3UScaledNs. Represents unsigned values of time and time interval in units of 2^{-16} ns (see 6.4.3.2).

j) Ieee8021ASV3PTPInstanceIdentifier. Entity of a single time-aware system that executes gPTP in one gPTP domain (see 7.2.1 and 8.1).

k) Ieee8021ASV3Timestamp. Value of Ieee8021ASV2Timestamp is equal to the remainder obtained upon dividing the respective timestamp, expressed in units of 2^{-16} ns, by 2^{48} (see 14.10.54).

15.6 IEEE 802.1AS MIB module[21,22]

In the following MIB module definitions, if any discrepancy between the DESCRIPTION text and the corresponding definition in any other part of this standard occurs, the definitions outside this subclause take precedence.

[21] Copyright release for MIBs: Users of this standard may freely reproduce the MIBs contained in this subclause so that they can be used for their intended purpose.

[22] An ASCII version of this MIB module can be obtained from the IEEE 802.1 website at https://www.ieee802.org/1/pages/MIBS.html.

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```
IEEE8021-AS-V3-MIB DEFINITIONS ::= BEGIN
-- ============================================================
-- MIB for support of 802.1AS Timing and Synchronization in
-- IEEE 802.1Q Bridged Local Area Networks
-- ============================================================

IMPORTS
    MODULE-IDENTITY, OBJECT-TYPE, Unsigned32, Integer32, Counter32
        FROM SNMPv2-SMI              -- [RFC2578]
    TEXTUAL-CONVENTION, TruthValue, RowStatus, TimeStamp
        FROM SNMPv2-TC               -- [RFC2579]
    MODULE-COMPLIANCE, OBJECT-GROUP  -- [RFC2580]
        FROM SNMPv2-CONF
    SnmpAdminString
        FROM SNMP-FRAMEWORK-MIB      -- [RFC3411]
    InterfaceIndexOrZero
        FROM IF-MIB                  -- [RFC2863]
    Float64TC
        FROM FLOAT-TC-MIB            -- [RFC6340]
    IEEE8021BridgePortNumber
        FROM IEEE8021-TC-MIB
    ;

ieee8021AsV3TimeSyncMib MODULE-IDENTITY
    LAST-UPDATED "202512100000Z" -- December 10, 2025
    ORGANIZATION "IEEE 802.1 Working Group"
    CONTACT-INFO
                "WG-URL: http://ieee802.org/1/
                WG-EMail: stds-802-1-l@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            445 Hoes Lane
            Piscataway, NJ 08854
            USA

    E-mail: stds-802-1-chairs@ieee.org"

    DESCRIPTION
        "The Management Information Base module for
         IEEE 802.1AS time-synchronization protocol.
         Copyright (C) IEEE (2025).
         This version of this MIB module is part of IEEE Std
         802.1AS-2025; see the standard itself for full legal
         notices."

    REVISION "202512100000Z" -- December 10, 2025
    DESCRIPTION
            "Published as part of IEEE Std 802.1AS-2025.
            Cross references updated."

    REVISION    "202402140000Z" -- February 14, 2024
    DESCRIPTION
        "This MIB module remains unchanged from
         IEEE8021-AS-V2-MIB except updating the terminology.

         Published as part of IEEE Std 802.1ASdr-2024
         This version of this MIB module is part of IEEE Std
```


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```
        802.1ASdr-2024; see the standard itself for full legal
        notices."

 ::= { iso(1) org(3) ieee(111)
        standards-association-numbers-series-standards (2)
        lan-man-stds (802) ieee802dot1 (1) ieee802dot1mibs (1) 40 }

ieee8021AsV3MIBObjects    OBJECT IDENTIFIER ::= {ieee8021AsV3TimeSyncMib 1}
ieee8021AsV3Conformance   OBJECT IDENTIFIER ::= {ieee8021AsV3TimeSyncMib 2}

-- ============================================================
-- Textual Conventions
-- ============================================================

Ieee8021AsV3ClockIdentity ::= TEXTUAL-CONVENTION
    DISPLAY-HINT
        "1x:"
    STATUS current
    DESCRIPTION
        "The Ieee8021AsV3ClockIdentity type identifies a PTP Instance.
         The clockIdentity attribute shall be as specified in
         IEEE Std 1588-2019."
    REFERENCE    "6.4.3.6, 8.5.2.2 and IEEE Std 1588-2019 7.5.2.2"
    SYNTAX OCTET STRING (SIZE (8))

Ieee8021AsV3GPtpProfileIdentifier ::= TEXTUAL-CONVENTION
    DISPLAY-HINT
        "1x:"
    STATUS current
    DESCRIPTION
        "The Ieee8021AsV3GPtpProfileIdentifier attribute is the
         profileIdentifier for this PTP profile."
    REFERENCE    "14.11.2, F.2 "
    SYNTAX OCTET STRING (SIZE (6))

Ieee8021AsV3ClockClassValue ::= TEXTUAL-CONVENTION
    STATUS       current
    DESCRIPTION
        "The Ieee8021AsV3ClockClassValue attribute denotes the traceability
        of the synchronized time distributed by a ClockTimeTransmitter when it is
         the Grandmaster PTP Instance.
         A more detailed description of clockClass can be found in
         IEEE Std 1588-2019."
    REFERENCE    "8.6.2.2 and IEEE Std 1588-2019 7.6.2.5"
    SYNTAX     INTEGER {
        primarySync(6),
        primarySyncLost(7),
        applicationSpecificSync(13),
        applicationSpecficSyncLost(14),
        primarySyncAlternativeA(52),
        applicationSpecificAlternativeA(58),
        primarySyncAlternativeB(187),
        applicationSpecficAlternativeB(193),
        defaultClock(248),
        timeReceiverOnlyClock(255)
        }

Ieee8021AsV3ClockAccuracyValue ::= TEXTUAL-CONVENTION
    STATUS       current
```


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```
    DESCRIPTION
        "The Ieee8021AsV3ClockAccuracyValue attribute indicates the
         expected time accuracy of a ClockTimeTransmitter.
         A more detailed description of clockAccuracy can be found in
         IEEE Std 1588-2019."
    REFERENCE    "8.6.2.3 and IEEE Std 1588-2019 7.6.2.6"
    SYNTAX     INTEGER {
        timeAccurateTo25ns(32),
        timeAccurateTo100ns(33),
        timeAccurateTo250ns(34),
        timeAccurateTo1us(35),
        timeAccurateTo2dot5us(36),
        timeAccurateTo10us(37),
        timeAccurateTo25us(38),
        timeAccurateTo100us(39),
        timeAccurateTo250us(40),
        timeAccurateTo1ms(41),
        timeAccurateTo2dot5ms(42),
        timeAccurateTo10ms(43),
        timeAccurateTo25ms(44),
        timeAccurateTo100ms(45),
        timeAccurateTo250ms(46),
        timeAccurateTo1s(47),
        timeAccurateTo10s(48),
        timeAccurateToGT10s(49),
        timeAccurateToUnknown(254)
        }

Ieee8021AsV3TimeSourceValue ::= TEXTUAL-CONVENTION
    STATUS       current
    DESCRIPTION
        "The Ieee8021AsV3TimeSourceValue is an information only
         attribute indicating the type of source of time used by a
         ClockTimeTransmitter. The value is not used in the selection of the
         Grandmaster PTP Instance. The values of TimeSource are
         given below and are specified in Table 8-2. These represent
         categories. For example, the GPS entry includes not only the
         GPS system of the U.S. Department of Defense but the European
         Galileo system and other present and future GNSSs.

         In the absence of a default value set by a user of this standard,
         the default value of timeSource shall be INTERNAL_OSCILLATOR.

         A more detailed description of timeSource can be found in
         IEEE Std 1588-2019.

         The following interpretation is placed on the value:
            0x10:  Atomic Clock,
            0x20:  GPS,
            0x30:  Terrestrial Radio,
            0x40:  PTP,
            0x50:  NTP,
            0x60:  Hand Set,
            0x90:  Other,
            0xA0:  Internal Oscillator "
    REFERENCE  "8.6.2.7, Table 8-2 and IEEE Std 1588-2019 7.6.2.8"
    SYNTAX      INTEGER {
            atomicClock(16),
            gps(32),
```


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```
            terrestrialRadio(48),
            ptp(64),
            ntp(80),
            handSet(96),
            other(144),
            internalOscillator(160)
           }

Ieee8021ASV3PtpTimeInterval ::= TEXTUAL-CONVENTION
    STATUS current
    DESCRIPTION
        "The Ieee8021ASV3PtpTimeInterval type represents time intervals
         in units of 2^-16 ns. Positive or negative time
         intervals outside the maximum range of this data type shall
         be encoded as the largest positive and negative values of
         the data type respectively.
         For example: 2.5 ns is expressed as:
         (hex) 0x0000 0000 0002 8000"
    REFERENCE "6.4.3.3"
SYNTAX OCTET STRING (SIZE (8))

Ieee8021ASV3PtpPortIdentity ::= TEXTUAL-CONVENTION
STATUS current
DESCRIPTION
        "The Ieee8021ASV3PtpPortIdentity type identifies a port of a
         PTP Instance.
         The first 8 octets within this value specifies the
         ClockIdentity.
         The last 2 octets within this value specifies the port number."
    REFERENCE   "6.4.3.7"
SYNTAX OCTET STRING (SIZE (10))

Ieee8021ASV3ScaledNs ::= TEXTUAL-CONVENTION
STATUS current
DESCRIPTION
        "The Ieee8021ASV3ScaledNs type represents signed values of
         time and time interval in units of 2^-16 ns.
         Positive or negative values of time or time interval outside the
         maximum range of this data type are encoded as the largest
         positive or negative value of the data type, respectively.
         For example: -2.5 ns is expressed as:
         (hex) 0xFFFF FFFF FFFF FFFD 8000"
    REFERENCE   "6.4.3.1"
SYNTAX OCTET STRING (SIZE (12))

Ieee8021ASV3UScaledNs ::= TEXTUAL-CONVENTION
STATUS current
DESCRIPTION
        "The Ieee8021ASV3UScaledNs type represents unsigned values of
         time and time interval in units of 2^-16 ns.
         Positive or negative values of time or time interval outside
         the maximum range of this data type are encoded as the largest
         positive or negative value of the data type, respectively.
         For example: 2.5 ns is expressed as:
         (hex) 0x0000 0000 0000 0000 0002 8000"
    REFERENCE   "6.4.3.2"
SYNTAX OCTET STRING (SIZE (12))

Ieee8021ASV3PTPInstanceIdentifier ::= TEXTUAL-CONVENTION
```


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```
DISPLAY-HINT "d"
STATUS current
DESCRIPTION
        "The entity of a single time-aware system that executes gPTP in
         one gPTP domain is called a PTP Instance. A time-aware system
         can contain multiple PTP Instances, which are each associated
         with a different gPTP domain. There are two types of
         PTP Instances, a PTP End Instance and a PTP Relay Instance."
    REFERENCE   "7.2.1"
SYNTAX Unsigned32

Ieee8021ASV3Timestamp ::= TEXTUAL-CONVENTION
STATUS current
DESCRIPTION
        "The value of Ieee8021ASV3Timestamp is equal to the remainder
         obtained upon dividing the respective timestamp, expressed
         in units of 2^-16 ns, by 2^48)."
    REFERENCE   "14.10.54, 14.18.26 and Table 14-12"
SYNTAX OCTET STRING (SIZE (6))

-- ================================================================
-- subtrees in the IEEE8021-AS-MIB
--
-- System Time-Aware Parameters/Capabilities for each instance of
-- gPTP domain. ieee8021AsV3InstanceListIndex that is of
-- ieee8021AsV3DomainIdentificationNumber object-type is used as Index.
--
-- ================================================================

-- ================================================================
-- The PTP Instance set is used to allow for dynamic creation and
-- deletion of PTP Instances and logical ports implementations that
-- support dynamic create/delete of devices.
-- ================================================================

ieee8021AsV3PtpInstanceTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3PtpInstanceEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table is used to allow for dynamic creation and deletion
         of PTP Instances and logical ports implementations that support
         dynamic create/delete of devices."
    REFERENCE   "14.1"
    ::= { ieee8021AsV3MIBObjects 1 }

ieee8021AsV3PtpInstanceEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3PtpInstanceEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry that specifies a PTP Instance."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3PtpInstanceTable 1 }

Ieee8021AsV3PtpInstanceEntry ::=
  SEQUENCE {
    ieee8021AsV3PtpInstance           Ieee8021ASV3PTPInstanceIdentifier,
    ieee8021AsV3PtpInstanceName       SnmpAdminString,
```


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```
    ieee8021AsV3PtpInstanceRowStatus   RowStatus
  }

ieee8021AsV3PtpInstance OBJECT-TYPE
    SYNTAX Ieee8021ASV3PTPInstanceIdentifier
    MAX-ACCESS not-accessible
    STATUS current
    DESCRIPTION
        "The entity of a single time-aware system that executes gPTP in
         one gPTP domain is called a PTP Instance. A time-aware system can
         contain multiple PTP Instances, which are each associated with
         a different gPTP domain. There are two types of PTP Instances,
         a PTP End Instance and a PTP Relay Instance."
    REFERENCE "7.2.1"
    ::= { ieee8021AsV3PtpInstanceEntry 1 }

ieee8021AsV3PtpInstanceName OBJECT-TYPE
    SYNTAX SnmpAdminString
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
    "Name for identification of a PTP Instance."
    DEFVAL { "" }
    ::= { ieee8021AsV3PtpInstanceEntry 2 }

ieee8021AsV3PtpInstanceRowStatus OBJECT-TYPE
    SYNTAX RowStatus
    MAX-ACCESS read-create
    STATUS current
    DESCRIPTION
        "This attribute is used to create and delete PTP Instances."
    REFERENCE "14.1"
    ::= { ieee8021AsV3PtpInstanceEntry 3 }

-- ================================================================
-- The Default data set represents native time capability of a time-
-- aware system and is consistent with respective IEEE 1588 data set.
-- ================================================================

ieee8021AsV3DefaultDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3DefaultDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Default Parameter Data Set represents the native capabilities
         of a PTP Instance, i.e., a PTP Relay Instance or a
         PTP End Instance."
    REFERENCE   "14.2"
    ::= { ieee8021AsV3MIBObjects 2 }

ieee8021AsV3DefaultDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3DefaultDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Default Data Set contains the profile Identifier for
         this instance of gPTP domain."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3DefaultDSTable 1 }
```


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```
Ieee8021AsV3DefaultDSEntry ::=
  SEQUENCE {
    ieee8021AsV3DefaultDSClockIdentity          Ieee8021AsV3ClockIdentity,
    ieee8021AsV3DefaultDSNumberPorts            Unsigned32,
    ieee8021AsV3DefaultDSClockQualityClockClass Ieee8021AsV3ClockClassValue,
    ieee8021AsV3DefaultDSClockQualityClockAccuracy
Ieee8021AsV3ClockAccuracyValue,
    ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance Unsigned32,
    ieee8021AsV3DefaultDSPriority1              Unsigned32,
    ieee8021AsV3DefaultDSPriority2              Unsigned32,
    ieee8021AsV3DefaultDSGmCapable              TruthValue,
    ieee8021AsV3DefaultDSCurrentUtcOffset       Integer32,
    ieee8021AsV3DefaultDSCurrentUtcOffsetValid  TruthValue,
    ieee8021AsV3DefaultDSLeap59                 TruthValue,
    ieee8021AsV3DefaultDSLeap61                 TruthValue,
    ieee8021AsV3DefaultDSTimeTraceable          TruthValue,
    ieee8021AsV3DefaultDSFrequencyTraceable     TruthValue,
    ieee8021AsV3DefaultDSPtpTimescale           TruthValue,
    ieee8021AsV3DefaultDSTimeSource             Ieee8021AsV3TimeSourceValue,
    ieee8021AsV3DefaultDSDomainNumber           Unsigned32,
    ieee8021AsV3DefaultDSSdoId                  Unsigned32,
    ieee8021AsV3DefaultDSExternalPortConfigurationEnabled TruthValue,
    ieee8021AsV3DefaultDSInstanceEnable         TruthValue
    }

ieee8021AsV3DefaultDSClockIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockIdentity of the PTP Instance.
         The clockIdentity attribute shall be as specified in
         IEEE Std 1588-2019."
    REFERENCE   "14.2.2 and IEEE Std 1588-2019 7.5.2.2"
    ::= { ieee8021AsV3DefaultDSEntry 1 }


ieee8021AsV3DefaultDSNumberPorts OBJECT-TYPE
    SYNTAX      Unsigned32(1..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The number of ports of the PTP Instance. For an end
         station the value is 1."
    REFERENCE   "14.2.3"
    ::= { ieee8021AsV3DefaultDSEntry 2 }


ieee8021AsV3DefaultDSClockQualityClockClass OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockClassValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockClass of the PTP Instance, which
         implements the clockClass specifications of 8.6.2.2."
    REFERENCE   "14.2.4.2"
    ::= { ieee8021AsV3DefaultDSEntry 3 }

```


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```
ieee8021AsV3DefaultDSClockQualityClockAccuracy OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockAccuracyValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockAccuracy of the PTP Instance, which
         implements the clockAccuracy specifications of 8.6.2.3."
    REFERENCE   "14.2.4.3"
    ::= { ieee8021AsV3DefaultDSEntry 4 }


ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the offsetScaledLogVariance of the PTP Instance,
         which implements the offsetScaledLogVariance specifications
         of 8.6.2.4."
    REFERENCE   "14.2.4.4"
    ::= { ieee8021AsV3DefaultDSEntry 5 }

ieee8021AsV3DefaultDSPriority1 OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the priority1 attribute of the PTP Instance."
    REFERENCE   "14.2.5"
    ::= { ieee8021AsV3DefaultDSEntry 6 }


ieee8021AsV3DefaultDSPriority2 OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the priority2 attribute of the PTP Instance."
    REFERENCE   "14.2.6"
    DEFVAL { 248 }
    ::= { ieee8021AsV3DefaultDSEntry 7 }


ieee8021AsV3DefaultDSGmCapable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is TRUE (1) if the PTP Instance is capable of being a
         Grandmaster PTP Instance, and FALSE (2) if the PTP Instance is
         not capable of being a Grandmaster PTP Instance."
    REFERENCE   "14.2.7"
    ::= { ieee8021AsV3DefaultDSEntry 8 }


ieee8021AsV3DefaultDSCurrentUtcOffset OBJECT-TYPE
    SYNTAX      Integer32(-32768..32767)
    UNITS       "seconds"
```


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```
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the offset between TAI and UTC, relative to
         the ClockTimeTransmitter entity of this PTP Instance. It is equal
         to the global variable sysCurrentUtcOffset.
         The value is in units of seconds.

         The default value is selected as follows:
             a)The value is the value obtained from a primary
               reference if the value is known at the time of
               initialization, else
             b)The value is the current IERS defined value of
               TAI - UTC (see IERS Bulletin C) when the PTP Instance
               is designed.currentUtcOffsetValid"
    REFERENCE   "14.2.8"
    ::= { ieee8021AsV3DefaultDSEntry 9 }


ieee8021AsV3DefaultDSCurrentUtcOffsetValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The default value is TRUE (1) if the value of
         ieee8021AsV3DefaultDSCurrentUtcOffset is known to be
         correct, otherwise it is set to FALSE (2)."
    REFERENCE   "14.2.9"
    ::= { ieee8021AsV3DefaultDSEntry 10 }


ieee8021AsV3DefaultDSLeap59 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A TRUE (1) value indicates that the last minute of the
         current UTC day, relative to the ClockTimeTransmitter entity of
         this PTP Instance, will contain 59 s. It is equal to the
         global variable sysLeap59.

         The value is selected as follows:
            a)The value is obtained from a primary reference if
              known at the time of initialization, else
            b)The value is set to FALSE (2)."
    REFERENCE   "14.2.10"
    ::= { ieee8021AsV3DefaultDSEntry 11 }


ieee8021AsV3DefaultDSLeap61 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A TRUE (1) value indicates that the last minute of the
         current UTC day, relative to the ClockTimeTransmitter entity of
         this PTP Instance, will contain 61 s. It is equal to the global
         variable sysLeap61.

```


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```
         The value is selected as follows:
             a)The value is obtained from a primary reference if
               known at the time of initialization, else
             b)The value is set to FALSE (2)."
    REFERENCE   "14.2.11"
    ::= { ieee8021AsV3DefaultDSEntry 12 }


ieee8021AsV3DefaultDSTimeTraceable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is set to TRUE (1) if the timescale and the value
         of currentUtcOffset, relative to the ClockTimeTransmitter entity of
         this PTP Instance, are traceable to a primary reference
         standard; otherwise the value is set to FALSE (2).
         It is equal to the global variable sysTimeTraceable.

         The value is selected as follows:
            a)If the time and the value of currentUtcOffset are
              traceable to a primary reference standard at the time of
              initialization, the value is set to TRUE (1), else
            b)The value is set to FALSE (2)."
    REFERENCE   "14.2.12"
    ::= { ieee8021AsV3DefaultDSEntry 13 }


ieee8021AsV3DefaultDSFrequencyTraceable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is set to TRUE (1) if the frequency determining the
         timescale of the ClockTimeTransmitter Entity of this PTP Instance is
         traceable to a primary standard; otherwise the value is set
         to FALSE (2). It is equal to the global variable
         sysFrequencyTraceable.

         The value is selected as follows:
            a)If the frequency is traceable to a primary reference
              standard at the time of initialization the value is set
              to TRUE (1), else
            b)The value is set to FALSE (2)."
    REFERENCE   "14.2.13"
    ::= { ieee8021AsV3DefaultDSEntry 14 }

ieee8021AsV3DefaultDSPtpTimescale OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is set to TRUE (1) if the clock timescale of the
         ClockTimeTransmitter Entity of this PTP Instance is PTP and
         FALSE (2) otherwise."
    REFERENCE   "14.2.14"
    ::= { ieee8021AsV3DefaultDSEntry 15 }

ieee8021AsV3DefaultDSTimeSource OBJECT-TYPE
```


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```
    SYNTAX      Ieee8021AsV3TimeSourceValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the source of time used by the
         Grandmaster PTP Instance clock."
    REFERENCE   "14.2.15"
    ::= { ieee8021AsV3DefaultDSEntry 16 }

ieee8021AsV3DefaultDSDomainNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the domain number of the gPTP domain for this
         instance of gPTP supported by the time-aware system."
    REFERENCE   "14.2.16"
    ::= { ieee8021AsV3DefaultDSEntry 17 }

ieee8021AsV3DefaultDSSdoId OBJECT-TYPE
    SYNTAX      Unsigned32(0..4095)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the sdoId of the gPTP domain for this instance
         of gPTP supported by the time-aware system.
         For compatibility with IEEE Std 1588, the range of the
         managed object is limited to 12 bits; in addition, only the
         single value 0x100 is specified in this standard for the
         gPTP domain of a PTP Instance."
    REFERENCE   "14.2.17"
    ::= { ieee8021AsV3DefaultDSEntry 18 }

ieee8021AsV3DefaultDSExternalPortConfigurationEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the externalPortConfigurationEnabled attribute
         of the PTP Instance."
    REFERENCE   "14.2.18"
    ::= { ieee8021AsV3DefaultDSEntry 19 }

ieee8021AsV3DefaultDSInstanceEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the instanceEnable attribute of the PTP Instance."
    REFERENCE   "14.2.19"
    ::= { ieee8021AsV3DefaultDSEntry 20 }

-- ================================================================
-- The Current data set represents this system's topological location
-- relative to the known Grandmaster PTP Instance.
-- This data set is consistent with respective IEEE 1588 data set.
-- ================================================================

ieee8021AsV3CurrentDSTable      OBJECT-TYPE
```


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```
    SYNTAX      SEQUENCE OF Ieee8021AsV3CurrentDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Current Parameter Data Set represents the position of a local
        system and other information, relative to the
        Grandmaster PTP Instance."
    REFERENCE   "14.3"
    ::= { ieee8021AsV3MIBObjects 3 }

ieee8021AsV3CurrentDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CurrentDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Current Data Set for a specific PTP Instance."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3CurrentDSTable 1 }

Ieee8021AsV3CurrentDSEntry ::=
  SEQUENCE {
    ieee8021AsV3CurrentDSStepsRemoved                   Unsigned32,
    ieee8021AsV3CurrentDSOffsetFromTimeTransmitter
Ieee8021ASV3PtpTimeInterval,
    ieee8021AsV3CurrentDSLastGmPhaseChange              Ieee8021ASV3ScaledNs,
    ieee8021AsV3CurrentDSLastGmFreqChange               Float64TC,
    ieee8021AsV3CurrentDSGmTimebaseIndicator            Unsigned32,
    ieee8021AsV3CurrentDSGmChangeCount                  Counter32,
    ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent        TimeStamp,
    ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent   TimeStamp,
    ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent    TimeStamp
    }

ieee8021AsV3CurrentDSStepsRemoved OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the number of gPTP communication paths
         traversed between this PTP Instance and the
         Grandmaster PTP Instance, as specified in 10.3.3."
    REFERENCE   "14.3.2"
    ::= { ieee8021AsV3CurrentDSEntry 1 }


ieee8021AsV3CurrentDSOffsetFromTimeTransmitter OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is an implementation-specific representation of
         the current value of the time difference between a timeReceiver
         and the Grandmaster Clock, as computed by the timeReceiver, and
         as specified in 10.2.10."
    REFERENCE   "14.3.3"
    ::= { ieee8021AsV3CurrentDSEntry 2 }

ieee8021AsV3CurrentDSLastGmPhaseChange OBJECT-TYPE
    SYNTAX      Ieee8021ASV3ScaledNs
```


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```
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the phase change that occurred on the most
         recent change in either Grandmaster PTP Instance or
         gmTimeBaseIndicator."
    REFERENCE   "14.3.4"
    ::= { ieee8021AsV3CurrentDSEntry 3 }

ieee8021AsV3CurrentDSLastGmFreqChange OBJECT-TYPE
    SYNTAX      Float64TC
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the frequency change that occurred on the most
         recent change in either Grandmaster PTP Instance or
         gmTimeBaseIndicator."
    REFERENCE   "14.3.5"
    ::= { ieee8021AsV3CurrentDSEntry 4 }

ieee8021AsV3CurrentDSGmTimebaseIndicator OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the value of timeBaseIndicator of the
         current Grandmaster PTP Instance."
    REFERENCE   "14.3.6"
    ::= { ieee8021AsV3CurrentDSEntry 5 }

ieee8021AsV3CurrentDSGmChangeCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This statistics counter tracks the number of times the
         Grandmaster PTP Instance has changed in a gPTP domain.
         This counter increments when the PortAnnounceInformation
         state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state
         or the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state."
    REFERENCE   "14.3.7"
    ::= { ieee8021AsV3CurrentDSEntry 6 }

ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent OBJECT-TYPE
    SYNTAX      TimeStamp
    UNITS       "0.01 seconds"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This timestamp takes the value of sysUpTime (see RFC3418) when
         the most recent Grandmaster PTP Instance change occurred in
         a gPTP domain.
         This timestamp is updated when the PortAnnounceInformation
         state machine enters the SUPERIOR_TIME_TRANSMITTER_PORT state or
         the INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state."
    REFERENCE   "14.3.8"
    ::= { ieee8021AsV3CurrentDSEntry 7 }
```


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```
ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent OBJECT-TYPE
    SYNTAX      TimeStamp
    UNITS       "0.01 seconds"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This timestamp takes the value of sysUpTime (see RFC3418)
         when the most recent change in Grandmaster Clock phase
         occurred, due to a change of either the
         Grandmaster PTP Instance or the Grandmaster Clock
         time base. This timestamp is updated when one of the
         following occurs:
            a)The PortAnnounceInformation state machine enters the
              SUPERIOR_TIME_TRANSMITTER_PORT state or the
              INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state, or
            b)The gmTimebaseIndicator managed object changes and the
              lastGmPhaseChange field of the most recently received
              Follow_Up information TLV is nonzero."
    REFERENCE   "14.3.9"
    ::= { ieee8021AsV3CurrentDSEntry 8 }

ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent OBJECT-TYPE
    SYNTAX      TimeStamp
    UNITS       "0.01 seconds"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This timestamp takes the value of sysUpTime (see RFC3418)
         when the most recent change in Grandmaster Clock frequency
         occurred, due to a change of either the Grandmaster PTP
         Instance or the Grandmaster Clock time base. This timestamp
         is updated when one of the following occurs:
            a)The PortAnnounceInformation state machine enters the
              SUPERIOR_TIME_TRANSMITTER_PORT state or the
              INFERIOR_TIME_TRANSMITTER_OR_OTHER_PORT state, or
            b)The gmTimebaseIndicator managed object changes and the
              lastGmFreqChange field of the most recently received
              Follow_Up information TLV is nonzero."
    REFERENCE   "14.3.10"
    ::= { ieee8021AsV3CurrentDSEntry 9 }

-- ================================================================
-- The Parent data set represents the upstream (toward
-- Grandmaster PTP Instance) system's timing parameters as measured
-- at this system.
-- This data set is consistent with the respective IEEE 1588 data set.
-- ================================================================
ieee8021AsV3ParentDSTable     OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3ParentDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Parent Parameter Data Set represents capabilities of the
         upstream system, toward the Grandmaster PTP Instance, as
         measured at a local system."
    REFERENCE   "14.4"
    ::= { ieee8021AsV3MIBObjects 4 }

ieee8021AsV3ParentDSEntry   OBJECT-TYPE
```


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```
    SYNTAX      Ieee8021AsV3ParentDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Parent Data Set for a specific PTP Instance."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3ParentDSTable 1 }

Ieee8021AsV3ParentDSEntry ::=
  SEQUENCE {
    ieee8021AsV3ParentDSParentClockIdentity     Ieee8021AsV3ClockIdentity,
    ieee8021AsV3ParentDSParentPortNumber        Unsigned32,
    ieee8021AsV3ParentDSCumulativeRateRatio      Integer32,
    ieee8021AsV3ParentDSGrandmasterIdentity     Ieee8021AsV3ClockIdentity,
    ieee8021AsV3ParentDSGrandmasterClockQualityclockClass
                                                Ieee8021AsV3ClockClassValue,
    ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy
                                                Ieee8021AsV3ClockAccuracyValue,
    ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar
                                                Unsigned32,
    ieee8021AsV3ParentDSGrandmasterPriority1    Unsigned32,
    ieee8021AsV3ParentDSGrandmasterPriority2    Unsigned32
    }

ieee8021AsV3ParentDSParentClockIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the first of the parentPortIdentity attribute
         for this instance of gPTP domain, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.4.2"
    ::= { ieee8021AsV3ParentDSEntry 1 }

ieee8021AsV3ParentDSParentPortNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the second of the parentPortIdentity attribute
         for this instance of gPTP domain, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.4.2"
    ::= { ieee8021AsV3ParentDSEntry 2 }

ieee8021AsV3ParentDSCumulativeRateRatio OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is an estimate of the ratio of the frequency of
         the Grandmaster Clock to the frequency of the LocalClock
         entity of this PTP Instance.
         CumulativeRateRatio is expressed as the fractional
         frequency offset multiplied by 2^41, i.e., the quantity
         (rateRatio - 1.0)(2^41), where rateRatio is computed by
         the PortSyncSyncReceive state machine."
    REFERENCE   "14.4.3"
```


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```
   ::= { ieee8021AsV3ParentDSEntry 3 }

ieee8021AsV3ParentDSGrandmasterIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockIdentity attribute of the
         Grandmaster PTP Instance."
    REFERENCE   "14.4.4"
    ::= { ieee8021AsV3ParentDSEntry 4 }

ieee8021AsV3ParentDSGrandmasterClockQualityclockClass OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockClassValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockClass of the Grandmaster PTP Instance."
    REFERENCE   "14.4.5.2"
    ::= { ieee8021AsV3ParentDSEntry 5 }

ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockAccuracyValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockAccuracy of the Grandmaster PTP Instance."
    REFERENCE   "14.4.5.3"
    ::= { ieee8021AsV3ParentDSEntry 6 }

ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar
                OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the offsetScaledLogVariance of the
         Grandmaster PTP Instance."
    REFERENCE   "14.4.5.4"
    ::= { ieee8021AsV3ParentDSEntry 7 }

ieee8021AsV3ParentDSGrandmasterPriority1 OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the priority1 attribute of the
         Grandmaster PTP Instance."
    REFERENCE   "14.4.6"
    ::= { ieee8021AsV3ParentDSEntry 8 }

ieee8021AsV3ParentDSGrandmasterPriority2 OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the priority2 attribute of the
         Grandmaster PTP Instance."
    REFERENCE   "14.4.7"
```


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```
    ::= { ieee8021AsV3ParentDSEntry 9 }

-- ================================================================
-- TimePropertiesDS represents the Grandmaster PTP Instance's
-- parameters, as measured at this system and are derived from
-- IEEE 802.1AS protocol.
-- This data set is consistent with respective IEEE 1588 data set.
-- ================================================================
ieee8021AsV3TimePropertiesDSTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3TimePropertiesDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Time Properties Parameter Data Set represents capabilities of
         the Grandmaster PTP Instance, as measured at a local system"
    REFERENCE   "14.5"
        ::= { ieee8021AsV3MIBObjects 5 }

ieee8021AsV3TimePropertiesDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3TimePropertiesDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Time Properties Data Set contains the profile Identifier for
         this instance of gPTP domain."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3TimePropertiesDSTable 1 }

Ieee8021AsV3TimePropertiesDSEntry ::=
  SEQUENCE {
    ieee8021AsV3TimePropertiesDSCurrentUtcOffset      Integer32,
    ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid TruthValue,
    ieee8021AsV3TimePropertiesDSLeap59                TruthValue,
    ieee8021AsV3TimePropertiesDSLeap61                TruthValue,
    ieee8021AsV3TimePropertiesDSTimeTraceable         TruthValue,
    ieee8021AsV3TimePropertiesDSFrequencyTraceable    TruthValue,
    ieee8021AsV3TimePropertiesDSPtpTimescale          TruthValue,
    ieee8021AsV3TimePropertiesDSTimeSource           Ieee8021AsV3TimeSourceValue
    }

ieee8021AsV3TimePropertiesDSCurrentUtcOffset OBJECT-TYPE
    SYNTAX      Integer32(-32768..32767)
    UNITS       "seconds"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is currentUtcOffset for the current
         Grandmaster PTP Instance. It is equal to the value  of
         the global variable currentUtcOffset. The value is in
         units of seconds."
    REFERENCE   "14.5.2"
    ::= { ieee8021AsV3TimePropertiesDSEntry 1 }

ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is currentUtcOffsetValid for the current
```


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```
         Grandmaster PTP Instance. It is equal to the global
         variable currentUtcOffsetValid."
    REFERENCE   "14.5.3"
    ::= { ieee8021AsV3TimePropertiesDSEntry 2 }

ieee8021AsV3TimePropertiesDSLeap59 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is leap59 for the current Grandmaster PTP Instance.
         It is equal to the global variable leap59."
    REFERENCE   "14.5.4"
    ::= { ieee8021AsV3TimePropertiesDSEntry 3 }

ieee8021AsV3TimePropertiesDSLeap61 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is leap61 for the current Grandmaster PTP Instance.
         It is equal to the global variable leap61."
    REFERENCE   "14.5.5"
    ::= { ieee8021AsV3TimePropertiesDSEntry 4 }

ieee8021AsV3TimePropertiesDSTimeTraceable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is timeTraceable for the current
         Grandmaster PTP Instance. It is equal to the global
         variable timeTraceable."
    REFERENCE   "14.5.6"
    ::= { ieee8021AsV3TimePropertiesDSEntry 5 }

ieee8021AsV3TimePropertiesDSFrequencyTraceable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is frequencyTraceable for the current
         Grandmaster PTP Instance. It is equal to the global
         variable frequencyTraceable."
    REFERENCE   "14.5.7"
    ::= { ieee8021AsV3TimePropertiesDSEntry 6 }

ieee8021AsV3TimePropertiesDSPtpTimescale OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is ptpTimescale for the current
         Grandmaster PTP Instance."
    REFERENCE   "14.5.8"
    ::= { ieee8021AsV3TimePropertiesDSEntry 7 }

ieee8021AsV3TimePropertiesDSTimeSource OBJECT-TYPE
    SYNTAX      Ieee8021AsV3TimeSourceValue
```


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```
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is timeSource for the current
         Grandmaster PTP Instance. It is equal to the global
         variable timeSource"
    REFERENCE   "14.5.9"
    ::= { ieee8021AsV3TimePropertiesDSEntry 8 }

-- ============================================================
-- The Path Trace Parameter Data set represents the current path
-- trace information available at the PTP Instance.
-- ============================================================

ieee8021AsV3PathTraceDSTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3PathTraceDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The pathTraceDS represents the current path trace information
         available at the PTP Instance."
    REFERENCE   "14.6"
    ::= { ieee8021AsV3MIBObjects 6 }

ieee8021AsV3PathTraceDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3PathTraceDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Path Trace Data Set for a specific PTP Instance."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3PathTraceDSTable 1 }

Ieee8021AsV3PathTraceDSEntry ::=
SEQUENCE {
    ieee8021AsV3PathTraceDSEnable                    TruthValue
    }

ieee8021AsV3PathTraceDSEnable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is TRUE.
         NOTE: This member is included for compatibility with
         IEEE Std 1588. In IEEE Std 1588, the path trace mechanism
         is optional, and the pathTraceDS.enable member is
         configurable (its value in IEEE Std 1588 is TRUE (1) or
         FALSE (2), depending on whether the path trace mechanism is
         operational or not operational, respectively. However, the
         pathTrace mechanism is mandatory in this standard, and the
         value of enable is always TRUE (1)."
    REFERENCE   "14.6.3"
    ::= { ieee8021AsV3PathTraceDSEntry 2 }

ieee8021AsV3PathTraceDSArrayTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3PathTraceDSArrayEntry
    MAX-ACCESS  not-accessible
    STATUS      current
```


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```
    DESCRIPTION
        "This object contains an array of ClockIdentity values contained
         in the pathTrace array, which represents the current path trace
         information, and which is carried in the path trace TLV per
         PTP Instance."
    REFERENCE   "14.6.2"
    ::= { ieee8021AsV3MIBObjects 7 }

ieee8021AsV3PathTraceDSArrayEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3PathTraceDSArrayEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Path Trace Data Set Table Array for a specific PTP Instance."
    INDEX { ieee8021AsV3PtpInstance, ieee8021AsV3PathTraceDSArrayIndex }
    ::= { ieee8021AsV3PathTraceDSArrayTable 1 }

Ieee8021AsV3PathTraceDSArrayEntry ::=
  SEQUENCE {
    ieee8021AsV3PathTraceDSArrayIndex     Unsigned32,
    ieee8021AsV3PathTraceDSArrayList      Ieee8021AsV3ClockIdentity
    }

ieee8021AsV3PathTraceDSArrayIndex OBJECT-TYPE
    SYNTAX      Unsigned32 (1..179)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Index of the Path Trace Data Set Array."
    REFERENCE   "10.3.9.23"
    ::= { ieee8021AsV3PathTraceDSArrayEntry 1 }

ieee8021AsV3PathTraceDSArrayList OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the array of ClockIdentity values contained
         in the pathTrace array, which represents the current
         path trace information, and which is carried in the path
         trace TLV."
    REFERENCE   "14.6.2"
    ::= { ieee8021AsV3PathTraceDSArrayEntry 2 }


-- ************************************************************
-- The Acceptable TimeTransmitter Table Parameter Data Set represents the
-- acceptable timeTransmitter table used when an EPON port is used by a PTP
-- Instance of a time-aware system.
-- ************************************************************

ieee8021AsV3AcceptableTimeTransmitterTableDSTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3AcceptableTimeTransmitterTableDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The acceptableTimeTransmitterTableDS represents the acceptable
timeTransmitter
         table used when an EPON port is used by a PTP Instance of a
```


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```
        time-aware system."
    REFERENCE   "14.7"
::= { ieee8021AsV3MIBObjects 8 }

ieee8021AsV3AcceptableTimeTransmitterTableDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterTableDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Acceptable TimeTransmitter Table Data Set represents the acceptable
timeTransmitter
         table used when an EPON port is used by a PTP Instance of a
         time-aware system."
    INDEX { ieee8021AsV3PtpInstance }
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSTable 1 }

Ieee8021AsV3AcceptableTimeTransmitterTableDSEntry ::=
  SEQUENCE {
                ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize
Unsigned32,
                ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize
Unsigned32
    }

ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize  OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the maximum size of the AcceptableTimeTransmitterTable.
         It is equal to the maxTableSize member of the
         AcceptableTimeTransmitterTable structure."
    REFERENCE   "14.7.2"
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSEntry 1 }

ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize  OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the actual size of the AcceptableTimeTransmitterTable.
         It is equal to the actualTableSize member of the
         AcceptableTimeTransmitterTable structure, i.e., the current number
         of elements in the acceptable timeTransmitter array. The actual
         table size is less than or equal to the max table size."
    REFERENCE   "14.7.3"
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSEntry 2 }

ieee8021AsV3AcceptableTimeTransmitterTableDSArrayTable    OBJECT-TYPE
    SYNTAX      SEQUENCE OF
Ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The acceptableTimeTransmitterTableDS represents the acceptable
timeTransmitter table
         used when an EPON port is used by a PTP Instance of a time-aware
         system."
    REFERENCE   "14.7"
```


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```
::= { ieee8021AsV3MIBObjects 9 }

ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Each element of this array is an AcceptableTimeTransmitter structure per
         PTP Instance."
    INDEX { ieee8021AsV3PtpInstance,
ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex }
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayTable 1 }

Ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry ::=
  SEQUENCE {
    ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex
Unsigned32,
    ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity
Ieee8021ASV3PtpPortIdentity,
    ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1       Unsigned32
    }

ieee8021AsV3AcceptableTimeTransmitterTableDSArrayIndex  OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Index of the Acceptable TimeTransmitter Table Data Set Array."
    REFERENCE   "14.7.4"
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 1 }

ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity  OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpPortIdentity
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The acceptablePortIdentity member is the PortIdentity of
         an acceptable timeTransmitter port."
    REFERENCE   "14.7.4"
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 2 }

ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1  OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The alternatePriority1 member contains an alternate value
         for the priority1 attribute of the acceptable timeTransmitter port."
    REFERENCE   "14.7.4"
    ::= { ieee8021AsV3AcceptableTimeTransmitterTableDSArrayEntry 3 }

-- ============================================================
-- The Port Parameter Data Set (portDS) represents PTP Port
-- time-aware capabilities for a PTP Instance of a time-aware
-- system.
-- ============================================================
ieee8021AsV3PortDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3PortDSEntry
    MAX-ACCESS  not-accessible
```


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```
    STATUS      current
    DESCRIPTION
        "For the single PTP Port of a PTP End Instance and for each
         PTP Port of a PTP Relay Instance , the portDS is maintained
         as the basis for making protocol decisions and providing
         values for message fields.
         The number of such data sets is the same as the value of
         defaultDS.numberPorts."
    REFERENCE   "14.10"
    ::= { ieee8021AsV3MIBObjects 10 }

ieee8021AsV3PortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3PortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "A list of objects pertaining to a PTP Port of a PTP Instance."
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3PortDSIndex }
    ::= { ieee8021AsV3PortDSTable 1 }

Ieee8021AsV3PortDSEntry ::=
     SEQUENCE {
         ieee8021AsV3BridgeBasePort          IEEE8021BridgePortNumber,
         ieee8021AsV3PortDSIndex         InterfaceIndexOrZero,
         ieee8021AsV3PortDSClockIdentity              Ieee8021AsV3ClockIdentity,
         ieee8021AsV3PortDSPortNumber                 Unsigned32,
         ieee8021AsV3PortDSPortState                   INTEGER,
         ieee8021AsV3PortDSPtpPortEnabled             TruthValue,
         ieee8021AsV3PortDSDelayMechanism             INTEGER,
         ieee8021AsV3PortDSIsMeasuringDelay            TruthValue,
         ieee8021AsV3PortDSAsCapable                   TruthValue,
         ieee8021AsV3PortDSMeanLinkDelay           Ieee8021ASV3PtpTimeInterval,
         ieee8021AsV3PortDSMeanLinkDelayThresh     Ieee8021ASV3PtpTimeInterval,
         ieee8021AsV3PortDSDelayAsym               Ieee8021ASV3PtpTimeInterval,
         ieee8021AsV3PortDSNbrRateRatio            Integer32,
         ieee8021AsV3PortDSInitialLogAnnounceInterval   Integer32,
         ieee8021AsV3PortDSCurrentLogAnnounceInterval   Integer32,
         ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval    TruthValue,
         ieee8021AsV3PortDSMgtSettableLogAnnounceInterval       Integer32,
         ieee8021AsV3PortDSAnnounceReceiptTimeout       Unsigned32,
         ieee8021AsV3PortDSInitialLogSyncInterval       Integer32,
         ieee8021AsV3PortDSCurrentLogSyncInterval       Integer32,
         ieee8021AsV3PortDSUseMgtSettableLogSyncInterval        TruthValue,
         ieee8021AsV3PortDSMgtSettableLogSyncInterval           Integer32,
         ieee8021AsV3PortDSSyncReceiptTimeout           Unsigned32,
         ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval Ieee8021ASV3UScaledNs,
         ieee8021AsV3PortDSInitialLogPdelayReqInterval Integer32,
         ieee8021AsV3PortDSCurrentLogPdelayReqInterval Integer32,
         ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval    TruthValue,
         ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval       Integer32,
         ieee8021AsV3PortDSInitialLogGptpCapableMessageInterval     Integer32,
         ieee8021AsV3PortDSCurrentLogGptpCapableMessageInterval     Integer32,
         ieee8021AsV3PortDSUseMgtSettableLogGptpCapableMessageInterval
TruthValue,
         ieee8021AsV3PortDSMgtSettableLogGptpCapableMessageInterval
Integer32,
         ieee8021AsV3PortDSInitialComputeNbrRateRatio     TruthValue,
         ieee8021AsV3PortDSCurrentComputeNbrRateRatio     TruthValue,
```


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```
        ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio     TruthValue,
        ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio     TruthValue,
        ieee8021AsV3PortDSInitialComputeMeanLinkDelay     TruthValue,
        ieee8021AsV3PortDSCurrentComputeMeanLinkDelay     TruthValue,
        ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay     TruthValue,
        ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay     TruthValue,
        ieee8021AsV3PortDSAllowedLostRsp        Unsigned32,
        ieee8021AsV3PortDSAllowedFaults              Unsigned32,
        ieee8021AsV3PortDSGPtpCapableReceiptTimeout     Unsigned32,
        ieee8021AsV3PortDSVersionNumber              Unsigned32,
        ieee8021AsV3PortDSNup                      Float64TC,
        ieee8021AsV3PortDSNdown                      Float64TC,
        ieee8021AsV3PortDSOneStepTxOper              TruthValue,
        ieee8021AsV3PortDSOneStepReceive             TruthValue,
        ieee8021AsV3PortDSOneStepTransmit            TruthValue,
        ieee8021AsV3PortDSInitialOneStepTxOper       TruthValue,
        ieee8021AsV3PortDSCurrentOneStepTxOper       TruthValue,
        ieee8021AsV3PortDSUseMgtSettableOneStepTxOper         TruthValue,
        ieee8021AsV3PortDSMgtSettableOneStepTxOper            TruthValue,
        ieee8021AsV3PortDSSyncLocked                 TruthValue,
        ieee8021AsV3PortDSPdelayTruncTST1          Ieee8021ASV3Timestamp,
        ieee8021AsV3PortDSPdelayTruncTST2          Ieee8021ASV3Timestamp,
        ieee8021AsV3PortDSPdelayTruncTST3          Ieee8021ASV3Timestamp,
        ieee8021AsV3PortDSPdelayTruncTST4          Ieee8021ASV3Timestamp,
        ieee8021AsV3PortDSMinorVersionNumber         Unsigned32
      }

ieee8021AsV3BridgeBasePort OBJECT-TYPE
    SYNTAX      IEEE8021BridgePortNumber
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the Bridge port number of the port for
         which this entry contains Bridge management information.
         For end stations, this port number shall be (1)."
    ::= { ieee8021AsV3PortDSEntry 1 }

ieee8021AsV3PortDSIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the gPTP interface group within
         the system for which this entry contains information. It
         is the value of the instance of the IfIndex object,
         defined in the IF-MIB, for the gPTP interface group
         corresponding to this port, or the value 0 if the port
         has not been bound to an underlying frame source and
         sink.

         For a given media port of a Bridge or an end station,
         there can be one or more PTP Port, and depends whether
         a media port supports point to point link (e.g. IEEE
         802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
         802.3 EPON) links on the media port."
    ::= { ieee8021AsV3PortDSEntry 2 }

ieee8021AsV3PortDSClockIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
```


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```
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the first of the portIdentity attribute
         of the local port, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.10.2"
    ::= { ieee8021AsV3PortDSEntry 3 }


ieee8021AsV3PortDSPortNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the second of the portIdentity attribute
         of the local port, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.10.2"
    ::= { ieee8021AsV3PortDSEntry 4 }

ieee8021AsV3PortDSPortState OBJECT-TYPE
    SYNTAX      INTEGER {
            disabledPort(3),
            timeTransmitterPort(6),
            passivePort(7),
            timeReceiverPort(9)
           }
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the value of the PTP Port state of this
         PTP Port (see Table 10-2) and is taken from the enumeration
         in Table 14-10. It is equal to the value of the global
         variable selectedState."
    REFERENCE   "14.10.3"
    ::= { ieee8021AsV3PortDSEntry 5 }

ieee8021AsV3PortDSPtpPortEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean ptpPortEnabled.
         Setting this managed object causes the Boolean ptpPortEnabled
         to have the same value."
    REFERENCE   "14.10.4"
    ::= { ieee8021AsV3PortDSEntry 6 }

ieee8021AsV3PortDSDelayMechanism OBJECT-TYPE
    SYNTAX      INTEGER {
            p2p(2),
            commonp2p(3),
            special(4)
           }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value indicates the mechanism for measuring mean
```


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```
        propagation delay and neighbor rate ratio on the link
        attached to this PTP Port, and is taken from the enumeration
        in Table 14-11. If the domain number is not 0, portDS.delay
        mechanism must not be P2P."
    REFERENCE   "14.10.5"
    ::= { ieee8021AsV3PortDSEntry 7 }

ieee8021AsV3PortDSIsMeasuringDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean isMeasuringDelay."
    REFERENCE   "14.10.6"
    ::= { ieee8021AsV3PortDSEntry 8 }


ieee8021AsV3PortDSAsCapable OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean asCapable."
    REFERENCE   "14.10.7"
    ::= { ieee8021AsV3PortDSEntry 9 }

ieee8021AsV3PortDSMeanLinkDelay OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable meanLinkDelay. It is an estimate of the current
         one-way propagation time on the link attached to this
         PTP Port, measured as specified for the respective medium.
         The value is zero for PTP Port attached to IEEE 802.3
         EPON links and for the timeTransmitter port of an IEEE 802.11 link,
         because one-way propagation delay is not measured on the
         latter and not directly measured on the former.
         NOTE: The underlying per-PTP Port, global variable
         meanLinkDelay is of type UScaledNS, which is a 96-Bit value.
         meanLinkDelay values that are larger than the maximum value
         that can be represented by the TimeInterval data type, i.e.,
         0xFFFF FFFF FFFF FFFF (where the units are 2 sup -16 ns), used
         for this managed object are set to this largest value."
    REFERENCE   "14.10.8"
    ::= { ieee8021AsV3PortDSEntry 10 }


ieee8021AsV3PortDSMeanLinkDelayThresh OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable meanLinkDelayThresh. It is the propagation time
         threshold above which a PTP Port is considered not capable of
         participating in the IEEE 802.1AS protocol.
         Setting this managed object causes the per PTP Port global
```


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```
        variable meanLinkDelayThresh to have the same value.
        NOTE: The underlying per-PTP Port, global variable
        meanLinkDelayThresh is of type UScaledNS, which is a 96-Bit
        value. meanLinkDelayThresh values that are larger than the
        maximum value that can be represented by the TimeInterval
        data type, i.e., 0xFFFF FFFF FFFF FFFF (where the units are
        2 sup -16 ns), used for this managed object are set to this
        largest value."
    REFERENCE   "14.10.9"
    ::= { ieee8021AsV3PortDSEntry 11 }

ieee8021AsV3PortDSDelayAsym OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the asymmetry in the propagation delay on
        the link attached to this PTP Port relative to the
        Grandmaster Clock time base, as defined in 10.2.5.9 and
        8.3. If propagation delay asymmetry is not modeled, then
        delayAsymmetry is 0.
        NOTE: The underlying per-port global variable delayAsymmetry
        is of type ScaledNS, which is a 96-Bit value.
        delayAsymmetry values that are larger than the maximum value
        that can be represented by the TimeInterval data type, i.e.,
        0x7FFF FFFF FFFF FFFF, (where the units are 2 sup -16 ns),
        used for this managed object are set to this largest value.
        delayAsymmetry values that are less than the minimum value
        that can be represented by the TimeInterval data type, i.e.,
        0x8000 0000 0000 0001 written in twos complement form (where
        the units are 2 sup -16 ns), used for this managed object are
        set to this smallest value."
    REFERENCE   "14.10.11 and 8.3"
    ::= { ieee8021AsV3PortDSEntry 12 }

ieee8021AsV3PortDSNbrRateRatio OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is an estimate of the ratio of the frequency of
        the LocalClock entity of the PTP Instance at the other end
        of the link attached to this PTP Port, to the frequency of
        the LocalClock entity of this PTP Instance. neighborRateRatio
        is expressed as the fractional frequency offset multiplied
        by 2^41, i.e., the quantity (neighborRateRatio -1.0)(2^41)."
    REFERENCE   "14.10.12"
    ::= { ieee8021AsV3PortDSEntry 13 }


ieee8021AsV3PortDSInitialLogAnnounceInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableLogAnnounceInterval is FALSE (2), the
        value is the logarithm to base 2 of the announce interval
        used when (a) the PTP Port is initialized, or (b) a message
        interval request TLV is received with the logAnnounceInterval
```


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```
         field set to 126."
    REFERENCE   "14.10.13"
    DEFVAL { 0 }
    ::= { ieee8021AsV3PortDSEntry 14 }

ieee8021AsV3PortDSCurrentLogAnnounceInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to the base 2 of the
         current announce interval."
    REFERENCE   "14.10.14"
    ::= { ieee8021AsV3PortDSEntry 15 }

ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the
         source of the announce interval. If the value is TRUE (1),
         the value of currentLogAnnounceInterval is set equal to the
         value of mgtSettableLogAnnounceInterval. If the value is
         FALSE (2), the value of currentLogAnnounceInterval is
         determined by the AnnounceIntervalSetting state machine. The
         default value of useMgtSettableLogAnnounceInterval is
         FALSE (2) for domain 0 and TRUE (1) for domains other than
         domain 0."
    REFERENCE   "14.10.15"
    ::= { ieee8021AsV3PortDSEntry 16 }

ieee8021AsV3PortDSMgtSettableLogAnnounceInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the announce interval
         used if useMgtSettableLogAnnounceInterval is TRUE (1).
         The value is not used if useMgtSettableLogAnnounceInterval is
         FALSE (2)."
    REFERENCE   "14.10.16"
    ::= { ieee8021AsV3PortDSEntry 17 }

ieee8021AsV3PortDSAnnounceReceiptTimeout OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the number of Announce message transmission
         intervals that a timeReceiver port waits without receiving an
         Announce message, before assuming that the timeTransmitter is no
         longer transmitting Announce messages and the BTCA needs
         to be run, if appropriate."
    REFERENCE   "14.10.17"
    DEFVAL { 3 }
    ::= { ieee8021AsV3PortDSEntry 18 }
```


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```
ieee8021AsV3PortDSInitialLogSyncInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableLogSyncInterval is FALSE (2), the
         value is the logarithm to base 2 of the sync interval used
         when (a) the PTP Port is initialized, or (b) a message
         interval request TLV is received with the logTimeSyncInterval
         field set to 126."
    REFERENCE   "14.10.18"
    ::= { ieee8021AsV3PortDSEntry 19 }

ieee8021AsV3PortDSCurrentLogSyncInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to the base 2 of the current
         time-synchronization transmission interval."
    REFERENCE   "14.10.19"
    ::= { ieee8021AsV3PortDSEntry 20 }

ieee8021AsV3PortDSUseMgtSettableLogSyncInterval OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the sync interval. If the value is TRUE (1), the value
         of currentLogSyncInterval is set equal to the value of
         mgtSettableLogSyncInterval. If the value of the managed
         object is FALSE (2), the value of currentLogSyncInterval is
         determined by the SyncIntervalSetting state machine. The
         default value of useMgtSettableLogSyncInterval is FALSE (2)
         for domain 0 and TRUE (1) for domains other than domain 0."
    REFERENCE   "14.10.20"
    ::= { ieee8021AsV3PortDSEntry 21 }

ieee8021AsV3PortDSMgtSettableLogSyncInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the sync interval
         if useMgtSettableLogSyncInterval is TRUE (1). The value is
         not used if useMgtSettableLogSyncInterval is FALSE (2)."
    REFERENCE   "14.10.21"
    ::= { ieee8021AsV3PortDSEntry 22 }

ieee8021AsV3PortDSSyncReceiptTimeout OBJECT-TYPE
    SYNTAX      Unsigned32(0..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the number of time-synchronization transmission
         intervals that a timeReceiver port waits without receiving
         synchronization information, before assuming that the timeTransmitter
         is no longer transmitting synchronization information and
```


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```
         that the BTCA needs to be run, if appropriate."
    REFERENCE   "14.10.22"
    DEFVAL { 3 }
    ::= { ieee8021AsV3PortDSEntry 23 }


ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval OBJECT-TYPE
    SYNTAX      Ieee8021ASV3UScaledNs
    UNITS       "2**-16 ns"
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable syncReceiptTimeoutTimeInterval. It is the time
         interval after which sync receipt timeout occurs if
         time-synchronization information has not been received during
         the interval."
    REFERENCE   "14.10.23"
    ::= { ieee8021AsV3PortDSEntry 24 }

ieee8021AsV3PortDSInitialLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the logarithm to base 2 of the Pdelay_Req
         message transmission interval used when (a) the PTP Port is
         initialized, or (b) a message interval request TLV is
         received with the logLinkDelayInterval field set to 126.
         For all other media, the value is 127."
    REFERENCE   "14.10.24"
    ::= { ieee8021AsV3PortDSEntry 25 }

ieee8021AsV3PortDSCurrentLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the logarithm to the base 2 of the current
         Pdelay_Req message transmission interval.
         For all other media, the value is 127."
    REFERENCE   "14.10.25"
    ::= { ieee8021AsV3PortDSEntry 26 }

ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the mean time interval between successive Pdelay_Req
         messages. If the value is TRUE (1), the value of
         currentLogPdelayReqInterval is set equal to the value of
         mgtSettableLogPdelayReqInterval. If the value of the managed
         object is FALSE (2), the value of currentLogPdelayReqInterval
```


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```
         is determined by the LinkDelayIntervalSetting state machine.
         The default value of useMgtSettableLogPdelayReqInterval is
         FALSE (2)."
    REFERENCE   "14.10.26"
    DEFVAL { false }
    ::= { ieee8021AsV3PortDSEntry 27 }

ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32 (-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the mean time
         interval between successive Pdelay_Req messages if
         useMgtSettableLogPdelayReqInterval is TRUE (1). The
         value is not used if useMgtSettableLogPdelayReqInterval
         is FALSE (2)."
    REFERENCE   "14.10.27"
    ::= { ieee8021AsV3PortDSEntry 28 }

ieee8021AsV3PortDSInitialLogGptpCapableMessageInterval OBJECT-TYPE
    SYNTAX      Integer32 (-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the gPTP capable
         message interval used when (a) the PTP Port is initialized,
         or (b) a gPtpCapableMessage interval request TLV is received
         with the logGptpCapableMessageInterval field set to 126."
    REFERENCE   "14.10.28"
    ::= { ieee8021AsV3PortDSEntry 29 }

ieee8021AsV3PortDSCurrentLogGptpCapableMessageInterval OBJECT-TYPE
    SYNTAX      Integer32 (-128..127)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to the base 2 of the current
         gPTP capable message interval."
    REFERENCE   "14.10.29"
    ::= { ieee8021AsV3PortDSEntry 30 }

ieee8021AsV3PortDSUseMgtSettableLogGptpCapableMessageInterval OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the gPTP capable message interval. If the value is
         TRUE (1), the value of currentLogGptpCapableMessageInterval
         is set equal to the value of
         mgtSettableLogGptpCapableMessageInterval. If the value of
         the managed object is FALSE (2), the value of
         currentLogGptpCapableMessageInterval is determined by the
         GptpCapableMessageIntervalSetting state machine.
         The default value of
         useMgtSettableLogGptpCapableMessageInterval is FALSE (2)."
    REFERENCE   "14.10.30"
    DEFVAL { false }
```


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```
    ::= { ieee8021AsV3PortDSEntry 31 }

ieee8021AsV3PortDSMgtSettableLogGptpCapableMessageInterval OBJECT-TYPE
    SYNTAX      Integer32 (-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the
         gPtpCapableMessageInterval if
         useMgtSettableLogGptpCapableMessageInterval is TRUE (1).
         The value is not used if
         useMgtSettableLogGptpCapableMessageInterval is FALSE (2)."
    REFERENCE   "14.10.31"
    ::= { ieee8021AsV3PortDSEntry 32 }

ieee8021AsV3PortDSInitialComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableComputeNeighborRateRatio is FALSE (2),
         then for full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the initial value of computeNeighborRateRatio."
    REFERENCE   "14.10.32"
    ::= { ieee8021AsV3PortDSEntry 33 }

ieee8021AsV3PortDSCurrentComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the current value of computeNeighborRateRatio."
    REFERENCE   "14.10.33"
    ::= { ieee8021AsV3PortDSEntry 34 }

ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the value of computeNeighborRateRatio. If the value is
         TRUE (1), the value of computeNeighborRateRatio is set equal
         to the value of mgtSettableComputeNeighborRateRatio. If the
         value of the managed object is FALSE (2), the value of
         currentComputeNeighborRateRatio is determined by the
         LinkDelayIntervalSetting state machine.
         The default value of useMgtSettableComputeNbrRateRatio is
         FALSE (2)."
    REFERENCE   "14.10.34"
    DEFVAL { false }
    ::= { ieee8021AsV3PortDSEntry 35 }

ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
```


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```
    STATUS      current
    DESCRIPTION
        "ComputeNeighborRateRatio is configured to this value if
         useMgtSettableComputeNeighborRateRatio is TRUE (1). The
         value is not used if useMgtSettableComputeNeighborRateRatio
         is FALSE (2)."
    REFERENCE   "14.10.35"
    ::= { ieee8021AsV3PortDSEntry 36 }

ieee8021AsV3PortDSInitialComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableComputeMeanLinkDelay is FALSE (2) then,
         for full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay,
         the value is the initial value of computeMeanLinkDelay."
    REFERENCE   "14.10.36"
    ::= { ieee8021AsV3PortDSEntry 37 }

ieee8021AsV3PortDSCurrentComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the current value of computeMeanLinkDelay."
    REFERENCE   "14.10.37"
    ::= { ieee8021AsV3PortDSEntry 38 }

ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the value of computeMeanLinkDelay. If the value is
         TRUE (1), the value of computeMeanLinkDelay is set equal to
         the value of mgtSettableComputeMeanLinkDelay. If the value
         of the managed object is FALSE (2), the value of
         currentComputeMeanLinkDelay is determined by the
         LinkDelayIntervalSetting state machine.
         The default value of useMgtSettableComputeMeanLinkDelay
         is FALSE (2)."
    REFERENCE   "14.10.38"
    DEFVAL { false }
    ::= { ieee8021AsV3PortDSEntry 39 }

ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "ComputeMeanLinkDelay is configured to this value if
         useMgtSettableComputeMeanLinkDelay is TRUE (1). The
         value is not used if useMgtSettableComputeMeanLinkDelay
         is FALSE (2)."
```


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```
    REFERENCE   "14.10.39"
    ::= { ieee8021AsV3PortDSEntry 40 }

ieee8021AsV3PortDSAllowedLostRsp OBJECT-TYPE
    SYNTAX      Unsigned32(1..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable allowedLostResponses. It is the number of Pdelay_Req
         messages without valid responses above which a PTP Port
         is considered to be not exchanging peer delay messages with
         its neighbor.
         Setting this managed object causes the  per-PTP Port global
         variable allowedLostResponses to have the same value."
    REFERENCE   "14.10.40 and 11.5.3"
    DEFVAL { 9 }
    ::= { ieee8021AsV3PortDSEntry 41 }

ieee8021AsV3PortDSAllowedFaults OBJECT-TYPE
    SYNTAX      Unsigned32(1..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP-Port global
         variable allowedFaults. It is the number of faults above
         which asCapable is set to FALSE (1), i.e., a PTP Port is
         considered not capable of interoperating with its
         neighbor via the IEEE 802.1AS protocol.
         Setting this managed object causes the per-PTP Port global
         variable allowedFaults to have the same value."
    REFERENCE   "14.10.41"
    DEFVAL { 9 }
    ::= { ieee8021AsV3PortDSEntry 42 }

ieee8021AsV3PortDSGPtpCapableReceiptTimeout OBJECT-TYPE
    SYNTAX      Unsigned32(1..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the number of transmission intervals that a
         PTP Port waits without receiving the gPTP-capable TLV, before
         assuming that the neighbor PTP Port is no longer invoking
         the gPTP protocol."
    REFERENCE   "14.10.42"
    DEFVAL { 9 }
    ::= { ieee8021AsV3PortDSEntry 43 }

ieee8021AsV3PortDSVersionNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..16)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value is set to versionPTP as specified in 10.6.2.2.4."
    REFERENCE   "14.10.43"
    ::= { ieee8021AsV3PortDSEntry 44 }

ieee8021AsV3PortDSNup OBJECT-TYPE
    SYNTAX      Float64TC
```


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```
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "For an OLT port of an IEEE 802.3 EPON link, the value is
         the effective index of refraction for the EPON upstream
         wavelength light of the optical path. The default value is
         1.46770 for 1 Gb/s upstream links, and 1.46773 for
         10 Gb/s upstream links.
         For all other PTP Ports, the value is 0."
    REFERENCE   "14.10.44"
    ::= { ieee8021AsV3PortDSEntry 45 }

ieee8021AsV3PortDSNdown OBJECT-TYPE
    SYNTAX      Float64TC
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "For an OLT port of an IEEE 802.3 EPON link, the value is
         the effective index of refraction for the EPON downstream
         wavelength light of the optical path. The default value is
         1.46805 for 1 Gb/s downstream links, and 1.46851 for
         10 Gb/s downstream links.
         For all other PTP Ports, the value is 0."
    REFERENCE   "14.10.45"
    ::= { ieee8021AsV3PortDSEntry 46 }

ieee8021AsV3PortDSOneStepTxOper OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable oneStepTxOper. Its value is TRUE (1) if the
         PTP Port is sending one-step Sync messages, and FALSE (2)
         if the PTP Port is sending two-step Sync and Follow_Up
         messages."
    REFERENCE   "14.10.46"
    ::= { ieee8021AsV3PortDSEntry 47 }

ieee8021AsV3PortDSOneStepReceive OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable oneStepReceive. Its value is TRUE (1) if the
         PTP Port is capable of receiving and processing one-step
         Sync messages."
    REFERENCE   "14.10.47"
    ::= { ieee8021AsV3PortDSEntry 48 }

ieee8021AsV3PortDSOneStepTransmit OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable oneStepTransmit. Its value is TRUE (1) if the
         PTP Port is capable of transmitting one-step Sync messages."
```


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```
    REFERENCE   "14.10.48"
    ::= { ieee8021AsV3PortDSEntry 49 }

ieee8021AsV3PortDSInitialOneStepTxOper OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableOneStepTxOper is FALSE (2), the value is
         used to initialize currentOneStepTxOper when the PTP Port is
         initialized. If useMgtSettableOneStepTxOper is TRUE (1),
         the value of initialOneStepTxOper is not used."
    REFERENCE   "14.10.49"
    DEFVAL { false }
    ::= { ieee8021AsV3PortDSEntry 50 }

ieee8021AsV3PortDSCurrentOneStepTxOper OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is TRUE (1) if it is desired, either via
         management or via a received Signaling message, that the
         PTP Port transmit one-step Sync messages. The value is
         FALSE (2) if it is not desired, either via management or via
         a received Signaling message, that the PTP Port transmit
         one-step  Sync messages.
         NOTE: The PTP Port will send one-step Sync messages only if
         currentOneStepTxOper and oneStepTransmit are both TRUE (1)."
    REFERENCE   "14.10.50"
    ::= { ieee8021AsV3PortDSEntry 51 }

ieee8021AsV3PortDSUseMgtSettableOneStepTxOper OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of currentOneStepTxOper. If the value is TRUE (1), the
         value of currentOneStepTxOper is set equal to the value of
         mgtSettableOneStepTxOper. If the value is FALSE (2), the
         value of currentOneStepTxOper is determined by the
         OneStepTxOperSetting state machine.
         The default value of useMgtSettableOneStepTxOper is TRUE (1)."
    REFERENCE   "14.10.51"
    DEFVAL     { true }
    ::= { ieee8021AsV3PortDSEntry 52 }

ieee8021AsV3PortDSMgtSettableOneStepTxOper OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableOneStepTxOper is TRUE (1),
         currentOneStepTxOper is set equal to the value of
         mgtSettableOneStepTxOper. The value of mgtSettableOneStepTxOper
         is not used if useMgtSettableOneStepTxOper is FALSE (2).
         The default value of mgtSettableOneStepTxOper is FALSE (2)
         for domains other than domain 0."
```


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```
    REFERENCE   "14.10.52"
    ::= { ieee8021AsV3PortDSEntry 53 }

ieee8021AsV3PortDSSyncLocked OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-PTP Port global
         variable syncLocked. Its value is TRUE (1) if the PTP Port
         will transmit a Sync as soon as possible after the timeReceiver
         PTP Port receives a Sync."
    REFERENCE   "14.10.53"
    ::= { ieee8021AsV3PortDSEntry 54 }

ieee8021AsV3PortDSPdelayTruncTST1 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         first value, T1, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t1 in
         Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp , expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange."
    REFERENCE   "14.10.54"
    ::= { ieee8021AsV3PortDSEntry 55 }

ieee8021AsV3PortDSPdelayTruncTST2 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         second value, T2, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t2 in
         Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp , expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange."
    REFERENCE   "14.10.54"
    ::= { ieee8021AsV3PortDSEntry 56 }

ieee8021AsV3PortDSPdelayTruncTST3 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
```


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```
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         third value, T3, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t3 in
         Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp , expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange."
    REFERENCE   "14.10.54"
    ::= { ieee8021AsV3PortDSEntry 57 }

ieee8021AsV3PortDSPdelayTruncTST4 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         fourth value, T4, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t4 in
         Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp , expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange."
    REFERENCE   "14.10.54"
    ::= { ieee8021AsV3PortDSEntry 58 }

ieee8021AsV3PortDSMinorVersionNumber OBJECT-TYPE
    SYNTAX      Unsigned32 (0..15)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value is set to minorVersionPTP as specified in 10.6.2.2.3."
    REFERENCE   "14.10.55"
    ::= { ieee8021AsV3PortDSEntry 59 }

-- ============================================================
-- The Description Port Parameter Data Set contains the
-- profileIdentifier for this PTP profile, as specified in F.2
-- ============================================================

ieee8021AsV3DescriptionPortDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3DescriptionPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The descriptionPortDS contains the profileIdentifier for
         this PTP profile, as specified in F.2."
    REFERENCE   "14.11"
    ::= { ieee8021AsV3MIBObjects 11 }
```


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```
ieee8021AsV3DescriptionPortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3DescriptionPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The descriptionPortDS contains the profileIdentifier for
         this PTP profile"
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3DescriptionPortDSAsIndex }
    ::= { ieee8021AsV3DescriptionPortDSTable 1 }

Ieee8021AsV3DescriptionPortDSEntry ::=
     SEQUENCE {
         ieee8021AsV3DescriptionPortDSAsIndex
                                           InterfaceIndexOrZero,
         ieee8021AsV3DescriptionPortDSProfileIdentifier
                                           Ieee8021AsV3GPtpProfileIdentifier }

ieee8021AsV3DescriptionPortDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the gPTP interface group within
         the system for which this entry contains information. It
         is the value of the instance of the IfIndex object,
         defined in the IF-MIB, for the gPTP interface group
         corresponding to this port, or the value 0 if the port
         has not been bound to an underlying frame source and
         sink.

         For a given media port of a Bridge or an end station,
         there can be one or more PTP Port, and depends whether
         a media port supports point to point link (e.g. IEEE
         802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
         802.3 EPON) links on the media port."
    REFERENCE   "IEEE Std 802.1AS Description Port Parameter DS Group
                 PTP Port Index"
    ::= { ieee8021AsV3DescriptionPortDSEntry 1 }

ieee8021AsV3DescriptionPortDSProfileIdentifier OBJECT-TYPE
    SYNTAX      Ieee8021AsV3GPtpProfileIdentifier
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the profileIdentifier for this PTP profile."
    REFERENCE   "14.11.2 and F.2"
    ::= { ieee8021AsV3DescriptionPortDSEntry 2 }

-- ============================================================
-- The Port Parameter Statistics Data Set provides counters
-- associated with PTP Port capabilities at a given PTP Instance.
-- ============================================================

ieee8021AsV3PortStatDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3PortStatDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
```


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```
    DESCRIPTION
        "The portStatisticsDS provides counters associated with PTP Port
         capabilities at a given PTP Instance."
    REFERENCE   "14.12"
    ::= { ieee8021AsV3MIBObjects 12 }

ieee8021AsV3PortStatDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3PortStatDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "Port Statistics Data Set provides counters associated with
         PTP Port capabilities at a given PTP Instance."
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3PortDSIndex }
    ::= { ieee8021AsV3PortStatDSTable 1 }

Ieee8021AsV3PortStatDSEntry ::=
     SEQUENCE {
         ieee8021AsV3PortStatRxSyncCount               Counter32,
         ieee8021AsV3PortStatRxOneStepSyncCount        Counter32,
         ieee8021AsV3PortStatRxFollowUpCount           Counter32,
         ieee8021AsV3PortStatRxPdelayRequestCount      Counter32,
         ieee8021AsV3PortStatRxPdelayRspCount     Counter32,
         ieee8021AsV3PortStatRxPdelayRspFollowUpCount  Counter32,
         ieee8021AsV3PortStatRxAnnounceCount           Counter32,
         ieee8021AsV3PortStatRxPtpPacketDiscardCount   Counter32,
         ieee8021AsV3PortStatSyncReceiptTimeoutCount    Counter32,
         ieee8021AsV3PortStatAnnounceReceiptTimeoutCount   Counter32,
         ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount Counter32,
         ieee8021AsV3PortStatTxSyncCount               Counter32,
         ieee8021AsV3PortStatTxOneStepSyncCount        Counter32,
         ieee8021AsV3PortStatTxFollowUpCount           Counter32,
         ieee8021AsV3PortStatTxPdelayRequestCount      Counter32,
         ieee8021AsV3PortStatTxPdelayRspCount     Counter32,
         ieee8021AsV3PortStatTxPdelayRspFollowUpCount Counter32,
         ieee8021AsV3PortStatTxAnnounceCount           Counter32
      }

ieee8021AsV3PortStatRxSyncCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time synchronization
         information is received."
    REFERENCE   "14.12.2"
    ::= { ieee8021AsV3PortStatDSEntry 1 }

ieee8021AsV3PortStatRxOneStepSyncCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a one-step Sync
         message is received."
    REFERENCE   "14.12.3"
    ::= { ieee8021AsV3PortStatDSEntry 2 }

```


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```
ieee8021AsV3PortStatRxFollowUpCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Follow_Up message
         is received."
    REFERENCE   "14.12.4"
    ::= { ieee8021AsV3PortStatDSEntry 3 }

ieee8021AsV3PortStatRxPdelayRequestCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Req message
         is received."
    REFERENCE   "14.12.5"
    ::= { ieee8021AsV3PortStatDSEntry 4 }

ieee8021AsV3PortStatRxPdelayRspCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp message
         is received."
    REFERENCE   "14.12.6"
    ::= { ieee8021AsV3PortStatDSEntry 5 }

ieee8021AsV3PortStatRxPdelayRspFollowUpCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp_Follow_Up
         message is received."
    REFERENCE   "14.12.7"
    ::= { ieee8021AsV3PortStatDSEntry 6 }

ieee8021AsV3PortStatRxAnnounceCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time an Announce message
         is received."
    REFERENCE   "14.12.8"
    ::= { ieee8021AsV3PortStatDSEntry 7 }

ieee8021AsV3PortStatRxPtpPacketDiscardCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a PTP message of the
         respective PTP Instance is discarded."
    REFERENCE   "14.12.9"
    ::= { ieee8021AsV3PortStatDSEntry 8 }
```


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```
ieee8021AsV3PortStatSyncReceiptTimeoutCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time sync receipt timeout
         occurs."
    REFERENCE   "14.12.10"
    ::= { ieee8021AsV3PortStatDSEntry 9 }

ieee8021AsV3PortStatAnnounceReceiptTimeoutCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time announce receipt timeout
         occurs."
    REFERENCE   "14.12.11"
    ::= { ieee8021AsV3PortStatDSEntry 10 }

ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time the value of the
         variable lostResponses exceeds the value of the variable
         allowedLostResponses."
    REFERENCE   "14.12.12"
    ::= { ieee8021AsV3PortStatDSEntry 11 }

ieee8021AsV3PortStatTxSyncCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time synchronization
         information is transmitted."
    REFERENCE   "14.12.13"
    ::= { ieee8021AsV3PortStatDSEntry 12 }

ieee8021AsV3PortStatTxOneStepSyncCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a one-step Sync
         message is transmitted."
    REFERENCE   "14.12.14"
    ::= { ieee8021AsV3PortStatDSEntry 13 }

ieee8021AsV3PortStatTxFollowUpCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Follow_Up message
         is transmitted."
```


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```
    REFERENCE   "14.12.15"
    ::= { ieee8021AsV3PortStatDSEntry 14 }

ieee8021AsV3PortStatTxPdelayRequestCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Req message
         is transmitted."
    REFERENCE   "14.12.16"
    ::= { ieee8021AsV3PortStatDSEntry 15 }

ieee8021AsV3PortStatTxPdelayRspCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp message
         is transmitted."
    REFERENCE   "14.12.17"
    ::= { ieee8021AsV3PortStatDSEntry 16 }

ieee8021AsV3PortStatTxPdelayRspFollowUpCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a
         Pdelay_Resp_Follow_Up message is transmitted."
    REFERENCE   "14.12.18"
    ::= { ieee8021AsV3PortStatDSEntry 17 }

ieee8021AsV3PortStatTxAnnounceCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time an Announce message is
         transmitted."
    REFERENCE   "14.12.19"
    ::= { ieee8021AsV3PortStatDSEntry 18 }

-- ================================================================
-- The Acceptable TimeTransmitter Port Parameter Data Ser represents the
-- capability to enable/disable the acceptable timeTransmitter table
-- feature on a PTP Port.
-- ================================================================

ieee8021AsV3AcceptableTimeTransmitterPortDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "For the single PTP Port of a PTP End Instance and for each
         PTP Port of a PTP Relay Instance, the acceptableTimeTransmitterPortDS
         contains the single member acceptableTimeTransmitterTableEnabled, which
         is used to enable/disable the Acceptable TimeTransmitter Table Feature.
         The number of such data sets is the same as the value of
```


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```
         defaultDS.numberPorts."
    REFERENCE   "14.13"
    ::= { ieee8021AsV3MIBObjects 13 }

ieee8021AsV3AcceptableTimeTransmitterPortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Acceptable TimeTransmitter Port Data Set represents the capability
         to enable/disable the acceptable timeTransmitter table feature on a
         PTP Port.
         For the single PTP Port of a PTP End Instance and for each
         PTP Port of a PTP Relay Instance, the acceptableTimeTransmitterPortDS
         contains the single member acceptableTimeTransmitterTableEnabled, which
         is used to enable/disable the Acceptable TimeTransmitter Table Feature.
         The number of such data sets is the same as the value of
         defaultDS.numberPorts."
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex }
    ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSTable 1 }

Ieee8021AsV3AcceptableTimeTransmitterPortDSEntry ::=
  SEQUENCE {
     ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex    InterfaceIndexOrZero,
      ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled         TruthValue
      }

ieee8021AsV3AcceptableTimeTransmitterPortDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Acceptable TimeTransmitter
         Port Table Data Set."
    REFERENCE   "14.13"
    ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSEntry 1 }

ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         acceptableTimeTransmitterTableEnabled."
    REFERENCE   "14.13.2"
    ::= { ieee8021AsV3AcceptableTimeTransmitterPortDSEntry 2 }

-- ============================================================
-- The External Port Configuration Port Data Set is used with
-- the external port configuration option to indicate the
-- desired state for the PTP Port.
-- ============================================================
ieee8021AsV3ExternalPortConfigurationPortDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3ExternalPortConfigurationPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The externalPortConfigurationPortDS contains the single member
```


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```
         desiredState, which indicates the desired state for the PTP Port.
         The number of such data sets is the same as the value of
         defaultDS.numberPorts."
    REFERENCE   "14.14"
    ::= { ieee8021AsV3MIBObjects 14 }

ieee8021AsV3ExternalPortConfigurationPortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ExternalPortConfigurationPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The externalPortConfigurationPortDS contains the single member
         desiredState, which indicates the desired state for the PTP Port.
         The number of such data sets is the same as the value of
         defaultDS.numberPorts."
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3ExternalPortConfigurationPortDSAsIndex }
    ::= { ieee8021AsV3ExternalPortConfigurationPortDSTable 1 }

Ieee8021AsV3ExternalPortConfigurationPortDSEntry ::=
  SEQUENCE {
     ieee8021AsV3ExternalPortConfigurationPortDSAsIndex    InterfaceIndexOrZero,
      ieee8021AsV3ExternalPortConfigurationPortDSDesiredState         INTEGER
      }

ieee8021AsV3ExternalPortConfigurationPortDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the External Port
         Configuration Port Table Data Set."
    REFERENCE   "14.14"
    ::= { ieee8021AsV3ExternalPortConfigurationPortDSEntry 1 }

ieee8021AsV3ExternalPortConfigurationPortDSDesiredState OBJECT-TYPE
    SYNTAX           INTEGER {
            disabledPort(3),
            timeTransmitterPort(6),
            passivePort(7),
            timeReceiverPort(9)
           }
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "When the value of defaultDS.externalPortConfigurationEnabled
         is TRUE (1), the value of
         externalPortConfigurationPortDS.desiredState is the desired
         state of the PTP Port. This member sets the value of the
         variable portStateInd. When a new value is written to the
         member by management, the variable rcvdPortStateInd is set
         to TRUE (1)."
    REFERENCE   "14.14.2"
    ::= { ieee8021AsV3ExternalPortConfigurationPortDSEntry 2 }

-- ============================================================
-- Asymmetry Measurement Mode Parameter Data Set
-- to enable/disable the feature on a PTP Port.
-- ============================================================
```


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```
ieee8021AsV3AsymMeasurementModeDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3AsymMeasurementModeDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The asymmetryMeasurementModeDS represents the capability to
         enable/disable the Asymmetry Compensation Measurement Procedure
         on a PTP Port (see Annex G). This data set is used instead of
         the cmldsAsymmetryMeasurementModeDS, when only domain 0 is
         present and CMLDS is not used."
    REFERENCE   "14.15"
    ::= { ieee8021AsV3MIBObjects 15 }

ieee8021AsV3AsymMeasurementModeDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3AsymMeasurementModeDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The asymmetryMeasurementModeDS represents the capability to
         enable/disable the Asymmetry Compensation Measurement Procedure
         on a PTP Port (see Annex G). This data set is used instead of
         the cmldsAsymmetryMeasurementModeDS, when only domain 0 is
         present and CMLDS is not used. "
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3AsymMeasurementModeDSAsIndex }
    ::= { ieee8021AsV3AsymMeasurementModeDSTable 1 }

Ieee8021AsV3AsymMeasurementModeDSEntry ::=
     SEQUENCE {
         ieee8021AsV3AsymMeasurementModeDSAsIndex      InterfaceIndexOrZero,
         ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode
TruthValue
      }

ieee8021AsV3AsymMeasurementModeDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Asymmetry Measurement
         Mode Data Set."
    REFERENCE   "14.15"
    ::= { ieee8021AsV3AsymMeasurementModeDSEntry 1 }

ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         asymmetryMeasurementMode. For full-duplex IEEE 802.3
         media, the value is TRUE (1) if an asymmetry measurement
         is being performed for the link attached to this PTP Port,
         and FALSE (2) otherwise. For all other media, the value
         shall be FALSE (2). Setting this managed object causes the
         Boolean  asymmetryMeasurementMode to have the same value.
         NOTE: If an asymmetry measurement is being performed for a
         link, asymmetryMeasurementMode must be TRUE (1) for the
```


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```
         PTP Ports at each end of the link."
    REFERENCE   "14.15.2"
    ::= { ieee8021AsV3AsymMeasurementModeDSEntry 2 }

-- ============================================================
-- The Common Services Port Parameter Data Set enables a
-- PTP Port of a PTP Instance to determine which port of the
-- respective common service corresponds to that PTP Port.
-- ============================================================
ieee8021AsV3CommonServicesPortDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3CommonServicesPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This member contains the port number
         of the CMLDS Link Port that corresponds to this PTP Port."
    REFERENCE   "14.16"
    ::= { ieee8021AsV3MIBObjects 16 }

ieee8021AsV3CommonServicesPortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CommonServicesPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This member contains the port number
        of the CMLDS Link Port that
         corresponds to this PTP Port."
    INDEX { ieee8021AsV3PtpInstance,
            ieee8021AsV3CommonServicesPortDSAsIndex }
    ::= { ieee8021AsV3CommonServicesPortDSTable 1 }

Ieee8021AsV3CommonServicesPortDSEntry ::=
  SEQUENCE {
      ieee8021AsV3CommonServicesPortDSAsIndex    InterfaceIndexOrZero,
      ieee8021AsV3CommonServicesPortDSCmldsLinkPortPortNumber         Unsigned32
      }

ieee8021AsV3CommonServicesPortDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Common Services Port
         Data Set."
    REFERENCE   "14.16"
    ::= { ieee8021AsV3CommonServicesPortDSEntry 1 }

ieee8021AsV3CommonServicesPortDSCmldsLinkPortPortNumber OBJECT-TYPE
    SYNTAX          Unsigned32 (0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the portNumber attribute of the
         cmldsLinkPortDS.portIdentity of the Link Port that
         corresponds to this PTP Port."
    REFERENCE   "14.16.2"
    ::= { ieee8021AsV3CommonServicesPortDSEntry 2 }

-- ============================================================
```


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```
-- The Common Mean Link Delay Service Default Parameter Data Set
-- describes the per-time-aware-system attributes of the Common
-- Mean Link Delay Service.
-- ============================================================

ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The cmldsDefaultDS describes the per-time-aware-system attributes
         of the Common Mean Link Delay Service."
    REFERENCE   "14.17"
    ::= { ieee8021AsV3MIBObjects 17 }

ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The cmldsDefaultDS describes the per-time-aware-system attributes
         of the Common Mean Link Delay Service."
    INDEX { ieee8021AsV3CmldsDefaultDSAsIndex }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSTable 1 }

Ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry ::=
   SEQUENCE {
      ieee8021AsV3CmldsDefaultDSAsIndex  InterfaceIndexOrZero,
      ieee8021AsV3CmldsDefaultDSClockIdentity   Ieee8021AsV3ClockIdentity,
      ieee8021AsV3CmldsDefaultDSNumberLinkPorts Unsigned32
      }


ieee8021AsV3CmldsDefaultDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Common Mean Link
         Delay Default Data Set."
    REFERENCE   "14.17"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 1 }

ieee8021AsV3CmldsDefaultDSClockIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the clockIdentity that will be used to
         identify the Common Mean Link Delay Service."
    REFERENCE   "14.17.2"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 2 }

ieee8021AsV3CmldsDefaultDSNumberLinkPorts OBJECT-TYPE
    SYNTAX      Unsigned32 (0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the number of Link Ports of the time-aware
```


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```
        system on which the Common Mean Link Delay Service is
        implemented. For an end station the value is 1."
    REFERENCE   "14.17.3"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSEntry 3 }

-- ============================================================
-- The Common Mean Link Delay Service Link Port Parameter Data Set
-- represents time-aware Link Port capabilities for the Common Mean
-- Link Delay Service of a Link Port of a time-aware system.
-- ============================================================

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSTable   OBJECT-TYPE
    SYNTAX     SEQUENCE OF Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "For every Link Port of the Common Mean Link Delay Service of a
        time-aware system, the cmldsLinkPortDS is maintained as the
        basis for making protocol decisions and providing values for
        message fields. The number of such data sets is the same as
        the value of cmldsDefaultDS.numberLinkPorts."
    REFERENCE   "14.18"
    ::= { ieee8021AsV3MIBObjects 18 }

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "For every Link Port of the Common Mean Link Delay Service of a
        time-aware system, the cmldsLinkPortDS is maintained as the
        basis for making protocol decisions and providing values for
        message fields. The number of such data sets is the same as
        the value of cmldsDefaultDS.numberLinkPorts."
    INDEX { ieee8021AsV3BridgeBasePort,
            ieee8021AsV3CmldsLinkPortDSAsIndex }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSTable 1 }

Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry ::=
   SEQUENCE {
     ieee8021AsV3CmldsLinkPortDSAsIndex     InterfaceIndexOrZero,
     ieee8021AsV3CmldsLinkPortDSClockIdentity       Ieee8021AsV3ClockIdentity,
     ieee8021AsV3CmldsLinkPortDSPortNumber             Unsigned32,
     ieee8021AsV3CmldsLinkPortDSCmldsLinkPortEnabled   TruthValue,
     ieee8021AsV3CmldsLinkPortDSIsMeasuringDelay       TruthValue,
     ieee8021AsV3CmldsLinkPortDSAsCapableAcrossDomains TruthValue,
     ieee8021AsV3CmldsLinkPortDSMeanLinkDelay       Ieee8021ASV3PtpTimeInterval,
     ieee8021AsV3CmldsLinkPortDSMeanLinkDelayThresh Ieee8021ASV3PtpTimeInterval,
     ieee8021AsV3CmldsLinkPortDSDelayAsym           Ieee8021ASV3PtpTimeInterval,
     ieee8021AsV3CmldsLinkPortDSNbrRateRatio      Integer32,
     ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval Integer32,
     ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval Integer32,
     ieee8021AsV3CmldsLinkPortDSUseMgtSettableLogPdelayReqInterval  TruthValue,
     ieee8021AsV3CmldsLinkPortDSMgtSettableLogPdelayReqInterval     Integer32,
     ieee8021AsV3CmldsLinkPortDSInitialComputeNbrRateRatio     TruthValue,
     ieee8021AsV3CmldsLinkPortDSCurrentComputeNbrRateRatio     TruthValue,
     ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio    TruthValue,
     ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio     TruthValue,
     ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay     TruthValue,
```


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```
    ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay     TruthValue,
    ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay
TruthValue,
    ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay     TruthValue,
    ieee8021AsV3CmldsLinkPortDSAllowedLostRsp   Unsigned32,
    ieee8021AsV3CmldsLinkPortDSAllowedFaults          Unsigned32,
    ieee8021AsV3CmldsLinkPortDSVersionNumber          Unsigned32,
    ieee8021AsV3CmldsLinkPortDSPdelayTruncTST1 Ieee8021ASV3Timestamp,
    ieee8021AsV3CmldsLinkPortDSPdelayTruncTST2 Ieee8021ASV3Timestamp,
    ieee8021AsV3CmldsLinkPortDSPdelayTruncTST3 Ieee8021ASV3Timestamp,
    ieee8021AsV3CmldsLinkPortDSPdelayTruncTST4 Ieee8021ASV3Timestamp,
    ieee8021AsV3CmldsLinkPortDSMinorVersionNumber     Unsigned32
     }

ieee8021AsV3CmldsLinkPortDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Comon Mean Link
         Delay Link Port Data Set."
    REFERENCE   "14.18"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 1 }

ieee8021AsV3CmldsLinkPortDSClockIdentity OBJECT-TYPE
    SYNTAX      Ieee8021AsV3ClockIdentity
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the first of the portIdentity attribute
         of the local port, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.18.2"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 2 }

ieee8021AsV3CmldsLinkPortDSPortNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..65535)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is the second of the portIdentity attribute
         of the local port, which is a set made of
         Ieee8021AsV3ClockIdentity and portNumber."
    REFERENCE   "14.18.2"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 3 }

ieee8021AsV3CmldsLinkPortDSCmldsLinkPortEnabled
 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         cmldsLinkPortEnabled."
    REFERENCE   "14.18.3"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 4 }

ieee8021AsV3CmldsLinkPortDSIsMeasuringDelay
 OBJECT-TYPE
```


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```
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         isMeasuringDelay."
    REFERENCE   "14.18.4"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 5 }

ieee8021AsV3CmldsLinkPortDSAsCapableAcrossDomains
 OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         asCapableAcrossDomains."
    REFERENCE   "14.18.5"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 6 }

ieee8021AsV3CmldsLinkPortDSMeanLinkDelay
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-port global
         variable meanLinkDelay. It is an estimate of the current
         one-way propagation time on the link attached to this Link
         Port, measured as specified for the respective medium. The
         value is zero for Link Ports attached to IEEE 802.3 EPON
         links and for the timeTransmitter port of an IEEE 802.11 link,
         because one-way propagation delay is not measured on the
         latter and not directly measured on the former.
         NOTE: The underlying per-port global variable meanLinkDelay is
         of type UScaledNS, which is a 96-Bit value. meanLinkDelay
         values that are larger than the maximum value that can be
         represented by the TimeInterval data type, i.e.,
         0xFFFF FFFF FFFF FFFF (where the units are 2 sup -16 ns), used
         for this managed object are set to this largest value."
    REFERENCE   "14.18.6"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 7 }

ieee8021AsV3CmldsLinkPortDSMeanLinkDelayThresh
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-Link-Port global
         variable meanLinkDelayThresh. It is the propagation time
         threshold above which a Link Port (and therefore any PTP Ports
         that use the CMLDS on this Link Port) is considered not
         capable of participating in the IEEE 802.1AS protocol.
         Setting this managed object causes the per-Link-Port global
         variable meanLinkDelayThresh to have the same value.
         NOTE: The underlying per-port global variable
         meanLinkDelayThresh is of type UScaledNS, which is a 96-Bit
         value. meanLinkDelayThresh values that are larger than the
```


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```
         maximum value that can be represented by the TimeInterval
         data type, i.e., 0xFFFF FFFF FFFF FFFF (where the units are
         2 sup -16 ns), used for this managed object are set to this
         largest value."
    REFERENCE   "14.18.7"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 8 }

ieee8021AsV3CmldsLinkPortDSDelayAsym
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3PtpTimeInterval
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the asymmetry in the propagation delay on the
         link attached to this Link Port relative to the local clock.
         If propagation delay asymmetry is not modeled, then
         delayAsymmetry is 0.
         NOTE: The underlying per-port global variable delayAsymmetry
         is of type ScaledNS, which is a 96-Bit value.
         delayAsymmetry values that are larger than the maximum value
         that can be represented by the TimeInterval data type, i.e.,
         0x7FFF FFFF FFFF FFFF, (where the units are 2 sup -16 ns),
         used for this managed object are set to this largest value.
         delayAsymmetry values that are less than the minimum value
         that can be represented by the TimeInterval data type, i.e.,
         0x8000 0000 0000 0001 written in twos complement form (where
        the units are 2 sup -16 ns), used for this managed object are
        set to this smallest value."
    REFERENCE   "14.18.9"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 9 }

ieee8021AsV3CmldsLinkPortDSNbrRateRatio
 OBJECT-TYPE
    SYNTAX      Integer32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The value is an estimate of the ratio of the frequency of the
         LocalClock entity of the time-aware system at the other end
         of the link attached to this Link Port, to the frequency of
         the LocalClock entity of this time-aware system.
         neighborRateRatio is expressed as the fractional frequency
         offset multiplied by 2^41, i.e., the quantity
         (neighborRateRatio -1.0)(2^41)."
    REFERENCE   "14.18.10"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 10 }

ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableLogPdelayReqInterval is FALSE (2) then,
         for full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         value is the logarithm to base 2 of the Pdelay_Req message
         transmission interval used when (a) the Link Port is
         initialized, or (b) a message interval request TLV is
         received with the logLinkDelayInterval field set to 126.
```


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```
         For all other media, the value is 127."
    REFERENCE   "14.18.11"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 11 }

ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32(-128..127)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay,
         the value is the logarithm to the base 2 of the current
         Pdelay_Req message transmission interval.
         For all other media, the value is 127."
    REFERENCE   "14.18.12"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 12 }

ieee8021AsV3CmldsLinkPortDSUseMgtSettableLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the sync interval and mean time interval between
         successive Pdelay_Req messages. If the value is TRUE (1),
         the value of currentLogPdelayReqInterval is set equal to
         the value of mgtSettableLogPdelayReqInterval. If the value
         of the managed object is FALSE (2), the value of
         currentLogPdelayReqInterval is determined by the
         LinkDelayIntervalSetting state machine."
    REFERENCE   "14.18.13"
    DEFVAL { false }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 13 }

ieee8021AsV3CmldsLinkPortDSMgtSettableLogPdelayReqInterval OBJECT-TYPE
    SYNTAX      Integer32 (-128..127)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is the logarithm to base 2 of the mean time
         interval between successive Pdelay_Req messages if
         useMgtSettableLogPdelayReqInterval is TRUE (1). The
         value is not used if useMgtSettableLogPdelayReqInterval
         is FALSE (2)."
    REFERENCE   "14.18.14"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 14 }

ieee8021AsV3CmldsLinkPortDSInitialComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableComputeNeighborRateRatio is FALSE (2),
         then for full-duplex IEEE 802.3 media and for CSN media
         that use the peer-to-peer delay mechanism to measure path
         delay, the value is the initial value of
         computeNeighborRateRatio.
         For all other media, the value is TRUE."
    REFERENCE   "14.18.15"
```


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```
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 15 }

ieee8021AsV3CmldsLinkPortDSCurrentComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay,
         the value is the current value of computeNeighborRateRatio.
         For all other media, the value is TRUE (1)."
    REFERENCE   "14.18.16"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 16 }

ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the value of computeNeighborRateRatio. If the value is
         TRUE (1), the value of computeNeighborRateRatio is set equal
         to the value of mgtSettablecomputeNeighborRateRatio. If
         the value of the managed object is FALSE (2), the
         value of currentComputeNeighborRateRatio is determined by
         the LinkDelayIntervalSetting state machine."
    REFERENCE   "14.18.17"
    DEFVAL { false }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 17 }

ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "computeNeighborRateRatio is configured to this value if
         useMgtSettableComputeNeighborRateRatio is TRUE (1). The
         value is not used if useMgtSettableComputeNeighborRateRatio
         is FALSE (2)."
    REFERENCE   "14.18.18"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 18 }

ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "If useMgtSettableComputeMeanLinkDelay is FALSE (2) then,
         for full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay,
         the value is the initial value of computeMeanLinkDelay.
         For all other media, the value is TRUE (1)."
    REFERENCE   "14.18.19"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 19 }

ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-only
    STATUS      current
```


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```
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the value is the current value of computeMeanLinkDelay.
         For all other media, the value is TRUE."
    REFERENCE   "14.18.20"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 20 }

ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The managed object is a Boolean that determines the source
         of the value of computeMeanLinkDelay. If the value is
         TRUE (1), the value of computeMeanLinkDelay is set equal
         to the value of mgtSettableComputeMeanLinkDelay. If the
         value of the managed object is FALSE (2), the value of
         currentComputeMeanLinkDelay is determined by the
         LinkDelayIntervalSetting state machine."
    REFERENCE   "14.18.21"
    DEFVAL { false }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 21 }

ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "computeMeanLinkDelay is configured to this value if
         useMgtSettableComputeMeanLinkDelay is TRUE (1). The value
         is not used if useMgtSettableComputeMeanLinkDelay is
         FALSE (2)."
    REFERENCE   "14.18.22"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 22 }

ieee8021AsV3CmldsLinkPortDSAllowedLostRsp
 OBJECT-TYPE
    SYNTAX      Unsigned32(1..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the per-Link-Port
         global variable allowedLostResponses. It is the number
         of Pdelay_Req messages without valid responses
         above which a Link Port is considered to be not
         exchanging peer delay messages with its neighbor.
         Setting this managed object causes the per-Link-Port global
         variable allowedLostResponses to have the same value."
    REFERENCE   "14.18.23"
    DEFVAL { 9 }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 23 }


ieee8021AsV3CmldsLinkPortDSAllowedFaults OBJECT-TYPE
    SYNTAX      Unsigned32(1..255)
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
```


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```
        "The value is equal to the value of the per-Link-Port global
         variable allowedFaults. It is the number of faults above
         which asCapableAcrossDomains is set to FALSE (2), i.e., a
         Link Port is considered not capable of interoperating
         with its neighbor via the IEEE 802.1AS protocol.
         Setting this managed object causes the per-Link-Port global
         variable allowedFaults to have the same value."
    REFERENCE   "14.18.24"
    DEFVAL { 9 }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 24 }

ieee8021AsV3CmldsLinkPortDSVersionNumber OBJECT-TYPE
    SYNTAX      Unsigned32(0..15)
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value is set to versionPTP as specified in 10.6.2.2.4."
    REFERENCE   "14.18.25"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 25 }

ieee8021AsV3CmldsLinkPortDSPdelayTruncTST1
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         first value, T1, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t1 modulo 2^32
         in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp, expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange.
         NOTE: This managed object is used with the Asymmetry
         Compensation Measurement Procedure, which is based on
         line-swapping."
    REFERENCE   "14.18.26"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 26 }

ieee8021AsV3CmldsLinkPortDSPdelayTruncTST2
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         second value, T2, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t1 modulo 2^32
         in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp, expressed in units
         of 2^-16 ns, by 2^48).
```


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```
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange.
         NOTE: This managed object is used with the Asymmetry
         Compensation Measurement Procedure, which is based on
         line-swapping."
    REFERENCE   "14.18.26"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 27 }

ieee8021AsV3CmldsLinkPortDSPdelayTruncTST3
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         third value, T3, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t1 modulo 2^32
         in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp, expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange.
         NOTE: This managed object is used with the Asymmetry
         Compensation Measurement Procedure, which is based on
         line-swapping."
    REFERENCE   "14.18.26"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 28 }

ieee8021AsV3CmldsLinkPortDSPdelayTruncTST4
 OBJECT-TYPE
    SYNTAX      Ieee8021ASV3Timestamp
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "For full-duplex IEEE 802.3 media and for CSN media that use
         the peer-to-peer delay mechanism to measure path delay, the
         fourth value, T4, of the four elements of this array is as
         described in Table 14-9. For all other media, the values are
         zero. This object corresponds to the timestamp t1 modulo 2^32
         in Figure 11-1, and expressed in units of 2^-16 ns (i.e., the
         value of this array element is equal to the remainder obtained
         upon dividing the respective timestamp, expressed in units
         of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the T1, T2,
         T3, T4 PdelayTruncTS are for the same, and most recently
         completed, peer delay message exchange.
         NOTE: This managed object is used with the Asymmetry
         Compensation Measurement Procedure, which is based on
         line-swapping."
    REFERENCE   "14.18.26"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 29 }

ieee8021AsV3CmldsLinkPortDSMinorVersionNumber OBJECT-TYPE
    SYNTAX      Unsigned32 (0..15)
```


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```
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "This value is set to minorVersionPTP as specified in
         10.6.2.2.3."
    REFERENCE   "14.18.27"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSEntry 30 }

-- ============================================================
-- The Common Mean Link Delay Service Link Port Parameter
-- Statistics Data Set provides counters associated with Link
-- Port capabilities at a given time-aware system.
-- ============================================================

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSTable  OBJECT-TYPE
    SYNTAX      SEQUENCE OF
Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "For every Link Port of the Common Mean Link Delay Service of a
         time-aware system, the following cmldsLinkPortStatisticsDS
         provides counters. The number of such statistics sets is the
         same as the value of cmldsDefaultDS.numberLinkPorts."
    REFERENCE   "14.19"
    ::= { ieee8021AsV3MIBObjects 19 }

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "For every Link Port of the Common Mean Link Delay Service of a
         time-aware system, the following cmldsLinkPortStatisticsDS
         provides counters. The number of such statistics sets is the
         same as the value of cmldsDefaultDS.numberLinkPorts."
    INDEX { ieee8021AsV3BridgeBasePort,
            ieee8021AsV3CmldsLinkPortStatDSIndex }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSTable 1 }

Ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry ::=
  SEQUENCE {
    ieee8021AsV3CmldsLinkPortStatDSIndex           InterfaceIndexOrZero,
    ieee8021AsV3CmldsLinkPortStatDSRxPdelayRequestCount            Counter32,
    ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspCount           Counter32,
    ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspFollowUpCount   Counter32,
    ieee8021AsV3CmldsLinkPortStatDSRxPtpPacketDiscardCount         Counter32,
    ieee8021AsV3CmldsLinkPortStatDSPdelayAllowedLostRspExceededCount  Counter32,
    ieee8021AsV3CmldsLinkPortStatDSTxPdelayRequestCount            Counter32,
    ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspCount           Counter32,
    ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspFollowUpCount   Counter32
      }

ieee8021AsV3CmldsLinkPortStatDSIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An index to identify an entry in the Common Mean Link
```


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```
         Port Statistics Data Set."
    REFERENCE   "14.19"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 1 }

ieee8021AsV3CmldsLinkPortStatDSRxPdelayRequestCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Req message is
         received."
    REFERENCE   "14.19.2"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 2 }

ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp message is
         received."
    REFERENCE   "14.19.3"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 3 }

ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspFollowUpCount OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp_Follow_Up
         message is received."
    REFERENCE   "14.19.4"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 4 }

ieee8021AsV3CmldsLinkPortStatDSRxPtpPacketDiscardCount
 OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a PTP message of the
         Common Mean Link Delay Service is discarded, caused by the
         occurrence of any of the conditions given in 14.17.5."
    REFERENCE   "14.19.5"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 5 }

ieee8021AsV3CmldsLinkPortStatDSPdelayAllowedLostRspExceededCount
 OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time the value of the variable
         lostResponses exceeds the value of the variable
         allowedLostResponses, in the RESET state of the
         MDPdelayReq state machine."
    REFERENCE   "14.19.6"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 6 }

```


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```
ieee8021AsV3CmldsLinkPortStatDSTxPdelayRequestCount
 OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Req message is
         transmitted."
    REFERENCE   "14.19.7"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 7 }

ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspCount
 OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp message is
         transmitted."
    REFERENCE   "14.19.8"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 8 }

ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspFollowUpCount
 OBJECT-TYPE
    SYNTAX      Counter32
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "A counter that increments every time a Pdelay_Resp_Follow_Up
         message is transmitted."
    REFERENCE   "14.19.9"
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSEntry 9 }

-- ============================================================
-- The Common Mean Link Delay Service Asymmetry Measurement Mode
-- Parameter Data Set represents the capability to enable/disable
-- the Asymmetry Compensation Measurement Procedure on a Link Port
-- (see Annex G).
-- ============================================================

ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSTable   OBJECT-TYPE
    SYNTAX      SEQUENCE OF
Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The Common Mean Link Delay Service Asymmetry Measurement Mode
         Parameter Data Set represents the capability to enable/disable
         the Asymmetry Compensation Measurement Procedure on a Link Port
         (see Annex G)."
    REFERENCE   "14.20"
    ::= { ieee8021AsV3MIBObjects 20 }

ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry   OBJECT-TYPE
    SYNTAX      Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table uses
```


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```
        ieee8021AsV3CmldsAsymmetryMeasurementModeDSAsIndex,
        and corresponds to
        ieee8021AsV3CommonMeanLinkDelayServiceAsymmetryMeasurementModeDSTable
        entry."
    INDEX { ieee8021AsV3BridgeBasePort,
            ieee8021AsV3CmldsAsymMeasurementModeDSAsIndex }
    ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSTable 1 }

Ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry ::=
     SEQUENCE {
 ieee8021AsV3CmldsAsymMeasurementModeDSAsIndex InterfaceIndexOrZero,
 ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode    TruthValue
      }

ieee8021AsV3CmldsAsymMeasurementModeDSAsIndex OBJECT-TYPE
    SYNTAX      InterfaceIndexOrZero
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This object identifies the gPTP interface group within
         the system for which this entry contains information. It
         is the value of the instance of the IfIndex object,
         defined in the IF-MIB, for the gPTP interface group
         corresponding to this port, or the value 0 if the port
         has not been bound to an underlying frame source and
         sink.

         For a given media port of a Bridge or an end station,
         there can be one or more PTP Port, and depends whether
         a media port supports point to point link (e.g. IEEE
         802.3 Ethernet) or point to multi-point (e.g. CSN, IEEE
         802.3 EPON) links on the media port."
    REFERENCE   "IEEE Std 802.1AS
CommonMeanLinkDelaySvcAsymMeasurementModeParamDS Group PTP Port Index"
  ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry 1 }

ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode
    OBJECT-TYPE
    SYNTAX      TruthValue
    MAX-ACCESS  read-write
    STATUS      current
    DESCRIPTION
        "The value is equal to the value of the Boolean
         asymmetryMeasurementMode (see G.3). For full-duplex
         IEEE 802.3 media, the value is TRUE (1) if an asymmetry
          measurement is being performed for the link attached to
         this Link Port, and FALSE (2) otherwise. For all other
         media, the value shall be FALSE (2) (see 10.2.5.2).
         Setting this managed object causes the Boolean
         allowedFaults to have the same value.
         NOTE: If an asymmetry measurement is being performed
         for a link, asymmetryMeasurementMode must be TRUE (1)
         for the Link Ports at each end of the link.
         There is one Common Mean Link Delay Service Asymmetry
         Measurement Mode Parameter Data Set Table for all PTP
         Instances, per Link Port."
    REFERENCE   "14.20.2"
  ::= { ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSEntry 2 }

```


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```
-- ****************************************************************
-- IEEE 802.1ASV3 MIB - Conformance Information
-- ****************************************************************
ieee8021AsV3Groups          OBJECT IDENTIFIER ::= { ieee8021AsV3Conformance 1 }
ieee8021AsV3Compliances     OBJECT IDENTIFIER ::= { ieee8021AsV3Conformance 2 }

-- ================================================================
-- units of conformance
-- ================================================================

ieee8021AsV3PtpInstanceGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3PtpInstanceName,
        ieee8021AsV3PtpInstanceRowStatus
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information for dynamic
         creation and deletion of PTP Instances and logical ports."
    ::= { ieee8021AsV3Groups 1 }

ieee8021AsV3DefaultDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3DefaultDSClockIdentity,
        ieee8021AsV3DefaultDSNumberPorts,
        ieee8021AsV3DefaultDSClockQualityClockClass,
        ieee8021AsV3DefaultDSClockQualityClockAccuracy,
        ieee8021AsV3DefaultDSClockQualityOffsetScaledLogVariance,
        ieee8021AsV3DefaultDSPriority1,
        ieee8021AsV3DefaultDSPriority2,
        ieee8021AsV3DefaultDSGmCapable,
        ieee8021AsV3DefaultDSCurrentUtcOffset,
        ieee8021AsV3DefaultDSCurrentUtcOffsetValid,
        ieee8021AsV3DefaultDSLeap59,
        ieee8021AsV3DefaultDSLeap61,
        ieee8021AsV3DefaultDSTimeTraceable,
        ieee8021AsV3DefaultDSFrequencyTraceable,
        ieee8021AsV3DefaultDSPtpTimescale,
        ieee8021AsV3DefaultDSTimeSource,
        ieee8021AsV3DefaultDSDomainNumber,
        ieee8021AsV3DefaultDSSdoId,
        ieee8021AsV3DefaultDSExternalPortConfigurationEnabled,
        ieee8021AsV3DefaultDSInstanceEnable
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Default
         Parameter Data Set representing the native capabilities of a
         PTP Instance, i.e., a PTP Relay Instance or a PTP End Instance."
    ::= { ieee8021AsV3Groups 2 }

ieee8021AsV3CurrentDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CurrentDSStepsRemoved,
        ieee8021AsV3CurrentDSOffsetFromTimeTransmitter,
        ieee8021AsV3CurrentDSLastGmPhaseChange,
        ieee8021AsV3CurrentDSLastGmFreqChange,
        ieee8021AsV3CurrentDSGmTimebaseIndicator,
        ieee8021AsV3CurrentDSGmChangeCount,
```


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```
        ieee8021AsV3CurrentDSTimeOfLastGmChangeEvent,
        ieee8021AsV3CurrentDSTimeOfLastGmPhaseChangeEvent,
        ieee8021AsV3CurrentDSTimeOfLastGmFreqChangeEvent
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Current
         Parameter Data Set representing the position of a local system
         and other information, relative to the Grandmaster PTP Instance."
    ::= { ieee8021AsV3Groups 3 }

ieee8021AsV3ParentDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3ParentDSParentClockIdentity,
        ieee8021AsV3ParentDSParentPortNumber,
        ieee8021AsV3ParentDSCumulativeRateRatio,
        ieee8021AsV3ParentDSGrandmasterIdentity,
        ieee8021AsV3ParentDSGrandmasterClockQualityclockClass,
        ieee8021AsV3ParentDSGrandmasterClockQualityclockAccuracy,
        ieee8021AsV3ParentDSGrandmasterClockQualityoffsetScaledLogVar,
        ieee8021AsV3ParentDSGrandmasterPriority1,
        ieee8021AsV3ParentDSGrandmasterPriority2
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Parent
         Parameter Data Set representing capabilities of the upstream
         system, toward the Grandmaster PTP Instance, as measured at
         a local system."
    ::= { ieee8021AsV3Groups 4 }

ieee8021AsV3TimePropertiesDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3TimePropertiesDSCurrentUtcOffset,
        ieee8021AsV3TimePropertiesDSCurrentUtcOffsetValid,
        ieee8021AsV3TimePropertiesDSLeap59,
        ieee8021AsV3TimePropertiesDSLeap61,
        ieee8021AsV3TimePropertiesDSTimeTraceable,
        ieee8021AsV3TimePropertiesDSFrequencyTraceable,
        ieee8021AsV3TimePropertiesDSPtpTimescale,
        ieee8021AsV3TimePropertiesDSTimeSource
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Time
         Properties Parameter Data Set representing capabilities of
         the Grandmaster PTP Instance, as measured at a local system."
    ::= { ieee8021AsV3Groups 5 }

ieee8021AsV3PathTraceDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3PathTraceDSEnable
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Path Trace
         Data Set representing the current path trace information
         available at the PTP Instance."
    ::= { ieee8021AsV3Groups 6 }
```


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```
ieee8021AsV3PathTraceDSArrayTableGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3PathTraceDSArrayList
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information of an array of
         ClockIdentity values contained in the pathTrace array,
         representing the current path trace information, and which is
         carried in the path trace TLV per PTP Instance."
    ::= { ieee8021AsV3Groups 7 }

ieee8021AsV3AcceptableTimeTransmitterTableDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3AcceptableTimeTransmitterTableDSMaxTableSize,
        ieee8021AsV3AcceptableTimeTransmitterTableDSActualTableSize
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Acceptable TimeTransmitter Table Data Set representing the acceptable
         timeTransmitter table used when an EPON port is used by a PTP Instance
         of a time-aware system."
    ::= { ieee8021AsV3Groups 8 }

ieee8021AsV3AcceptableTimeTransmitterTableDSArrayGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3AcceptableTimeTransmitterTableDSArrayPortIdentity,
        ieee8021AsV3AcceptableTTTableDSArrayAlternatePriority1
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Acceptable TimeTransmitter Table Array Data Set representing the
         acceptable timeTransmitter table used when an EPON port is used by a
         PTP Instance of a time-aware system."
    ::= { ieee8021AsV3Groups 9 }

ieee8021AsV3PortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3PortDSClockIdentity,
        ieee8021AsV3PortDSPortNumber,
        ieee8021AsV3PortDSPortState,
        ieee8021AsV3PortDSPtpPortEnabled,
        ieee8021AsV3PortDSDelayMechanism,
        ieee8021AsV3PortDSIsMeasuringDelay,
        ieee8021AsV3PortDSAsCapable,
        ieee8021AsV3PortDSMeanLinkDelay,
        ieee8021AsV3PortDSMeanLinkDelayThresh,
        ieee8021AsV3PortDSDelayAsym,
        ieee8021AsV3PortDSNbrRateRatio,
        ieee8021AsV3PortDSInitialLogAnnounceInterval,
        ieee8021AsV3PortDSCurrentLogAnnounceInterval,
        ieee8021AsV3PortDSUseMgtSettableLogAnnounceInterval,
        ieee8021AsV3PortDSMgtSettableLogAnnounceInterval,
        ieee8021AsV3PortDSAnnounceReceiptTimeout,
        ieee8021AsV3PortDSInitialLogSyncInterval,
        ieee8021AsV3PortDSCurrentLogSyncInterval,
```


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```
        ieee8021AsV3PortDSUseMgtSettableLogSyncInterval,
        ieee8021AsV3PortDSMgtSettableLogSyncInterval,
        ieee8021AsV3PortDSSyncReceiptTimeout,
        ieee8021AsV3PortDSSyncReceiptTimeoutTimeInterval,
        ieee8021AsV3PortDSInitialLogPdelayReqInterval,
        ieee8021AsV3PortDSCurrentLogPdelayReqInterval,
        ieee8021AsV3PortDSUseMgtSettableLogPdelayReqInterval,
        ieee8021AsV3PortDSMgtSettableLogPdelayReqInterval,
        ieee8021AsV3PortDSInitialLogGptpCapableMessageInterval,
        ieee8021AsV3PortDSCurrentLogGptpCapableMessageInterval,
        ieee8021AsV3PortDSUseMgtSettableLogGptpCapableMessageInterval,
        ieee8021AsV3PortDSMgtSettableLogGptpCapableMessageInterval,
        ieee8021AsV3PortDSInitialComputeNbrRateRatio,
        ieee8021AsV3PortDSCurrentComputeNbrRateRatio,
        ieee8021AsV3PortDSUseMgtSettableComputeNbrRateRatio,
        ieee8021AsV3PortDSMgtSettableComputeNbrRateRatio,
        ieee8021AsV3PortDSInitialComputeMeanLinkDelay,
        ieee8021AsV3PortDSCurrentComputeMeanLinkDelay,
        ieee8021AsV3PortDSUseMgtSettableComputeMeanLinkDelay,
        ieee8021AsV3PortDSMgtSettableComputeMeanLinkDelay,
        ieee8021AsV3PortDSAllowedLostRsp,
        ieee8021AsV3PortDSAllowedFaults,
        ieee8021AsV3PortDSGPtpCapableReceiptTimeout,
        ieee8021AsV3PortDSVersionNumber,
        ieee8021AsV3PortDSNup,
        ieee8021AsV3PortDSNdown,
        ieee8021AsV3PortDSOneStepTxOper,
        ieee8021AsV3PortDSOneStepReceive,
        ieee8021AsV3PortDSOneStepTransmit,
        ieee8021AsV3PortDSInitialOneStepTxOper,
        ieee8021AsV3PortDSCurrentOneStepTxOper,
        ieee8021AsV3PortDSUseMgtSettableOneStepTxOper,
        ieee8021AsV3PortDSMgtSettableOneStepTxOper,
        ieee8021AsV3PortDSSyncLocked,
        ieee8021AsV3PortDSPdelayTruncTST1,
        ieee8021AsV3PortDSPdelayTruncTST2,
        ieee8021AsV3PortDSPdelayTruncTST3,
        ieee8021AsV3PortDSPdelayTruncTST4,
        ieee8021AsV3PortDSMinorVersionNumber
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on PTP Port
         related variables in a time-aware Bridge or for a time-aware
         end station."
    ::= { ieee8021AsV3Groups 10 }

ieee8021AsV3DescriptionPortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3DescriptionPortDSProfileIdentifier
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Description Port Data Set containing the profileIdentifier for
         this PTP profile, as specified in F.2."
    ::= { ieee8021AsV3Groups 11 }

ieee8021AsV3PortStatIfGroup OBJECT-GROUP
```


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```
    OBJECTS {
        ieee8021AsV3PortStatRxSyncCount,
        ieee8021AsV3PortStatRxOneStepSyncCount,
        ieee8021AsV3PortStatRxFollowUpCount,
        ieee8021AsV3PortStatRxPdelayRequestCount,
        ieee8021AsV3PortStatRxPdelayRspCount,
        ieee8021AsV3PortStatRxPdelayRspFollowUpCount,
        ieee8021AsV3PortStatRxAnnounceCount,
        ieee8021AsV3PortStatRxPtpPacketDiscardCount,
        ieee8021AsV3PortStatSyncReceiptTimeoutCount,
        ieee8021AsV3PortStatAnnounceReceiptTimeoutCount,
        ieee8021AsV3PortStatPdelayAllowedLostRspExceededCount,
        ieee8021AsV3PortStatTxSyncCount,
        ieee8021AsV3PortStatTxOneStepSyncCount,
        ieee8021AsV3PortStatTxFollowUpCount,
        ieee8021AsV3PortStatTxPdelayRequestCount,
        ieee8021AsV3PortStatTxPdelayRspCount,
        ieee8021AsV3PortStatTxPdelayRspFollowUpCount,
        ieee8021AsV3PortStatTxAnnounceCount
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the Port
         Statistics Data Set providing counters associated with PTP Port
         capabilities at a given PTP Instance."
    ::= { ieee8021AsV3Groups 12 }

ieee8021AsV3AcceptableTimeTransmitterPortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3AcceptableTTPortDSAcceptableTTTableEnabled
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information for the single
         PTP Port of a PTP End Instance and for each PTP Port of a
         PTP Relay Instance."
    ::= { ieee8021AsV3Groups 13 }

ieee8021AsV3ExternalPortConfigurationPortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3ExternalPortConfigurationPortDSDesiredState
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         External Port Configuration Port Data Set containing the
         single member desiredState, which indicates the desired state
         for the PTP Port."
    ::= { ieee8021AsV3Groups 14 }

ieee8021AsV3AsymMeasurementModeDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3AsymMeasurementModeDSAsymMeasurementMode
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Asymmetry Measurement Mode Data Set representing the capability
         to enable/disable the Asymmetry Compensation Measurement
```


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```
         Procedure on a Link Port (see Annex G)."
    ::= { ieee8021AsV3Groups 15 }

ieee8021AsV3CommonServicesPortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CommonServicesPortDSCmldsLinkPortPortNumber
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Common Services Port Data Set."
    ::= { ieee8021AsV3Groups 16 }

ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CmldsDefaultDSClockIdentity,
        ieee8021AsV3CmldsDefaultDSNumberLinkPorts
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         CMLDs Default Data Set describing the per-time-aware-system
         attributes of the Common Mean Link Delay Service."
    ::= { ieee8021AsV3Groups 17 }

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CmldsLinkPortDSClockIdentity,
        ieee8021AsV3CmldsLinkPortDSPortNumber,
        ieee8021AsV3CmldsLinkPortDSCmldsLinkPortEnabled,
        ieee8021AsV3CmldsLinkPortDSIsMeasuringDelay,
        ieee8021AsV3CmldsLinkPortDSAsCapableAcrossDomains,
        ieee8021AsV3CmldsLinkPortDSMeanLinkDelay,
        ieee8021AsV3CmldsLinkPortDSMeanLinkDelayThresh,
        ieee8021AsV3CmldsLinkPortDSDelayAsym,
        ieee8021AsV3CmldsLinkPortDSNbrRateRatio,
        ieee8021AsV3CmldsLinkPortDSInitialLogPdelayReqInterval,
        ieee8021AsV3CmldsLinkPortDSCurrentLogPdelayReqInterval,
        ieee8021AsV3CmldsLinkPortDSUseMgtSettableLogPdelayReqInterval,
        ieee8021AsV3CmldsLinkPortDSMgtSettableLogPdelayReqInterval,
        ieee8021AsV3CmldsLinkPortDSInitialComputeNbrRateRatio,
        ieee8021AsV3CmldsLinkPortDSCurrentComputeNbrRateRatio,
        ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeNbrRateRatio,
        ieee8021AsV3CmldsLinkPortDSMgtSettableComputeNbrRateRatio,
        ieee8021AsV3CmldsLinkPortDSInitialComputeMeanLinkDelay,
        ieee8021AsV3CmldsLinkPortDSCurrentComputeMeanLinkDelay,
        ieee8021AsV3CmldsLinkPortDSUseMgtSettableComputeMeanLinkDelay,
        ieee8021AsV3CmldsLinkPortDSMgtSettableComputeMeanLinkDelay,
        ieee8021AsV3CmldsLinkPortDSAllowedLostRsp,
        ieee8021AsV3CmldsLinkPortDSAllowedFaults,
        ieee8021AsV3CmldsLinkPortDSVersionNumber,
        ieee8021AsV3CmldsLinkPortDSPdelayTruncTST1,
        ieee8021AsV3CmldsLinkPortDSPdelayTruncTST2,
        ieee8021AsV3CmldsLinkPortDSPdelayTruncTST3,
        ieee8021AsV3CmldsLinkPortDSPdelayTruncTST4,
        ieee8021AsV3CmldsLinkPortDSMinorVersionNumber
    }
    STATUS      current
    DESCRIPTION
```


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```
        "A collection of objects providing information for every
         Link Port of the Common Mean Link Delay Service of a
         time-aware system."
    ::= { ieee8021AsV3Groups 18 }

ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CmldsLinkPortStatDSRxPdelayRequestCount,
        ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspCount,
        ieee8021AsV3CmldsLinkPortStatDSRxPdelayRspFollowUpCount,
        ieee8021AsV3CmldsLinkPortStatDSRxPtpPacketDiscardCount,
        ieee8021AsV3CmldsLinkPortStatDSPdelayAllowedLostRspExceededCount,
        ieee8021AsV3CmldsLinkPortStatDSTxPdelayRequestCount,
        ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspCount,
        ieee8021AsV3CmldsLinkPortStatDSTxPdelayRspFollowUpCount
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information for every
         Link Port Statistics of the Common Mean Link Delay Service of a
         time-aware system."
    ::= { ieee8021AsV3Groups 19 }

ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSGroup OBJECT-GROUP
    OBJECTS {
        ieee8021AsV3CmldsAsymMeasurementModeDSAsymMeasurementMode
    }
    STATUS      current
    DESCRIPTION
        "A collection of objects providing information on the
         Common Mean Link Delay Service Asymmetry Measurement Mode
         Parameter Data Set representing the capability to enable/disable
         the Asymmetry Compensation Measurement Procedure on a Link Port
         (see Annex G)."
    ::= { ieee8021AsV3Groups 20 }

-- ================================================================
-- compliance statements
-- ================================================================

ieee8021AsV3Compliance MODULE-COMPLIANCE
    STATUS      current
    DESCRIPTION
        "The compliance statement for devices supporting
         IEEE Std 802.1AS-2020."

    MODULE -- this module

    GROUP ieee8021AsV3PtpInstanceGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3DefaultDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CurrentDSGroup
    DESCRIPTION
        "Implementation of this group is optional."
```


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```
    GROUP ieee8021AsV3ParentDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3TimePropertiesDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3PathTraceDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3PathTraceDSArrayTableGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3AcceptableTimeTransmitterTableDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3AcceptableTimeTransmitterTableDSArrayGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3PortDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3DescriptionPortDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3PortStatIfGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3AcceptableTimeTransmitterPortDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3ExternalPortConfigurationPortDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3AsymMeasurementModeDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CommonServicesPortDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CommonMeanLinkDelayServiceDefaultDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CommonMeanLinkDelayServiceLinkPortDSGroup
    DESCRIPTION
```


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```
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CommonMeanLinkDelayServiceLinkPortStatDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    GROUP ieee8021AsV3CommonMeanLinkDelayServiceAsymMeasurementModeDSGroup
    DESCRIPTION
        "Implementation of this group is optional."

    ::= { ieee8021AsV3Compliances 1 }

END

```


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16. Media-dependent layer specification for CSN

16.1 Overview

Accurate synchronized time is distributed throughout a gPTP domain through time measurements between adjacent PTP Relay Instances or PTP End Instances in a packet network. Time is communicated from the root of the synchronization tree (i.e., the Grandmaster PTP Instance) toward the leaves of the synchronization tree (i.e., from leaf-facing "timeTransmitter" ports to root-facing "timeReceiver" ports) through measurements made across the links connecting the PTP Instances. While the semantics of time transfer are consistent across the time-aware packet network, the method for communicating synchronized time from a timeTransmitter port to its immediate downstream link partner varies depending on the type of link interconnecting the two PTP Instances.

This clause specifies the protocol that provides accurate synchronized time across links of a *coordinated shared network* (CSN) as part of a packet network.

16.2 Coordinated shared network characteristics

A CSN is a contention-free, time-division, multiplexed-access network of devices sharing a common medium and supporting reserved bandwidth based on priority or flow (QoS). One of the nodes of the CSN acts as the network coordinator, granting transmission opportunities to the other CSN nodes of the network. A CSN physically is a shared medium, in that a CSN node has a single physical port connected to the half-duplex medium, but is logically a fully connected one-hop mesh network, in that every CSN node can transmit frames to every other CSN node over the shared medium.

A CSN supports two types of transmission: unicast transmission for point-to-point (CSN node-to-node) transmission and multicast/broadcast transmission for point-to-multipoint (CSN node-to-other/all-nodes) transmission. Figure 16-1 illustrates a CSN acting as a backbone for PTP Instances.

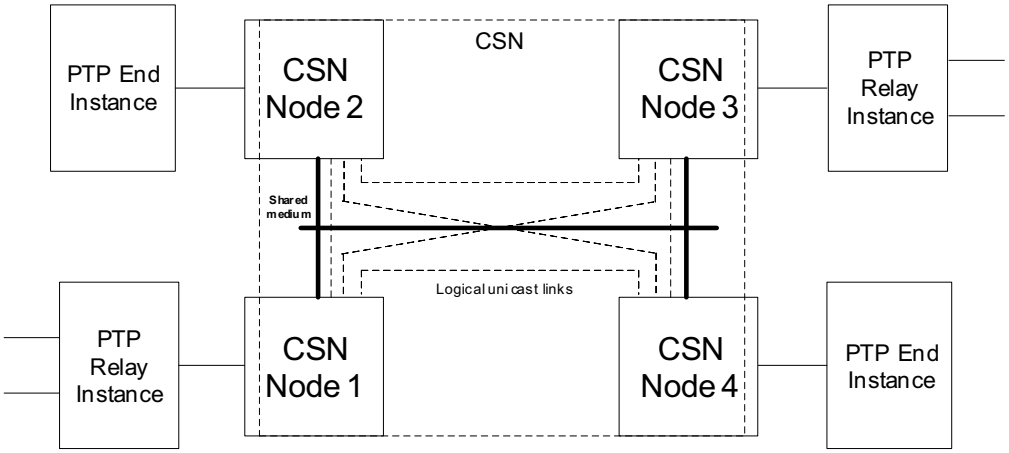


Figure 16-1—Example of CSN backbone

NOTE—In this clause, the term *node* is used to refer to a CSN node (i.e., it does not refer to a PTP Relay Instance or PTP End Instance). A CSN node is a 2-port PTP Relay Instance that forwards data packets between a segment external to the CSN (which can connect to an upstream or downstream PTP Instance) and the CSN, all at the data link layer. Nonetheless, to avoid confusion the term *node* is usually preceded by *CSN*, except when it is obvious that CSN nodes are being referenced.

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16.3 Layering for CSN links

One PortSync entity and one MD entity are together associated with each CSN logical port (CSN node-to-node link) as illustrated in Figure 16-2. The PortSync entitiy is described in 10.1.2. The MD entity translates media-independent primitives to MD primitives as necessary for communicating synchronized time over the CSN links. The CSN MD entity shall implement the MDSyncSendSM and MDSyncReceiveSM states machines of 11.2.13 and 11.2.14.

The CSN MD entity either implements the MDPdelayReq and MDPdelayResp state machines of 11.2.18 and 11.2.19 to measure the propagation delay on a CSN link, or measures it through a CSN-native method and populates the variables described in 16.4.3.3.

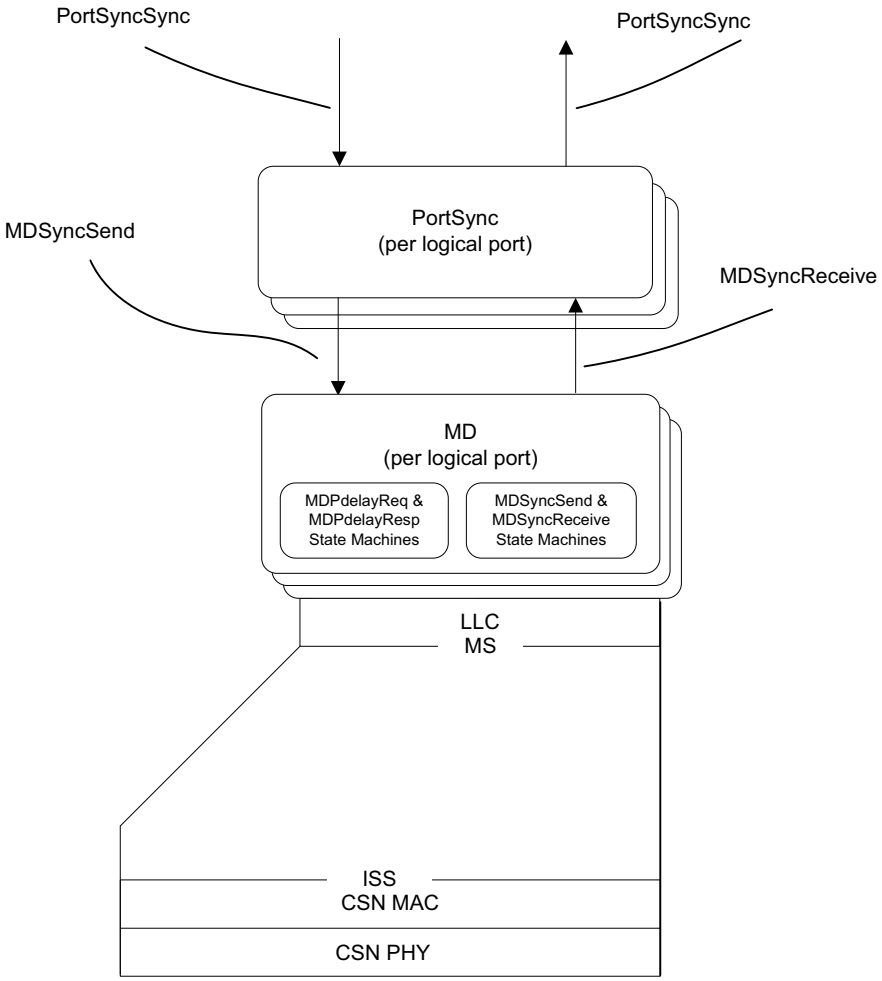


Figure 16-2—Media-dependent and lower entities in CSN nodes

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16.4 Path delay measurement over a CSN backbone

16.4.1 General

The path delay over a CSN backbone is calculated for the following path types:

a) Between the upstream PTP Relay Instance and the ingress CSN node
b) Between the ingress and egress CSN nodes
c) Between the egress CSN node and the downstream PTP Instance (PTP Relay Instance or PTP End Instance)

To maintain the synchronization, residence time on each PTP Instance and the propagation delay between PTP Instances is measured, requiring precise timestamping on both CSN node ingress and egress ports as illustrated in Figure 16-4 (the numbered paths in Figure 16-4 refer to the numbered paths in Figure 16-3). In Figure 16-4, ti_1 is the syncEventEgressTimestamp for the Sync message at the upstream PTP Relay Instance, ti_2 is the syncEventIngressTimestamp for the Sync message at the ingress CSN time-aware node, te_1 is the syncEventEgressTimestamp for the Sync message at the egress CSN time-aware node, and te_2 is the syncEventIngressTimestamp for the Sync message at the downstream PTP Relay Instance or PTP End Instance.

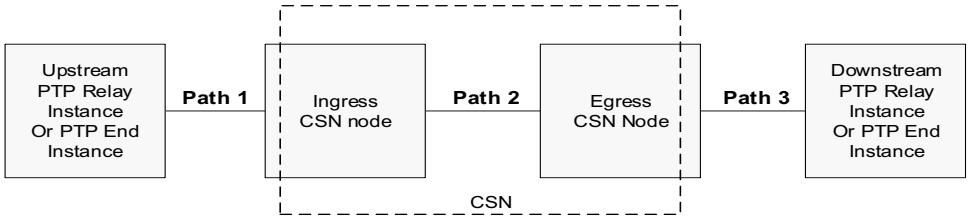


Figure 16-3—Path types over CSN as IEEE 802.1AS backbone

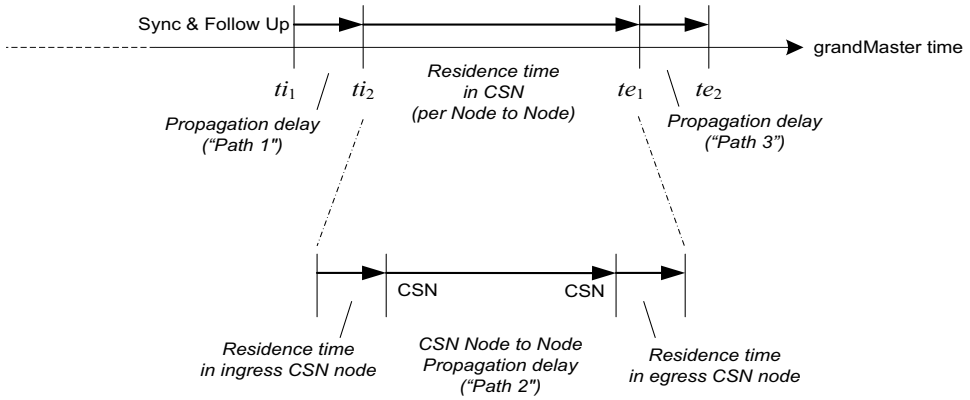


Figure 16-4—Propagation delay and residence time over a CSN backbone

16.4.2 Path delay measurement between CSN node and neighbor PTP Instance

The path delay measurement between a CSN node and a neighbor PTP Instance is made as specified for the respective medium. This path delay measurement is made for the link between the CSN node and the neighbor PTP Instance.

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16.4.3 Path delay measurement between CSN nodes

16.4.3.1 General

The path delay between the two nodes of a CSN is the propagation delay for the logical link that connects those two nodes. The method of measuring the path delay between two CSN nodes has two variations, which are described in 16.4.3.2 and 16.4.3.3. The specific method to be used for a specific link technology is specified in 16.6.

16.4.3.2 Path delay measurement without network clock reference

Each CSN node has a free-running local clock. The path delay measurement uses the peer-to-peer delay mechanism protocol, messages, and state machines described in Clause 11 for FDE links, as illustrated by Figure 16-5. The criteria of 11.2.16 for determining whether the peer-to-peer delay mechanism is transport-specific peer-to-peer delay mechanism or the CMLDS apply here.

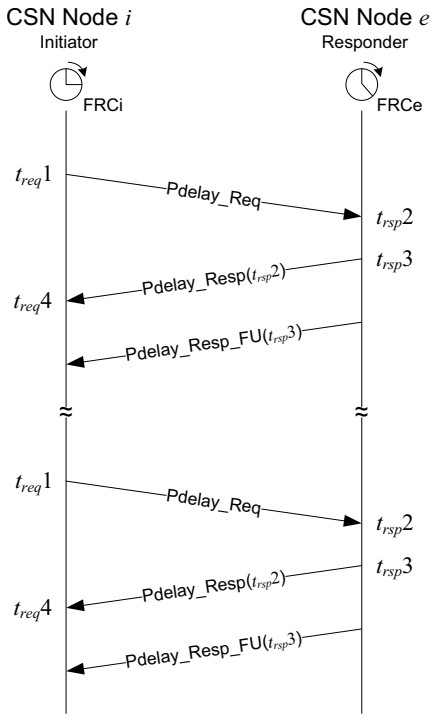


Figure 16-5—CSN node-to-node path delay measurement

The computation of the neighborRateRatio and meanLinkDelay between two CSN nodes is done using the timestamps at the initiator and information conveyed in the successive Pdelay_Resp and Pdelay_Resp_Follow_Up messages. Any scheme that uses this information is acceptable, as long as the performance requirements of B.2.4 are met. As one example, the neighborRateRatio is computed as the ratio between a time interval measured by the local clock of the responder and its associated time interval measured by the local clock of the initiator, using a set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages and a second set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages some number of Pdelay_Req message transmission intervals later, i.e., as shown in Equation (16-1).

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$$\frac{(t_{rsp}3)_N - (t_{rsp}3)_0}{(t_{req}4)_N - (t_{req}4)_0} \quad (16\text{-}1)$$

where

$(t_{rsp}3)_k$ is the time relative to the local clock of the responder that the kth Pdelay_Resp message is sent
$(t_{req}4)_k$ is the time relative to the local clock of the initiator that the kth Pdelay_Resp message is received
N is the number of Pdelay_Req message transmission intervals separating the first set of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages and the second set

The successive sets of received Pdelay_Resp and Pdelay_Resp_Follow_Up messages are indexed from 0 to N with the first set indexed 0. The meanLinkDelay between the PTP Instance and the CSN node is computed as shown in Equation (16-2).

$$\frac{(t_{req}4 - t_{req}1)r - (t_{rsp}3 - t_{rsp}2)}{2} \quad (16\text{-}2)$$

where

r is equal to neighborRateRatio
$t_{req}1$ is the time relative to the local clock of the initiator that the Pdelay_Req message for this message exchange is sent
$t_{rsp}2$ is the time relative to the local clock of the responder that the Pdelay_Req message for this message exchange is received
$t_{rsp}3$ is the time relative to the local clock of the responder that the Pdelay_Resp message for this message exchange is sent
$t_{req}4$ is the time relative to the local clock of the initiator that the Pdelay_Resp message for this message exchange is received

NOTE—The difference between mean propagation delay relative to the Grandmaster Clock time base and relative to the time base of the CSN node at the other end of the attached link (i.e., the responder CSN node) is usually negligible. To see this, note that the former can be obtained from the latter by multiplying the latter by the ratio of the Grandmaster Clock frequency to the frequency of the LocalClock entity of the CSN node at the other end of the link. This ratio differs from 1 by 200 ppm or less. For example, for a worst-case frequency offset of the LocalClock entity of the CSN node at the other end of the link, relative to the Grandmaster Clock, of 200 ppm, and a measured propagation time of 100 ns, the difference in mean propagation delay relative to the two time bases is 20 ps.

Although the propagation delay between two CSN nodes is constant, a Pdelay_Req message is still sent periodically by each CSN node to each other active CSN node to measure the neighborRateRatio between the node and each other node. Each CSN node shall implement the state machines described in 11.2.18 and 11.2.19.

16.4.3.3 Native CSN path delay measurement

Some CSN technologies feature a native mechanism that provides a path delay measurement with accuracy similar to the accuracy the peer delay protocol provides. For these CSNs, the path delay can be provided using the native measurement method rather than using the Pdelay protocol defined in 11.2.18 and 11.2.19. Such a situation is described in more detail as follows. The CSN MD entity populates the following per-PTP Port and MD-entity global variables (described respectively in 10.2.5 and 11.2.12) as indicated:

— asCapable (10.2.5.1) is set to TRUE.
— neighborRateRatio (10.2.5.7) is set to the value provided by the native CSN measurement.
— meanLinkDelay (10.2.5.8) is set to the value provided by the native CSN measurement.
— computeNeighborRateRatio (10.2.5.10) is set to FALSE.

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— computeMeanLinkDelay (10.2.5.11) is set to FALSE.
— isMeasuringDelay (11.2.12.6) is set to TRUE to indicate that the CSN MD entity is measuring path delay (in this case, using its internal mechanism).
— domainNumber (8.1) is set to the domainNumber of this gPTP domain.

16.4.3.4 Intrinsic CSN path delay measurement

If the CSN features a native mechanism that causes each CSN node's local clock to be fully synchronized to the local clocks of other nodes of the CSN such that the synchronized CSN time complies with the requirements specified in B.1, the CSN nodes need not implement the path delay mechanism but rather can treat the path delay as part of the residence time of the distributed system. The propagation of the Sync messages in this case is described in 16.5.3.

16.5 Synchronization messages

16.5.1 General

The CSN shall propagate synchronized time over the CSN to CSN end stations and to downstream non-CSN links, using Sync (and, if the message is two-step, the associated Follow_Up) messages, as illustrated in Figure 16-6.

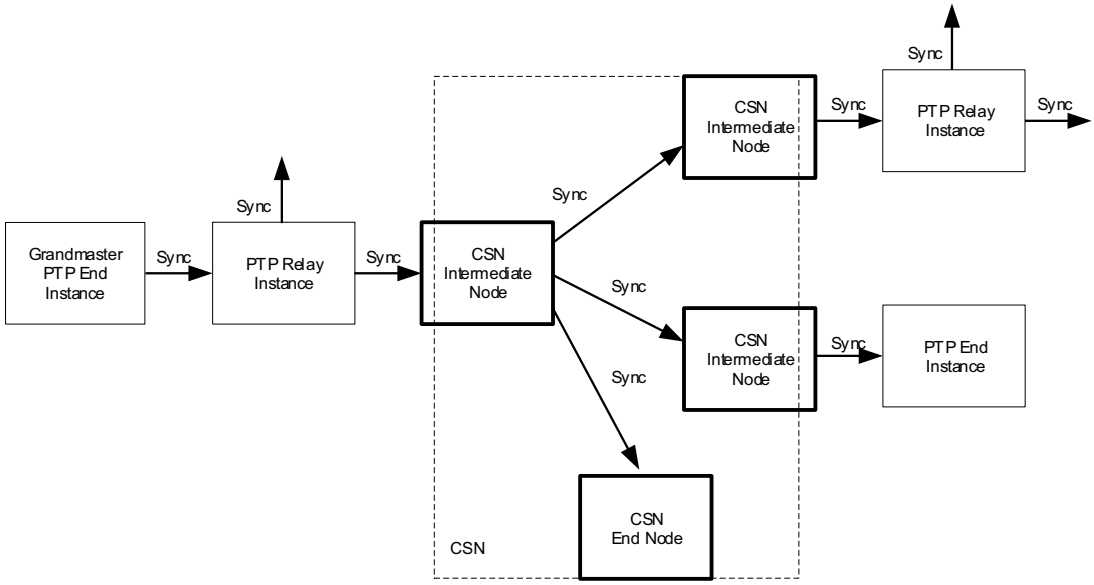


Figure 16-6—IEEE 802.1AS Sync message propagation over the CSN backbone

Once the path delays have been measured (a) between the upstream PTP Relay Instance or PTP End Instance and the ingress CSN node, (b) between the CSN nodes, and (c) between the egress CSN node and the downstream PTP Relay Instance or PTP End Instance, the CSN backbone can propagate the synchronization information received at its boundary nodes.

As with path delay measurements, various CSN technologies choose various methods for propagating time. These methods are described in 16.5.2 and 16.5.3.

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16.5.2 Synchronization message propagation on CSN without network reference clock

If the CSN does not feature a native mechanism that synchronizes the CSN node local clocks to each other or to a reference (i.e., to synchronize the CSN time in compliance with the requirements specified in B.1), the CSN local clocks at CSN ingress and CSN egress nodes are considered independent, free-running clocks.

In this case, synchronization over the CSN links uses the Sync and Follow_Up protocol (or only Sync if optional one-step processing is used), messages, and state machines specified for FDE links in 10.2.8, 10.2.12, 11.2.13, and 11.2.14. Individual CSN link technologies may specify media-dependent encapsulation of gPTP event messages. See Table 16-2 for selection of options per link technology.

One PortSync and one MD entity is instantiated per logical port (i.e., per CSN link). A CSN node behaves equivalently to a PTP Instance. Sync and, in the two-step case, Follow_Up messages are either transmitted using unicast on each link or broadcasted. However, if Sync and Follow_Up messages are broadcasted, the following apply:

— The Sync and Follow_Up messages are broadcast with the same port number used to broadcast Announce messages.
— All PortSync/MD entity pairs except one set their logSyncInterval attribute (see 10.7.2.3) to 127 so that they do not generate any Sync messages.
— A dynamic selection of the MD entity that broadcasts the Sync message is needed (a CSN node can dynamically leave the CSN). The dynamic selection algorithm is implementation specific and out of the scope of this standard.

16.5.3 Synchronization message propagation on a CSN with network reference clock

16.5.3.1 General

If the CSN features a native mechanism that allows the CSN node's local clocks to be fully synchronized to each other in a way that complies with the requirements specified in B.1, it is possible to simplify the path delay mechanism, as described below. This method is an alternative to 16.5.2. Individual CSN link technologies may specify media-dependent encapsulation of gPTP event messages. See Table 16-2 (in 16.6.1) for selection of options per link technology.

Sync messages are timestamped when they are:

— Received at the ingress CSN node's PTP Port (syncEventIngressTimestamp); and
— Transmitted at the egress CSN node's PTP Port (syncEventEgressTimestamp).

The elapsed time between the egress and ingress timestamps is computed as the CSN residence time. In this scheme, the Sync message handling is split between the MDSyncSendSM state machine in the CSN ingress node and the MDSyncReceiveSM state machine in the CSN egress node as described in 16.5.3.2 and 16.5.3.3.

The reference plane for a CSN port and the path delay measurement method are specific to the type of CSN technology, and are defined in Table 16-2.

16.5.3.2 CSN ingress node

16.5.3.2.1 General

The CSN ingress node timestamps Sync messages received from the upstream PTP Instance and computes the upstreamTxTime as described in item i) of 11.2.13.2.1.

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In addition, the setFollowUp function of the MDSyncSendSM state machine [see item a) in 11.2.14.2.3] is modified as follows:

a) The quantity *rateRatio* × *(syncEventEgressTimestamp* – *upstreamTxTime)* is not added to the followUpCorrectionField of the Follow_Up message (or the Sync message in the one-step case) to be transmitted by the ingress to the egress CSN node.

b) The CSN TLV (see 16.5.3.2.2) is appended to the Follow_Up message (or to the Sync message in the one-step case) transmitted by the ingress to the egress CSN node.

16.5.3.2.2 CSN TLV

16.5.3.2.2.1 General

The fields of the CSN TLV are specified in Table 16-1 and 16.5.3.2.2.2 through 16.5.3.2.2.10. This TLV is a standard organization extension TLV for the Follow_Up message (or the Sync message in the one-step case), as specified in 14.3 of IEEE Std 1588-2019. This TLV is not allowed to occur before the Follow_Up information TLV (see 11.4.4.3).

Table 16-1—CSN TLV

Bits								Octets	Offset from start of TLV
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
organizationId								3	4
organizationSubType								3	7
upstreamTXTime								12	10
neighborRateRatio								4	22
meanLinkDelay								12	26
delayAsymmetry								12	38
domainNumber								1	50

16.5.3.2.2.2 tlvType (Enumeration16)

The value of the tlvType field is 0x3.

NOTE—This value indicates the TLV is a vendor and standard organization extension TLV, as specified in 14.3.2.1 and Table 50 of IEEE Std 1588-2019. The tlvType is specified in that standard as ORGANIZATION_EXTENSION with a value of 0x3.

16.5.3.2.2.3 lengthField (UInteger16)

The value of the length is 47.

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16.5.3.2.2.4 organizationId (Octet3)

The value of organizationId is 00-80-C2.

16.5.3.2.2.5 organizationSubType (Enumeration24)

The value of organizationSubType is 3.

16.5.3.2.2.6 upstreamTxTime (UScaledNs)

The computed upstreamTxTime value is described in item i) of 11.2.13.2.1.

16.5.3.2.2.7 neighborRateRatio (Integer32)

The neighborRateRatio value is described in 10.2.5.7.

16.5.3.2.2.8 meanLinkDelay (UScaledNs)

The meanLinkDelay value is described in 10.2.5.8.

16.5.3.2.2.9 delayAsymmetry (ScaledNs)

The delayAsymmetry value is described in 10.2.5.9.

16.5.3.2.2.10 domainNumber (UInteger8)

This parameter is the domainNumber of this gPTP domain.

16.5.3.3 CSN egress node

The CSN egress port sets neighborRateRatio to 1 and meanLinkDelay to 0 for its CSN port.

The CSN egress port modifies the function setMDSyncReceiveMDSR of the MDSyncReceiveSM state machine [item i) in 11.2.13.2.1] for the port attached to the CSN link entity to extract upstreamTxTime, meanLinkDelay, and neighborRateRatio from the respective fields of the CSN TLV in the Follow_Up message (or Sync message in the one-step case) received from the CSN ingress node.

The CSN egress port also modifies the ClockTimeReceiverSync state machine (see 10.2.13) to get the upstreamTxTime, meanLinkDelay, neighborRateRatio, and delayAsymmetry values from the respective fields of the CSN TLV in the Follow_Up message (or Sync message in the one-step case) received from the CSN ingress node.

The CSN egress node removes the CSN TLV from the Follow_Up message (or Sync message in the one-step case) received from the CSN ingress node and transmits the message to the downstream PTP Instance.

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16.6 Specific CSN requirements

16.6.1 General

The reference plane for a CSN port is specific to the type of CSN technology and is defined in Table 16-2.

Table 16-2—Definitions and option selections per link technology

CSN link technology	Reference plane	Path delay measurement	gPTP event message encapsulation
Multimedia over Coax Alliance (MoCA) v2.0	The first bit of an Event message crossing to and from the MoCA CTC domain.	MoCA Ranging Protocol (method of either 16.4.3.3 or 16.4.3.4).	Encapsulated in control frames as described in the MoCA MAC/PHY Specification v2.0.
ITU-T G.hn (SG15)	The first bit of an Event message crossing the A-interface (see 16.6.3).	Peer-to-peer mechanism as defined in 16.4.3.2.	None.

16.6.2 MoCA-specific behavior

The MoCA network is a CSN. The non-MoCA port of a CSN time-aware node behaves as a PTP Port, which might or might not use the MoCA channel time clock (CTC) for timestamping event messages. If the non-MoCA port of the CSN time-aware node uses a different clock than the MoCA CTC, then the CSN time-aware node reconciles the non-CTC timestamp with the CTC time.

Sync messages shall be timestamped using the CTC when they cross:

— The MoCA ingress node's timestamp reference plane (syncEventIngressTimestamp); and
— The egress CSN node's reference plane (syncEventEgressTimestamp).

The elapsed time between the egress and ingress timestamps, syncEventIngressTimestamp – syncEventIngressTimestamp, is computed as the MoCA residence time.

The MoCA port whose PTP Port state is TimeTransmitterPort propagates the Sync and Follow_Up messages (or only the Sync message in the one-step case) as described in 16.5.3. The CSN TLV values of the Follow_Up message sent over the MoCA network are computed using the LocalClock, i.e., the MoCA CTC.

IEEE 802.1AS frames shall be transmitted over the MoCA network as MoCA control frames as described in the MoCA MAC/PHY Specification v2.0.

NOTE—The CTC is specified in the MoCA MAC/PHY Specification v2.0.

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16.6.3 ITU-T G.hn-specific behavior

A port of a PTP Instance that includes one or more ITU-T G.hn ports shall behave as defined in 16.3, 16.4, and 16.5; however, for aspects where more than one behavior or option is described, the system behaves as defined in Table 16-2.

ITU-T G.hn defines a 32-bit timestamp (which is placed in the TSMP field of an ITU-T G.hn encapsulation header). This timestamp, as described in Table 16-2, is captured each time an Event message is transmitted or received by an ITU-T G.hn port and is used for all gPTP event messages, including SYNC and PDELAY messages.

16.7 Grandmaster PTP Instance capability

Each CSN Node may be grandmaster capable to allow a CSN node to act as the Grandmaster PTP Instance either for a homogeneous CSN or for a heterogeneous network.

The Announce messages are either sent via unicast on each link or broadcasted. However, if Announce messages are broadcasted, the Announce message shall use the same port number as used by the Sync and Follow_Up messages (or only the Sync message in the one-step case) by a single PortSync/MD entity pair on the PTP Port. This is accomplished by setting the logAnnounceInterval attribute (see 10.7.2.2) to 127 for all but one PortSync/MD entity pair so that they do not generate any Announce messages.

16.8 CSN clock and node performance requirements

The CSN clock performance complies with the requirements specified in B.1. The CSN node performance complies with the requirements specified in B.2.4 and should follow the recommendations of B.2.2 and B.2.3.

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17. YANG data model

YANG (IETF RFC 7950) is a data modeling language used to model configuration data and state data for remote network management protocols. YANG-based remote network management protocols include the Network Configuration Protocol (NETCONF) (IETF RFC 6241 [B18]) and RESTCONF (IETF RFC 8040 [B21]). Each remote network management protocol uses a specific encoding on the wire, such as XML or JSON. A YANG module specifies the organization and rules for the management data, and a mapping from YANG to the specific encoding enables the data to be understood correctly by both client (e.g., network manager) and server (e.g., PTP Instances).

This clause specifies the YANG data model for IEEE Std 802.1AS.

This clause:

a) Introduces the organization of the data models, including the relationship with other standards (17.1)
b) Provides an overview of the hierarchy of the data models using a representation similar to the Unified Modeling Language (UML) (17.2)
c) Summarizes the structure of the YANG data model (17.3)
d) Reviews security considerations (17.4)
e) Provides schema trees as overviews of the YANG modules (17.5)
f) Specifies the YANG modules ()

17.1 YANG framework

The YANG framework applies hierarchy in the following areas:

a) The Uniform Resource Name (URN), as specified in IEEE Std 802.
b) The YANG objects form a hierarchy of configuration and operational data structures that define the YANG model.

Clause 14 specifies the information model for management of this standard. The data model for a specific management mechanism is derived from the information model. Since YANG-based protocols are an example of a management mechanism, the YANG data model of this clause is derived from Clause 14.

NOTE—The MIB modules specified in Clause 15 were also derived from Clause 14. Consequently, the capabilities and structure of the YANG data models are aligned with those represented by the MIB. However the YANG data model has not been derived from the MIB, and there has been no attempt to include data or modeling constructs that might appear in the MIB but not in the information model.

The information model in Clause 14 is organized as a hierarchy of data sets. Each data set contains one or more related members (items of data that can be read or written). In the context of YANG, each data set is represented as a YANG “container,” and each member is represented as a YANG “leaf.”

17.1.1 Relationship to the IEEE Std 1588 data model

The YANG data models specified in this standard are based on, and augment, those specified in IEEE Std 1588. In particular the ieee802-dot1as-gptp.yang module imports the ieee1588-ptp-tt module as a whole, augmenting that module as necessary to meet the requirements of this standard. The ieee802-dot1as-hs.yang module imports the ieee1588-ptp-tt and ieee802-dot1as-gptp modules as a whole, augmenting those modules as necessary to meet the requirements of this standard. The ieee802dot1as-ftt.yang module imports the ieee1588-tt and ieee802-dot1as-gptp modules as a whole, augmenting those modules as necessary to meet the requirements of this standard.

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Some of the data sets in Clause 14 (e.g., defaultDS) are derived from IEEE Std 1588, and some of the data sets are unique to IEEE Std 802.1AS (i.e., not derived from IEEE Std 1588). For each data set in Clause 14 that is derived from IEEE Std 1588, a portion of the members are derived from IEEE Std 1588, and the remaining members are unique to IEEE Std 802.1AS. For the members that are derived from IEEE Std 1588, the specifications in both standards are analogous (same name, data type, semantics, etc.).

The YANG data model for IEEE Std 1588-2019 is published as amendment IEEE Std 1588e-2024. The YANG module of IEEE Std 1588e (ieee1588-ptp-tt.yang) contains the hierarchy (tree) of data sets and their members.

The YANG modules of this clause (ieee802-dot1as-gptp.yang, ieee802-dot1as-hs.yang, and ieee802-dot1as-ftt.yang) use the YANG "import" statement to import the YANG module of IEEE Std 1588e. This effectively uses the IEEE Std 1588 YANG tree as the foundation of the IEEE Std 802.1AS YANG tree. By importing the tree and its data set containers, all members from Clause 14 that are derived from IEEE Std 1588 are also imported.

The core of the YANG module for IEEE Std 802.1AS consists of YANG "augment" statements, used to add members to the tree that are unique for IEEE Std 802.1AS.

NOTE— IETF RFC 8575 [B24] is the standard YANG data model for IEEE Std 1588-2008. The YANG data model of IEEE Std 1588e is effectively a newer version of RFC 8575. Therefore, the YANG module of RFC 8575 is not imported by the YANG module of this clause.

17.2 IEEE 802.1AS YANG data model

This clause uses a UML-like representation to provide an overview of the hierarchy of the IEEE Std 802.1AS YANG data model.

A representation of the management model is provided in Figure 17-1 through Figure 17-4. The purpose of the diagrams is to express the model design in a concise manner. The structure of the representation shows the name of the object followed by a list of properties for the object. The properties indicate their type and accessibility. The representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype used to encode the object in either MIB or YANG. In the representation, a box with a white background represents information that comes from sources outside of this IEEE standard. A box with a gray background represents objects that are defined by this IEEE standard.

NOTE 1—OMG® UML 2.5[23] [B34] conventions together with C++ language constructs are used in this clause as a representation to convey model structure and relationships.

NOTE 2—This standard specifies YANG for Clause 14 of this standard. There are optional features in the YANG module of IEEE Std 1588 that are not specified in Clause 14, and are therefore not shown in the figures of this subclause. If optional IEEE Std 1588 YANG features are implemented, conformance is specified by IEEE Std 1588.

For all figures, Clause 14 data that is imported from the ieee1588-ptp-tt.yang module is shown in white, and Clause 14 data in augments of ieee802-dot1as-gptp.yang is shown in gray.

Figure 17-1 provides an overview of the IEEE Std 802.1AS YANG tree. The top level instance list provides the list of one or more PTP Instances, each with data sets. For each PTP Instance, port-ds-list provides the list of one or more PTP Ports, each with data sets. The common-services apply to all PTP Instances, including the CMLDS.

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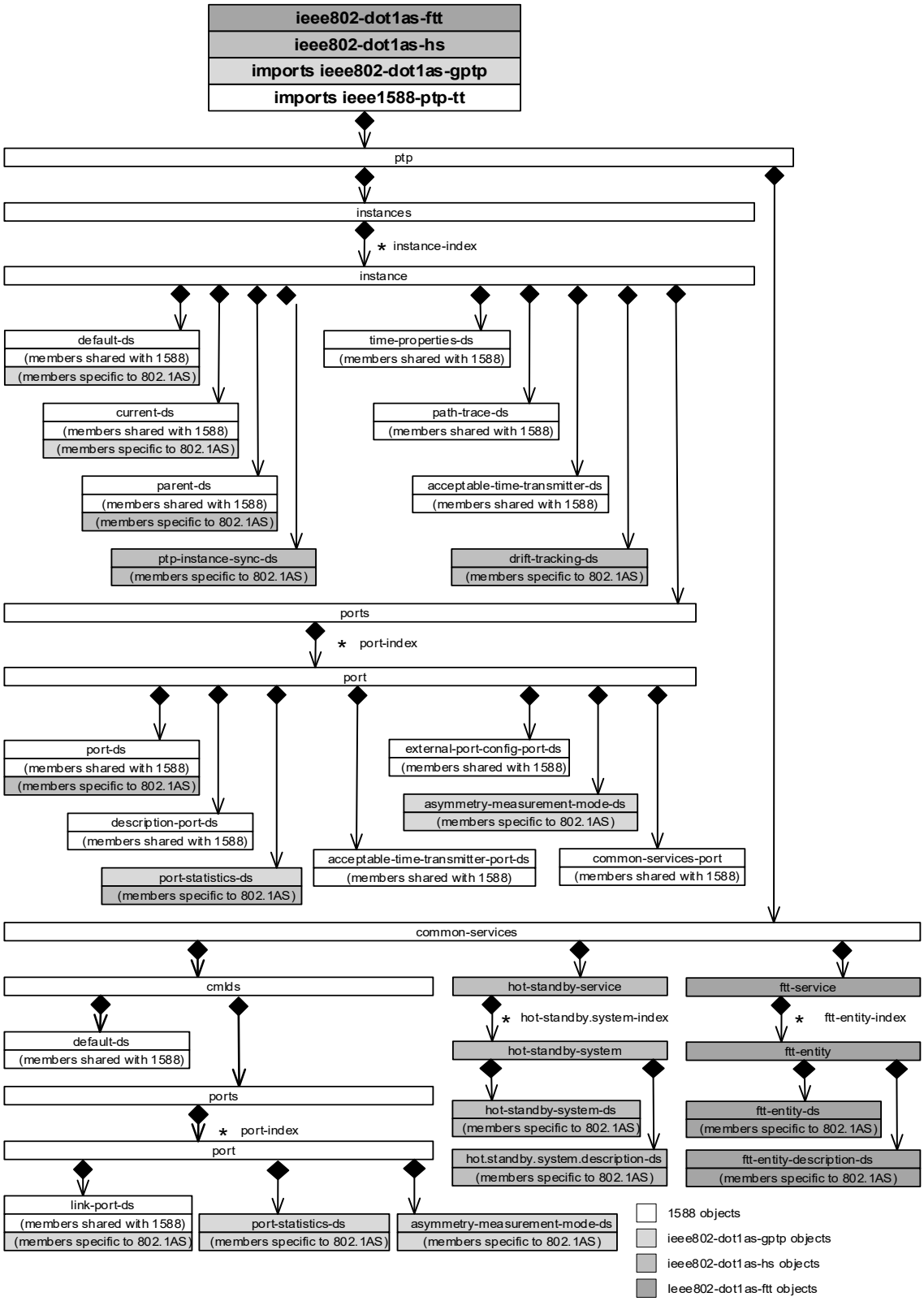


Figure 17-1—Overview of YANG tree

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Figure 17-2 provides detail for the data sets of each PTP Instance, including each data set member.

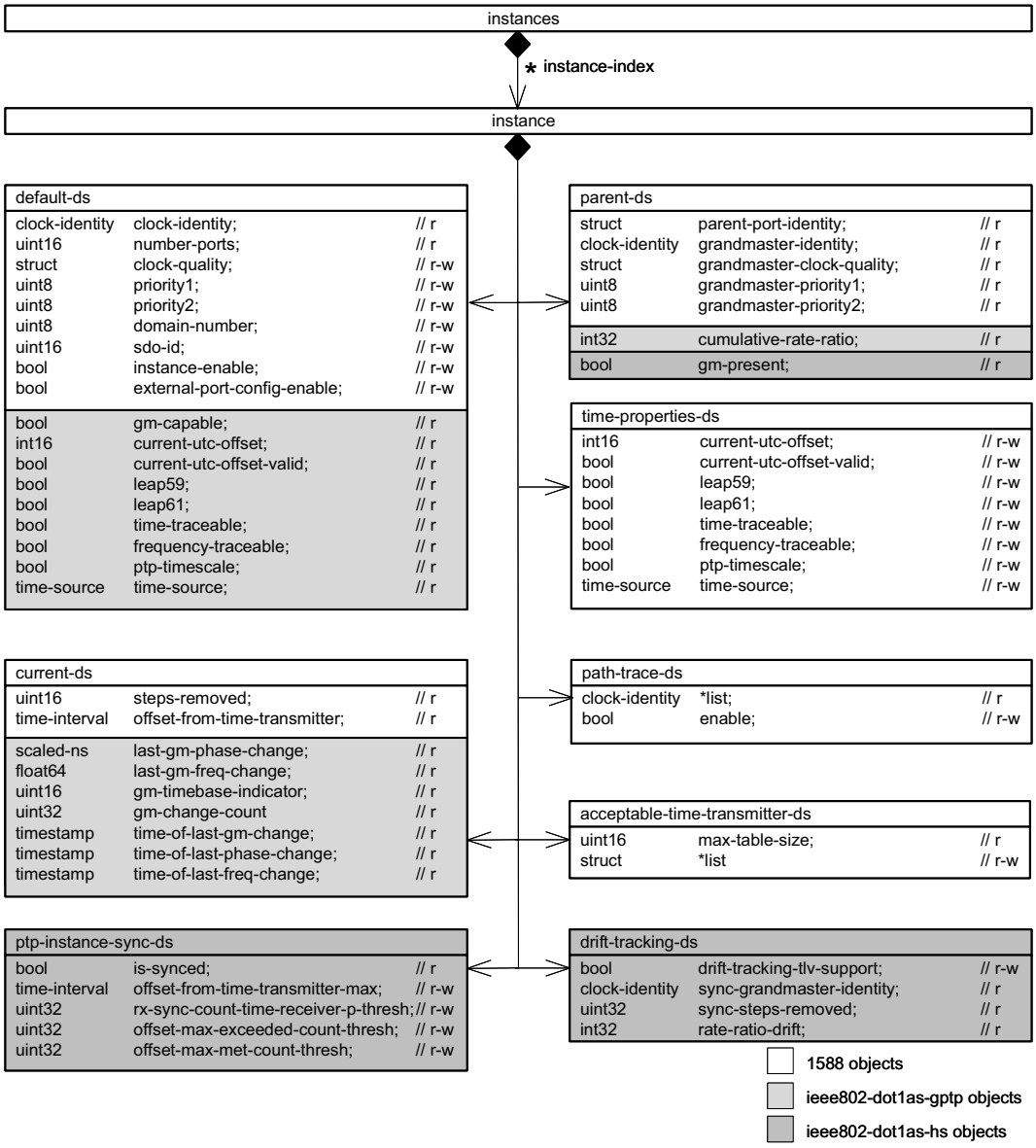


Figure 17-2—PTP Instance detail

Figure 17-3 provides detail for the data sets of each PTP Port, including each data set member.

NOTE 3—Subclause 14.10.4 specifies ptpPortEnabled (ptp-port-enabled), which is provided in YANG as the semantically equivalent node in ieee1588-ptp-tt named port-enable (in port-ds of Figure 17-3). Subclause 14.10.16 specifies mgtSettableLogAnnounceInterval (mgt-settable-log-announce-interval), which is provided in YANG as the semantically equivalent node in ieee1588-ptp-tt named log-announce-interval (in port-ds of Figure 17-3). Subclause 14.10.21 specifies mgtSettableLogSyncInterval (mgt-settable-log-sync-interval), which is provided in YANG as the semantically equivalent node in ieee1588-ptp-tt named log-sync-interval (in port-ds of Figure 17-3).

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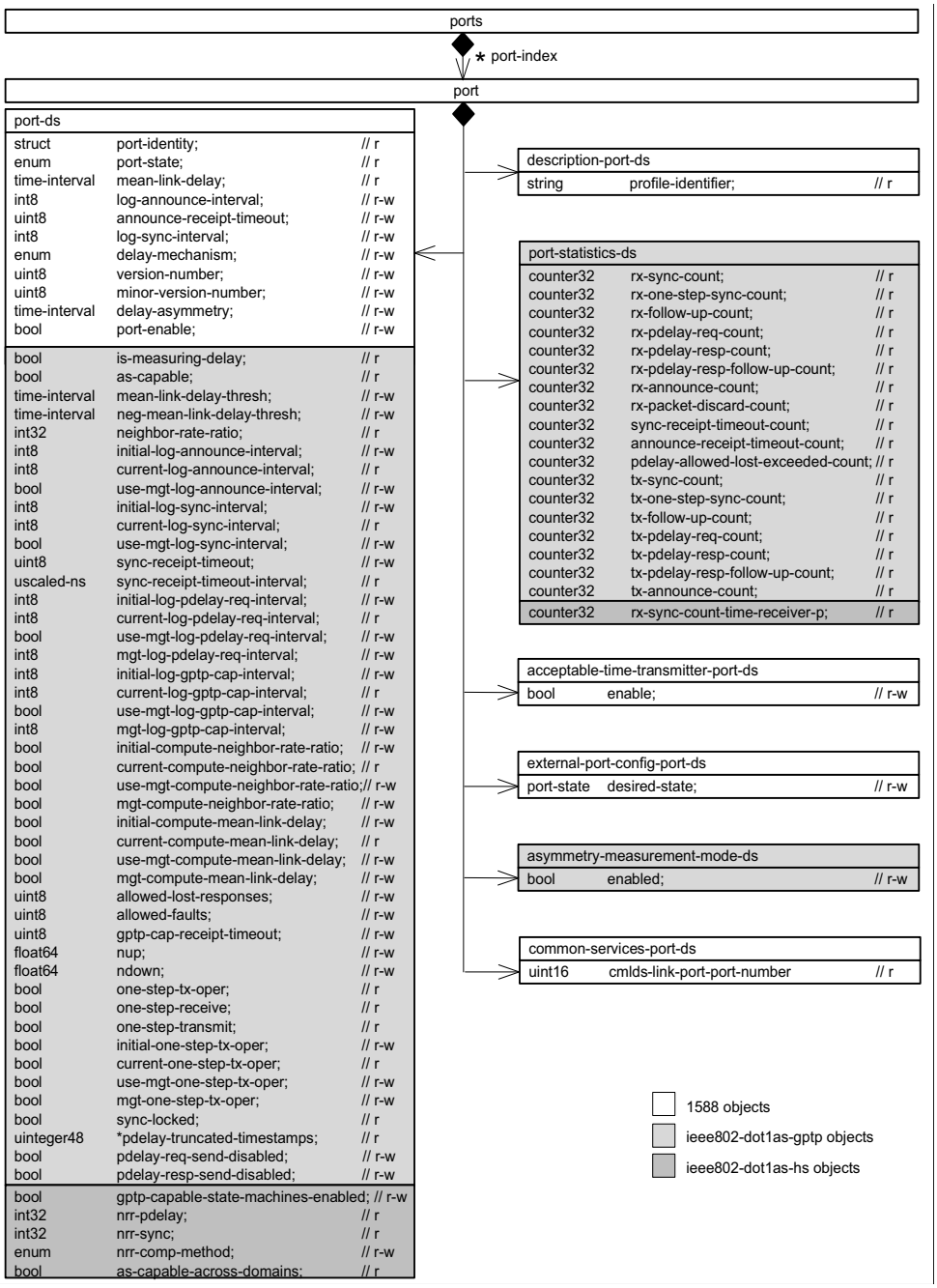


Figure 17-3—PTP Port detail

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Figure 17-4 provides detail for the common services, including each data set member. The CMLDS has a data set for the service itself (default-ds), and data sets for each Link Port. The Hot Standby Service has data sets for each HotStandbySystem. The FTT service has data sets for each FTT entity.

NOTE 4—Subclause 14.18.10 specifies nrrPdelay (nrr-pdelay), which is provided in YANG as the semantically equivalent node in ieee1588-ptp-tt named scaled-neighbor-rate-ratio (in link-port-ds of Figure 17-4).

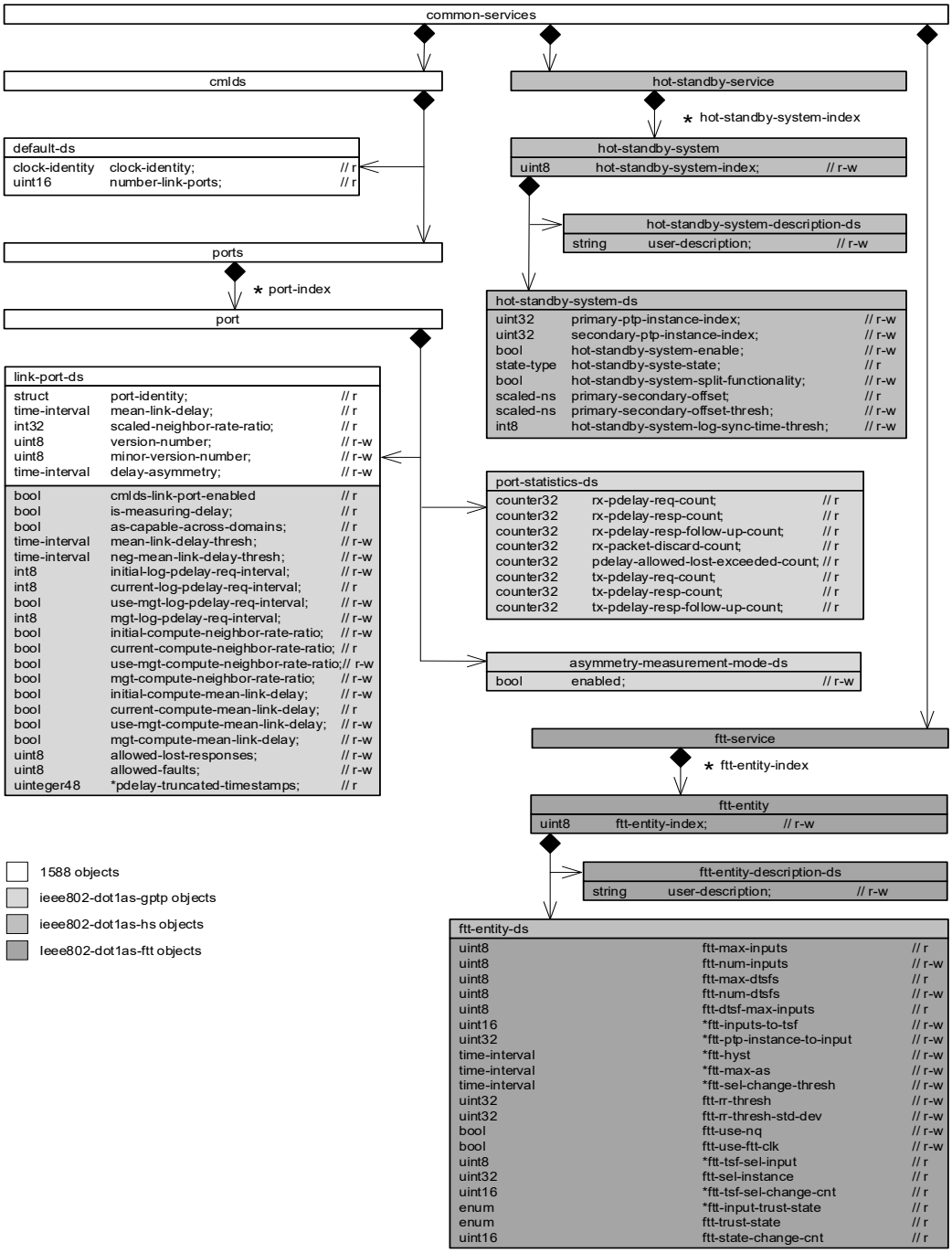


Figure 17-4—Common services detail

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17.3 Structure of the YANG data model

The YANG data model specified by this standard uses the YANG modules summarized in Table 17-1.

In the YANG module definitions, if any discrepancy between the “description” text and the corresponding definition in any other part of this standard occurs, the definitions outside this clause (Clause 17) take precedence.

Table 17-1—Summary of the YANG modules

Module	Managed functionality	YANG specification notes
ietf-yang-types	Type definitions	IETF RFC 6991—Common YANG Data Types.
ieee1588-ptp-tt	Clause 14	IEEE Std 1588e—MIB and YANG Data Models. it imports this YANG module as its foundational tree, including a subset of members from Clause 14.
ieee802-dot1as-gptp	Clause 14	The YANG module of this clause uses YANG augments to add members from Clause 14 that are unique to IEEE Std 802.1AS.
ieee802-dot1as-hs	Clause 14	Hot standby—YANG Data Model. The YANG module of this clause uses YANG augments to add members from Clause 14 that are unique to hot standby and drift tracking features.
ieee802-dot1as-ftt	Clause 14	Fault-tolerant timing with time integrity—YANG Data Model. The YANG module of this clause uses YANG augments to add members from Clause 14 that are unique to IEEE Std 802.1ASed.

17.4 Security considerations

The YANG modules specified in this document define a schema for data that is designed to be accessed via network management protocols such as NETCONF (IETF RFC 6241 [B18]) and RESTCONF (IETF RFC 8040 [B21]). NETCONF and RESTCONF protocols provide the means to secure communication between client and server, using secure transport layers such as Secure Shell (SSH) (IETF RFC 6242 [B19]) and Transport Layer Security (TLS) (IETF RFC 7589 [B20]).

It is the responsibility of a system’s implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those principals (users) that have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

The Network Configuration Access Control Model (IETF RFC 8341 [B23]) provides the means to restrict access for particular NETCONF or RESTCONF users to a pre-configured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data sets in this YANG module that contain writable data nodes (IETF RFC 8040 [B21]), such as:

```
/ptp-tt/instances/instance/default-ds
/ptp-tt/instances/instance/path-trace-ds
/ptp-tt/instances/instance/acceptable-time-transmitter-ds
/ptp-tt/instances/instance/ports/port/port-ds
```


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```
/ptp-tt/instances/instance/ports/port/acceptable-time-transmitter-port-ds
/ptp-tt/instances/instance/ports/port/external-port-config-port-ds
/ptp-tt/instances/instance/ports/port/asymmetry-measurement-mode-ds
/ptp-tt/common-services/cmlds/ports/port/link-port-ds
/ptp-tt/common-services/cmlds/ports/port/asymmetry-measurement-mode-ds
```

Write operations (e.g., edit-config) to these data nodes without proper protection can have a negative effect on network operations. Specifically, an inappropriate configuration of them can adversely impact a PTP synchronization network. For example, loss of synchronization on a clock, accuracy degradation on a set of clocks, or even breakdown of a whole synchronization network.

17.5 YANG schema tree definitions

The schema trees in this clause are provided as an overview of the YANG modules in . The symbols and their meaning are specified in YANG Tree Diagrams (IETF RFC 8340 [B22]).

17.5.1 Tree diagram for ieee802-dot1as-gptp.yang

```
module: ieee802-dot1as-gptp

  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:default-ds:
    +--ro gm-capable?                 boolean
    +--ro current-utc-offset?         int16
    +--ro current-utc-offset-valid?   boolean
    +--ro leap59?                     boolean
    +--ro leap61?                     boolean
    +--ro time-traceable?             boolean
    +--ro frequency-traceable?        boolean
    +--ro ptp-timescale?              boolean
    +--ro time-source?                identityref
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:current-ds:
    +--ro last-gm-phase-change?        scaled-ns
    +--ro last-gm-freq-change?         float64
    +--ro gm-timebase-indicator?       uint16
    +--ro gm-change-count?             yang:counter32
    +--ro time-of-last-gm-change?      yang:timestamp
    +--ro time-of-last-phase-change?   yang:timestamp
    +--ro time-of-last-freq-change?    yang:timestamp
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
    +--ro cumulative-rate-ratio?   int32
  augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port
            /ptp-tt:port-ds:
    +--ro is-measuring-delay?                    boolean
    +--ro as-capable?                            boolean
    +--rw mean-link-delay-thresh?                ptp-tt:time-interval
    +--rw neg-mean-link-delay-thresh?            ptp-tt:time-interval
    +--ro neighbor-rate-ratio?                   int32
    +--rw initial-log-announce-interval?         int8
    +--ro current-log-announce-interval?         int8
    +--rw use-mgt-log-announce-interval?         boolean
    +--rw initial-log-sync-interval?             int8
    +--ro current-log-sync-interval?             int8
    +--rw use-mgt-log-sync-interval?             boolean
    +--rw sync-receipt-timeout?                  uint8
    +--ro sync-receipt-timeout-interval?         uscaled-ns
    +--rw initial-log-pdelay-req-interval?       int8
    +--ro current-log-pdelay-req-interval?       int8
```


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```
  +--rw use-mgt-log-pdelay-req-interval?       boolean
  +--rw mgt-log-pdelay-req-interval?           int8
  +--rw initial-log-gptp-cap-interval?         int8
  +--ro current-log-gptp-cap-interval?         int8
  +--rw use-mgt-log-gptp-cap-interval?         boolean
  +--rw mgt-log-gptp-cap-interval?             int8
  +--rw initial-compute-neighbor-rate-ratio?   boolean
  +--ro current-compute-neighbor-rate-ratio?   boolean
  +--rw use-mgt-compute-neighbor-rate-ratio?   boolean
  +--rw mgt-compute-neighbor-rate-ratio?       boolean
  +--rw initial-compute-mean-link-delay?       boolean
  +--ro current-compute-mean-link-delay?       boolean
  +--rw use-mgt-compute-mean-link-delay?       boolean
  +--rw mgt-compute-mean-link-delay?           boolean
  +--rw allowed-lost-responses?                uint8
  +--rw allowed-faults?                        uint8
  +--rw gptp-cap-receipt-timeout?              uint8
  +--rw nup?                                   float64
  +--rw ndown?                                 float64
  +--ro one-step-tx-oper?                      boolean
  +--ro one-step-receive?                      boolean
  +--ro one-step-transmit?                     boolean
  +--rw initial-one-step-tx-oper?              boolean
  +--ro current-one-step-tx-oper?              boolean
  +--rw use-mgt-one-step-tx-oper?              boolean
  +--rw mgt-one-step-tx-oper?                  boolean
  +--ro sync-locked?                           boolean
  +--ro pdelay-truncated-timestamps*           uinteger48
  +--rw pdelay-req-send-disabled?              boolean
  +--rw pdelay-resp-send-disabled?             boolean
augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
  +--rw port-statistics-ds
     +--ro rx-sync-count?                        yang:counter32
     +--ro rx-one-step-sync-count?               yang:counter32
     +--ro rx-follow-up-count?                   yang:counter32
     +--ro rx-pdelay-req-count?                  yang:counter32
     +--ro rx-pdelay-resp-count?                 yang:counter32
     +--ro rx-pdelay-resp-follow-up-count?       yang:counter32
     +--ro rx-announce-count?                    yang:counter32
     +--ro rx-packet-discard-count?              yang:counter32
     +--ro sync-receipt-timeout-count?           yang:counter32
     +--ro announce-receipt-timeout-count?       yang:counter32
     +--ro pdelay-allowed-lost-exceeded-count?   yang:counter32
     +--ro tx-sync-count?                        yang:counter32
     +--ro tx-one-step-sync-count?               yang:counter32
     +--ro tx-follow-up-count?                   yang:counter32
     +--ro tx-pdelay-req-count?                  yang:counter32
     +--ro tx-pdelay-resp-count?                 yang:counter32
     +--ro tx-pdelay-resp-follow-up-count?       yang:counter32
     +--ro tx-announce-count?                    yang:counter32
augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port:
  +--rw asymmetry-measurement-mode-ds
     +--rw enabled?   boolean
augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports
          /ptp-tt:port/ptp-tt:link-port-ds:
  +--ro cmlds-link-port-enabled?               boolean
  +--ro is-measuring-delay?                    boolean
  +--ro as-capable-across-domains?             boolean
  +--rw mean-link-delay-thresh?                ptp-tt:time-interval
```


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```
   +--rw neg-mean-link-delay-thresh?            ptp-tt:time-interval
   +--rw initial-log-pdelay-req-interval?       int8
   +--ro current-log-pdelay-req-interval?       int8
   +--rw use-mgt-log-pdelay-req-interval?       boolean
   +--rw mgt-log-pdelay-req-interval?           int8
   +--rw initial-compute-neighbor-rate-ratio?   boolean
   +--ro current-compute-neighbor-rate-ratio?   boolean
   +--rw use-mgt-compute-neighbor-rate-ratio?   boolean
   +--rw mgt-compute-neighbor-rate-ratio?       boolean
   +--rw initial-compute-mean-link-delay?       boolean
   +--ro current-compute-mean-link-delay?       boolean
   +--rw use-mgt-compute-mean-link-delay?       boolean
   +--rw mgt-compute-mean-link-delay?           boolean
   +--rw allowed-lost-responses?                uint8
   +--rw allowed-faults?                        uint8
   +--ro pdelay-truncated-timestamps*           uinteger48
 augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports
           /ptp-tt:port:
   +--rw port-statistics-ds
      +--ro rx-pdelay-req-count?                  yang:counter32
      +--ro rx-pdelay-resp-count?                 yang:counter32
      +--ro rx-pdelay-resp-follow-up-count?       yang:counter32
      +--ro rx-packet-discard-count?              yang:counter32
      +--ro pdelay-allowed-lost-exceeded-count?   yang:counter32
      +--ro tx-pdelay-req-count?                  yang:counter32
      +--ro tx-pdelay-resp-count?                 yang:counter32
      +--ro tx-pdelay-resp-follow-up-count?       yang:counter32
 augment /ptp-tt:ptp/ptp-tt:common-services/ptp-tt:cmlds/ptp-tt:ports
           /ptp-tt:port:
   +--rw asymmetry-measurement-mode-ds
      +--rw enabled?   boolean
```

17.5.2 Tree diagram for ieee802-dot1as-hs.yang

```
module: ieee802-dot1as-hs

 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:parent-ds:
   +--ro gm-present?   boolean
 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance:
   +--rw ptp-instance-sync-ds
   |  +--ro is-synced?                              boolean
   |  +--rw offset-from-time-transmitter-max?       ptp-tt:time-interval
   |  +--rw rx-sync-count-time-receiver-p-thresh?   uint32
   |  +--rw offset-max-exceeded-count-thresh?       uint32
   |  +--rw offset-max-met-count-thresh?            uint32
   +--rw drift-tracking-ds
      +--rw drift-tracking-tlv-support?   boolean
      +--ro sync-grandmaster-identity?    ptp-tt:clock-identity
      +--ro sync-steps-removed?           uint32
      +--ro rate-ratio-drift?             int32
 augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port
           /ptp-tt:port-ds:
   +--rw gptp-capable-state-machines-enabled?   boolean
   +--ro nrr-pdelay?                            int32
   +--ro nrr-sync?                              int32
   +--rw nrr-comp-method?                       nrr-comp-method-type
   +--ro as-capable-across-domains?             boolean
```


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```
augment /ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance/ptp-tt:ports/ptp-tt:port
          /dot1as-gptp:port-statistics-ds:
  +--ro rx-sync-count-time-receiver-p?   yang:counter32
augment /ptp-tt:ptp/ptp-tt:common-services:
  +--rw hot-standby-service {hot-standby}?
     +--rw hot-standby-system* [hot-standby-system-index]
        +--rw hot-standby-system-index            uint8
        +--rw hot-standby-system-ds
        |  +--rw primary-ptp-instance-index?               uint32
        |  +--rw secondary-ptp-instance-index?             uint32
        |  +--rw hot-standby-system-enable?                boolean
        |  +--ro hot-standby-system-state?
        |  |       hot-standby-system-state-type
        |  +--rw hot-standby-system-split-functionality?   boolean
        |  +--ro primary-secondary-offset?
        |  |       dot1as-gptp:scaled-ns
        |  +--rw primary-secondary-offset-thresh?
        |  |       dot1as-gptp:scaled-ns
        |  +--rw hot-standby-system-log-sync-time-thresh?  int8
        +--rw hot-standby-system-description-ds
           +--rw user-description?   string
```

17.5.3 Tree diagram for ieee802-dot1as-ftt.yang

```
module: ieee802-dot1as-ftt

  augment /ptp-tt:ptp/ptp-tt:common-services:
    +--rw ftt-service {ftt}?
       +--rw ftt-entity* [ftt-entity-index]
          +--rw ftt-entity-index            uint8
          +--rw ftt-entity-ds
          |  +--ro ftt-max-inputs?             uint8
          |  +--rw ftt-num-inputs?             uint8
          |  +--ro ftt-max-dtsfs?              uint8
          |  +--rw ftt-num-dtsfs?              uint8
          |  +--ro ftt-dtsf-max-inputs?        uint8
          |  +--rw ftt-inputs-to-tsf* [tsf-instance-number]
          |  |  +--rw tsf-instance-number    uint8
          |  |  +--rw assigned-inputs?       uint16
          |  +--rw ftt-ptp-instance-to-input* [ftt-input-index]
          |  |  +--rw ftt-input-index       uint8
          |  |  +--rw ptp-instance-index?   ptp-instance-ref
          |  +--rw ftt-hyst* [ftt-input-index]
          |  |  +--rw ftt-input-index    uint8
          |  |  +--rw hyst-list* [ftt-input-index]
          |  |     +--rw ftt-input-index    uint8
          |  |     +--rw hyst-value?        ptp-tt:time-interval
          |  +--rw ftt-max-as* [ftt-input-index]
          |  |  +--rw ftt-input-index    uint8
          |  |  +--rw max-as-list* [ftt-input-index]
          |  |     +--rw ftt-input-index    uint8
          |  |     +--rw max-as-value?      ptp-tt:time-interval
          |  +--rw ftt-sel-change-thresh* [tsf-instance-number]
          |  |  +--rw tsf-instance-number    uint8
          |  |  +--rw change-threshold?      ptp-tt:time-interval
          |  +--rw ftt-rr-thresh?              uint32
          |  +--rw ftt-rr-thresh-std-dev?      uint32
```


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```
|  +--rw ftt-use-nq* [tsf-instance-number]
|  |  +--rw tsf-instance-number    uint8
|  |  +--rw use-nq?                boolean
|  +--rw ftt-use-ftt-clk?             boolean
|  +--ro ftt-tsf-sel-input* [tsf-instance-number]
|  |  +--ro tsf-instance-number         uint8
|  |  +--ro selected-ftt-input-index?   uint8
|  +--ro ftt-sel-instance?            uint32
|  +--ro ftt-tsf-sel-change-cnt* [tsf-instance-number]
|  |  +--ro tsf-instance-number    uint8
|  |  +--ro sel-change-cnt?        uint16
|  +--ro ftt-input-trust-state* [ftt-input-index]
|  |  +--ro ftt-input-index          uint8
|  |  +--ro ftt-input-trust-state?   ftt-input-trust-state-type
|  +--ro ftt-trust-state?             ftt-output-trust-state-type
|  +--ro ftt-state-change-cnt?        uint16
+--rw ftt-entity-description-ds
   +--rw user-description?   string
```

17.6 YANG modules[24, 25]

The ieee1588-ptp-tt.yang YANG module[26] specified by IEEE Std 1588e serves as the foundation of the YANG modules specified in this clause.

17.6.1 Module ieee802-dot1as-gptp.yang

```
module ieee802-dot1as-gptp {
  yang-version "1.1";
  namespace urn:ieee:std:802.1AS:yang:ieee802-dot1as-gptp;
  prefix dot1as-gptp;

  import ietf-yang-types {
    prefix yang;
  }
  import ieee1588-ptp-tt {
    prefix ptp-tt;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
              Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              445 Hoes Lane
              Piscataway, NJ 08854
```

[24] Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

[25] ASCII versions of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

[26] An ASCII version of the IEEE Std 1588 YANG module can be obtained from https://github.com/YangModels/yang/tree/main/standard/ieee/published/1588.

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```
            USA

    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that control timing and synchronization for
     time sensitive applications, as specified in Clause 14 of
     IEEE Std 802.1AS.

     Copyright (C) IEEE (2026). This version of this YANG module is
     part of IEEE Std 802.1AS; see the standard itself for full
     legal notices.";

  revision 2026-06-04 {
    description
      "Published as part of IEEE Std 802.1ASds-2026.";
    reference
      "IEEE Std 802.1AS - Timing and Synchronization for
       Time-Sensitive Applications: IEEE Std 802.1AS-2025.
       IEEE Std 802.1ASed-2026, IEEE Std 802.1ASds-2026,
       IEEE Std 1588 - IEEE Standard for a Precision Clock
       Synchronization Protocol for Networked Measurement and
       Control Systems: IEEE Std 1588-2019, IEEE Std 1588g-2022,
       IEEE Std 1588e-2024.";
  }
  typedef scaled-ns {
    type string {
      pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
    }
    description
      "The IEEE Std 802.1AS ScaledNs type represents signed values
       of time and time interval in units of 2^16 ns, as a signed
       96-bit integer. Each of the 12 octets is represented as a
       pair of hexadecimal characters, using uppercase for a letter.
       Octets are separated by a dash character. The most
       significant octet is first.";
    reference
      "6.4.3.1 of IEEE Std 802.1AS";
  }
  typedef uscaled-ns {
    type string {
      pattern "[0-9A-F]{2}(-[0-9A-F]{2}){11}";
    }
    description
      "The IEEE Std 802.1AS UScaledNs type represents unsigned
       values of time and time interval in units of 2^16 ns, as an
       unsigned 96-bit integer. Each of the 12 octets is represented
       as a pair of hexadecimal characters, using uppercase for a
       letter. Octets are separated by a dash character. The most
       significant octet is first.";
    reference
      "6.4.3.2 of IEEE Std 802.1AS";
  }
  typedef float64 {
    type string {
      pattern "[0-9A-F]{2}(-[0-9A-F]{2}){7}";
    }
    description
      "The IEEE Std 802.1AS Float64 type represents IEEE Std 754
       binary64. Each of the 8 octets is represented as a pair of
```


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```
      hexadecimal characters, using uppercase for a letter. Octets
      are separated by a dash character. The most significant octet
      is first.";
  reference
    "6.4.2 of IEEE Std 802.1AS";
}
typedef uinteger48 {
  type uint64 {
    range "0..281474976710655";
  }
  description
    "48-bit unsigned integer data type.";
  reference
    "6.4.2 of IEEE Std 802.1AS";
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:default-ds" {
  description
    "Augment IEEE Std 1588 defaultDS.";
  leaf gm-capable {
    type boolean;
    config false;
    description
      "The value is true if the time-aware system is capable of
       being a Grandmaster, and false if the time-aware system is
       not capable of being a Grandmaster.";
    reference
      "14.2.7 of IEEE Std 802.1AS";
  }
  leaf current-utc-offset {
    when
      "../current-utc-offset-valid='true'";
    type int16;
    config false;
    description
      "Offset from UTC (TAI - UTC). The offset is in units of
       seconds. This leaf applies to the ClockTimeTransmitter
       entity (i.e., local only, unrelated to a remote GM).";
    reference
      "14.2.8 of IEEE Std 802.1AS";
  }
  leaf current-utc-offset-valid {
    type boolean;
    config false;
    description
      "The value of current-utc-offset-valid shall be true if the
       value of current-utc-offset is known to be correct,
       otherwise it shall be false. This leaf applies to the
       ClockTimeTransmitter entity (i.e., local only, unrelated to
       a remote GM).";
    reference
      "14.2.9 of IEEE Std 802.1AS";
  }
  leaf leap59 {
    type boolean;
    config false;
    description
      "If the timescale is PTP, a true value for leap59 shall
```


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```
      indicate that the last minute of the current UTC day
      contains 59 seconds. If the timescale is not PTP, the value
      shall be false. This leaf applies to the
      ClockTimeTransmitter entity (i.e., local only, unrelated to
      a remote GM).";
    reference
      "14.2.10 of IEEE Std 802.1AS";
  }
  leaf leap61 {
    type boolean;
    config false;
    description
      "If the timescale is PTP, a true value for leap61 shall
       indicate that the last minute of the current UTC day
       contains 61 seconds. If the timescale is not PTP, the value
       shall be false. This leaf applies to the
       ClockTimeTransmitter entity (i.e., local only, unrelated to
       a remote GM).";
    reference
      "14.2.11 of IEEE Std 802.1AS";
  }
  leaf time-traceable {
    type boolean;
    config false;
    description
      "The value of time-traceable shall be true if the timescale
       is traceable to a primary reference; otherwise, the value
       shall be false. This leaf applies to the
       ClockTimeTransmitter entity (i.e., local only, unrelated to
       a remote GM).";
    reference
      "14.2.12 of IEEE Std 802.1AS";
  }
  leaf frequency-traceable {
    type boolean;
    config false;
    description
      "The value of frequency-traceable shall be true if the
       frequency determining the timescale is traceable to a
       primary reference; otherwise, the value shall be false.
       This leaf applies to the ClockTimeTransmitter entity
       (i.e., local only, unrelated to a remote GM).";
    reference
      "14.2.13 of IEEE Std 802.1AS";
  }
  leaf ptp-timescale {
    type boolean;
    config false;
    description
      "If ptp-timescale is true, the timescale of the
       ClockTimeTransmitter entity is PTP, which is the elapsed
       time since the PTP epoch measured using the second defined
       by International Atomic Time (TAI). If ptp-timescale is
       false, the timescale of the ClockTimeTransmitter entity is
       ARB, which is the elapsed time since an arbitrary epoch.
       This leaf applies to the ClockTimeTransmitter entity
       (i.e., local only, unrelated to a remote GM).";
    reference
      "14.2.14 of IEEE Std 802.1AS";
```


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```
  }
  leaf time-source {
    type identityref {
      base ptp-tt:time-source;
    }
    config false;
    description
      "The source of time used by the Grandmaster Clock. This leaf
       applies to the ClockTimeTransmitter entity (i.e., local
       only, unrelated to a remote GM).";
    reference
      "14.2.15 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:current-ds" {
  description
    "Augment IEEE Std 1588 currentDS.";
  leaf last-gm-phase-change {
    type scaled-ns;
    config false;
    description
      "Phase change that occurred on the most recent change in
       either the Grandmaster PTP Instance or
       gm-timebase-indicator leaf.";
    reference
      "14.3.4 of IEEE Std 802.1AS";
  }
  leaf last-gm-freq-change {
    type float64;
    config false;
    description
      "Frequency change that occurred on the most recent change in
       either the Grandmaster PTP Instance or
       gm-timebase-indicator leaf.";
    reference
      "14.3.5 of IEEE Std 802.1AS";
  }
  leaf gm-timebase-indicator {
    type uint16;
    config false;
    description
      "The timeBaseIndicator of the current Grandmaster PTP
       Instance.";
    reference
      "14.3.6 of IEEE Std 802.1AS";
  }
  leaf gm-change-count {
    type yang:counter32;
    config false;
    description
      "This statistics counter tracks the number of times the
       Grandmaster PTP Instance has changed in a gPTP domain.";
    reference
      "14.3.7 of IEEE Std 802.1AS";
  }
  leaf time-of-last-gm-change {
    type yang:timestamp;
```


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```
    config false;
    description
      "System time when the most recent Grandmaster Clock change
       occurred in a gPTP domain. This leaf's type is YANG
       timestamp, which is based on system time. System time is an
       unsigned integer in units of 10 milliseconds, using an
       epoch defined by the implementation (typically time of
       boot-up).";
    reference
      "14.3.8 of IEEE Std 802.1AS";
  }
  leaf time-of-last-phase-change {
    type yang:timestamp;
    config false;
    description
      "System time when the most recent change in Grandmaster
       Clock phase occurred. This leaf's type is YANG timestamp,
       which is based on system time. System time is an unsigned
       integer in units of 10 milliseconds, using an epoch defined
       by the implementation (typically time of boot-up).";
    reference
      "14.3.9 of IEEE Std 802.1AS";
  }
  leaf time-of-last-freq-change {
    type yang:timestamp;
    config false;
    description
      "System time when the most recent change in Grandmaster
       Clock frequency occurred. This leaf's type is YANG
       timestamp, which is based on system time. System time is an
       unsigned integer in units of 10 milliseconds, using an
       epoch defined by the implementation (typically time of
       boot-up).";
    reference
      "14.3.10 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:parent-ds" {
  description
    "Augment IEEE Std 1588 parentDS.";
  leaf cumulative-rate-ratio {
    type int32;
    config false;
    description
      "Estimate of the ratio of the frequency of the Grandmaster
       Clock to the frequency of the LocalClock entity of this PTP
       Instance. cumulative-rate-ratio is expressed as the
       fractional frequency offset multiplied by 2^41, i.e., the
       quantity (rateRatio - 1.0)(2^41).";
    reference
      "14.4.3 of IEEE Std 802.1AS";
  }
}

augment
  "/ptp-tt:ptp"+
  "/ptp-tt:instances"+
```


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```
  "/ptp-tt:instance"+
  "/ptp-tt:ports"+
  "/ptp-tt:port"+
  "/ptp-tt:port-ds" {
  description
    "Augment IEEE Std 1588 portDS.

     14.10.4 of IEEE Std 802.1AS specifies ptpPortEnabled
     (ptp-port-enabled), which is provided in YANG as the
     semantically equivalent node in ieee1588-ptp-tt named
     port-enable (in port-ds).

     14.10.16 of IEEE Std 802.1AS specifies
     mgtSettableLogAnnounceInterval (mgt-log-announce-interval),
     which is provided in YANG as the semantically equivalent node
     in ieee1588-ptp-tt named log-announce-interval (in port-ds).
     In the context of IEEE Std 802.1AS, log-announce-interval
     cannot be used unless use-mgt-log-announce-interval is true.

     14.10.21 of IEEE Std 802.1AS specifies
     mgtSettableLogSyncInterval (mgt-log-sync-interval), which is
     provided in YANG as the semantically equivalent node in
     ieee1588-ptp-tt named log-sync-interval (in port-ds). In the
     context of IEEE Std 802.1AS, log-sync-interval cannot be used
     unless use-mgt-log-sync-interval is true.";
  leaf is-measuring-delay {
    type boolean;
    config false;
    description
      "Boolean that is true if the port is measuring PTP Link
       propagation delay.";
    reference
      "14.10.6 of IEEE Std 802.1AS";
  }
  leaf as-capable {
    type boolean;
    config false;
    description
      "Boolean that is true if and only if it is determined that
       this PTP Instance and the PTP Instance at the other end of
       the link attached to this port can interoperate with each
       other via the IEEE Std 802.1AS protocol.";
    reference
      "10.2.5.1 of IEEE Std 802.1AS
       14.10.7 of IEEE Std 802.1AS";
  }
  leaf mean-link-delay-thresh {
    type ptp-tt:time-interval;
    description
      "Propagation time threshold for mean-link-delay, above which
       a port is not considered capable of participating in the
       IEEE Std 802.1AS protocol.";
    reference
      "14.10.9 of IEEE Std 802.1AS";
  }
  leaf neg-mean-link-delay-thresh {
    type ptp-tt:time-interval;
    description
      "The negative propagation time threshold for
```


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```
      mean-link-delay, below which a port is not considered
      capable of participating in the IEEE Std 802.1AS
      protocol.";
  reference
    "14.10.10 of IEEE Std 802.1AS";
}
leaf neighbor-rate-ratio {
  type int32;
  config false;
  description
    "Estimate of the ratio of the frequency of the LocalClock
     entity of the PTP Instance at the other end of the link
     attached to this PTP Port, to the frequency of the
     LocalClock entity of this PTP Instance. neighbor-rate-ratio
     is expressed as the fractional frequency offset multiplied
     by 2^41, i.e., the quantity (rateRatio - 1.0)(2^41).";
  reference
    "14.10.12 of IEEE Std 802.1AS";
}
leaf initial-log-announce-interval {
  type int8;
  description
    "When use-mgt-log-announce-interval is false (i.e., change
     with Signaling message), this is the logarithm to base 2 of
     the announce interval used when the port is initialized.";
  reference
    "14.10.13 of IEEE Std 802.1AS";
}
leaf current-log-announce-interval {
  type int8;
  config false;
  description
    "Logarithm to base 2 of the current announce interval.";
  reference
    "14.10.14 of IEEE Std 802.1AS";
}
leaf use-mgt-log-announce-interval {
  type boolean;
  description
    "Boolean that determines the source of the announce
     interval. If the value is true, the announce interval
     (current-log-announce-interval) is set equal to the value
     of mgt-log-announce-interval. If the value is false, the
     announce interval is determined by the
     AnnounceIntervalSetting state machine (i.e., changed with
     Signaling message).";
  reference
    "14.10.15 of IEEE Std 802.1AS";
}
leaf initial-log-sync-interval {
  type int8;
  description
    "When use-mgt-log-sync-interval is false (i.e., change with
     Signaling message), this is the logarithm to base 2 of the
     sync interval used when the port is initialized.";
  reference
    "14.10.18 of IEEE Std 802.1AS";
}
leaf current-log-sync-interval {
```


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```
    type int8;
    config false;
    description
      "Logarithm to base 2 of the current sync interval.";
    reference
      "14.10.19 of IEEE Std 802.1AS";
  }
  leaf use-mgt-log-sync-interval {
    type boolean;
    description
      "Boolean that determines the source of the sync interval.
       If the value is true, the sync interval
       (current-log-sync-interval) is set equal to the value of
       mgt-log-sync-interval. If the value is false, the sync
       interval is determined by the SyncIntervalSetting
       state machine (i.e., changed with Signaling message).";
    reference
      "14.10.20 of IEEE Std 802.1AS";
  }
  leaf sync-receipt-timeout {
    type uint8;
    description
      "Number of sync intervals that a timeReceiver port waits
       without receiving synchronization information, before
       assuming that the timeTransmitter is no longer transmitting
       synchronization information and that the BTCA needs to be
       run, if appropriate.";
    reference
      "14.10.22 of IEEE Std 802.1AS";
  }
  leaf sync-receipt-timeout-interval {
    type uscaled-ns;
    config false;
    description
      "Time interval after which sync receipt timeout occurs if
       time-synchronization information has not been received
       during the interval.";
    reference
      "14.10.23 of IEEE Std 802.1AS";
  }
  leaf initial-log-pdelay-req-interval {
    type int8;
    description
      "When use-mgt-log-pdelay-req-interval is false (i.e., change
       with Signaling message), this is the logarithm to base 2 of
       the Pdelay_Req transmit interval used when the port is
       initialized.";
    reference
      "14.10.24 of IEEE Std 802.1AS";
  }
  leaf current-log-pdelay-req-interval {
    type int8;
    config false;
    description
      "Logarithm to base 2 of the current Pdelay_Req transmit
       interval.";
    reference
      "14.10.25 of IEEE Std 802.1AS";
  }
```


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```
leaf use-mgt-log-pdelay-req-interval {
  type boolean;
  description
    "Boolean that determines the source of the Pdelay_Req
     transmit interval. If the value is true, the Pdelay_Req
     transmit interval (current-log-pdelay-req-interval) is set
     equal to the value of mgt-log-pdelay-req-interval. If the
     value is false, the Pdelay_Req transmit interval is
     determined by the LinkDelayIntervalSetting state machine
     (i.e., changed with Signaling message).";
  reference
    "14.10.26 of IEEE Std 802.1AS";
}
leaf mgt-log-pdelay-req-interval {
  type int8;
  description
    "Logarithm to base 2 of the Pdelay_Req transmit interval,
     used if use-mgt-log-pdelay-req-interval is true. This value
     is not used if use-mgt-log-pdelay-req-interval is false.";
  reference
    "14.10.27 of IEEE Std 802.1AS";
}
leaf initial-log-gptp-cap-interval {
  type int8;
  description
    "When use-mgt-log-gptp-cap-interval is false (i.e., change
     with Signaling message), this is the logarithm to base 2 of
     the gPTP capable message interval used when the port is
     initialized.";
  reference
    "14.10.28 of IEEE Std 802.1AS";
}
leaf current-log-gptp-cap-interval {
  type int8;
  config false;
  description
    "Logarithm to base 2 of the current gPTP capable message
     interval.";
  reference
    "14.10.29 of IEEE Std 802.1AS";
}
leaf use-mgt-log-gptp-cap-interval {
  type boolean;
  description
    "Boolean that determines the source of the gPTP capable
     message interval. If the value is true, the gPTP capable
     message interval (current-log-gptp-cap-interval) is set
     equal to the value of mgt-gptp-cap-req-interval. If the
     value is false, the gPTP capable message interval is
     determined by the GptpCapableMessageIntervalSetting state
     machine (i.e., changed with Signaling message).";
  reference
    "14.10.30 of IEEE Std 802.1AS";
}
leaf mgt-log-gptp-cap-interval {
  type int8;
  description
    "Logarithm to base 2 of the gPTP capable message interval,
     used if use-mgt-log-gptp-cap-interval is true. This value
```


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```
      is not used if use-mgt-log-pdelay-req-interval is false.";
    reference
      "14.10.31 of IEEE Std 802.1AS";
  }
  leaf initial-compute-neighbor-rate-ratio {
    type boolean;
    description
      "When use-mgt-compute-neighbor-rate-ratio is false
       (i.e., change with Signaling message), this is the initial
       value of computeNeighborRateRatio.";
    reference
      "14.10.32 of IEEE Std 802.1AS";
  }
  leaf current-compute-neighbor-rate-ratio {
    type boolean;
    config false;
    description
      "Current value of computeNeighborRateRatio.";
    reference
      "14.10.33 of IEEE Std 802.1AS";
  }
  leaf use-mgt-compute-neighbor-rate-ratio {
    type boolean;
    description
      "Boolean that determines the source of
       computeNeighborRateRatio. If the value is true,
       computeNeighborRateRatio is set equal to the value of
       mgt-compute-neighbor-rate-ratio. If the value is false,
       computeNeighborRateRatio is determined by the
       LinkDelayIntervalSetting state machine (i.e., changed with
       Signaling message).";
    reference
      "14.10.34 of IEEE Std 802.1AS";
  }
  leaf mgt-compute-neighbor-rate-ratio {
    type boolean;
    description
      "Value of computeNeighborRateRatio, used if
       use-mgt-compute-neighbor-rate-ratio is true. This value is
       not used if use-mgt-compute-neighbor-rate-ratio is false.";
    reference
      "14.10.35 of IEEE Std 802.1AS";
  }
  leaf initial-compute-mean-link-delay {
    type boolean;
    description
      "When use-mgt-compute-mean-link-delay is false (i.e., change
       with Signaling message), this is the initial value of
       computeMeanLinkDelay.";
    reference
      "14.10.36 of IEEE Std 802.1AS";
  }
  leaf current-compute-mean-link-delay {
    type boolean;
    config false;
    description
      "Current value of computeMeanLinkDelay.";
    reference
      "14.10.37 of IEEE Std 802.1AS";
```


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```
  }
  leaf use-mgt-compute-mean-link-delay {
    type boolean;
    description
      "Boolean that determines the source of computeMeanLinkDelay.
       If the value is true, computeMeanLinkDelay is set equal to
       the value of mgt-compute-mean-link-delay. If the value is
       false, computeMeanLinkDelay is determined by the
       LinkDelayIntervalSetting state machine (i.e., changed with
       Signaling message).";
    reference
      "14.10.38 of IEEE Std 802.1AS";
  }
  leaf mgt-compute-mean-link-delay {
    type boolean;
    description
      "Value of computeMeanLinkDelay, used if
       use-mgt-compute-mean-link-delay is true. This value is not
       used if use-mgt-compute-mean-link-delay is false.";
    reference
      "14.10.39 of IEEE Std 802.1AS";
  }
  leaf allowed-lost-responses {
    type uint8;
    description
      "Number of Pdelay_Req messages for which a valid response is
       not received, above which a port is considered to not be
       exchanging peer delay messages with its neighbor.";
    reference
      "14.10.40 of IEEE Std 802.1AS";
  }
  leaf allowed-faults {
    type uint8;
    description
      "Number of faults above which asCapable is set to false.";
    reference
      "14.10.41 of IEEE Std 802.1AS";
  }
  leaf gptp-cap-receipt-timeout {
    type uint8;
    description
      "Number of transmission intervals that a port waits without
       receiving the gPTP-capable TLV, before assuming that the
       neighbor port is no longer invoking the gPTP protocol.";
    reference
      "14.10.42 of IEEE Std 802.1AS";
  }
  leaf nup {
    type float64;
    description
      "For an OLT port of an IEEE Std 802.3 EPON link, this value
       is the effective index of refraction for the EPON upstream
       wavelength light of the optical path.";
    reference
      "14.10.44 of IEEE Std 802.1AS";
  }
  leaf ndown {
    type float64;
    description
```


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```
      "For an OLT port of an IEEE 802.3 EPON link, this value is
       the effective index of refraction for the EPON downstream
       wavelength light of the optical path.";
    reference
      "14.10.45 of IEEE Std 802.1AS";
  }
  leaf one-step-tx-oper {
    type boolean;
    config false;
    description
      "This value is true if the port is sending one-step Sync
       messages, and false if the port is sending two-step Sync
       and Follow_Up messages.";
    reference
      "14.10.46 of IEEE Std 802.1AS";
  }
  leaf one-step-receive {
    type boolean;
    config false;
    description
      "This value is true if the port is capable of receiving and
       processing one-step Sync messages.";
    reference
      "14.10.47 of IEEE Std 802.1AS";
  }
  leaf one-step-transmit {
    type boolean;
    config false;
    description
      "This value is true if the port is capable of transmitting
       one-step Sync messages.";
    reference
      "14.10.48 of IEEE Std 802.1AS";
  }
  leaf initial-one-step-tx-oper {
    type boolean;
    description
      "When use-mgt-one-step-tx-oper is false (i.e., change with
       Signaling message), this is the initial value of
       current-one-step-tx-oper.";
    reference
      "14.10.49 of IEEE Std 802.1AS";
  }
  leaf current-one-step-tx-oper {
    type boolean;
    config false;
    description
      "This value is true if the port is configured to transmit
       one-step Sync messages, either via management
       (mgt-one-step-tx-oper) or Signaling. If both
       current-one-step-tx-oper and one-step-transmit are true,
       the port transmits one-step Sync messages
       (i.e., one-step-tx-oper true).";
    reference
      "14.10.50 of IEEE Std 802.1AS";
  }
  leaf use-mgt-one-step-tx-oper {
    type boolean;
    description
```


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```
        "Boolean that determines the source of
         current-one-step-tx-oper. If the value is true,
         current-one-step-tx-oper is set equal to the value of
         mgt-one-step-tx-oper. If the value is false,
         current-one-step-tx-oper is determined by the
         OneStepTxOperSetting state machine (i.e., changed with
         Signaling message).";
      reference
        "14.10.51 of IEEE Std 802.1AS";
    }
    leaf mgt-one-step-tx-oper {
      type boolean;
      description
        "If use-mgt-one-step-tx-oper is true,
         current-one-step-tx-oper is set equal to this value. This
         value is not used if use-mgt-one-step-tx-oper is false.";
      reference
        "14.10.52 of IEEE Std 802.1AS";
    }
    leaf sync-locked {
      type boolean;
      config false;
      description
        "This value is true if the port will transmit a Sync as soon
         as possible after the timeReceiver port receives a Sync
         message.";
      reference
        "14.10.53 of IEEE Std 802.1AS";
    }
    leaf-list pdelay-truncated-timestamps {
      type uinteger48;
      config false;
      description
        "For full-duplex IEEE Std 802.3 media, and CSN media that
         use the peer-to-peer delay mechanism to measure path delay,
         the values of the four elements of this leaf-list
         correspond to the timestamps t1, t2, t3, and t4, listed in
         that order. Each timestamp is expressed in units of
         2^-16 ns (i.e., the value of each array element is equal to
         the remainder obtained upon dividing the respective
         timestamp, expressed in units of 2^-16 ns, by 2^48).
         At any given time, the timestamp values stored in the array
         are for the same, and most recently completed, peer delay
         message exchange. For each timestamp, only 48-bits are
         valid (the upper 16-bits are always zero).";
      reference
        "14.10.54 of IEEE Std 802.1AS";
}
    leaf pdelay-req-send-disabled {
      type boolean;
      description
        "A boolean that is administratively set to TRUE
         if Pdelay_Req messages are not transmitted by this port.
         The default value for this variable shall be FALSE.";
      reference
        "14.10.61 of IEEE Std 802.1AS";
    }
    leaf pdelay-resp-send-disabled {
      type boolean;
```


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```
    description
      "A boolean that is administratively set to TRUE
       if Pdelay_Resp messages are not transmitted by this port.
       The default value for this variable shall be FALSE.";
    reference
      "14.10.62 of IEEE Std 802.1AS";
  }
}

augment
  "/ptp-tt:ptp"+
  "/ptp-tt:instances"+
  "/ptp-tt:instance"+
  "/ptp-tt:ports"+
  "/ptp-tt:port" {
  description
    "Augment to add port-statistics-ds to IEEE Std 1588 PTP
     Port.";
  container port-statistics-ds {
    description
      "Provides counters associated with the port of the PTP
       Instance.";
    reference
      "14.12 of IEEE Std 802.1AS";
    leaf rx-sync-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time synchronization
         information is received.";
      reference
        "14.12.2 of IEEE Std 802.1AS";
    }
    leaf rx-one-step-sync-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a one-step Sync
         message is received.";
      reference
        "14.12.3 of IEEE Std 802.1AS";
    }
    leaf rx-follow-up-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Follow_Up message is
         received.";
      reference
        "14.12.4 of IEEE Std 802.1AS";
    }
    leaf rx-pdelay-req-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Req message
         is received.";
      reference
        "14.12.5 of IEEE Std 802.1AS";
```


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```
    }
    leaf rx-pdelay-resp-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Resp message
         is received.";
      reference
        "14.12.6 of IEEE Std 802.1AS";
    }
    leaf rx-pdelay-resp-follow-up-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a
         Pdelay_Resp_Follow_Up message is received.";
      reference
        "14.12.7 of IEEE Std 802.1AS";
    }
    leaf rx-announce-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time an Announce message is
         received.";
      reference
        "14.12.8 of IEEE Std 802.1AS";
    }
    leaf rx-packet-discard-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a PTP message of the
         respective PTP Instance is discarded.";
      reference
        "14.12.9 of IEEE Std 802.1AS";
    }
    leaf sync-receipt-timeout-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a sync receipt timeout
         occurs.";
      reference
        "14.12.10 of IEEE Std 802.1AS";
    }
    leaf announce-receipt-timeout-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time an announce receipt
         timeout occurs.";
      reference
        "14.12.11 of IEEE Std 802.1AS";
    }
    leaf pdelay-allowed-lost-exceeded-count {
      type yang:counter32;
      config false;
      description
```


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```
        "Counter that increments every time the value of the
         variable lostResponses exceeds the value of the variable
         allowedLostResponses, in the RESET state of the
         MDPdelayReq state machine.";
      reference
        "14.12.12 of IEEE Std 802.1AS";
    }
    leaf tx-sync-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time synchronization
         information is transmitted.";
      reference
        "14.12.13 of IEEE Std 802.1AS";
    }
    leaf tx-one-step-sync-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a one-step Sync
         message is transmitted.";
      reference
        "14.12.14 of IEEE Std 802.1AS";
    }
    leaf tx-follow-up-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Follow_Up message is
         transmitted.";
      reference
        "14.12.15 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-req-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Req message
         is transmitted.";
      reference
        "14.12.16 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-resp-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a Pdelay_Resp message
         is transmitted.";
      reference
        "14.12.17 of IEEE Std 802.1AS";
    }
    leaf tx-pdelay-resp-follow-up-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time a
         Pdelay_Resp_Follow_Up message is transmitted.";
      reference
```


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```
        "14.12.18 of IEEE Std 802.1AS";
    }
    leaf tx-announce-count {
      type yang:counter32;
      config false;
      description
        "Counter that increments every time an Announce message is
         transmitted.";
      reference
        "14.12.19 of IEEE Std 802.1AS";
    }
  }
}

augment
  "/ptp-tt:ptp"+
  "/ptp-tt:instances"+
  "/ptp-tt:instance"+
  "/ptp-tt:ports"+
  "/ptp-tt:port" {
  description
    "Augment to add asymmetry-measurement-mode-ds to IEEE Std 1588
     PTP Port.";
  container asymmetry-measurement-mode-ds {
    description
      "Represents the capability to enable/disable the Asymmetry
       Compensation Measurement Procedure on a PTP Port. This data
       set is used instead of the CMLDS
       asymmetry-measurement-mode-ds when only a single PTP
       Instance is present (i.e., CMLDS is not used).";
    reference
      "14.15 of IEEE Std 802.1AS
       Annex G of IEEE Std 802.1AS";
    leaf enabled {
      type boolean;
      description
        "For full-duplex IEEE Std 802.3 media, the value is true
         if an asymmetry measurement is being performed for the
         link attached to this PTP Port, and false otherwise. For
         all other media, the value shall be false.";
      reference
        "14.15.2 of IEEE Std 802.1AS";
    }
  }
}

augment
  "/ptp-tt:ptp"+
  "/ptp-tt:common-services"+
  "/ptp-tt:cmlds"+
  "/ptp-tt:ports"+
  "/ptp-tt:port"+
  "/ptp-tt:link-port-ds" {
  description
    "Augment IEEE Std 1588 cmldsLinkPortDS.";
  leaf cmlds-link-port-enabled {
    type boolean;
    config false;
    description
```


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```
      "Boolean that is true if both delay-mechanism is common-p2p
       and the value of ptp-port-enabled is true, for at least one
       PTP Port that uses the CMLDS; otherwise, the value is
       false.";
    reference
      "11.2.18.1 of IEEE Std 802.1AS
       14.18.3 of IEEE Std 802.1AS";
  }
  leaf is-measuring-delay {
    type boolean;
    config false;
    description
      "This leaf is analogous to is-measuring-delay for a PTP
       Port, but applicable to this Link Port.";
    reference
      "14.18.4 of IEEE Std 802.1AS";
  }
  leaf as-capable-across-domains {
    type boolean;
    config false;
    description
      "This leaf is true when all PTP Instances (domains) for this
       Link Port detect proper exchange of Pdelay messages.";
    reference
      "11.2.2 of IEEE Std 802.1AS
       14.18.5 of IEEE Std 802.1AS";
  }
  leaf mean-link-delay-thresh {
    type ptp-tt:time-interval;
    description
      "Propagation time threshold for mean-link-delay, above which
       a Link Port is not considered capable of participating in
       the IEEE Std 802.1AS protocol.";
    reference
      "14.18.7 of IEEE Std 802.1AS";
  }
  leaf neg-mean-link-delay-thresh {
    type ptp-tt:time-interval;
    description
      "The negative propagation time threshold for
       mean-link-delay, below which a Link Port is not considered
       capable of participating in the IEEE Std 802.1AS
       protocol.";
    reference
      "14.18.8 of IEEE Std 802.1AS";
  }
  leaf initial-log-pdelay-req-interval {
    type int8;
    description
      "This leaf is analogous to initial-log-pdelay-req-interval
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.11 of IEEE Std 802.1AS";
  }
  leaf current-log-pdelay-req-interval {
    type int8;
    config false;
    description
      "This leaf is analogous to current-log-pdelay-req-interval
```


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```
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.12 of IEEE Std 802.1AS";
  }
  leaf use-mgt-log-pdelay-req-interval {
    type boolean;
    description
      "This leaf is analogous to use-mgt-log-pdelay-req-interval
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.13 of IEEE Std 802.1AS";
  }
  leaf mgt-log-pdelay-req-interval {
    type int8;
    description
      "This leaf is analogous to mgt-log-pdelay-req-interval for
       a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.14 of IEEE Std 802.1AS";
  }
  leaf initial-compute-neighbor-rate-ratio {
    type boolean;
    description
      "This leaf is analogous to
       initial-compute-neighbor-rate-ratio for a PTP Port, but
       applicable to this Link Port.";
    reference
      "14.18.15 of IEEE Std 802.1AS";
  }
  leaf current-compute-neighbor-rate-ratio {
    type boolean;
    config false;
    description
      "This leaf is analogous to
       current-compute-neighbor-rate-ratio for a PTP Port, but
       applicable to this Link Port.";
    reference
      "14.18.16 of IEEE Std 802.1AS";
  }
  leaf use-mgt-compute-neighbor-rate-ratio {
    type boolean;
    description
      "This leaf is analogous to
       use-mgt-compute-neighbor-rate-ratio for a PTP Port, but
       applicable to this Link Port.";
    reference
      "14.18.17 of IEEE Std 802.1AS";
  }
  leaf mgt-compute-neighbor-rate-ratio {
    type boolean;
    description
      "This leaf is analogous to mgt-compute-neighbor-rate-ratio
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.18 of IEEE Std 802.1AS";
  }
  leaf initial-compute-mean-link-delay {
    type boolean;
    description
```


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```
      "This leaf is analogous to initial-compute-mean-link-delay
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.19 of IEEE Std 802.1AS";
  }
  leaf current-compute-mean-link-delay {
    type boolean;
    config false;
    description
      "This leaf is analogous to current-compute-mean-link-delay
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.20 of IEEE Std 802.1AS";
  }
  leaf use-mgt-compute-mean-link-delay {
    type boolean;
    description
      "This leaf is analogous to use-mgt-compute-mean-link-delay
       for a PTP Port, but applicable to this Link Port.";
    reference
      "14.18.21 of IEEE Std 802.1AS";
  }
  leaf mgt-compute-mean-link-delay {
    type boolean;
    description
      "This leaf is analogous to mgt-compute-mean-link-delay for a
       PTP Port, but applicable to this Link Port.";
    reference
      "14.18.22 of IEEE Std 802.1AS";
  }
  leaf allowed-lost-responses {
    type uint8;
    description
      "This leaf is analogous to allowed-lost-responses for a PTP
       Port, but applicable to this Link Port.";
    reference
      "14.18.23 of IEEE Std 802.1AS";
  }
  leaf allowed-faults {
    type uint8;
    description
      "This leaf is analogous to allowed-faults for a PTP Port,
       but applicable to this Link Port.";
    reference
      "14.18.24 of IEEE Std 802.1AS";
  }
  leaf-list pdelay-truncated-timestamps {
    type uinteger48;
    config false;
    description
      "This leaf is analogous to pdelay-truncated-timestamps for a
       PTP Port, but applicable to this Link Port.";
    reference
      "14.18.26 of IEEE Std 802.1AS";
  }
}

augment
  "/ptp-tt:ptp"+
```


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```
  "/ptp-tt:common-services"+
  "/ptp-tt:cmlds"+
  "/ptp-tt:ports"+
  "/ptp-tt:port" {
  description
    "Augment to add port-statistics-ds to IEEE Std 1588 Link
     Port.";
  container port-statistics-ds {
    description
      "This container is analogous to port-statistics-ds for a PTP
       Port, but applicable to this Link Port.";
    reference
      "14.19 of IEEE Std 802.1AS";
    leaf rx-pdelay-req-count {
      type yang:counter32;
      config false;
      description
        "This leaf is analogous to rx-pdelay-req-count for a PTP
         Port, but applicable to this Link Port.";
      reference
        "14.19.2 of IEEE Std 802.1AS";
    }
    leaf rx-pdelay-resp-count {
      type yang:counter32;
      config false;
      description
        "This leaf is analogous to rx-pdelay-resp-count for a PTP
         Port, but applicable to this Link Port.";
      reference
        "14.19.3 of IEEE Std 802.1AS";
    }
    leaf rx-pdelay-resp-follow-up-count {
      type yang:counter32;
      config false;
      description
        "This leaf is analogous to rx-pdelay-resp-follow-up-count
         for a PTP Port, but applicable to this Link Port.";
      reference
        "14.19.4 of IEEE Std 802.1AS";
    }
    leaf rx-packet-discard-count {
      type yang:counter32;
      config false;
      description
        "This leaf is analogous to rx-packet-discard-count for a
         PTP Port, but applicable to this Link Port.";
      reference
        "14.19.5 of IEEE Std 802.1AS";
    }
    leaf pdelay-allowed-lost-exceeded-count {
      type yang:counter32;
      config false;
      description
        "This leaf is analogous to
         pdelay-allowed-lost-exceeded-count for a PTP Port, but
         applicable to this Link Port.";
      reference
        "14.19.6 of IEEE Std 802.1AS";
    }
```


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```
      leaf tx-pdelay-req-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-req-count for a
           PTP Port, but applicable to this Link Port.";
        reference
          "14.19.7 of IEEE Std 802.1AS";
      }
      leaf tx-pdelay-resp-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-resp-count for a
           PTP Port, but applicable to this Link Port.";
        reference
          "14.19.8 of IEEE Std 802.1AS";
      }
      leaf tx-pdelay-resp-follow-up-count {
        type yang:counter32;
        config false;
        description
          "This leaf is analogous to tx-pdelay-resp-follow-up-count
           for a PTP Port, but applicable to this Link Port.";
        reference
          "14.19.9 of IEEE Std 802.1AS";
      }
    }
  }

  augment
    "/ptp-tt:ptp"+
    "/ptp-tt:common-services"+
    "/ptp-tt:cmlds"+
    "/ptp-tt:ports"+
    "/ptp-tt:port" {
    description
      "Augment to add asymmetry-measurement-mode-ds to IEEE
       Std 1588 Link Port.";
    container asymmetry-measurement-mode-ds {
      description
        "This container is analogous to
         asymmetry-measurement-mode-ds for a PTP Port, but
         applicable to this Link Port.";
      reference
        "14.20 of IEEE Std 802.1AS";
      leaf enabled {
        type boolean;
        description
          "This leaf is analogous to
           asymmetry-measurement-mode-ds.enabled for a PTP Port, but
           applicable to this Link Port.";
        reference
          "14.20.2 of IEEE Std 802.1AS";
      }
    }
  }
}
```


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17.6.2 Module ieee802-dot1as-hs.yang

```
module ieee802-dot1as-hs {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AS:yang:ieee802-dot1as-hs";
  prefix dot1as-hs;

  import ietf-yang-types {
    prefix yang;
  }
  import ieee1588-ptp-tt {
    prefix ptp-tt;
  }
  import ieee802-dot1as-gptp {
    prefix dot1as-gptp;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://ieee802.org/1/
     WG-EMail: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
              Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              445 Hoes Lane
              Piscataway, NJ 08854
              USA

     E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that control hot standby systems as
     specified in Clause 18 of IEEE Std 802.1AS-2025 and
     drift tracking features as specified in 11.4.4.4 of
     IEEE Std 802.1AS-2025.

     References in this YANG module to IEEE Std 802.1AS are to
     IEEE Std 802.1AS-2025.

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     This version of this YANG module is part of IEEE Std
     802.1AS-2025; see the standard itself for full legal
     notices.";

  revision 2025-12-10 {
    description
      "Published as part of IEEE Std 802.1AS-2025.";
    reference
      "IEEE Std 802.1AS - Timing and Synchronization for
       Time-Sensitive Applications: IEEE Std 802.1AS-2025.";
  }

  feature hot-standby {
    description
      "This feature indicates that the device supports the
       hot-standby functionality.";
  }
```


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```
typedef hot-standby-system-state-type {
  type enumeration {
    enum init {
      value 0;
      description
        "Initialization after the HotStandbySystem powers on and
         is enabled. In this state, the system is waiting for
         both PTP Instances to synchronize.";
    }
    enum redundant {
      value 1;
      description
        "Both PTP Instances are synchronized according to the
         requirements of the respective application or profile
         standard. Time synchronization is redundant.";
    }
    enum not-redundant {
      value 2;
      description
        "One PTP Instance is synchronized, and the other
         PTP Instance is faulted (not synchronized). Time
         synchronization continues to meet the requirements of
         the respective application or profile standard. Time
         synchronization is not redundant.";
    }
    enum out-of-sync {
      value 3;
      description
        "The HotStandbySystem is adjusting phase/frequency of
         its local time using the data stored while the system
         was in the REDUNDANT or NOT_REDUNDANT state, but the
         local time will eventually drift relative to other
         time-aware systems. During OUT_OF_SYNC state, time
         synchronization might not meet the requirements of the
         respective application or profile standard.";
    }
  }
  description
    "Possible states of the hot-standby system.";
}

typedef nrr-comp-method-type {
  type enumeration {
    enum sync {
      value 0;
      description
        "If the value is Sync and driftTrackingTlvSupport is
         TRUE, neighborRateRatio is populated with the value of
         nrrSync whenever a new value of nrrSync is computed.";
    }
    enum pdelay {
      value 1;
      description
        "If the value is Pdelay or if driftTrackingTlvSupport
         is FALSE, neighborRateRatio is populated with the value
         of nrrPdelay whenever a new value of nrrPdelay is
         computed.";
    }
```


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```
  }
  description
    "typdedef for nrrCompMethod.";
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:parent-ds" {
  description
    "Augment IEEE Std 1588 parentDS.";
  leaf gm-present {
    type boolean;
    config false;
    description
      "The value of gmPresent is set equal to the value of the
       global variable gmPresent.  This parameter indicates to the
       ClockTarget whether a Grandmaster PTP Instance is
       present.";
    reference
      "14.4.8 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance" {
  description
    "Augment IEEE Std 1588 instanceList.";
  container ptp-instance-sync-ds {
    description
      "The ptpInstanceSyncDS describes the synchronization status
       of the PTP Instance.";
    reference
      "14.8 of IEEE Std 802.1AS";
    leaf is-synced {
      type boolean;
      config false;
      description
        "The value of the global variable isSynced.";
      reference
        "14.8.2 of IEEE Std 802.1AS";
    }
    leaf offset-from-time-transmitter-max {
      type ptp-tt:time-interval;
      description
        "The value is the threshold for
         offsetFromTimeTransmitter, below which the PTP Instance
         is considered to be synchronized.";
      reference
        "14.8.3 of IEEE Std 802.1AS";
    }
    leaf rx-sync-count-time-receiver-p-thresh {
      type uint32;
      description
        "The value of rxSyncCountTimeReceiverPThresh is the
         threshold for rxSyncCountTimeReceiverP, above which
         the PTP Instance is considered to be synchronized.";
      reference
        "14.8.4 of IEEE Std 802.1AS";
    }
    leaf offset-max-exceeded-count-thresh {
      type uint32;
```


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```
      description
        "The value of offsetMaxExceededCountThresh is the
         threshold for the number of consecutive exceedances of
         offsetFromTimeTransmitterMax by
         offsetFromTimeTransmitter, at which isSynced is no
         longer TRUE.";
      reference
        "14.8.5 of IEEE Std 802.1AS";
    }
    leaf offset-max-met-count-thresh {
      type uint32;
      description
        "The value of offsetMaxMetCountThresh is the threshold
         for the number of consecutive occurrences of
         offsetFromTimeTransmitter being within
         offsetFromTimeTransmitterMax, at which isSynced is
         changed to TRUE if it currently is FALSE.";
      reference
        "14.8.6 of IEEE Std 802.1AS";
    }
  }
  container drift-tracking-ds {
    description
      "The driftTrackingDS contains managed objects that are used
       to enable or disable the optional Drift_Tracking TLV.";
    reference
      "14.9 of IEEE Std 802.1AS";
    leaf drift-tracking-tlv-support {
      type boolean;
      description
        "The value of driftTrackingTlvSupport indicates whether
         the Drift_Tracking TLV is enabled or disabled.";
      reference
        "14.9.2 of IEEE Std 802.1AS";
    }
    leaf sync-grandmaster-identity {
      type ptp-tt:clock-identity;
      config false;
      description
        "The value of syncGrandmasterIdentity as determined by
         the MDSyncSend state machine.";
      reference
        "14.9.3 of IEEE Std 802.1AS";
    }
    leaf sync-steps-removed {
      type uint32;
      config false;
      description
        "The value of syncStepsRemoved as determined by the
         MDSyncSend state machine.";
      reference
        "14.9.4 of IEEE Std 802.1AS";
    }
    leaf rate-ratio-drift {
      type int32;
      config false;
      description
        "This value is equal to (RRdrift - 1.0) x (2^41),
         truncated to the next smaller signed integer, where
```


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```
          RRdrift is the measured estimate of the rate of change
          per second of the ratio of the frequency of the
          Grandmaster Clock to the frequency of the local clock
          entity in the PTP Instance that sends the message.";
        reference
          "14.9.5 of IEEE Std 802.1AS";
      }
    }
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:ports/ptp-tt:port/ptp-tt:port-ds" {
    description
      "Augment IEEE Std 1588 commonServices.";
    leaf gptp-capable-state-machines-enabled {
      type boolean;
      description
        "A Boolean that is used to enable or disable the
         GptpCapableTransmit, GptpCapableReceive, and
         GptpCapableIntervalSetting state machines.";
      reference
        "14.10.56 of IEEE Std 802.1AS";
    }
    leaf nrr-pdelay {
      type int32;
      config false;
      description
        "The value is an estimate of the ratio of the frequency of
         the LocalClock entity of the time-aware system at the
         other end of the link attached to this Link Port, to the
         frequency of the LocalClock entity of this time-aware
         system. nrrPdelay is expressed as the fractional frequency
         offset stored in the global variable nrrPdelay multiplied
         by 2^41, i.e., the quantity (nrrPdelay - 1.0)(2^41).";
      reference
        "14.10.57 of IEEE Std 802.1AS";
    }
    leaf nrr-sync {
      type int32;
      config false;
      description
        "The value is an estimate of the ratio of the frequency of
         the LocalClock entity of the time-aware system at the
         other end of the link attached to this Link Port, to the
         frequency of the LocalClock entity of this time-aware
         system. nrrSync is expressed as the fractional frequency
         offset stored in the global variable nrrSync multiplied by
         2^41, i.e., the quantity (nrrSync - 1.0)(2^41).";
      reference
        "14.10.58 of IEEE Std 802.1AS";
    }
    leaf nrr-comp-method {
      type nrr-comp-method-type;
      description
        "An Enumeration that takes on the values sync and pdelay to
         indicate the source of the value of neighborRateRatio.";
      reference
        "14.10.59 of IEEE Std 802.1AS";
    }
```


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```
  leaf as-capable-across-domains {
    type boolean;
    config false;
    description
      "This leaf is true when this PTP Port detects proper
       exchange of Pdelay messages.";
    reference
      "14.10.60 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
      + "/ptp-tt:ports/ptp-tt:port"
      + "/dot1as-gptp:port-statistics-ds" {
  description
    "Augment IEEE Std 802.1AS PortStatisticsDS.";
  leaf rx-sync-count-time-receiver-p {
    type yang:counter32;
    config false;
    description
      "This counter increments whenever time synchronization
       information is received on a PTP Port when its port
       state is TimeReceiverPort.";
    reference
      "14.12.20 of IEEE Std 802.1AS";
  }
}

augment "/ptp-tt:ptp/ptp-tt:common-services" {
  description
    "Augment IEEE Std 1588 commonServices.

     IEEE Std 802.1AS-2025 specifies nrrPdelay
     (nrr-pdelay), which is provided in YANG as the
     semantically equivalent node in ieee1588-ptp-tt named
     scaled-neighbor-rate-ratio (in link-port-ds).";
  container hot-standby-service {
    if-feature "hot-standby";
    description
      "The hotStandbyService structure contains the
       hotStandbySystemList, which is a list of instances of the
       Hot Standby Service.";
    reference
      "14.21 of IEEE Std 802.1AS";
    list hot-standby-system {
      key "hot-standby-system-index";
      description
        "List of instances of the Hot Standby Service";
      leaf hot-standby-system-index {
        type uint8;
        description
          "Index for the hot-standby system.";
      }
      container hot-standby-system-ds {
        description
          "The hotStandbySystemDS describes the attributes of the
           respective instance of the Hot Standby Service.";
        reference
          "14.21 of IEEE Std 802.1AS";
```


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```
        leaf primary-ptp-instance-index {
          type uint32;
          description
            "The value of primaryPtpInstanceIndex is the index of
             the primary PTP Instance associated with this
             hotStandbySystem instance.";
          reference
            "14.21.2 of IEEE Std 802.1AS";
        }
        leaf secondary-ptp-instance-index {
          type uint32;
          description
            "The value of secondaryPtpInstanceIndex is the index
             of the secondaryPTP Instance associated with this
             hotStandbySystem instance.";
          reference
            "14.21.3 of IEEE Std 802.1AS";
        }
        leaf hot-standby-system-enable {
          type boolean;
          description
            "The value is the hotStandbySystemEnable attribute of
             the HotStandbySystem entity.";
          reference
            "14.21.4 of IEEE Std 802.1AS";
        }
        leaf hot-standby-system-state {
          type hot-standby-system-state-type;
          config false;
          description
            "The value of hotStandbySystemState is the state of
             the hotStandbySystem, i.e., the value of the global
             variable hotStandbySystemState.";
          reference
            "14.21.5 of IEEE Std 802.1AS";
        }
        leaf hot-standby-system-split-functionality {
          type boolean;
          description
            "If the value is TRUE, the optional split
             functionality is used. If the value is FALSE, the
             optional split functionality is not used.";
          reference
            "14.21.6 of IEEE Std 802.1AS";
        }
        leaf primary-secondary-offset {
          type dot1as-gptp:scaled-ns;
          config false;
          description
            "The absolute value of the difference between the
             clockTimeTransmitterTimes of the primary and
             secondary PTP Instances.";
          reference
            "14.21.7 of IEEE Std 802.1AS";
        }
        leaf primary-secondary-offset-thresh {
          type dot1as-gptp:scaled-ns;
          description
            "The threshold for
```


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```
            hotStandbySystemDS.primarySecondaryOffset, above
            which the hotStandbySytemState transitions from
            REDUNDANT to NOT_REDUNDANT, or does not transition
            from NOT_REDUNDANT or OUT_OF_SYNC to REDUNDANT even
            if other conditions for these transitions are
            satisfied.";
          reference
            "14.21.8 of IEEE Std 802.1AS";
        }
        leaf hot-standby-system-log-sync-time-thresh {
          type int8;
          description
            "The value of hotStandbySystemLogSyncTimeThresh is
             the logarithm to base 2 of the time interval, in
             seconds, after which the hotStandbySystem
             transitions from the OUT_OF_SYNC state to either the
             NOT-REDUNDANT or REDUNDANT state, or from the
             NOT_REDUNDANT to the REDUNDANT state, if all other
             conditions for the respective transition are met.
             The value -128 means that the transition time is
             zero, i.e., the transition occurs immediately.";
          reference
            "14.21.9 of IEEE Std 802.1AS";
        }
      }
      container hot-standby-system-description-ds {
        description
          "The hotStandbySystemDescriptionDS contains descriptive
           information for the respective instance of the Hot
           Standby Service.";
        reference
          "14.22 of IEEE Std 802.1AS";
        leaf user-description {
          type string {
            length "0..128";
          }
          description
            "Configurable description of the hot standby system.";
          reference
            "14.22.2 of IEEE Std 802.1AS";
        }
      }
    }
  }
}

```

17.6.3 Module ieee802-dot1as-ftt.yang

```
module ieee802-dot1as-ftt {
  yang-version 1.1;
  namespace "urn:ieee:std:802.1AS:yang:ieee802-dot1as-ftt";
  prefix dot1as-ftt;

  import ieee1588-ptp-tt {
    prefix ptp-tt;
  }
```


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```
organization
  "IEEE 802.1 Working Group";
contact
  "WG-URL: http://www.ieee802.org/1/
   WG-EMail: stds-802-1-L@ieee.org
   Contact: IEEE 802.1 Working Group Chair
   Postal: C/O IEEE 802.1 Working Group
   IEEE Standards Association
   445 Hoes Lane
   Piscataway, NJ 08854
   USA
   E-mail: stds-802-1-chairs@ieee.org";
description
  "Management objects that control fault-tolerant timing with
   time integrity (FTT) entities as specified in
   IEEE Std 802.1ASed-2026.

   References in this YANG module are to IEEE Std 802.1AS-2025
   as amended by IEEE Std 802.1ASed-2026.

   Copyright (C) IEEE (2026).
   This version of this YANG module is part of IEEE Std 802.1AS;
   see the standard itself for full legal notices.";

revision 2026-02-11 {
  description
    "Published as part of IEEE Std 802.1ASed-2026.
     Initial version.";
  reference
    "IEEE Std 802.1AS - Timing and Synchronization for
     Time-Sensitive Applications: IEEE Std 802.1AS-2025,
     IEEE Std 802.1ASed-2026.";
}

feature ftt {
  description
    "This feature indicates that the device supports the FTT
     service.";
}

typedef ptp-instance-ref {
  type leafref {
    path "/ptp-tt:ptp/ptp-tt:instances/ptp-tt:instance"
       + "/ptp-tt:instance-index";
  }
  description
    "This type is used by data models that need to reference
     interfaces.";
}

typedef ftt-output-trust-state-type {
  type enumeration {
    enum NOT-TRUSTED {
      value 0;
      description
        "Integrity of time synchronization function is not
         achieved.";
    }
```


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```
    enum TIME-TRUSTED {
      value 1;
      description
        "Integrity of time synchronization is achieved.";
    }
    enum FREQ-TRUSTED {
      value 2;
      description
        "Only the integrity of the rate-of-change of time is
         achieved.";
    }
    enum INIT {
      value 3;
      description
        "System is waiting to be invoked and has not determined a
         trust state.";
    }
  }
  description
    "The fttOutputTrustState type is an enumerated value that
     holds the trust state of the FTT entity.";
  reference
    "20.3.2.1 of IEEE Std 802.1AS.";
}

typedef ftt-input-trust-state-type {
  type enumeration {
    enum INACTIVE {
      value 0;
      description
        "The FTT input is not considered to be providing time
         values.";
    }
    enum ACTIVE {
      value 1;
      description
        "The FTT input is actively providing time values.";
    }
    enum TRUSTED {
      value 2;
      description
        "The FTT input is actively providing trusted time
         values.";
    }
    enum UNTRUSTED {
      value 3;
      description
        "The FTT input is actively providing untrusted time
         values.";
    }
  }
  description
    "The fttInputTrustState type is an enumerated value that
     holds the trust state of an FTT input.";
  reference
    "20.3.2.2 of IEEE Std 802.1AS.";
}
```


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```
typedef uint48 {
  type uint64 {
    range "0..281474976710655";
  }
  description
    "Unsigned 48-bit integer.";
}

grouping fault-tolerant-timing-group {
  description
    "Management of a single FTT entity.";
  reference
    "14.23 of IEEE Std 802.1AS.";
  leaf ftt-max-inputs {
    type uint8 {
      range "0..16";
    }
    config false;
    description
      "Maximum number of input ClockTarget interfaces available
       on the FTT entity.";
    reference
      "14.23.2 of IEEE Std 802.1AS.";
  }
  leaf ftt-num-inputs {
    type uint8;
    description
      "The number of enabled input ClockTarget interfaces on
       the FTT entity.";
    reference
      "14.23.3 of IEEE Std 802.1AS.";
  }
  leaf ftt-max-dtsfs {
    type uint8 {
      range "0..126";
    }
    config false;
    description
      "Maximum number of DTSF instances available in the FTT
       entity.";
    reference
      "14.23.4 of IEEE Std 802.1AS.";
  }
  leaf ftt-num-dtsfs {
    type uint8;
    description
      "Number of DTSF instances enabled in the FTT entity.";
    reference
      "14.23.5 of IEEE Std 802.1AS.";
  }
  leaf ftt-dtsf-max-inputs {
    type uint8 {
      range "0..16";
    }
    config false;
    description
      "Maximum number of input ClockTarget interfaces available
       on each DTSF instance in the FTT entity.";
    reference
```


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```
      "14.23.6 of IEEE Std 802.1AS.";
  }
  list ftt-inputs-to-tsf {
    key "tsf-instance-number";
    description
      "Mapping of FTT entity input ClockTarget interface index
       numbers to TSF instance numbers.";
    reference
      "14.23.7 of IEEE Std 802.1AS.";
    leaf tsf-instance-number {
      type uint8;
      description
        "The TSF instance number that the FTT entity's input
         index number is connected to.";
      reference
        "20.2.3 of IEEE Std 802.1AS.";
    }
    leaf assigned-inputs {
      type uint16;
      description
        "The bit map of FTT inputs that are assigned to the TSF
         instance. The most-significant bit corresponds to input
         index 16, the least-significant bit corresponds to input
         index 1. The number of inputs assigned to any DTSF
         instance (i.e., TSF instance with tsf-instance-number of
         1 or higher) cannot be greater than fttDtsfMaxInputs.";
      reference
        "14.23.7 of IEEE Std 802.1AS.";
    }
  }
  list ftt-ptp-instance-to-input {
    key "ftt-input-index";
    description
      "Mapping of PTP Instance index numbers to FTT entity input
       index numbers.";
    reference
      "14.23.8 of IEEE Std 802.1AS.";
    leaf ftt-input-index {
      type uint8;
      description
        "The FTT entity's input index number.";
      reference
        "20.2.1 of IEEE Std 802.1AS.";
    }
    leaf ptp-instance-index {
      type ptp-instance-ref;
      description
        "The instanceIndex number of the PTP Instance that is
         connected to the FTT input.";
      reference
        "14.1.1 of IEEE Std 802.1AS-2020.";
    }
  }
  list ftt-hyst {
    key "ftt-input-index";
    description
      "Hysteresis values to be added to fttMaxAs for pairs of
       input ClockTarget interfaces.";
    reference
```


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```
      "14.23.9 of IEEE Std 802.1AS.";
    leaf ftt-input-index {
      type uint8;
      description
        "The first input index number.";
      reference
        "20.2.1 of IEEE Std 802.1AS.";
    }
    list hyst-list {
      key "ftt-input-index";
      description
        "Hysteresis values for pairs of input ClockTarget
         interfaces.";
      reference
        "14.23.9 of IEEE Std 802.1AS.";
      leaf ftt-input-index {
        type uint8;
        description
          "The second input index number.";
        reference
          "20.2.1 of IEEE Std 802.1AS.";
      }
      leaf hyst-value {
        type ptp-tt:time-interval;
        default "0";
        description
          "Hysteresis value to be added to fttMaxAs for the pair
           of input interfaces.";
        reference
          "14.23.9 of IEEE Std 802.1AS.";
      }
    }
  }
  list ftt-max-as {
    key "ftt-input-index";
    description
      "Maximum expected skew between pairs of input ClockTarget
       interfaces.";
    reference
      "14.23.10 of IEEE Std 802.1AS.";
    leaf ftt-input-index {
      type uint8;
      description
        "The first input index number.";
      reference
        "20.2.1 of IEEE Std 802.1AS.";
    }
    list max-as-list {
      key "ftt-input-index";
      description
        "Maximum expected skew between pairs of input ClockTarget
         interfaces.";
      reference
        "14.23.10 of IEEE Std 802.1AS.";
      leaf ftt-input-index {
        type uint8;
        description
          "The second input index number.";
        reference
```


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```
          "20.2.1 of IEEE Std 802.1AS.";
      }
      leaf max-as-value {
        type ptp-tt:time-interval;
        default "0";
        description
          "Maximum expected skew between the pair of input
           interfaces when those time values are not faulty.";
        reference
          "14.23.10 of IEEE Std 802.1AS.";
      }
    }
  }
  list ftt-sel-change-thresh {
    key "tsf-instance-number";
    description
      "Time difference change threshold used by TSF instances.";
    reference
      "14.23.11 of IEEE Std 802.1AS.";
    leaf tsf-instance-number {
      type uint8;
      description
        "The TSF instance number.";
      reference
        "20.2.3 of IEEE Std 802.1AS.";
    }
    leaf change-threshold {
      type ptp-tt:time-interval;
      description
        "The TimeInterval threshold used by the TSF instance.";
      reference
        "14.23.11 of IEEE Std 802.1AS.";
    }
  }
  leaf ftt-rr-thresh {
    type uint32;
    default "0";
    description
      "Maximum rate-ratio offset magnitude deemed acceptable for
       FREQ_TRUSTED state.";
    reference
      "14.23.12 of IEEE Std 802.1AS.";
  }
  leaf ftt-rr-thresh-std-dev {
    type uint32;
    default "0";
    description
      "Maximum standard deviation of rate-ratio offset deemed
       acceptable for FREQ_TRUSTED state.";
    reference
      "14.23.13 of IEEE Std 802.1AS.";
  }
  list ftt-use-nq {
    key "tsf-instance-number";
    description
      "Specifies whether a TSF instance uses the NQ input when no
       trusted time is available.
       If ftt-use-nq is TRUE, the use of the NQ input is enabled.
       If ftt-use-nq is FALSE, the use of the NQ input is not
```


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```
       enabled. The default value is implementation-specific.";
    reference
      "14.23.14 and 20.2.4 of IEEE Std 802.1AS.";
    leaf tsf-instance-number {
      type uint8;
      description
        "The TSF instance number.";
      reference
        "20.2.3 of IEEE Std 802.1AS.";
    }
    leaf use-nq {
      type boolean;
      description
        "Whether the TSF instance uses the NQ input.";
      reference
        "14.23.14 of IEEE Std 802.1AS.";
    }
  }
  leaf ftt-use-ftt-clk {
    type boolean;
    description
      "Defines whether the FTT_CLK frequency is used as a
       reference for time integrity.
       If fttUseFTTClk is TRUE, the FTT_CLK frequency is used as
       a reference for checking time integrity.
       If fttUseFTTClk is FALSE, the FTT_CLK frequency is not
       used as a reference for checking time integrity.
       The default value is implementation-specific.";
    reference
      "14.23.15 of IEEE Std 802.1AS.";
  }
  list ftt-tsf-sel-input {
    key "tsf-instance-number";
    config false;
    description
      "The index number of the FTT input selected by each TSF
       instance.";
    reference
      "14.23.16 and 20.3.2.3 of IEEE Std 802.1AS.";
    leaf tsf-instance-number {
      type uint8;
      description
        "The TSF instance number.";
      reference
        "20.2.3 of IEEE Std 802.1AS.";
    }
    leaf selected-ftt-input-index {
      type uint8;
      description
        "FTT input index that is selected by the TSF instance.";
      reference
        "14.23.16 of IEEE Std 802.1AS.";
    }
  }
  leaf ftt-sel-instance {
    type uint32;
    config false;
    description
      "Instance index of the PTP Instance selected as the trusted
```


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```
        time source.";
      reference
        "14.23.17 of IEEE Std 802.1AS.";
    }
    list ftt-tsf-sel-change-cnt {
      key "tsf-instance-number";
      config false;
      description
        "The number of times each TSF has changed its selection.";
      reference
        "14.23.18 of IEEE Std 802.1AS.";
      leaf tsf-instance-number {
        type uint8;
        description
          "The TSF instance number.";
        reference
          "20.2.3 of IEEE Std 802.1AS.";
      }
      leaf sel-change-cnt {
        type uint16;
        description
          "The number of times the TSF instance has changed its
           selection.";
        reference
          "14.23.18 of IEEE Std 802.1AS.";
      }
    }
    list ftt-input-trust-state {
      key "ftt-input-index";
      config false;
      description
        "Trust state of all FTT inputs.";
      reference
        "14.23.19 and 20.3.2.2 of IEEE Std 802.1AS.";
      leaf ftt-input-index {
        type uint8;
        description
         "The FTT input index.";
        reference
          "20.2.1 of IEEE Std 802.1AS.";
      }
      leaf ftt-input-trust-state {
        type ftt-input-trust-state-type;
        description
          "Trust state of the FTT input.";
        reference
          "14.23.19 of IEEE Std 802.1AS.";
      }
    }
    leaf ftt-trust-state {
      type ftt-output-trust-state-type;
      default "NOT-TRUSTED";
      config false;
      description
        "The trust state of the FTT entity.";
      reference
        "14.23.20 and 20.3.2.1 of IEEE Std 802.1AS.";
    }
    leaf ftt-state-change-cnt {
```


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```
    type uint16;
    default "0";
    config false;
    description
      "Number of times the FTT entity has changed its trust
       state.";
    reference
      "14.23.21 of IEEE Std 802.1AS.";
  }
}

augment "/ptp-tt:ptp/ptp-tt:common-services" {
  description
    "Augment IEEE Std 1588 commonServices with ftt-service.";
  container ftt-service {
    if-feature "ftt";
    description
      "The FTT service structure contains the fttEntityList,
       which is a list of FTT entity instances in the FTT
       service.";
    reference
      "14.1.1 of IEEE Std 802.1AS.";
    list ftt-entity {
      key "ftt-entity-index";
      description
        "Indexed list of FTT entities in the FTT service.";
      leaf ftt-entity-index {
        type uint8;
        description
          "Index of the FTT entity instance.";
      }
      container ftt-entity-ds {
        description
          "The fttEntityDS describes the attributes of the
           respective FTT entity instance in the FTT service.";
        reference
          "14.23 of IEEE Std 802.1AS.";
        uses fault-tolerant-timing-group;
      }
      container ftt-entity-description-ds {
        description
          "The fttEntityDescriptionDS contains descriptive
           information for the respective FTT entity instance
           in the FTT service.";
        reference
          "14.24 of IEEE Std 802.1AS.";
        leaf user-description {
          type string {
            length "0..128";
          }
          description
            "Configuration description of the FTT entity
             instance.";
          reference
            "14.24.2 of IEEE Std 802.1AS.";
        }
      }
    }
  }
```


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```
  }
}


```


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18. Provisions for hot standby on the media-independent layer

18.1 General

Hot standby extends the media-dependent layer by the following:

— A function that transforms the synchronized times of two generalized precision time protocol (gPTP) domains into one synchronized time for use by applications;
— A function that directs the synchronized time of one gPTP domain into a different gPTP domain; and
— Mechanisms that determine whether a gPTP domain has sufficient quality to be used for hot standby.

For time synchronization using hot standby, two distinct domains are statically configured in the network and the best timeTransmitter clock algorithm (BTCA) is disabled for these domains. When hot standby is used for redundancy, a time-aware system operates two PTP Instances simultaneously, each in its own domain. One of the domains is considered the primary domain, and the other domain is considered the secondary domain. A time-aware system that has a primary domain and a secondary domain available uses the primary domain via the associated PTP Instance. If the primary domain becomes unavailable (e.g., due to temporary or permanent failure of a physical link) and the secondary domain is still available to the time-aware system, the time-aware system begins using the secondary domain.

Examples of hot standby are shown in 7.3.3 of this standard.

18.2 Overview

Figure 18-1 provides a model of hot standby for time synchronization.

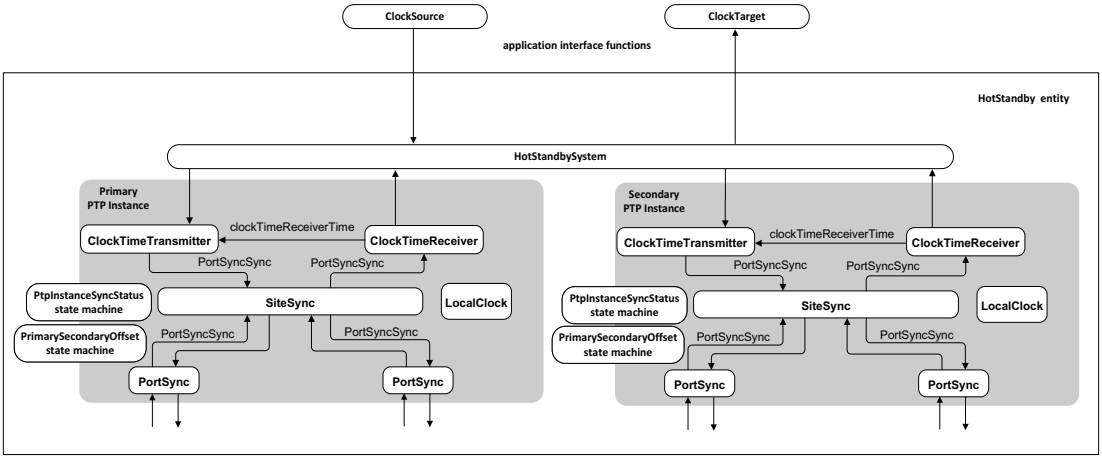


Figure 18-1—Model for hot standby entity in a time-aware system, and its interfaces to higher-layer applications

Each PTP Instance has a corresponding PtpInstanceSyncStatus state machine (see 18.4). The PtpInstanceSyncStatus state machine monitors its PTP Instance to determine whether it is synchronized according to the requirements of the respective application or profile standard.

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There is one HotStandbySystem entity, which interacts with the primary and secondary PTP Instances via the HotStandbySystem state machine (see 18.5) in order to provide a single value of time to the ClockTarget (see Clause 9). This single value of time is based on the redundant values of time provided by either the primary PTP Instance, or the secondary PTP Instance, or both. There is one HotStandbySystem for both PTP Instances that use hot standby. The primary and secondary PTP Instances use the same timescale (see 8.2).

The ClockSource entity is used to transfer the source of time when at least one of the PTP Instances is grandmaster-capable. The ClockSource and its application interfaces are analogous to the ClockSource specified in Clause 9. The ClockSource might be present if the PTP Instance is capable of becoming a Grandmaster PTP Instance.

There is one LocalClock entity for each PTP Instance. Therefore, Figure 18-1 shows two LocalClock entities.

NOTE—The LocalClock entities can be traceable to the same oscillator or to different oscillators.

18.3 PTP Instance configuration

PTP Instances that support hot standby are configured as follows:

a) The two PTP Instances are enabled using domainNumbers as specified by the respective application or profile standard;

b) For both PTP Instances, externalPortConfigurationEnabled is set to TRUE;

c) If a PTP Instance (primary or secondary) is grandmaster, externalPortConfigurationPortDS.desiredState is configured to TimeTransmitterPort or PassivePort for all PTP Ports (portNumber 1 and higher); otherwise, externalPortConfiguration.desiredState is configured to TimeReceiverPort for one PTP Port, and is configured to TimeTransmitterPort or PassivePort for other PTP Ports; and

d) Each PTP Instance shall have a corresponding PtpInstanceSyncStatus state machine (see 18.4).

18.4 PtpInstanceSyncStatus state machine

This state machine operates within the PTP Instance, to determine whether it is synchronized according to the requirements of the respective application or profile standard (see 3.25).

18.4.1 PtpInstanceSyncStatus state machine global variables

The following variables are used in the state diagram in Figure 18-2 (in 18.4.4):

18.4.1.1 isSynced: The synchronization status of the PTP Instance. The variable is a Boolean, and is TRUE if synchronization is sufficient according to the operation of the PtpInstanceSyncStatus state machine and FALSE otherwise. An application or profile standard can set the offsetFromTimeTransmitterMax (see 18.4.1.3), rxSyncCountTimeReceiverPThresh (see 18.4.1.5), offsetMaxExceededCountThresh (see 18.4.1.7), and offsetMaxMetCountThresh (see 18.4.1.9) according to its requirements.

18.4.1.2 offsetFromTimeTransmitter: The value of the managed object currentDS.offsetFromTimeTransmitter (see 14.3.3).

18.4.1.3 offsetFromTimeTransmitterMax: The value of the managed object ptpInstanceSyncDS.offsetFromTimeTransmitterMax (see 14.8.3)

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18.4.1.4 rxSyncCountTimeReceiverP: The value of the managed object portStatisticsDS.rxSyncCountTimeReceiverP (see 14.12.20).

18.4.1.5 rxSyncCountTimeReceiverPThresh: The value of the managed object ptpInstanceSyncDS.rxSyncCountTimeReceiverPThresh (see 14.8.4)

18.4.1.6 offsetMaxExceededCount: The current number of consecutive exceedances of offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter (see 18.4.1.2). The data type of offsetMaxExceededCount is UInteger32.

18.4.1.7 offsetMaxExceededCountThresh: The threshold (see 14.8.5) for the number of consecutive exceedances of offsetFromTimeTransmitterMax (see 18.4.1.3) by offsetFromTimeTransmitter (see 18.4.1.2), at which isSynced for the PTP Instance is no longer TRUE (see 18.4.3.3).

18.4.1.8 offsetMaxMetCount: The current number of consecutive occurrences of offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax (see 18.4.1.3). The data type of offsetMaxMetCount is UInteger32.

18.4.1.9 offsetMaxMetCountThresh: The threshold (see 14.8.6) for the number of consecutive occurrences of offsetFromTimeTransmitter (see 18.4.1.2) being within offsetFromTimeTransmitterMax (see 18.4.1.3), at which isSynced for the PTP Instance is changed from FALSE to TRUE (see 18.4.3.3).

18.4.2 PtpInstanceSyncStatus state machine local variables

18.4.2.1 lastRxSyncCountTimeReceiverP: Holds the last value of rxSyncCountTimeReceiverP (see 18.4.1.4) that the state machine read for the TimeReceiverPort, as part of monitoring for a sync event message timeout.

18.4.3 PtpInstanceSyncStatus state machine functions

18.4.3.1 isGptpCapable(): This function returns a Boolean value that is TRUE when ptpPortEnabled (see 10.2.5.13) and asCapable (see 10.2.5.1) are TRUE, as determined by the function portIsCapable, for at least one PTP Port (i.e., port for which portNumber is not zero).

```
isGptpCapable()
{
        if (!instanceEnable)
                return (FALSE);
        for (int i = 1; i <= numberPorts; i++)
        // see 8.6.2.8 for numberPorts
        {
                if (portIsCapable (i))
                        return (TRUE);
        }
        return (FALSE);
}
```


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18.4.3.2 isGm(): This function determines if the PTP Instance is a Grandmaster PTP Instance (TRUE) or not (FALSE) by searching the selectedState array (see 10.2.4.20) for absence or presence of at least one value equal to TimeReceiverPort (see Table 10-2).

```
isGm()
{
        for (int i = 1; i <= numberPorts; i++)
        // see 8.6.2.8 for numberPorts
        {
          if (selectedState[i] == TimeReceiverPort)
            return (FALSE);
        }
        if (priority1 == 255)
            return (FALSE);

        return (TRUE);
}
```

18.4.3.3 portIsCapable(j): This function returns a Boolean value that is TRUE if the PTP Port whose portList index is j is enabled and asCapable is TRUE, and FALSE otherwise.

```
portIsCapable (j)
{
        if (ptpPortEnabled && asCapable)
                        return (TRUE);
                else
                        return (FALSE);
}
```

18.4.4 State diagram

The PtpInstanceSyncStatus state machine shall implement the function specified by the state diagram in Figure 18-2, the variables specified in 18.4.1 and 18.4.2, and the functions specified in 18.4.3.

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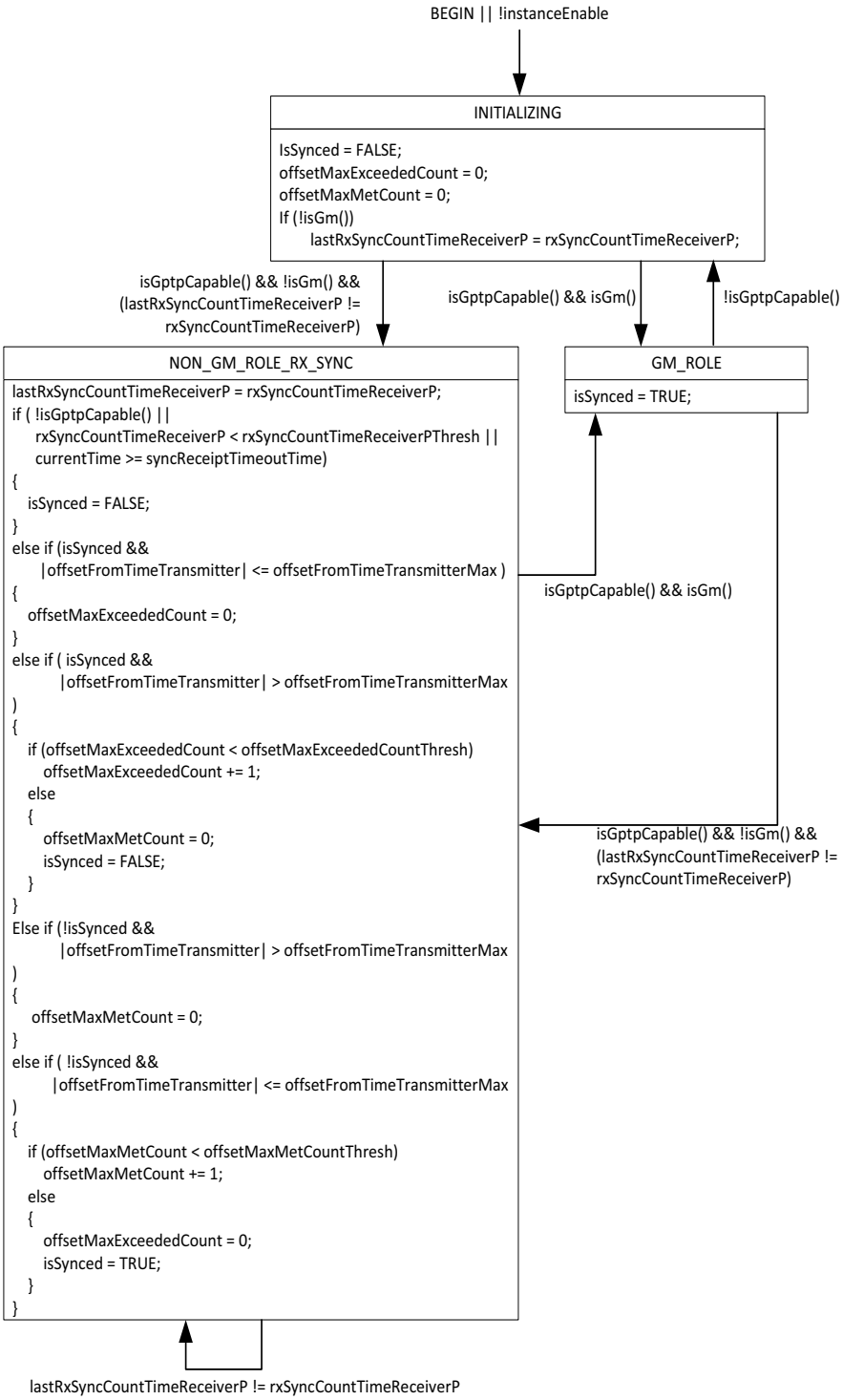


Figure 18-2—PtpInstanceSyncStatus state machine

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18.5 HotStandbySystem state machine

This state machine interacts with the primary and secondary PTP Instances in order to provide a single redundant time to the application.

18.5.1 HotStandbySystem state machine global variables

18.5.1.1 hotStandbySystemState: State of the HotStandbySystem. The variable is an enumeration that takes one of the following values:

a) INIT: Initialization after the HotStandbySystem powers on and is enabled. In this state, the system is waiting for both PTP Instances to synchronize.

b) REDUNDANT: Both PTP Instances are synchronized according to the requirements of the respective application or profile standard (see 3.25). Time synchronization is redundant.

NOTE—A PTP Instance is implicitly considered to be synchronized if it is a Grandmaster PTP Instance (see 18.4.3).

c) NOT_REDUNDANT: One PTP Instance is synchronized, and the other PTP Instance is faulted (not synchronized). Time synchronization continues to meet the requirements of the respective application or profile standard (see 3.25). Time synchronization is not redundant.

d) OUT_OF_SYNC: The HotStandbySystem can adjust phase/frequency of its local time using the data stored while the system was in the REDUNDANT or NOT_REDUNDANT state, but the local time will eventually drift relative to other time-aware systems. During OUT_OF_SYNC state, time synchronization might not meet the requirements of the respective application or profile standard (see 3.25).

18.5.1.2 hotStandbySystemEnable: A Boolean variable used to enable/disable the HotStandbySystem.

18.5.1.3 hotStandbySystemLogSyncTimeThresh: The value of the managed object hotStandbySystemLogSyncTimeThreshold (see 14.21.9) of the hotStandbySystemDS. The data type for hotStandbySystemLogSyncTimeThreshold is Integer8.

18.5.1.4 primarySecondaryOffset: The value of the managed object hotStandbySystemDS.primarySecondaryOffset (see 14.21.7), which is the absolute value of the difference between the clockTimeReceiverTime variables (see 10.2.4.3) of the primary and secondary PTP Instances.

18.5.2 HotStandbySystem state machine local variables

18.5.2.1 primary: Reference to the data sets of the PTP Instance configured to use the primary domainNumber as specified by the respective application or profile standard (see 3.25). This references an element of the instanceList specified in 14.1.

18.5.2.2 secondary: Reference to the data sets of the PTP Instance configured to use the secondary domainNumber as specified by the respective application or profile standard (see 3.25). This references an element of the instanceList specified in 14.1.

18.5.2.3 TEMP: A temporary variable used to reduce clutter in the state diagram (see Figure 18-3). The data type for TEMP is Integer16.

18.5.2.4 transitionTime: The value of currentTime (see 10.2.4.12) after which the hotStandbySystem transitions from the OUT_OF_SYNC state to either the NOT_REDUNDANT or REDUNDANT state, or from the NOT_REDUNDANT to the REDUNDANT state, if all other conditions for the respective transition are met. The data type for transitionTime is UScaledNs.

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18.5.2.5 primarySecondaryOffsetThresh: The value of the managed object hotStandbySystemDS.primarySecondaryOffsetThresh, which is the threshold for primarySecondaryOffset (see 18.5.1.4), above which the hotStandbySytemState transitions from REDUNDANT to NOT_REDUNDANT, or stays in NOT_REDUNDANT or OUT_OF_SYNC even if other conditions for transitioning to REDUNDANT are satisfied.

18.5.3 State diagram

The HotStandbySystem state machine shall implement the function specified by the state diagram in Figure 18-3, the variables specified in 18.5.1 and 18.5.2, and the state requirements specified in 18.6.

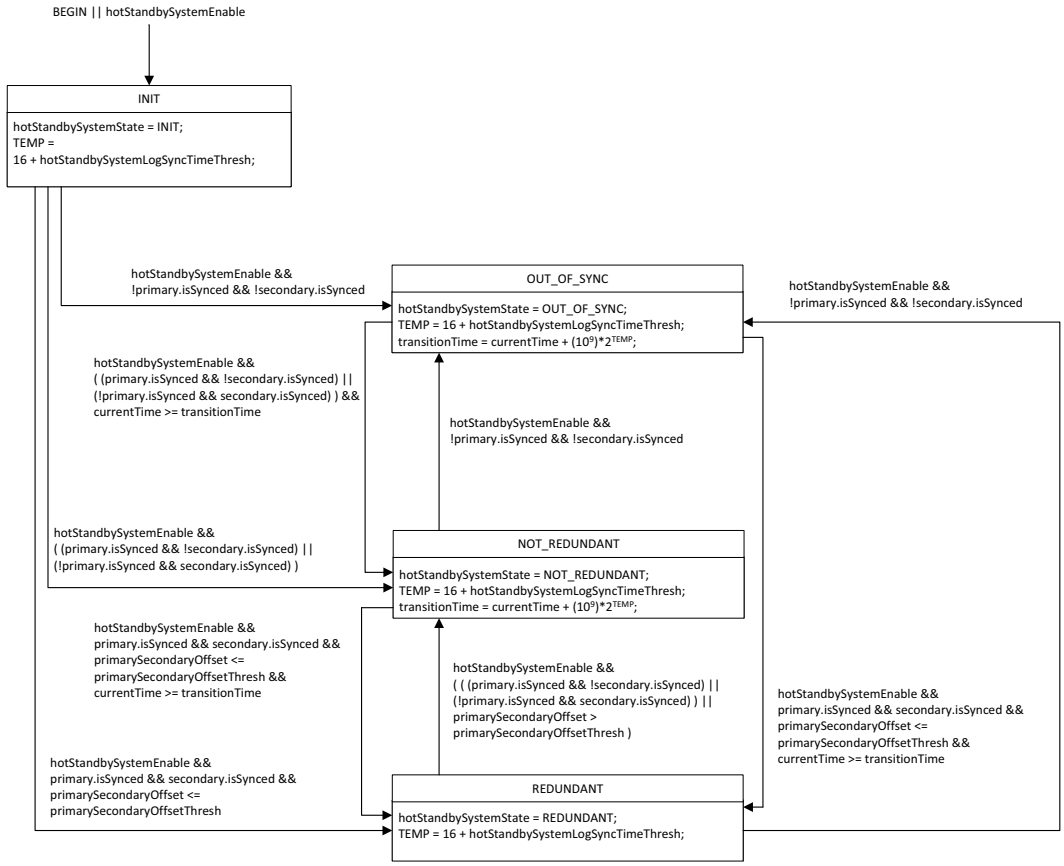


Figure 18-3—HotStandbySystem state machine

18.6 Information transfers of the HotStandbySystem entity associated with the HotStandbySystem state machine

18.6.1 Primary grandmaster

When the primary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort state), the HotStandbySystem entity shall transfer phase, frequency, clockSourceTimeBaseIndicator (see 10.2.4.8), clockSourceLastGmPhaseChange (see 10.2.4.10), and clockSourceLastGmFreqChange (see 10.2.4.11) from the ClockSource to the ClockTimeTransmitter of the primary PTP Instance

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(see Figure 18-1). If no external source of time is implemented, the ClockSource is equivalent to the LocalClock of this PTP Instance.

When the primary PTP Instance is grandmaster, there is no requirement for the HotStandbySystem entity to receive time from the ClockTimeReceiver of the secondary PTP Instance.

18.6.2 Secondary grandmaster

18.6.2.1 HotStandbySystem in REDUNDANT state

When the secondary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort state), and the hotStandbySystemState is REDUNDANT, the HotStandbySystem entity shall transfer phase, frequency, clockSourceTimeBaseIndicator (see 10.2.4.8), clockSourceLastGmPhaseChange (see 10.2.4.10), and clockSourceLastGmFreqChange (see 10.2.4.11) from the ClockTimeReceiver of the primary PTP Instance to the ClockTimeTransmitter of the secondary PTP Instance (see Figure 18-1). By using phase from the primary PTP Instance, the secondary grandmaster can maintain continuity in the event of a fault in the primary domain.

18.6.2.2 HotStandbySystem in NOT_REDUNDANT state

When the secondary PTP Instance is a Grandmaster PTP Instance (i.e., no PTP Port in TimeReceiverPort state), and the hotStandbySystemState is NOT_REDUNDANT, which means that isSynced for the primary PTP Instance is FALSE and isSynced for the secondary PTP Instance is TRUE, the HotStandbySystem entity shall transfer phase, frequency, clockSourceTimeBaseIndicator (see 10.2.4.8), clockSourceLastGmPhaseChange (see 10.2.4.10), and clockSourceLastGmFreqChange (see 10.2.4.11) from the ClockSource to the ClockTimeTransmitter of the secondary PTP Instance (see Figure 18-1). If no external source of time is implemented, the ClockSource is equivalent to the LocalClock of this PTP Instance.

The secondary PTP Instance is considered synchronized if it is a Grandmaster PTP Instance (see 18.4.4 and Figure 18-2). Since the hotStandbySystemState is NOT_REDUNDANT, this means that either the primary PTP Instance is not synchronized or primarySecondaryOffset exceeds primarySecondaryOffsetThresh. However, in the former case primarySecondaryOffset can exceed primarySecondaryOffsetThresh because in this case the primary and secondary PTP Instances receive timing from different sources. The result is that the HotStandbySystemState remains in the NOT_REDUNDANT state, even if the primary PTP Instance subsequently becomes synchronized again by the primary Grandmaster PTP Instance, because primarySecondaryOffset continues to exceed primarySecondaryOffsetThresh. This situation does not occur if it is the primary PTP Instance that is synchronized and the secondary PTP Instance that is not synchronized due to some fault condition. When the fault condition is cleared, the secondary Grandmaster PTP Instance ClockTimeTransmitter would receive timing from the primary PTP Instance ClockTimeReceiver (see 18.6.2.1) and, as a result, it would become synchronized again and the hotStandbySystemState would change to REDUNDANT.

The situation described in the previous paragraph is avoided if the optional split functionality (see 18.6.4) is used. In this case, timing is transferred directly from the secondary SiteSync entity to the primary SiteSync entity. This will synchronize the primary PTP Instance to the secondary PTP Instance; hence, the primarySecondaryOffset changes to within primarySecondaryOffsetThresh. The HotStandbySystemState changes to REDUNDANT and the secondary Grandmaster PTP Instance begins receiving timing from the primary PTP Instance. However, even if the optional split functionality is not used, an application can choose to re-initialize the secondary PTP Instance after the primary PTP Instance becomes synchronized. If this is done, the secondary Grandmaster PTP Instance receives timing from the primary PTP Instance ClockTimeReceiver after initialization and hotStandbySystemState is REDUNDANT.

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NOTE—If the secondary PTP Instance is re-initialized after the primary PTP Instance becomes synchronized, this can cause a jump in time when the secondary PTP Instance synchronizes to the primary PTP Instance.

On initialization, if the primary PTP Instance is synchronized, the secondary PTP Instance shall receive timing from the primary PTP Instance. When the conditions as specified the HotStandbySystem state machine (see Figure 18-3) are satisfied, the hotStandbySystemState machine transitions to the REDUNDANT state.

18.6.3 TimeReceiver HotStandbySystem

18.6.3.1 General

A HotStandbySystem entity is said to be a TimeReceiver HotStandbySystem if and only if neither PTP Instance is a Grandmaster PTP Instance.

18.6.3.2 HotStandbySystem in REDUNDANT state

When the HotStandbySystem entity is a timeReceiver (i.e., neither PTP Instance is a grandmaster), and hotStandbySystemState is REDUNDANT, the HotStandbySystem entity shall transfer phase, frequency, gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange (see 10.2.4.17) from the ClockTimeReceiver of the primary PTP Instance to the ClockTarget (application) if a ClockTarget is present.

18.6.3.3 HotStandbySystem in NOT_REDUNDANT state

18.6.3.3.1 Transfer of synchronized time to the ClockTarget

When the HotStandbySystem entity is a timeReceiver, and the HotStandbySystemState is in the NOT_REDUNDANT state, the HotStandbySystem entity shall transfer phase, frequency, gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange (see 10.2.4.17) from the ClockTimeReceiver of the synchronized PTP Instance (i.e., isSynced == TRUE) to the ClockTarget if a ClockTarget is present.

18.6.4 Split functionality

The HotStandbySystem entity may provide an optional split functionality. If the managed object hotStandbySystemDS.hotStandbySystemSplitFunctionality is set to TRUE, the split functionality is invoked. If the managed object hotStandbySystemDS.hotStandbySystemSplitFunctionality is set to FALSE, the split functionality is not invoked. The split functionality shall provide an interworking function (IWF) that transfers time synchronization information from the primary PTP Instance to the secondary PTP Instance when isSynced is FALSE (e.g., the timeReceiver port of the PTP Instance is in failure condition, see 18.4.1.1) for the secondary PTP Instance and isSynced is TRUE (see 18.4.1.1) for the primary PTP Instance, or from the secondary PTP Instance to the primary PTP Instance when isSynced is FALSE for the primary PTP Instance and isSynced is TRUE for the secondary PTP Instance. The IWF provides the most recently received PortSyncSync structure at the timeReceiver port of the synchronized PTP Instance (isSynced == TRUE, see 18.4.1.1) to the unsynchronized PTP Instance (isSynced == FALSE) SiteSyncSync entity, in order to generate a PortSyncSync structure that is used by the timeTransmitter ports of the unsynchronized PTP Instance, as follows:

a) The domainNumber of the received PortSyncSync structure is changed by the IWF from the synchronized PTP Instance domainNumber to the unsynchronized PTP Instance domainNumber;

b) The localPortNumber of the received PortSyncSync structure is changed by the IWF to the portNumber of the unsynchronized PTP Instance timeReceiver port; and

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c) All other members (i.e., syncReceiptTimeoutTime, followUpCorrectionField, sourcePortIdentity, logMessageInterval, preciseOriginTimestamp, upstreamTxTime, rateRatio, gmTimeBaseIndicator, lastGmPhaseChange, and lastGmFreqChange) of the received PortSyncSync structure of the synchronized PTP Instance are provided to the unsynchronized PTP Instance SiteSync entity unchanged.

d) The flags leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource of received Announce messages on the timeReceiver port of the unsynchronized PTP Instance are ignored. The IWF provides the values of these corresponding members of the timePropertiesDS of the synchronized PTP Instance to the unsynchronized PTP Instance. These values are then copied to:

   i) The respective members of the timePropertiesDS of the unsynchronized PTP Instance;
   ii) The unsynchronized PTP Instance global variables leap61, leap59, currentUtcOffsetValid, ptpTimescale, timeTraceable, and frequencyTraceable, and the fields currentUtcOffset and timeSource, respectively; and
   iii) The unsynchronized PTP Instance global variables annLeap61, annLeap59, annCurrentUtcOffsetValid, annPtpTimescale, annTimeTraceable, and annFrequencyTraceable, and the fields annCurrentUtcOffset and annTimeSource, respectively.

e) The messagePriorityVector (see 10.3.4 and 10.3.5) information of received Announce messages on the timeReceiver port of the unsynchronized PTP Instance is ignored. The IWF provides the values of the corresponding members of the synchronized PTP Instance parentDS, i.e., grandmasterIdentity, grandmasterClockQuality.clockClass, grandmasterClockQuality.clockAccuracy, grandmasterClockQuality.offsetScaledLogVariance, grandmasterPriority1, grandmasterPriority2, which are copied to:

   i) The corresponding members of the parentDS of the unsynchronized PTP Instance; and
   ii) The gmPriorityVector of the unsynchronized PTP Instance.

The IWF also provides the value of the stepsRemoved member of the synchronized PTP Instance currentDS, which is copied to:

   iii) The stepsRemoved member of the currentDS of the unsynchronized PTP Instance; and
   iv) The gmPriorityVector of the unsynchronized PTP Instance.

NOTE 1—With the above, the secondary/primary PTP Instance state machines operate as though the time synchronization information had been received from the secondary/primary PTP Instance timeReceiver port. The SiteSync entity of the secondary/primary PTP Instance transfers the timing information to the PortSync entity of each timeTransmitter port of the secondary/primary PTP Instance. Each PortSyncSync state machine computes rateRatio, which now is relative to the primary/secondary PTP Instance GM. Each MDSyncSend state machine computes the fields of transmitted Sync and, in the two-step case, Follow_Up messages. The copied syncReceiptTimeout time is less than currentTime because sync receipt timeout has not occurred at the primary/secondary PTP Instance.

NOTE 2—The split functionality is used to transfer time synchronization information from the PTP Instance for which isSynced is TRUE to the PTP Instance for which isSynced is FALSE. It is not meant to cover the case where isSynced is FALSE for both the primary and secondary PTP Instances.

18.6.5 Both PTP Instances have isSynced FALSE

When isSynced is FALSE for both the primary and secondary PTP Instances (i.e., both are not synchronized), the HotStandbySystemState is OUT_OF_SYNC and the HotStandbySystem entity shall transfer phase, frequency, gmTimeBaseIndicator (see 10.2.4.15), lastGmPhaseChange (see 10.2.4.16), and lastGmFrequencyChange (see 10.2.4.17) from the ClockTimeReceiver of the PTP Instance for which isSynced was TRUE, before the HotStandbySystemState transitioned to the OUT_OF_SYNC state, to the ClockTarget if a ClockTarget is present.

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When the HotStandbySystemState is OUT_OF_SYNC, time synchronization performance is not required to meet the respective application or profile standard (see 3.25) requirements. Nevertheless, in order to mitigate LocalClock frequency drift, the PTP Instance slowly adjusts the frequency of its LocalClock using the data stored while isSynced for that PTP Instance was TRUE and the HotStandbySystemState was in the REDUNDANT or NOT_REDUNDANT state.

18.7 PrimarySecondaryOffset state machine

This state machine updates the value of the global variable primarySecondaryOffset (see 18.5.1.4) whenever a new value of clockTimeReceiverTime (see 10.2.4.3) is computed for either the primary or secondary PTP Instance. A new value of clockTimeReceiverTime is computed when the global variable rcvdPSSyncCSS (see 10.2.4.29) or the global variable rcvdLocalClockTickCSS (see 10.2.4.30) is set to TRUE.

18.7.1 State diagram

The PrimarySecondaryOffset state machine shall implement the function specified by the state diagram in Figure 18-4.

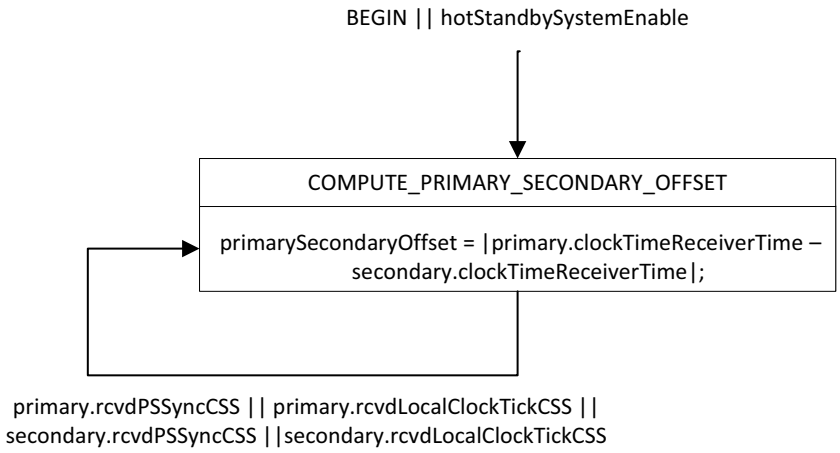


Figure 18-4—PrimarySecondaryOffset state machine

18.8 Transition of hotStandbySystemState from REDUNDANT to NOT_REDUNDANT

The time synchronization provided to the application can fulfill the time synchronization requirements of the respective application or profile standard (see 3.25) during a HotStandbySystemState change from REDUNDANT to NOT_REDUNDANT (or NOT_REDUNDANT to REDUNDANT).

NOTE 1—The secondary grandmaster is responsible for maintaining continuity (no “jump” in time) when a fault occurs in the primary grandmaster.

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NOTE 2—The primary domain global variables lastGmPhaseChange and lastGmFrequencyChange are propagated throughout the secondary domain, and can be used to compensate for any phase or frequency jump that occurs when the HotStandbySystemState becomes NOT_REDUNDANT.

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19. Media-dependent layer specification for IEEE Std 802.3 Clause 4 Media Access Control (MAC) operating in half-duplex

19.1 Overview

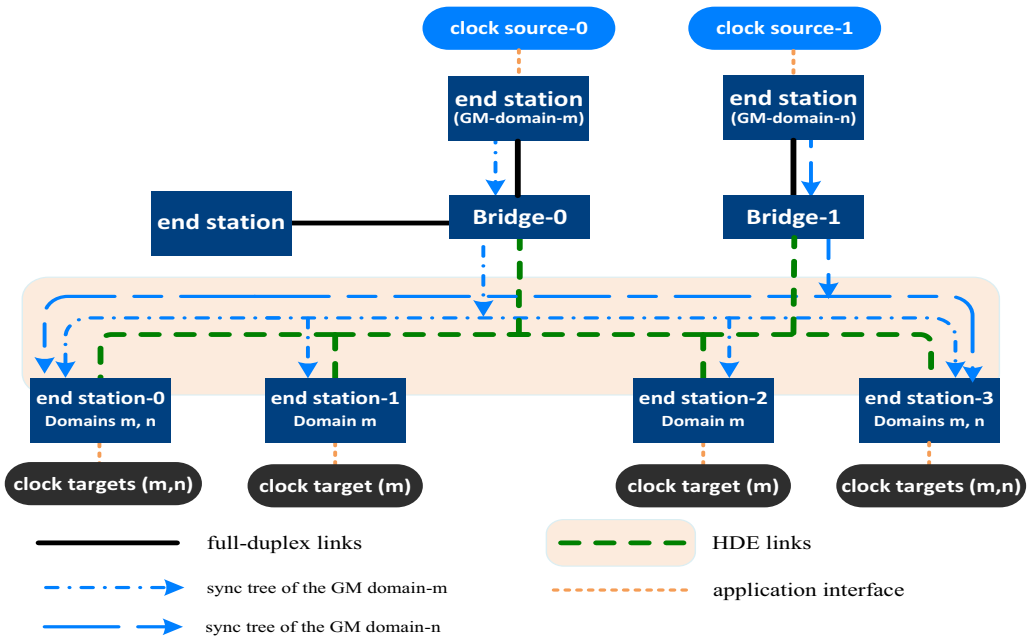
19.1.1 General

This clause specifies the media-dependent layer that provides synchronized time across links using the IEEE Std 802.3 Clause 4 MAC operating in half-duplex mode.

19.1.1.1 Half-duplex Ethernet (HDE) characteristics

The IEEE Std 802.3 Clause 4 MAC can operate in either full-duplex or half-duplex mode. When this MAC is operating in full-duplex, its media-dependent specification for gPTP is covered in Clause 11. Clause 19 is used when the IEEE Std 802.3 Clause 4 MAC is operating in half-duplex. This mode necessitates additional managed object settings and frame processing due to the effects of the shared media this mode supports, which are described in this clause.

Figure 19-1 shows an example of time-aware network using HDE links.



Note 1: Bridge-0 and Bridge-1 bridges in this figure are each an example of a time-aware system that contains a PTP Relay Instance, and the end stations with clock targets are examples of time-aware systems that contain PTP End Instances.
Note 2: GM denotes Grandmaster PTP Instance

Figure 19-1—Time-aware network example of end stations connected to bridges via HDE links

When using HDE links, the use of "neighbor" refers to a device or to devices (end station or Bridge) that is/are the intended recipient(s) of a transmitted PTP message. The intended recipient(s) of a PTP message from a gPTP timeTransmitter port is/are one or more gPTP timeReceiver port(s). The intended recipient of a PTP message from a gPTP timeReceiver port is the gPTP timeTransmitter port, regardless of whether or not

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other gPTP timeTransmitter ports received that message (e.g., if all ports are connected via a shared medium). In the example in Figure 19-1, PTP messages are exchanged between a gPTP timeTransmitter port of a Bridge (e.g., Bridge-0) and the four gPTP timeReceiver ports of the four end stations via a shared medium.

NOTE—For example, this standard supports 10BASE-T1S, as specified in Clause 147 of IEEE Std 802.3, utilizing Physical Layer Collision Avoidance (PLCA), as specified in Clause 148 of IEEE Std 802.3.

19.1.1.2 Overview of the major differences and restrictions of using HDE links

This clause uses Clause 11 in its entirety with the following differences (complete details follow starting in 19.1.2):

a) The peer delay initiator is restricted to timeReceivers only (see 19.1.2).
b) External port configuration mode is the only mode supported (see 19.8).

The following features of this standard have not been qualified for use with HDE links:

c) One-step time transport (see 19.1.3 and 19.2.15).
d) CMLDS (see 19.2.16).
e) The use of Signaling messages (see 19.8).
f) Hot Standby (see 19.8).

19.1.2 Propagation delay measurement over HDE links

The measurement of propagation delay on an HDE link using the peer-to-peer delay mechanism is illustrated in Figure 11-1 and is described in 11.1.2, with the following differences:

a) The peer delay initiator is restricted to each timeReceiver port.
b) The timeTransmitter port does not initiate the peer-to-peer delay mechanism.

Therefore, pdelayReqSendDisabled and pdelayRespSendDisabled are set as follows:

c) pdelayReqSendDisabled is set to FALSE when the PTP Port state is TimeReceiverPort and is otherwise set to TRUE.
d) pdelayRespSendDisabled is set to FALSE when the PTP Port state is TimeTransmitterPort and is otherwise set to TRUE.

Other configuration values for item c) and item d) are not intended for use with HDE links by this standard.

19.1.3 Transport of time-synchronization information

HDE links use two-step time transport as specified in 11.1.3. The variables oneStepReceive (see 19.2.12.10), oneStepTransmit (see 19.2.12.11), and oneStepTxOper (see 19.2.12.12) are set to FALSE for HDE links. Other configuration values for oneStepTransmit and oneStepTxOper are not intended for use with HDE links by this standard.

NOTE—The transport of time-synchronization information by a PTP Instance, using Sync and Follow_Up messages, is illustrated in Figure 11-2.

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19.1.4 Model of operation

The model for a PTP Instance of a time-aware system with FDE links is shown in Figure 11-3. This clause reuses Figure 11-3 (as its structure is unchanged for this clause), where all references to Clause 11 in Figure 11-3 are to be replaced by references to Clause 19 (this clause). The presence of one HDE MD entity per PTP Port is assumed.

A general, media-independent description of the generation of timestamps is given in 8.4.3. A more specific description for PTP event messages is given in 11.3.2.1. A PTP event message is timestamped relative to the LocalClock entity when the message timestamp point (see 3.17) crosses the timestamp measurement plane (see 3.33). The timestamp is corrected for any ingressLatency or egressLatency (see 8.4.3) to produce a timestamp relative to the reference plane (see 3.26). The corrected timestamp value is provided to the MD entity.

The MD entity behavior and detailed state machines specific to FDE links, which are described in 11.2, are reused for HDE links subject to the conditions defined in 19.2.

19.2 State machines for HDE links

19.2.1 General

The state machines for HDE links are described in 11.2.1 except that there is one instance of the MDPdelayReq state machine per PTP Port, there is one instance of the MDPdelayResp state machine per port, and the LinkDelayIntervalSetting state machine is not used.

19.2.2 Determination of asCapable and asCapableAcrossDomains

As specified in 11.2.2.

NOTE—For FDE links, asCapableAcrossDomains is one global variable for all domains per link port. For HDE links, asCapableAcrossDomains is a global variable per PTP Instance (i.e., per domain), the name asCapableAcrossDomains has been kept for backwards compatibility with existing implementations.

19.2.3 Use of MAC Control PAUSE operation

As specified in 11.2.3.

19.2.4 Use of priority-based flow control

As specified in 11.2.4.

19.2.5 Use of link aggregation

As specified in 11.2.5.

19.2.6 DL-UNITDATA.request

This service primitive is as specified in 11.2.7.

19.2.7 DL-UNITDATA.indication

This service primitive is as specified in 11.2.8.

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19.2.8 MDTimestampReceive

This data structure is as specified in 11.2.9.

19.2.9 MDSyncReceive

This data structure is as specified in 10.2.2.2.

19.2.10 MDSyncSend

This structure is as specified in 10.2.2.1.

19.2.11 Overview of MD entity global variables

The overview of MD entity global variables is given in 11.2.12.

19.2.12 MD entity global variables

19.2.12.1 currentLogPdelayReqInterval: This variable is as specified in 11.2.13.1. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.2 initialLogPdelayReqInterval: This variable is as specified in 11.2.13.2. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.3 pdelayReqInterval: This variable is as specified in 11.2.13.3. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.4 allowedLostResponses: This variable is as specified in 11.2.13.4. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.5 allowedFaults: This variable is as specified in 11.2.13.5. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.6 isMeasuringDelay: This variable is as specified in 11.2.13.6. However, for HDE, the port measures propagation delay if it is receiving Pdelay_Resp and Pdelay_Resp_Follow_Up messages from the requested port with the requested clockIdentity. Also, for HDE, there is one instance of this variable per PTP Port.

19.2.12.7 meanLinkDelayThresh: This variable is as specified in 11.2.13.7. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.8 negMeanLinkDelayThresh: This variable is as specified in 11.2.13.8. However, for HDE, there is one instance of this variable per PTP Port.

19.2.12.9 syncSequenceId: This variable is as specified in 11.2.13.9.

19.2.12.10 oneStepReceive: This variable is as specified in 11.2.13.10. However, for HDE, this variable is FALSE.

19.2.12.11 oneStepTransmit: This variable is as specified in 11.2.13.11. However, for HDE, this variable is FALSE.

19.2.12.12 oneStepTxOper: This variable is as specified in 11.2.13.12. However, for HDE, this variable is FALSE.

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19.2.12.13 asCapableAcrossDomains: This variable is as specified in 11.2.13.13.

19.2.12.14 nrrPdelay: This variable is as specified in 11.2.13.14.

19.2.12.15 nrrSync: This variable is as specified in 11.2.13.15.

19.2.12.16 nrrCompMethod: This variable is as specified in 11.2.13.16.

19.2.13 MDSyncReceiveSM state machine

The MDSyncReceiveSM state machine is as specified in 11.2.14.

19.2.14 MDSyncSendSM state machine

The MDSyncSendSM state machine is as specified in 11.2.15.

19.2.15 OneStepTxOperSetting state machine

This state machine is not used for HDE links.

19.2.16 Common Mean Link Delay Service (CMLDS)

CMLDS has not been qualified for use with HDE links.

HDE links use the transport-specific peer-to-peer delay mechanism for each domain. Therefore, if the time-aware system implements multiple domains on an HDE link, each domain uses the transport-specific peer-to-peer delay mechanism on that link regardless of its domain number.

If multiple timeTransmitter ports are present on an HDE link, they are in different gPTP domains. CMLDS cannot be used because, in general, the timeTransmitter ports can be on physical ports of different time-aware systems (i.e., different Bridges). In this case, both meanLinkDelay and neighborRateRatio between a PTP End Instance in one of the domains and the timeTransmitter it is communicating with can be different from meanLinkDelay and neighborRateRatio between a PTP End Instance in another domain on the same end station and the timeTransmitter that PTP End Instance is communicating with. For instance, in Figure 19-1 the meanLinkDelay and neighborRateRatio between the PTP End Instance over domain-m and over domain-n at end station-0 are computed using timestamps from two distinct time aware systems, GM-domain-m and GM-domain-n, respectively; and hence does not use CMLDS.

19.2.17 Common Mean Link Delay Service (CMLDS) global variable

The Common Mean Link Delay Service (CMLDS) global variable is as specified in 11.2.18.

19.2.18 MDPdelayReq state machine

The MDPdelayReq state machine is as specified in 11.2.19.

The variable pdelayReqSendDisabled is set per 19.1.2.

Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmittetd with domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4). Any received Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages are ignored by the state machines if the domainNumber in the received message does not match the gPTP domain associated with the PTP Port.

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19.2.19 MDPdelayResp state machine

The MDPdelayResp state machine is as specified in 11.2.20.

The variable pdelayRespSendDisabled is set per 19.1.2.

Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted with the domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism (see 19.4). Any received Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up messages are ignored by the state machines if the domainNumber in the received message does not match the gPTP domain associated with the PTP Port.

19.2.20 LinkDelayIntervalSetting state machine

This state machine is not used for HDE links.

19.3 Message attributes

Message attributes are as specified in 11.3.

19.4 Message formats

Message formats are as specified in 11.4, except for 11.4.2.4, where the Pdelay_Req, Pdelay_Resp, and Pdelay_Resp_Follow_Up messages shall be transmitted with the domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism. Any Pdelay_Req, Pdelay_Resp, or Pdelay_Resp_Follow_Up message received with a domainNumber other than the domainNumber of the HDE gPTP domain used by the transport-specific peer delay mechanism shall be ignored. The domainNumber for all other PTP messages is as specified in 10.6.2.2.6.

19.5 Protocol timing characterization

19.5.1 General

This subclause specifies timing attributes for the media-dependent sublayer specified in this clause.

19.5.2 Message transmission intervals

19.5.2.1 General interval specification

The general message transmission interval specification is as specified in 11.5.2.1.

19.5.2.2 Pdelay_Req message transmission interval

The Pdelay_Req message transmission interval is as specified in 11.5.2.2. The variable useMgtSettableLogPdelayReqInterval is set to TRUE. Other configuration values for useMgtSettableLogPdelayReqInterval are not intended for use with HDE links by this standard.

19.5.2.3 Sync message transmission interval default value

The Sync message transmission interval default value is as specified in 11.5.2.3.

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19.5.3 allowedLostResponses

The variable allowedLostResponses is as specified in 11.5.3.

19.5.4 allowedFaults

The variable allowedFaults is as specified in 11.5.4.

19.6 Control of computation of neighborRateRatio

The control of computation of neighborRateRatio is as specified in 11.6.

19.7 Control of computation of meanLinkDelay

The control of computation of meanLinkDelay is as specified in 11.7.

19.8 HDE links settings and configuration

This subclause provides settings and configurations that are specific for HDE links, other configuration values are not intended for use with HDE links by this standard.

a) The per PTP Instance global variable externalPortConfigurationEnabled is set to TRUE for HDE links.

b) Both GptpCapableTransmit and GptpCapableReceive state machines are disabled for HDE links, and therefore gptpCapableStateMachinesEnabled is set to FALSE (see 10.4.1). According to 10.4.1, if the managed object gptpCapableStateMachinesEnabled is FALSE, the global variable neighborGptpCapable for the port (see 10.2.5.16) is set to TRUE.

c) The variables useMgtSettableLogSyncInterval and useMgtSettableLogAnnounceInterval are set to TRUE. As a consequence, the SyncIntervalSetting and the AnnounceIntervalSetting state machines are not used for HDE links.

d) The GptpCapableIntervalSetting state machine is not used for HDE links. The variable useMgtSettableLogGptpCapableMessageInterval is set to TRUE.

Hot Standby (see Clause 18) has not been qualified for use with HDE links.

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20. Fault-tolerant timing with time integrity

20.1 Overview

20.1.1 General

To achieve fault-tolerant timing with time integrity, this standard defines an FTT entity. The concepts used by the FTT entity are described in 20.1.2. An overview of the FTT entity is given in 20.1.3 and details on the FTT entity are given in 20.2.

20.1.2 Fault-tolerant time synchronization concepts

20.1.2.1 Availability of time

The continuous availability of time is enhanced by redundancy. For gPTP, this redundancy can be implemented by using multiple time domains, multiple time distribution paths, and multiple PTP Instances in Bridges and end stations.

20.1.2.2 Trust and integrity of time

For this standard, a trusted time is one that passes a specified criterion that identifies it as being within acceptable bounds. This establishment of trust gives integrity to the time.

For gPTP, trust, and hence integrity, is established through the comparison of the times coming from independent time sources and the observation that they match within the specified criterion. See H.4 for an example of such a criterion.

20.1.2.3 Time distribution

Time distribution, per this standard, is the process of distributing the time from time source nodes (GMs) to timeReceivers (PTP Instances) in Bridges and end stations over gPTP communication paths.

20.1.2.4 Dependent times

Dependent times share one or more common time-influencing components, which includes a common GM, continuously synchronized GMs, GMs that share a common (continuously connected) ClockSource, or a common PTP Relay Instance.

Because dependent times share one or more common influencers, they do not, on their own, enable end-to-end integrity checking of the time synchronization function. However, they can be used to improve the availability of a given time source and can provide partial integrity checks. For example, an application that receives timing from a single GM through more than one redundant synchronization tree (and PTP Instances) has increased availability of that GM's time and can check the integrity of the synchronization trees by comparing the time received from the respective PTP Instances. However, because the time originates from a single GM, the integrity of that GM's time cannot be confirmed and, thus, end-to-end integrity of the time synchronization function is not achieved.

Dependent times can be identified by checking if they meet any of the following criteria:

— They have the same gPTP domainNumber. This indicates that gPTP messages from the same PTP GM are received by two PTP End Instances, on two distinct physical ports, that are serviced by the FTT entity.

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— They have different gPTP domainNumbers but the same gmTimeBaseIndicator. This indicates that the gPTP messages come from different PTP GMs that share the same ClockSource.
— They have gPTP domainNumbers that are defined by a management entity to be dependent. For example, the management entity could define gPTP time inputs at an FTT entity as dependent if their respective gPTP communication paths share a common PTP Relay Instance or a common Bridge.

20.1.2.5 Independent times

Independent times do not share any common time-influencing components with each other and, therefore, deliver independent time values. Independent gPTP times are, by definition, from different domains.

Because independent times do not share any common influencers, they can enable end-to-end integrity checking of the time synchronization function, provided their times track sufficiently closely. The clock sources of GMs providing independent times need to be aligned to each other in a manner that is resilient to faults (see J.1). Because the independent times provide common time, the inherent redundancy can also be used to improve the availability of the time synchronization function.

When a set of dependent times are used in combination with a set of independent times, the dependent times can be reduced to a single independent time and used to achieve end-to-end integrity of the time synchronization function. This operation is performed by the FTT entity. See H.3 for a discussion on balancing availability and integrity with FTT entities.

20.1.3 Model of operation for fault-tolerant timing with time integrity

An FTT entity enables fault-tolerant timing with time integrity. The FTT entity is located between a ClockTarget entity (Clause 9) and one or more PTP Instances of a time-aware system. The FTT entity manages the selection of a time source from amongst two or more PTP Instances to support increased availability and integrity. The input to an FTT entity, hereafter called simply FTT input is the ClockTargetEventCapture.result (see 9.3.3) from the PTP Instances connected to the FTT entity. The output of an FTT entity, hereafter called simply FTT output, is the ClockTargetEventCapture.result of the PTP Instance selected by the FTT entity. The FTT entity also supports the use of a single PTP Instance (single input) but, with a single input, it does not provide any enhancements for increased availability or for integrity. The effect of the FTT entity's selection is limited to the time-aware system in which it operates. Figure 20-1 illustrates the FTT entity operating with three PTP Instances.

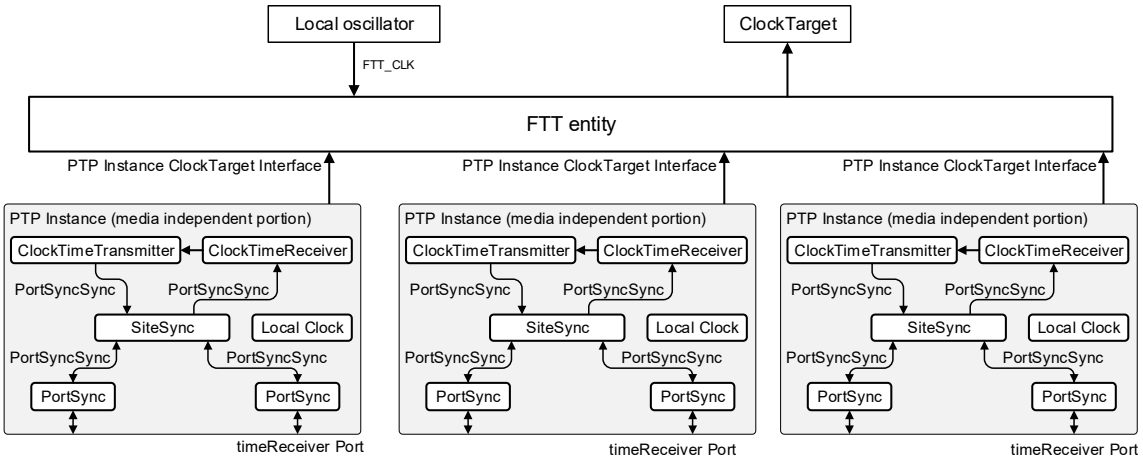


Figure 20-1—FTT entity in operation

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20.1.3.1 Assumptions

The following list provides the detailed assumptions and goals for the FTT entity:

— A fault-tolerant network (with time integrity) and its configuration are static during normal operation.
— All gPTP ports are configured using the external port configuration provision (i.e., the BTCA is not used).
— There is no administrative reconfiguration during run-time in the event of faults.
— While operation with one PTP Instance (single input) is supported by the FTT entity, two or more PTP Instances (multiple inputs) are required for fault-tolerant timing with time integrity.
— gPTP times are recognized as being dependent or independent as defined in 20.1.2.4 and 20.1.2.5, respectively.

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20.2 FTT entity

A functional block diagram of the FTT entity is shown in Figure 20-2.

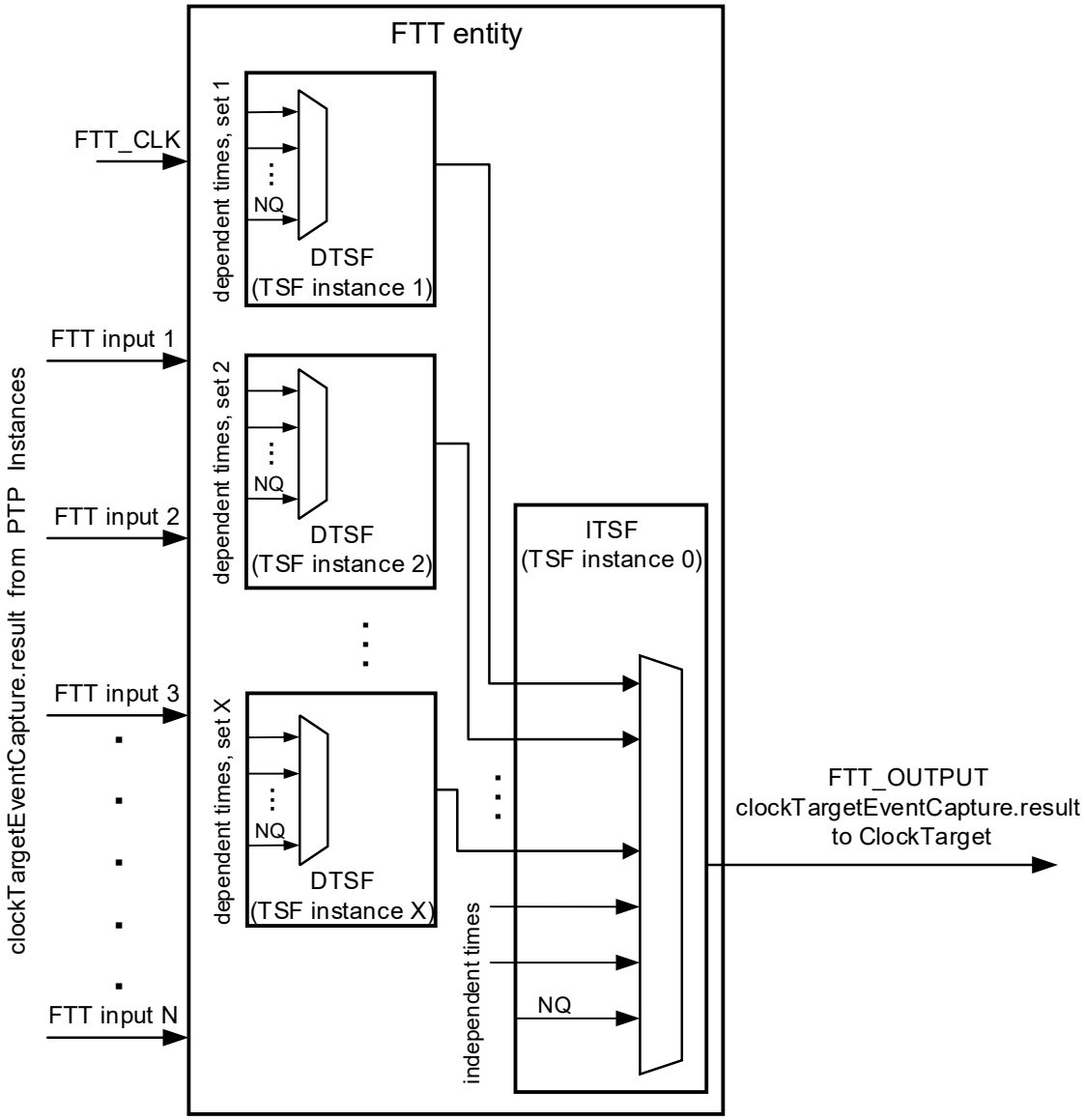


Figure 20-2—FTT entity functional block diagram

20.2.1 FTT input interfaces

The input interfaces to the FTT entity, hereafter referred to as FTT inputs, are the ClockTargetEventCapture interface of the PTP Instances connected to the FTT entity. They provide time values from the PTP Instances to the time selection function (TSF, see 20.2.3) instances of the FTT entity. The FTT inputs are indexed (FTT input index) for ease of reference. Each input to the FTT entity is also accompanied by the index number of the corresponding PTP Instance. The managed object fttPtpInstanceToInput provides the mapping between the PTP Instance index and the FTT input index.

The FTT inputs provide either dependent times or independent times to the FTT entity. FTT inputs that provide dependent times are referred to as dependent inputs and FTT inputs that provide independent times are referred to as independent inputs.

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The FTT entity uses the ClockTargetEventCapture interface type (see 9.3). The use of other ClockTarget interface types with the FTT entity is discussed in H.5.

20.2.2 FTT clock

An FTT entity can use a free running clock (FTT_CLK) as an additional source of timing. If fttUseFTTClk is TRUE, the FTT entity uses a free-running clock (FTT_CLK) as a potential source of timing information for establishing trust (20.3.2). FTT_CLK is independent of the time values being received by the PTP Instances that are connected to the FTT entity. The health and trust of FTT_CLK is outside the scope of this standard.

20.2.3 Time selection function

An FTT entity includes one or more TSF instances, each referenced by a TSF instance number. Each TSF instance analyzes its set of inputs (ClockTarget interfaces), determines which corresponding time values can be trusted and selects one of the trusted time values using a time selection algorithm. The output of the TSF instance is set to the selected input and the corresponding PTP Instance.

The inputs to a TSF may provide either dependent times or independent times.

One TSF instance in the FTT entity is used for selecting a time from a set of independent times (see 20.1.2.5). This TSF instance is called the independent time selection function (ITSF) and is assigned the TSF instance number of 0.

If any of the inputs provide dependent times to the FTT entity (see 20.1.2.4), then one TSF instance is used to select a time for each set of dependent times. These TSF instances are called dependent time selection functions (DTSFs) and are assigned TSF instance numbers from 1 to fttNumDtsfs (see 14.23.5).

Therefore, an FTT entity includes zero or more DTSF instances and exactly one ITSF instance. The TSF instance number is used to uniquely reference a particular TSF instance.

The time-index selection algorithm for the TSF is the mid-value time selection algorithm (MVTSA, see 20.4). The MVTSA shall be supported if the FTT entity is supported.

The TSF processes are described in 20.3.3.4.

20.2.4 Not qualified (NQ) input

When none of the inputs to a TSF instance are determined to be trusted and the TSF instance's fttUseNQ parameter is TRUE, the TSF instance selects its NQ input as the TSF output. The NQ input has an input index of 255. The NQ input contains the ClockTargetEventCapture interface parameters from any one (selection is implementation specific) of the inputs being processed, but with the isSynced status and the gmPresent status set to FALSE, to indicate the untrusted condition.

NOTE—Because the time provided by the NQ input is, by definition, not trusted, the values of its other parameters (aside from isSynced and gmPresent) have no functional impact.

All parameters in the NQ time are listed below:

- domainNumber = the domainNumber from the selected input
- timeReceiverTimeCallback = the timeReceiverTimeCallback value from the selected input
- isSynced = FALSE
- gmPresent = FALSE
- errorCondition = the errorCondition value from the selected input

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- clockPeriod = the clockPeriod value from the selected input
- timeReceiverCallbackPhase = the timeReceiverCallbackPhase value invoked by the ClockTarget entity
- grandmasterIdentity = the grandmasterIdentity value from the selected input
- gmTimeBaseIndicator = the gmTimeBaseIndicator value from the selected input
- lastGmPhaseChange = the lastGmPhaseChange value from the selected input
- lastGmFreqChange = the lastGmFreqChange value from the selected input

20.2.5 FTT output interface

The FTT entity's output is the ClockTargetEventCatpure.result from the PTP Instance selected by the ITSF instance in the FTT entity. The FTT entity accompanies the output ClockTargetEventCatpure.result with the index number of the selected PTP Instance.

The FTT entity provides a fault-tolerant time to a ClockTarget entity via an output ClockTargetEventCapture application interface identified as FTT_OUTPUT. The index number of the PTP Instance associated with the output ClockTargetEventCapture application interface is available via the fttSelInstance management object (see 14.23.17).

20.3 FTT state machine

20.3.1 General

The FTT entity state machine specified in 20.3 interacts with all the TSFs (instantiated in the FTT entity's DTSF instances and in the FTT entity's ITSF instance) to find and select a trusted input, if one exists, to present as the FTT entity's output.

The TSF process is described in 20.3.3.4. The mid-value time selection algorithm (MVTSA) used by the TSF instances is described in 20.4.

20.3.2 FTT state machine global variables

20.3.2.1 fttTrustState:

The trust state of the FTT entity. The value is taken from the enumeration in Table 20-1.

Table 20-1—fttTrustState enumeration

Value	State	Specification
0	NOT-TRUSTED	The input selected by the FTT entity does not meet the specified criteria to establish trust and the conditions to use the FTT_CLK are not satisfied. In this state, integrity of time synchronization function is not achieved.
1	TIME-TRUSTED	The FTT entity has selected an input whose time value meets the specified criterion to establish trust. In this state, integrity of time synchronization is achieved.
2	FREQ-TRUSTED	The FTT entity has selected an input whose time value does not meet the specified criteria to establish trust, but the conditions for using the FTT_CLK to maintain a trusted status are satisfied. In this state, only the integrity of the rate-of-change of time is achieved.
3	INIT	Initialization after the FTT entity powers on and is enabled. In this state, the system is waiting to be invoked and has not determined a trust state.

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20.3.2.2 fttInputTrustState:

An Enumeration2 array of length fttNumInputs. fttInputTrustState[*j*], where *j* is a value from 1 to fttNumInputs, is the trust state of the FTT input (see 20.2.1) with index *j*. The trust state of the FTT input takes a value from the enumeration in Table 20-2.

Table 20-2—fttInputTrustState enumeration

Value	State	Specification
0	INACTIVE	Either the isSynced or the gmPresent value of the FTT input is FALSE. In this state, the FTT input is not considered to be providing time values.
1	ACTIVE	Both the isSynced and gmPresent values of the FTT input are TRUE. In this state, the FTT input is actively providing time values but has not yet determined to be TRUSTED or UNTRUSTED.
2	TRUSTED	The FTT input is ACTIVE and the provided time values meet the criteria to establish trust (see 20.4.1). In this state, the FTT input is providing trusted time values.
3	UNTRUSTED	The FTT input is ACTIVE and the provided time values do not meet the criteria to establish trust (see 20.4.1). In this state, the FTT input is providing untrusted time values.

20.3.2.3 fttTsfSelInput:

An UInteger8 array of length fttNumDtsfs + 1 (see 14.23.5). fftTsfSelInput[*j*] is the index number of the FTT input selected by TSF instance with instance number *j*. The values range from 1 to fttNumInputs and the value of 255, which represents the NQ input (see 20.2.4).

20.3.2.4 itsfSelInput:

The index number of the FTT input selected by the ITSF instance. The values range from 1 to fttNumInputs and the value of 255, which represents the NQ input (see 20.2.4). The data type for itsfSelInput is UInteger8.

20.3.2.5 itsfTrustState:

Enumerated trust state of the FTT input selected by the ITSF instance. It is the value of fttInputTrustState[itsfSelInput] (see 20.3.2.2).

20.3.2.6 rRatioOff:

Offset of the ratio between the rate of the time provided on the output of the ITSF instance to the rate of the FTT entity's local clock, FTT_CLK. The value is expressed as a fractional frequency offset multiplied by 2^{41}, that is:

$$rRatioOff = (|\text{rate of time at the output of ITSF} / \text{rate of FTT_CLK} - 1.0|) \times 2^{41}$$

The magnitude of this offset is one of the considerations used by the FTT entity's state machine to determine whether it enters the FREQ_TRUSTED state or the NOT_TRUSTED state. See 20.3.4. The data type for rRatioOff is unsigned 32-bit integer.

NOTE—The rate measurement period is application dependent.

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20.3.2.7 rRatioSdOff:

The standard deviation of the offset of the ratio between the rate of the time provided on the output of the ITSF instance to the rate of the FTT entity's local clock, FTT_CLK. It is one of the considerations used by the FTT entity's state machine to determine whether it enters the FREQ_TRUSTED state or the NOT_TRUSTED state. See 20.3.4. The data type for rRatioSdOff is unsigned 32-bit integer.

NOTE—The rate measurement period is application dependent.

20.3.2.8 fttCtecPtr:

An array of pointers of length fttNumInputs. fttCtecPtr[*j*] is the pointer to a structure whose members contain the values of the parameters of a ClockTargetEventCapture.result primitive from the FTT input with index *j*.

20.3.3 FTT state machine functions

20.3.3.1 setCtec(N):

Creates a structure whose parameters contain the fields of ClockTargetEventCapture.result (see 9.3.3), populates the fields with values from the FTT input (of index N), and returns a pointer to this structure.

20.3.3.2 updateItsfInputs():

Appends the inputs selected by all enabled DTSFs, fttTsfSelInput[1:fttNumDtsfs] to the ITSF input set, fttInputsToTsf[0].

20.3.3.3 updateStateChangeCnt(newState):

The updateStateChangeCnt procedure updates the fttStateChangeCnt (see 14.23.21) based on the input argument (newState) and the current state of the fttTrustState object as follows:

If the newState is different from the fttTrustState, then increment fttStateChangeCnt by 1. The fttStateChangeCnt rolls over to 0 if the count is incremented when its current value is 65 535. The default value is 0.

20.3.3.4 tsf(tsfNum):

Selects one of the inputs from the set of FTT inputs mapped to the TSF with instance number tsfNum via the global variable fttInputsToTsf and sets the output of the TSF to the global variable fttTsfSelInput (14.23.16), as follows:

a) If the number of inputs to the TSF is 1, then the only available input, fttInputsToTsf[tsfNum], is set as the selected input in the fttTsfSelInput[tsfNum] variable.

b) If the number of inputs to the TSF is more than 1, call the MVTSA with the TSF instance number (tsfNum) and set the selected input, fttTsfSelInput[tsfNum], to the returned value (input index) from MVTSA if all of the following conditions are met:

   — MVTSA returns a different input than the previously selected input to the TSF, fttTsfSelInput[tsfNum].
   — The difference in time value of the newly selected input and the previously selected input is greater than the fttSelChangeThresh defined for this instance of TSF or the trust state of the newly selected and the previously selected inputs differ from each other.

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If a new input is selected by the TSF as per item b), increment fttTsfSelChangeCnt[tsfNum] by 1.

NOTE—If an iteration of the TSF function does not result in the selection of a (new) input as per items a) or b), the selected input of the corresponding DTSF or ITSF does not change.

The time selection function is invoked by each DTSF instance and by the ITSF instance to select an input from the set of available inputs. The TSF function utilizes a time selection algorithm to select one of the inputs and sets the fttTsfSelInput object for the corresponding TSF instance. The time selection algorithm, MVTSA, is described in 20.4.

20.3.3.5 resetTsfStatistics()

The resetTsfStatistics procedure sets the fttTsfSelChangeCnt value to zero for all TSF instances.

20.3.4 FTT state diagram

The FTT entity state machine shall implement the function specified by the state diagram in Figure 20-3, the variables specified in 20.3.2, and the functions specified in 20.3.3.

The FTT_INITIALIZE state of the FTT entity state machine resets the fttTsfSelChangeCnt and fttStatusChangeCnt management objects to zero and the fttTrustState global variable to INIT. This prepares these objects for further updates by this state machine. The state machine transitions unconditionally to the WAIT_INVOKE state.

The WAIT_INVOKE state of the FTT entity state machine waits for a ClockTargetEventCapture.invoke event from the ClockTarget. When the event is received, the state machine transitions to the GET_INPUTS state.

The GET_INPUTS state of the FTT state machine simultaneously sends a ClockTargetEventCapture.invoke event to all the PTP Instances connected to the FTT entity. Once all the PTP Instances respond by providing their ClockTargetEventCapture.result, this state transitions to the SAVE_CLKTARGET_RSLT state.

The SAVE_CLKTARGET_RSLT state calls the setCtec procedure for each of the FTT entity’s input ClockTargetEventCapture interfaces. For each FTT input, the setCtec procedure returns a pointer to the structure containing the fields of the ClockTargetEventCapture.result from the corresponding PTP Instance. These pointers are saved in the fttCtecPtr global variable, indexed by the FTT input index.

If any DTSF instances are enabled in the FTT entity, the state machine transitions to the DTSF_SELCT state. If no DTSF instances are enabled in the FTT entity, the state machine transitions to the ITSF_SELECT state.

The DTSF_SELECT state calls the TSF function for every enabled DTSF instance in the FTT entity. This causes every DTSF instance to run its selection procedure and produce an updated output that is saved in the fttTsfSelInput object. After all DTSFs complete their selection process, the updateItsfInput function is called to add the inputs selected by each DTSF to the ITSF input set. The state machine transitions unconditionally to the ITSF_SELECT state.

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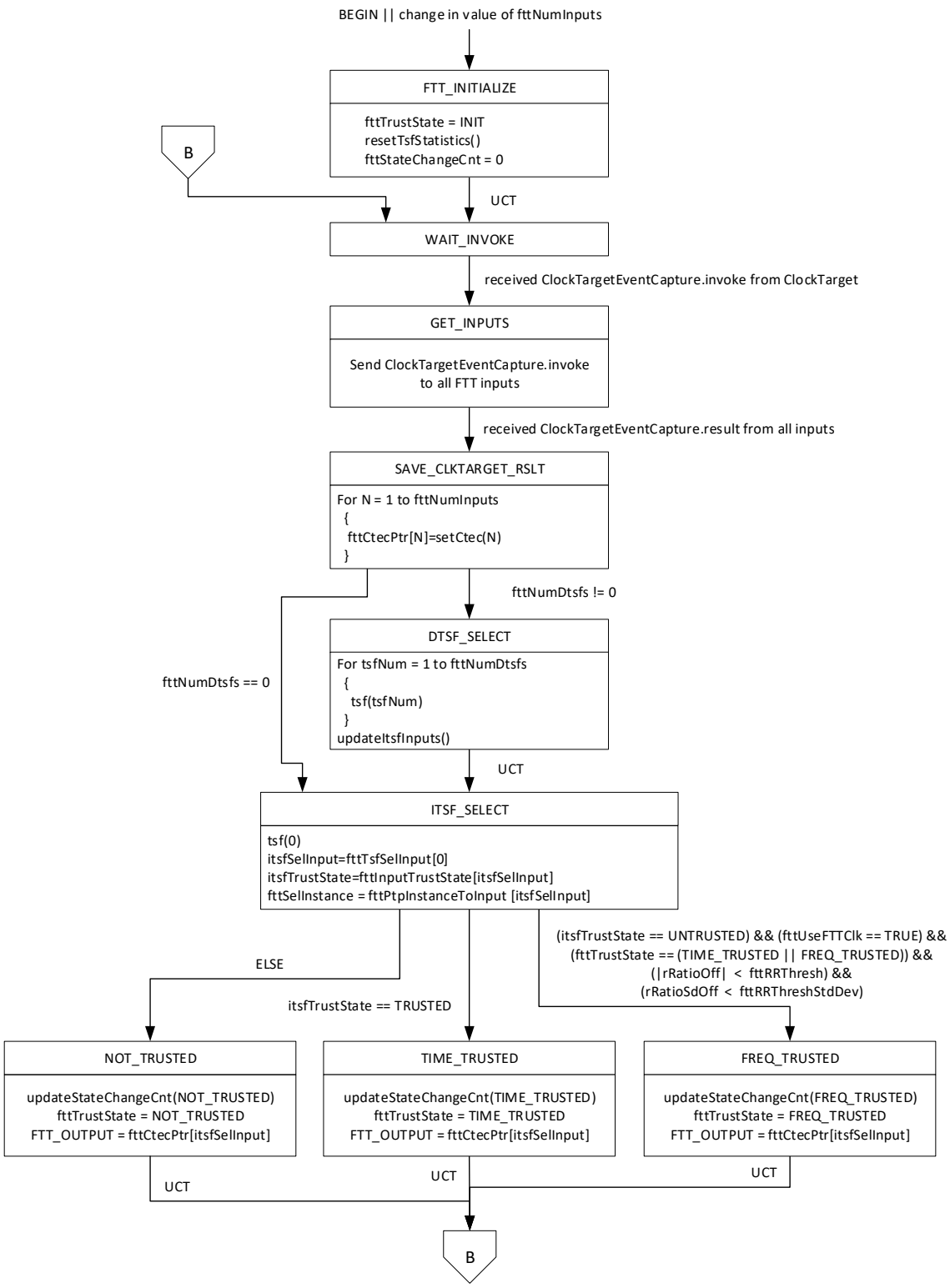


Figure 20-3—FTT entity state machine

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The ITSF_SELECT state calls the TSF function for the ITSF instance, which causes the ITSF instance to run its selection procedure and produce an updated output, and then performs corresponding updates to the itsfSelInput variable and the itsfTrustState and fttSelInstance management objects. If the trust state of the ITSF instance's selected input is TRUSTED, the state machine transitions to the TIME_TRUSTED state. If the trust state of the ITSF instance's selected input is NOT_TRUSTED but the conditions for using the FTT_CLK to maintain a trusted status are satisfied (i.e., the FTT entity was previously in the TIME_TRUSTED state or the FREQ_TRUSTED state and the time from its previously selected input is progressing at a rate and with a rate variation that are within configured thresholds), the state machine transitions to the FREQ_TRUSTED state. Otherwise, the state machine transitions to the NOT_TRUSTED state.

The NOT_TRUSTED, TIME_TRUSTED, and FREQ_TRUSTED states all call the updateStateChangeCnt procedure to update the ftttStateChangeCnt object, set the fttTrustState object to the corresponding state, and update the ClockTargetEventCapture.result output of the FTT entity, on FTT_OUTPUT, with the contents of the selected input ClockTargetEventCapture interface. The state machine transitions unconditionally to the WAIT_INVOKE state.

The FREQ_TRUSTED state calls the updateStateChangeCnt procedure to update the ftttStateChangeCnt object, sets the fttTrustState object to FREQ_TRUSTED, and updates the ClockTargetEventCapture.result output of the FTT entity, on FTT_OUTPUT, with the contents of the selected input ClockTargetEventCapture interface. The state machine transitions unconditionally to the WAIT_INVOKE state.

The NOT_TRUSTED state calls the updateStateChangeCnt procedure to update the ftttStateChangeCnt object, sets the fttTrustState object to NOT_TRUSTED, and updates the ClockTargetEventCapture.result output of the FTT entity, on FTT_OUTPUT, with the contents of either the NQ time or the selected input ClockTargetEventCapture interface. The state machine transitions unconditionally to the WAIT_INVOKE state.

20.4 Mid-value time selection algorithm

The mid-value time selection algorithm is used to determine which inputs have trusted time values and then find, amongst those trusted inputs, the input with the mid-value time. The MVTSA is the default algorithm called by the TSF Function (20.3.3.4).

The MVTSA looks at all possible combinations of input pairs to determine which pairs have time values that satisfy their corresponding specified maximum accepted skew magnitude threshold, fttMaxAs (see 14.23.10). An input with a time value that satisfies the corresponding fttMaxAs threshold with one or more other inputs is considered to be trusted. The input that has the mid-value time amongst all the trusted inputs is selected as the output of the MVTSA. If the number of trusted time indexes is even, the selected time index is the one of the middle two entries with the smaller index value.

MVTSA(tsfNum): Selects the mid-value time from trusted inputs of a TSF and returns the index of the selected input or the index of NQ input by executing the following three procedures (20.4.1, 20.4.2, and 20.4.3) steps in order.

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20.4.1 setInputTrustState(tsfNum):

Determines the trust state of each FTT input assigned to the TSF with index tsfNum via the global variable fttInputsToTsf and sets the state value for the corresponding input in the global variable fttInputTrustState as follows:

a) If both the isSynced and gmPresent value of an input are TRUE, the trust state for the corresponding input is set to ACTIVE in global variable fttInputTrustState. Otherwise, the trust state of the input is set to INACTIVE.

b) A pair-wise comparison of time values from all ACTIVE inputs is conducted to determine if the inputs are TRUSTED or UNTRUSTED. The trust state of an ACTIVE input, with FTT input index of *x*, is set to TRUSTED in fttInputTrustState if one of the following two conditions is TRUE and is set to UNTRUSTED if both of the following two conditions are FALSE.

   1) The fttInputTrustState[*x*] was previously UNTRUSTED and its current time value is within the maximum expected skew, fttMaxAs[*x*][*y*], of at least one other FTT input's (index *y*) time value.

   2) The fttInputTrustState[*x*] was previously TRUSTED and its current time value is within the maximum expected skew plus hysteresis (fttMaxAs[*x*][*y*] + fttHyst[*x*][*y*]) of at least one other FTT input's (index *y*) time value.

NOTE 1—The hysteresis in item b) enables the use of one skew level to assert the trust status and another skew level to de-assert the trust status. This can prevent superfluous changes in the trust state of two inputs (with input index *x* and *y*) when the time values of those inputs are close to the fttMaxAs[*x*][*y*] threshold. The value of this hysteresis is to be determined by the user of the FTT entity.

NOTE 2—The method for storing the previous trust state of FTT inputs is implementation dependent.

20.4.2 setSelectionPool():

Creates a candidate pool of time values to be considered for selection by the selectionFunction procedure based on the fttInputTrustState established by the setInputTrustState procedure, as follows:

a) If any of the inputs were determined to be TRUSTED, then the selection pool consists of the time values of all the inputs that were determined to be TRUSTED.

b) If none of the inputs were determined to be TRUSTED and fttUseNQ[tsfNum] is TRUE, then the selection pool consists of only the time value from its NQ input (see 20.2.4).

c) If none of the inputs were determined to be TRUSTED and fttUseNQ[tsfNum] is FALSE and at least one input was determined to be ACTIVE, the selection pool consists of the time values from all the inputs that were determined to be ACTIVE.

d) If none of the inputs were determined to be TRUSTED and fttUseNQ[tsfNum] is FALSE and none of the inputs were determined to be ACTIVE, the selection pool consists of the time values from all the inputs.

NOTE—When fttUseNQ[tsfNum] is FALSE [items c) and d)], the corresponding TSF instance does not select its NQ input even in the absence of a trusted or active input. This allows a TSF instance to select either an untrusted active or inactive input as its output when no trusted or active inputs are found.

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20.4.3 selectionFunction():

Selects one of the time values from the selection pool to be returned by the MVTSA function as follows:

- All the time values in the selection pool are ordered from smallest (i.e., earliest) to largest (i.e., latest).
- If the ordered list of time values contains an odd number of candidates, the middle value is selected.
- If the ordered list of time values contains an even number of candidates, one of the two middle values is selected based on the FTT input index associated with the input providing the corresponding time value. The candidate with the lower input index number is selected.
- If the selection pool has a single entry, the lone value is selected.

The MVTSA returns the index number of the FTT input whose time value is selected by the selectionFunction procedure.

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Annex A

(normative)

Protocol Implementation Conformance Statement (PICS) proforma[27]

A.1 Introduction

The supplier of a protocol implementation that is claimed to conform to this standard shall complete the following PICS proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. The PICS can have a number of uses, including the following:

a) By the protocol implementer, as a checklist to reduce the risk of failure to conform to the standard through oversight;

b) By the supplier and acquirer—or potential acquirer—of the implementation, as a detailed indication of the capabilities of the implementation, stated relative to the common basis for understanding provided by the standard PICS proforma;

c) By the user—or potential user—of the implementation, as a basis for initially checking the possibility of interworking with another implementation (note that, while interworking can never be guaranteed, failure to interwork can often be predicted from incompatible PICS);

d) By a protocol tester, as the basis for selecting appropriate tests against which to assess the claim for conformance of the implementation.

A.2 Abbreviations and special symbols

A.2.1 Status symbols

M	mandatory
O	optional
O.*n*	optional, but support of at least one of the group of options labeled by the same numeral *n* is required
X	prohibited
pred:	conditional-item symbol, including predicate identification (see A.3.4)
¬	logical negation, applied to a conditional item's predicate

A.2.2 General abbreviations

N/A	not applicable
PICS	Protocol Implementation Conformance Statement

[27] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

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A.3 Instructions for completing the PICS proforma

A.3.1 General structure of the PICS proforma

The first part of the PICS proforma, implementation identification and protocol summary, is to be completed as indicated with the information necessary to identify fully both the supplier and the implementation.

The main part of the PICS proforma is a fixed-format questionnaire, divided into several subclauses, each containing a number of individual items. Answers to the questionnaire items are to be provided in the rightmost column, either by simply marking an answer to indicate a restricted choice (usually Yes or No), or by entering a value or a set or range of values. (Note that there are some items where two or more choices from a set of possible answers can apply; all relevant choices are to be marked.)

Each item is identified by an item reference in the first column. The second column contains the question to be answered; the third column records the status of the item—whether support is mandatory, optional, or conditional (see also A.3.4). The fourth column contains the reference or references to the material that specifies the item in the main body of this standard, and the fifth column provides the space for the answers.

A supplier might also provide (or be required to provide) further information, categorized as either Additional Information or Exception Information. When present, each kind of further information is to be provided in a further subclause of items labeled *Ai* or *Xi*, respectively, for cross-referencing purposes, where *i* is any unambiguous identification for the item (e.g., simply a numeral). There are no other restrictions on its format and presentation.

A completed PICS proforma, including any Additional Information and Exception Information, is the Protocol Implementation Conformation Statement for the implementation in question.

NOTE—Where an implementation is capable of being configured in more than one way, a single PICS might be able to describe all such configurations. However, the supplier has the choice of providing more than one PICS, each covering some subset of the implementation's configuration capabilities, in case that makes for easier and clearer presentation of the information.

A.3.2 Additional Information

Items of Additional Information allow a supplier to provide further information intended to assist the interpretation of the PICS. It is not intended or expected that a large quantity will be supplied, and a PICS can be considered complete without any such information. Examples might be an outline of the ways in which a (single) implementation can be set up to operate in a variety of environments and configurations, or information about aspects of the implementation that are outside the scope of this standard but that have a bearing upon the answers to some items.

References to items of Additional Information may be entered next to any answer in the questionnaire and may be included in items of Exception Information.

A.3.3 Exception Information

It occasionally happens that a supplier will wish to answer an item with mandatory status (after any conditions have been applied) in a way that conflicts with the indicated requirement. No preprinted answer will be found in the Support column for this; instead, the supplier shall write the missing answer into the Support column, together with an *Xi* reference to an item of Exception Information, and shall provide the appropriate rationale in the Exception item itself.

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An implementation for which an Exception item is required in this way does not conform to this standard.

NOTE—A possible reason for the situation described above is that a defect in this standard has been reported, a correction for which is expected to change the requirement not met by the implementation.

A.3.4 Conditional status

A.3.4.1 Conditional items

The PICS proforma contains a number of conditional items. These are items for which both the applicability of the item itself, and its status if it does apply—mandatory or optional—are dependent upon whether or not certain other items are supported.

Where a group of items is subject to the same condition for applicability, a separate preliminary question about the condition appears at the head of the group, with an instruction to skip to a later point in the questionnaire if the Not Applicable (N/A) answer is selected. Otherwise, individual conditional items are indicated by a conditional symbol in the Status column.

A conditional symbol is of the form "**pred:** S" where **pred** is a predicate as described in A.3.4.2, and S is a status symbol, M or O.

If the value of the predicate is true (see A.3.4.2), the conditional item is applicable, and its status is indicated by the status symbol following the predicate; the answer column is to be marked in the usual way. If the value of the predicate is false, the N/A answer is to be marked.

A.3.4.2 Predicates

A predicate is one of the following:

a) An item-reference for an item in the PICS proforma; the value of the predicate is true if the item is marked as supported, and is false otherwise;

b) A predicate-name, for a predicate defined as a Boolean expression constructed by combining item references using the Boolean operator "or"; the value of the predicate is true if one or more of the items is marked as supported;

c) The logical negation symbol "¬" prefixed to an item-reference or predicate-name; the value of the predicate is true if the value of the predicate formed by omitting the "¬" symbol is false, and vice versa.

Each item whose reference is used in a predicate or predicate definition, or in a preliminary question for grouped conditional items, is indicated by an asterisk in the Item column.

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A.4 PICS proforma for IEEE Std 802.1AS-2025

A.4.1 Implementation identification

Supplier	
Contact point for queries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) of machines and/or operating system names	

NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirement for full identification.

NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).

A.4.2 Protocol summary, IEEE Std 802.1AS

Identification of protocol specification	IEEE Std 802.1AS-2025, IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications
Identification of amendments and corrigenda to the PICS proforma which have been completed as part of the PICS	Amd. : Corr. : Amd. : Corr. :
Have any Exception items been required? (See A.3.3: the answer Yes means that the implementation does not conform to IEEE Std 802.1AS-2025)	No [ ] Yes [ ]

Date of Statement	

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A.5 Major capabilities

Item	Feature	Status	References	Support
DOMADD	Does the time-aware system support one or more PTP Instances with domainNumber in the range 1 to 127?	O	item e) of 5.4.2, 8.1	Yes [ ] No [ ]
DOM0	Does the time-aware system support one or more PTP Instances with domainNumber 0?	O	item e) of 5.4.2, 8.1	Yes [ ] No [ ]
MINTA	Does the PTP Instance support at least one PTP Port with minimal requirements?	M	10.2.13, item c) of 5.4, A.7	Yes [ ]
BTCX	Does any PTP Instance with domainNumber in the range of 1 to 127 implement the best timeTransmitter clock algorithm?	DOMADD:O	10.2.13, item g) of 5.4.3, 10.3, A.9	Yes [ ] No [ ]
SIG	Does the PTP Instance transmit Signaling messages?	M	item h) of 5.4.1, 10.4, 10.6.4, A.8	Yes [ ]
GMCAP	Is the PTP Instance capable of acting as a Grandmaster PTP Instance?	O	10.2.13, item c) of 5.4.2, 10.1.3, A.10	Yes [ ] No [ ]
BRDG	Does the PTP Instance act as a PTP Relay Instance on two or more PTP Ports?	O	item d) of 5.4.2, 5.4.4	Yes [ ] No [ ]
MITT	Does the PTP Instance support media-independent timeTransmitter functionality on at least one PTP Port?	GMCAP or BRDG:M	item b) of 5.4.2, A.11	Yes [ ] N/A [ ]
MIPERF	Does the PTP Instance support the performance requirements?	M	10.2.13, item j) of 5.4, A.12	Yes [ ]
EXT	Does the PTP Instance support external port configuration?	O	item f) of 5.4.2, A.21	Yes [ ] No [ ]
MDFDPP	Does the PTP Instance support media-dependent full-duplex point-to-point functionality on one or more PTP Ports?	O.1	5.5, Clause 11, A.6, A.13	Yes [ ] No [ ]
MDDOT11	Does the PTP Instance support media-dependent IEEE 802.11 link functionality on one or more PTP Ports?	O.1	5.6, Clause 12, A.6, A.14	Yes [ ] No [ ]
MDEPON	Does the PTP Instance support IEEE 802.3 Passive Optical Networking (EPON)?	O.1	5.7, Clause 13, A.6, A.15	Yes [ ] No [ ]
MDGHN	Does the PTP Instance support media-dependent ITU-T G.hn functionality on one or more PTP Ports?	O.1	item b) of 5.8, 16.6.3, A.18	Yes [ ] No [ ]
MDMOCA	Does the PTP Instance support media-dependent MoCA functionality on one or more PTP Ports?	O.1	item b) of 5.8, 16.6.2, A.17	Yes [ ] No [ ]
MDCSN	Does the PTP Instance support media-dependent CSN functionality on one or more PTP Ports?	MDGHN or MDMOCA:M	5.8, Clause 16, A.6, A.16	Yes [ ] No [ ]

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A.5 Major capabilities *(continued)*

Item	Feature	Status	References	Support
MGT	Is management of the PTP Instance supported?	O	item j) of 5.4.2, Clause 14	Yes [ ] No [ ]
RMGT	Is a remote management protocol supported?	MGT: O	item k) of 5.4.2, A.19	Yes [ ] No [ ]
APPL	Does the PTP Instance support one or more of the application interfaces?	O	item i) of 5.4.2, Clause 9, A.20	Yes [ ] No [ ]
HOTSTDBY	Does the time-aware system support hot standby as specified in Clause 18?	O	Clause 18, 14.8, item m) of 5.4.2, 9.3.3.4, 9.4.3.4, 9.5.3.4, 9.6.2.6	Yes [ ] No [ ]
DRFTRK	Does the PTP Instance support the Drift_Tracking TLV as specified in 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.13, 11.2.14, and 11.4.4?	O	item n) of 5.4.2, 10.2.4.26, 10.2.4.27, 10.2.4.28, 11.2.13, 11.2.14, and 11.4.4	Yes [ ] No [ ]
MDHDE	Does the PTP Instance support media-dependent HDE link functionality on one or more PTP Ports?	O.1	5.9, Clause 11, Clause 19, A.6, A.13	Yes [ ] No [ ]
FTT Entity	Does the time-aware system implement the FTT entity as specified in 20.2 through 20.4?	O	5.3, 9.3, 14.23, 17.6.3, 20.1.3	Yes [ ] No [ ]

A.6 Media access control methods

Item	Feature	Status	References	Support
MAC-IEEE-802.3	Which MAC methods are implemented in conformance with the relevant MAC standards?	O:2	11.1 19.1	Yes [ ] No [ ] Yes [ ] No [ ]
MAC-IEEE-802.11		O:2	12.1	Yes [ ] No [ ]
MAC-1	Has a PICS been completed for each of the MAC methods implemented as required by the relevant MAC Standards?	M		Yes [ ]
MAC-2	Do all the MAC methods implemented support the MAC Timing aware Service as specified?	M	Clause 11, Clause 12, Clause 13 Clause 19	Yes [ ]

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A.7 Minimal time-aware system

Item	Feature	Status	References	Support
MINTA-1	Do all PTP Instances of the device implement the functionality specified by the SiteSyncSync state machine in Figure 10-3 in compliance with the requirements of 10.2.7?	M	item f) of 5.4, 10.2.7	Yes [ ]
MINTA-2	Do all PTP Instances of the device implement the functionality specified by the PortSyncSyncReceive state machine in Figure 10-4 on each PTP Port in compliance with the requirements of 10.2.8?	M	item d) of 5.4	Yes [ ]
MINTA-3	Do all PTP Instances of the device implement the functionality specified by the ClockTimeReceiverSync state machine in Figure 10-9 in compliance with the requirements of 10.2.13?	M	10.2.13, item e) of 5.4	Yes [ ]
MINTA-4	For all PTP Instances of the device, does the PTP Port sending a Signaling message that contains a message interval request TLV adjust its syncReceiptTimeoutTimeInterval of this PTP Instance in compliance with the requirements of 10.6.4.3.7 and Table 10-16?	SIG:M	10.6.4.3.7	Yes [ ] N/A [ ]
MINTA-5	Is the clockIdentity constructed in compliance with the requirements of 8.5.2.2?	M	8.5.2.2	Yes [ ]
MINTA-6	Is the domainNumber for all transmitted messages in the range 0 through 127, in compliance with the requirements of 8.1?	M	8.1	Yes [ ]
MINTA-7	Is the majorSdoId 0x1 and the minorSdoId 0x0 for all transmitted gPTP domain messages?	M	8.1	Yes [ ]
MINTA-8	Is the domainNumber for at least one of the gPTP domains supported by the time-aware system, in compliance with the requirements of 8.1?	M	8.1	Yes [ ]
MINTA-9	Is the IEEE 802.1AS time of domain 0 measured relative to the PTP epoch in compliance with the requirements of 8.2.2?	M	8.2.2	Yes [ ]
MINTA-10	If path delay asymmetry is modeled by this PTP Instance does it comply with the requirements of 8.3?	O	8.3	Yes [ ] No [ ] N/A [ ]
MINTA-11	Do all derived data types that are transmitted in IEEE 802.1AS messages and headers comply with 6.4.4?	M	6.4.4	Yes [ ]
MINTA-12	Is the granularity of the local clock 40 ns or better in compliance with the requirements of B.1.2?	M	B.1.2	Yes [ ]
MINTA-13	Is the frequency of the local clock relative to TAI ±100 ppm in compliance with the requirements of B.1.1?	M	B.1.1	Yes [ ]

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A.7 Minimal time-aware system *(continued)*

Item	Feature	Status	References	Support
MINTA-14	Does the PTP Instance ignore non-propagating TLVs of Announce and Signaling messages that it cannot parse, and attempt to parse the next TLV, in compliance with the requirements of 10.6.1?	M	10.6.1	Yes [ ]
MINTA-15	Does the PTP Instance support the state machines related to signaling gPTP capability?	M	item g) of 5.4.1, 10.4.3	Yes [ ]
MINTA-16	For receive of all messages and for transmit of all messages except Announce, does the PTP Instance support the message requirements?	M	item h) of 5.4.1, 10.5, 10.6, 10.7	Yes [ ]
MINTA-17	Does the PTP Instance support the gPTP requirements specified in Clause 8, including the PTP Instance attributes?	M	item a) of 5.4, Clause 8, 8.6.2	Yes [ ]
MINTA-18	Does the PTP Instance support the requirements for time-synchronization state machines?	M	item b) of 5.4	Yes [ ]
MINTA-19	Does the PTP Instance implement the path trace TLV (i.e., process this TLV when received in an Announce message, and attach this TLV to a transmitted Announce message unless the TLV would cause the maximum frame size to be exceeded)?	M	10.3.11, 10.3.13, 10.3.14, 10.3.16	Yes [ ]
MINTA-20	Does the PTP Instance forward TLVs as required?	M	10.6.1	Yes [ ]
MINTA-21	Do the PTP Ports of this PTP Instance implement the functionality of the GptpCapableIntervalSetting state machine in compliance with the requirements of 10.4.4 and Figure 10-23?	MINTA-15:O	10.4.4, item h) of 5.4.2, item h) of 5.4.1	Yes [ ] No [ ]
MINTA-22	Does the PTP Instance implement the PtpInstanceSyncStatus state machine of 18.4 and the PtpInstanceStatusDS of 14.8?	¬HOTSTDBY: O	18.4, 14.8	Yes [ ] No [ ]

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A.8 Signaling

Item	Feature	Status	References	Support
SIG-1	Do the sequence numbers of Signaling messages comply with the requirements of 10.5.7?	SIG:M	10.5.7	Yes [ ]
SIG-2	Does the Signaling message body comply with the requirements of 10.6.4.1 and Table 10-13?	SIG:M	10.6.4.1	Yes [ ]
SIG-3	Does the Signaling message header comply with the requirements of 10.6.2?	SIG:M	10.6.2	Yes [ ]
SIG-4	Are all Signaling message reserved fields equal to 0 in compliance with the requirements of 10.6.1?	SIG:M	10.6.1	Yes [ ]
SIG-5	Is the destination MAC address for all Signaling messages equal to 01:80:C2:00:00:0E in compliance with the requirements of 10.5.3?	SIG:M	10.5.3	Yes [ ]
SIG-6	Is the EtherType for all Signaling messages equal to 88-F7 in compliance with the requirements of 10.5.4?	SIG:M	10.5.4	Yes [ ]
SIG-7	Does the message interval request TLV for Signaling messages comply with the requirements in 10.6.4.3?	BTC-23:M, MITT-16:M, MDFDPP-5:M	10.6.4.3	Yes [ ]

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A.9 Best timeTransmitter clock

Item	Feature	Status	References	Support
BTC-1	Does the PTP Instance implement the functionality specified by the PortAnnounceReceive state machine in Figure 10-14 on each PTP Port in compliance with the requirements of 10.3.11?	DOM0 or BTCX:M	10.3.11	Yes [ ]
BTC-2	Does the PTP Instance implement the functionality specified by the PortAnnounceInformation state machine in Figure 10-15 on each PTP Port in compliance with the requirements of 10.3.12?	DOM0 or BTCX:M	10.3.12	Yes [ ]
BTC-3	Does the PTP Instance implement the functionality specified by the PortStateSelection state machine in Figure 10-16 on each PTP Port in compliance with the requirements of 10.3.13? NOTE—There is one instance of the PortStateSelection state machine for the PTP Instance, for each gPTP domain. Some of the PortStateSelection state machine computations are performed for each PTP Port, and some of the computations are performed for the PTP Instance as a whole (and all the computations are performed for each gPTP domain).	DOM0 or BTCX:M	10.3.13	Yes [ ]
BTC-4	If the value of clockA's SystemIdentity is less than that of clockB, is clockA selected as Grandmaster PTP Instance in compliance with the requirements of 10.3.2?	DOM0 or BTCX:M	10.3.2	Yes [ ]
BTC-5	Does the value of priority1 comply with the requirements of 8.6.2.1?	DOM0 or BTCX:M	8.6.2.1	Yes [ ]
BTC-6	Does the value of clockClass comply with the requirements of 8.6.2.2?	DOM0 or BTCX:M	8.6.2.2	Yes [ ]
BTC-7	Does the value of priority2 comply with the requirements of 8.6.2.5?	DOM0 or BTCX:M	8.6.2.5	Yes [ ]
BTC-8	Does the value of clockAccuracy comply with requirements of 8.6.2.3?	DOM0 or BTCX:M	8.6.2.3	Yes [ ]
BTC-9	Does the value of offsetScaledVariance comply with the requirements of 8.6.2.4?	DOM0 or BTCX:M	8.6.2.4	Yes [ ]
BTC-10	Does the value of timeSource comply with requirements of 8.6.2.7 and Table 8-2?	DOM0 or BTCX:M	8.6.2.7	Yes [ ]
BTC-11	Is the PTP Port number equal to 1 in compliance with the requirements of 8.5.2.3?	(DOM0 or BTCX) and ¬BRDG:M	8.5.2.3	Yes [ ] N/A [ ]

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A.9 Best timeTransmitter clock *(continued)*

Item	Feature	Status	References	Support
BTC-12	Are the PTP Ports numbered 1 through N for each of N PTP Ports in compliance with the requirements of 8.5.2.3?	DOM0 or BTCX:M	8.5.2.3	Yes [ ]
BTC-13	Does the clockIdentity field comply with the requirements of 8.5.2.2?	DOM0 or BTCX:M	8.5.2.2	Yes [ ]
BTC-14	When no grandmaster-capable PTP Instance is available does the behavior of the PTP Instance comply with the requirements of 10.2.13.2, i.e., the clockTimeReceiverTime should be provided by the local clock?	DOM0 or BTCX:M	10.2.13.2	Yes [ ]
BTC-15	Does the value of announceReceiptTimeout comply with the requirements of 10.7.3.2?	DOM0 or BTCX:M	10.7.3.2	Yes [ ]
BTC-16	Does the TimeReceiverPort remove the PTP Port from the BTC selection after announceReceiptTimeout expires in compliance with the requirements of 10.7.3.2?	DOM0 or BTCX:M	10.7.3.2	Yes [ ]
BTC-17	Does the value of syncReceiptTimeout comply with the requirements of 10.7.3.1?	DOM0 or BTCX:M	10.7.3.1	Yes [ ]
BTC-18	Does the TimeReceiverPort remove the PTP Port from the BTC selection after syncReceiptTimeout expires in compliance with 10.7.3.1?	DOM0 or BTCX:M	10.7.3.1	Yes [ ]
BTC-19	Does the PTP Port sending a message interval request Signaling message adjust its announceReceiptTimeoutTimeInterval in compliance with the requirements of 10.6.4.3.8 and Table 10-17?	SIG:M	10.6.4.3.8	Yes [ ]
BTC-20	If the PTP Instance implements the ClockSourceTime interface, does the value of lastGmPhaseChange comply with the requirements of 9.2.2 and 6.4.3.3?	O	9.2.2	Yes [ ] No [ ]
BTC-21	Does the transmitted timing information comply with the requirements of 10.3.1, including specifications for externalPortConfigurationEnabled value of false?	GMCAP:M	10.3.1	Yes [ ] N/A [ ]
BTC-22	Does the PTP Instance implement BTCA requirements that are not listed in the preceding BTC rows?	DOM0 or BTCX:M	10.3.2, 10.3.3 10.3.4, 10.3.5, 10.3.6, 10.3.8, 10.3.10	Yes [ ]
BTC-23	Do the TimeTransmitterPorts of this PTP Instance implement the functionality of the AnnounceIntervalSetting state machine in compliance with the requirements of 10.3.17 and Figure 10-20?	BTCX:O	10.3.17, item g) of 5.4.2, item h) of 5.4.1	Yes [ ] No [ ]

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A.10 Grandmaster-capable PTP Instance

Item	Feature	Status	References	Support
	If GMCAP not supported, mark N/A.			N/A [ ]
GMCAP-1	Does the PTP Instance implement the functionality specified by the ClockTimeTransmitterSyncSend state machine in compliance with the requirements of 10.2.9 and Figure 10-5?	GMCAP:M	10.2.9	Yes [ ]
GMCAP-2	Does the PTP Instance implement the functionality specified by the ClockTimeTransmitterSyncOffset state machine in compliance with the requirements of 10.2.10 and Figure 10-6?	GMCAP:M	10.2.10	Yes [ ]
GMCAP-3	Does the device implement the functionality specified by the ClockTimeTransmitterSyncReceive state machine in compliance with the requirements of 10.2.11 and Figure 10-7?	GMCAP:M	10.2.11	Yes [ ]

A.11 Media-independent timeTransmitter

Item	Feature	Status	References	Support
	If MITT not supported, mark N/A.			N/A [ ]
MITT-1	Does the PTP Instance implement the functionality of the AnnounceIntervalSetting state machine in compliance with the requirements of 10.3.17 and Figure 10-20 on each PTP Port?	MITT:M	10.3.17	Yes [ ]
MITT-2	Does the PTP Instance implement the functionality of the PortSyncSyncSend state machine in compliance with the requirements of 10.2.9 and Figure 10-8 on each PTP Port?	MITT:M	10.2.9	Yes [ ]
MITT-3	Does the PTP Instance implement the functionality of the PortAnnounceTransmit state machine in compliance with the requirements of 10.3.16 and Figure 10-19 on each PTP Port?	MITT:M	10.3.16	Yes [ ]
MITT-4	Does the destination MAC address of all Announce messages equal 01:80:C2:00:00:0E?	MITT:M	10.5.3	Yes [ ]
MITT-5	Does the EtherType of all Announce messages equal 88-F7?	MITT:M	10.5.4	Yes [ ]
MITT-6	Do the sequence numbers of Announce messages comply with the requirements of 10.5.7?	MITT:M	10.5.7	Yes [ ]

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A.11 Media-independent timeTransmitter *(continued)*

Item	Feature	Status	References	Support
MITT-7	Does the Announce message header comply with 10.6.2?	MITT:M	10.6.2	Yes [ ]
MITT-8	Does the Announce message body comply with the requirements in 10.6.3.1 and Table 10-11?	MITT:M	10.6.3.1	Yes [ ]
MITT-9	Are all Announce message reserved fields equal to 0?	MITT:M	10.6.1	Yes [ ]
MITT-10	If it is not otherwise specified, is the logAnnounceInterval equal to zero or within the allowed range?	MITT:M	10.7.2.1	Yes [ ]
MITT-11	Does the value of currentUtcOffset comply with the requirements of 8.2.3?	MITT:M	8.2.3	Yes [ ]
MITT-12	Do the values of the leap59, leap61, and currentUtcOffsetValid flags comply with the requirements of 10.3.8?	MITT:M	10.3.8	Yes [ ]
MITT-13	Does this PTP Instance ensure that messages that traverse it or originate from it are not transmitted with VLAN tags in compliance with the requirements of 11.3.3?	MITT:M	11.3.3	Yes [ ]
MITT-14	Is the computation of cumulative rateRatio in accordance with 10.2.8.3?	MITT:M	10.2.8.3	Yes [ ] N/A [ ]
MITT-15	For transmit of the Announce message, does the PTP Instance support the message requirements?	MITT:M	10.5, 10.6, 10.7	Yes [ ]
MITT-16	Do the TimeTransmitterPorts of this PTP Instance implement the functionality of the SyncIntervalSetting state machine in compliance with the requirements of 10.2.14 and Figure 10-10?	MITT:O	10.2.14, item b) 3) of 5.4.2, item h) of 5.4.1	Yes [ ] No [ ]

A.12 Media-independent performance requirements

Item	Feature	Status	References	Support
MIPERF-1	Does the PTP Instance comply with the performance requirements of B.1?	M	B.1	Yes [ ]
MIPERF-2	Does the PTP Instance comply with the performance requirements of B.2.4?	M	B.2.4	Yes [ ]
MIPERF-3	Does the PTP Instance comply with the performance recommendations of B.2.2?	O	B.2.2	Yes [ ] No [ ]
MIPERF-4	Does the PTP Instance comply with the performance recommendations of B.2.3?	O	B.2.3	Yes [ ] No [ ]

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A.13 Media-dependent, FDE link

Item	Feature	Status	References	Support
MDFDPP-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 11.2.13 and Figure 11-6?	MDFDPP:M	11.2.13	Yes [ ]
MDFDPP-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 11.2.14 and Figure 11-7?	MITT and MDFDPP:M	11.2.14	Yes [ ]
MDFDPP-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 11.2.18 and Figure 11-9?	MDFDPP:M	11.2.18	Yes [ ]
MDFDPP-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 11.2.19 and Figure 11-10?	MDFDPP:M	11.2.19	Yes [ ]
MDFDPP-5	Does this port implement the functionality of the LinkDelayIntervalSetting state machine in compliance with the requirements of 11.2.20 and Figure 11-11?	MDFDPP:O	11.2.20, item l) of 5.5, item h) of 5.4.1	Yes [ ] No [ ]
MDFDPP-6	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes [ ]
MDFDPP-7	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MITT and MDFDPP:M	11.3.2.1	Yes [ ]
MDFDPP-8	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes [ ]
MDFDPP-9	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 11.3.2.1 and 11.3.9?	MDFDPP:M	11.3.2.1	Yes [ ]
MDFDPP-10	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 11.3.3?	MDFDPP:M	11.3.3	Yes [ ]
MDFDPP-11	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 11.3.4 [01-80-C2-00-00-0E]?	MDFDPP:M	11.3.4	Yes [ ]
MDFDPP-12	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 11.3.4?	MDFDPP:M	11.3.4	Yes [ ]

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A.13 Media-dependent, FDE link *(continued)*

Item	Feature	Status	References	Support
MDFDPP-13	Do all media-dependent messages transmitted on this port use an EtherType specified in Table 11-4 [88-F7]?	MDFDPP:M	11.3.5	Yes [ ]
MDFDPP-14	Does the header of all the media-dependent messages on this port comply with the requirements of 11.4.2 and Table 10-7?	MDFDPP:M	11.4.2	Yes [ ] N/A [ ]
MDFDPP-15	Does the body of Sync messages sent on this PTP Port comply with the requirements of 11.4.3, Table 11-8, and Table 11-9?	MDFDPP:M	11.4.3	Yes [ ]
MDFDPP-16	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 11.4.4, (lastGmPhaseChange), and Table 11-10?	MDFDPP:M	11.4.4	Yes [ ]
MDFDPP-17	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 11.4.5 and Table 11-13?	MDFDPP:M	11.4.5	Yes [ ]
MDFDPP-18	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 11.4.6 and Table 11-14?	MDFDPP:M	11.4.6	Yes [ ]
MDFDPP-19	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 11.4.7 and Table 11-15?	MDFDPP:M	11.4.7	Yes [ ]
MDFDPP-20	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 11.4.1?	MDFDPP:M	11.4.1	Yes [ ]
MDFDPP-21	Do the Sync message sequence numbers comply with the requirements of 11.3.8?	MITT and MDFDPP:M	11.3.8	Yes [ ] N/A [ ]
MDFDPP-22	Do the Pdelay_Req message sequence numbers comply with the requirements of 11.3.8?	MDFDPP:M	11.3.8	Yes [ ]
MDFDPP-23	Does the Pdelay mean request transmission interval comply with the requirements of 11.5.2.2?	MDFDPP:M	11.5.2.2	Yes [ ]
MDFDPP-24	Does the Sync mean transmission interval comply with the requirements of 11.5.2.3?	MDFDPP:M	11.5.2.3	Yes [ ]
MDFDPP-25	Does the full-duplex point-to-point media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 11.2.2?	MDFDPP:M	11.2.2	Yes [ ]
MDFDPP-26	Does the device’s use of flow control comply with the requirements of 11.2.3 and 11.2.4?	MDFDPP:M	11.2.3, 11.2.4	Yes [ ]

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A.13 Media-dependent, FDE link *(continued)*

Item	Feature	Status	References	Support
MDFDPP-27	Does the PTP Instance or CMLDS consider the PTP Port or Link Port, respectively, to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 11.5.3?	MDFDPP:M	11.5.3	Yes [ ]
MDFDPP-28	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 11.4.1?	MDFDPP:M	11.4.1	Yes [ ]
MDFDPP-29	Does the time-aware system initialize meanLinkDelayThresh as specified in 11.2.2?	MDFDPP:M	11.2.2	Yes [ ]
MDFDPP-30	Does this port of the time-aware system support asymmetry measurement mode (see Annex G for informative description)?	MDFDPP:O	14.15, 14.20, 10.2.5, 10.2.8, 10.3.12, 10.3.13, 10.3.15, 10.3.16, 11.2.13, 11.2.14, 11.2.19	Yes [ ] No [ ]
MDFDPP-31	Does this PTP Port support one-step receive?	MDFDPP:O	11.2.13	Yes [ ] No [ ]
MDFDPP-32	Does this PTP Port support one-step transmit?	MDFDPP:O	11.2.14	Yes [ ] No [ ]
MDFDPP-33	Does this PTP Port implement the functionality of the OneStepTxOperSetting state machine in compliance with the requirements of 11.4, 11.2.15, and Figure 11-8?	MDFDPP:O	11.4, 11.2.15	Yes [ ] No [ ]
MDFDPP-34	Does this port support propagation delay averaging?	MDFDPP:O	11.2.18.3.4	Yes [ ] No [ ]
MDFDPP-35	If the time-aware system implements more than one domain, does the time-aware system provide CMLDS?	MDFDPP:M	11.2.16.1	Yes [ ]
MDFDPP-36	If the time-aware system implements only one domain, does the time-aware system provide CMLDS?	MDFDPP:O	11.2.16.2	Yes [ ] No [ ]
MDFDPP-37	Does this port support two-step capability on receive?	MDFDPP:M	11.2.13, item c) of 5.5	Yes [ ]
MDFDPP-38	Does this port support two-step capability on transmit?	MDFDPP:M	11.2.14, item d) of 5.5	Yes [ ]

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A.14 Media-dependent IEEE 802.11 link

Item	Feature	Status	References	Support
MDDOT11-1	Does the IEEE 802.11 MAC implement the timeTransmitter port functionality in compliance with the requirements of 12.5.1?	MDDOT11 and MITT:M	item d) of 5.6, 12.5.1	Yes [ ]
MDDOT11-2	Does the IEEE 802.11 MAC implement the timeReceiver port functionality in compliance with the requirements of 12.5.2?	MDDOT11:M	item a), item b), and item d) of 5.6, 12.5.2	Yes [ ]
MDDOT11-3	Does the IEEE 802.11 MAC determine the value of asCapable in compliance with the requirements of 12.4?	MDDOT11:M	12.4	Yes [ ]
MDDOT11-4	Does the IEEE 802.11 MAC determine the value of mean time interval between synchronization messages in compliance with the requirements of 12.8?	MDDOT11 and MITT:M	12.8	Yes [ ]
MDDOT11-5	Does the IEEE 802.11 MAC support the use of the VendorSpecific information element of 12.7 to carry end-to-end link-independent timing information?	MDDOT11:M	12.7	Yes [ ]
MDDOT11-6	Does the IEEE 802.11 MAC implement fine timing measurement as a timeTransmitter port?	MDDOT11-1:O	item c) and item e) of 5.6, 12.5.1	Yes [ ] No [ ]
MDDOT11-7	Does the IEEE 802.11 MAC implement fine timing measurement as a timeReceiver port?	MDDOT11-2:O	item c) and item e) of 5.6, 12.5.2	Yes [ ] No [ ]

A.15 Media-dependent IEEE 802.3 EPON link

Item	Feature	Status	References	Support
MDEPON-1	Does the TIMESYNC message format comply with the requirements of 13.3 and Table 13-1?	MDEPON:M	13.3	Yes [ ]
MDEPON-2	Does the PTP Instance implement the functionality specified by the requester state machine in compliance with the requirements of 13.8.1 and Figure 13-3?	MDEPON and MITT:M	13.8.1.4	Yes [ ]
MDEPON-3	Does the PTP Instance implement the functionality specified by the responder state machine in compliance with the requirements of 13.8.2 and Figure 13-4?	MDEPON:M	13.8.2.4	Yes [ ]
MDEPON-4	Does the TIMESYNC message transmission interval comply with the requirements of 13.9.1 and 13.9.2?	MDEPON:M	13.9.1, 13.9.2	Yes [ ]

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A.15 Media-dependent IEEE 802.3 EPON link *(continued)*

Item	Feature	Status	References	Support
MDEPON-5	Does the implementation of best timeTransmitter selection comply with the requirements of 13.1.3?	MDEPON:M	13.1.3	Yes [ ]
MDEPON-6	Does the determination of the value of asCapable comply with the requirements of 13.4?	MDEPON:M	13.4	Yes [ ]

A.16 Media-dependent CSN link

Item	Feature	Status	References	Support
MDCSN-1	Does the PTP Instance implement the functionality of the MDSyncSendSM state machine in compliance with 11.2.14?	MDCSN and MITT:M	11.2.14	Yes [ ]
MDCSN-2	Does the PTP Instance implement the functionality of the MDSyncReceiveSM state machine in compliance with 11.2.13?	MDCSN:M	11.2.13	Yes [ ]
MDCSN-3	Does the PTP Instance calculate path delay in compliance with the requirement of 16.4?	MDCSN:M	16.4.1, 16.4.2, 16.4.3	Yes [ ]
MDCSN-4	Does the PTP Instance propagate synchronized time in compliance with the requirements of 16.5?	MDCSN:M	16.5.2, 16.5.3	Yes [ ]
MDCSN-5	Does the PTP Instance act as Grandmaster PTP Instance in compliance with the requirements of 16.7?	GMCAP and MDCSN:M	16.7	Yes [ ]
MDCSN-6	Does the PTP Instance comply with the performance requirements of 16.8?	GMCAP and MDCSN:M	16.8	Yes [ ]

A.17 Media-dependent MoCA link

Item	Feature	Status	References	Support
MDMOCA-1	Does the MoCA MD entity propagate Sync messages in compliance with the requirements of 16.6.2?	MDMOCA:M	16.6.2	Yes [ ]

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A.18 Media-dependent ITU-T G.hn link

Item	Feature	Status	References	Support
MDGHN-1	Does the GHN MD entity propagate Sync messages in compliance with the requirements of 16.6.3?	MDGHN:M	16.6.3	Yes [ ]

A.19 Remote management

Item	Feature	Status	References	Support
	If item RMGT is not supported, mark N/A.			N/A[ ]
RMGT-1	What management protocol standard(s) or specification(s) are supported?	RMGT:M	item k) 1) of 5.4.2	
RMGT-2	What standard(s) or specification(s) for managed object definitions and encodings are supported for use by the remote management protocol?	RMGT:M	item k) 2) of 5.4.2	
RMGT-3	If the Simple Network Management Protocol (SNMP) is listed in RMGT-2, is the IEEE 8021-AS-MIB module fully supported (per its MODULE-COMPLIANCE)?	RMGT:O	item k) 3) of 5.4.2, Clause 15	Yes [ ] No [ ] N/A [ ]
RMGT-4	If a remote management protocol that supports YANG is listed in RMGT-2, is the YANG data model ieee802-dot1as-gptp of Clause 17 supported?	RMGT:O	item k) 4) of 5.4.2, Clause 17	Yes [ ] No [ ]
RMGT-5	If a remote management protocol that supports YANG is listed in RMGT-2, and if hot standby is supported, is the YANG data model ieee802-dot1as-hs of Clause 17 supported?	RMGT:O	item k) 5) of 5.4.2, Clause 17, Clause 18	Yes [ ] No [ ] N/A [ ]
RMGT-6	If a remote management protocol that supports YANG is listed in RMGT-2, and if the FTT entity is supported, is the YANG data model ieee802-dot1as-ftt of Clause 17 supported?	RMGT:O	5.3, 17, 20.1.3	Yes [ ] No [ ] N/A [ ]

A.20 Application interfaces

Item	Feature	Status	References	Support
	If item APPL is not supported, mark N/A.			N/A[ ]
APPL-1	What application interfaces(s) are supported?	APPL:M	item i) of 5.4.2	

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A.21 External port configuration

Item	Feature	Status	References	Support
	If item EXT is not supported, mark N/A.			NA[ ]
EXT-1	Does the PTP Instance support the specifications for externalPortConfigurationEnabled value of true?	EXT:M	10.3.1 19.8	Yes [ ]
EXT-2	Does the PTP Instance support the PortAnnounceInformationExt state machine?	EXT:M	10.3.14	Yes [ ]
EXT-3	Does the PTP Instance support the PortStateSettingExt state machine?	EXT:M	10.3.15	Yes [ ]

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A.22 Media-dependent HDE link

Item	Feature	Status	References	Support
MDHDE-1	Does this PTP Port implement the functionality of the MDSyncReceiveSM state machine in compliance with the requirements of 19.2.13, 11.2.13, and Figure 11-6?	MDHDE:M	19.2.13, 11.2.13	Yes [ ]
MDHDE-2	Does this PTP Port implement the functionality of the MDSyncSendSM state machine in compliance with the requirements of 19.2.14, 11.2.14, and Figure 11-7?	MIMSTR and MDHDE:M	19.2.14, 11.2.14	Yes [ ]
MDHDE-3	Does this port implement the functionality of the MDPdelayRequest state machine in compliance with the requirements of 19.1.2, 19.2.18, 11.2.18, and Figure 11-9?	MDHDE:M	19.1.2, 19.2.18, 11.2.18	Yes [ ]
MDHDE-4	Does this port implement the functionality of the MDPdelayResponse state machine in compliance with the requirements of 19.1.2, 19.2.19, 11.2.19, and Figure 11-10?	MDHDE:M,	19.2.19 19.1.2 11.2.19	Yes [ ]
MDHDE-5	Does this PTP Port timestamp Sync messages on ingress with respect to the LocalClock in compliance with 19.3, 11.3.2.1, and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]
MDHDE-6	Does this PTP Port timestamp Sync messages on egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1, and 11.3.9?	MIMSTR and MDHDE:M	19.3, 11.3.2.1	Yes [ ]
MDHDE-7	Does this port timestamp Pdelay_Req messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1, and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]
MDHDE-8	Does this port timestamp Pdelay_Resp messages on ingress and egress with respect to the LocalClock in compliance with the requirements of 19.3, 11.3.2.1, and 11.3.9?	MDHDE:M	19.3, 11.3.2.1	Yes [ ]

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A.22 Media-dependent HDE link *(continued)*

Item	Feature	Status	References	Support
MDHDE-9	Are all IEEE 802.1AS messages on this port sent without a Q-tag in compliance with the requirements of 19.3 and 11.3.3?	MDHDE:M	19.3, 11.3.3	Yes [ ]
MDHDE-10	Do all media-dependent messages transmitted on this port use a destination MAC address taken from Table 11-3 in compliance with the requirements of 19.3 and 11.3.4 [01-80-C2-00-00-0E]?	MDHDE:M	19.3, 11.3.4	Yes [ ]
MDHDE-11	Do all media-dependent messages transmitted on this port use a source MAC address that is assigned to that port in compliance with the requirements of 19.3 and 11.3.4?	MDHDE:M	19.3, 11.3.4	Yes [ ]
MDHDE-12	Do all media-dependent messages transmitted on this port us an EtherType specified in Table 11-4 [88-F7]?	MDHDE:M	19.3, 11.3.5	Yes [ ]
MDHDE-13	Does the header of all the media-dependent messages on this port comply with the requirements of 19.4, 11.4.2, and Table 10-7?	MDHDE:M	19.4, 11.4.2	Yes [ ] N/A [ ]
MDHDE-14	Does the body of Sync messages sent on this PTP Port comply with the requirements of 19.4, 11.4.3, Table 11-8, and Table 11-8?	MDHDE:M	19.4, 11.4.3	Yes [ ]
MDHDE-15	Does the body of Follow_Up messages sent on this PTP Port comply with the requirements of 19.4, 11.4.4, 6.4.3.3 (lastGmPhaseChange), and Table 11-10?	MDHDE:M	19.4, 11.4.4, 6.4.3.3	Yes [ ]
MDHDE-16	Does the body of Pdelay_Req messages sent on this port comply with the requirements of 19.4, 11.4.5, and Table 11-13?	MDHDE:M	19.4, 11.4.5	Yes [ ]
MDHDE-17	Does the body of Pdelay_Resp messages sent on this port comply with the requirements of 19.4, 11.4.6, and Table 11-14?	MDHDE:M	19.4, 11.4.6	Yes [ ]
MDHDE-18	Does the body of Pdelay_Resp_Follow_Up messages sent on this port comply with the requirements of 19.4, 11.4.7, and Table 11-15?	MDHDE:M	19.4, 11.4.7	Yes [ ]
MDHDE-19	Are all reserved fields in media-dependent messages sent on this port set to 0 in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes [ ]
MDHDE-20	Do the Sync message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MIMSTR and MDHDE:M	19.3, 11.3.8	Yes [ ] N/A [ ]
MDHDE-21	Do the Pdelay_Req message sequence numbers comply with the requirements of 19.3 and 11.3.8?	MDHDE:M	19.3, 11.3.8	Yes [ ]

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A.22 Media-dependent HDE link *(continued)*

Item	Feature	Status	References	Support
MDHDE-22	Does the Pdelay mean request transmission interval comply with the requirements of 19.5.2.2 and 11.5.2.2?	MDHDE:M	19.5.2.2, 11.5.2.2	Yes [ ]
MDHDE-23	Does the Sync mean transmission interval comply with the requirements of 19.5.2.3 and 11.5.2.3?	MDHDE:M	19.5.2.3, 11.5.2.3	Yes [ ]
MDHDE-24	Does the HDE media-dependent layer set the asCapable global variable in the media-independent PortSync entity in compliance with the requirements of 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes [ ]
MDHDE-25	Does the PTP Instance consider the PTP Port or Link Port to not be exchanging Pdelay messages when a valid response is not received in compliance with the requirements of 19.5.3 and 11.5.3?	MDHDE:M	19.5.3, 11.5.3	Yes [ ]
MDHDE-26	Does the PTP Instance ignore TLVs, of PTP messages, that it cannot parse and attempt to parse the next TLV, in compliance with the requirements of 19.4 and 11.4.1?	MDHDE:M	19.4, 11.4.1	Yes [ ]
MDHDE-27	Does the time-aware system initialize meanLinkDelayThresh as specified in 19.2.2 and 11.2.2?	MDHDE:M	19.2.2, 11.2.2	Yes [ ]
MDHDE-28	Does this port support propagation delay averaging?	MDHDE:O	19.2.18, 11.2.18.3.4	Yes [ ] No [ ]
MDHDE-29	Does this port support two-step capability on receive?	MDHDE:M	19.2.13, 11.2.13, item c) of 5.5	Yes [ ]
MDHDE-30	Does this port support two-step capability on transmit?	MDHDE:M	19.2.14, 11.2.15, item d) of 5.5	Yes [ ]
MDHDE-31	Is the transport-specific peer-to-peer delay mechanism supported?	MDHDE:M	19.2.16	Yes [ ]

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Annex B

(normative)

Performance requirements

B.1 LocalClock requirements

B.1.1 Frequency accuracy

The fractional frequency offset of the LocalClock relative to the TAI frequency (see ISO 80000-3:2006 and Annex C) shall be within ±100 ppm.

B.1.2 Time measurement granularity

The granularity with which the LocalClock measures time shall be less than or equal to 40/(1 – 0.0001) ns.

B.1.3 Noise generation

B.1.3.1 Jitter generation

The jitter generation of the free-running LocalClock shall not exceed 2 ns peak-to-peak, when measured over a 60 s measurement interval using a band-pass filter that consists of the following low-pass and high-pass filters:

a) High-pass filter: first-order characteristic (i.e., 0 dB gain peaking), 20 dB/decade roll-off, and 3 dB bandwidth (i.e., corner frequency) of 10 Hz

b) Low-pass filter: maximally-flat (i.e., Butterworth) characteristic, 60 dB/decade roll-off, and 3 dB bandwidth equal to the Nyquist rate of the LocalClock entity (i.e., one-half the nominal frequency of the LocalClock entity)

B.1.3.2 Wander generation

Wander generation is specified using the Time Deviation (TDEV) parameter. The corresponding values of the Allan Deviation (ADEV) and PTP Deviation (PTPDEV) are given for information; the former is also useful in describing the wander generation of clocks and oscillators, and the latter is related to the offsetScaledLogVariance attribute (see 8.6.2.4). Information on ADEV and TDEV is contained in ITU-T G.810 [B30] and IEEE Std 1139™-1999 [B11]. Information on Allan Deviation and PTP Variance (PTP Deviation is the square root of PTP Variance) is contained in 7.6.3 of IEEE Std 1588-2019.

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TDEV, denoted $\sigma_x(\tau)$, is estimated from a set of measurements, as shown in Equation (B-1).

$$\sigma_x(\tau) = \sqrt{\frac{1}{6n^2(N-3n+1)} \sum_{j=1}^{N-3n+1} \left[\sum_{i=j}^{n+j-1} (x_{i+2n} - 2x_{i+n} + x_i) \right]^2}, n = 1, 2, \ldots \left\lfloor \frac{N}{3} \right\rfloor \quad \text{(B-1)}$$

where

τ is $n\tau_0$ = observation interval
τ_0 is the sampling interval
N is the total number of samples [$(N-1)\tau_0$ = measurement interval]
$\lfloor y \rfloor$ denotes the floor function, i.e., the greatest integer less than or equal to y
x_i is the measured phase (time) error at the ith sampling time [the units of x_i and $\sigma_x(\tau)$ are the same]

ADEV, denoted $\sigma_y(\tau)$, is estimated from a set of measurements, as shown in Equation (B-2).

$$\sigma_y(\tau) = \sqrt{\frac{1}{2n^2\tau_0^2(N-2n)} \sum_{i=1}^{N-2n} (x_{i+2n} - 2x_{i+n} + x_i)^2}, n = 1, 2, \ldots \left\lfloor \frac{N-1}{2} \right\rfloor \quad \text{(B-2)}$$

where the notation is the same as defined above for TDEV.

PTPDEV, denoted $\sigma_{PTP}(\tau)$, is estimated from a set of measurements, as shown in Equation (B-3).

$$\sigma_{PTP}(\tau) = \sqrt{\frac{1}{6(N-2n)} \sum_{i=1}^{N-2n} (x_{i+2n} - 2x_{i+n} + x_i)^2}, n = 1, 2, \ldots \left\lfloor \frac{N-1}{2} \right\rfloor \quad \text{(B-3)}$$

where the notation is the same as defined above for TDEV.

TDEV, ADEV, and PTPDEV are second-order statistics on the phase error. All three statistics are functions of second differences of the phase error. As a result, these statistics are not affected by a constant frequency offset. This behavior is desired, because these statistics are used here to constrain noise generation.

TDEV for the LocalClock entity shall not exceed the mask of Table B-1 and Figure B-1, when measured using:

a) A measurement interval that is at least 120 s (i.e., at least 12 times the longest observation interval),
b) A low-pass filter with 3 dB bandwidth of 10 Hz, first-order characteristic, and 20 dB/decade roll-off, and
c) A sampling interval τ_0 that does not exceed 1/30 s.

Table B-1—Wander generation TDEV requirement for LocalClock entity

TDEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
5.0τ ns	$0.05 \le \tau \le 10$ s
No requirement	$\tau > 10$ s

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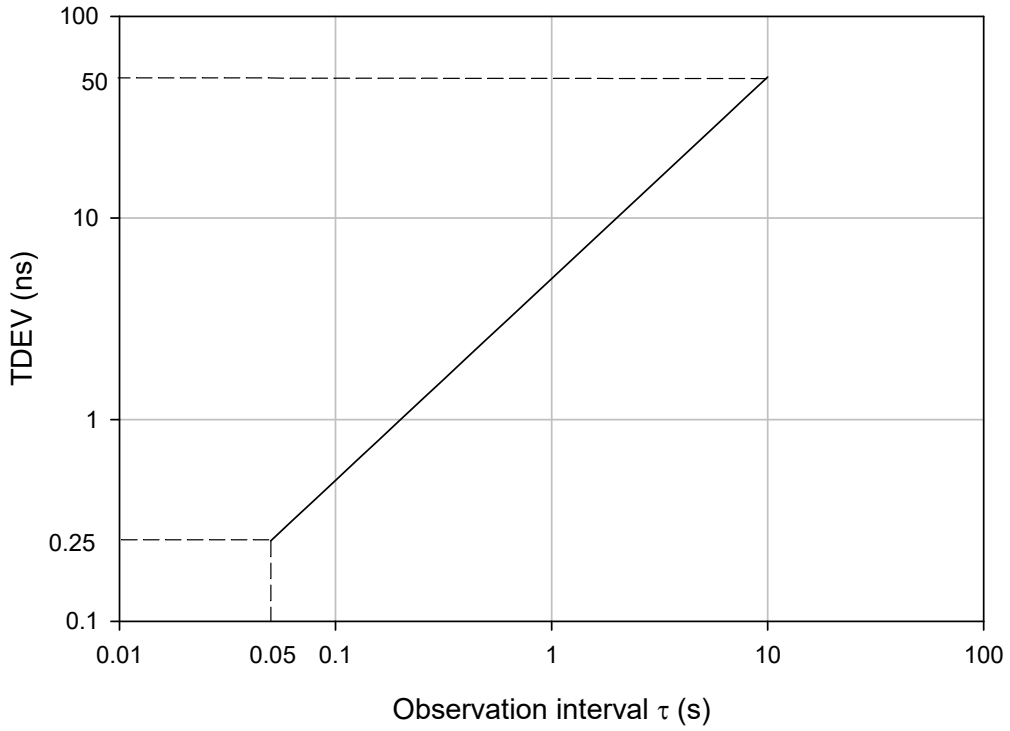


Figure B-1—Wander generation (TDEV) requirement for LocalClock entity

The ADEV limit that corresponds to the TDEV requirement of Table B-1 and Figure B-1 is shown in Table B-2 and Figure B-2, respectively.

Table B-2—ADEV limit corresponding to wander generation requirement of Table B-1

ADEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
1.054×10^{-8}	$0.05 \leq \tau \leq 10$ s
No requirement	$\tau > 10$ s

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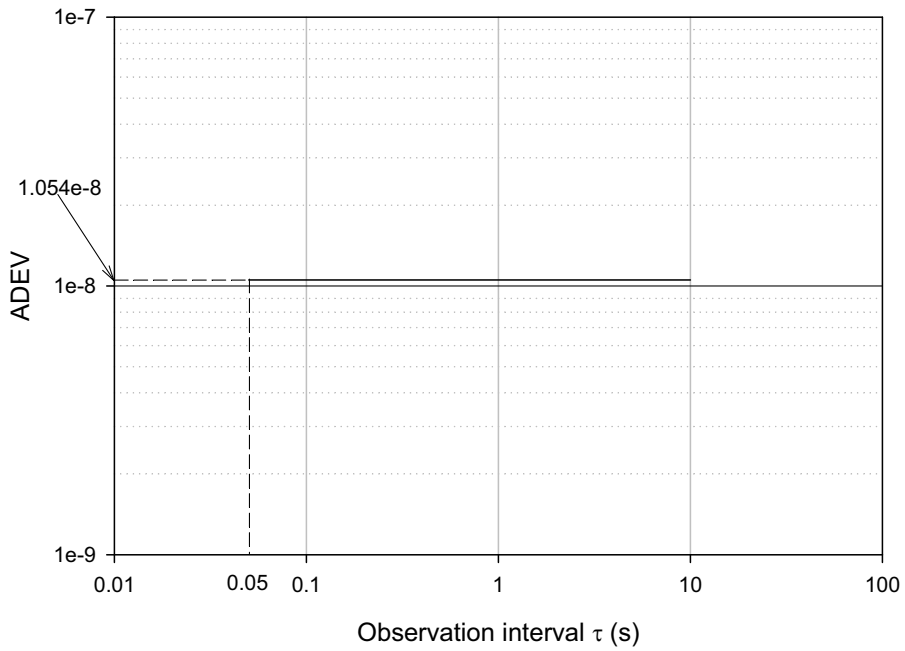


Figure B-2—ADEV limit corresponding to wander generation requirement of Figure B-1

The PTPDEV limit that corresponds to the TDEV requirement of Table B-1 and Figure B-1 is shown in Table B-3 and Figure B-3, respectively.

Table B-3—PTPDEV limit corresponding to wander generation requirement of Table B-1

PTPDEV limit	Observation interval τ
No requirement	$\tau < 0.05$ s
6.08τ ns	$0.05 \leq \tau \leq 10$ s
No requirement	$\tau > 10$ s

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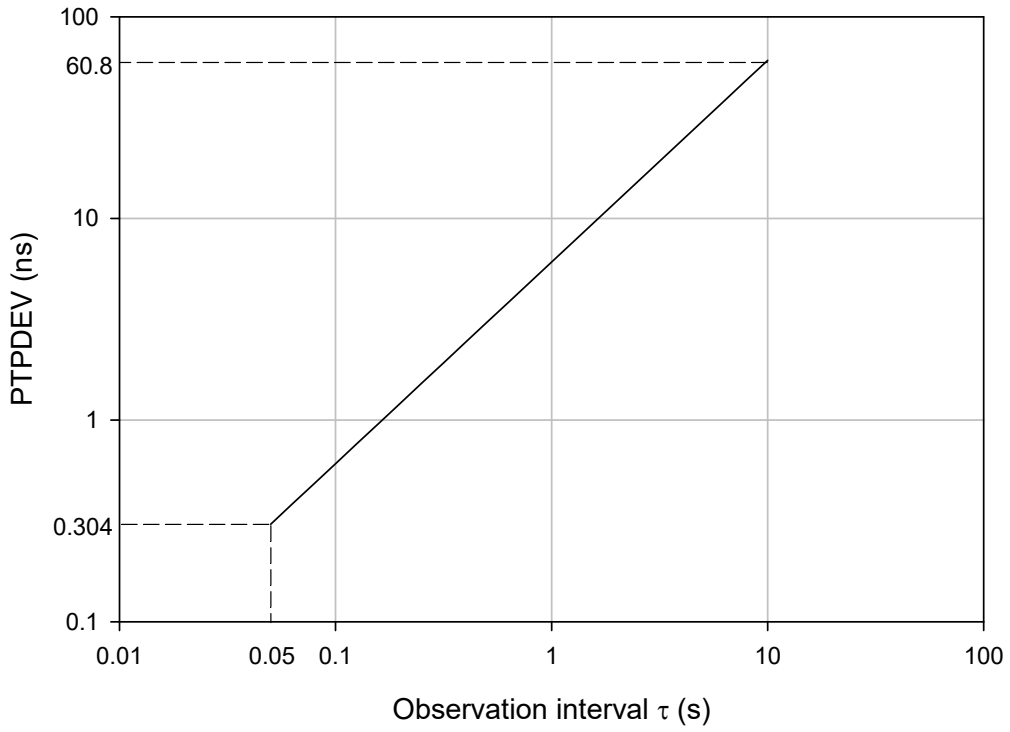


Figure B-3—PTPDEV limit corresponding to wander generation requirement of Figure B-1

B.2 PTP Instance requirements

B.2.1 General

In order to achieve the accuracy goals, certain constraints are placed on the responsiveness and accuracy of PTP Instances.

B.2.2 Residence time

The residence time (see 3.32) of a PTP Instance, measured relative to the TAI second (see 8.2), should be less than or equal to 10 ms.

Any error in the measured frequency offset relative to the Grandmaster Clock (i.e., error in accumulated rateRatio) results in an error in the transported synchronized time that is equal to the frequency offset error multiplied by the residence time (see 7.4.3 and 11.1.3).

B.2.3 Pdelay turnaround time

The pdelay turnaround time is the duration of the interval between the receipt of a Pdelay_Req message by a port of a time-aware system, and the sending of the corresponding Pdelay_Resp message.

The pdelay turnaround time of a time-aware system, measured relative to the TAI second (see 8.2), should be less than or equal to 10 ms.

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A nonzero pdelay turnaround time and any error in the measured frequency offset between the peer delay initiator and peer delay responder (i.e., error in nrrPdelay) results in an error in the measured mean propagation delay. This in turn results in an error in the transported synchronized time (see 11.1.2 for more details).

NOTE—While a larger value of pdelay turnaround time can result in worse time-synchronization performance, the peer delay protocol will still operate as long as the peer delay initiator receives Pdelay_Resp and Pdelay_Resp_Follow_Up within a time interval since sending Pdelay_Req that is less than the current Pdelay_Req message transmission interval (see 11.2.18.4 and 11.5.2.2).

B.2.4 Measurement of rate ratio

This standard requires the measurement of rate ratio or, equivalently, frequency offset, in several subclauses (see 10.2.10, 10.2.11, 11.2.18, 12.5.2, 16.4.2, and 16.4.3.2). The error inherent in any scheme used to measure rate ratio shall not exceed ±0.1 ppm.

NOTE—This requirement is consistent with a rate ratio measurement made by measuring the default Pdelay_Req message transmission interval (the nominal interval duration is 1 s; see 11.5.2.2) relative to the clocks whose rate ratio is desired, assuming the clocks meet the time measurement granularity requirement of B.1.2 (i.e., no worse than 40 ns).

B.3 End-to-end time-synchronization performance

Assuming that the requirements of this standard and of standards referenced for each medium are met, any two PTP Instances separated by six or fewer PTP Instances (i.e., seven or fewer hops) will be synchronized to within 1 microsecond peak-to-peak of each other during steady-state operation (i.e., each PTP Instance receives time-synchronization information every sync interval).

B.4 End-to-end jitter and wander performance

The requirements of this standard and standards referenced by this standard ensure that the synchronized time at a PTP Instance that is separated from the Grandmaster PTP Instance by six or fewer PTP Instances (i.e., seven or fewer hops) will, when filtered by a reference endpoint filter with rolloff of 20 dB/decade, gain peaking that does not exceed 0.1 dB, and bandwidth that does not exceed the value given in each entry of Table B-4, have maximum time interval error (MTIE) (see ITU-T G.810 [B30]) that does not exceed the MTIE for that entry of Table B-4, and jitter that does not exceed the peak-to-peak jitter of Table B-4 when measured through the corresponding high-pass jitter measurement filter given in Table B-4.

NOTE—For example, the endpoint filter can be of the following form:

$$y_k = a_1 y_{k-1} + a_2 y_{k-2} + \ldots + a_n y_{k-n} + b_0 x_k + b_1 x_{k-1} + \ldots + b_n x_{k-n}$$

where the x_k are the unfiltered synchronized time values, the y_k are the filtered synchronized time values, and the a_k and b_k are filter coefficients. The a_k and b_k are chosen such that the filter has desired bandwidth and gain peaking that does not exceed 0.1 dB. The preceding equation is a general infinite impulse response (IIR) digital filter. Simplified forms, e.g., a second order IIR filter obtained by setting $n = 2$, or a finite impulse response (FIR) filter obtained by setting the a_k to zero are possible.

Mask 1 of Table B-4 and Figure B-4 corresponds to consumer digital audio applications. Mask 2 of Table B-4 and Figure B-4 corresponds to professional digital audio applications. Mask 1 is derived from the requirements given in IEC 60958-3 [B9]. Mask 2 is derived from the requirements given in IEC 60958-4 [B10], AES3-2009 [B1], and AES11-2009 [B2]. Garner [B6] describes the methodology for deriving MTIE from the various jitter and synchronization requirements and presents the detailed derivations for masks 1 and 2.

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Table B-4—Maximum endpoint filter bandwidths needed to meet respective MTIE masks and peak-to-peak jitter limits

Endpoint filter maximum bandwidth (Hz)	Corresponding MTIE mask of Figure B-4 that is not exceeded	Corresponding jitter high-pass measurement filter (Hz)	Corresponding peak-to-peak jitter that is not exceeded (ns)
10	Mask 2 (Figure B-4, Table B-6)	8000	10.2
1	Mask 1 (Figure B-4, Table B-5)	200	11.1

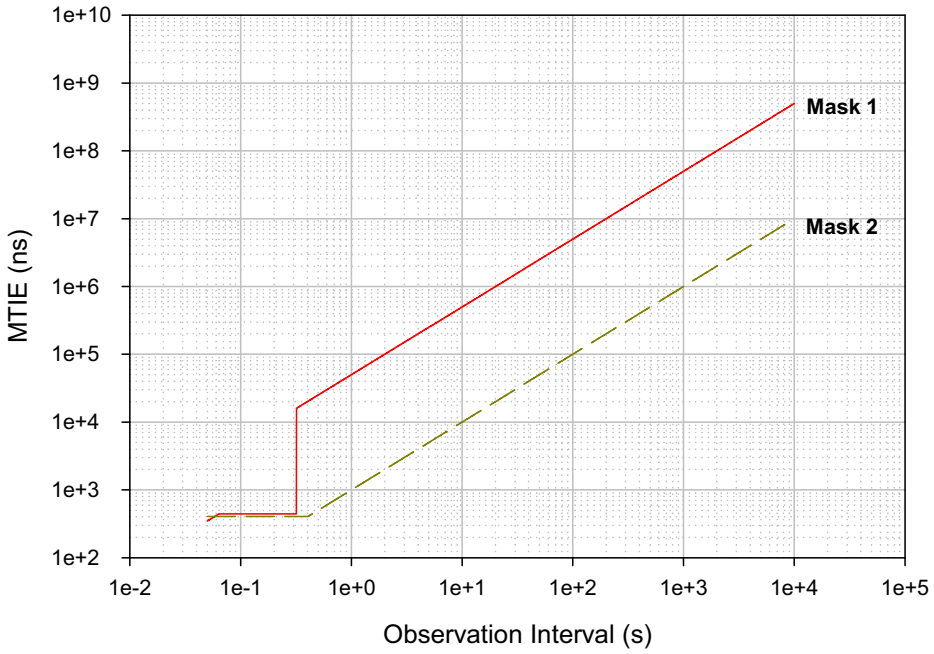


Figure B-4—MTIE masks met for maximum endpoint filter bandwidths of Table B-4

Table B-5—Breakpoints for Mask 1

Observation interval S (s)	MTIE (ns)
$0.05 \le S < 0.0637$	6954.8S
$0.0637 \le S < 0.3183$	443
$0.3183 \le S \le 10000$	50000S

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Table B-6—Breakpoints for Mask 2

Observation interval S (s)	MTIE (ns)
$0.05 \leq S < 0.4069$	407
$0.4069 \leq S < 10000$	1000S

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Annex C

(informative)

Timescales and epochs

C.1 Overview

A more detailed discussion of many of the topics in this annex can be found in Allan et al. [B3].

For historical reasons, time is specified in a variety of ways as listed in Table C-1. GPS, PTP, and TAI times are based on values yielded by atomic clocks and advance on each second. NTP and UTC times are similar, but are occasionally adjusted by one leap second, to account for differences between the atomic clocks and the rotation time of the earth.

Table C-1—Timescale parameters

	Timescale				
Parameter	**GPS**	**PTP**	**TAI**	**NTP**	**UTC**
Approximate epoch[a]	1980-01-06 1999-08-22 2019-04-07	1970-01-01	No epoch defined	1900-01-01	No epoch defined
Representation	weeks.seconds	seconds	YYYY-MM-DD hh:mm:ss	seconds	YYYY-MM-DD hh:mm:ss
Rollover (years)	19.7	≈8 900 000	10 000	≈136	10 000
NOTE 1—After 1972-01-01 00:00:00 TAI, TAI and UTC differ by only integer seconds. NOTE 2—The following represent the same instant in time: 1958-01-01 00:00:00 TAI and 1958-01-01 00:00:00 UTC.					

[a] Each approximate epoch occurs at 00:00:00 on the respective date.

GPS global positioning satellite
NTP network time protocol
PTP IEEE 1588 precision time protocol
TAI International Atomic Time (from the French term *Temps Atomique International*)
UTC Coordinated Universal Time [The abbreviation is a compromise between the English phrase *coordinated universal time* (CUT) and the French phrase *temps universel coordonné* (TUC).]

C.2 TAI and UTC

TAI and UTC are international standards for time based on the SI second (see Allan et al. [B3], IAU [B37], and Petit and Luzum [B36]). The SI second is the duration of 9 192 631 770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium 133 atom.[28] TAI is implemented by a suite of atomic clocks and forms the timekeeping basis for other timescales in common use. The rate at which UTC time advances is normally identical to the rate of

[28] See ITU-R TF.460-6 [B29], Jekeli [B32], Service de la Rotation Terrestre [B38], SI [B25], IAU [B37], and Petit and Luzum [B36] for more details on UTC, TAI, and the SI second.

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TAI. An exception is an occasion when UTC is modified by adding or subtracting exactly one whole leap second. The TAI frequency is the frequency of a signal whose period is the TAI second.

The TAI and UTC timescales were introduced as of 1958-01-01, and the times 1958-01-01 00:00:00 TAI and 1958-01-01 00:00:00 UTC represent the same instant in time. However, this instant in time is not an epoch for TAI and UTC. An epoch, in the sense of the origin of a timescale (see 8.2.2), is not defined for either TAI or UTC. TAI and UTC are expressed in the form YYYY-MM-DD hh:mm:ss, rather than as elapsed time since an epoch; here, YYYY-MM-DD denotes the date, and hh:mm:ss denotes the time in each day.

Prior to 1972-01-01, corrections to the offset between UTC and TAI were made by applying fractional-second corrections to UTC and corrections to the rate at which UTC advanced relative to the rate at which TAI advanced. After 1972-01-01, leap-second corrections are applied to UTC by inserting or deleting second(s) at the end of the last minute of preferably the last day of June or December. Also after 1972-01-01, UTC and TAI advance at the same rate. As of 2006-01-01, TAI and UTC times differed by +33 s (i.e., TAI time minus UTC time equals +33 s for 2006-01-01).

In computer networks, the common POSIX-based time conversion algorithms are typically used to produce the correct ISO 8601:2004 [B26] printed representations for both TAI and UTC.

The PTP epoch is set such that a direct application of the POSIX algorithm to a PTP timescale timestamp converts the PTP timestamp to the ISO 8601:2004 [B26] printed representation of TAI. PTP also distributes the current offset between TAI and UTC in the currentUtcOffset field of Announce messages. Except during leap seconds, subtracting currentUtcOffset from a PTP timestamp and then applying the POSIX algorithm results in the ISO 8601:2004 printed representation of UTC. Conversely, except during leap seconds, applying the inverse POSIX algorithm and adding currentUtcOffset converts from the ISO 8601:2004 printed form of UTC to the form required to generate a PTP timestamp.

For example, at 0h 2 January 1972 TAI, the value of PTP Instance Time was 63 158 400. At this time currentUtcOffset was 10. The POSIX algorithm applied to the value (63 158 400 – 10) gives a value of 1972-01-01 23:59:50 (10 s before 0h 2 January 1972 UTC). The value of PTP Instance Time on 0h 2 January 1972 TAI is computed by observing that PTP Instance Time = 0 on 0h 1 January 1970 TAI, i.e., Modified Julian Day (MJD) 40 587 (see Petit and Luzum [B36]). On 0h 2 January 1972 TAI, MJD = 41 318. Thus, PTP Instance Time on 0h 2 January 1972 TAI is 0 + 86 400 × (41 318 – 40 587). Note that if this calculation were done for a day in which a leap second occurred, a more complex algorithm would be required to ensure that, for the duration of the leap second, the ISO 8601:2004 [B26] print form seconds value was 60 for a positive leap second.

International standards specify that if a correction to UTC relative to TAI is required, the leap second occurs at the last second of the UTC day, preferably at the end of June 30 or December 31. For a negative leap second, the last minute of the designated day has only 59 seconds. Negative leap seconds have never occurred and are unlikely to occur in the future. For a positive leap second, the last minute of the designated day has 61 seconds.

Although a negative leap second is unlikely to occur, if such a correction becomes necessary, the ISO 8601:2004 [B26] printed representation would appear as follows for a hypothetical negative leap second on 30 June 1972 UTC:

1972-06-30 23:59:57, 1972-06-30 23:59:58, 1972-07-01 00:00:00

For the positive leap second that actually occurred on 30 June 1972 UTC, the ISO 8601:2004 [B26] printed representation appeared as follows:

1972-06-30 23:59:59, 1972-06-30 23:59:60, 1972-07-01 00:00:00

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Note the 23:59:60 notation to indicate the added second.

The update semantics for leap second updates in PTP are discussed in B.2.2 of IEEE Std 1588-2019.

C.3 NTP and GPS

Two standard time sources are of particular interest in implementing PTP Instances: NTP and GPS. Both NTP and GPS systems are expected to provide time references for calibration of the grandmaster-supplied PTP time.

NTP represents seconds as a 32-bit unsigned integer that rolls-over every 2^{32} s ≈ 136 year, with the first such rollover occurring in the year 2036. The precision of NTP systems is usually in the millisecond range.

NTP is a widely used protocol for synchronizing computer systems. NTP is based on sets of servers, to which NTP clients synchronize. These servers themselves are synchronized to time servers that are traceable to international standards.

NTP version 4 provides the current UTC time and warning flags indicating that a leap second will be inserted at the end of the current UTC day. The NTP clock effectively stops for one second when the leap second is inserted.

GPS time comes from a global positioning satellite system, GPS, maintained by the U.S. Department of Defense. The precision of the GPS system is usually in the 10 ns to 100 ns range. GPS system transmissions represent the time as {*weeks*, *secondsInWeek*}, the number of weeks since the GPS epoch and the number of seconds since the beginning of the current week.

GPS provides a leap seconds offset and warning flags marking the introduction of a leap second correction (see IS-GPS-200J [B28]). UTC and TAI times can be computed solely based the information contained in the GPS transmissions.

GPS timing receivers generally manage the epoch transitions (1024-week rollovers), providing the correct time (YYYY-MM-DD hh:mm:ss) in TAI and/or UTC timescales, and often also local time; in addition to providing the raw GPS week, second of week, and leap-second information.

C.4 Timescale conversions

Previously discussed representations of time can be readily converted to/from PTP *time* based on a constant offset and the distributed *utcOffset* value, as specified in Table C-2. Within Table C-2, all variables represent integers; “/” and “%” represent an integer divide and remainder operation, respectively.

NOTE—Some of the conversions in Table C-2 will not give the correct result when the current second is a leap second.

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Table C-2—Timescale conversions

ta		PTP value tb
Name	**Format**	
GPS	weeks:seconds	tb = ta.seconds + 315 964 819 + (gpsRollovers × 1024 + ta.weeks) × (7 × DAYSECS);
		ta.weeks = (tb – 315 964 819) / (7 × DAYSECS) – gpsRollovers × 1024; ta.seconds = (tb – 315 964 819) % (7 × DAYSECS);
TAI	date{YYYY,MM,DD}:time{hh,mm,ss}	tb = DateToDays("1970-01-01", ta.date) × DAYSECS + ((ta.time.hh × 24) + ta.time.mm) × 60) + ta.time.ss;

		secs = tb % DAYSECS;
		ta.date = DaysToDate("1970-01-01", tb / DAYSECS);
		ta.time.hh = secs / 3600;
		ta.time.mm = (secs % 3600) / 60;
		ta.time.ss = (secs % 60);

NTP	seconds	tb = (ta + utcOffset) – 2 208 988 800;
		ta = (tb – utcOffset) + 2 208 988 800;
UTC	date{YYYY,MM,DD}:time{hh,mm,ss}	tb = DateToDays("1970-01-01", ta.date) × DAYSECS + ((ta.time.hh × 24) + ta.time.mm) × 60) + ta.time.ss + utcOffset;

		tc = tb – utcOffset;
		secs = tc % DAYSECS;
		ta.date = DaysToDate("1970-01-01", tc/DAYSECS);
		ta.time.hh = secs / 3600;
		ta.time.mm = (secs % 3600)/60;
		ta.time.ss = (secs % 60);

gpsRollovers Currently equals 1; changed from 0 to 1 between 1999-08-15 and 1999-08-22
DAYSECS The number of seconds within a day: (60 × 60 × 24)
utcOffset The difference TAI – UTC, in seconds [i.e., currentUtcOffset (see 8.2.3)]
DateToDays For arguments DateToDays(*past*, *present*), returns days between *past* and *present* dates
DaysToDate For arguments DaysToDate(*past*, *days*), returns the current date, *days* after the *past* date

C.5 Time zones and GMT

The term *Greenwich Mean Time* (GMT) once referred to mean solar time at the Royal Observatory in Greenwich, England. GMT now commonly refers to the timescale UTC; or the UK winter time zone (Western European Time, WET). Such GMT references are, strictly speaking, incorrect but nevertheless quite common. The following representations correspond to the same instant of time:

18:07:00 (GMT), commonplace usage	13:07:00 (Eastern Standard Time, EST)
18:07:00 (UTC)	01:07 PM (Eastern Standard Time, EST)
18:07:00 (Western European Time, WET)	10:07:00 (Pacific Standard Time, PST)
06:07 PM (Western European Time, WET)	10:07 AM (Pacific Standard Time, PST)

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Annex D

(informative)

Reserved for future use

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Annex E

(informative)

Reserved for future used

(For information on media-dependent layer specification for CSN, see Clause 16.)

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Annex F

(informative)

PTP profile included in this standard

F.1 General

The specification in this standard of synchronized time transport over a FDE link includes a PTP profile. The information contained in a PTP profile is described in 20.3 of IEEE Std 1588-2019. This annex summarizes the PTP profile for transport of timing over FDE links. This PTP profile is also used in the transport of timing over HDE links (see Clause 19), and in the transport of timing over CSN when a CSN clock reference is not present (see Clause 16). This PTP profile is not used in the transport of timing over IEEE 802.11 links and IEEE 802.3 EPON links; both these transports use native timing mechanisms to assist in the synchronized time transport.

F.2 Identification

The identification values for this PTP profile (see 20.3.3 of IEEE Std 1588-2019) are as follows:

PTP Profile:
IEEE 802.1AS PTP profile for transport of timing
Profile Name: IEEE 802.1AS PTP profile
profileNumber: 0
primaryVersion: 2
revisionNumber: 0
profileIdentifier: 00-80-C2-00-02-00

NOTE—In the above profileIdentifier (see 20.3.3 of IEEE Std 1588-2019):

a) 00-80-C2 (first three octets) is the OUI owned by the IEEE 802.1 Working Group (it identifies the organization that specifies the PTP profile);
b) 00 (fourth octet) is a number that identifies the PTP profile, among all profiles specified by the organization that owns the OUI (or, in general, OUI or CID) of item a);
c) 02 (fifth octet) is the primary version of this PTP profile; and
d) 00 (sixth octet) is the revisionNumber of this PTP profile.

This profile is specified by the IEEE 802.1 Working Group of the IEEE 802 LAN/MAN Standards Committee.

A copy can be obtained by ordering IEEE Std 802.1AS-2025 from the IEEE Standards Organization.[29]

This PTP profile is a revision of the PTP profile included in IEEE Std 802.1AS-2011, i.e., major changes have been made to the profile relative to Version 1.0. Therefore, the primaryVersion is changed to 2, with revisionNumber 0.

[29] IEEE publications are available from The Institute of Electrical and Electronics Engineers (https://standards.ieee.org).

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F.3 PTP attribute values

The ranges and default values for time-aware system and PTP Instance attributes covered by this profile are as follows:

a) A domain whose domainNumber is 0 is present. A domain whose domainNumber is in the range 1 through 127 can be present (see 8.1).
b) The default logAnnounceInterval (see 10.7.2.2) is 0. The value 127 is supported.
c) The default logSyncInterval (see 11.5.2.3) is –3. The value 127 is supported.
d) The default logPdelayReqInterval (see 11.5.2.2) is 0. The value 127 is supported.
e) The default announceReceiptTimeout (see 10.7.3.2) is 3.
f) The default values of priority1, for different media, are specified in 8.6.2.1, Table 8-1. The value of priority1 for a PTP Instance that is not grandmaster-capable is 255.
g) The default value of priority2 is 248 (see 8.6.2.5).

The default observation interval for offsetScaledLogVariance is equal to the default sync interval, i.e., 0.125 s (see 8.6.2.4).

F.4 PTP options

a) The BTCA of this standard is the default BTCA according to the specifications of 9.3 of IEEE Std 1588-2019.
b) The following options of 17.7 of IEEE Std 1588-2019 are invoked:
    1) The FAULTY state is not used.
    2) The UNCALIBRATED state is not used.
    3) The LISTENING state is not used.
    4) The PRE_TIME_TRANSMITTER state, and PRE_TIME_TRANSMITTER qualification are not used.
    5) The foreign timeTransmitter feature is not used.
c) The management mechanism is the mechanism specified in Clause 14, Clause 15, and Clause 17.
d) The path delay mechanism is the peer-to-peer delay mechanism (see Clause 11).
e) The transport mechanism is FDE, and it uses attribute values described in Annex E of IEEE Std 1588-2019 for IEEE 802.3 Ethernet. Specifically, the address, EtherType, and subtype are specified in 11.3.4, 11.3.5, and 11.3.6.
f) A PTP Instance that contains one PortSync and one MD entity is an Ordinary Clock. A PTP Instance that contains more than one PortSync and more than one MD entity is a Boundary Clock.
g) Each port of a time-aware system measures the frequency offset of its neighbor, at the other end of the attached link, relative to itself (see Clause 11). The frequency offset, relative to the Grandmaster Clock, is accumulated in a standard organization TLV that is attached to the Follow_Up message if the PTP Port is two-step and the Sync message if the PTP Port is one-step (see 11.4.4.3). The standard organization TLV also carries information on Grandmaster Clock traceability and phase and frequency change due to the most recent Grandmaster PTP Instance change. The physical adjustment of the frequency of the LocalClock entity (i.e., physical syntonization) is allowed but not required.

NOTE 1—This feature is similar to the cumulative frequency transfer method specified in 16.10 of IEEE Std 1588-2019; however, it is not the same feature and uses a different TLV from the one used in the IEEE 1588 feature. This feature existed in the 2011 edition of the present standard and is retained for backward compatibility and also because the TLV of this standard carries additional information that is not carried in the IEEE 1588 feature [see item g) in this subclause].

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h) A standard organization TLV is defined to allow a PTP Port to signal to its neighbor PTP Port that it is capable of invoking gPTP (see 10.4 and 10.6.4.4).

i) The path trace feature of 16.2 of IEEE Std 1588-2019 is used (see 10.6.3.2.8).

j) A standard organization TLV is defined that allows a port of a time-aware system to request that its neighbor slow down or speed up the rate at which it sends Sync/Follow_Up, peer delay, and/or Announce messages (see 10.6.4.3).

k) The acceptable timeTransmitter table feature of IEEE Std 1588-2019 is used with IEEE 802.3 EPON links to ensure that the OLT is timeTransmitter and ONUs are timeReceivers.

NOTE 2—This feature is used with EPON links and therefore could be considered to be outside the PTP profile (because EPON links are not part of the PTP profile). It is included here because it is one of the optional features described in IEEE Std 1588-2019.

l) The profile isolation feature of IEEE Std 1588-2019 is not explicitly used; however, the PTP profile specified in this standard is isolated from other PTP profiles because it uses sdoId 0x100 (see 8.1).

m) A time-aware system can have more than one gPTP domain (see 8.1).

n) The Common Mean Link Delay Service specified in 16.6 of IEEE Std 1588-2019 is required if more than one domain is implemented and is optional if one domain (with domainNumber 0) is implemented (see 11.2.16).

o) The security mechanism of 16.14 of IEEE Std 1588-2019 and security annex (Annex P) of IEEE Std 1588-2019 are not used.

p) The external port configuration feature of 17.6 of IEEE Std 1588-2019 is optional in the present standard.

q) Except for items g) through p) in this subclause, the optional features of Clause 16 and Clause 17 and the optional annexes of IEEE Std 1588-2019 are not used.

F.5 LocalClock and PTP Instance performance requirements

The LocalClock performance requirements are as specified in B.1. The PTP Instance performance requirements are as specified in B.2.

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Annex G

(informative)

The Asymmetry Compensation Measurement Procedure based on line-swapping

G.1 Introduction

This annex describes the Asymmetry Compensation Measurement Procedure based on the line-swapping method. The entire procedure is controlled by the Network Management System (NMS), except that the line-swapping is manually operated. This annex is intended to address the delay asymmetry due to length differences between transmit and receive fibers for long fiber runs.

NOTE—When a port is put into asymmetry measurement mode, it is put into this mode on all domains. The per-port global variable asymmetryMeasurement mode is common to, and accessible by, all domains (see 10.2.5.2).

G.2 Pre-conditions for measurement

The following pre-conditions should be met to both improve the accuracy of the measurement and make the measurement procedure more convenient:

a) The measurement environment, including the testing nodes (i.e., the time-aware systems at the endpoint of the link whose asymmetry is being compensated) and related nodes (i.e., nodes in the paths between the testing nodes and the Grandmaster PTP Instance), should enable gPTP and the BTCA so that the test can be made for each link without changing the configuration.

b) The testing nodes should have redundant paths for synchronization so that they remain synchronized to the Grandmaster Clock during the asymmetry compensation measurement.

G.3 Measurement procedure

The assumed measurement environment is shown in Figure G-1. Before the measurement starts, every node has enabled syntonization [i.e., by measuring nrrPdelay (see 11.2.12.14) for each port of each time-aware system and accumulating the respective values over the synchronization tree paths to obtain rateRatio (see 10.2.8.1.4) relative to the Grandmaster Clock[30]] and time synchronization. The testing link is between Port2 on node ACC1 and Port1 on node ACC2.

The criteria of 11.2.16 for determining whether the peer-to-peer delay mechanism is the transport-specific peer-to-peer delay mechanism or CMLDS apply here.

[30] This standard neither requires nor prohibits syntonization (see 3.37) at the physical layer, e.g, using Synchronous Ethernet as the physical layer. If frequency is syntonized at the physical layer, the respective nrrPdelay and rateRatio values are expected to be close to 1.

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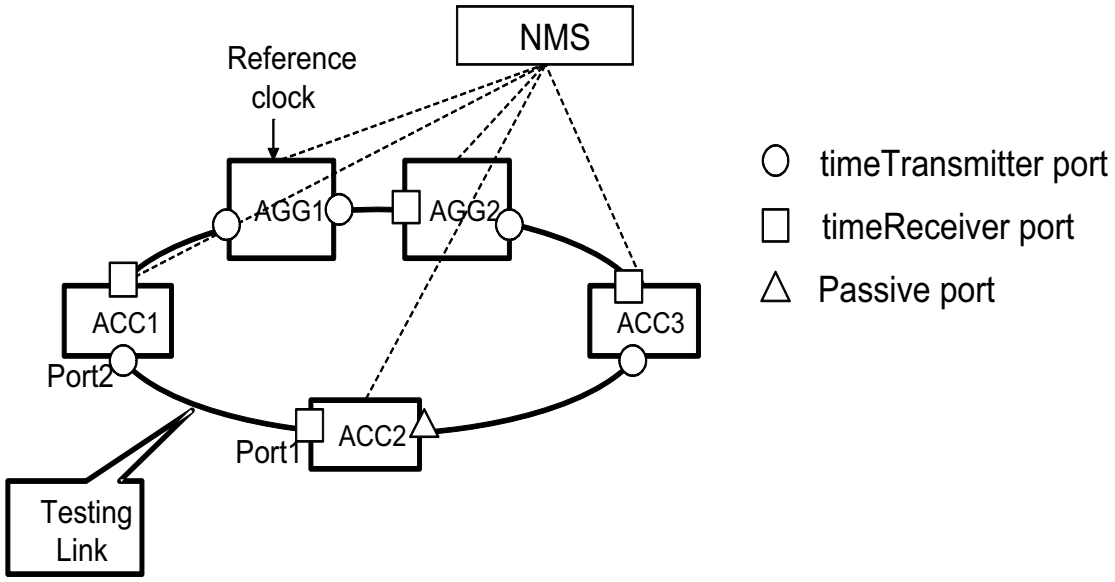


Figure G-1—Asymmetry Compensation Measurement Procedure

The measurement procedure is as follows:

a) The NMS puts Port1 of ACC2 and Port2 of ACC1 into asymmetry measurement mode through the MIB (i.e., by setting the managed object asymmetryMeasurementMode for each port to TRUE). These two ports will not affect the PTP calculations of either node when the ports are in asymmetry measurement mode. If synchronization flowed over the link connecting these ports prior to their being put into asymmetry measurement node, the BTCA will result in a reconfiguration of the synchronization tree so that both ACC1 and ACC2 remain synchronized.

b) Port1 of ACC2 will send Pdelay_Req messages periodically using the peer delay measurement mechanism and receive Pdelay_Resp and Pdelay_Resp_Follow_Up messages. For each set of messages, ACC2 should save t3 [the pdelayRespEventEgressTimestamp (see 11.3.2.1), carried in the Pdelay_Resp_Follow_Up message] and t4 (the pdelayRespEventIngressTimestamp, taken when the Pdelay_Resp message timestamp point crosses the reference plane at Port1 of ACC2 on reception).

c) The NMS reads and saves multiple sets of (t3, t4) from ACC2 through the MIB. It is necessary that each set of (t3, t4) be from the same measurement, i.e., from the same peer delay message exchange. The number of (t3, t4) sets can be decided as required and is outside the scope of gPTP.

d) The tester manually exchanges the transmit and receive fibers of Port2 of ACC1 and Port1 of ACC2. Then the tester waits until the port status and protocol status become stable again.

e) Port1 of ACC2 will again make periodic peer delay measurements and save each set of measurement values (t3', t4') (the primes are used to denote measurements that have occurred after the transmit and receive fibers have been exchanged).

f) The NMS reads and saves the multiple sets of (t3', t4') from ACC2 through the MIB. Then the NMS can compute the delay asymmetry, in units of time, as (t4' – t4) × nrrPdelay – (t3' – t3). The NMS can use multiple sets of (t3, t4, t3', t4') to compute average values to get a more accurate result. The averaging method can be decided as required and is outside the scope of gPTP. If ACC1 and ACC2 are frequency synchronized, then nrrPdelay is 1, and the delay asymmetry is (t4' – t3') – (t4 – t3).

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g) Based on the above result, the NMS sets the asymmetry value for Port1 of ACC2 and Port2 of ACC1 through the MIB.

h) After the NMS sets Port 1 of ACC2 and Port2 of ACC1 into normal mode (i.e., by setting the managed object asymmetryMeasurementMode for each port to FALSE), the delay asymmetry measurement of the testing link is completed. ACC1 and ACC2 will use the computed delay asymmetry as compensation in the PTP calculations.

It is also possible to compute the asymmetry ratio, i.e., the ratio of the delay on the receive fiber at ACC2 (Delay_rx_fiber, the delay of the Pdelay_Resp) to the delay on the transmit fiber at ACC2 (Delay_tx_fiber, the delay of the Pdelay_Req), both after line swapping. In this case, the NMS should collect multiple sets of (t1, t2, t3, t4) and (t1', t2', t3', t4') before and after line-swapping, respectively, where t1 is the pdelayReqEventEgressTimestamp (taken when the Pdelay_Req message timestamp point crosses the reference plane at Port1 of ACC2 on transmission), t2 is the pdelayReqEventIngressTimestamp (carried in the requestReceiptTimestamp field of the Pdelay_Resp message), and t3 and t4 are as given above. The NMS can compute the delay on the receive fiber as $[(t4' - t1) \times nrrPdelay - (t3' - t2)] /2$ and the delay on the transmit fiber as $[(t4 - t1') \times nrrPdelay - (t3 - t2')]/2$. Then the asymmetry ratio is as follows:

$$Delay_rx_fiber/Delay_tx_fiber = [(t4' - t1) \times nrrPdelay - (t3' - t2)] / [(t4 - t1') \times nrrPdelay - (t3 - t2')]$$

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Annex H

(informative)

Fault-tolerant time synchronization

H.1 Introduction

It is important for some time-sensitive applications (e.g., aerospace, automotive, and industrial automation) to consider fault tolerance and integrity for the time synchronization function, to enable reliable and trustworthy system behavior. For example, an aerospace network is typically expected to tolerate one or more arbitrary faults in Bridges, end stations, links, and GMs while maintaining continuous availability and integrity for the time synchronization function.

The typical timing requirement scenarios are summarized below:

a) **High availability timing:** This scenario requires availability of the time for a high percentage of a service period. A fault can introduce an interruption to this availability, but this interruption is short in duration relative to the corresponding service period.

b) **Fault-tolerant timing:** This scenario requires continuous availability of the time, even in the presence of a fault or, in some scenarios, multiple faults.

c) **Fault-tolerant timing with time integrity:** This scenario requires continuous availability of a trusted time, even in the presence of a fault or, in some scenarios, multiple faults.

Features of gPTP, as defined in this standard, can be used to support all three of the above timing requirement scenarios. The use of these features can enable system designers to meet their applications' timing requirements for assured systems.

For the high availability timing scenario, this standard supports multiple time distribution paths and potential reorganization of the synchronization trees, via the BTCA, to overcome faults. The BTCA is, however, not compatible with some applications that have stringent fault-tolerance requirements. First, a reorganization of the synchronization tree to overcome a fault takes time to complete. Depending on the size of the network and where the fault occurred, there might be a high degree of variability in the fault recovery time of the BTCA. Second, many fault-sensitive applications use engineered networks with static configuration and deterministic behavior and, thus, cannot use the BTCA.

For the fault-tolerant timing scenario, this standard supports multiple time domains, multiple GMs, multiple time distribution paths, multiple PTP Instances per port in Bridges and end stations, and the external port configuration mode (10.3.1.3), which allows static time distribution paths to be established. These features provide active redundancy, enabling the redundant function to take over when the original function fails.

For the fault-tolerant timing with time integrity scenario, this standard supports the features listed under the fault-tolerant timing scenario and adds support for the FTT entity functionality. The FTT entity uses multiple domains with multiple (potentially redundant) synchronization trees to provide configurable and deterministic fault recovery behavior, to determine the trustworthiness of the times that it is receiving, and to select a trusted time for use by its clock target.

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H.2 Fault-tolerance claims for the FTT entity

The fault-tolerance and integrity establishment claims that can be made for the FTT entity are given in this annex. The concepts upon which these claims are based are described in 20.1.2.2.

Table H-1—Fault-tolerance claims of the FTT entity

Fault condition in FTT inputs	Fault detection	Fault tolerance
Independent times with uncorrelated faults	Yes, for all cases	Yes, if there is at least one pair of non-faulty independent FTT inputs
Independent times with correlated[a] faults	Yes, if there are $(2n + 1)$ independent FTT inputs for n faults	Yes, if there are $(2n + 1)$ independent FTT inputs for n faults
Dependent times with correlated[b] faults	Yes, detected at ITSF	Yes, if there are $(2n + 1)$ independent inputs to the ITSF for n independent faults

[a] Independent times with correlated faults is a special case used for illustration purposes only. In practice, this condition would only happen by chance and would not be maintained long-term.
[b] This scenario assumes the failure occurs on the time-influencing component that is common to the FTT inputs that carry the dependent times.

For the first fault condition (i.e., independent times with uncorrelated faults) in Table H-1, any faulty FTT input, x, would not match within the fttMaxAs[x][y] skew limit between it and any other FTT input, y, regardless of whether FTT input y is faulty or non-faulty. Thus, any faulty FTT input can be identified. Even if there is just one pair of non-faulty FTT inputs amongst many faulty FTT inputs, this pair alone would be deemed to be trustworthy and one FTT input from the pair would be selected by the FTT entity for its output.

For the second fault condition (i.e., independent times with correlated faults) in Table H-1, the exceptional circumstance of a faulty FTT input having an error that matches the error of another faulty FTT input is considered. While this circumstance can happen by chance, this correlated failure would not be expected to be maintained given the FTT inputs do not share any common time influencing components. Under this circumstance, more non-faulty FTT inputs are needed to override the faulty FTT inputs that have a correlated error.

The third fault condition (i.e., dependent times with correlated faults) in Table H-1 is a typical failure scenario amongst dependent times because they share a common time-influencing component. These dependent times are processed by a DTSF. The DTSF produces an output with the correlated fault that is used by the ITSF. At the ITSF level, this correlated fault is detected when the time value at the DTSF output is compared to other time values at the ITSF inputs.

H.3 Balancing availability and integrity

The following compromises to fault-tolerant timing and time integrity are common in many implementations:

— Use of a common clock source for all GMs.
— Use of only two domains instead of three or more.

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— An insufficient number of independent paths from the GMs to the FTT entities.

For the first listed compromise, some applications only need the time-aware components of the local system to be synchronized to the arbitrary timescale of the local system and the integrity of this timescale does not need to be protected. In this scenario, a single clock source running off a local oscillator might be sufficient to satisfy the time agreement and preservation requirements of the system and be used as the source for all the GMs in the system. An example network topology with this compromise is discussed in J.2.2.

For the second listed compromise, some applications have a fail-stop requirement instead of a fail-operational requirement. A fail-operational requirement means the application has to continue operating correctly even if a fault (or up to *N* faults) has been detected. A fail-stop requirement means the application stops operation once a fault is detected. Only two independent times are needed to satisfy a fail-stop requirement, and this can be achieved using two domains. Example network topologies with this type of compromise are discussed in J.2.1, J.2.2, and J.2.3.

For the third listed compromise, it can be difficult and perhaps even impossible to create independent paths from all GMs to every FTT entity in a network. For these networks, the FTT entity can still support increased availability and integrity scenario for fail-stop applications if there are at least two independent paths from two GMs to the FTT entity (see the discussion about Figure J-5 in J.2.3). Otherwise, if there are no independent paths from any of the GMs, then the FTT entity is only able to support increased availability but not integrity (see 20.1.2.2). Example network topologies with this compromise are discussed in J.2.3 and J.2.4.

The availability and integrity scenarios supported by the FTT entity are listed below:

— Multiple time inputs, all of which are independent:
  — This scenario operates with multiple domains all processed by the ITSF.
  — In this mode, the FTT entity supports increased availability and integrity.
— Multiple time inputs, some of which are dependent with each other and some of which are independent with each other:
  — This scenario operates with multiple domains.
  — The dependent time inputs are processed together in a DTSF.
  — The independent time inputs and the outputs of every DTSF are processed by the ITSF.
  — In this mode, the FTT entity supports increased availability and integrity.
— Multiple time inputs, all of which are dependent with each other:
  — This scenario operates with a single domain or with multiple domains.
  — All time inputs share a common time-influencing component and, thus, can suffer from a common-mode failure.
  — In this mode, the FTT entity supports increased availability and potential time distribution integrity, but not GM integrity.
— Single time input:
  — This scenario operates with a single time input and a single domain.
  — In this mode, the FTT entity does not support increased availability or integrity.

The application is expected to be aware of the availability and integrity limitations based on the scenario in which the FTT entity is operating.

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H.4 Time error accumulation example

H.4.1 General

As gPTP time is distributed through a network from a GM to PTP Relay Instances and/or PTP Instances, time error accumulates. Some of the reasons for this time error accumulation are listed below:

— Timing errors at the GM (TEgm)
— Timing errors at intermediate PTP Relay Instances (TErly)
— Timing errors at the PTP End Instance (TEend)
— Link asymmetry between PTP Instances (TElnk)

The above time errors are illustrated in Figure H-1.

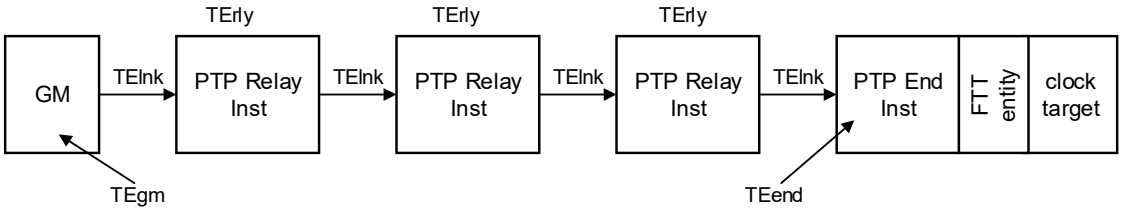


Figure H-1—Time error accumulation across a network

It is possible to determine the potential maximum absolute value of each of the above time errors and, thus, the maximum potential time error at the PTP End Instance. This result, maxAccumTE (see H.4.2), if available, can be used by the FTT entity in its selection process for a trusted gPTP time to present to the ClockTarget.

NOTE—The potential maximum absolute values of TEgm, TErly, TEend, and TElnk are expected to be calculated or measured prior to operational usage. This could be done at the design, characterization, or certification phase. The specific methods and procedures to do this are not defined in this standard.

The ENHANCED_ACCURACY_METRICS TLV from IEEE Std 1588a-2023 [B13] can be used to accumulate the maximum constant and dynamic time errors of each PTP Instance and the connecting links, on a hop-by-hop basis, in the path from the GM to, but not including, the final PTP End Instance. This TLV is carried in Announce messages.

Time skew between two time distribution paths is shown in Figure H-2. The time error contributors across each path are the same as shown in Figure H-1, but the contributors on each path are marked with the path's identifying suffix, x or y.

The various time skew results are described in H.4.2, H.4.3, H.4.4, and H.4.5.

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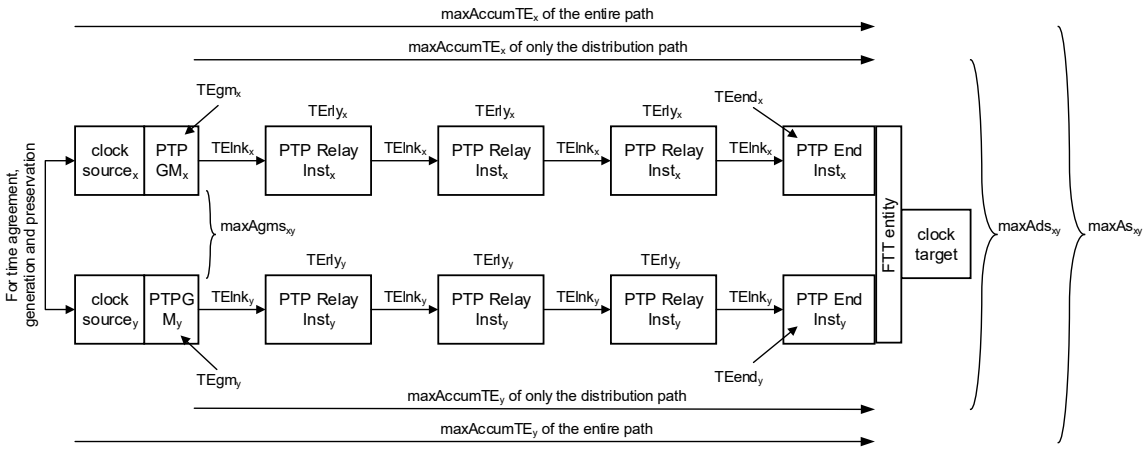


Figure H-2—Time skew across two time distribution paths

H.4.2 maxAccumTE$_x$

For the list of reasons for time error accumulation given in H.4.1, the parameter maxAccumTEx is the maximum non-faulty accumulated time error magnitude for the time distributed on path x, from its GM (TEgm), through all intermediate PTP Relay Instances (TErly) and the corresponding links (TElnk), to the PTP End Instance (TEend) that is connected to the FTT entity.

$$maxAccumTE_x = max(|TEgm_x|) + \sum max(|TErly_x|) + \sum max(|TE\ln k_x|) + max(|TEend_x|)$$

Where Σ is the summation symbol and represents a summation of all instances of the term adjacent to it.

H.4.3 maxAgms$_{xy}$

The parameter maxAgms$_{xy}$ is the maximum accepted time skew magnitude between two non-faulty PTP GMs, GM$_x$ and GM$_y$. This value is equal to the worst-case time error magnitude between the two GMs when they are not faulty.

$$maxAgms_{xy} = max(|TEgm_x|) + max(|TEgm_y|)$$

H.4.4 maxAds$_{xy}$

The parameter maxAds$_{xy}$ is the maximum accepted distribution skew magnitude between the time of two non-faulty times, distributed on path x and path y. This value is equal to the worst-case time error magnitude between the two times, from the perspective of the FTT entity, resulting from their distribution paths when they are not faulty.

$$maxAds_{xy} = \sum max(|TErly_x|) + \sum max(|TE\ln k_x|) + max(|TEend_x|) + \sum max(|TErly_y|) + \sum max(|TE\ln k_y|) + max(|TEend_y|)$$

H.4.5 maxAs$_{xy}$

The parameter maxAs$_{xy}$ is the maximum accepted skew magnitude between two non-faulty times, distributed on path x and path y. This value is equal to the worst-case time error magnitude between two

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synchronized times, from the perspective of the FTT entity, when they are not faulty. This value can be used as a criterion to determine the trustworthiness of the times being compared.

$$maxAs_{xy} = maxAgms_{xy} + maxAds_{xy}$$
$$= maxAccumTE_x + maxAccumTE_y$$

System integrators have to design their TSN network, which supports fault-tolerant timing and time integrity, such that synchronization-dependent nodes can withstand a drift equal to the magnitude of this maximum accepted skew.

H.5 Alternate ClockTarget interfaces

This annex discusses concepts for interworking alternate ClockTarget interfaces (e.g., ClockTargetTriggerGenerate and ClockTargetClockGenerator interfaces from 9.4 and 9.5, respectively) from a PTP Instance to a ClockTargetEventCapture interface (see 9.3) of the FTT entity and from the ClockTargetEventCapture interface of the FTT entity to a ClockTarget entity.

Figure H-3 shows an interworking function between PTP Instances, the FTT entity, and the ClockTarget entity, where the PTP Instances and the ClockTarget entity do not use the ClockTargetEventCapture interface. The interworking function has multiple proxy timers, each of which holds the PTP time that is recovered from a corresponding PTP Instance using the information provided by its alternate ClockTarget interface. These proxy timers are used as sources of time for the ClockTargetEventCapture interfaces to the FTT entity. The FTT entity's fttTrustState (see 14.23.20) and fttSelInstance (see 14.23.17) management objects are then used to select which proxy timer's time, or the NQ input time (see 20.2.4) if trust is not found, is sent to a functional block that generates the alternate ClockTarget interface to the ClockTarget entity.

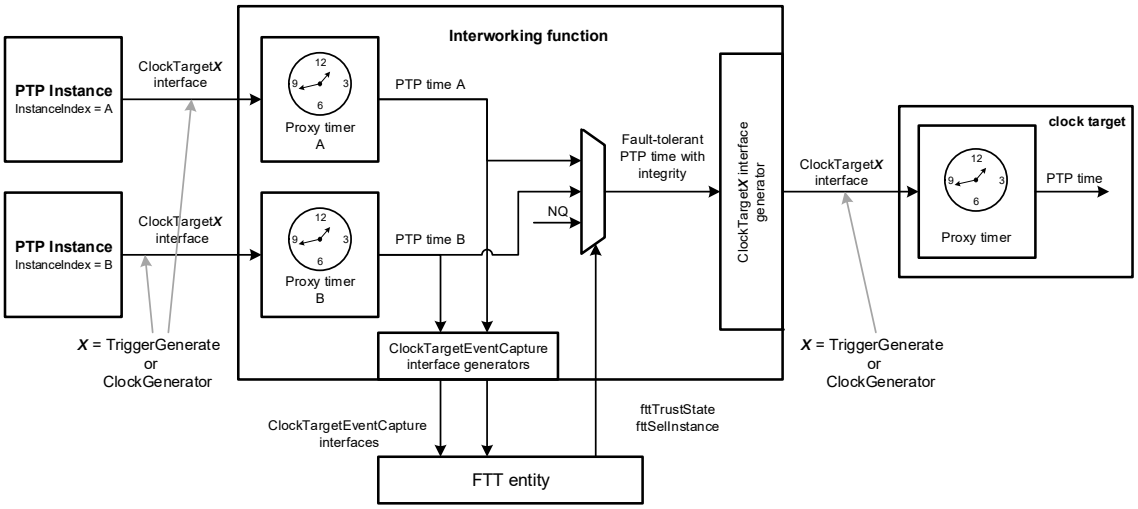


Figure H-3—Conceptual interworking function to alternate ClockTarget interfaces for the FTT entity

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Annex I

(informative)

Time agreement generation and preservation

I.1 Overview

Time agreement generation and preservation is the process by which multiple time source nodes (e.g., clock sources for GMs) come to an agreement on the time and maintain that agreement in the presence of both faults and oscillator drift. This process preserves both the collective accuracy and relative precision of the resulting set of GMs. Time agreement generation and preservation is done in a manner that is resilient to faults as described in I.2. This annex provides time agreement generation and preservation examples.

I.2 Fault models

These implementations need to be tolerant to an occurrence or some occurrences of the following fault types, which include Byzantine faults. All implementations have to reduce the possibility and/or probability of occurrence for all of these fault types.

a) Value domain faults
   - Manifest: symmetric and asymmetric
   - Omissive: symmetric and asymmetric
   - Transmissive: symmetric and asymmetric

b) Time domain faults

Value domain faults are faults where the value of the information received by the intended user is incorrect.

Time domain faults are faults where the information is received by the intended user at an incorrect/unexpected time (e.g., inadvertent activations, out-of-order messages). These faults can sometimes be transformed into value-domain faults. For example, the reception of a correct value at the wrong time can be treated as an incorrect value at the right time.

Manifest faults are faults where the intended user of the information can discern that the information is incorrect (e.g., checksum error). These faults can be transformed into omissive faults.

Omissive faults are faults where the information is not received by the intended user, but the intended user might not be aware of this omission.

Transmissive faults are faults where incorrect information is received by the intended user, but the intended user can only discern that this information is incorrect by comparison with redundant sources.

Symmetric faults are faults that are the same for all the intended users of the information.

Asymmetric faults are faults that are different for some of or all the intended users of the information.

Byzantine faults are asymmetric. It is commonly accepted (see Doley et al. [B4]) that at least $2N + 1$ clock sources and $3N + 1$ components (technically, fault-containment regions, see Hodson et al. [B8]) are required to achieve tolerance for N Byzantine faults. This would be sufficient to achieve tolerance for N faults of any

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of the types listed above. The examples shown in this annex have four components, each a clock source, and, thus, can tolerate one Byzantine fault.

It is not necessary for an FTT entity to have four input times to achieve time integrity for a Clock Target because, once time agreement is established at the GMs, only $2N + 1 = 3$ components are required to tolerate $N = 1$ Byzantine faults during time distribution. This is because no convergence operations are performed (i.e., no feedback) for time distribution and, thus, it does not have Byzantine manifestations.

I.3 Assumptions

Assumptions related to implementations of a time agreement generation and preservation function.

a) Fault-free oscillators have a specified bounded rate of drift with respect to time.
b) For initial synchronization, the cluster of clocks being subject to agreement generation and preservation satisfy the initial precision (i.e., the time source nodes all have the same unit of time, within a defined relative difference).
c) For preservation of synchronization (in the presence of faults):
    1) Each synchronization period begins with the time source nodes satisfying initial precision.
    2) During the synchronization period, the time source nodes can drift within the bounds of the known oscillator drift rate.
    3) During the synchronization period, the time source nodes exchange their time with other time source nodes (a small error can be introduced with this communication).
    4) Convergence function: The time source nodes compute a local adjustment using some fault-tolerant average of the readings from the other time source nodes.
    5) This convergence function needs to satisfy the following, within some specified number of faults:
        — Precision enhancement: If the time source nodes satisfy initial precision at the beginning of the period, the adjusted values after computing and adjusting using the fault-tolerant average will again satisfy initial precision at the beginning of the next period (this implies some convergence property due to the computation of the fault-tolerant average).
        — Accuracy preservation: The computed averages, when applied, constrain the cluster of synchronized clocks to a bounded rate of drift with respect to an ideal time reference.

I.4 PTP-based time agreement generation and preservation

A PTP-based time agreement generation and preservation example is shown in Figure I-1.

In this example, PTP sync messages are broadcast from a PTP port, which is configured to be in the PASSIVE state (see Lamport and Melliar-Smith [B33]), from each time agreement generation and preservation (TAGP) entity through a high integrity PTP message distributor with PTP Transparent Clock (see Lamport and Melliar-Smith [B33]) capabilities to a PTP Port, which is also configured to be in the PASSIVE state, on all the other TAGP entities. These PTP sync messages and their corresponding departure times and arrival times are used by each TAGP entity to compare the frequency and time of its clock source against those of the other clock sources and to syntonize the frequency and synchronize the time of its clock source.

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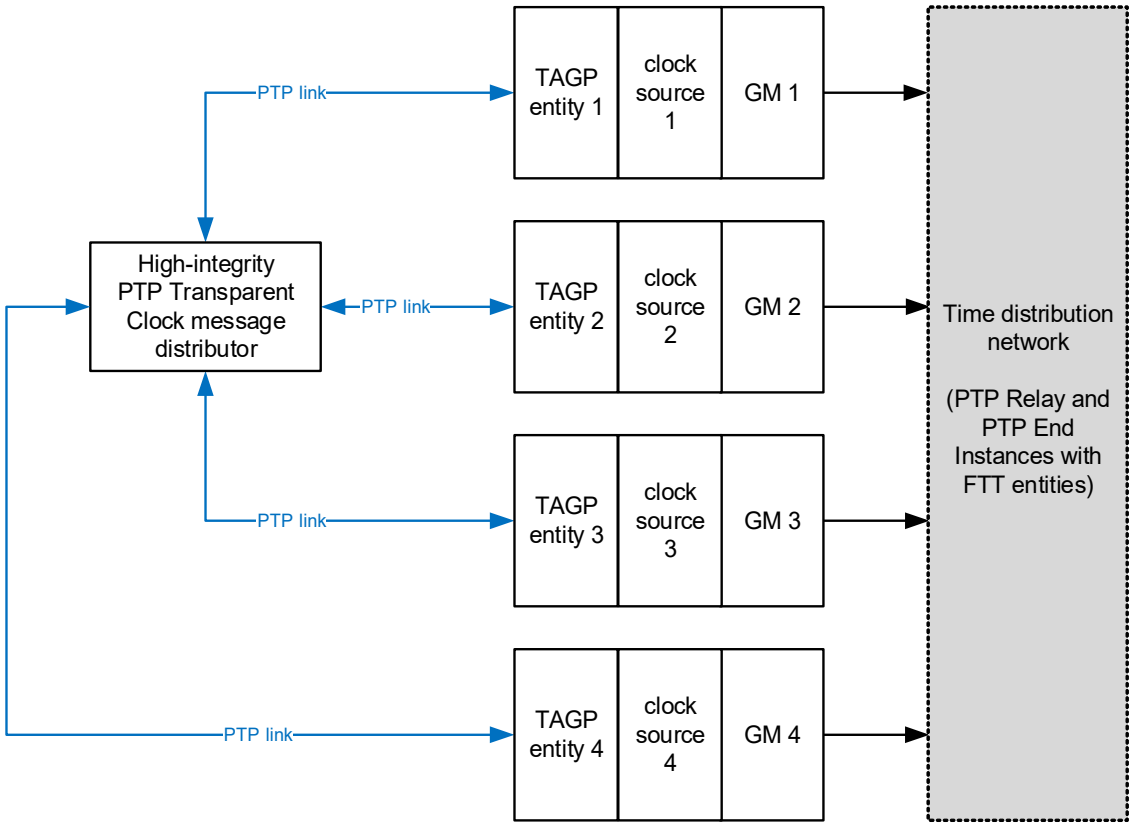


Figure I-1—PTP-based time agreement generation and preservation example with high-integrity message distributor

The integrity of the PTP message distributor is critical in this example because it is a common potential source of failure in the conveyance of the PTP information between the TAGP entities. This PTP message distributor needs to reduce its probability of creating false information while distributing the PTP sync messages. This could be done by adding redundancy and error checking to the PTP message distributor's timestamping, correctionField updating, and broadcasting functions.

Another PTP-based time agreement generation and preservation example is shown in Figure I-2.

In the example of Figure I-2, point-to-point links are used to convey PTP sync messages from one TAGP entity to another TAGP entity. In comparison with the example of Figure I-1, there is no high-integrity message distributor but more PTP ports (also in PASSIVE state) are needed at each TAGP entity. Like in the example of Figure I-1, the PTP sync messages and their corresponding departure times and arrival times are used by each TAGP entity to compare the frequency and time of its clock source against those of the other clock sources and to syntonize the frequency and synchronize the time of its clock source.

For initial time agreement generation, it might be acceptable to assume that no fault exists in the time agreement functionality. Thus, each TAGP entity could use the PTP sync message timestamps to compare the frequency offsets of all the clock sources (including its own), and then tune its clock source's frequency to match the average of the frequencies of all clock sources. After syntonization of all clock sources is achieved, each TAGP entity can perform a similar compare, average, and alignment process to synchronize their times.

Once initial time agreement generation is achieved, the TAGP entities switch to a time agreement preservation mode. In this mode, each TAGP entity could continuously use the PTP sync message

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timestamps to compare the frequency and time offsets of all the clock sources (including its own), determine which match within an expected range, exclude those (including its own clock source) that do not meet this expectation, and then tune its clock source's frequency and/or time to match the average of the of all the non-excluded clock sources.

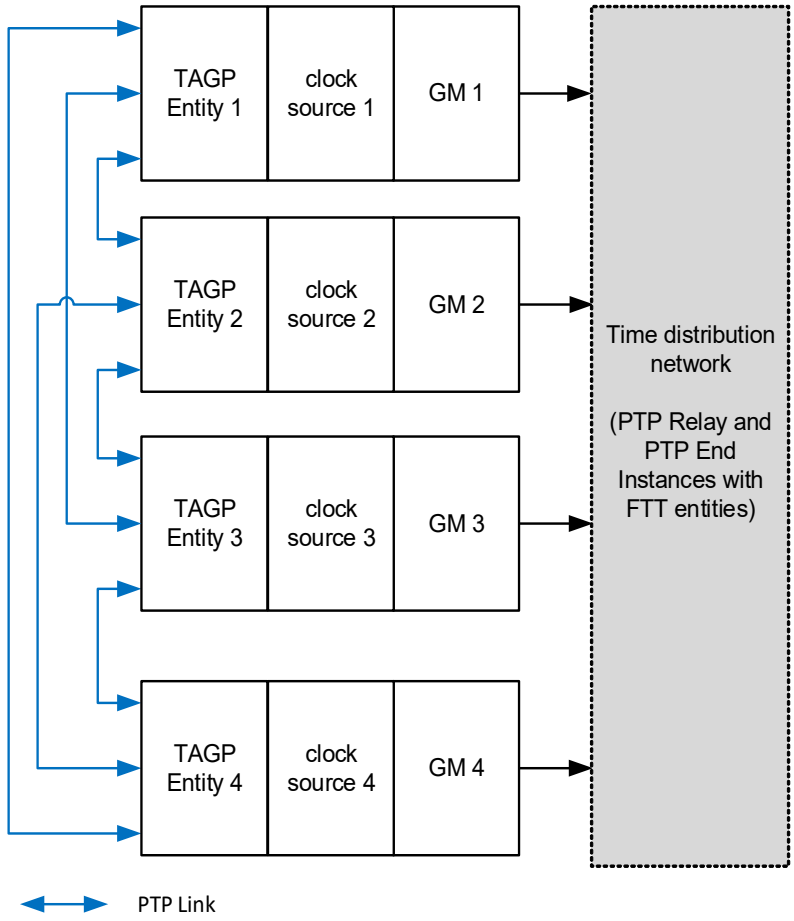


Figure I-2—PTP-based time agreement generation and preservation example without high-integrity message distributor

I.5 Other time agreement generation and preservation methods

Time agreement generation and preservation could be implemented without use of PTP.

Pulse per second (PPS) interfaces could be used instead of PTP interfaces to convey clock source frequency and time between the TAGP entities. A typical PPS interface includes the PPS event signal, which conveys the occurrence of a PPS event, and a UART interface, which conveys the time that corresponds to the PPS event.

Another method is to use GNSS as clock sources at each of the redundant GMs since the time provided by a GNSS has already achieved time agreement and preservation. Thus, its use as the clock source for a GM comes with this established benefit. However, the potential loss of GNSS signaling due to environmental conditions or jamming has to be considered for this use case.

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Annex J

(informative)

FTT in systems (examples)

J.1 Clock domain management

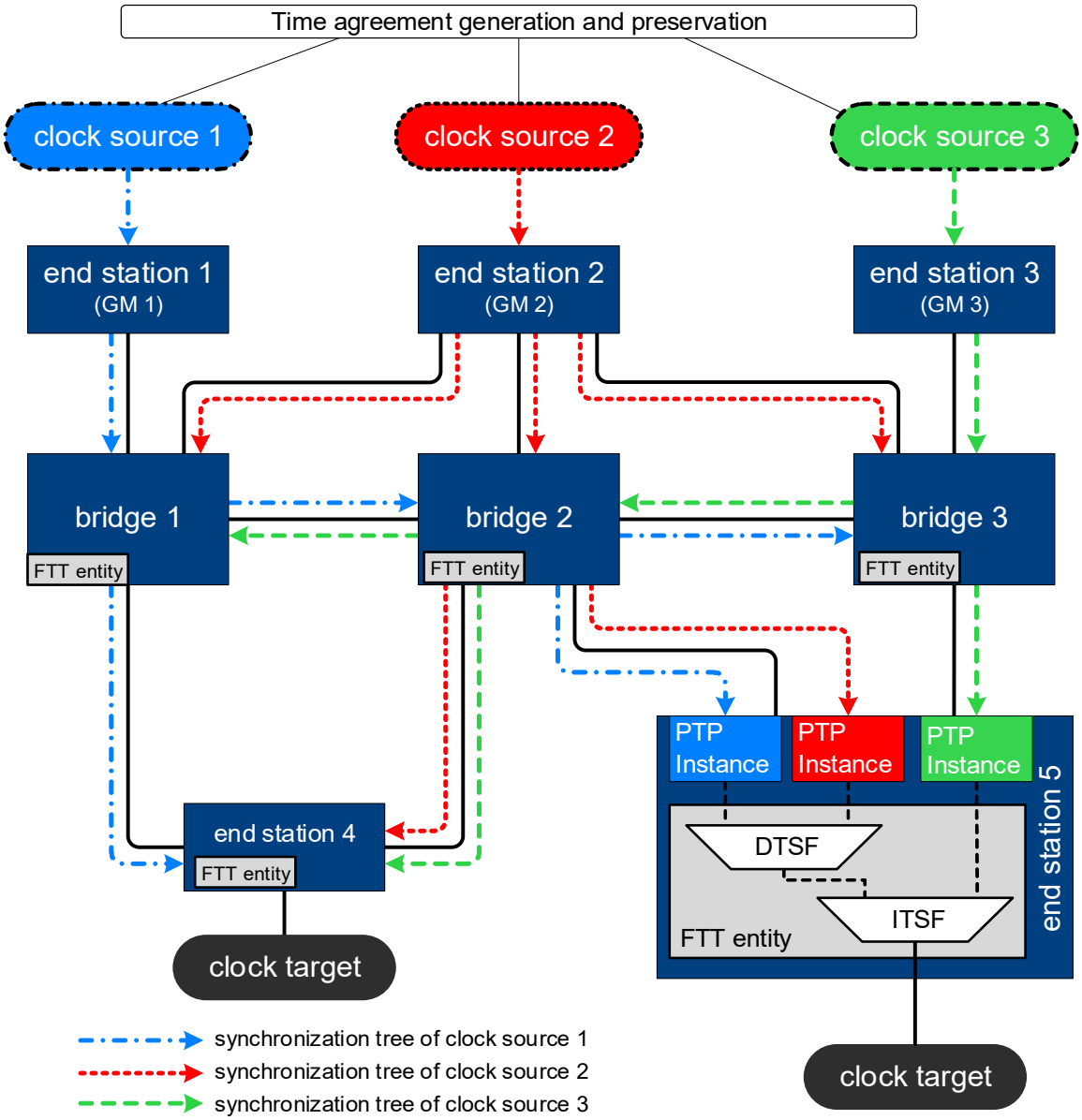
The FTT entity defined in this standard uses multiple domains with multiple (potentially redundant) synchronization trees to provide configurable and deterministic fault recovery behavior and to support time integrity.

To illustrate how the FTT entity works in an example application, Figure J-1 shows a simple network using three clock sources distributing time through multiple time synchronization trees to Bridges and end stations that implement FTT entities to select a time source from the multiple domains. The existence of multiple overlapping timing domains presents a problem for the forwarding of scheduled traffic because each port can only forward traffic using one clock reference. The three clock sources in the example are therefore assumed to share a common time through implementation of a time agreement and preservation function, (see I.1), to allow the forwarding of time-aware traffic across domains. The time agreement and preservation function is implementation-specific and is not specified in this standard. The clock sources in the example figure synchronize three GMs in end stations 1, 2, and 3, that distribute time through the network on separate, but overlapping, synchronization trees. Bridges 1, 2, and 3 distribute time from the three GMs through the network to all Bridges and end stations where FTT entities select the best available clock to generate a local clock target that is used to support the enhancements for scheduled traffic (8.6.8.4 of IEEE Std 802.1Q-2022).

End station 5 shows three PTP Instances receiving time through the network and providing clock target interfaces to an FTT entity consisting of one DTSF instance and one ITSF instance. Clock sources 1 and 2 arrive at end station 5 through a common Bridge device, bridge 2, and are therefore considered to be dependent and require a DTSF instance to select the best clock source to provide this as an independent input to the ITSF instance along with the output of the PTP Instance that uses clock source 3. The time from clock source 3 arrives at end station 5 through a path that is independent from the path taken by clock sources 1 and 2, and can therefore be considered to be independent from these. The output of the ITSF instance is then used to generate the local clock target for end station 5.

As shown in the figure, each of the Bridges in the network and end station 4 also include FTT entities to generate local clock targets. FTT entities can be configured differently in each device in the network depending upon the relationships of the available clock sources to one another, and select a clock source to generate the local clock target that is used to support generation and forwarding of time-aware traffic. To simplify the figure, end stations 1, 2, and 3 are not shown as recipients of the synchronization trees from the other two clock sources and do not show FTT entities; however, in a real network application they might be expected to receive the other time signals and to contain FTT entities to support fault-tolerant traffic generation. Once the three clock sources have established time agreement and each of the FTT entities have selected the best clock at each of network nodes, then all of the clock targets will be synchronized and time-aware traffic from any end point in the network can be forwarded synchronously to any other end point through the available Bridges.

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NOTE 1 — The methods to synchronize the ClockSources of all GMs are not specified in this standard.
NOTE 2 — All the “bridges” in this figure are examples of time-aware systems that contain Relay Instances and an FTT entity. The methods used to terminate time from PTP Relay Instances is not specified in this standard.
NOTE 3 — All the end stations in this figure are examples of time-aware systems that contain PTP End Instances and an FTT entity.

Figure J-1—Multiple domains with FTT entities

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Under normal operating conditions, each of the nodes in the example network will receive time from each of the three domains and generate a local clock target by selecting the best clock source according to the configured FTT entity attributes. In the case of a fault in the network, an input becomes UNTRUSTED at a Bridge or at an end station and the corresponding FTT entity selects the best available time from its remaining inputs. For example, if the link between bridges 2 and 3 were to fail, the synchronization tree from clock source 3 would not propagate to bridges 1 and 2 or to end station 4, and the synchronization tree from clock source 1 would not propagate to bridge 3. In this case, bridge 3 would still receive times from clock sources 2 and 3, and end station 4 would still receive times from clock sources 1 and 2. The FTT entities at bridge 3 and end station 4 could still select between these inputs.

This simple example does not address time integrity. See H.1 for a discussion of integrity and availability.

J.2 FTT entity operation in example network topologies

The use of FTT entities in examples of commonly used network topologies is shown in this annex, with details on the potential enhancements to time availability and time integrity.

J.2.1 *N* × point-to-point networks

An example *N* × point-to-point network topology is shown in Figure J-2. This network topology provides *N* independent inputs to the FTT entity via *N* PTP instances.

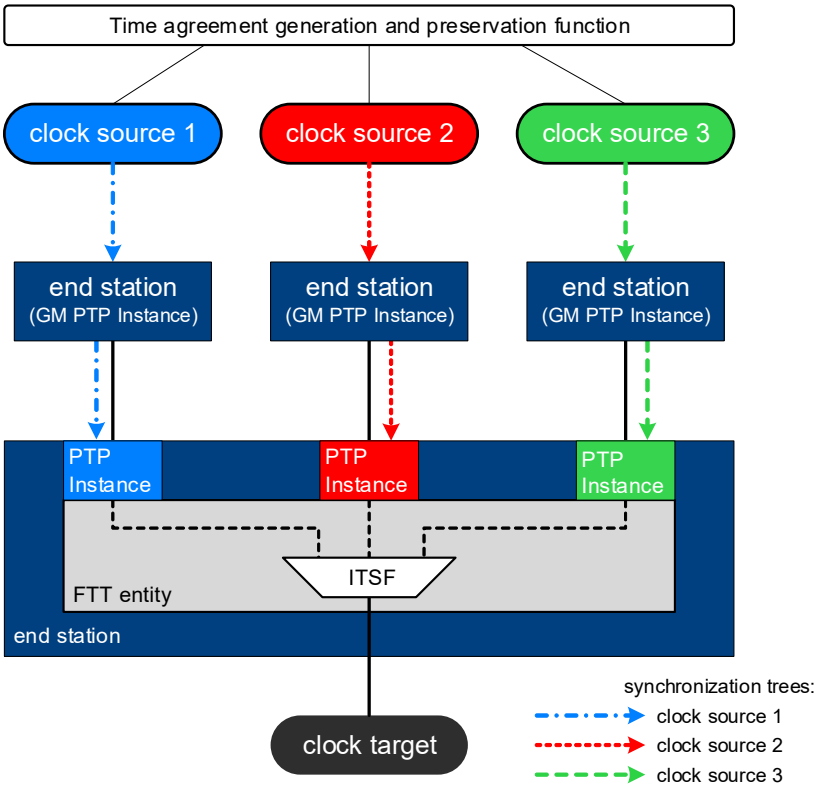


Figure J-2—FTT entity in example 3 × point-to-point network

In a scenario with *N* = 2, time availability with time integrity is achieved when there are no faults. Thus, a 2 × point-to-point network topology would be suitable for fail-stop applications, which terminate operation when any failure is detected in the time integrity.

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In the scenario of Figure J-2, with $N = 3$, time availability with time integrity is achieved when there is up to one fault. Thus, a 3 × point-to-point network topology would be suitable for fail-operational applications that can continue running when there is up to one fault.

J.2.2 Dual-homed network

An example dual-homed network is shown in Figure J-3. It provides two dependent inputs from one GM, through two PTP Instances, to the FTT entity. Because the two time inputs are dependent, they pass through a DTSF instance in the FTT entity. Because there are no independent time inputs to the FTT entity, the DTSF instance's output is the only connection to the ITSF instance.

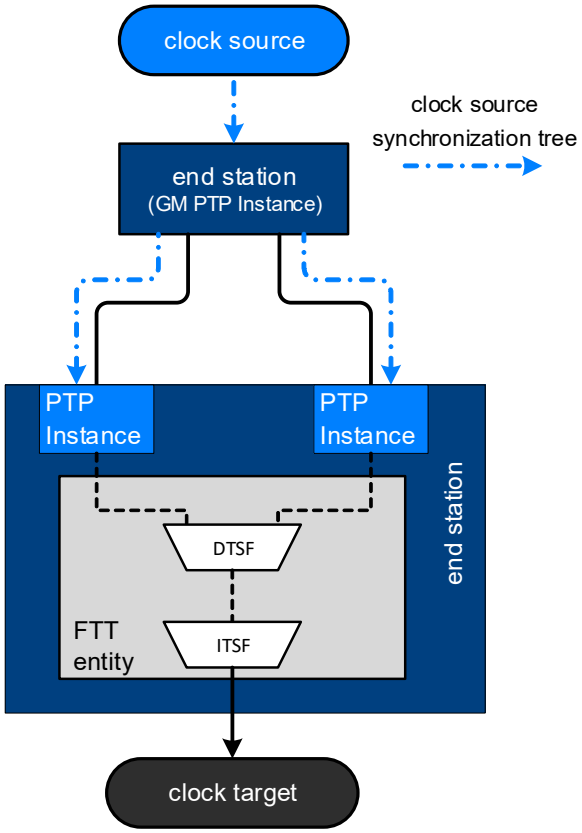


Figure J-3—FTT entity in example dual-homed network

Because there is just one GM in this network topology, the FTT entity supports increased time availability and time distribution integrity, but not GM integrity. Also, because there are just two time distribution paths, the limited integrity is lost if one of these two time distribution paths becomes faulty. Thus, the dual-homed network topology would be suitable for fail-stop applications in which the integrity of time distribution is important, but the integrity of the GM itself is not important.

J.2.3 Dual-star network

In dual redundant star network topologies, every end station is connected independently, in a star fashion, to a Bridge and to a redundant Bridge. A failure of any connection or of any Bridge is overcome by using the second connection or the second Bridge.

The example dual-star network shown in Figure J-4 provides two independent inputs, one from each GM through their associated PTP Instances, to each FTT entity. With just two independent inputs to each FTT

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entity, this dual-star network topology can only detect a fault, but cannot overcome the fault, which makes it suitable for fail-stop applications (as opposed to fail-operational applications).

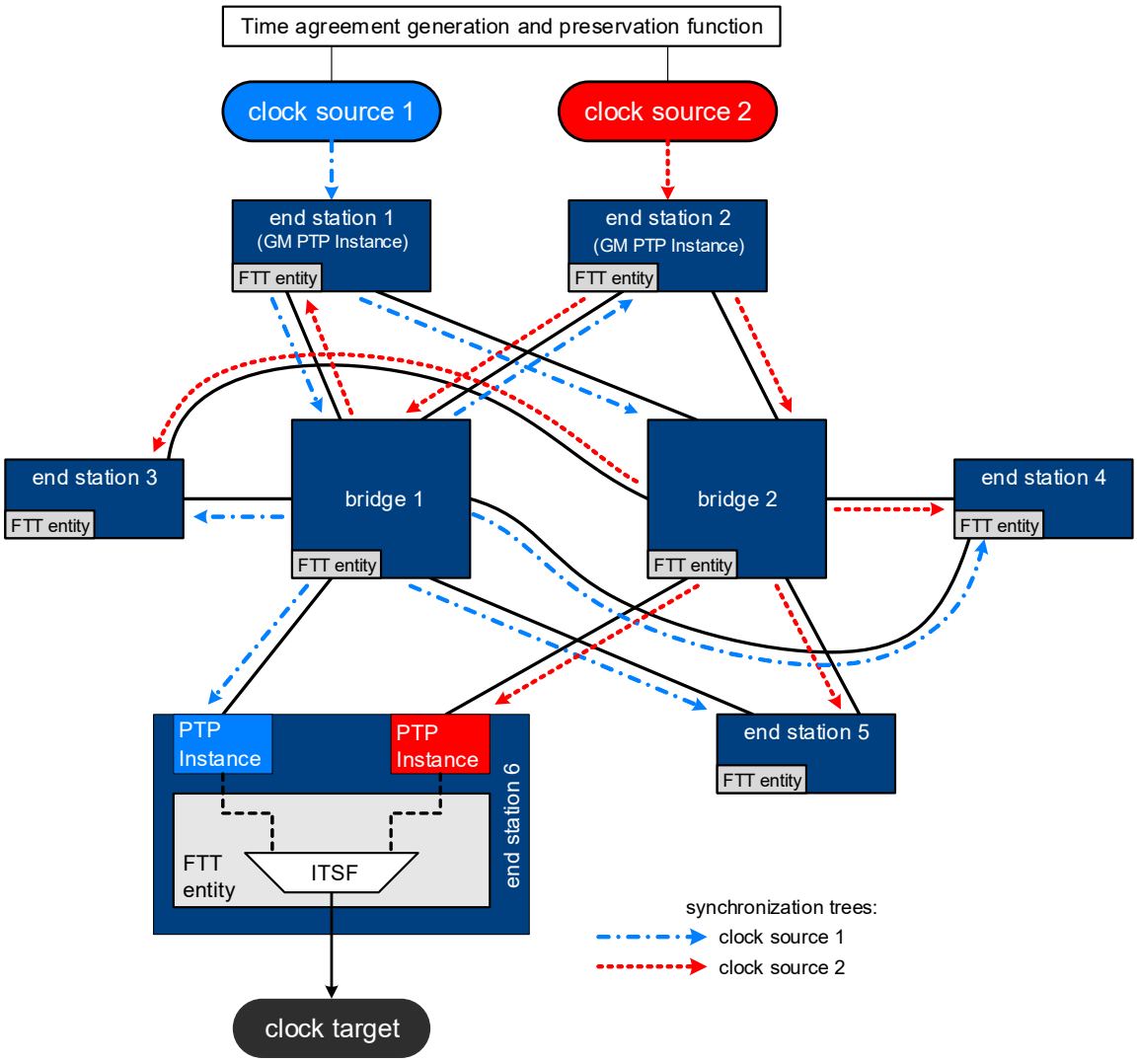


Figure J-4—FTT entity in example dual-star network with fail-stop time integrity

In contrast, the example dual-star network shown in Figure J-5 provides three independent inputs from each of the three GMs to the FTT entity of the two Bridges, two independent inputs from each of two GMs to the FTT entity of any end station, and two dependent inputs from a third GM to the FTT entity of end stations 3, 5, and 6. Even though the two dependent inputs originating from clock source 3's GM to the FTT entity of any end station have more than one dependency (i.e., the common clock source 3 and either bridge 1 or bridge 2), the FTT instance's DTSF is configured to only use the dependency on clock source 3 to determine the trustworthiness of these inputs for the following reasons:

a) Using clock source 3 as the dependent element results in the use of one DTSF instance, with two inputs, and generates three input times to the ITSF instance.
   - If the DTSF instance determines that the time values from the two inputs from clock source 3 match and, thus, can be trusted, then there is no need to pass clock source 3 through a DTSF instance with either clock source 1 or with clock source 2 as this is done in the ITSF.
   - If the DTSF instance determines that the two arriving times from clock source 3 do not match and, thus, cannot be trusted, then clock source 3 does not provide an applicable input to the ITSF

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instance and is removed from contention as a trustworthy source of time to the ClockTarget entity. With this result, there is again no need to pass clock source 3 through a DTSF instance with either clock source 1 or with clock source 2.

— The result is three inputs to the ITSF, which enables trust determination when there is up to one fault.

b) Using bridge 1 and bridge 2 as the dependent element results in the use of two DTSF instances, each with two inputs. This results in just two inputs to the ITSF instance. Not finding trust in either DTSF instance results in an inability to determine trust at the ITSF instance. This configuration can detect a fault but cannot find trust when there is one fault in the system.

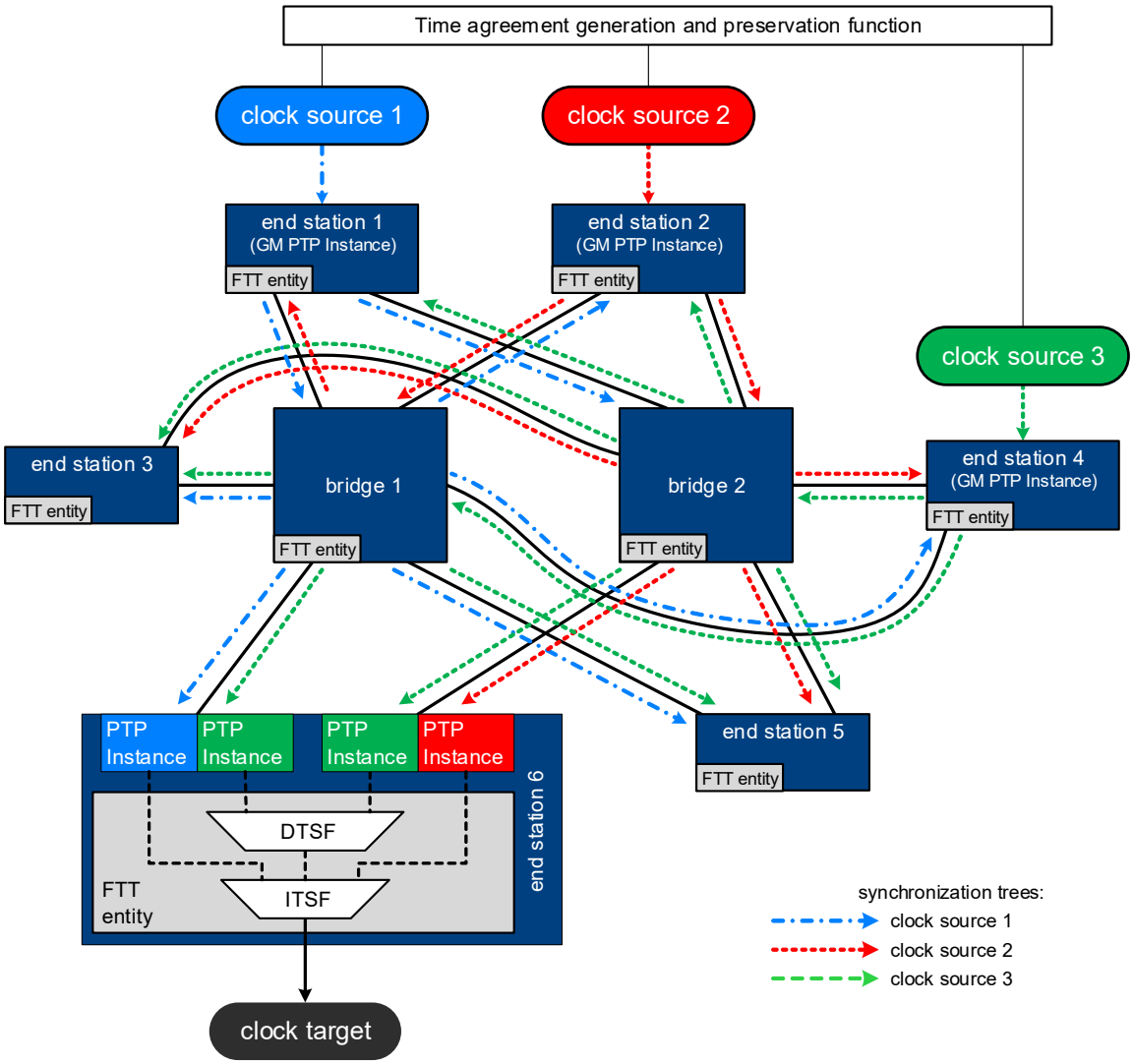


Figure J-5—FTT entity in example dual-star network with fail-operational time integrity

This dual-star network topology is suitable for fail-operational applications that can continue running when there is up to one fault in the timing function.

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J.2.4 Ring network

A basic ring network topology provides a low fan-out network with availability recovery for a single fault. However, the following fundamental characteristics of the ring network topology impede the FTT entity's goals of maintaining time availability and integrity during fault conditions:

— The low fan-out characteristic of a ring can prevent an FTT entity from receiving all independent inputs and, thus, impedes its ability to achieve enhanced availability and integrity (see 20.1.3).
— The forwarding path of traffic, which includes PTP messages, is rearranged based on the location of a fault in the ring. This could lead to delay or loss of PTP messages, which in turn could affect a PTP timeReceiver's ability to retain a good PTP time.

The Ethernet ring protection mechanisms defined in ITU-T Recommendation G.8032/Y.1344 [B31] use a mechanism called the ring protection link (RPL) to introduce a break in the ring by blocking frames. This break prevents the formation of loops in the ring, and it determines which direction around the ring a frame takes to get to its destination. When a failure introduces another break in the ring, the nodes on both sides of this break are reconfigured to block frames and the RPL is reconfigured so it no longer blocks frames, which opens a new path for frames to get around the ring.

The ring network shown in Figure J-6 provides two dependent inputs and one independent input to the FTT entities at every Bridge in the ring. This is possible because the break in the ring (due to the RPL) does not require any Bridge to receive all three domains from the same ring direction. For simplicity, the end stations that connect to these Bridges and the fault-tolerance of their respective timing functions are not considered in this example.

The same ring network is shown in Figure J-7 with a repositioned break in the ring, resulting from a fault. With this repositioned break, this ring network can provide two dependent inputs and one independent input to the FTT entity in only four of the six Bridges. The other two Bridges are provided with three dependent inputs (they all share at least one common Bridge on the ring) to their FTT entity. As a result, these two Bridges cannot achieve time integrity (see 20.1.3).

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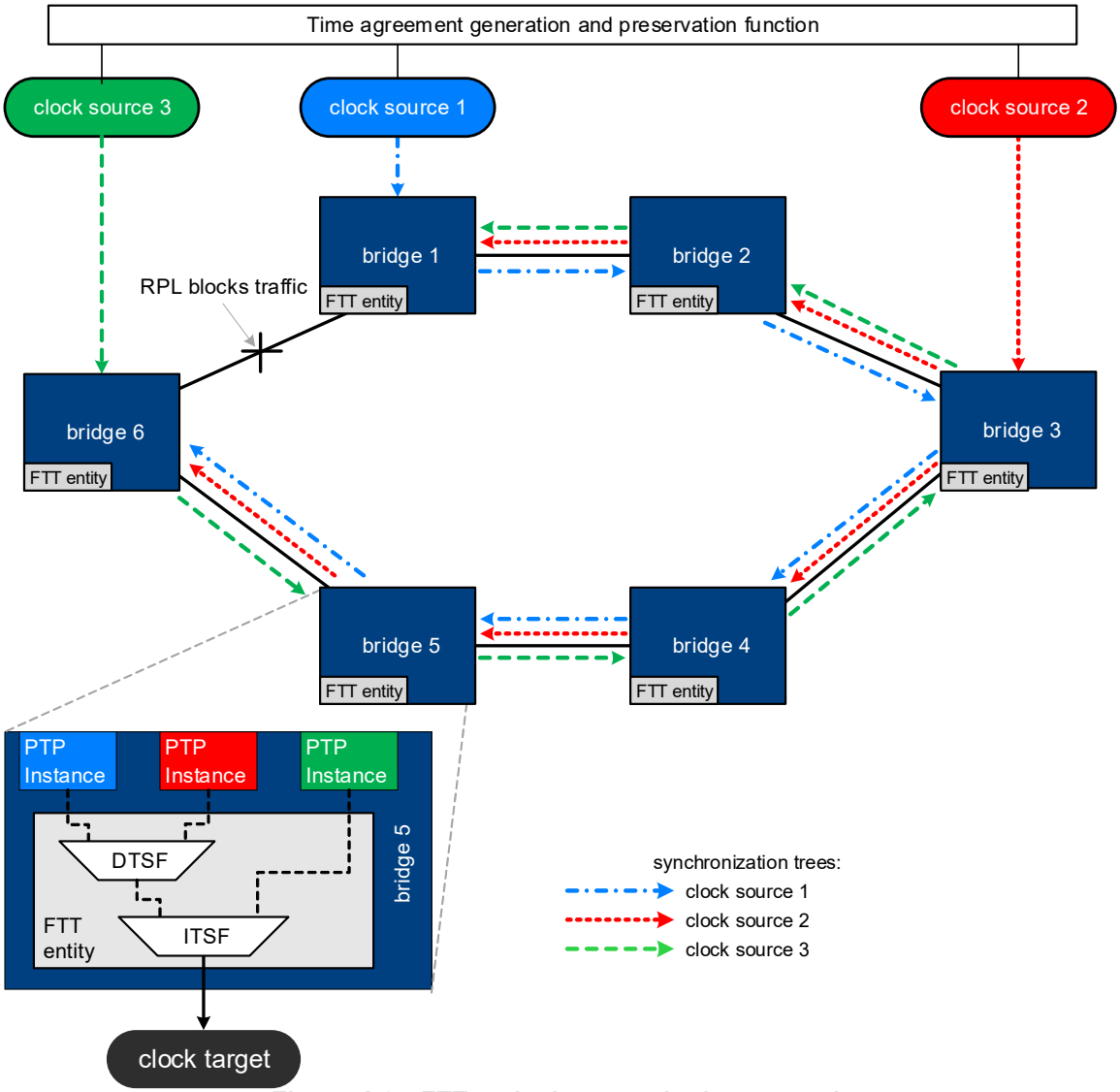


Figure J-6—FTT entity in example ring network

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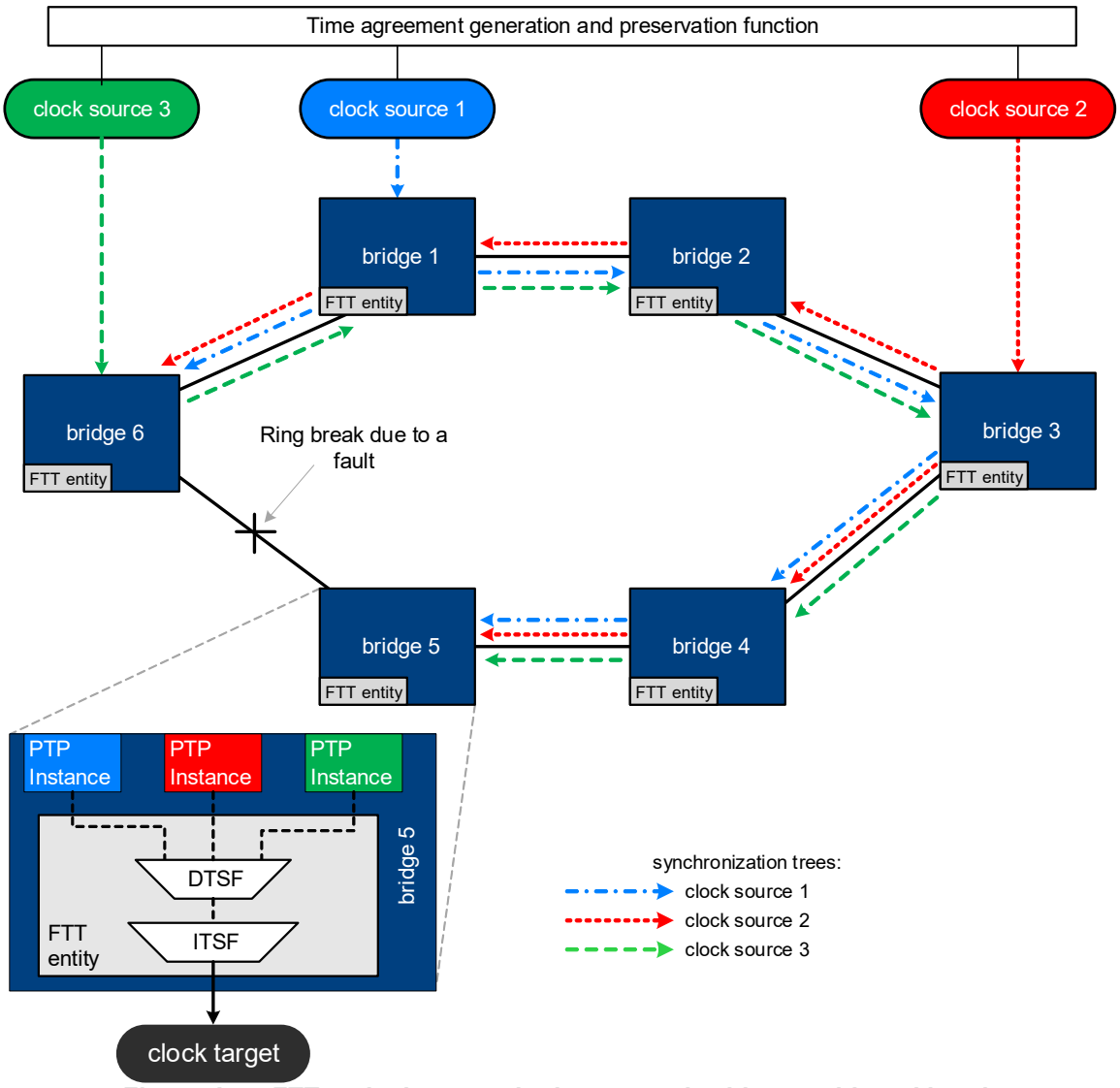


Figure J-7—FTT entity in example ring network with repositioned break

J.2.5 Mesh networks

A mesh network topology connects every network element directly to every other network element. The failure of one element does not adversely affect the ability of another working element to communicate with any other working element in the mesh.

Figure J-8 shows a mesh network of Bridges with three GMs. For simplicity, the end stations that connect to these Bridges are not shown.

Because every Bridge has a direct connection to each GM, its FTT entity can receive an independent input from each of the three GMs (this is shown for only one Bridge in Figure J-8). Thus, this network topology would be suitable for fail-operational applications that can continue running when there is up to one fault in the time integrity function.

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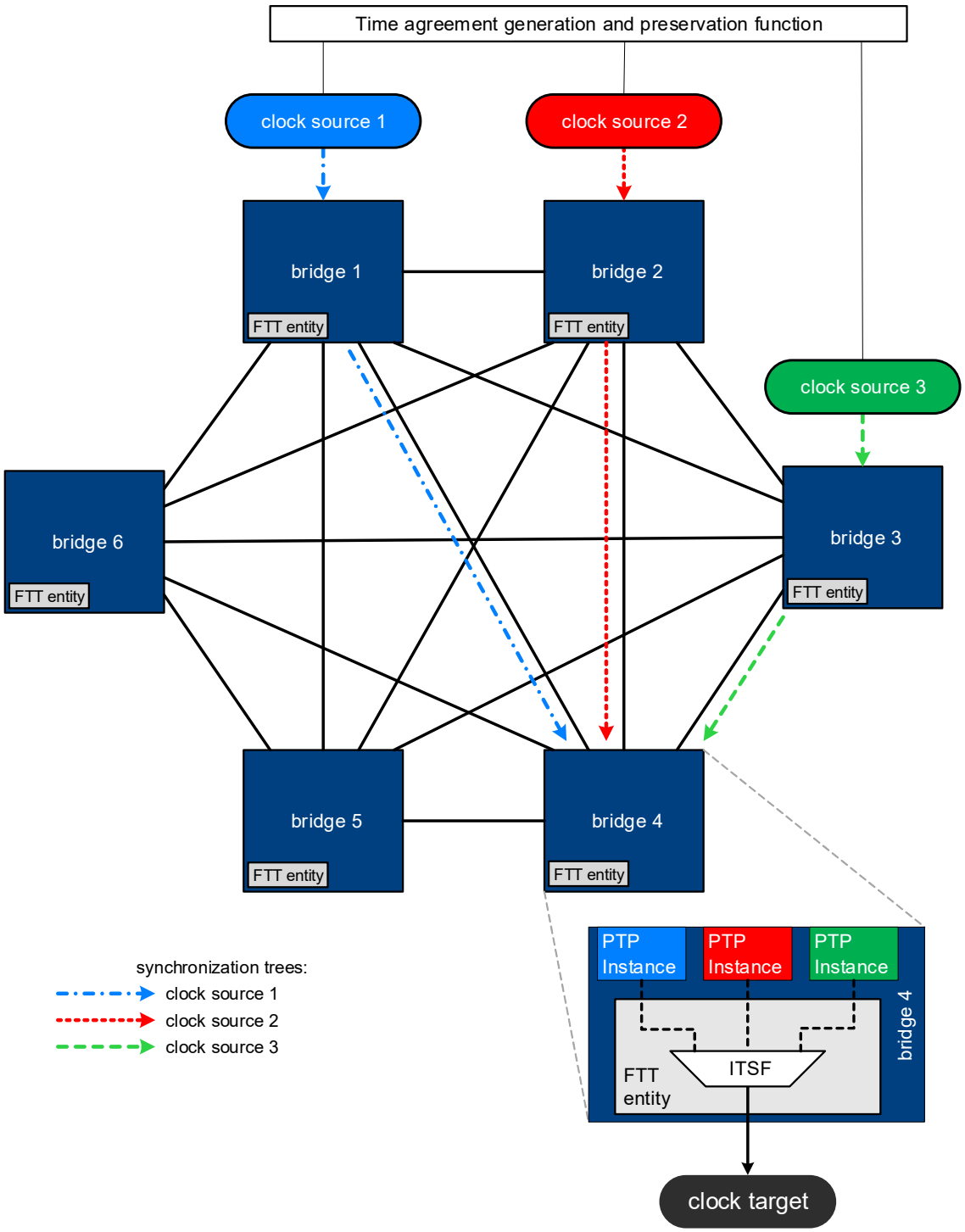


Figure J-8—FTT entity in example mesh network

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Annex K

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[50] Available at (https://www.iers.org/SharedDocs/Publikationen/EN/IERS/Publications/tn/TechnNote36/tn36.pdf).
[51] (https://www.usno.navy.mil/USNO).

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